SHOULDER & ELBOW — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations · 60 of 60 topics –

#	Chapter	Section index
1	AC Arthritis 4	1 / 60
2	Acromioclavicular Joint 9	2 / 60
3	Acromioclavicular Joint Injury 13	3 / 60
4	Adhesive Capsulitis (Frozen Shoulder) 23	4 / 60
5	Avascular Necrosis of the Shoulder 33	5 / 60
6	Biceps Subluxation 39	6 / 60
7	Biceps Tendonitis 44	7 / 60
8	Brachial Neuritis (Parsonage-Turner Syndrome) 50	8 / 60
9	Calcific Tendonitis 57	9 / 60
10	Distal Biceps Avulsion 63	10 / 60
11	Distal Clavicle Osteolysis 75	11 / 60
12	Elbow Anatomy & Biomechanics 79	12 / 60
13	Elbow Arthritis 86	13 / 60
14	Elbow Arthroscopy: Indications & Approach 94	14 / 60
15	Elbow Physical Exam 101	15 / 60
16	Elbow Stiffness and Contractures 109	16 / 60
17	Glenohumeral Arthritis (Shoulder Arthritis) 115	17 / 60
18	Glenohumeral Internal Rotation Deficit (GIRD) 123	18 / 60
19	Glenohumeral Joint Anatomy, Stabilizer, and Biomechanics 132	19 / 60
20	Internal Impingement 141	20 / 60
21	Lateral Epicondylitis (Tennis Elbow) 148	21 / 60
22	Lateral Ulnar Collateral Ligament Injury (PLRI) 156	22 / 60
23	Little League Elbow 164	23 / 60
24	Little Leaguer's Shoulder 170	24 / 60
25	Luxatio Erecta (Inferior Glenohumeral Joint Dislocation) 177	25 / 60
26	Medial Epicondylitis (Golfer's Elbow) 181	26 / 60
27	Medial Ulnar Collateral Ligament Injury 186	27 / 60

Shoulder & Elbow

#	Chapter	Section index
28	Multidirectional Shoulder Instability (MDI) 197	28 / 60
29	Neuropathic (Charcot) Joint of Shoulder 206	29 / 60
30	Olecranon Stress Fracture 212	30 / 60
31	Os Acromiale 216	31 / 60
32	Osteochondritis Dissecans of Elbow 221	32 / 60
33	Pectoralis Major Rupture 227	33 / 60
34	Phases of Throwing 234	34 / 60
35	Posterior Labral Tear 240	35 / 60
36	Posterior Shoulder Instability & Dislocation 246	36 / 60
37	Quadrilateral Space Syndrome 256	37 / 60
38	Reverse Shoulder Arthroplasty 262	38 / 60
39	Revision Total Elbow Arthroplasty 279	39 / 60
40	Rotator Cuff Arthropathy 286	40 / 60
41	Rotator Cuff Tears 295	41 / 60
42	Scapular Winging 319	42 / 60
43	Scapulothoracic Dyskinesis 327	43 / 60
44	Shoulder Arthrodesis 330	44 / 60
45	Shoulder Arthroscopy: Indications & Approach 333	45 / 60
46	Shoulder Exam 339	46 / 60
47	Shoulder Hemiarthroplasty 347	47 / 60
48	Shoulder Imaging 353	48 / 60
49	Shoulder Periprosthetic Fracture 358	49 / 60
50	Shoulder Prosthetic Joint Infection 364	50 / 60
51	SLAP Lesion 372	51 / 60
52	Sternoclavicular Joint 380	52 / 60
53	Subacromial Impingement 382	53 / 60
54	Suprascapular Neuropathy 389	54 / 60
55	Thoracic Outlet Syndrome 396	55 / 60
56	Total Elbow Arthroplasty 404	56 / 60
57	Total Shoulder Arthroplasty 414	57 / 60
58	Traumatic Anterior Shoulder Instability (TUBS) 432	58 / 60
59	Triceps Rupture 459	59 / 60
60	Valgus Extension Overload (Pitcher's Elbow) 464	60 / 60

Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer

– – Active space – –

AC ARTHRITIS

– Shoulder & Elbow · 1 / 60 –

1 Basics

Epidemiology

Frequency	**most common cause of AC joint pain**
Demographics	↑ with age; can start by **2nd decade**
Classic patient	**weight-lifters** + overhead throwing athletes (constant heavy overhead activity)
Risk factors	trauma · post-traumatic (**clavicle fracture, AC instability**) · **distal clavicle osteolysis** · inflammatory (RA) · post-infectious (septic arthritis)

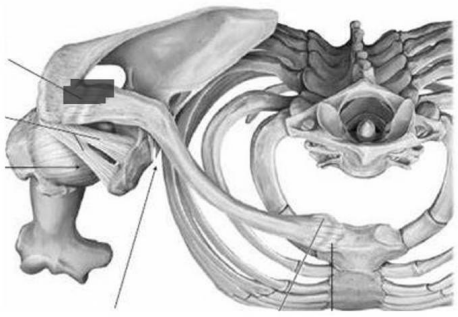
Pathophysiology

Mechanism	**large axial loads through a small contact area → repetitive microtrauma** → OA (**same mechanism as distal clavicle osteolysis**)
Aggravators	articular disc degeneration + oblique joint surface orientation → ↑ contact forces

Anatomy – AC joint & ligaments

AC joint **diarthrodial**; scapula ↔ clavicle; **fibrocartilaginous intra-articular disc** (analogous to knee meniscus); small surface, high axial + rotational loads

Ligament	Stability provided	Injury →
AC ligaments	**Anterior–posterior (horizontal)**	AP instability
Posterior + superior AC ligaments	**Most important** AC stabilisers	**Posterior instability** of distal clavicle
Anterior + superior AC ligaments	Anterior	**Anterior instability** (one OB explanation) · **no instability** (another OB explanation) – read the options
CC ligaments (conoid + trapezoid)	**Superior–inferior (vertical)**	**Superior instability**



Axial view – AC joint ligaments; posterior + superior AC ligaments highlighted

Not AC stabilisers	**coracohumeral ligament** (rotator interval) · **coracoacromial ligament** (anterior acromion; released in subacromial decompression) · deltotrapezial fascia (only elevated in open DCE)
Rare instabilities	**inferior** (CC ligaments resist superior displacement) · **rotational** (rotation is normal baseline load)

3 Presentation

Symptoms	activity-related **superior shoulder pain**; overhead activity, **pressing (bench press, push-up)**, lying on affected side
Palpation	**direct AC joint tenderness**; prominent distal clavicle (osteophytes)
Provocative	**cross-body adduction test** (most classic) · **O'Brien active compression** (90° forward flexion) → AC pain



Cross-body adduction test

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray – Zanca view (15° cephalic tilt; 10–15° in TO/OB)	**Best view**; osteophytes, **joint space narrowing**, **distal clavicle osteolysis**
X-ray caveat	Changes often present **without symptoms** (poor clinical correlation)
MRI	↑ signal + **oedema** in AC joint; shows associated cuff / LHB pathology

– – Active space – –



Zanca view – AC osteophytes and joint space narrowing

Diagnostic injection

AC joint injection (corticosteroid / local anaesthetic, e.g. bupivacaine)	**diagnostic + therapeutic**; relief confirms AC source
Accuracy	joint hard to access, **often missed → ultrasound ↑ accuracy**; most get **no long-term relief**

5 Management

- **Symptomatic AC arthritis**
  - First line → **activity modification (avoid pressing) + PT (shoulder girdle) + NSAIDs**
  - Persistent → **corticosteroid injection (US-guided)**
  - Failed nonop, severe symptoms → **distal clavicle excision (Mumford)**
    - Arthroscopic → **resect 0.5–1 cm, evaluate + treat GH pathology**
    - Open → **meticulous deltotrapezial fascia repair**

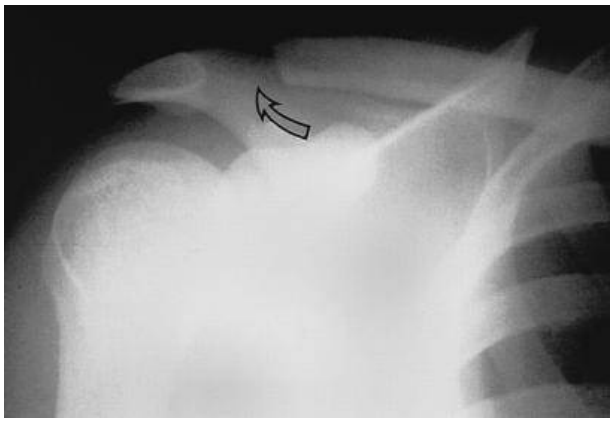
Distal clavicle excision – preserving posterosuperior capsule

Mumford procedure distal clavicle resection, arthroscopic (burr) or open; indication = **severe symptoms, failed nonop**

Resection amount	Source
0.5–1.0 cm (lateral clavicle)	Orthobullets, OB
4–8 mm may suffice arthroscopically (preserves superior AC ligaments → less bone than open)	OB (Nuber)
>1–1.5 cm = aggressive → **instability**	Orthobullets, OB

Structure to preserve	**posterosuperior joint capsule / posterior + superior AC ligaments** → prevents **AP (posterior) instability**
Over-resection	AC or CC ligament damage → instability; even correct length + posterosuperior damage → AP instability

Under-resection persistent pain



Arthroscopic distal clavicle excision

Arthroscopic vs open distal clavicle excision

Outcomes	similar; choice = surgeon preference
Only proven arthroscopic advantage	**evaluation of the glenohumeral joint** + treat associated pathology (**rotator cuff, LHB, labrum / SLAP**)
Missed pathology with open	**15 missed SLAP lesions** in failed open DCE (Berg) → do **diagnostic arthroscopy even with open DCE**
Not proven arthroscopic advantages	lower complication rate · lower infection rate · ↓ surgical time · preservation of CC or inferior AC ligaments
Other arthroscopic pluses (OB)	direct view of clavicle end, less soft-tissue violation, shorter rehab, quicker return
Open DCE	requires **meticulous deltotrapezial fascia repair**

6 Complications

Complication	Cause
AP instability	**Over-resection >1–1.5 cm** · sacrificing **posterior + superior AC ligaments**
Superior–inferior instability	Iatrogenic; resection compromising **CC ligaments**
Persistent pain	**Incomplete resection** – most commonly **posterior-superior distal clavicle**
Heterotopic ossification	
Stiffness, infection	Causes of failed DCE (OB)
Deltoid dehiscence	Inadequate **deltotrapezial fascia repair** after **open** DCE

Failed DCE management conservative, revision, or **ligament reconstruction**

– – Active space – –

X vs Y

	AC ligaments	CC ligaments
Components	Superior, posterior, anterior, inferior	**Conoid + trapezoid**
Plane	**Anterior–posterior**	**Superior–inferior**
Injured by	Aggressive DCE / debridement	Over-aggressive resection (iatrogenic)
Instability	**Posterior** (post + sup injured)	**Superior**

	Arthroscopic DCE	Open DCE
GH evaluation	**Yes**	No (add scope)
Key closure step	–	**Deltotrapezial fascia repair**
Specific risk	Under-resection (visualisation)	**Deltoid dehiscence**, missed SLAP

How it's asked

"Weightlifter, superior shoulder pain with bench press — best X-ray view?"	→ **Zanca (15° cephalic tilt)**
"Weightlifter, persistent popping after arthroscopic DCE, posterosuperior ligaments damaged — instability direction?"	→ **posterior** ✗ Wrong options: anterior · inferior · rotational · superior = CC ligaments
"Structure most important to preserve during arthroscopic DCE?"	→ **posterosuperior joint capsule** ✗ Wrong options: coracohumeral · coracoacromial · anterosuperior capsule · deltotrapezial insertion
"Only documented benefit of arthroscopic over open DCE?"	→ **ability to evaluate the glenohumeral joint** ✗ Wrong options: lower complications · ↓ surgical time · CC / inferior AC ligament preservation
"Persistent pain after DCE — commonest cause?"	→ **incomplete resection of posterior-superior distal clavicle**
"Deltoid weakness after open DCE?"	→ **inadequate deltotrapezial fascia repair**
"How much clavicle to resect arthroscopically?"	→ **0.5–1 cm** (4–8 mm may suffice; >1–1.5 cm → instability)

ACROMIOCLAVICULAR JOINT

– Shoulder & Elbow · 2 / 60 –

1 Basics

Osteology & articular disc

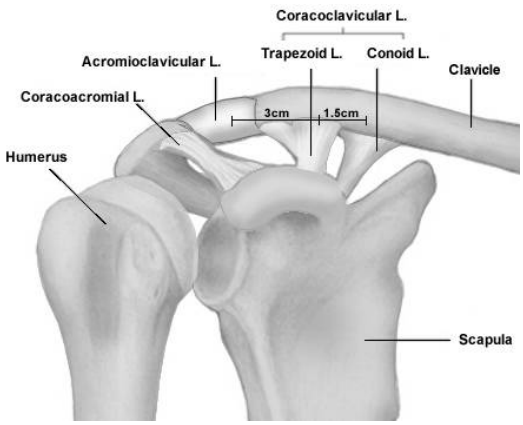
Joint type	**diarthrodial**; capsule + intra-articular synovium + articular cartilage interface
Intra-articular disc	**fibrocartilaginous**; usually **meniscoid-like** · complete disc **very rare** (TO); variable size/shape
Cartilage (TO)	hyaline → **fibrocartilage**: **acromial side by age 17** · **clavicular side by age 24**

Motion

Clavicle	rotates **40–50° posteriorly** with shoulder elevation
Through AC joint	only **8°** of rotation; remainder = **scapular rotation + SC joint motion**
Joint itself	**gliding motions only**; most motion from the bones, not through the joint

Stability – AC vs CC ligaments

Restraint	Direction	Key facts
AC ligaments / capsule (superior, inferior, anterior, posterior)	**Horizontal (AP)**	**Superior strongest**, then **posterior**; constrain posterior clavicular rotation; small-displacement vertical stability (TO)
Trapezoid (CC; **anterolateral**)	Horizontal + vertical loads	Inserts **3 cm** from clavicle end; **>3× thicker at clavicular end** (TO); origin **posterior half of coracoid dorsum** (TO)
Conoid (CC; **posteromedial**)	**Vertical**	Inserts **4.5 cm** from clavicle end, **posterior border**; **more important** CC ligament; clavicular insertion **~2× wider/ thicker** than coracoid → **inverted cone** (TO); origin more posterior on **coracoid base** (TO)
Deltotrapezial fascia, capsule, deltoid, trapezius	Additional	Secondary stabilisers



Coracoclavicular ligaments – trapezoid and conoid

– – Active space – –

Normal CC distance	**11–13 mm** (superior coracoid → inferior clavicle)

Biomechanics of the AC ligaments (TO)

AP vs vertical (TO)	AC capsule + ligaments give **3× more control in AP** than vertical direction
Thickness (TO)	**1–5 mm**; inner osseous insertion **3–4 mm from joint line**, peripheral fibres farther
Superior ligament	**most robust**; principal horizontal stabiliser; **~50%** of AP strength (OB); **thickest posteriorly**
Posterior ligament	adds **~25%** of AP strength (OB)
Two-bundle model (TO)	**superoposterior bundle** resists **posterior** clavicular translation · thin **anteroinferior bundle** limits **anterior** displacement

5 Management

Distal clavicle excision – posterosuperior AC ligaments

Mumford procedure	(arthroscopic) distal clavicle resection for **AC arthrosis**; resection length **debated**
Must preserve	**superior + posterior AC ligaments / capsule** → prevents post-op **AP (posterior) instability**
Over-resection sign	popping + pain + **mild posterior instability**, no deformity on X-ray (e.g. 8 mm excised in weightlifter) → **posterior + superior AC ligaments compromised**
✗ Wrong option: anterior + superior AC	opposite answer; these resist **anterior** translation
✗ Conoid / trapezoid	**vertical** restraints; injured in **grade III or higher** separations → superior clavicular migration
✗ Wrong option: coracohumeral ligament	not an AC stabiliser

AC separation – reduction & stabilisation

AC separation	Feature	Treatment
Type III	–	**Controversial**
Type V	Clavicle elevated **100–300%** + **extensive soft-tissue stripping**	**Surgical AC reduction + stabilisation**

– – Active space – –



Type V AC separation – marked clavicular prominence

✗ Type V distractors sling · NSAIDs + rapid return · arthroscopic partial claviculectomy · acromionectomy

X vs Y

	AC ligaments	CC ligaments (conoid + trapezoid)
Plane	**Horizontal / AP**	**Vertical**
Strongest component	**Superior** (then posterior)	**Conoid** (more important)
Threatened by	**Distal clavicle excision** (too much)	**Grade ≥III** AC separation
Failure →	**Posterior instability**, popping	**Superior clavicle migration**

	Trapezoid	Conoid
Position	**Anterolateral**	**Posteromedial**
Distance from clavicle end	**3 cm**	**4.5 cm**
Shape (TO)	>3× thicker at clavicular end	**Inverted cone**
Role	Horizontal + vertical	**Vertical** (primary)

How it's asked

"Structure responsible for AP stability of AC joint?"	→ **posterosuperior part of capsule** (TO) ✗ Wrong options: anteroinferior capsule · trapezoid · conoid
"Arthroscopic distal clavicle excision – preserve what to prevent AP instability?"	→ **superior + posterior AC capsule** ✗ Wrong options: trapezoid · conoid · coracohumeral
"Popping + posterior clavicle instability after Mumford – what was compromised?"	→ **posterior + superior AC ligaments** ✗ Wrong options: anterior + superior
"Quarterback, clavicle tenting under skin (type V) – management?"	→ **AC reduction + stabilisation** ✗ Wrong options: sling · NSAIDs · claviculectomy · acromionectomy
"Primary vertical stabiliser of AC joint?"	→ **conoid ligament**

– – Active space – –

"Strongest AC ligament?"	→ **superior** (~50% AP strength), then posterior (~25%)

– – Active space – –

ACROMIOCLAVICULAR JOINT INJURY

– Shoulder & Elbow · 3 / 60 –

1 Basics

Epidemiology

Frequency	**9% of shoulder girdle injuries · 40–50% of shoulder injuries in sports** (OB)
Demographics	♂ **>** ♀, **athletes**, young
AKA	"shoulder separation" = disruption of AC ± CC ligaments

Anatomy

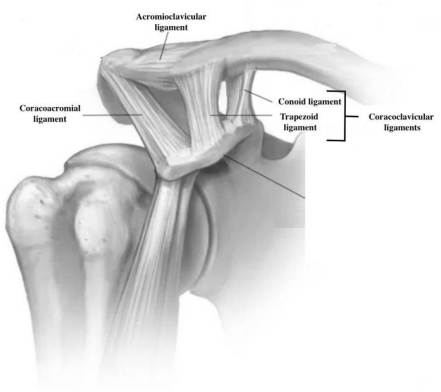
AC joint	**diarthrodial**; medial acromion + lateral clavicle; oblique surface
Intra-articular disc	**fibrocartilaginous** (≈ knee meniscus); involutes, **gone by age 40**
Motion	mainly **gliding**; clavicle rotates **40–50° posteriorly** with elevation but only **~8° through AC joint** (synchronous scapuloclavicular motion)
Dynamic stabilisers	**anterior deltoid + trapezius** (deltotrapezial fascia)

AC vs CC ligament function (horizontal vs vertical stability)

Ligament	Controls	Key fact
AC ligaments (sup, inf, ant, post)	**Horizontal = anterior–posterior** stability	**Posterior + superior (posterosuperior) most important** → preserve in distal clavicle resection
Conoid (CC)	**Vertical = superior–inferior**	**Most important for vertical stability**; **conoid tubercle, posteromedial**; inserts **4.5 cm** medial to lateral clavicle edge
Trapezoid (CC)	Vertical	**Trapezoid tubercle, anterolateral**; inserts **3 cm** medial to lateral edge
Coracoacromial (CA)	–	**Spared** in AC injuries

✗ Ligament traps ✗ Wrong options: anterosuperior AC = vertical · conoid = horizontal · **inferior AC** = no significant horizontal role · trapezoid "most posteromedial" (that is conoid) · **anterior + inferior AC** most important · trapezoid at 4.5 cm

– – Active space – –



AC, CA and CC (conoid, trapezoid) ligaments

Normal CC distance **11–13 mm** (top of coracoid → undersurface of clavicle)

2 Mechanism

Mechanism **direct blow to the shoulder** (fall onto **adducted** shoulder); bike falls, contact sport

3 Presentation

Symptoms pain over AC joint ± referred to **trapezius**; visible step-off

Exam AC/lateral clavicle tenderness; abnormal contour vs other side

Stability testing **horizontal (AP) = AC ligaments** → cross-body adduction; **vertical (sup–inf) = CC ligaments**

Horizontal instability (ISAKOS IIIB) may need more aggressive treatment

Type IV clue **skin tenting + posterior fullness**

Type V clue **severe shoulder droop, not corrected by shrug**

Type VI clue rare, associated injuries, **paraesthesias**

Provocative tests

Test	Technique	Positive =
O'Brien (active compression)	**Resisted flexion at 90° flexion, 10° adduction, full IR**	**Superficial pain → AC joint · deep pain → SLAP**
Cross-body adduction	Arm adducted across chest	AC joint pain
✗ Wrong option: Jobe / empty can	Resisted abduction 90° scapular plane, pronated	Supraspinatus tear
✗ Wrong option: apprehension	Supine abduction + ER	Anterior instability
✗ Wrong option: belly press	Palm on abdomen, resisted IR	Subscapularis tear
✗ Wrong option: Neer	Passive forward elevation in IR	Subacromial impingement

Persistent AC pain after nonop type III (6 wk) — **O'Brien / cross-body adduction** → if pain + instability → **AC + CC reconstruction**

– – Active space – –

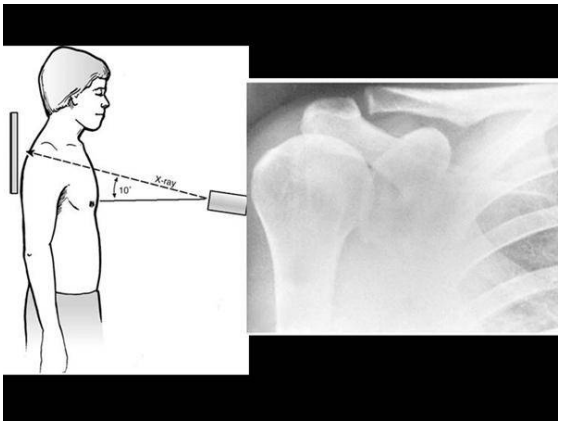
4 Diagnosis & Classification

Key radiology findings

View	Technique	Use
Bilateral AP AC joints	**1/3 penetration** of standard shoulder AP	Compare **CC distance** with contralateral (III vs V)
Axillary lateral	–	**Required to diagnose type IV (posterior)**
Zanca	**10–15° cephalad tilt, 50% penetrance**	**Most accurate view of AC joint**
Cross-body adduction (Basmania)	Scapular Y with cross-body adduction stress	Horizontal stress view
Alexander	Scapular Y centred on AC joint, **15° cranial tilt**	**Posterior displacement** (also superior)
Weighted stress views	–	Rarely used; II vs III
CT	–	Degree/direction of displacement; often unnecessary
✗ Wrong option: **Serendipity**	**40° cephalic tilt**	**SC joint**
✗ Wrong option: Stryker notch · West Point · supraspinatus outlet · Velpeau	–	Not for AC joint

Penetration — AC joint needs **1/3–1/2** of glenohumeral exposure → standard AP **over-penetrates (dark)** AC joint

Zanca tilt — **10–15°** (Orthobullets, OB) · **15°** (one OB explanation)

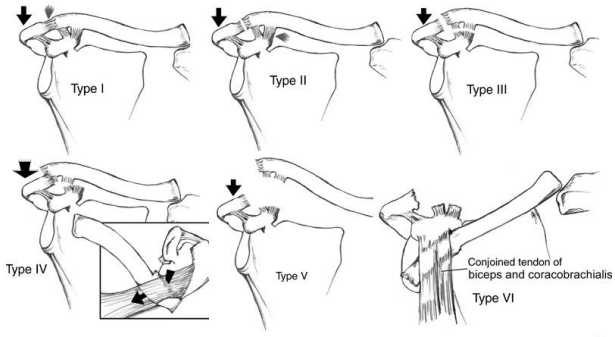


Zanca view technique

– – Active space – –

Rockwood classification & ligament injury pattern

Type	AC lig	CC lig	Exam	Radiograph	Reducible	Treatment
I	Sprain	Normal	Tender, **no instability**	Normal	Yes	Sling
II	**Torn**	**Sprain**	**Horizontal instability**	AC disrupted; **CC ↑ < 25%**	Yes	Sling
III	Torn	**Torn**	–	**CC ↑ 25–100%**	Yes	**Controversial**
IIIA (ISAKOS)	Torn	Torn	**Vertical instability, no horizontal**	CC ↑ 25–100%	Yes	–
IIIB (ISAKOS)	Torn	Torn	**Vertical + horizontal instability**	CC ↑ 25–100%	Yes	**May need more aggressive**
IV	Torn	Torn	**Skin tenting, posterior fullness**	Clavicle **posterior through trapezius** on **axillary**	**No**	Surgery
V	Torn	Torn	**Severe droop**, not improved by shrug	**CC ↑ > 100%**	**No**	Surgery
VI	Torn	Torn	Rare; associated injuries; paraesthesias	**Inferior** – **subacromial or subcoracoid**, tethered by **conjoint tendon**	**No**	Surgery



Rockwood classification types I–VI

Type III CC threshold	**25–100%** (Orthobullets, OB) · **20–100%** (one OB explanation) – read the options
Type I trap	written in one OB explanation as "AC sprain + CC sprain, no instability" (Orthobullets: CC normal)
Types III–VI	**both AC + CC torn**; IV–VI = **irreducible**
Type V irreducible	clavicle **buttonholed through deltotrapezial fascia** (most common culprit)
✗ Interposed-structure traps	conjoint tendon (coracoid tip) · CC ligaments (ruptured, not incarcerated) · CA ligament
Type IV and trapezius	clavicle **pierces trapezial fascia** → muscle innervated by **CN XI (spinal accessory)**

✗ Nerve traps axillary (deltoid, teres minor) · CN VII (face, platysma) · pectoral nerves · suprascapular (SS, IS)

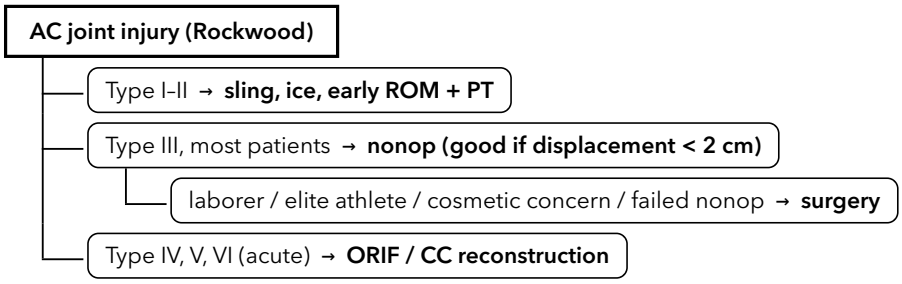
ISAKOS consensus Rockwood misses nuances → added **biomechanical stability (IIIA vs IIIB)**

Differential (mimics)

Mimic	Distinguishing feature
Base of coracoid fracture	Superiorly displaced clavicle but **normal CC distance** (CC ligs intact)
Distal clavicle fracture Neer 2A	Ligaments stay on distal fragment; medial fragment displaces
Distal clavicle fracture IIB (OB)	Fracture **between conoid + trapezoid → only conoid** disrupted; distal fragment CC distance intact
Paediatric distal / medial clavicle physeal injury	Physis

AC separation vs IIB clavicle fracture AC separation = **conoid + trapezoid** both injured (CC ↑ 25–100%)

5 Management



Low-grade injuries – sling & early motion

Nonop regimen brief sling (**2–4 wk** for I–III, OB), ice, activity modification, **early ROM**, PT; ± steroid injection

Timeline (nonop) functional motion by **6 wk** · normal activity at **12 wk**

20% ↑ CC distance = **type II → sling + early PT**
✗ Wrong options: CC reconstruction · AC capsule repair · retrograde pinning · distal clavicle excision

58-yo accountant, non-dominant, II vs III, no AP translation **sling + early ROM**
✗ Wrong options: Weaver-Dunn · CC recon

Nonop complications AC arthritis, chronic subluxation/instability

Operative criteria

Indication	Detail
Acute type IV, V, VI	Recent studies: no functional difference op vs nonop even for high grade (IV–V may be treated nonop, OB)
Acute type III	**Laborers, elite athletes, cosmetic concerns**
Chronic type III	**Failed nonop**
Contraindications	Poor rehab compliance; skin problems over approach

– – Active space – –

Timing acute stabilisation usually **within first 2 wk** (OB) · surgery **after 6 wk failed nonop = same outcome as immediate** (Orthobullets)

Old teaching acute → ORIF, chronic → CC reconstruction (no longer supported)

Type III injuries – operative vs nonoperative

Outcome	Operative	Nonoperative
Complications	**Higher** (hardware commonest)	Lower
Return to work/sport	**Slower**	**Faster**
Early function (6 wk, 3 mo)	Worse	**Better**
Function / PROs / QoL at 1 yr	**Equal**	**Equal**
Return to athletics at 1 yr	Equal	Equal
Cosmesis	**Better** (only consistent benefit)	Poorer
Radiographic alignment	**Better** (hook plate, COTS RCT)	Worse
Disability at 2 yr	**Both improve significantly**	**Both improve significantly**
ROM / strength / endurance	No difference	No difference
Secondary surgery	Higher (hardware removal)	Lower
Cost	Higher	Lower

Early-score wording nonop "higher DASH at 6 wk & 3 mo" (Orthobullets; ↑ DASH = worse) · nonop **better** early scores/function at 6 wk (OB) — read the options

Spencer systematic review surgery **not better**, **higher complication rate**, **longer recovery**

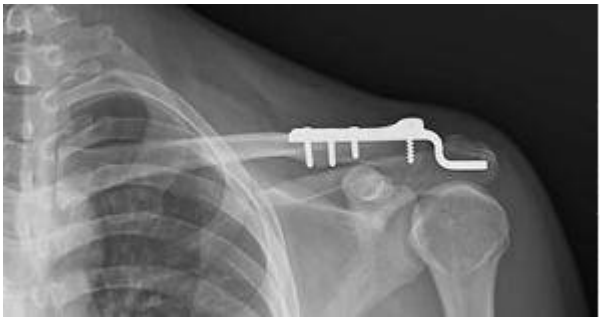
Canadian (COTS) RCT hook plate: **better radiographic alignment**, **not clinically superior**; nonop better early scores

Surgical techniques

– – Active space – –

Technique	Pros	Cons / complications
CC screw (Bosworth) – clavicle → coracoid, sup → inf	Rigid	Out of favour; screw too long → critical structures; **routine removal 8–12 wk** (breakage from clavicle–scapula motion); **failure at coracoid thread purchase**
CC suture fixation – around/through clavicle + around coracoid base (or anchors)	**No hardware failure/migration**	Weaker than screw; **careful passage under coracoid** (NV structures); **suture erosion → distal clavicle fracture**
AC pin (Phemister) – wire across AC joint	Percutaneous	**High pin migration** (catastrophic); rarely done
Hook plate	Rigid	**2nd surgery** for removal; **acromial erosion**, hook pullout
Modified Weaver-Dunn – **distal clavicle excision + CA ligament transfer** to clavicle	–	**CA ligament only 20% as strong** as CC; failure without internal fixation
Free tendon graft CC reconstruction – figure-of-8 around coracoid, clavicle tunnels	**Closest to native CC strength**	Graft morbidity; needs internal fixation backup
Cortical flip button (Dog Bone) ± arthroscopy	–	–

Grafts autograft **palmaris longus, semitendinosus** · allograft **tibialis anterior**



Hook plate fixation

Post-op rehab sling **6 wk, no ROM** · full activity at **6 months**

AC joint cyst (Geyser sign) (TO)

Entity (TO) **synovial cyst of AC joint**; uncommon; elderly; chronic shoulder pain, ↓ ROM, large lump, cosmetic concern

Type (TO)	Cause	Treatment (TO)
Type 1	**AC joint chronic arthropathy**	**Excision + lateral clavicle resection**
Type 2 (**most common**)	**Chronic rotator cuff tear**	Mild/moderate pain + reparable cuff → **distal clavicle resection + cuff repair** · severe pain + irreparable cuff (elderly) → **shoulder arthroplasty (RSA > hemi)**

Mechanism (TO) chronic cuff tear → **superior humeral head migration** → erodes **inferior AC capsule** → fluid escapes into AC joint + subcutaneous tissue

– – Active space – –

Geyser sign (TO) fluid jet through AC joint on **MRI / arthrogram / USG**; MRI reliably shows cyst type

(✗ Wrong options: **Distractors (TO)** – AC exostosis) · SLAP type 4 · biceps sheath GCT



Geyser sign on MRI

6 Complications

Complication	Fact
Residual AC pain	**30–50%**
AC arthritis	**More common after surgery** than nonop (nonop: no articular contact → no arthritic symptoms; surgery: cartilage injury + incongruity)
Hardware failure	CC screw breakage/pullout **at coracoid**
Pin migration	AC pinning (Phemister)
Coracoid fracture	**Coracoid tunnel drilling**
Distal clavicle fracture	Suture erosion
Acromial erosion	Hook plate
Neurological injury	**Musculocutaneous nerve** closest to coracoid (✗ Wrong options: axillary (runs at inferior subscapularis, distant))

Surgical complications (Weaver-Dunn AP translation, AC arthritis)

Failed modified Weaver-Dunn persistent symptoms from **anterior–posterior (horizontal) translation**
✗ Wrong options: sup–inf translation · inadequate resection · over-reduction · SC instability

Prevention **anatomic** repair/reconstruction **preserving distal clavicle** → less horizontal instability

CC screw statement traps ✗ Wrong options: "more migration than AC pinning" · "fails at clavicle" (coracoid) · "avoid removal" (remove at 8–12 wk) · "axillary nerve"

X vs Y

	Type III	Type V
CC distance ↑	**25–100%**	**> 100%**
Reducible	Yes	**No** (deltotrapezial fascia)
Shrug	Corrects	**Does not correct droop**
Treatment	Controversial / mostly nonop	Surgery

	Type IV	Type VI
Direction	**Posterior** (through trapezius)	**Inferior** (subacromial/subcoracoid)
View	**Axillary lateral** (Alexander)	AP
Clue	Posterior fullness, skin tenting	Rare, paraesthesias, associated injuries

	Zanca	Serendipity
Joint	**AC**	**SC**
Tilt	**10–15° cephalad**	**40° cephalic**

How it's asked

"Type V on X-ray – ligaments torn?"	→ **AC + CC** ✗ Wrong options: SC · CA
"CC distance 13 mm contralateral, type III – which labelled ligaments torn?"	→ **AC + conoid + trapezoid** ✗ Wrong options: CA ligament (spared)
"Type II AC injury?"	→ **AC torn, CC sprain, horizontal instability** ✗ Wrong options: no instability = type I · CC torn = III
"IIB distal clavicle fracture vs AC separation?"	→ **fracture = conoid only; AC separation = conoid + trapezoid**
"Posterior clavicle displacement after fall over handlebars?"	→ **type IV**
"Type V, irreducible, skin tenting – interposed structure?"	→ **deltotrapezial fascia** ✗ Wrong options: conjoint · CC · CA
"Type IV, posterior fullness – nerve to muscle involved?"	→ **CN XI** (trapezius)
"Most important for horizontal stability?"	→ **posterosuperior AC ligament**
"Most important for vertical stability?"	→ **conoid**
"Trapezoid inserts how far from lateral clavicle?"	→ **3 cm** (conoid 4.5 cm)
"Best extra view for AC joint?"	→ **Zanca** ✗ Wrong options: Stryker · West Point · outlet · Velpeau · Serendipity = SC

– – Active space – –

"Type III – surgery vs nonop?"	→ **higher complication rate** with surgery
"Type III non-athlete – only benefit of fixation?"	→ **improved cosmesis**
"Type III, 19-yo wants surgery – true?"	→ **surgery ↑ complications + nonop faster return** ✗ Wrong options: better function
"Type III nonop vs op?"	→ **nonop better function at 6 wk**; equal at 1 yr
"Hook plate RCT – true?"	→ **both groups improve significantly by 2 yr** ✗ Wrong options: op faster return · op worse cosmesis · equal alignment
"Type III nonop, still painful at 6 wk – exam?"	→ **O'Brien: resisted flexion at 90° flexion, 10° adduction, full IR**
"20% ↑ CC distance, symptomatic – treatment?"	→ **sling + early PT**
"58-yo accountant, non-dominant AC separation, no AP translation?"	→ **sling + early ROM**
"Persistent symptoms after modified Weaver-Dunn?"	→ **anterior–posterior translation**
"CC screw fixation – true post-op?"	→ **AC arthritis more likely than nonop**
"Elderly, large shoulder lump, MRI Geyser sign?"	→ **AC joint cyst** (TO)
"Commonest AC cyst type?"	→ **type 2 – chronic cuff tear** (TO)
"Geyser sign, severe pain, irreparable cuff, elderly?"	→ **arthroplasty (RSA > hemi)** (TO)

ADHESIVE CAPSULITIS (FROZEN SHOULDER)

– Shoulder & Elbow · 4 / 60 –

1 Basics

Epidemiology

Prevalence (TO)	**2–5%** of population · incidence in **diabetics 10–36%**
Demographics	♀ **>** ♂, **40–60 y**; **nondominant** arm most often; **minorities > Caucasians** ✗ Wrong options: Caucasian ↑ risk
Bilateral	**age <50** and **diabetes** → ↑ risk of bilateral disease; **sequential, not simultaneous**

Risk factors & associations (diabetes, thyroid disease)

Risk factors (TO)	**female**, **age >40**, **preceding trauma**, **HLA-B27 positivity**, **prolonged immobilisation** of GH joint
Associations (TO)	**diabetes**, **thyroid disease**, **cerebrovascular disease**, **coronary artery disease**, **autoimmune disease**, **Dupuytren's**
Other associations	atherosclerotic disease, **cervical disc disease**, **history of MI**, **Crohn's**, **breast cancer treatment / axillary dissection**, previous surgery
Diabetes (both types)	**most common association**; stiffness may be **first manifestation of DM** → work up · often **bilateral** · **worse outcomes regardless of treatment** (nonop + surgical)
Insulin-dependent (TO)	**more severe** limitation + **more resistant** to nonop than non-insulin diabetics
Risk ↑ within diabetics	**older age (>50)** · **duration of DM** · **autonomic neuropathy** · **history of MI**
Not risk factors in diabetics	**type of diabetes** (↑ in **both type I and II**) · diabetic regimen (insulin vs non-insulin) · glycaemic control
Thyroid	autoimmune aetiology; **hypothyroidism** (↑ TSH) classic
✗ Distractors	**menopause** (no association) · **steroid use** (steroids treat it) · epilepsy · renal disease · prior dislocations

– – Active space – –

Genes & lab markers

Test	Finding in AC	✗ Distractor meaning
TSH	↑ (hypothyroidism)	↑ **free T4** = hyperthyroidism (wrong option)
HbA1c (hemoglobin A1C)	↑ (diabetes)	–
HLA-B27 (TO)	Positivity = risk factor	–
Uric acid	Normal	↑ = gout
ESR / CRP	Normal	↑ = infection / inflammatory arthropathy
RF / anti-CCP	Normal	↑ = RA
ANA	Normal	+ = lupus

Workup metabolic panel + endocrine labs (**TSH, HbA1c**) before choosing treatment

Pathophysiology – fibroblastic proliferation & reduced capsular volume

Forms **primary/idiopathic** (most) · **post-traumatic** (proximal humerus fracture, immobilisation) · **post-surgical** (RCR, axillary dissection)

Pathoanatomy inflammation (subsynovial layer, TO) → **fibroblastic proliferation** of capsule → thickening, fibrosis, adherence of capsule to itself + humerus → **mechanical block** · **↓ intracapsular volume**

Cell type **fibroblasts / myofibroblasts**
✗ Wrong options: adipocytes · granulocytes · lymphocytes · monocytes

Collagen abundant **type III collagen**

Essential lesion **coracohumeral ligament (CHL)** + **rotator interval capsule**

Why ER lost first (TO) changes start in **subsynovial layer of capsule in rotator interval**

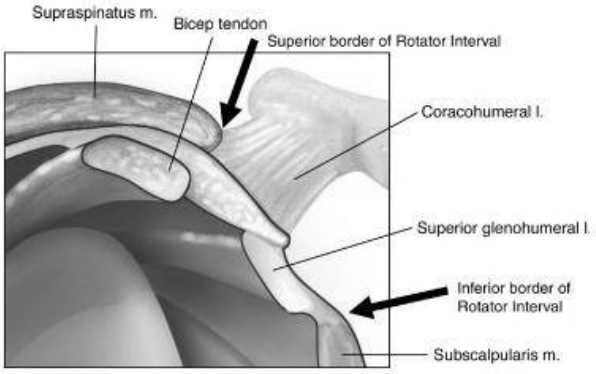
✗ Wrong options (other processes) isolated posterior capsular tightness = **GIRD** · ↓ blood supply = **AVN** · degenerative tendon tear = **cuff tear** (active loss only)

Rotator interval anatomy

Boundary	Structure
Superior	**Anterior edge of supraspinatus**
Inferior	**Superior edge of subscapularis**
Medial (base)	**Coracoid process**
Lateral (apex)	**Transverse humeral ligament**
Contents	**LHB tendon, SGHL, CHL, RI capsule**

✗ RI traps LHB = content · AIGHL + **axillary pouch** lie **inferior** to RI · MGHL is not the inferior border (subscapularis is)
✗ Wrong options: superior border

RI clinical **closed** for instability · **contracted** in AC → **released**

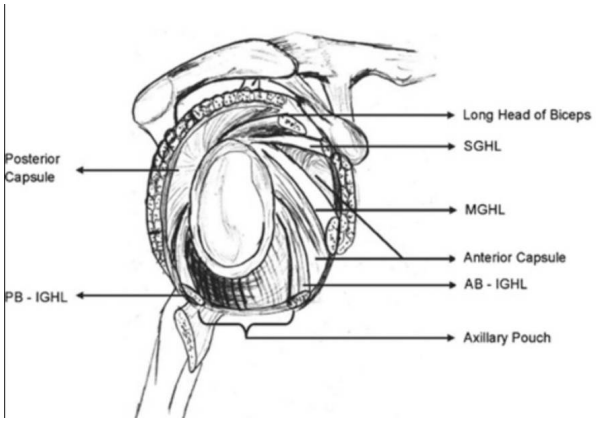


Rotator interval borders and contents

Capsuloligamentous restraints

Structure	Restraint	Release effect
RI (SGHL + CHL + capsule)	**Inferior translation + ER in adduction**	↑ **ER at side** + inferior translation
SGHL	**Inferior translation at 0° abduction**	–
MGHL	**Anterior (± posterior) translation at ~45° abduction** + ER	↑ anterior translation at 45°
AB-IGHL	**Anterior + inferior translation at 90° abduction + max ER**	↑ anteroinferior translation at 90°
Posterior capsule	IR + **cross-body adduction**	↑ **IR + cross-body adduction**

Normal role check reins at extremes of motion; IGHL complex = anterior band + axillary fold + posterior band



Capsular structures from glenoid side

3 Presentation

Clinical diagnosis – loss of active and passive motion

Symptoms **insidious** diffuse pain **preceding** loss of motion · **night pain / difficulty sleeping** · ADL difficulty (reaching overhead)

Exam **symmetric loss of both active AND passive ROM** · **tethered endpoint** · pain throughout arc or at terminal motion (stage-dependent)

– – Active space – –

First movement lost (TO) **external rotation**
✗ Wrong options: IR · flexion · abduction

Most common + most sensitive **passive ER deficit vs contralateral** shoulder

Provocative tests impingement, biceps, SLAP often **positive** (false); cuff testing limited by stiffness

Diagnosis **clinical** (normal X-ray + painful active + passive loss); MRI not necessary

✗ Wrong option: exam pattern	Diagnosis
Active < passive (e.g. active FF 100°, passive 150°)	**Weakness** – cuff tear
Pain 0–30° active abduction, full passive	Supraspinatus tendinosis/tear
Empty can + pain/weakness, full passive	Rotator cuff tear
Speed's +, full ROM	Biceps tendinopathy
Isolated IR loss (posterior capsule)	GIRD

Clinical stages

Stage	Features	Duration
1 Freezing / painful	Gradual diffuse pain at rest + motion, night pain	**6 wk – 9 mo**
2 Frozen / stiff	Pain lessens; ↓ ROM affecting ADLs	**4 – 9 mo or more**
3 Thawing / resolution	Pain gone; gradual return of motion (less than normal)	**5 – 26 mo**

Total duration self-limited; average **3–6 mo**, up to **18 mo – 2 y** (OB) – read the options

4 Diagnosis & Classification

Key radiology findings

X-ray views **AP in neutral rotation**, **scapular-Y**, **axillary lateral** (± AP in IR/ER) · **mandatory** to exclude **OA, posterior dislocation, hardware**

X-ray finding normal or **disuse osteopenia**; no high-riding head

MRI (± arthrography) not needed for diagnosis; excludes other pathology · **loss of axillary recess** (capsular contracture) · thickened CHL

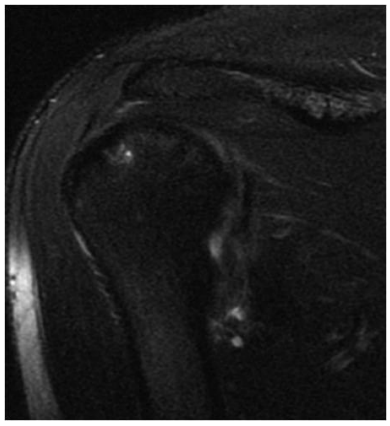
MR arthrogram **↓ intracapsular volume**

MRI sign of AC (TO)	Value
Axillary recess thickening	**≥1.3 cm** (TO)
Anterior capsule thickening	**>3.5 mm** + hyperintensity
Synovial thickness	**>4 mm**
RI	Soft tissue thickening + signal change; scarring
Biceps anchor	Encased by abnormal soft tissue
CHL	**Thickened**
Subcoracoid triangle sign	**Obliteration of fat** in subcoracoid space
Inferior capsular fold	**Obliterated**; capsule hypervascular

NOT seen (TO) **widening** of axillary pouch (pouch is contracted/obliterated)

CHL on MRI thickened (Orthobullets) · **not better visualised** than normal (OB wrong option) — read the options

Incidental partial cuff tear common in **asymptomatic >60 y** → still treat AC with stretching



Coronal MRI, loss of axillary recess

✗ MRI distractor	Diagnosis
Fluid extravasation down humerus	**HAGL**
Hypertrophied synovium, papillary projections	**PVNS**
Massive retracted tear	Cuff tear

Arthroscopic stages

Stage	Finding
1	**Patchy, fibrinous synovitis**
2	**Capsular contraction + fibrinous adhesions**
3	Increasing contraction, **synovitis resolving**
4	**Severe contraction**

5 Management

Adhesive capsulitis

- Early (freezing) → **reassurance + gentle stretching ± IA steroid + NSAIDs**
  (treat diabetes / thyroid)
- Failed nonop (3–6 mo) → **MUA and/or arthroscopic capsular release**
  - diabetic / post cuff or labral repair → **release (MUA 50% fail)**
- Residual IR / cross-body loss → **add posterior capsular release**

Nonoperative management – stretching, injection & natural history

First-line **PT + NSAIDs ± IA steroid injection** ± heat/cryotherapy; often effective, **majority** succeed

– – Active space – –

PT program	**gentle, pain-free stretching + moist heat**; supervised; **3–6 mo**; stress **motion, not strengthening**
Home exercise program	**as helpful as organised therapy**
IA steroid	↓ inflammation → allows stretching; **earlier** return of ROM, **no long-term difference** (OB, TO) · stage II → **IA steroid + stretching**
PT detail (TO)	stretching then strengthening = mainstay **regardless of stage**; **avoid abduction initially** (impingement)

(✗ Wrong options: **Wrong early** – **sling** (↑ contracture, prolonged recovery)) · **immediate aggressive therapy** (counterproductive unless pain controlled) · **PRP** (no evidence) · MUA / arthroscopy before conservative trial · cuff strengthening

Distension arthrography	rarely performed
Natural history	**self-limited**; most resolve without surgery
Most common outcome after nonop stretching	**decreased range of motion vs contralateral shoulder** ✗ Wrong options: limiting pain · recurrence (low) · surgery (rare, **>90% satisfied**)

Stretching exercise mechanics

ER wand exercise (arm adducted, elbow flexed)	stretches **anterior-superior capsule, SGHL, CHL, IGHL** · **LEAST stretch on posterior capsule**
Most commonly contracted	**anterior-superior capsule (RI)** → limits **ER in adduction**

Operative criteria

Factor	Threshold / rule
Failed therapy before release	**3 mo** (Orthobullets) · **3–6 mo** (OB) · **≥6 mo** (OB) · **4–6 mo** compliant therapy (TO) – read the options
Indication	**Progressive loss of motion** despite prolonged PT
MUA contraindication	**Diabetics (50% failure)** · **after RCR or labral repair** · freezing phase (controversial)
Arthroscopy (TO)	Reserve for **recalcitrant** cases (✗ Wrong options: RCT superiority over MUA)

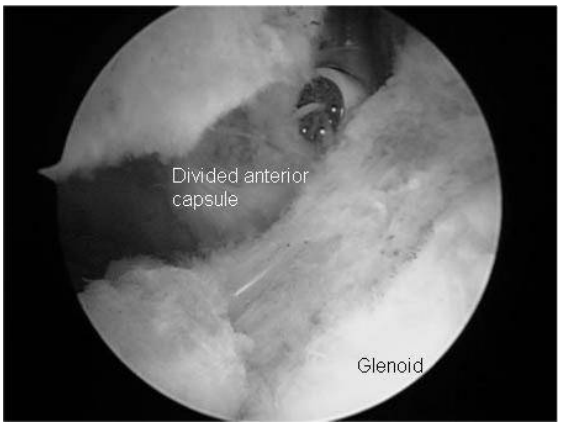
Manipulation under anaesthesia

Technique	**document ROM** first · **full muscle paralysis** · indwelling regional catheter · **steady force** · **early PT**
FEAR sequence (TO)	**F**lexion, **E**xtension, **A**bduction, adduction, external + internal **R**otation
Post-MUA (TO)	**20 ml 0.5% bupivacaine + corticosteroid** IA
Results (TO)	**up to 94%** subjective relief, sustained long term

Arthroscopic capsular release – structures & outcomes

Portals	slightly **higher** than normal (contracted capsule)
Rotator interval release	from **anterior biceps to superior subscapularis** → CHL visualised + released → restores **ER at side**

Released	**SGHL, MGHL, IGHL, CHL** (circumferential) · **not coracoacromial ligament** (= landmark for complete RI release)
Posterior capsular release	↑ **IR + cross-body adduction**
Subacromial	**bursectomy + adhesion release**; **no acromioplasty**
Sequence	MUA before or after release; TO: release before manipulation allows inspection + precise capsulotomy; **anterior release always ± posterior** (posterior: early benefit, similar outcome)
Advantage over open	**spares subscapularis**; releases intra-articular + subacromial adhesions
Not done	completion + repair of small partial tear (that is for **>6 mm or 50%**) · **capsular plication** (instability) · **distal clavicle resection** (AC arthritis)
Outcome	gains in ROM + function **maintained at long-term follow-up**



Arthroscopic anterior capsular release

6 Complications

Complication	Detail / prevention
Residual / recurrent stiffness	**Most common** after release → **immediate ROM + PT**, **no sling**, ± CPM, regional analgesia
Axillary nerve injury	**Inferior capsular release** → stay **close to glenoid rim**; rare
Infection, instability, impingement	Less common
MUA (overzealous, osteoporotic bone)	**Proximal humerus fracture**, dislocation, **cuff + labral tears**, **brachial plexus palsy**

Prognosis	self-limited · **worse in diabetics**

– – Active space – –

X vs Y

	Adhesive capsulitis	Cuff tear	GIRD	GH OA / posterior dislocation
Active ROM	↓	↓	↓ IR	↓
Passive ROM	**↓ (= active)**	**Normal**	↓ IR only	↓
First / key loss	**ER**	Strength (weakness)	**IR** (posterior capsule)	–
X-ray	Normal / osteopenia	± high-riding head	Normal	**Diagnostic** (why X-ray is mandatory)

✗ Fibroblastic distractor	Diagnosis
>50 y, giant cell arteritis, bilateral shoulder stiffness	**PMR**
Podagra, acute shoulder pain	**Gout** (crystal)
Ulnar drift	**RA**
+ANA	**Lupus**

How it's asked

"Known risk factor for adhesive capsulitis?"	**→ diabetes mellitus** ✗ Wrong options: menopause · steroid use · renal disease · epilepsy
"Risk factor in diabetics EXCEPT"	**→ type of diabetes** ✗ Wrong options: age >50, duration, MI, autonomic neuropathy
"55 F, stiff shoulder, no PMH – lab most likely abnormal?"	**→ ↑ TSH** ✗ Wrong options: uric acid, ESR, anti-CCP, dislocations
"Basic blood work – which elevated?"	**→ HbA1c** ✗ Wrong options: free T4, RF, CRP, lactic acid
"True statement about AC?"	**→ age <50 + diabetes → bilateral disease** ✗ Wrong options: Caucasian · CHL better seen · release usually needed · flexion lost first
"NOT seen in adhesive capsulitis (MRI)?"	**→ widening and thickening of axillary pouch** (TO)
"MR arthrogram in diabetic stiff shoulder?"	**→ ↓ intracapsular volume** ✗ Wrong options: PVNS · HAGL extravasation

"Fibroblastic proliferative process — which patient?"	→ **50 F hypothyroid, loss of active + passive motion** ✗ Wrong options: PMR, gout, RA, lupus
"Pathophysiology + other finding?"	→ **fibroblastic proliferation of capsule; HbA1c 11.7%** ✗ Wrong options: posterior capsule = GIRD
"Cell line responsible?"	→ **fibroblasts**
"First movement hampered?"	→ **ER** (TO)
"Uncontrolled diabetic, 3 mo shoulder pain — exam?"	→ **symmetric loss of active + passive ROM**
"Most common outcome after stretching program?"	→ **↓ ROM vs contralateral**
"Early stiffening stage, 1 mo — treatment?"	→ **reassurance + gentle stretching** ✗ Wrong options: aggressive PT, PRP, MUA, arthroscopy
"Diabetic, 10 wk, greatest ER loss — treatment?"	→ **PT + medical management** ✗ Wrong options: cuff repair · posterior release · RSA
"Recent-onset AC, 2 wk — next step?"	→ **home exercise program** ✗ Wrong options: sling
"4 mo diffuse pain, stage II — management?"	→ **IA corticosteroid + stretching** ✗ Wrong options: sling · aggressive stretching
"Global stiffness + MRI partial supraspinatus tear, >60 y?"	→ **stretching regimen** ✗ Wrong options: strengthening · acromioplasty / repair
"Mitigate most common complication of arthroscopic release?"	→ **immediate ROM + PT** (stiffness)
"Release increases inferior translation + ER in adduction?"	→ **rotator interval (A)** ✗ Wrong options: MGHL · AIGHL
"Release improves cross-body adduction?"	→ **posterior capsule (H)**
"Failed nonop, 2 mm articular tear — surgery includes?"	→ **rotator interval release** ✗ Wrong options: tear completion, plication, SAD, DCR
"NOT released in AC arthroscopic release?"	→ **coracoacromial ligament**
"Expectation counselling before release?"	→ **improvements maintained long term** ✗ Wrong options: axillary nerve most common · posterior capsule most common stiffness · stiffness rare
"Arthroscopic release structure (figure)?"	→ **rotator interval (D)** ✗ Wrong options: subscapularis, supraspinatus, biceps, infraspinatus

– – Active space – –

"Rotator interval superior border?"	→ **anterior edge of supraspinatus** ✗ Wrong options: LHB · AIGHL · MGHL · axillary pouch
"ER wand exercise — LEAST stretch?"	→ **posterior capsule**

AVASCULAR NECROSIS OF THE SHOULDER

– Shoulder & Elbow · 5 / 60 –

1 Basics

Pathogenesis & diagnosis (corticosteroid risk)

Pathogenesis interrupted blood supply → **cellular death of subchondral bone** → resorption + remodelling → **subchondral collapse** → incongruity → arthritis

✗ Pathogenesis distractors cell-mediated synovial hypertrophy = **RA** · humoral response after infection, HLA-B27 = **Reiter's** · hyperuricaemic crystals = **gout** · bacterial PMN cartilage destruction = **septic arthritis**

Most common cause **corticosteroids** (asthma, Crohn's flares, SLE, leukaemia chemotherapy, graft-versus-host disease); also **sickle cell**, chronic **alcohol**

Young patients high-dose steroids in chemotherapy regimens → ↑ demand for arthroplasty in the young

vs hip humeral head AVN **less common than hip**; same pathophysiology (genetic predisposition + risk factors such as steroids); treatment similar to hip, based on **subchondral collapse**

Workup proximal humerus AVN on X-ray → also get **hip radiographs**

ASEPTIC mnemonic	Cause
A	Alcohol, AIDS
S	**Steroids (most common)**, Sickle cell, SLE
E	Erlenmeyer flask (**Gaucher**)
P	Pancreatitis
T	Trauma
I	Idiopathic / Infection
C	Caisson's (the bends)

Gaucher (wrong option) **large bony infarcts** + severe pain; less common cause than chronic steroids

Post-traumatic AVN

Proximal humerus injury	AVN rate
4-part fracture-dislocation	**~100%**
Displaced 4-part	~45%
3-part	~14%
Valgus-impacted 4-part	**~11%**

– – Active space – –

Anatomy – humeral head blood supply

Anterior humeral circumflex	**ascending branch** runs **lateral to LHB tendon in bicipital groove → arcuate artery** = intraosseous continuation, penetrates head · **35%** ✗ Wrong options: injure when plating
Posterior humeral circumflex	**main supply** per most current literature · **65%**

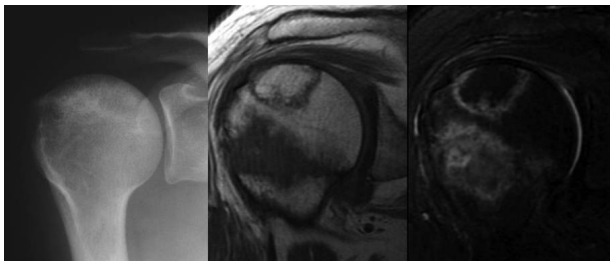
3 Presentation

Symptoms	**insidious** shoulder pain, often no inciting event · ↓ motion · crepitus · weakness
Exam	limited ROM (pain throughout arc) · crepitus · cuff + deltoid weakness (cuff strength may be normal)

4 Diagnosis & Classification

Key radiology findings

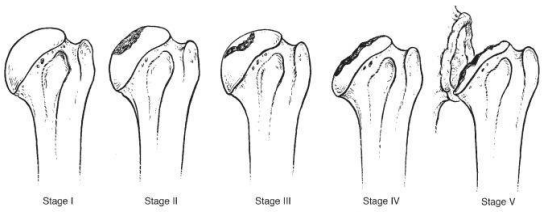
X-ray	5 views; best seen on **neutral rotation AP** · **normal at onset** · osteolytic lesion first at **superior-middle humeral head** (most common initial site)
Stage signs (X-ray)	**"snowcap" sign** = sclerosis, contour preserved (II) · **crescent sign** = subchondral fracture (III) · flattening/collapse, loss of sphericity (IV) · glenoid changes (V)
MRI	**preferred** modality · **~100% sensitive** · early = **↓ subchondral signal on T1 and T2** (oedema) · variegated pattern · metaphyseal oedema common/normal



Cruess stage II: snowcap sclerosis with preserved contour (X-ray, T1, T2 fat-sat)

Cruess classification

Stage	Radiographic findings	Treatment
I	**Normal X-ray, MRI changes only**	Core decompression
II	**Sclerosis** (wedged, mottled), osteopenia; snowcap	Core decompression
III	**Crescent sign** = subchondral fracture	Resurfacing or hemiarthroplasty
IV	**Flattening and collapse** (glenoid spared)	Hemiarthroplasty (or resurfacing)
V	Degenerative changes **extend to glenoid**	**TSA**



Cruess stages I–V

Ficat-Arlet hip staging adapted to the shoulder (used in core decompression series)

5 Management

Humeral head AVN

- First line → **analgesia, activity modification (no overhead/manual labour), PT**
- Precollapse (Cruess I–II) → **core decompression ± arthroscopy**
- Collapse, glenoid + cuff intact (III–IV) → **hemiarthroplasty**
  - focal III defect + enough epiphyseal bone → **resurfacing**
- Glenoid involved (V) → **TSA**
- Older, cuff pathology → **reverse TSA**

Operative criteria

Procedure	Indication	Not when
Core decompression (+ arthroscopy to confirm cartilage)	**Precollapse Cruess I–II**	Subchondral fracture/flattening (post-collapse = **contraindicated**)
Resurfacing hemiarthroplasty	**Stage III focal chondral defect + sufficient epiphyseal bone**; least bone removal	Large defect · global head involvement
Hemiarthroplasty	**III–IV**: collapse, **glenoid intact, cuff intact**; young → preserves glenoid bone stock for later TSA	**Irreparable cuff tear** · glenoid changes
TSA	**Stage V** (glenoid involved)	Young + intact glenoid (glenoid loosening)
Reverse TSA	Older + cuff pathology	**Intact cuff**, young

– – Active space – –

Treatment by stage – core decompression vs resurfacing/hemiarthroplasty

Core decompression ↓ subchondral marrow pressure → **neovascularisation** → slows progression · ± bone graft into void · percutaneous small-diameter technique (Harreld: no arthroplasty at 32 mo, 95% improved)

Core decompression (LaPorte, ~10 y; Ficat stage)	Success
I	**94%**
II	**88%**
III	70%
IV	14%

Stage I–II outcome good/excellent **88–100%** at 5–10 y → **~90% successful long term** · better in **non-steroid** than steroid cases
✗ Wrong options: 50% satisfied · most need arthroplasty by 10 y · unpredictable · not indicated

Conflict on core decompression effective for I–II (Mont, LaPorte; LaPorte recommends I–III; Harreld tries it with arthroscopy in III) · role in precollapse "**highly debated**" and drilling did **not** alter progression (L'insalata) · Sarris: I–II **conservative**, surgery when persistently symptomatic + collapse — read the options

Arthroscopic debridement / release no sustained benefit; AVN keeps progressing; role unknown

Most reliable, durable treatment after collapse **prosthetic arthroplasty → hemiarthroplasty**
✗ Wrong options: arthroscopic debridement · arthrodesis · resection arthroplasty · cortisone injection

Arthrodesis indications brachial plexus injury · infantile paralysis · deltoid + cuff insufficiency · chronic infection · failed revision arthroplasty · refractory instability · tumour resection bone loss

Stemless resurfacing preserves metaphyseal bone · Levy & Copeland 285 cementless · Hawi (Eclipse) 9 y, no loosening

"Ream and run" HA + glenoid reaming → unnecessary with intact glenoid cartilage

Arthroplasty outcomes HA + TSA both safe/effective · 10-y survivorship **HA 88.9%, TSA 80%**; **TSA better internal rotation** (Wang) · RSA ≈ TSA when cuff compromised · AVN arthroplasty has **↑ complications** vs other diagnoses (bone quality, comorbidity)

Prognosis related to **stage**; stage + lesion size predict progression

6 Complications

Hemiarthroplasty **glenoid cartilage wear** over time → 45% unsatisfactory (Smith, steroid AVN); still used in young active III–IV

Glenoid component in young **loosening** risk → avoid TSA/RTSA when glenoid spared

X vs Y

	AVN	Calcific tendinitis	OA	Gaucher
Onset	Insidious, steroid user	**Severe acute pain**	Long-standing symptoms	Severe pain
Imaging	**Subchondral** ↓ T1/T2 signal, collapse, fragmentation	**Hypointense deposit in cuff soft tissue**	No subchondral collapse/ fragmentation	**Large bony infarcts**

Stage	Core decompression	Hemiarthroplasty	TSA
Cruess	I–II (precollapse)	III–IV (collapse, glenoid spared)	V (glenoid)
Cuff	–	Must be intact	Intact
Key trap	Not after collapse	Not with irreparable cuff or glenoid cysts	Not in young with intact glenoid

How it's asked

"Asthmatic, 20 y steroids, humeral head collapse — pathogenesis?"	→ **cellular death of subchondral bone after vascular interruption** ✗ Wrong options: RA · Reiter's · gout · septic
"Crohn's on prednisone, atraumatic pain, MRI subchondral lesion — diagnosis?"	→ **osteonecrosis** ✗ Wrong options: Gaucher · OA · cuff tendinopathy · calcific tendinitis
"Steroid user, snowcap sclerosis, contour preserved — surgery?"	→ **core decompression** ✗ Wrong options: HA · RTSA with intact cuff · resurfacing · TSA
"Leukaemia + high-dose cortisone, collapse — most reliable pain relief?"	→ **hemiarthroplasty** ✗ Wrong options: debridement · arthrodesis · resection · injection
"39F, collapse + loss of sphericity, glenoid spared, failed PT — procedure?"	→ **resurfacing hemiarthroplasty** ✗ Wrong options: arthroscopic release · core decompression post-collapse · TSA/RTSA glenoid loosening
"GVHD steroid AVN — best case for hemiarthroplasty?"	→ **subchondral fracture + intact cuff** ✗ Wrong options: sclerosis = decompression · irreparable cuff · glenoid cysts = TSA
"SLE patient had HA — preop imaging?"	→ **isolated humeral head collapse** (Cruess IV) ✗ Wrong options: glenoid changes = V · osteopenia / MRI-only / mottled sclerosis = I–II
"Leukaemia, flattening + collapse, glenoid intact, cuff normal?"	→ **hemiarthroplasty** ✗ Wrong options: TSA · core decompression · ream and run · RTSA

– – Active space – –

"Alcoholic, Cruess II – core decompression outcome?"	**→ ~90% successful long term**
"Humeral head AVN on X-ray – what else to image?"	**→ hips**

BICEPS SUBLUXATION

– Shoulder & Elbow · 6 / 60 –

1 Basics

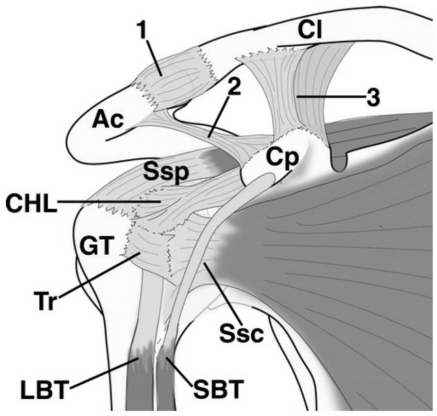
Association with subscapularis tears

Definition	LHBT subluxes/dislocates **medially** out of the bicipital groove → recognised cause of **anterior shoulder pain**
Most common associated pathology	**subscapularis tear** (even **partial**); biceps subluxation is **almost always** associated with a subscapularis tear
Most important medial restraint	**subscapularis tendon** = most important restraint to medial subluxation/ dislocation of LHBT
Walch series	dislocated LHBT associated with partial/complete **subscapularis tear in 96%**
Other cause	**disruption of the biceps sling (pulley)**
Not the cause	**subacromial impingement** does not cause LHBT instability · isolated degenerative tendinosis (instability is from the subscapularis tear) · SLAP

Biceps pulley stabilisers (coracohumeral & transverse humeral ligaments)

Origin	LHBT arises from **supraglenoid tubercle + superior labrum** → intra-articular → bicipital groove
Function	**dynamic stabiliser**; shoulder flexion, abduction

Biceps sling / pulley (intra-articular)	Rotator interval tissue?
Coracohumeral ligament (CHL)	**Yes**
Superior glenohumeral ligament (SGHL)	**Yes**
Subscapularis fibres	Contributes to sling
Supraspinatus (anterior fibres)	Contributes to sling



Biceps sling: CHL, SGHL, subscapularis, supraspinatus over LHBT

Transverse humeral ligament (THL)	stabilises LHBT in the groove **at the level of the greater + lesser tuberosities** (distal to pulley); **not rotator interval tissue**

– – Active space – –

Bicipital groove length 81–87 mm · width 7.7–11.6 mm · depth 3.7–6 mm · **bottleneck ~2/3 down** from top → THL may constrain groove → anterior pain; tenodesis **below the groove** may be preferred

✗ Distractor	Why wrong
Coracoacromial ligament (CAL)	Not rotator interval; no LHBT stability at tuberosities
IGHL	Not rotator interval; not an LHBT stabiliser
AIGHL	Not associated with biceps subluxation (injured in anterior dislocation)
Superior labrum / infraspinatus / posterior labrum	Not associated with **medial** biceps subluxation

Imaging finding = injured structure MRI medial biceps subluxation → **CHL disruption** (sling)

Anterosuperior cuff tear sling disrupted → LHBT subluxes medially → **signals subscapularis tear** (often missed on MRI)

3 Presentation

Symptoms **anterior** shoulder pain ± **clicking**

Yergason test anterior pain with **resisted supination**, arm at side, elbow flexed 90°

Speed test anterior pain with **resisted forward flexion**, shoulder 90° flexed, elbow extended, palm up

Palpable click with **abduction + external rotation** as tendon subluxes/dislocates out of groove

Always test subscapularis strength **lift-off** (Gerber), belly press, push-off

Lift-off test (Gerber) back of hand held off mid-lumbar spine in IR; hand falls or patient extends elbow/shoulder to compensate → **subscapularis tear**

✗ Exam distractor	Tests
Drop arm	**Supraspinatus**
External rotation lag	**Infraspinatus**
Hornblower's	**Teres minor**
Kim test	**Posteroinferior labral tear**

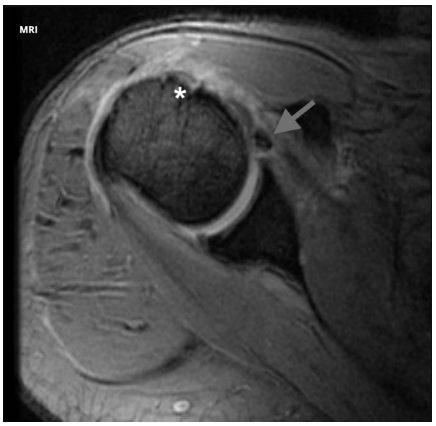
4 Diagnosis & Classification

Key radiology findings

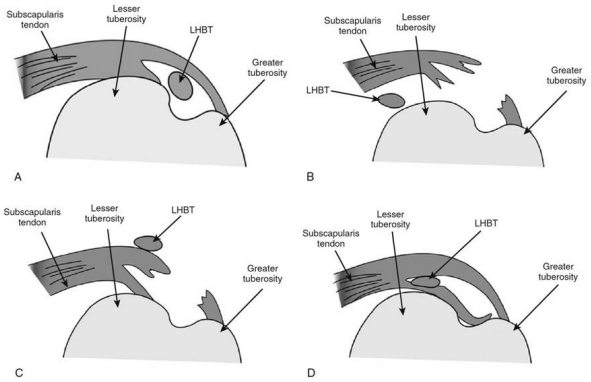
MRI ↑ **T2 signal** + LHBT **displaced out of bicipital groove** (axial: on lesser tuberosity → fully medial); coincides with **subscapularis tear**

MRI limitation misses subscapularis tears unless tear extends **≥½ cranio-caudal** length → small tears missed → test lift-off

LHBT subluxation on MRI as predictor of full-thickness subscapularis tear (Shi)	sensitivity **82%** · specificity **80%** · PPV **35%** · **NPV 97%** → value lies in **NPV** (no subluxation → full-thickness subscapularis tear unlikely); correlates with tear severity
Ultrasound	**dynamic test** of biceps instability
Arthroscopy	↓ **biceps–glenoid angle** associated with subluxation (**87°** vs **90°**; Koh) → statistically significant but **not clinically relevant**



Axial MRI: LHBT dislocated medial to empty groove



LHBT position relative to subscapularis and tuberosities

5 Management

Biceps subluxation

- Initial → **NSAIDs + PT + steroid injection**
  (injection near, not into, the tendon)
- Refractory → **arthroscopic/open tenotomy or tenodesis**
  - subscapularis tear → **+ subscapularis repair**

Nonoperative

First line	**NSAIDs, PT strengthening, activity modification, steroid injection** (adjacent to tendon, **not into it**)

Operative

Indication	**refractory** symptoms

– – Active space – –

Options	arthroscopic vs open **biceps tenotomy vs tenodesis ± subscapularis repair**; instability can be tested intra-op
Acute traumatic medial LHBT instability (Maier)	open subscapularis repair + LHBT stabilisation = equal function to tenotomy/tenodesis, **better cosmesis, less cramping**
THL release	routine release at tenotomy/tenodesis may ↓ post-op anterior pain

Biceps tenotomy for irreparable cuff tear

Indication large/massive **irreparable** cuff tear, **pain** main complaint, **good active elevation** (e.g. 120°), **normal articular surfaces → biceps tenotomy**

Alternative	When
Latissimus dorsi transfer	Severe **ER deficit** + deficient **teres minor**
Reverse TSA	**Cuff tear arthropathy + pseudoparalysis**
✗ Wrong option: SLAP repair	Contraindicated in this age group

X vs Y

	Biceps sling (pulley)	Transverse humeral ligament
Level	Intra-articular, rotator interval	Bicipital groove at **tuberosities**
Components	CHL, SGHL + subscapularis, supraspinatus	Single ligament, superficial roof of groove
Rotator interval tissue	**CHL + SGHL = yes**	**No**

	Tenotomy for massive cuff tear	Latissimus transfer	Reverse TSA
Key feature	Pain, elevation preserved, no arthritis	ER deficit, teres minor deficient	Arthropathy + pseudoparalysis

How it's asked

"62F, anterior pain, + Yergason/Speed, MRI LHBT on lesser tuberosity then dislocated – true statement?"	**→ biceps pathology due to partial subscapularis tear** ✗ Wrong options: SLAP repair · isolated tendinosis · impingement
"MR arthrogram, biceps out of groove – arthroscopy will show?"	**→ subscapularis tear + biceps subluxation** ✗ Wrong options: THL disruption · absent CHL · short head rupture
"Skier, MRI dislocated biceps tendon – most important exam?"	**→ lift-off test** ✗ Wrong options: drop arm · ER lag · Hornblower · Kim
"Post-dislocation, MRI medial biceps subluxation – injured structure?"	**→ coracohumeral ligament** ✗ Wrong options: AIGHL · superior labrum · infraspinatus · posterior labrum

"Stabilises LHBT at tuberosities, not rotator interval?"	→ **transverse humeral ligament** ✗ Wrong options: CHL · SGHL = interval · CAL · IGHL
"Main diagnostic value of LHBT subluxation on MRI?"	→ **negative predictive value (97%)**
"Massive irreparable cuff tear, FE 120°, normal joint, MRI biceps subluxed?"	→ **biceps tenotomy** ✗ Wrong options: latissimus transfer · reverse TSA
"Most important medial restraint of LHBT?"	→ **subscapularis tendon**

– – Active space – –

– – Active space – –

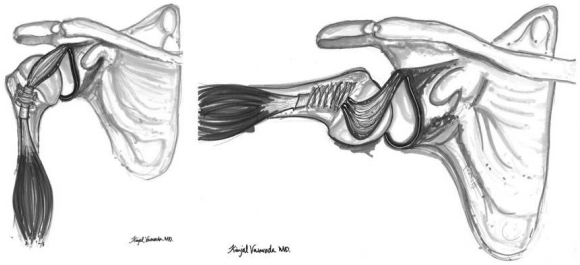
BICEPS TENDONITIS

– Shoulder & Elbow · 7 / 60 –

1 Basics

Pathoanatomy

Nature	**tendinosis** (degenerative) > true inflammation; inflammatory, traumatic or degenerative causes
Associations	usually **secondary** to other shoulder pathology: **subacromial impingement**, **bicipital groove stenosis**, **rotator cuff tears – esp. subscapularis**
Hourglass biceps	chronic inflammation → **hypertrophied intra-articular tendon** → **incarcerated in joint** during motion, esp. **elevation**



Hourglass biceps incarcerated in the joint

Long head of biceps anatomy – labral attachment, rotator interval & pulley

Origin (TO)	**supraglenoid tubercle + superior labrum**; most commonly **50% labrum / 50% tubercle**
Insertion	with short head → **radial tuberosity** · action: **elbow flexion + forearm supination** · **no shoulder ROM/strength contribution**
Synovial status (TO)	**intra-articular but extrasynovial**; synovial sheath communicates with GH joint → extends into groove

Tuoheti classification – LHB attachment on labrum (TO)

Type	Attachment	Frequency
1	Completely posterior	–
2	**Predominantly posterior**	**Most common**
3	Anterior + posterior	–
4	Anterior	**Least common**

Rotator interval (TO)

– – Active space – –

Feature	Structure
Superior border	**Supraspinatus**
Inferior border	**Subscapularis**
Medial edge (base)	**Coracoid process**
Ceiling	Joint capsule
Floor	Articular surface of humeral head
Reinforced	**CHL laterally**, **SGHL medially**
Contents	**LHB** (on the base, covered by capsule), CHL, SGHL, RI capsule

✗ Boundary trap (TO) **CHL is NOT a boundary** (content/reinforcement)

Shape (TO) triangular **space** (3-D), not 2-D area

Jost 4 layers (TO) 1 superficial CHL fibres (coracoid → greater + lesser tuberosities) · 2 CHL blended with cuff tendons · 3 deep CHL fibres · 4 **SGHL + capsule** mesh

CHL (TO) arises **outside joint**, lateral base of **coracoid** → inserts **greater + lesser tuberosities** → **spans bicipital groove** · superficial layer covers articular surface of supra/infraspinatus · **deep layer thicker** → greater tuberosity

Biceps pulley & stabilisers (TO)

Biceps pulley (TO) **SGHL + CHL** fuse along lateral half of RI with **subscapularis + supraspinatus** fibres → cradle intra-articular LHB

LHB portion	Stabilisers (TO)
Intra-articular	**Labrum + capsuloligamentous structures**, **CHL**, **SGHL**, **supraspinatus**, **subscapularis**
Extra-articular	**Transverse humeral ligament**, **pectoralis major** muscle/tendon, **bicipital groove**

✗ Stabiliser traps (TO) **transverse humeral ligament = extra-articular** (✗ Wrong options: intra-articular) · CHL/SGHL = **intra-articular** stabilisers (✗ Wrong options: extra-articular)

Rotator cable vs crescent (TO)

	Rotator cable	Rotator crescent
Structure	**Thick bundle** of fibres	**Thin, crescent-shaped sheet** of cuff
Location	Avascular zone of CHL; **perpendicular to supraspinatus fibres**	**Distal** supraspinatus + infraspinatus insertions
Relationship	Proximal margin of crescent	Bounded proximally by cable

"Suspension bridge" (TO) cable + crescent span supraspinatus + infraspinatus insertions · swapping the two definitions = classic trap

3 Presentation

Symptoms **anterior shoulder pain** ± radiating down biceps; mimics cuff/impingement pain

– – Active space – –

Groove tenderness over **bicipital groove**; worse with arm **internally rotated 10°**

Test	Manoeuvre	Positive
Speed	Forward elevate against resistance, **elbow extended + forearm supinated**	Pain in bicipital groove; **also + in SLAP**
Yergason	Active **supination against resistance**, **elbow 90°**, forearm **pronated**	Pain in bicipital groove
Popeye	Distal bulge of biceps muscle	**Rupture**



Popeye deformity

4 Diagnosis & Classification

Key radiology findings

US **thickened tendon** within bicipital groove

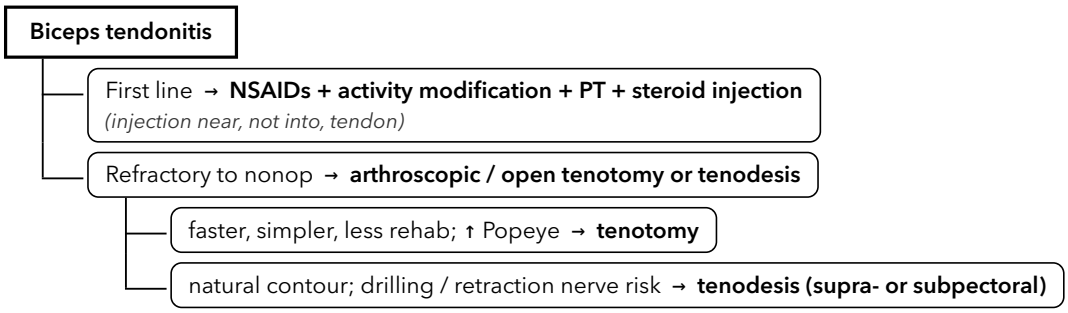
MRI evaluates **concurrent pathology** (subscapularis, cuff) · **thickening + tenosynovitis** of proximal LHB · **↑ T2 signal around tendon** · longitudinal split tears



Axial MRI, fluid around LHB in groove

Diagnostic US-guided injection groove steroid → temporary relief supports LHB as pain generator (e.g. 90% relief × 4 wk)

5 Management



Nonoperative

Regimen NSAIDs, **PT strengthening**, activity modification, **steroid injection** (in proximity, **not into tendon**)

Tenodesis vs tenotomy

Only proven difference **Popeye (cosmetic) deformity ↑ with tenotomy**

No difference **strength**, **pain (VAS)**, **ASES / patient-reported outcomes**, **shoulder + elbow ROM**, **supination** (level 1 evidence)

Cramping tenodesis **possibly ↓ arm cramping** (surgeon belief) · **not shown in high-level studies** – read the options

Tenotomy pros faster, simpler, **less rehab**, **faster return** to activity

Tenodesis pros natural muscle **contour**; but extra risks: nerve injury (**bicortical drilling**, **musculocutaneous retraction**)

Bite study (OB) tenotomy vs suprapectoral tenodesis with cuff repair: differences small; non-inferiority **statistically inconclusive** on Constant-Murley at 1 y

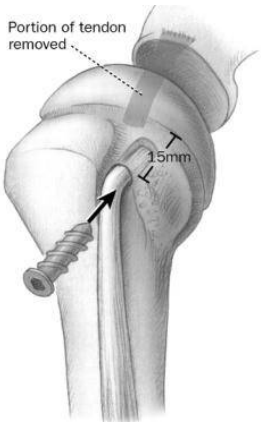
Tenodesis technique – suprapectoral vs subpectoral, arthroscopic vs open

	Arthroscopic suprapectoral	Open subpectoral
Fixation site	**Proximal** in groove	**Distal** groove, at **pectoralis major insertion**
Rationale	–	Removes tendon from groove (eliminates **groove pain**)
Residual / groove pain	**Similar**	**Similar**
Popeye, ROM, strength, PROMs, revision	**Similar** (revision very low)	**Similar**
Nerve injury	Lower (**unicortical** drilling)	**↑ transient neuropraxia** · **axillary n.** (bicortical drilling) · **medial NV** structures (retraction)

Complications overall complication rates **similar** (OB) · **subpectoral ↑ transient neuropraxia** (OB) – read the options

Groove pain Orthobullets lists tenodesis → possible "**groove pain**"; recent prospective studies → **no difference** supra vs subpectoral

– – Active space – –



Subpectoral tenodesis

Postoperative function after tenodesis

Rehab	full active shoulder ROM; **avoid active supination with elbow 90°** + **resisted elbow flexion** until healed
Long-term	**no change in shoulder or elbow strength** ✗ Wrong options: ↓ elbow flexion · ↓ supination · ↓ shoulder abduction/ elevation

6 Complications

Procedure	Complication
Tenotomy	**Popeye deformity** · ? cramping
Tenodesis	Groove pain · **nerve injury** (axillary, musculocutaneous, medial NV) · fixation failure

X vs Y

	Speed	Yergason
Elbow	**Extended**	**Flexed 90°**
Forearm	Supinated	Pronated → **resisted supination**
Resisted motion	**Forward elevation**	**Supination**
Also + in	**SLAP**	–

	Tenotomy	Tenodesis
Popeye	↑	↓
Strength / pain / PROMs / ROM	Same	Same
Rehab / return	**Faster**	Slower
Nerve risk	–	Drilling / retraction

How it's asked

"Tenodesis vs tenotomy — possibly lower incidence of?"	→ **arm cramping** ✗ Wrong options: elbow flexion weakness · stiffness · shoulder weakness
"True comparing tenodesis to tenotomy?"	→ **↓ Popeye deformity with tenodesis** ✗ Wrong options: strength, pain, supination, ROM differences
"Primary outcome difference tenodesis vs tenotomy?"	→ **cosmetic deformity** ✗ Wrong options: ASES, ROM, strength, VAS
"Photographer worried about strength + deformity (figure D)?"	→ **tenotomy ↑ cosmetic deformity, not inferior in PROMs** ✗ Wrong options: tenodesis superior
"Suprapectoral vs subpectoral — true?"	→ **similar residual anterior shoulder pain** ✗ Wrong options: ROM, cosmesis, nerve, PROMs differences
"Subpectoral differs from suprapectoral in?"	→ **higher rate of nerve injury** ✗ Wrong options: PROMs, strength, revision, ROM
"Arthroscopic suprapectoral vs prior open subpectoral — counselling?"	→ **equivalent patient-reported outcomes**
"Post-op avoid active supination at 90° — long-term function?"	→ **no change in shoulder or elbow strength**
"Rotator interval boundaries EXCEPT"	→ **CHL** (TO) ✗ Wrong options: coracoid, subscapularis, supraspinatus
"True about rotator interval structures?"	→ **intra-articular LHB is extrasynovial** (TO) ✗ Wrong options: cable/crescent swapped · CHL/SGHL extra-articular
"NOT a stabiliser of intra-articular LHB?"	→ **transverse humeral ligament** (TO)
"Tuoheti — most common LHB attachment?"	→ **predominantly posterior** (TO)
"Speed test also positive in?"	→ **SLAP lesion**

– – Active space – –

BRACHIAL NEURITIS (PARSONAGE-TURNER SYNDROME)

– Shoulder & Elbow · 8 / 60 –

1 Basics

Epidemiology

Incidence	**1.6–3/100,000/y** (Orthobullets text) · **1–30/100,000/y** (Orthobullets INA vs HNA table); underdiagnosed
Demographics	♂ **>** ♀ (1.5:1 to 11.5:1) · **middle age (4th decade)**, 20–60 y, mean 41 (range 3–81)
Side	unilateral · **bilateral 10–30%** (16% simultaneous)
Synonyms	**neuralgic amyotrophy** (NA) · idiopathic neuralgic amyotrophy (**INA**) · Parsonage-Turner syndrome (**PTS**) · acute brachial neuritis/plexitis · idiopathic brachial plexus neuropathy

Nerves involved

Distribution	any brachial plexus nerve/branch · **outside plexus in 17%** · usually **upper plexus + >1 nerve**
Most common nerves	**long thoracic** (→ **medial scapular winging**) · **suprascapular** · **axillary** · **musculocutaneous** · radial (Orthobullets, OB)
Others	cervical roots, **AIN**, PIN, LABC
Muscles	deltoid, supraspinatus, infraspinatus, **serratus anterior**, biceps



Medial scapular winging (long thoracic nerve) in brachial neuritis

Pathophysiology

Aetiology	unknown; multifactorial → **inflammatory response in brachial plexus**
Autoimmune	↑ blastogenic lymphocyte activity vs plexus tissue · ↑ **complement-fixing antibodies to peripheral nerve myelin**
Biomechanical	**mobile upper trunk** → wear of **blood–nerve barrier** → immune factors reach nerve (strenuous exercise, manual labour)
Operative finding	**hourglass-like constrictions** of nerves, **no external compression**

Risk factors / antecedent events

Trigger	Frequency	Examples
Viral infection (most common)	**25–55%**	EBV, VZV, Coxsackie B, **parvovirus B19**, CMV, mumps, smallpox, HIV; "flu-like illness", URTI
Bacterial infection	–	Leptospira, TB, Yersinia, Salmonella, Borrelia
Immunisation	**15%**	Tetanus, hepatitis B; **influenza vaccine** (OB stem)
Perioperative / peripartum	**14%**	Elective surgery
Strenuous exercise	**8%**	Cross-training, heavy upper-body work
Drugs	–	Abacavir, streptokinase, **heroin**, infliximab, **interferon**
Iatrogenic	–	**Interscalene block**, surgery, LP, irradiation; burns

Viral illness figure — **25–55%** (Orthobullets, OB) · **25–50%** (one OB explanation) – read the options

Genes & lab markers

HNA gene — **autosomal dominant** · **SEPT9 (septin 9)** on **chromosome 17q24** (cytoskeletal protein in glial cells) · ~200 families worldwide

Routine labs — **CBC + ESR usually normal**

Inconsistent labs — ↑ liver enzymes · **+ antiganglioside antibodies** · + ANA

CSF — usually normal; ± mildly ↑ protein, slight pleocytosis, oligoclonal bands

3 Presentation

Diagnosis – acute pain followed by weakness

Phase	Features	Duration
I – Pain	**Sudden, severe, unrelenting shoulder pain** (90%); radiates to proximal arm/neck; **wakes from sleep**; burning; often atraumatic	**Days–weeks** (♂ 45 d vs ♀ 23 d)
II – Painless flaccid paralysis	Weakness as pain subsides: within **24 h (33%) to 4 wk (80%)**; **patchy paresis** (muscles of same nerve unequally affected)	–
III – Recovery	Slow, steady motor return; length ∝ length of pain phase	**6–18 months**

Key exam pattern — **inability to actively abduct** with **full passive motion** · **normal cervical spine exam** (–ve Spurling)

Pain behaviour — **not worsened by motion or palpation** · **not relieved by subacromial lidocaine** (neuropathic)

Motor — **severe weakness of ER + abduction** (SS, IS, deltoid) · hypotonia, **areflexia** (LMN) · **atrophy**, fasciculations · **medial winging** (serratus anterior)

Sensory — changes in **78%**: paraesthesia/hypoaesthesia over deltoid, lateral arm, radial forearm; often unnoticed · sensation may be **normal** (OB stems)

Autonomic (15%) — trophic skin, temperature dysregulation, ↑ sweating, altered nail/hair

– – Active space – –

Upper trunk pattern pain → weakness + atrophy + scapular winging after viral illness = **upper trunk neuralgic amyotrophy**
✗ Wrong options: isolated LTN palsy (doesn't explain whole picture)

4 Diagnosis & Classification

Diagnosis **clinical**; often a **diagnosis of exclusion**; tests rule out mimics; **EMG confirms**

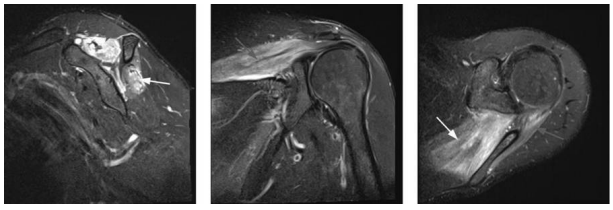
Key radiology findings

X-ray shoulder / C-spine **normal**; rule out **calcific tendinitis** (also severe incapacitating pain)

Chest X-ray baseline to exclude compressive **Pancoast tumour** (OB)

MRI timing	Sequence	Finding
Early	**T2 / STIR**	**Diffuse ↑ signal (oedema)** in muscles of plexus (SS, IS, deltoid, teres minor most common)
Late	**T1**	**Fatty infiltration + atrophy**

MRI also excludes cuff tear (intact cuff) · SSN notch/spinoglenoid **cyst** · brain MS plaques · cervical disc



T2 MRI – diffuse muscle oedema in brachial neuritis

EMG findings

EMG compatible findings in **96%**; confirms diagnosis + **monitors recovery** (serial)

Timing	EMG
Early – **3–4 wk** (Orthobullets) · **first 3 wk** (OB)	**Acute denervation**: **positive sharp waves + fibrillation potentials** in **both peripheral nerve AND nerve root distributions**
Late – **3–4 months** (Orthobullets) · **as early as 6 wk** (OB)	Chronic denervation + **early reinnervation**: **polyphasic MUPs** (= **resolving** PTS)

Sensory NCS less useful than EMG · **↓ amplitude (axonal loss)**, **preserved conduction velocity + distal latency** · may be normal (doesn't exclude)

✗ EMG distractor	Diagnosis
Normal EMG	PTS shows denervation
Fibrillations from **spinoglenoid notch** compression	SSN compression (cyst on MRI)
Sharp waves + fibs **deltoid + biceps only**	**C5 root** compression
Polyphasic MUPs, early reinnervation	**Resolving** PTS (not early)

Classification – INA vs HNA

Factor	INA (idiopathic)	HNA (hereditary)
Incidence	1–30/100,000/y	Very rare (~200 families)
Gene	–	**Septin 9 (chr 17q24)**, **AD**
Age at onset	Middle age (20–60)	**Young** (childhood/adolescence; mean 28)
Recurrence	**Rare** (~1.5 episodes)	**Frequent** (3.5 episodes)
Appearance	Normal	**Dysmorphic**: short stature, **hypotelorism**, cleft palate, facial asymmetry, skin folds
Nerves outside plexus	Uncommon (**17%**)	Common (**56%**): **lower cranial nerves** (VII, IX, X, XI, XII), lumbosacral plexus, phrenic, recurrent laryngeal
Severity / outcome	Better	More severe paresis, worse outcome

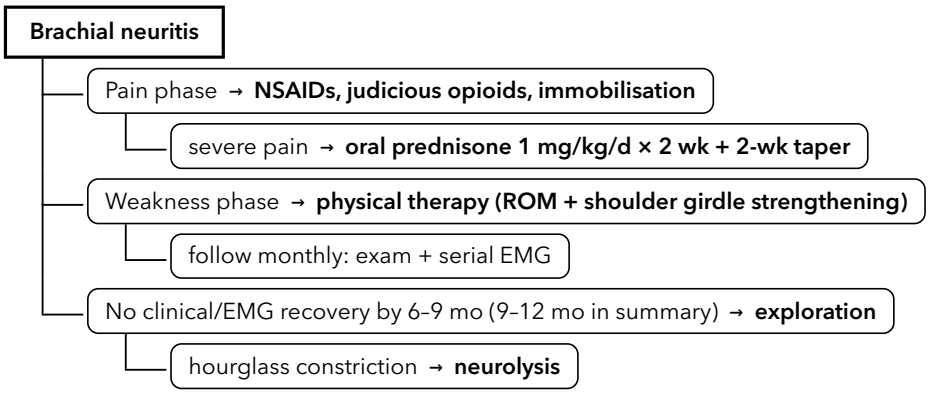
Differential diagnosis

Mimic	Distinguishing feature
Cervical radiculopathy (C5/C6–7, disc)	Single root dermatome/myotome; pain from neck; **↑ with movement**; abnormal neck exam/Spurling (normal neck exam here makes it unlikely)
Rotator cuff tear / impingement	Pain persists with weakness; impingement signs; **relieved by subacromial lidocaine**; MRI shows tear; does not explain winging
Calcific tendinitis	Calcification on X-ray; no acute severe weakness
Adhesive capsulitis (frozen shoulder)	**Loss of passive + active** motion
Entrapment neuropathy (SSN)	**Single nerve** distribution on EMG (SS + IS only)
Poliomyelitis	No (excluded by acute severe weakness pattern, OB)
ALS	**UMN + LMN** signs, **painless, no sensory loss**, cranial nerve signs
Idiopathic hypertrophic brachial neuritis	**Painless**; plexus hypertrophy on MRI; **demyelinating** NCS (slow velocity, prolonged latency)
Multiple sclerosis	Brain MRI demyelination

Exclusion logic (OB) loss of **active** motion excludes radiculopathy + impingement · **normal MRI** excludes cuff tear · **full passive motion** excludes adhesive capsulitis · normal C-spine exam excludes disc disease

– – Active space – –

5 Management



Nonoperative

Mainstay	**observation + pain control**; self-limiting
Corticosteroids	early, severe pain; **2 wk prednisone 1 mg/kg/d + 2 wk taper** · faster pain resolution, **no change in progression/prognosis** · OB: may shorten symptoms/accelerate recovery
Physical therapy	once pain abated; ROM + strengthening; reduce traction on nerves · reverses atrophy · **observation alone = observation + PT at 1 y**

Operative criteria

Indication	Threshold
Exploration / neurolysis / graft / transfer	**No regeneration on exam + EMG by 6–9 months** (Orthobullets treatment) · **9–12 months** (Orthobullets summary) – read the options

Procedure	Note
Neurolysis	**Superior** to neurorrhaphy + grafting; LTN microneurolysis, ulnar transposition, Guyon/radial tunnel/carpal tunnel release
Neurorrhaphy / grafting	Excise diseased segment
Nerve transfer	–
Split pectoralis major transfer	**Serratus anterior paralysis** (winging)

***✗* Wrong options: decompression / arthroscopic repair**	not indicated (no compressive lesion, cuff intact)

Prognosis

Overall — **good**; **gradual return of strength without long-term functional deficit**; slow · residual weakness may last months–years

Time	Recovery
1 month	**66%** some motor recovery
1 year	**35%** recovered (excellent 36%)
2 years	Excellent **75%**
3 years	**~90%** full strength, no pain (89%)
Up to 8 years	Full strength in some

Poor prognosis — **female** · **lower trunk** involvement (upper trunk = best) · **persistent pain + no motor recovery by 3 months** · **hereditary** cases

No effect — **age**

Recurrence — **rare** in idiopathic (common in HNA)

X vs Y

Prognosis statement (OB distractor)	Disease
Gradual return of strength, no long-term deficit	✓ **PTS**
Decompression improves strength	**SSN notch cyst**
Immunomodulators ↓ relapses	**Multiple sclerosis**
Arthroscopic repair best for active patient	**Rotator cuff tear**
Progressive weakness → **respiratory failure**	**ALS** (pure motor; no ↓ sensory amplitudes)

	Brachial neuritis	Cervical radiculopathy	Rotator cuff tear
Pain onset	Acute severe → resolves before weakness	Neck → arm	Persists with weakness
Pain with motion	**No**	**Yes**	Yes
Subacromial lidocaine	No relief	–	Relief
Distribution	**Multiple roots + peripheral nerves**, patchy	One root	Cuff only, no winging
EMG	**Root + peripheral nerve** denervation	Root	Normal

How it's asked

"Acute atraumatic burning shoulder pain, can't abduct, full passive ROM, normal C-spine + MRI – diagnosis?"

→ **brachial neuritis**
✗ Wrong options: C6–7 radiculopathy · impingement · cuff tear · adhesive capsulitis

– – Active space – –

"Flu-like illness 1 wk earlier, then burning pain → weakness, normal X-rays?"	→ **brachial neuritis** ✗ Wrong options: calcific tendinitis · poliomyelitis · cervical disc disease · impingement
"URTI, 3 mo pain + weakness, atrophy, medial winging, flaccid — diagnosis?"	→ **upper trunk neuralgic amyotrophy** ✗ Wrong options: isolated LTN palsy · ALS · frozen shoulder · cuff tear
"Post-influenza-vaccine pain → weakness, T2 oedema SS/IS — expected EMG?"	→ **acute denervation in both peripheral nerve + nerve root distributions with sharp waves + fibrillations**
"EMG showing polyphasic MUPs / early reinnervation?"	→ **resolving PTS** (late)
"Sharp waves + fibs in deltoid + biceps only?"	→ **C5 root compression**
"Cross-training athlete, viral illness, pain → weakness, ↓ sensory amplitudes, normal brain/C-spine MRI — prognosis?"	→ **gradual return of strength without long-term deficits** ✗ Wrong options: decompression · immunomodulators · arthroscopic repair · respiratory failure
"Most common antecedent event?"	→ **viral illness (25–55%)**
"Most commonly involved nerve / cause of winging?"	→ **long thoracic nerve**
"Gene in hereditary neuralgic amyotrophy?"	→ **SEPT9, chromosome 17q24, AD**
"Poor prognostic factors?"	→ **female, lower trunk, no recovery by 3 months, hereditary** ✗ Wrong options: age
"Indication for surgery?"	→ **no EMG/clinical recovery by 6–9 (9–12) months → neurolysis**

CALCIFIC TENDONITIS

– Shoulder & Elbow · 9 / 60 –

1 Basics

Epidemiology

Definition	calcification + tendon degeneration **near rotator cuff insertion** → pain + ↓ ROM; **unknown etiology**
Age	**30–60 y**
Sex	♀ > ♂
Location	**supraspinatus** most common · supraspinatus and/or **infraspinatus** (OB)
Risk factors	endocrine: **diabetes**, **hypothyroidism**

Genes & lab markers

Crystal	**calcium hydroxyapatite** (Orthobullets, OB) · "calcium carbonate apatite" (one OB explanation) – read the options
Histology	hydroxyapatite deposits surrounded by **fibroblasts, macrophages, giant cells, lymphocytes**
Molecular pathways	**osteopontin** · **cathepsin K** · **transglutaminase 2**
✗ Wrong option: calcium pyrophosphate	**positively birefringent rhomboid/rod** crystals → CPPD / **pseudogout**
✗ Wrong option: calcium oxalate	**kidney stones**, urine; excess in **ethylene glycol (antifreeze)** poisoning → renal failure
Process	**active, cell-mediated** ✗ Wrong options: purely degenerative; Uhthoff & Loehr

Phases of disease (resorptive phase pain)

Stage (Uhthoff)	Phase	Pathology	Pain
Precalcific	–	**Fibrocartilaginous metaplasia**; **tenocytes → chondrocytes**	**Pain-free**
Calcific	**Formative**	**Cell-mediated** calcific deposits	±
	Resting	**No inflammation / vascular infiltration**	±
	Resorptive	**Phagocytic resorption + vascular infiltration**	**Most painful**
Postcalcific	–	Calcium resorbed + **tissue remodelling**	–

✗ Resorptive distractors	"vascular infiltration, painless" (it is painful) · lowest inflammation (= resting) · cell-mediated deposits (= formative) · fibrocartilaginous metaplasia (= precalcific)

– – Active space – –

Fibrocartilaginous metaplasia	only in **calcific tendonitis** among shoulder pathologies ✗ Wrong options: cuff arthropathy · biceps tenosynovitis · chronic cuff tear · neuropathic arthropathy
Surgery timing	avoid surgery in **resorptive** phase (Uhthoff & Loehr) · surgery for refractory cases in **formative** phase

3 Presentation

History	similar to **subacromial impingement**
Symptoms	**atraumatic pain** (worst in resorptive phase) · **catching**, crepitus · mechanical block · pain with overhead activity / active elevation
Exam	supraspinatus fossa **atrophy** · **↓ active ROM** · scapular dyskinesia · ± ↓ cuff strength · positive **impingement signs**

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
Radiographs	**Gold standard / mainstay**; monitor progression; location, density, extent, delineation
AP, supraspinatus outlet, axillary	**Supraspinatus** calcification
Internal rotation view	**Infraspinatus + teres minor**
External rotation view	**Subscapularis**
Deposit site	**1–1.5 cm** from supraspinatus insertion (Orthobullets) · **1–2 cm**, zone of ↓ vascularity, can be multifocal (OB) – read the options
CT	Rarely required; 3D anatomy
MRI	**Low signal on all sequences** (globular, T1 + STIR/T2 hypointense); limited diagnostic use; refractory pain → concomitant cuff tear
Ultrasound	Deposits **hyperechoic**; quantify extent; **guides needle lavage**
MR arthrography alone	Poor sensitivity + inter-rater agreement (Zubler)

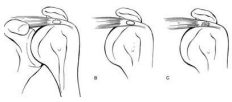


Calcific deposit over supraspinatus insertion

✗ Wrong options (next tests after failed barbotage) bone scan · MR arthrogram · EMG (no added value)

Imaging ✗ distractors no acromial spur (✗ Wrong options: impingement) · no tear (✗ Wrong options: partial/full-thickness tear) · **adhesive capsulitis = clinical diagnosis**, no consistent X-ray findings

Gartner & Heyer classification

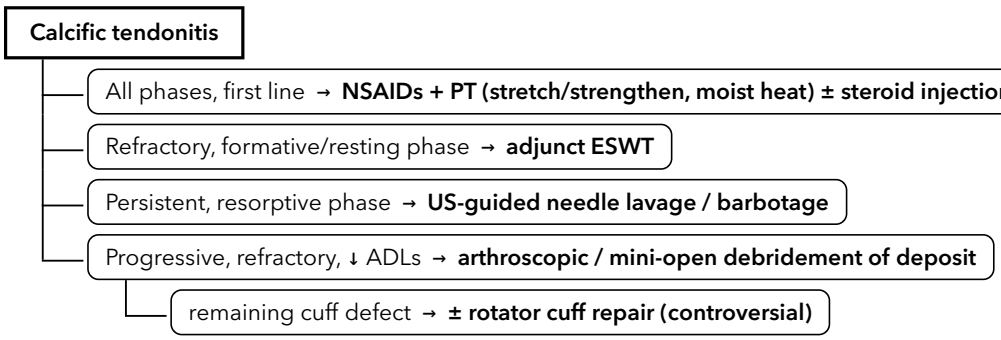
Type	Appearance	Phase	Image
I	**Well circumscribed, dense**	**Formative**	
II	Soft contour/dense **or** sharp/transparent	–	
III	**Translucent, cloudy**, no clear circumscription	**Resorptive**	

Radiographic types (one OB explanation) Type I = **fluffy, fleecy, poorly defined** → acute pain, formative phase · Type II = **discrete homogeneous, well defined** → subacute/chronic – conflicts with Gartner (Orthobullets: type I well circumscribed) – read the options

Mole et al. classification

Type	Appearance
A	Dense, **homogeneous**, sharp contours
B	Dense, **segmented**, sharp contours
C	**Heterogeneous, soft** contours
D	**Dystrophic** calcification at cuff **insertion**

5 Management



Treatment – conservative, barbotage & arthroscopic debridement

Nonop outcome symptoms resolve **60–70% at 6 months**; most improve spontaneously

Failure more likely **bilateral or large** deposits · deposit under **anterior third of acromion** · deposit **medial to acromion**

Steroid injection common but **controversial**; variable relief

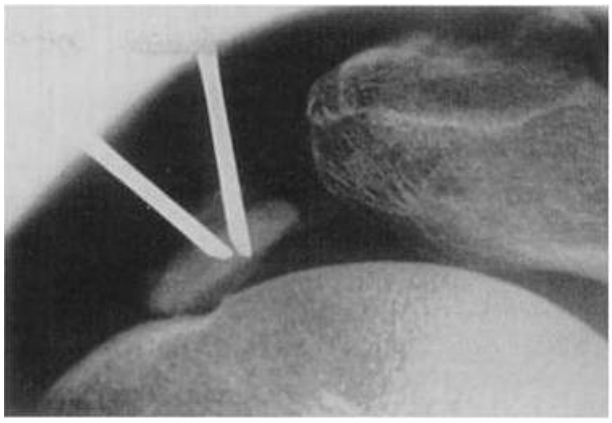
– – Active space – –

Formative phase, 4-month mild pain
moist heat + stretching first
✗ Wrong options: aspiration (deposit **granular** → hard to aspirate) · ESWT (role poorly defined, OB) · **low-dose radiation** (malignant transformation risk) · arthroscopic debridement (only after failed nonop)

ESWT
adjunct; refractory **formative + resting** phases; **dose-dependent**: high-energy > low-energy for outcome scores + resorption, but also > procedural pain + local reaction (ecchymosis)

Needle lavage
2 needles (inflow saline ± anaesthetic + aspiration outflow); indicated for **resorptive phase**; better outcomes **Gartner II/III vs I**

Barbotage
needle breaks up deposit → **corticosteroid injection**

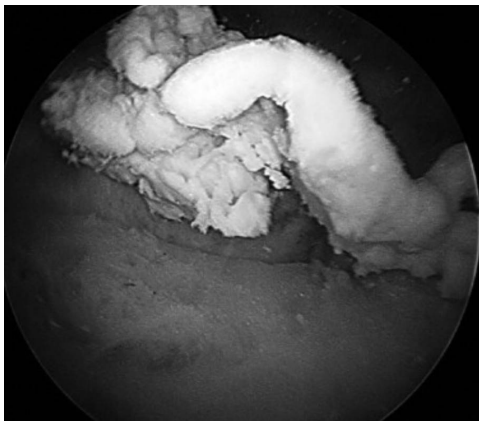


US-guided two-needle lavage

Operative criteria

Indication for surgical decompression of deposit
Progression of symptoms
Refractory to nonoperative treatment (incl. failed barbotage)
Interference with ADLs / work

Surgery
arthroscopic or mini-open removal (shaver or spinal needle) ± SAD ± RCR; **paste-like ("toothpaste") calcium** expressed from cuff



Arthroscopic paste-like calcium expressed from cuff

After failed barbotage
arthroscopic debridement
✗ Wrong options: isolated RCR without debridement (won't resolve symptoms)

RCR after debridement — **controversial**; depends on remaining tissue integrity + defect size · debridement without RCR safe: **3.7%** cuff tears at 7 y (El Shewy)

Deposit with osseous involvement — same results as tendon-only (Seyahi)

Debridement ± subacromial decompression

SAD + debridement vs debridement alone	Result
Time to pain-free activity	**↑ (longer)** with SAD (Marder)
Pain at midterm	No difference
ROM	No difference
Recurrence	No difference
Cuff tears	**↑ in operated shoulder vs contralateral** regardless (Balke, 6 y)

Debridement + SAD — 91% good/satisfactory at 26 mo (Ark); SAD gives **no apparent benefit**

Return to work — longer with SAD and/or RCR

6 Complications

List — **recurrence** · **persistent pain** · **stiffness** (operative risk) · **iatrogenic cuff injury** · ↑ cuff tears after surgery

X vs Y

	Formative phase	Resting phase	Resorptive phase
Process	Cell-mediated deposition	No inflammation / vascular infiltration	Phagocytosis + vascular infiltration
Pain	±	±	**Most severe**
Deposit (Gartner)	**Type I** dense, well circumscribed	–	**Type III** cloudy, translucent
Treatment niche	**ESWT**; conservative; aspiration difficult (granular)	**ESWT**	**Needle lavage**; avoid surgery

	Hydroxyapatite	CPP	Calcium oxalate
Disease	**Calcific tendonitis**	**Pseudogout / CPPD**	Kidney stones, ethylene glycol
Crystal	Deposits ringed by fibroblasts, macrophages, giant cells	**Positively birefringent rhomboid/rod**	Urine crystals
Colchicine	**no role**	–	–

– – Active space – –

How it's asked

"Calcific deposit on X-ray — structure abnormal at arthroscopy?"	→ **supraspinatus** ✗ Wrong options: infraspinatus · labrum · biceps · GH cartilage
"35-y ♀, insidious pain, low-signal globule in supraspinatus on MRI?"	→ **calcific tendinitis** ✗ Wrong options: impingement · adhesive capsulitis · cuff tear
"Fibrocartilaginous metaplasia — which image?"	→ **AP X-ray with supraspinatus calcification** ✗ Wrong options: cuff arthropathy · biceps tenosynovitis · neuropathic arthropathy
"Best description of resorptive phase?"	→ **highest pain levels** ✗ Wrong options: painless vascular infiltration
"Formative-phase deposit, mild 4-month pain — initial treatment?"	→ **moist heat + stretching** ✗ Wrong options: aspiration · ESWT · radiation · arthroscopy
"Debrided material pathology + next step?"	→ **calcium hydroxyapatite; debridement ± cuff repair** ✗ Wrong options: CPP · oxalate · IV colchicine
"Failed PT, NSAIDs and US barbotage — next?"	→ **arthroscopic debridement** ✗ Wrong options: isolated RCR · bone scan · MR arthrogram · EMG
"Adding SAD to debridement — effect?"	→ **longer time to pain-free activity** ✗ Wrong options: better pain · ROM · fewer tears · ↓ recurrence
"Most painful phase?"	→ **resorptive**
"View for subscapularis calcification?"	→ **external rotation AP**

DISTAL BICEPS AVULSION

– Shoulder & Elbow · 10 / 60 –

1 Basics

Epidemiology

Frequency rare; **~10% of all biceps ruptures**

Demographics **dominant elbow (86%)** of **men (93%)** in their **40s** (30–50 y)

Risk factors **anabolic steroids** · **smoking (7.5× risk)** · hypovascularity · intrinsic degeneration · mechanical impingement in space available for tendon

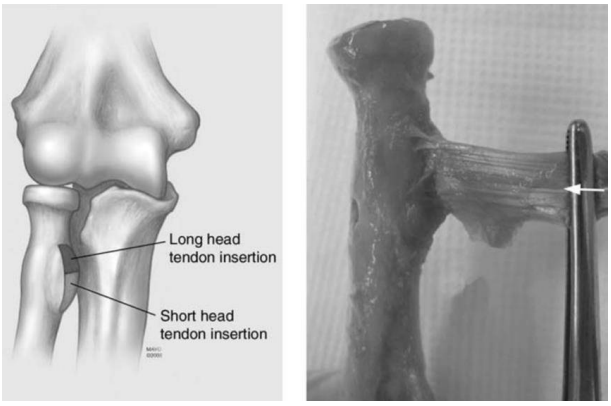
Patterns complete avulsion · **partial** (on **radial side** of tuberosity footprint) · **intersubstance muscle transection** (rope wrapped round arm, **tug-of-war**)

Rare association **median nerve compression**

Footprint anatomy

Insertion **radial (bicipital) tuberosity**; innervation **musculocutaneous only**
✗ Wrong options: dual musculocutaneous + radial

	Short head	Long head
Origin	**Coracoid**	Supraglenoid tubercle / superior labrum
Footprint	**Distal + anterior** (thin sliver)	**Proximal + posterior** (oval; **apex** of tuberosity)
Better at	**Elbow flexion**	**Supination** (furthest from axis of rotation)



Long head inserts proximal, short head distal on radial tuberosity

Biceps function **primary supinator**; lesser elbow flexor · **brachialis = primary elbow flexor** (inserts **coronoid + ulnar tuberosity**, not radial tuberosity)

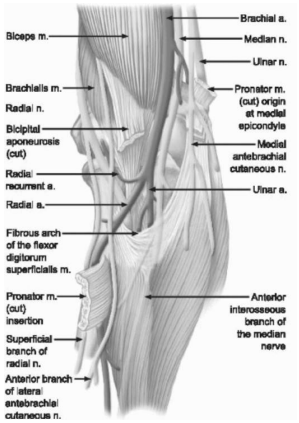
Annular ligament anterior radial head, PRUJ stability; does not attach to radial tuberosity

Lacertus fibrosus (bicipital aponeurosis) from **medial side of distal short head**, distal to elbow crease → obliquely distal + medial across cubital fossa → protects **brachial artery + median nerve** → continuous with flexor-origin fascia; **may be mistaken for intact tendon**; intact lacertus → ↓ deficit in low-demand patient

– – Active space – –

Antecubital fossa

Medial → lateral	**median nerve** (most medial) · **brachial artery** · **biceps tendon** · **radial nerve** (most lateral)
Borders	**pronator teres** ulnarly · **brachioradialis** radially
Radial recurrent artery	**superficial** to biceps tendon, crosses near tuberosity
Superficial radial nerve	**lateral (radial) + superficial** to biceps tendon; radial nerve bifurcates (PIN + SRN) **at level of antecubital fossa / radiocapitellar joint**
LABCN	terminal sensory branch of **musculocutaneous (lateral cord)** → exits **between biceps + brachialis** lateral to musculotendinous junction → pierces fascia, **lateral to biceps tendon, deep to cephalic vein** → superficial along **brachioradialis** → volar-lateral forearm sensation



Antecubital fossa: radial recurrent artery, SRN, LABCN

Nerve course (✗ Wrong options: distractors for LABCN)	Nerve
Branches 5–8 cm distal to lateral epicondyle, **between 2 heads of PT**, along volar FDP → pronator quadratus	**AIN**
From radial n. at radiocapitellar joint, **through supinator (arcade of Frohse)**, around **radial neck** → dorsal wrist capsule	**PIN**
Deep to brachioradialis, lateral to radial artery, pierces fascia **7–9 cm proximal to wrist** → snuffbox, dorsoradial hand	**SRN**
With brachial artery between **PT + biceps tendon**, between **FDS + FDP**, emerges between FDS + FPL → carpal tunnel	**Median**

Radial nerve motor order (Abrams)	BR → ECRL → supinator → ECRB → EDC → ECU → EDQ → APL → EPL → EPB → **EIP last**

2 Mechanism

Mechanism – eccentric contraction

Eccentric contraction	**muscle lengthens during contraction**; flexed elbow forced into extension ("**flexed elbow unacceptably challenged**")
Contributors	**vascular watershed** · mechanical attrition (abrasion during pronosupination)

Partial tear mechanism pronation → **sharp radial margin of tuberosity impinges tendon** → released in supination; rupture through pre-existing degeneration on **radial aspect**

✗ Wrong option: contraction type	Definition
Isometric	Constant **length**
Isokinetic	Constant **speed**
Isotonic	Constant **force**
Concentric	Muscle **shortens** (biceps curl)

3 Presentation

History painful **"pop"** as elbow eccentrically loaded (catching falling object, lifting furniture/engine, deadlift, curls)

Hallmark **weakness + pain in supination** > flexion

Inspection **"reverse Popeye"** (proximal retraction of muscle belly), contour change, **antecubital / medial ecchymosis**, **palpable defect**

Complete vs partial tendon **absent** in complete · **palpable but painful** in partial (otherwise similar picture)

Diagnosis – hook test, MRI, partial tears

Hook test (O'Driscoll) (TO) patient looks at palm, **shoulder elevated, elbow 90°, forearm supinated** → intact tendon = cord-like structure finger hooks around; torn → **brachialis is flat** → nothing to hook

Hook technique index finger hooks **lateral edge** of tendon, **lateral → medial**; intact/partial → finger 1 cm beneath tendon

Accuracy **sensitivity + specificity 100%** → **most sensitive exam** for complete rupture

False "normal" hook **partial tear** · **intact lacertus fibrosus** (hooking **medial → lateral** catches lacertus) · underlying **brachialis**

Partial tears hook test **normal but painful**; hook test not useful for partial tears



Hook test: finger hooks lateral edge of tendon

Ruland biceps squeeze test elbow **60–80° flexion**, forearm slightly **pronated**, squeeze biceps belly → **no supination = positive** (like Thompson); **sensitivity 96%**

Other exam (OB) **biceps crease interval test**, **passive forearm pronation** test

– – Active space – –

✗ Wrong option: milking manoeuvre tests **UCL**, not biceps

4 Diagnosis & Classification

Key radiology findings

X-ray usually **normal**; occasionally **fleck/avulsion** from radial tuberosity

MRI distinguishes **complete vs partial**, muscle vs tendon, **degree of retraction**; **FABS position** (elbow **flexion**, shoulder **abduction**, forearm **supination**) ↑ sensitivity

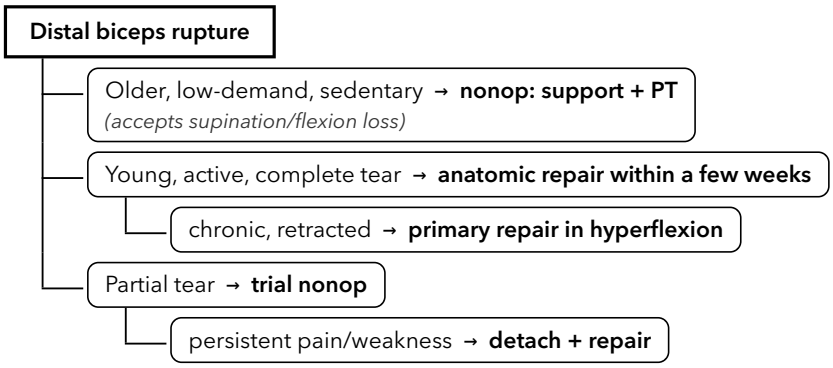
MRI when? **needed for partial tear** (palpable but painful tendon) · not always needed for complete tear with conclusive exam

Partial tear MRI ↑ signal + ↑ **diameter** of tendon · **avulsive marrow oedema** in tuberosity · **bicipitoradial bursa distension**

Ultrasound can confirm diagnosis in office

Clinical finding	Next step
Abnormal hook, retraction, ecchymosis	Clinical diagnosis → repair
Palpable tendon, painful hook	**MRI** (✗ Wrong options: CT · immobilise 3 wk · exploration · early PT)
Partial tear, failed ≥6 mo nonop	**Detachment + repair** of biceps tendon

5 Management



Nonoperative outcome – loss of supination strength

Greatest loss **forearm supination**
✗ Wrong options: elbow flexion (brachialis compensates; flexion returns near normal)

Nonop deficit	Orthobullets	OB explanations
Sustained supination (endurance)	**50%**	50%
Supination strength	**40%**	40% (up to 40%)
Flexion strength	**30%**	**30%** · **10%** (one OB) · **~40% sustained flexion** (one OB)
Grip	15%	–

Flexion loss value **30%** (Orthobullets, most OB) · **10%** (one OB) · **40% sustained** (one OB) – read the options

Also with nonop **delayed return to activity**
✗ Wrong options: chronic elbow pain · instability (terrible triad / anteromedial coronoid) · lateral pain on resisted wrist extension (lateral epicondylitis)

Activity hardest **using a screwdriver** (supination)

✗ Wrong option: activity affected	Injury
Pushing up from chair	**LUCL** tear
Bench press	**Pectoralis major** tear
Wrist extension	ECRB/ECRL (common extensor)
Elbow extension	**Triceps** tear

Nonop EXCEPT **LABCN neuritis** (exclusively surgical)

Nonop indications **older, low-demand, sedentary**, willing to sacrifice function · not healthy middle-aged with dominant-arm tear

Operative criteria

Indication	Detail
Young, healthy, active	Don't want to sacrifice function (**main aim = restore supination strength**)
Partial tear failing nonop	Persistent pain + weakness
Subacute / chronic	**Direct repair** (no allograft) usually possible; may need **hyperflexion**
Timing	**Within a few weeks**; delay → harder primary repair, **more extensile approach**

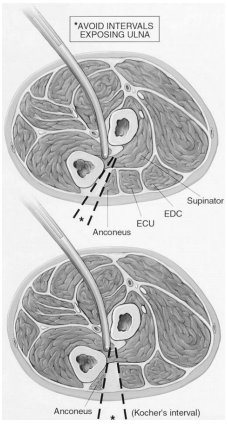
Not surgery indications cosmesis · preventing elbow degeneration

Repair outcome anatomic repairs **high success**; **loss of terminal extension** common after repair
✗ Wrong options: high re-rupture

– – Active space – –

Surgical anatomy – BR/PT interval & recurrent radial artery

	Anterior single-incision	Dual-incision (Boyd-Anderson)
Rationale	Developed to ↓ **HO + synostosis** of 2-incision	Developed to avoid **radial nerve / PIN** injury
Interval	**Brachioradialis (radial n.) / pronator teres (median n.)** – not PT/FCR	Same anterior + **posterolateral: ECU / EDC**
✗ Wrong interval	–	**Kocher (ECU/anconeus)**, **anconeus/ulna** → expose ulna → HO/synostosis
Forearm	**Maximal supination** protects PIN (✗ Wrong options: pronation)	Hemostat through IO space along **medial border of tuberosity** → pierces anconeus → tents skin
LABCN injury	**Higher**	Lower
HO / synostosis	Lower (↓ periosteal stripping) – **not eliminated**	**Higher**
Repair	–	**More anatomic** (better footprint exposure; ulnar placement)
Supination strength	Similar	Similar or slightly better



Avoid intervals exposing the ulna (ECU/anconeus, Kocher)

Single-incision steps limited antecubital incision → BR lateral, PT medial → **identify LABCN** (exits between biceps + brachialis) and retract laterally → **recurrent radial vessels** coagulated/ligated or retracted → protect PIN
✗ Wrong options: forceful lateral retraction, keep supinated

Recurrent radial artery **crosses field immediately deep to BR/PT interval, near tuberosity**; large, may bleed profusely → **ligate** if cannot retract → "flash of bright red blood" with deep retractor

✗ Artery distractor	Location
Anterior interosseous a.	From ulnar a., along **IOM**
Anterior recurrent ulnar a.	From ulnar a., proximal + superficial to brachialis insertion
Recurrent interosseous a.	From posterior interosseous a., **posterior forearm**
Inferior ulnar collateral a.	From brachial a., **medial elbow**
Ulnar a.	Far medial, between **FDS + FDP**

Approach vs effectiveness single vs two-incision: **no proven difference** in effectiveness

Post-op immobilise **110° flexion + moderate supination**

Fixation techniques

Technique	Approach	Key step
Bone tunnel	2-incision	Acorn reamer slot; **2–3 × 2-mm holes 1 cm apart** through far (lateral) radius; sutures tied over bone bridge
Suture anchors	Single	2 anchors; **distal anchor tied first** (brings tendon out to length) → maximises contact
Intraosseous (interference) screw	Single	Whip-stitch sutures through bioabsorbable tenodesis screw
Suspensory cortical button	Single	Ream anterior cortex, smaller hole far cortex, **flip button** on far cortex, tension-slide tendon in

Load	Force
Elbow 90°, no load	**50 N** (repair must withstand)
Elbow 90°, 1 kg load	112 N
Tendon rupture	200 N

Strength order **suture button (400 N) > suture anchor (380) > bone tunnel (310) > interference screw (230)**

Strongest **button + interference screw combination** · **weakest** – **interference screw alone**
✗ Wrong options: bone tunnels weakest · button alone strongest

Repair technique – anatomic (ulnar) footprint & chronic retracted tears

Insertion point repair to **ulnar aspect of tuberosity** → ↑ **supination lever arm (windlass effect)**; not more distal (off tuberosity); no flexion advantage

Chronic retracted (e.g. 5 wk, 6 cm) **primary fixation in hyperflexion is well tolerated**: **no flexion contracture / ROM loss**

Allograft augmentation only if tendon **cannot reach tuberosity even in hyperflexion**; less consistent results; not "necessary" for retraction

Chronic traps ✗ Wrong options: two-incision required (may need extensile approach) · add interference screw to button if tendon reaches · nonop equivalent

6 Complications

LABCN injury – most common complication (single-incision)

LABCN neuropraxia **most common complication overall** and **most common nerve** injured, either approach; **more common with single-incision**

Cause **aggressive / prolonged retraction**; not related to fixation type (anchor/screw/button) or timing
✗ Wrong options: direct laceration

Deficit numbness **volar-lateral (radial) forearm**

– – Active space – –

Origin **lateral cord → musculocutaneous → LABCN** (labelled musculocutaneous on plexus diagrams)

LABCN incidence	Source
9%	Orthobullets, OB
5–7%	OB (Cohen)
Nearly one-quarter	OB

Resolution **3–6 mo** (Orthobullets, most OB) · **by 3 mo** (one OB) · by 6 mo, **up to 2 y** (one OB) – read the options; **observe → gradual resolution**; permanent deficit/neuroma rare (neuroma excision + tension-free repair, traction resection, transposition)

PIN injury – two-incision technique & forearm positioning

PIN **most commonly injured motor nerve** (**1–2%**; 5% Cohen); resolves 3–6 mo

Mechanism **radially based retractors** around radial neck · prolonged traction · **Beath pin out dorsal forearm** · **entrapment by cortical button** on posterior radius

Protect PIN **maximal supination** + limit forceful lateral retraction

PIN palsy **weak wrist extension with radial deviation** (ECRL intact – radial n. proximal; ECU out – PIN) · weak thumb / **index finger extension**

PIN risk by approach **single-incision higher** (Orthobullets, OB) · **two-incision higher** (other OB explanations) · **similar** (recent meta-analysis, OB) – read the options

Complication rates (OB)

Minor (~20%)	Rate	Major (~4%)	Rate
LABCN neuropraxia	**9%**	**PIN neuropraxia**	**1–2%**
HO	3–4%	Re-rupture	1–2%
SRN palsy	**2–3%**	Deep infection	1%
Superficial infection	1%	**Radioulnar synostosis**	1%
Stiffness	1%	Ulnar / AIN injury	**<0.1%**

Nerve order LABCN > SRN > PIN > median > AIN (OB)

2nd most common nerve **SRN** = 2nd most common **cutaneous/sensory** (Orthobullets, OB) · **PIN** = 2nd most common nerve (one OB) – read the options

SRN injury numbness **dorsoradial hand / dorsal thumb**; lateral + superficial to tendon

Re-rupture **<4–5%** (OB) · **1–2%** (OB) → weak supination > flexion

HO when **IOM + ulnar periosteum disrupted**; risk of **2-incision**; often asymptomatic; some advise prophylaxis

Synostosis loss of pronation/supination; avoid exposing **ulnar periosteum** and dissection between radius + ulna; **more with 2-incision**; ↓ with intermuscular (ECU/EDC) interval

Proximal radius fracture large tunnels or tunnels **proximal to tuberosity**

Suture rupture bone tunnel method

Expanding hematoma brachial artery laceration / radial recurrent bleeding — very uncommon

– – Active space – –

Post-op deficit	Nerve
Lateral volar forearm numbness	**LABCN**
Dorsal thumb / dorsoradial hand numbness	**SRN**
Weak wrist / thumb / index extension	**PIN**
Medial volar forearm numbness	MABCN
Can't flex middle finger	Median
Can't flex thumb IP	AIN
Weak finger abduction/adduction	Ulnar

X vs Y

Proximal long-head biceps rupture

	Proximal LHB rupture	Distal biceps rupture
Deformity	**Popeye** deformity	**Reverse Popeye** (belly retracts proximally)
Hook test	–	**Abnormal** (no cord)
Main deficit	–	**Supination**

✗ Photo distractors brachial plexus palsy · pectoralis major tear · normal biceps

	Complete tear	Partial tear
Hook test	Abnormal	**Normal but painful**
Tendon	Absent	Palpable
Location	Off tuberosity	**Radial side** of footprint
MRI	Not always needed	**Needed**
Treatment	Repair (active)	Trial nonop → repair if fails

	LABCN	PIN
Type	Sensory (most common overall)	**Motor** (most common motor)
Approach	**Single-incision**	Disputed (see above)
Protection	Gentle retraction, retract laterally	**Supination**, no radial retractors

How it's asked

"Nonop distal biceps rupture – greatest loss?"	→ **forearm supination** ✗ Wrong options: elbow flexion (brachialis)
"Main indication for repair in a 45-year-old coach?"	→ **restore supination strength** ✗ Wrong options: cosmesis · prevent degeneration

– – Active space – –

"Nonop management results in all EXCEPT"	→ **lateral antebrachial neuritis**
"Partial tear MRI, nonop — outcome?"	→ **decreased supination strength** ✗ Wrong options: instability · chronic pain
"Activity hardest after nonop?"	→ **using a screwdriver** ✗ Wrong options: bench press = pec · chair push-up = LUCL
"Hook test diagnoses?"	→ **biceps tear** (TO)
"Partial tear, normal hook, failed 6 mo nonop — next?"	→ **detach + repair biceps** ✗ Wrong options: radial tunnel release · EMG
"Palpable tendon, painful hook — next step?"	→ **MRI scan** ✗ Wrong options: CT · immobilise + re-examine
"Partial tears typically occur in which part of the footprint?"	→ **radial** side
"True of the injured structure (MRI distal biceps)?"	→ **inability to hook a cord-like structure is most sensitive** ✗ Wrong options: LH distal · dual innervation · primary flexor · high re-rupture
"Orange area: distal + anterior on tuberosity?"	→ **short head of biceps** ✗ Wrong options: long head = proximal/posterior · lacertus = medial
"Healthy 44-year-old, positive hook, weak supination — management?"	→ **distal biceps repair** ✗ Wrong options: sling + PT · allograft · UCL reconstruction
"Contraction causing rupture?"	→ **muscle lengthens during contraction** (eccentric)
"Nerve injured in superficial dissection, single anterior incision?"	→ **LABCN**
"Most likely deficit after two-incision repair?"	→ **numbness lateral volar forearm** (LABCN)
"Most common complication after button repair?"	→ **LABCN irritation** ✗ Wrong options: re-rupture · synostosis · PIN
"True about surgical treatment of distal biceps rupture?"	→ **most common complication = injury to terminal branch of musculocutaneous nerve** ✗ Wrong options: button alone strongest · bone tunnels weakest
"Course of nerve in commonest neurologic complication?"	→ **pierces biceps fascia, lateral to tendon, deep to cephalic vein, superficial along BR**

"Burning radial forearm pain 3 mo after single-incision reconstruction — sequela?"	→ **gradual resolution**
"True of LABCN injury?"	→ **highest risk after single-incision** ✗ Wrong options: laceration · permanent · less common than PIN
"Sensory nerve from lateral cord injured — true?"	→ **more common single- than double-incision** ✗ Wrong options: resolves 9–12 mo · second most common
"Commonest nerve injured — brachial plexus origin?"	→ **lateral cord**
"Labelled plexus: branch of which nerve most at risk, single incision?"	→ **musculocutaneous (A)**
"Altered lateral forearm sensation after repair — contributing factor?"	→ **single-incision approach** ✗ Wrong options: fixation type · timing
"Dorsoradial hand numbness after single-incision repair — nerve relation to tendon?"	→ **lateral (radial)** (SRN)
"True of complications of distal biceps repair?"	→ **LABCN = most common complication of single-incision** ✗ Wrong options: ECU/EDC ↑ HO · screw highest load · synostosis eliminated
"Most common sensory + motor nerve deficits?"	→ **lateral forearm numbness + weak wrist extension** (LABCN + PIN)
"Arm kept maximally supinated during footprint prep protects?"	→ **PIN → wrist extension weakness**
"Most common complication, findings?"	→ **volar/radial forearm numbness**
"True of single-incision technique?"	→ **higher rate of sensory nerve injury** ✗ Wrong options: PT/FCR interval · pronation · more anatomic
"True of management (positive hook)?"	→ **single-incision uses BR/PT interval** ✗ Wrong options: anconeus/ulna interval · more flexion than supination loss
"Vessel most at risk in volar approach?"	→ **recurrent radial artery**
"Deep retractor in BR/PT interval, bright red blood?"	→ **recurrent radial artery** ✗ Wrong options: inferior ulnar collateral · recurrent interosseous
"5 weeks, 6 cm retraction — true?"	→ **primary fixation in hyperflexion well tolerated** ✗ Wrong options: allograft necessary · two-incision required
"Resident's plan: button on radial side of tuberosity — improve by?"	→ **move insertion more ulnar** ✗ Wrong options: more distal · anchors
"More ulnar insertion improves?"	→ **forearm supination**

– – Active space – –

"Photo best showing long head biceps rupture?"

→ **Popeye deformity of upper arm**
✗ Wrong options: abnormal hook = distal

– – Active space – –

DISTAL CLAVICLE OSTEOLYSIS

– Shoulder & Elbow · 11 / 60 –

1 Basics

Epidemiology

Demographics	**20s**, mostly ♂; **weightlifters** (bench press, push-ups), overhead athletes, wrestlers
Risk factor	history of **trauma** (acute injury) or **repetitive microtrauma**
Adolescents (OB)	mean age 15.9 y in one series; overhead sports / weightlifting

Anatomy

Clavicle	S-shaped; **last bone to fuse** (medial physis fuses early 20s)
AC joint	**diarthrodial** with **fibrocartilage meniscus**

Ligament	Stability
AC ligaments (esp. posterosuperior capsule)	**Horizontal**
CC ligaments	**Vertical**

2 Mechanism

Pathoanatomy	**repetitive stress + micro-fractures** of distal clavicle → **osteopenia + bone resorption**
Load	bench press / manoeuvres with **elbows posterior to torso** → traction through AC joint

3 Presentation

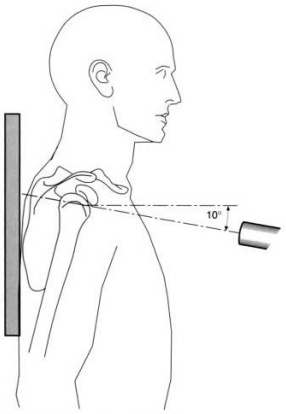
Pain	**distal clavicle / anterosuperior shoulder**; **insidious**; ↑ with repetitive loading (bench press, push-ups); mimics AC arthritis
Exam	**point tenderness at AC joint / distal clavicle** · **pain with cross-body adduction** · pain without weakness on resisted forward flexion + adduction · **full ROM**

4 Diagnosis & Classification

Key radiology findings

Views	**AP clavicle** + **Zanca view (15° cephalad tilt)** (Orthobullets) · **10–15°** (TO, OB) – read the options

– – Active space – –



Zanca view technique

X-ray (distal clavicle only, not acromion)	**cysts**, **osteopenia**, **resorption + erosion**, **tapering** of distal clavicle, **AC joint widening**



Distal clavicle osteolysis, AP

MRI	**↑ T2 signal + bone marrow oedema** in distal clavicle / AC joint (fluid-sensitive)
Bone scan	**↑ uptake** distal clavicle; **earlier than X-ray changes**
Follow-up MRI (OB)	mean **5 mm AC widening**

5 Management

- **Distal clavicle osteolysis**
  - First line → **rest + NSAIDs + PT + modify lifting technique**
    - persistent → **corticosteroid injection (diagnostic + therapeutic; US-guided)**
  - Failed nonop → **distal clavicle excision (open or arthroscopic)**
    - address cuff + LHB pathology at same time

Nonoperative

Activity modification	avoid aggravating lifts; **narrow hand grip**; **stop bar descent 4–6 cm above chest**; avoid elbows behind torso
Steroid injection	**diagnostic + therapeutic**; **more accurate with US**
Success (OB)	**93%** respond to conservative treatment (adolescent series)

Distal clavicle excision

Indication	persistent symptoms after **failed nonop** (rest, PT, cortisone)
Mumford (arthroscopic)	resect **only 0.5–1 cm**; **too much → AC instability**
Open vs arthroscopic	surgeon preference; arthroscopic **comparable pain relief**, ↓ morbidity, fewer restrictions, **quicker return**, better cosmesis, **evaluates GH joint** + treats concomitant pathology
Open	meticulous **deltoid-trapezial fascia** repair
Results (OB)	Auge: **10 weightlifters**, limited arthroscopic resection → **return to sport 3.2 d**, weight training **9.1 d** · Zawadsky: 22 excellent, 16 good, **3 failures – all traumatic aetiology**



After arthroscopic distal clavicle excision

6 Complications

Horizontal instability	**violating posterosuperior capsule / AC ligaments** during excision
AC instability	**over-resection** (beyond 0.5–1 cm)

X vs Y

	Distal clavicle osteolysis	AC separation	SLAP / biceps (wrestler)	Cuff tear
Key clue	AC tenderness, cross-body pain, **no weakness**, full ROM	**Instability**	Common in wrestlers, but exam ≠ AC	**Weakness**
Imaging	Lysis, cysts, tapering; marrow oedema	–	Labral pathology	Tendon tear
Surgery	**Distal clavicle excision**	**CC ligament reconstruction ± DCE**	SLAP repair / tenodesis	Cuff repair

– – Active space – –

How it's asked

"Bodybuilder, cross-body pain, AC tenderness, X-ray lysis — most successful treatment?"	→ **arthroscopic resection of the distal clavicle** ✗ Wrong options: GH injection · periscapular strengthening · labral repair · capsular release
"Wrestler, failed rest + PT + cortisone, AC tenderness, MRI marrow oedema — next step?"	→ **distal clavicle excision** ✗ Wrong options: SLAP repair · subpectoral tenodesis · CC reconstruction
"Best early imaging before X-ray changes?"	→ **bone scan**
"Complication of excessive / posterosuperior capsular resection?"	→ **horizontal AC instability**
"How much clavicle to resect?"	→ **0.5–1 cm**

– – Active space – –

ELBOW ANATOMY & BIOMECHANICS

– Shoulder & Elbow · 12 / 60 –

1 Basics

Joints & function

3 articulations	**ulnohumeral** (hinge) · **radiocapitellar / radiohumeral** (pivot) · **proximal radioulnar**
Function	lever arm positioning the hand · fulcrum for forearm lever

Range of motion & load sharing

Parameter	Value
Total flexion–extension	**0–150°**
Functional flexion–extension	**30–130°**
Functional pronation/supination	**50° each**
Carrying angle (valgus)	♂ **5–10°** · ♀ **10–15°** (anatomy note) · ~11° ♂ / ~13° ♀ (exam note) · 6–11° (UCL note); ↓ with flexion
Axial load, extended elbow	**40% ulnohumeral · 60% radiohumeral**

Osteology

Spiral groove	posterior humeral shaft; **radial nerve**; **~13 cm** proximal to trochlear articular surface
Distal flare	medial + lateral epicondyles; flare = half of elbow joint · **medial + lateral columns**
Trochlea	spool-shaped, **medial** · **capitellum** – lateral
Sublime tubercle	medial **coronoid** process; distal attachment of **anterior bundle of MCL**

– – Active space – –

Arthrology

Feature	Value
Articular surface **anterior tilt**	**~30°** relative to humeral shaft
Valgus	**6°**
Rotation	**5° internal**
Axis of flexion	Centre of **trochlea + capitellum**; passes through **anteroinferior medial epicondyle**
Radial head cartilage	Covers **~240°**; **lateral 120° bare** (safe zone for radial head fixation)
Capsule	**Max distension 70–80° flexion**; anterior capsule inserts **6 mm distal to coronoid tip → coronoid intra-articular**
Coronoid fossa	Receives coronoid tip in deep flexion
Coronoid tip	**Buttress against posterior dislocation**

Muscles of the elbow

Flexors **biceps** (radial tuberosity) · **brachialis** (inserts **11 mm distal to coronoid tip**) · brachioradialis

Extensor triceps

Collateral ligaments – MCL anterior bundle & LCL complex

Stabiliser tier	Structures
Primary static (3)	**Ulnohumeral joint (coronoid) · MCL (anterior bundle) · LCL complex**
Secondary static	**Radiocapitellar joint · capsule · flexor + extensor origins**
Dynamic	Muscles crossing elbow: **anconeus, brachialis, triceps, biceps** → compression

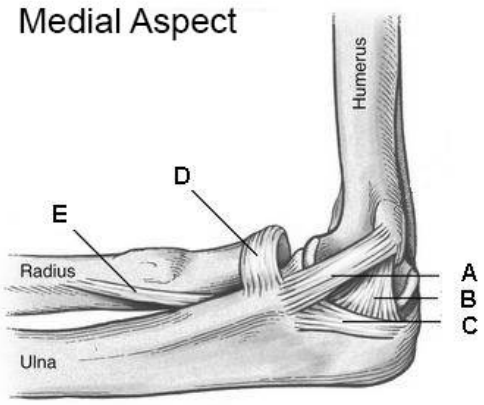
Coronoid loss of **≥50% height** → instability

MCL anterior, posterior, transverse bundles; resists **valgus + distraction**; origin **anteroinferior medial epicondyle** → insertion **sublime tubercle**

MCL bundle	Role
Anterior	**Most important valgus restraint** (prime medial stabiliser); esp. in flexion (Morrey)
Posterior (posterior oblique)	**Floor of cubital tunnel · primary valgus restraint in maximal flexion** · secondary medial stabiliser · tension ↑ **60–120° flexion** → contracture **limits flexion**
Transverse	**No role** in valgus stability or ROM

2nd most important valgus restraint **radial head**
✗ Wrong options: ulnotrochlear articulation

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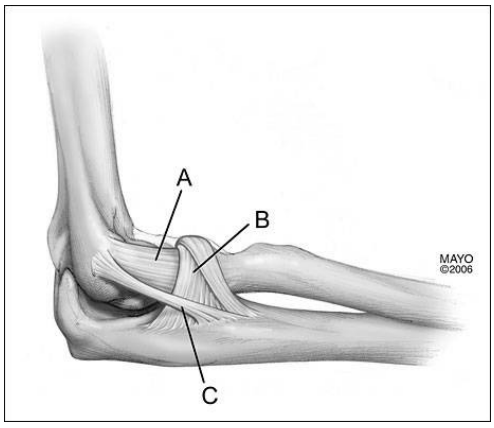


MCL – A anterior bundle, B posterior oblique band, C transverse, D annular, E oblique cord

LCL complex — **RCL** · **LUCL** · **accessory collateral ligament** · **annular ligament**; whole complex = **capsular thickening**; arises from **isometric point on lateral capitellum**

LCL component	Key fact
RCL	**Originates at axis of rotation for flexion**
LUCL	**Primary restraint to varus + external rotation** through full arc; origin **posterior lateral epicondyle** → insertion **crista supinatoris** (✗ Wrong options: sublime tubercle)
Accessory collateral	± substantial lateral stability (with RCL)
Annular	**PRUJ** stability

LCL repair — optimal stability with **appropriately tensioned** repair



LCL complex – A RCL, B annular, C LUCL

Not LCL complex — ulnohumeral joint (primary but not LCL) · radiocapitellar joint (secondary) · anterior bundle UCL · **transverse ligament (MCL complex)**

LUCL injury — abnormal **supination of ulna** on humerus → radial head subluxes **posteriorly** → **PLRI** (external rotatory subluxation of ulna + valgus on trochlea)

LCL disruption — **uniform** in dislocations + fracture-dislocations; commonest pattern = **avulsion from distal humerus** (McKee)

Radiocapitellar joint — important valgus constraint; radial head = **~30% of valgus stability**, most at **0–30° flexion / pronation**

– – Active space – –

Capsule greatest stability contribution **in extension** · anterior capsule resists varus-valgus **in extension, not flexion** (Morrey)

Nerves of the elbow

Nerve	Origin	At the elbow
Musculocutaneous	Lateral cord	Between **biceps + brachialis** (supplies both); exits laterally distal to biceps tendon → **LABC deep to cephalic vein**
Radial	Posterior cord	**Triangular interval** (teres major, long head triceps, humerus) → spiral groove **13 cm** above trochlea → pierces **lateral IM septum 7.5 cm** above trochlea (middle/distal third) → between **brachialis + brachioradialis** → superficial to capsule at radiocapitellar joint
Median	Medial + lateral cords	With brachial artery (lateral → medial); superficial to brachialis; **no branches in arm**
Ulnar	Medial cord	Medial to brachial artery → pierces medial IM septum (**arcade of Struthers**) → cubital tunnel; **no branches in arm**; first FCU branch **distal to elbow**

Blood supply

Brachial artery medial in arm → enters cubital fossa → splits into **radial + ulnar** arteries

Cubital fossa contents lateral → medial: **biceps tendon, brachial artery, median nerve** · borders: **brachioradialis** (lateral), **pronator teres** (medial), distal humerus (proximal)

Branches superior/inferior ulnar collateral · nutrient/muscular · supratrochlear

2 Mechanism

Axis of forearm rotation

Flexion–extension axis centre of trochlea; line through isometric points of capitellum + trochlea

Pronation–supination axis **centre of radial head → fovea of ulnar head** (capitellum → radial head → distal ulna); forms a **cone**

✗ Axis distractors radial head–radial styloid · radial head–ulnar styloid · coronoid–sigmoid notch · coronoid–radial styloid

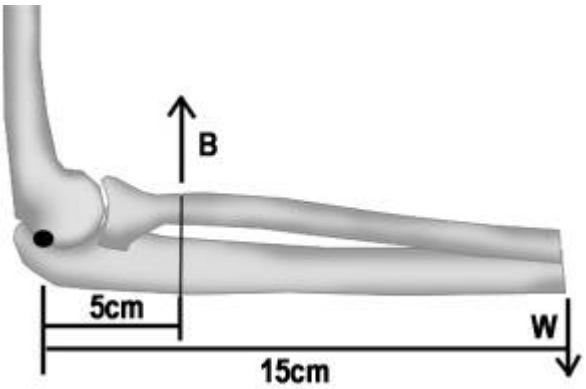
Forearm rotation radiocapitellar + PRUJ + DRUJ + IOM + forearm muscles · **pronators**: pronator teres + quadratus · **supinators**: biceps + supinator

Biomechanics calculation – moment arms & anchor strength

Joint reaction force large; **short, inefficient lever arms** (biceps inserts near COR) → degenerative change · static loads ≈ body weight · **dynamic loads > body weight**

Free body $\Sigma M = 0$ → **B × 5 cm = W × 15 cm → B = 3W**

– – Active space – –



Free body diagram – biceps 5 cm, weight 15 cm

Anchor problem arm 20 N at 15 cm, biceps 5 cm → biceps force **60 N** × safety factor 3 = **180 N** minimum pullout

3 Presentation

Olecranon bursitis (student's elbow)

Stem (TO) direct blow posterior elbow + **olecranon spur** → continued sport → **painless swelling with effusion** over elbow

Answer (TO) key **not provided by source**; module label = **olecranon bursitis (student's elbow)** · options also list miner's elbow, little leaguer's elbow, golfer's elbow – read the options

5 Management

Posterior bundle of MCL – release for flexion

Release for flexion loss **posterior (oblique) band of MCL + posterior capsule**; flexion 89° → **127°** (Wada; medial incision, ulnar nerve freed)

Limited extension **anterior capsulectomy**

✗ Release distractors **anterior band MCL** (→ valgus instability) · transverse band (no ROM gain) · **annular ligament** (never) · oblique cord of IOM

Arthrodesis position

Situation	Position
Unilateral	**90° flexion**, **0–7° valgus**
Bilateral	One at **110°** (feeding) · other at **65°** (perineal hygiene)

Diagnostic injection

Soft spot triangle of **lateral epicondyle + olecranon + radial head** → intra-articular injection

Complex instability (OB)

Simple dislocation closed reduction + short immobilisation; recurrent instability **uncommon**

– – Active space – –

Complex / persistent instability fix or replace bony elements + ligament repair ± **hinged fixation**

X vs Y

	MCL (anterior bundle)	LCL complex (LUCL)
Resists	**Valgus** + distraction	**Varus + external rotation**
Origin	**Anteroinferior medial epicondyle**	**Posterior lateral epicondyle** / isometric point capitellum
Insertion	**Sublime tubercle**	**Crista supinatoris**
Failure	Valgus instability	**PLRI**
Backup	**Radial head** (2nd valgus restraint)	–

	Posterior bundle MCL	Anterior bundle MCL
Valgus role	Primary only in **max flexion**	**Most important overall**
Release for stiffness?	**Yes – restores flexion**	**No –** valgus instability
Cubital tunnel	**Floor**	–

How it's asked

"Primary stabiliser against valgus (labelled figure)?"	→ **anterior bundle MCL** ✗ Wrong options: posterior bundle = secondary · transverse · annular · biceps
"Structure A on lateral ligament figure?"	→ **radial collateral ligament** (B annular, C LUCL; transverse = MCL complex)
"Sublime tubercle is the insertion of?"	→ **anterior bundle MCL**
"True statement about elbow stability?"	→ **RCL originates at axis of flexion** ✗ Wrong options: LUCL on sublime tubercle · anterior bundle primary in max flexion · ulnotrochlear 2nd valgus restraint
"Varus + external rotation instability – which primary stabilisers incompetent?"	→ **LUCL, accessory collateral, annular, RCL** ✗ Wrong options: ulnohumeral · radiocapitellar · transverse ligament
"Release to restore flexion?"	→ **posterior oblique band MCL (B)** ✗ Wrong options: anterior band · annular
"Axis of forearm rotation?"	→ **radial head → ulnar head**
"Anchor pullout, 20 N at 15 cm, biceps 5 cm, safety 3?"	→ **180N**
"Posterior elbow blow, olecranon spur, painless effusion?"	→ **student's elbow** (olecranon bursitis; key not provided, TO)
"Optimal unilateral arthrodesis?"	→ **90° flexion**

"Safe zone for radial head fixation?" → **lateral 120° non-articular**

– – Active space – –

– – Active space – –

ELBOW ARTHRITIS

– Shoulder & Elbow · 13 / 60 –

1 Basics

Types & epidemiology

3 types **osteoarthritis (primary)** · **post-traumatic** · **inflammatory**

	Primary OA	Post-traumatic	Inflammatory (RA)
Frequency	**Rare** (symptomatic ~**2%**)	**2nd most common**	**Most common / most prevalent** elbow arthritis (historically)
Patient	**Middle-aged ♂ laborer**, **♂:♀ 4:1**, 20–70 y (**avg 50**), **dominant arm**	**Younger** than other types	Older ♀; elbow involved in **20–50%** of RA
Cause	Strenuous manual labour	Non-op radial head fracture, (fracture-)dislocation, traumatic instability	RA, psoriatic, SLE, PVNS
Joint	**Radiocapitellar** preferentially; **ulnohumeral relatively spared**	Whole joint or focal ulnohumeral/radiocapitellar	Global; symmetric narrowing
Hallmark	**Osteophytes, capsular contracture, loose bodies**	Incongruity → altered load; stiffness, instability, malunion/ nonunion	Synovitis, **osteopenia**, ligament attenuation, instability

Secondary OA causes post-traumatic · **OCD** · **synovial osteochondromatosis** · **MUCL insufficiency / valgus extension overload**

Genes & lab markers

RA pathology **pannus → hyperplastic synovium → collagen + proteoglycan degradation** → chronic synovitis, ligament attenuation, **periarticular osteopenia**, contracture

RA trap synovial proliferation from pannus, **not rheumatoid factor mutation**

RA pathoanatomy fixed flexion contracture · instability · **ulnar** (or less often radial) neuropathy · erosions · cysts · deformity · progressive bone resorption

Stabilisers (relevance to surgery)

Primary **anterior band MCL** (**anterior oblique fibres** most important; valgus + distraction) · **LCL** · olecranon–coronoid–trochlea congruity

Secondary **radial head** (most important secondary; **30%** valgus stability; **0–30° flexion + pronation**) · capsule (**distraction in full extension**) · anconeus + lateral capsule (varus)

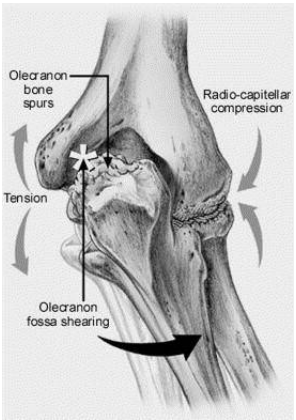
3 Presentation

OA symptoms	pain at **end range** · **loss of terminal extension** · painful **locking/catching** (loose bodies) · night pain unusual ✗ Wrong options: mid-range
OA exam	forearm rotation preserved early · **ulnar neuropathy in up to 50%**
RA	**hand + wrist precede elbow** · fixed flexion contracture · ligamentous incompetence · ± ulnar neuropathy · rheumatoid nodule · **evaluate C-spine in all RA**

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
X-ray	**AP + lateral**; **C-spine X-rays for RA before surgery**
OA osteophytes	**Coronoid process + fossa** · **radial head + fossa** · **olecranon tip + posteromedial olecranon fossa**
Joint space	Narrowing; **ulnohumeral relatively preserved** in OA
Loose bodies	**Underestimated on plain X-ray**
RA	**Periarticular erosions + cysts**, osteopenia; **osteophytes not prominent** (✗ Wrong options: characteristic)
CT	Surgical planning; better defines **osteophytes + loose bodies**



Olecranon osteophyte impinging in olecranon fossa

Larsen grading (RA)

Stage	Radiographs
1	Soft tissues involved; **near-normal** radiographs
2	**Periarticular erosions + mild cartilage loss**; soft-tissue swelling, osteopenia
3	**Marked joint space narrowing**
4	Advanced erosions **penetrating subchondral plate**
5	Advanced damage, **loss of articular contour**

– – Active space – –

TEA threshold **Larsen 3–5** in patients >65 y

5 Management

Elbow arthritis – failed NSAIDs, injection, splint, activity modification

- Mild–moderate OA, arc >90°, painful end-range, mechanical symptoms → **arthroscopic OCA**
- Severe contracture (e.g. 40–90°), prior ulnar transposition → **open: O-K or column procedure**
- Young (<65), high demand, END-STAGE (OA/RA/PTA), stable → **interposition arthroplasty**
  - isolated radiocapitellar → **anconeus interposition / radial head replacement**
- >65, low demand, severe (Larsen 3–5) → **semiconstrained TEA**

Primary OA – osteocapsular arthroplasty (arthroscopic vs open) & patient selection

Nonop NSAIDs · cortisone injection · resting splints · activity modification → **mild–moderate** symptoms

OCA removal of osteophytes (coronoid, olecranon, fossae) + loose bodies + **capsular release**

Best OCA candidate **<65 y**, active, **arc >90°**, **mild–moderate** OA, **painless mid-arc + painful end-range**

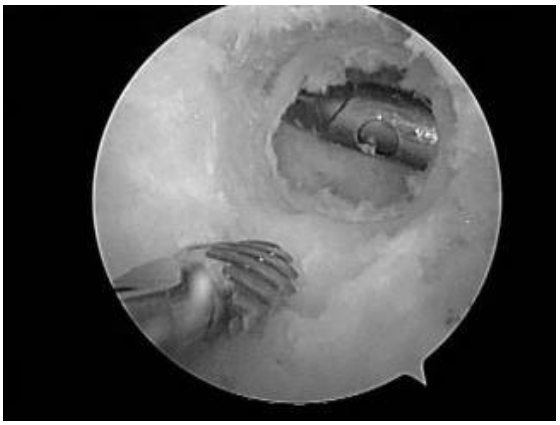
Arthroscopic debridement / release	
Indications	**Mechanical symptoms (loose bodies)** · capsular contracture · **bony block** to motion · **>90° motion** preferred
Contraindications (relative)	**Prior ulnar nerve transposition** (nerve location unknown) · **severe contracture / arthrofibrosis** · prior extensive open procedures
Not a CI	Prior **olecranon osteotomy** · ulnar nerve **subluxation** (✗ Wrong options: absolute CI; identify pre-op, esp. prior submuscular/intramuscular transposition)
Complications	**Neurologic** (**ulnar** palsy commonest, **radial** 2nd – anterolateral portal) · **synovial fistula** · recurrent stiffness

Arthroscopic vs open OCA	**Arthroscopic**
Post-op pain	↓
HO / ankylosis	↓ (✗ Wrong options: equal)
Access	**Easy anterior + posterior**, better cartilage view
Reliability	**Pain relief** more reliable (open → more predictable **ROM** gain)

O'Driscoll & Blonna 8 steps posterior view → remove debris + fat pad → osteophytes → posterior capsule release → anterior view → strip anterior capsule → coronoid osteophytes/reshape → excise anterior capsule; **4.8% transient nerve injury** (502 pts)

Elbow arthroscopy indications septic I&D · synovectomy (inflammatory) · debridement + loose body removal (OA) · contracture release · OCD capitellum · lateral epicondylitis release

Advanced radiocapitellar OA	consider **radial head replacement**
Avoid radial head resection in laborer	↑ load on arthritic ulnohumeral joint; doesn't fix ROM or end-range impingement
No PT / dynamic splints	no role initially in advanced painful arthritis with **fixed** contracture · no static splinting when arc functional + complaint is pain
Fusion	severe functional loss; rare (unmanageable **sepsis**, **Charcot**)



Outerbridge–Kashiwagi olecranon fossa fenestration

Outerbridge–Kashiwagi (ulnohumeral arthroplasty)	open **posterior** approach; **burr fenestration of olecranon fossa** → release posterior impingement + anterior access; younger pts with ↓ ROM; **decompress ulnar nerve if pre-op flexion contracture**; failure = missed anterior / peripheral (medial + lateral olecranon) osteophytes
Column procedure	medial or lateral open capsular release + bony resection for extrinsic contracture; **lateral most common** (access to all 3 joints) via lateral supracondylar ridge · anterior arthrotomy **ECRL/common extensor interval, stay anterior to LUCL** · posterior: elevate triceps · **go medial to gain flexion** (excise **posterior band MCL**)
Open release	elderly regain motion **more reliably than <20 y** (Strubel & Cohen)

Ulnar nerve – neuritis & decompression when gaining flexion

Risk	severely limited pre-op motion: **flexion <100°** · **extensor lag / max extension >60°** → **post-op ulnar nerve dysfunction**
Rule	any motion-restoring procedure (OCA, O-K, column, interposition) in such patients → **concomitant ulnar nerve decompression ± transposition**
Not radial neuritis	radial tension ↓ with gained extension
Not column procedure add-ons	release anterior band MUCL (primary valgus) · release LUCL (primary varus) · radial head resection (rotation full + painless) · **TFL interposition** (reserved **<50 y**)

Young post-traumatic arthritis – interposition arthroplasty

Indication	**young, high-demand, END-STAGE** OA / RA / **post-traumatic** who would otherwise get TEA if older; **no lifting restrictions**; needs **preserved bone stock + inherent stability**
Technique	burr distal humerus + proximal ulna congruent → graft over distal humerus (**Achilles allograft**, **autogenous TFL**, dermal/fascial) → ligament repair → **hinged distracting ex-fix** (offloads graft, early ROM, protects repairs)

– – Active space – –

Advantage can **convert to TEA** later (successful; Blaine)

Interposition contraindication	Type
Elbow instability / chronic instability without bone stock for ligament reconstruction	**Absolute**
Active infection	**Absolute**
Elderly **>60 y** (TEA candidates)	Relative
Bone loss: **destructive JRA**, **distal humerus osteonecrosis**	→ prosthetic replacement
Not CI	Young age · post-traumatic · inflammatory · previous debridement

Anconeus interposition arthritis **isolated to radiocapitellar** joint

Failed interposition (68 y, pain + instability) **convert to TEA**
✗ Wrong options: revision interposition · hemiarthroplasty (ulnar side arthritic) · arthrodesis

Rheumatoid elbow – TEA & synovectomy

Synovectomy open or arthroscopic; **young RA, joint space preserved**; most improve

Radial head resection + synovectomy only if synovitis **encircles radial head/neck** + painful crepitus on rotation + **PIN signs** (can't extend thumb)

Complication of radial head resection **rapid progression of ulnohumeral arthritis**
✗ Wrong options: loss of pronation/supination · longitudinal instability with intact IOM · ulnar palsy (less likely)

End-stage RA **semiconstrained TEA** (concomitant ligamentous instability)
✗ Wrong options: constrained (**high early loosening**) · arthroscopy (incomplete, unpredictable)

TEA outcome RA vs OA **equal or better** for RA
✗ Wrong options: poorer

Operative criteria – TEA

TEA	
Indications	**>65 y**, severe arthritis (**Larsen 3–5**) · complex distal humerus fracture, elderly, poor bone · distal humerus non/malunion, elderly low demand · post-traumatic arthritis (lower demand) · **failed interposition**
Contraindications	**Active <65 y** · **infection** · **Charcot joint**
Restriction	**Lifelong 5–10 lb lifting** limit

TEA design	Feature
Unconstrained	Needs **competent collaterals**, soft-tissue envelope, good bone
Semiconstrained	**"Sloppy hinge"** → some varus-valgus + rotational laxity → ↓ bone-cement stress → ↓ loosening; **best longevity + function**
Constrained	**High early loosening**

Elbow arthrodesis position (TO)

– – Active space – –

Arthrodesis	Position
Unilateral (TO)	**90° flexion + neutral to slight valgus** (✗ Wrong options: varus · 110°)
Bilateral (TO)	One **110°** (feeding) · other **65°** (perineal hygiene)

6 Complications

Total elbow arthroplasty complications

Rate up to **43%** (Orthobullets) · up to **25%** (OB) – read the options

Bushing wear **most common mode of failure of semiconstrained TEA** at midterm, esp. **post-traumatic indication** (younger, non-adherence to restrictions, unstable injury, deformity, contracture)
✗ Wrong options: gender · smoking · cement · age



Bushing wear in semiconstrained TEA

Infection / wound risk factors prior elbow surgery · prior infection (**S. epidermidis**) · psychiatric comorbidity · **RA** · wound drainage · reoperation · poor skin (**long-term steroids**); two-stage revision → poor survival

Aseptic loosening **linked** implants · **post-traumatic** OA

Instability **unlinked** implants

Others **ulnar neuritis** (iatrogenic or after motion restored without decompression) · **triceps avulsion/insufficiency** · fracture · mechanical failure

– – Active space – –

X vs Y

	Arthroscopic OCA	Interposition arthroplasty	TEA
Stage	**Mild–moderate**	**End-stage**	**End-stage**
Age / demand	Active, <65	**Young, high demand**	**>65, low demand**
Motion	Arc **>90°**	Any; needs stability	Any
Key CI	**Prior ulnar transposition**, severe contracture	**Instability**, infection	**Active <65**, infection, **Charcot**
Restriction	–	**None**	**5–10 lb lifetime**

	Elbow OA	Elbow RA
X-ray	**Osteophytes**, loose bodies, ulnohumeral spared	**Erosions, cysts, osteopenia**, symmetric narrowing
Symptom	**End-range** pain	Diffuse pain, effusion
Instability	No	**Ligament attenuation**
End-stage op	OCA / interposition / TEA	**Semiconstrained TEA**; synovectomy early

How it's asked

"Plumber, advanced OA, lacks 45° extension, flexes to 110° – initial treatment?"	→ **osteophyte resection + capsular release** ✗ Wrong options: PT/dynamic splints · TEA · fusion · radial head resection
"Carpenter, mid-range locking, arc 35–130°, failed injection?"	→ **arthroscopic capsular release + loose body removal + osteophyte decompression** ✗ Wrong options: interposition · radial head arthroplasty · TEA
"Firefighter 58, arc 30–130°, pain – most predictable pain relief?"	→ **arthroscopic debridement** ✗ Wrong options: open debridement = ROM · static splint · interposition
"True of arthroscopic vs open OCA?"	→ **less post-op pain** ✗ Wrong options: equal HO · limited access · radial palsy commonest · subluxation absolute CI
"Relative CI to arthroscopic debridement in post-traumatic OA?"	→ **prior ulnar nerve transposition** ✗ Wrong options: loose bodies · contracture · bony block · olecranon osteotomy
"Who benefits most from arthroscopic OCA?"	→ **50-y ♂, painful limited end-range** ✗ Wrong options: RA destruction · end-stage · elderly pain throughout

"Best OCA candidate?"	→ **51-y mailman, mild OA, arc 30–135°** ✗ Wrong options: 40–90° contracture → open · prior transposition · end-stage → interposition
"OCA for 50–90° arc — specific complication?"	→ **ulnar neuritis** ✗ Wrong options: radial neuritis · implant wear
"Lateral column procedure, pre-op arc 30–90° — add?"	→ **ulnar nerve decompression ± transposition** ✗ Wrong options: release anterior MUCL · LUCL · radial head resection · TFL
"Distraction interposition arthroplasty best for?"	→ **41-y laborer, post-traumatic arthritis** ✗ Wrong options: destructive JRA · osteonecrosis · 70-y RA
"Failed interposition at 68 y with pain + instability?"	→ **convert to TEA**
"Contraindication to interposition arthroplasty?"	→ **elbow instability** ✗ Wrong options: young age · PTA · inflammatory · prior debridement
"40-y inflammatory end-stage, stable, arc 75°?"	→ **Achilles interposition** ✗ Wrong options: anconeus = isolated radiocapitellar · TEA
"Burred surfaces + Achilles allograft + hinged ex-fix — best candidate?"	→ **20-y severe post-traumatic arthritis** ✗ Wrong options: active infection · chronic instability · elderly → TEA
"61-y RA, end-stage destruction?"	→ **semiconstrained TEA** ✗ Wrong options: constrained (early loosening)
"RA elbow pathology — true?"	→ **proteoglycan + collagen destruction** ✗ Wrong options: osteophytes · worse TEA results · unconstrained · RF mutation
"RA synovectomy + radial head resection — likely complication?"	→ **rapid ulnohumeral arthritis**
"Bushing wear at 6 y — risk factor?"	→ **indication for TEA (post-traumatic)** ✗ Wrong options: gender · smoking · cement · age
"Optimal unilateral elbow arthrodesis?"	→ **90° flexion + slight valgus** (TO)

– – Active space – –

– – Active space – –

ELBOW ARTHROSCOPY: INDICATIONS & APPROACH

– Shoulder & Elbow · 14 / 60 –

1 Basics

Indications

Source	Indications
Orthobullets	**Loose body removal**, osteophyte debridement, **synovectomy**, **capsular release (stiffness)**, **OCD capitellum**, lateral epicondylitis, **septic arthritis** debridement
TO	Loose bodies · **OCD capitellum** · osteochondral lesions radial head · osteophytes humerus/olecranon · **RA (partial synovectomy)** · stiff elbow (debridement, lysis of adhesions) · **tennis elbow release** · evaluation of painful elbow · **inflamed olecranon bursa** · **intra-articular fractures**

Osteocapsular arthroplasty — young patient with degenerative elbow where **TEA inappropriate**, or significant contracture

Advantages — better articular visualisation, ↓ post-op pain, **faster recovery**

Disadvantages — technically demanding; **high risk of NV injury** (proximity to joint)

Contraindications (prior ulnar nerve transposition)

Source	Contraindications
Orthobullets	Prior trauma, surgical scarring, **previous ulnar nerve transposition**
TO	**Bony ankylosis** · **severe fibrous ankylosis** (scope can't enter safely) · **active infection** · surgery altering anatomy e.g. **anterior ulnar nerve transposition (relative)**
OB	**Ulnar neuropathy with prior submuscular transposition** = contraindication

Not contraindications (OB) — **Panner disease** · loose body ulnohumeral joint · **s/p radial head ORIF** · elbow stiffness · **osteoarthritis**

Ulnar nerve subluxation — **not an absolute contraindication**; identify pre-op (esp. prior submuscular/intramuscular transposition)

Capsular anatomy & distension (TO)

Capsule (TO) — attached to articular margins; fibres blend with **annular ligament**; anterior → above coronoid + radial fossae; posterior → above olecranon fossa

Tension (TO) — taut **anteriorly in extension**, **posteriorly in flexion**

Maximum distension (TO) — **70–80° flexion**; capacity **25–30 mL**
✗ Wrong options: 90–120° · 10° · 120–150°

5 Management

Positioning & joint distension (TO)

Positions	**supine** (suspended), **prone**, **lateral decubitus**
Most commonly used (TO)	**lateral**, elbow hanging freely on support (**flexed**)
All positions = elbow flexed (TO)	**elbow extended, prone** = not a position
Anaesthesia	**general** (muscle relaxation, tolerates prone/lateral) · regional: **no immediate post-op nerve check**, poor tolerance of position

Portal technique

Distend first	**fully distend joint through lateral soft spot** → NV structures move away from trocar
Nick and spread	skin only with blade, **hemostat spread** to capsule, **blunt trocar**
Elbow position	**anterior portals at 90° flexion** · posterior portals in some extension
Kit	standard **30° scope** · tourniquet · **solid cannulas** → maintain distension, ↓ extravasation ✗ Wrong options: trephinated
Landmarks	olecranon, both epicondyles, radiocapitellar joint, **ulnar nerve**; **mark before insufflation** (distension distorts)
Order	distend → viewing portal (medial or lateral) → working portal under **direct vision + needle localisation**
First portal	**anterolateral** after distension (TO, OB) · establishing a **medial portal first** (before swelling) may be advantageous (Orthobullets) — read the options
Viewing vs working (TO)	**anterolateral = viewing**; **proximal anteromedial = most common working portal** + most commonly used medial portal
Posteromedial portal	**avoided** (lies over **ulnar nerve**)

Factors that ↓ NV injury (TO)

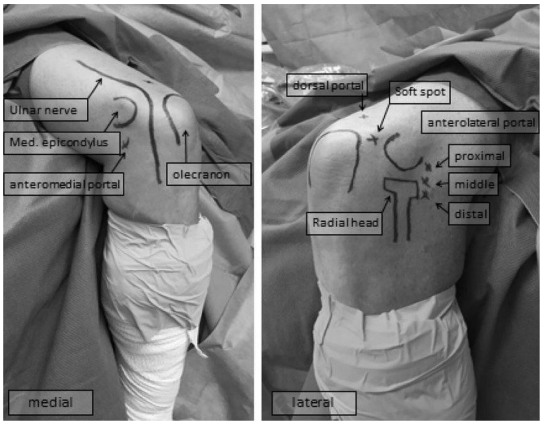
✓ **Manoeuvre (TO)**	**Protects**
90° flexion + joint distension	Radial nerve moves anteriorly
Pronation	**PIN**
Keep lateral portal more proximal	**Radial nerve / PIN**
Avoid deep knife; hemostat + blunt trocar	Lateral + posterior antebrachial cutaneous
Draw landmarks, **palpate ulnar nerve**	Ulnar
Low-flow infusion 40–45 mL/min or gravity (✗ Wrong options: pressurised)	Compartment syndrome
Instrument tips always in view; no suction near nerves	All

NOT a protective manoeuvre (TO)	**more distal lateral portal** (risk ↑ with distal placement)

– – Active space – –

Portals & nerves at risk

Portal	Location	Use	Nerves at risk
Proximal anterolateral	**1–2 cm proximal, 1 cm anterior** to lateral epicondyle	Views anteromedial joint; **safest anterolateral** (TO)	**Radial** (↓ risk more proximally)
Mid anterolateral (TO)	Just proximal, ~1 cm anterior to radiocapitellar joint	–	–
Distal anterolateral	**1–3 cm distal, 1 cm anterior** to lateral epicondyle; anterior to radial head/neck	1st portal supine; radial head, medial elbow, coronoid, trochlea, brachialis insertion, coronoid fossa	**PIN** + **lateral antebrachial cutaneous**; **radial at maximum risk** (TO) → rarely used
Direct lateral (soft spot)	Triangle **olecranon – radial head – lateral epicondyle**	**Initial distension**; posterior compartment (capitellum, radial head, radioulnar joint)	**Relatively safe**; LABC
Anteromedial (direct medial)	**2 cm anterior + 2 cm distal** to medial epicondyle; through flexor-pronator	Augments proximal AM; medial recess; place under direct vision	**MABC + median** (+ basilic vein, TO)
Proximal anteromedial (superomedial)	**2 cm proximal** to medial epicondyle, **anterior to IM septum**	Entire anterior compartment, radial head, capitellum, coronoid, trochlea; **most common working portal** (TO)	**MABC**; ulnar (7 mm, posterior to septum) + median (OB, TO)
Straight posterior (transtriceps)	**3 cm proximal** to olecranon, midline through triceps	Partial extension; **olecranon osteophytes**, posteromedial loose bodies	**Posterior antebrachial cutaneous**; ulnar **15–25 mm** away (OB)
Posterolateral	**2–3 cm proximal** to olecranon, **just lateral to triceps** (centre of anconeus triangle)	**20–30° flexion** (relax triceps); **best posterior compartment**, radiocapitellar (**OCD capitellum**), olecranon fossa	PABC, medial brachial cutaneous, ulnar (Orthobullets) · PABC + **lateral brachial cutaneous** (TO)
✗ Wrong option: **posteromedial**	Medial to triceps	**Not used**	**Ulnar nerve** – unacceptably high



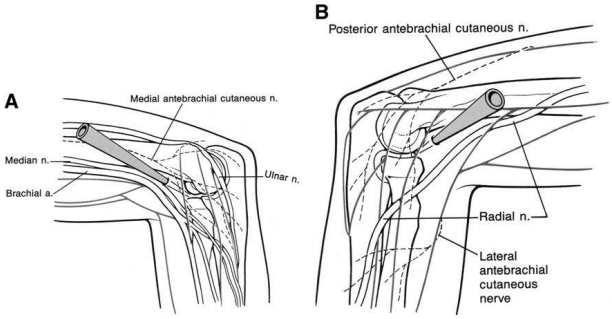
Standard elbow arthroscopy portals marked

– – Active space – –

Proximal anterolateral location	**1–2 cm proximal** (Orthobullets) · **2 cm proximal** (TO, OB) · **3 cm proximal** (one OB explanation) — read the options
Distal anterolateral location	**1–3 cm distal** (Orthobullets) · **2–3 / 3 cm distal** (TO) · **1–2 cm distal** (OB)
Straight posterior location	**3 cm proximal** (Orthobullets, OB) · **2–3 cm proximal, 2 cm medial to posterolateral** (TO)
Posterior compartment workhorses	**straight posterior + posterolateral**, both **3 cm proximal to olecranon tip** ✗ Wrong options: any posteromedial portal

Nerve distances (OB)

Portal	Nerve – distance
Anterolateral (joint line)	**Radial 4.8 mm** · PABC 12.6 mm
Proximal anterolateral	**Radial 4–7 mm** · PIN 1–14 mm
Anteromedial	**MABC 1 mm · median 7 mm** (one OB) · MABC 8.9 · median 12.9 · ulnar 22.1 mm (another OB)
Proximal anteromedial	MABC 1 mm · **median 12 mm** · ulnar 7 mm (posterior to septum)
Posterolateral	Radial 4.8 mm (3–8) (OB)
Straight posterior	Ulnar **~20 mm (15–25)**



Nerves at risk from medial (A) and lateral (B) portals

Radial nerve & anterior capsulectomy

Radial nerve course	on the **anterior capsule at mid-capitellum** (OB), runs longitudinally **medial to capitellum** → at risk in capsular excision; seen near probe when viewing radiocapitellar joint from **anteromedial** portal
Protective layer	**brachialis (~4 mm)** between capsule and radial nerve **proximal to radial neck** ✗ Wrong options: anconeus · brachioradialis · biceps · pronator teres
Danger zone	distal capsule **over radial head/neck** → transect capsule **≥2–3 cm proximal to radial head**

- - Active space - -

6 Complications

Nerve injury

Rate **nerve palsy 1–5%**; neurologic = **most common** complication, mostly **transient**

Greatest risk **rheumatoid arthritis**, **elbow contracture**

Ranking	Answer
Most common complication / nerve (TO, OB)	**Transient (sensory) ulnar nerve palsy**
2nd	**Radial nerve palsy** (Orthobullets, OB) · **persistent wound drainage**, then transient sensory radial palsy (TO)
Most common permanent nerve palsy (TO)	**PIN**, then ulnar
Least involved nerve (TO)	**Median**

Mechanism **direct** (trocars, instruments, no blunt dissection → neuromas) · **indirect** (**compartment syndrome** from aggressive distension/extravasation; **local anaesthetic extravasation** – transient)

Deficit → nerve → portal

Deficit	Nerve	Portal
Wrist drop: no wrist + MCP extension	**Radial nerve proper**	**Proximal anterolateral**
Weak wrist extension + **ulnar deviation** (lost ECRL)	Radial nerve proper	Proximal anterolateral
No finger (MCP) extension, partial wrist extension; can't extend thumb IP; wrist extends in **radial deviation**; no paraesthesia	**PIN** (ECRL spared)	**(Distal) anterolateral**
Paraesthesia dorsal 5th digit, intrinsic weakness, ↓ finger ab/adduction	**Ulnar**	**Posteromedial**
Paraesthesia **medial volar forearm**	MABC	Anteromedial
Can't flex wrist	Median	Anteromedial
Sensory deficit only	Superficial radial	–

PIN palsy EPL (thumb IP extension) + finger extensors; **ECRL** innervated **proximal to bifurcation** → spared

✗ PIN traps index abduction = **1st DI (ulnar)** · APB = **median** (APL = PIN) · "wrist extension + radial deviation lost" = radial nerve proper

– – Active space – –



PIN palsy after anterolateral portal: finger drop, partial wrist extension

Other complications

HO / ankylosis	**less than open**; minimise bleeding
Infection	**sinus tract (posterolateral portal)**; ↑ risk with **intra-articular steroid at time of arthroscopy**; possibly ↑ with **inflammatory arthritis**

Infection prevention – avoid intra-articular steroid

✓ **Prevent infection**	**avoid intra-op steroid injection**
✗ **Traps**	avoid posterior midline portals · avoid scope in OA · absorbable subcuticular suture only (no difference) · restrict ROM 3–4 wk (↑ stiffness)

X vs Y

	Proximal anterolateral	**Distal anterolateral**
Location	1–2 cm **proximal**, 1 cm anterior to LE	1–3 cm **distal**, 1 cm anterior to LE
Nerve	**Radial nerve proper** (wrist drop)	**PIN** (finger + thumb IP extension), LABC
Safety	**Safer** → more commonly used	Riskier → rarely used

	Anteromedial	**Proximal anteromedial**
Location	2 cm anterior + 2 cm distal to ME	2 cm proximal to ME, anterior to septum
Nerves	**MABC (1 mm) + median (7 mm)**	MABC (1 mm), ulnar (7 mm), median (12 mm)
Role	Augment PAMP, medial recess	**Most common working portal**

How it's asked

"First portal in elbow arthroscopy?"	→ **anterolateral** (TO)
"Most common nerve involved?"	→ **ulnar** (TO)
"Most common complication?"	→ **transient sensory ulnar nerve palsy** (TO)
"Most common permanent nerve palsy?"	→ **PIN** (TO)
"Least involved nerve?"	→ **median** (TO)

– – Active space – –

"NOT a manoeuvre to avoid nerve injury?"	→ **more distal lateral portal** (TO) ✗ Wrong options: pronation + 90° · distension · low flow
"Most commonly used working portal?"	→ **proximal anteromedial** (TO)
"NOT a position for elbow arthroscopy?"	→ **elbow extended, prone** (TO)
"Maximum elbow distension at?"	→ **70–80° flexion** (TO)
"Portal generally not used?"	→ **posteromedial** (ulnar nerve)
"Shaver through anterolateral portal, finger drop — structure?"	→ **PIN**
"Structure nearest probe, radiocapitellar joint from anteromedial portal?"	→ **radial nerve**
"Greatest risk with proximal anterolateral portal?"	→ **radial nerve**
"Two safest posterior compartment portals?"	→ **straight posterior + posterolateral, 3 cm proximal to olecranon**
"Portal 2 cm proximal, 1 cm anterior to LE — deficit if nerve injured?"	→ **loss of wrist + MCP extension**
"Can't extend thumb IP post-op — portal?"	→ **anterolateral** ✗ Wrong options: direct lateral = LABC · posterolateral = PABC
"Portal 2 cm anterior + 2 cm distal to ME — risk?"	→ **MABC + median**
"AL, AM and posteromedial portals used — likely deficit?"	→ **paraesthesia dorsal 5th digit** (ulnar)
"Protects radial nerve in anterior capsulectomy?"	→ **brachialis**
"Prior radial head ORIF, AL portal — iatrogenic finding?"	→ **can't extend thumb IP** ✗ Wrong options: wrist extension + radial deviation = radial proper
"Contraindication to elbow arthroscopy?"	→ **prior submuscular ulnar nerve transposition** ✗ Wrong options: Panner · loose body · stiffness · s/p radial head ORIF
"Prevent infection after elbow arthroscopy?"	→ **avoid intra-op steroid injection**

– – Active space – –

ELBOW PHYSICAL EXAM

– Shoulder & Elbow · 15 / 60 –

1 Basics

Exam sequence & normal values

Sequence inspection → palpation → ROM → stability → motor → sensory → vascular → provocative tests

Parameter	Normal	Functional
Carrying angle (full extension)	**~11° ♂ · ~13° ♀** (exam note) · ♂ 5–10° / ♀ 10–15° (anatomy note); ↑ in throwers	–
Flexion–extension	0–140°	**30–130°**
Pronation	75°	**50°**
Supination	85°	**50°**

Pronation–supination test with shoulders fully adducted + elbow at 90°; compare both sides, active + passive

End point **soft = effusion / capsular tightness** · **firm = mechanical block** (loose body, fracture, osteophyte)

Throwers loss of full extension even **without pathology**

Motor – muscle & root

Movement	Muscle	Root
Elbow flexion, **full supination**	Brachialis + biceps	C5–C6
Elbow flexion, **neutral** (thumb to ceiling)	**Brachioradialis**	C6
Elbow extension	Triceps	C7
Pronation	Flexor-pronator mass	C7–C8
Supination	Biceps (primarily)	C6
Wrist extension	ECRL, ECRB, ECU	C6–C8
Wrist flexion	FCR, FCU	C6–C8
Finger/thumb extension	EDC, EPL	C7–C8
Finger/thumb flexion	FDS, FDP, FPL	C7–C8
Intrinsics	Lumbricals, interossei	T1

Brachial artery anterior elbow, **medial to biceps tendon**; also check radial + ulnar pulses

– – Active space – –

3 Presentation

Inspection

Effusion elbow held at **70–80° flexion** (= max capsular distension) · fullness of **soft spot** (radial head + lateral epicondyle + olecranon)

Cubital tunnel **1st dorsal interosseous / 1st web atrophy** · ring/small clawing (more with **Guyon's** – unopposed FDP)

Distal biceps rupture **medial ecchymosis** + swelling · proximal retraction of belly = **reverse Popeye sign**

Palpation – point of maximal tenderness

Condition	Tender point
MCL injury	**MCL origin, just inferior to medial epicondyle**; palpate at **50–70° flexion** (flexor-pronator mass moves anterior)
Valgus extension overload	**Posteromedial olecranon**
Lateral epicondylitis	ECRB insertion, **few mm distal to tip of lateral epicondyle**
Radial tunnel syndrome	**3–5 cm distal to lateral epicondyle** (more distal than tennis elbow)
Medial epicondylitis	**5–10 mm distal + anterior** to medial epicondyle; swelling/warmth if inflamed

Radial head best palpated while rotating forearm pronation → supination

Ulnar nerve subluxation over medial epicondyle, extension → flexion · hypermobility in **33% of adults**
✗ Wrong options: necessarily CuTS

Snapping medial triceps snaps over medial epicondyle on **resisted extension from full flexion** (differentiate from ulnar nerve)

4 Diagnosis & Classification

Stability – MCL (valgus)

Valgus stress test flex **20–30°** (**unlocks olecranon**) + **externally rotate humerus** + valgus → tests **MCL**

Test	Technique	Positive	Accuracy
Milking maneuver	**Pull patient's thumb**, forearm **supinated**, elbow **90°**; supine or seated/standing	Apprehension, instability or **pain at MCL origin**	Sens **87.5%**, **NPV 100%**
Moving valgus stress	Same position + valgus while ranging full arc; shoulder **fully ER**	Pain at MCL origin **between 70° and 120°** (early acceleration 70° · late cocking 120°)	**Sens 100%, spec 75%**

– – Active space – –



Milking maneuver

Valgus extension overload — **pain with forced elbow extension** · valgus in terminal extension reproduces pain · **varus loading ↓ pain**

Stability – LCL / posterolateral rotatory instability

Test	Technique	Positive / accuracy
Lateral pivot-shift	Supine, arm overhead, shoulder fully ER, **forearm supinated + valgus**, extension → flexion	Pain/apprehension or **clunk at 40°** (radiocapitellar dislocated); 2nd clunk with ↑ flexion (triceps reduces radial head); **more reliable under anaesthesia**
PLR drawer	Supine, elbow **40°**, forearm supinated, AP force over lateral proximal forearm	Apprehension or **skin dimple** (radial head posterior subluxation)
Chair push-up	Push up from chair, forearm **supinated**	Unable / apprehension; sens **87.5%** (**100% with prone push-up**)
Prone push-up	Push-up, forearm supinated	Unable; sens 87.5% (100% with chair test)
Table-top relocation	Press-up on table edge, elbow lateral, forearm supinated; 3 parts	Pain in parts 1 + 3, **relief with thumb on radial head (part 2)** = PLRI

Table-top (wrong option) — pain in **all 3 parts** = intra-articular **radial head fracture**

– – Active space – –

Provocative tests – nerve

Condition	Test / sign	Finding
Cubital tunnel	**Froment**	Thumb **IP flexion by FPL (AIN)** on key pinch → compensates adductor pollicis (ulnar)
Cubital tunnel	**Jeanne**	Thumb **MCP hyperextension** + adduction by **EPL (radial)** on key pinch
Cubital tunnel	**Wartenberg**	Persistent **small finger abduction** + extension; weak intrinsics + **unopposed EDM**
Cubital tunnel	**Masse**	Flat palmar arch, lost ulnar hand elevation (weak **ODQ**, ↓ small MCP flexion)
Cubital tunnel	**Tinel**	Over cubital tunnel, **elbow extended**
Cubital tunnel	**Elbow flexion test**	Symptoms with flexion **>60 s**
Radial tunnel	**Resisted long finger extension** · **resisted supination** (elbow + wrist extended) · **passive pronation + wrist flexion**	Pain at radial tunnel (weakness from pain); supinator stretch → **250 mmHg** (normal 50)
PIN syndrome	Resisted supination ↑ pain · **normal tenodesis** (vs extensor tendon rupture)	Weak finger MCP extension; **wrist extends in radial deviation** (ECRL intact, ECU absent)
AIN syndrome	**OK sign** (FDP index + FPL)	**Can't make OK**; weak **PQ** = weak resisted pronation with **elbow maximally flexed**
AIN vs FPL rupture (RA)	**Tenodesis**	Intact tendons: passive wrist extension flexes thumb IP + index DIP

Cubital tunnel motor **weak pinch** (up to **70%** lost; adductor pollicis)

Pronator syndrome – site-specific tests

Pronator syndrome **Tinel proximal anterior forearm**, not at wrist · wrist-flexion provocation as in CTS

Resisted test	Compression site
Elbow flexion + forearm supination	**Bicipital aponeurosis (lacertus)**
Forearm pronation, elbow extended	**Two heads of pronator teres**
FDS contraction to middle finger	**FDS fibrous arch**

Sensory

Cutaneous nerves medial, lateral, posterior antebrachial cutaneous · ulnar · median · superficial radial

Cubital tunnel ↓ 2PD **small + ulnar ring** · **↓ dorsal ulnar hand** = proximal to wrist (dorsal branch arises **5 cm proximal** to wrist)

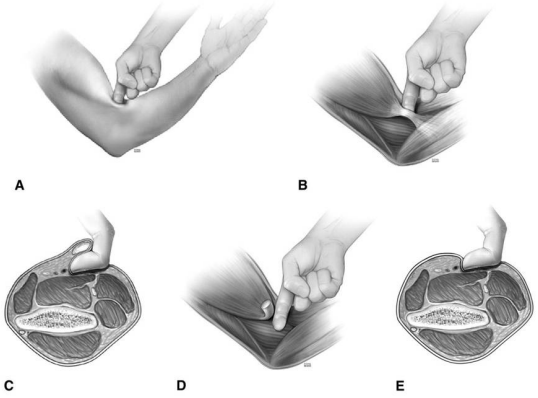
Pronator syndrome ↓ sensation in **palmar cutaneous branch** territory (arises **4–5 cm proximal** to carpal tunnel) · not in CTS

Provocative tests – tendon

Condition	Test	Key fact
Triceps rupture	**Modified Thompson squeeze**	Prone, forearm hanging; squeeze triceps → **no elbow extension** = complete tear (triceps + lateral expansion)
Distal biceps	**Hook test**	Elbow 90°, **full supination**, hook **lateral** edge; intact = finger **1 cm beneath tendon**; **sens + spec 100%**
Distal biceps	**Ruland biceps squeeze**	Elbow **60–80°**, slight pronation; squeeze belly → **no supination** = positive; **sens 96%**
Distal biceps	**Biceps crease interval (BCI)**	Crease → cusp distance **>6 cm or 1.2× contralateral**; sens 92%, spec 100%
Distal biceps	**Passive forearm pronation**	Belly moves proximal on supination, distal on pronation (intact)
Distal biceps	Strength	**Loss of supination > flexion**

Hook test false positives — **partial tear** · **intact lacertus fibrosus** · **underlying brachialis**

Biceps triad — hook + passive pronation + BCI in sequence → **100% sens + 100% spec** for complete rupture



Hook test



Biceps crease interval

Lateral epicondylitis — pain on resisted wrist extension (elbow extended, pronated) · **Maudsley = resisted middle finger extension** (ECRB) · maximal wrist flexion · passive wrist flexion in pronation

Chair test — elbow extended, forearm pronated, shoulder flexed, lift chair → lateral elbow pain = **lateral epicondylitis**

Medial epicondylitis — pain on **resisted pronation + wrist flexion** · **no instability/apprehension** on valgus stress or milking (vs MCL)

– – Active space – –

5 Management

Olecranon bursitis – aspiration & compression

Cause — **repeated minor trauma** (leaning on elbow, desk work) > single injury → inflamed lining secretes fluid



Olecranon bursitis

Aseptic olecranon bursitis

- First line → **nonoperative (more effective + safer than surgery)**
- Aspiration chosen → **posterolateral needle, aspirate alone**
  - then → **compressive wrap (bursal walls adhere)**

Aspiration ✓ — **alone** (no steroid) + **compressive dressing** → walls adhere, prevents reaccumulation

Approach — **posterolateral**
✗ Wrong options: posteromedial → ulnar nerve

Avoid steroid — **↑ septic bursitis** (local immunosuppression); no outcome benefit · **recurrence occurs with or without steroid**
✗ Wrong options: "steroid ↑ recurrence"

LUCL rupture — from repeated steroid injections for **lateral epicondylitis**, not bursa aspiration

Finding	Septic	Aseptic
Tenderness	88%	36%
Erythema/cellulitis	83%	27%
Warmth	84%	56%
Trauma / skin lesion	50%	25%
Fever	38%	**0%**

Septic vs aseptic — clinical + lab data **overlap** → hard to distinguish

X vs Y

	Lateral epicondylitis	Radial tunnel syndrome
Tender point	**Few mm distal** to lateral epicondyle (ECRB)	**3–5 cm distal**
Provocation	Maudsley, chair test, resisted wrist extension	Resisted long finger extension, resisted supination

	MCL injury	Medial epicondylitis
Tender point	**Just inferior** to medial epicondyle	**5–10 mm distal + anterior**
Milking / valgus stress	**Pain, apprehension, instability**	**No** instability or apprehension
Provocation	Moving valgus stress 70–120°	Resisted pronation + wrist flexion

	Pronator syndrome	Carpal tunnel syndrome
Tinel	**Proximal forearm**	Wrist
Palmar cutaneous branch sensation	↓	**Spared**

	Cubital tunnel	Guyon's canal
Clawing	Less	**More** (unopposed FDP)
Dorsal ulnar hand	↓	Spared

How it's asked

"Olecranon bursa aspiration – true statement?"	→ **compressive wrap after aspiration alone promotes bursal wall adherence** ✗ Wrong options: steroid ↑ recurrence · posteromedial · similar infection risk with steroid · LUCL rupture
"Steroid injected into olecranon bursa – risk?"	→ **↑ septic bursitis**
"Needle approach for olecranon bursa?"	→ **posterolateral** (avoid ulnar nerve)
"Position to palpate MCL origin?"	→ **50–70° flexion**, just inferior to medial epicondyle
"Elbow position to test valgus stability?"	→ **20–30° flexion** (unlocks olecranon), humerus ER
"Moving valgus stress test positive arc?"	→ **70–120°**
"Clunk at 40° on pivot-shift?"	→ **dislocated radiocapitellar joint** (PLRI)
"Table-top test, pain in all 3 parts?"	→ **radial head fracture** ✗ Wrong options: PLRI
"Hook test false positive?"	→ **partial tear / intact lacertus / brachialis**
"Tenderness 3–5 cm distal to lateral epicondyle?"	→ **radial tunnel syndrome**

– – Active space – –

"Elbow held at 70–80° flexion?"	→ **effusion** (max capsular volume)
"Wrist extends into radial deviation?"	→ **PIN palsy** (ECRL intact, ECU lost)
"Weak resisted pronation with elbow fully flexed?"	→ **pronator quadratus (AIN)**
"Resisted pronation, elbow extended reproduces median symptoms?"	→ **pronator teres heads**
"Numbness thenar eminence + forearm Tinel?"	→ **pronator syndrome** ✗ Wrong options: CTS (palmar cutaneous spared)
"Firm end point to flexion?"	→ **mechanical block** ✗ Wrong options: effusion = soft

– – Active space – –

ELBOW STIFFNESS AND CONTRACTURES

– Shoulder & Elbow · 16 / 60 –

1 Basics

Functional range of motion

Motion	Functional range
Flexion–extension	**30°–130° = 100° arc** (Orthobullets, OB) · **90° arc** (one OB explanation) – read the options
Pronation–supination	**50°–50°**

Tolerance **30° loss of extension well tolerated · flexion loss causes more dysfunction than extension loss**

Causes

Aetiology **OA**, **trauma (fractures)**, surgery, **CP**, **TBI**, **burns**, prolonged immobilisation, recurrent haemarthroses, congenital (**arthrogryposis**, **congenital radial head dislocation**)

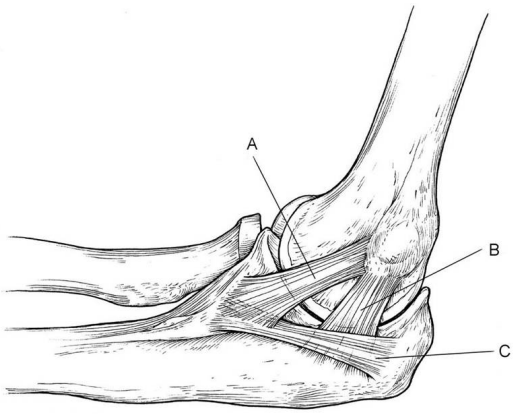
Type	Causes
Intrinsic (intra-articular)	Joint incongruity, synovitis, **loose bodies**, intra-articular fracture, malunion, OCD, post-traumatic arthritis (**coronoid + olecranon tip osteophytes**, radiocapitellar narrowing)
Extrinsic (extra-articular)	Burn **eschar**, **HO**, **capsular adhesions/contracture**, ligament contracture (**posterior oblique MCL**)
Mixed	Late effect of intrinsic → extrinsic stiffness

Elbow ligaments

Ligament	Key fact
Anterior bundle MUCL (anterior oblique)	**Isometric**; **primary valgus + distraction stabiliser**; highest tensile strength; **Tommy John** reconstruction · do not release → **valgus instability**
Posterior bundle MUCL (posterior oblique)	**Dorsal to axis of rotation**; lengthens **9 mm between 60° and 120°** flexion · contracture → **loss of flexion · release → ↑ flexion without valgus instability**
Transverse band MUCL	Release **does not improve ROM**
Radial collateral ligament	Do not release
LUCL	Do not release → **PLRI** (pivot shift)
Annular ligament	–

Anterior capsule contracture **loss of extension**

– – Active space – –



Medial ligaments: A anterior band, B posterior band, C transverse

Ulnar nerve	close to joint → at risk when contracted / during flexion gain

3 Presentation

Pain	**mid-arc pain → intra-articular** pathology · **pain at extremes** → extrinsic contracture (bony impingement + soft-tissue stretch)
Function	↓ motion limits **ADLs**
Exam	**scars**, skin quality, inflammation · flexion/extension + pronation/supination vs contralateral · **median, radial, ulnar nerve** function
Flexion <90–100°	**posterior band of MCL contracted** → release (+ consider ulnar nerve decompression/transposition)

4 Diagnosis & Classification

Key radiology findings

X-ray	**AP, lateral, oblique**; **serial X-rays if HO**



Heterotopic ossification, elbow

CT	**loose bodies**, nonunion, joint incongruity, abnormal bony anatomy; HO planning
MRI	rarely indicated
HO maturity	**well-corticated margins + no change on serial X-rays** · **labs not needed** · matures **12–18 mo** (OB) → **no bone scan** years after injury

5 Management

- **Elbow stiffness**
  - Most / contracture <40° → **NSAIDs + supervised PT (active + passive ROM)**
    - failed PT + flexion contracture >30° or flexion <130° → **static progressive splint**
  - Plateau 4–6 mo, extrinsic, congruent joint → **capsular release ± posterior band MCL**
  - Intrinsic arthritis → **choose by extent + demand**
    - confined to olecranon fossa → **osteophyte excision + capsular release**
    - diffuse, young high-demand → **distraction interposition arthroplasty**
    - diffuse, elderly low-demand → **TEA**

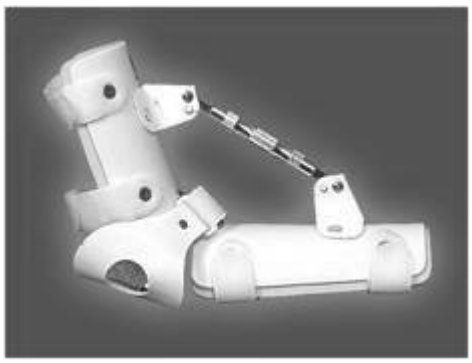
Static progressive splinting

Indications after **failed PT** with **flexion contracture >30°** or **flexion <130°**

Best initial option (OB) **supervised therapy + static progressive (turnbuckle) splinting** over **6 mo** (6–12 mo) → greatest ↑ **DASH** + functional ROM

Not indicated **mechanical block** (esp. pronation/supination) · **ankylosis** (extensive HO) · functional arc already (e.g. 10–140°) · routinely after arthroscopic loose body removal

✗ Alternative	Why wrong
Self-directed exercise	**Poor results**
Supervised therapy alone	First line, but response **variable, often unsatisfactory**
CPM	Limited role in **established** contractures; no end-range gain
Closed manipulation (MUA)	**Worsens stiffness**: tearing, **haemarthrosis**, more fibrosis



Turnbuckle static progressive splint

– – Active space – –

Operative criteria

Procedure	Indication
Capsular release ± posterior band MCL	**Extrinsic** contracture + **normal joint congruency** (most predictable) · arthritis (less predictable) · ankylosis → less predictable
Osteophyte excision (+ capsular release)	**Intrinsic** contracture, **arthritis confined to olecranon fossa**; bony block to terminal motion → remove **coronoid, coronoid fossa, olecranon, olecranon fossa**
Distraction interposition arthroplasty	Intrinsic, **diffuse arthritis**, **young high-demand**
TEA	Intrinsic, diffuse arthritis, **elderly low-demand** · high failure in young · **permanent 5-lb lifting limit**
Musculocutaneous neurectomy	**Neurogenic** contracture, **flexion deformity <90°**
Timing of release	**4–6 mo** post-injury/surgery once ROM **plateaued** despite splinting/therapy
HO excision	At **maturity**

Contraindications to release — **Charcot elbow**, **neurologic elbow disorder** · **poor skin** (relative → rotational flap)

Release – posterior bundle of MCL for flexion

Flexion gain — capsular release + **posterior oblique band of MUCL**
✗ Wrong options: anterior band · transverse · RCL · LUCL

Flexion <100° preop — consider **ulnar nerve transposition/decompression** (↓ nerve injury)

Arthroscopic vs open release (prior ulnar transposition, HO)

Arthroscopic — technically demanding · posterior: olecranon fossa/osteophyte debridement + posterior capsular release (**no suction medially** – ulnar nerve) · anterior: coronoid fossa debridement + anterior capsulotomy/capsulectomy

Nerves at risk with portals — **radial (most)** > ulnar > median (stiffness note) · overall most common = **transient ulnar**, most common permanent = **PIN** (TO, arthroscopy MCQs) – read the options

Relative contraindications to arthroscopic release — **prior ulnar nerve transposition** (variable nerve position) · **HO** · obesity · severe loss of pronation–supination · muscular contracture (**CP**)

Not contraindications age >40, gender, **osteophytes / arthritis** (unpredictable outcome only), heavy labour

Open approach	Key points
Lateral column (Morrey)	Lateral or posterior skin incision · elevate **ECRL + BR** off anterior supracondylar ridge, triceps posteriorly · brachialis off anterior capsule · **robust anterior access** → capsular release + **HO excision**; coronoid + olecranon tip/fossa, radiocapitellar joint
Medial "over the top" (Hotchkiss)	Extrinsic contracture, **MCL calcification**, **baseline ulnar nerve symptoms** · with ulnar nerve decompression/transposition · **release posterior band MCL** · work anterior to flexor-pronator mass
Combined medial + lateral	Single **posterior incision** → both columns

Rehabilitation & outcomes

Post-op **regional block**, **CPM through full ROM**, compressive dressing, active + active-assist ROM, extension / dynamic / static progressive splinting as needed

Compliance critical

Outcome variable; **retain ~two-thirds of motion gained** at surgery

6 Complications

Complication	Detail
Heterotopic ossification	Prophylaxis **low-dose radiation** or **indomethacin** · no radiation with **acute fracture** (nonunion risk)
Transient ulnar neuropraxia	After release (nerve close to contracted joint)
Ulnar nerve damage	Transpose if **preop flexion <100°**
Recurrent contracture	–

X vs Y

	Posterior band MUCL	Anterior band MUCL
Length change	**+9 mm (60°→120°)**	**Isometric**
Contracture →	**Loss of flexion**	–
Release →	**↑ flexion**, stable	**Valgus instability**
Other	–	Valgus restraint in throwing; Tommy John; strongest

	Arthroscopic release	Open release
Best for	Osteophytes, capsular contracture, loose bodies	**Extensive HO**, **prior ulnar transposition**, severe contracture
Risk	**Radial nerve** (portals) > ulnar > median	Ulnar neuropraxia / damage

– – Active space – –

How it's asked

"Static progressive turnbuckle splint most appropriate?"	→ **3 mo post distal humerus ORIF, arc 45–100°, plateaued** ✗ Wrong options: radial head block to supination · 10–140° arc · ankylosis with HO · after arthroscopic loose body removal
"2.5 mo after fracture-dislocation fixation, arc 75°, DASH 45 — best initial?"	→ **supervised therapy + static progressive splinting** ✗ Wrong options: self-directed · CPM · MUA
"True about posterior oblique MCL?"	→ **released to gain flexion in post-traumatic contracture** ✗ Wrong options: strongest · Tommy John · valgus restraint = anterior bundle · pivot shift = LUCL
"Lacks 40° extension, flexion to 80° — release capsule plus?"	→ **posterior band MCL (B)** ✗ Wrong options: anterior band → valgus instability · LUCL → PLRI
"Relative contraindication to arthroscopic release?"	→ **prior ulnar nerve transposition** ✗ Wrong options: age, gender, osteophytes, labour
"Terrible triad 5 y ago, arc 60–95°, extensive HO — treatment?"	→ **open capsular release + HO excision via lateral column approach** ✗ Wrong options: bone scan · arthroscopy · downsize radial head · interposition arthroplasty
"Nerve most at risk with elbow arthroscopy portals?"	→ **radial**
"Functional elbow ROM?"	→ **30–130° (100° arc)**; pronation/supination **50–50°**

GLENOHUMERAL ARTHRITIS (SHOULDER ARTHRITIS)

– Shoulder & Elbow · 17 / 60 –

1 Basics

Epidemiology

Incidence ↑ with age; mostly **>60 y** · ♀ **>** ♂

Dislocation as risk **56%** of primary anterior dislocators have arthrosis at **25 y**

Aetiology

Type	Key point
Primary OA	No known cause · irreversible cartilage loss + **hypertrophic subchondral bone** · flattened head, osteophytes, cysts, **posterior subluxation + posterior glenoid wear**
Post-traumatic	Healed articular incongruity (humeral fractures, chronic dislocations)
Dislocation arthropathy	↑ with **age >25 y** at dislocation + **posterior** dislocation · **not related to number of dislocations**
Post-capsulorrhaphy	Over-tightening (e.g. **Putti-Platt**) pushes head one way → eccentric wear; presents 5th decade
Cuff tear arthropathy	Torn cuff → head migrates up → abrasion against acromion
RA	Affects **90%** of RA patients · **central glenoid wear + medialisation** · cuff tear 25–50%
CPPD / gout	Pyrophosphate (pseudogout) / urate crystals
AVN	Proximal humerus fracture: **35% 3-part, 90% 4-part** (this topic's Orthobullets note) · chronic dislocation · cuff repair · steroid, EtOH, haemoglobinopathy, Gaucher
Chondrolysis	After arthroscopy: **radiofrequency**, **continuous intra-articular anaesthetic infusion**, bioabsorbable anchors, contrast · **fewer osteophytes** than OA
Neuropathic	Charcot

Cuff tears with OA **5–10%** (Orthobullets) · full-thickness supraspinatus **<10%** in primary OA (OB) · CTA excluded by definition
✗ Wrong options: 30–55% = asymptomatic >60 y population

Anatomy

Version glenoid **~3° retroverted** · humerus **20–30° retroverted**

Stabilisers static: GH ligaments, labrum · dynamic: cuff, rotator interval, biceps

2 Mechanism

Posterior wear + subluxation posterior capsule stretched · **anterior capsule + subscapularis contract → loss of ER**

– – Active space – –

Walch B vs C — **B**: posterior subluxation **causes** posterior wear · **C**: glenoid retroversion causes wear → subluxation later

3 Presentation

Symptoms — pain with motion, often none at rest · difficulty sleeping · **loss of ER** (anterior capsule contracture)

Exam — ↓ ER, FF, IR · crepitus · catching/squeaking · examine cuff carefully

Subscapularis tear — **weak belly-press** + **↑ passive ER** (90° at side)

Outcome scores — ASES · Constant · Oxford

4 Diagnosis & Classification

Key radiology findings

Views — **true AP (Grashey)** + AP + **axillary lateral** (axillary/CT to judge posterior glenoid wear)

Cause	X-ray hallmark
Primary OA	JSN, sclerosis, cysts, **inferior "goat's beard" osteophyte**, **posterior glenoid wear**, fixed posterior subluxation
Dislocation arthropathy	Large osteophytes, hardware
RA	**Osteopenia**, marginal erosions, **few osteophytes**, **central wear + medialisation**, ↓ acromiohumeral distance
CPPD	**Chondrocalcinosis**
Gout	Usually unremarkable; erosions after repeat attacks
AVN	Normal early → central head resorption → **crescent sign** → collapse
CTA	**Superior migration**, ↓ acromiohumeral interval, acromial erosion, superior glenoid loss, **"acetabularisation"** of CA arch, osteopenia



Primary OA with inferior goat's beard osteophyte

CT — **glenoid morphology** + templating · underestimates full-thickness RCT and fatty infiltration vs MRI

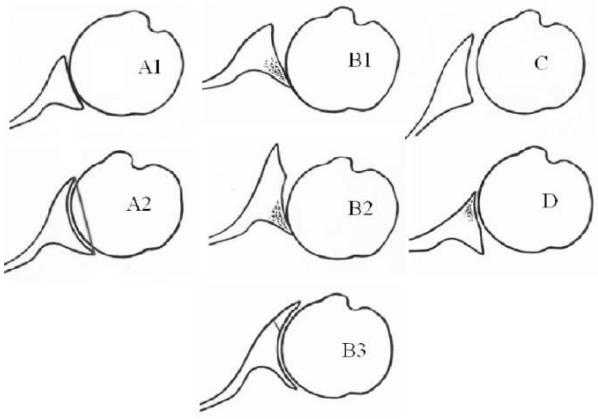
MRI — cuff assessment · **less accurate than CT** for glenoid type · **Goutallier IV** fatty infiltration = irreparable cuff

Glenoid wear patterns (Walch, posterior wear, medialisation in RA)

Most common OA wear — **posterior** (or central)
✗ Wrong options: anterior · superior · inferior

Inflammatory (RA/JRA) vs OA — **concentric erosion + medialisation of joint line** · severe erosion → glenoid resurfacing not always feasible → some favour HA
✗ Wrong options: inferior osteophyte · posterior subluxation · sclerotic glenoid · posterior wear = OA

Walch	Description
A (59%)	**Concentric**, head centred · **A1** minor central erosion · **A2** deep central erosion (rim line transects head)
B	**Biconcave**, asymmetric wear, **posterior subluxation** · **B0** pre-arthritic subluxation · **B1** (17%) posterior JSN, osteophytes, sclerosis, no bone loss · **B2** (15%) posterior rim erosion, **retroversion <25°** · **B3 monoconcave**, posterior wear, subluxation **≥70%** or retroversion **>15°**
C (9%)	**Dysplastic**, **retroversion >25°** · **C1** >25° regardless of erosion · **C2** biconcave + posterior bone loss + posterior translation · subtypes **"Lazy J"** + **"Delta"**
D	**Anteversion** or **anterior** subluxation (<40%)



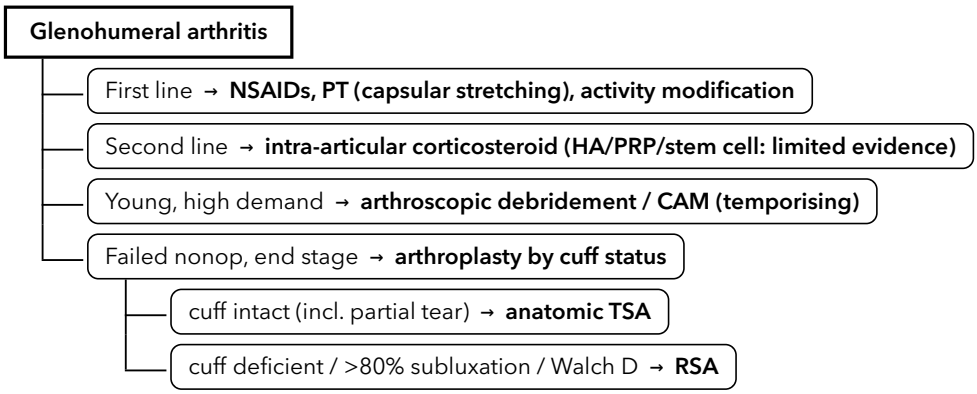
Walch classification

B2 trap — biconcave + posterior subluxation + **10°** retroversion = B2 · same with 25° = **C2** · deep concentric = A2 · monoconcave + 70% subluxation = B3 · anteversion = D

B2 outcomes — poorer after arthroplasty · anatomic TSA → instability/early glenoid loosening if retroversion **>27°** or **80%** posterior subluxation → **reverse** often better in older/severe deformity

– – Active space – –

5 Management



Operative criteria

Procedure	Indications	Contraindications / cautions
Anatomic TSA	**Intact cuff** (partial tear OK), glenoid wear, posterior subluxation, failed nonop	**Deficient deltoid or cuff** · active infection · **Charcot** · young high-demand (glenoid loosening)
Hemiarthroplasty	Young · **RA with irreparable cuff / poor glenoid bone stock** · **AVN without glenoid involvement**	Early failure, poor pain/function in OA (not routine)
Ream-and-run	Young, **intact cuff, no inflammatory arthropathy**	–
RSA	**Irreparable/large cuff tear** · CTA · OA/RA with major glenoid pathology · age · **failed arthroplasty** · complex fracture	Higher complication rate than TSA
Arthroscopic debridement	Mild–moderate OA without structural change · loose bodies · small AVN lesions · synovial chondromatosis	Advanced degeneration
Arthrodesis	Paralysis · recurrent infection · poor deltoid/soft tissue · brachial plexus palsy · failed repair of instability	–

Anatomic TSA vs hemiarthroplasty – indications & cuff status

TSA vs HA
TSA: **better pain relief + function** · HA: **↑ revision rate**
✗ Wrong options: better pain · ↑ blood loss · ↑ instability · ↑ infection

End-stage OA, cuff strong
TSA = most predictable pain relief
✗ Wrong options: RSA unnecessary · arthroscopic osteophyte removal · interposition · HA

Partial articular cuff tear (35%)
good candidate for anatomic TSA (± repair)

Not anatomic TSA candidates
52 y powerlifter (glenoid loosening → HA) · prior aTSA with high-riding humerus (cuff tear → **convert to RSA**) · Walch B **85%** subluxation (RSA if **>80%**) · elderly **Walch D** (RSA)

Tight anterior structures
ER **<40–45°** under EUA → **subscapularis lengthening + anterior capsule release**

TSA outcomes — 10-y survival **92–95%** · complications: **glenoid/humeral loosening**, infection, fracture, nerve injury, cuff tear · **↑ PJI if within 3 months of arthroscopy** · vs RSA: **↑ IR** + **↑ glenoid loosening**

Reverse shoulder arthroplasty – cuff deficiency, post-instability arthropathy, RA

Irreparable cuff + OA (Goutallier IV) — NOT appropriate = **arthroscopic cuff repair** · appropriate = NSAIDs/injection, PT, activity modification, then HA or RSA after failed nonop

Elderly, can't elevate arm (CTA), failed nonop — **RSA** most reliable (Delta III: Constant +42, +65° elevation, **96%** minimal/no pain) · HA = historical standard, non-uniform results

OA + complete subscapularis tear — **RSA**
✗ Wrong options: anatomic TSA: subscap repair fails

Post-Bankart arthropathy + supraspinatus fatty atrophy + posterior subluxation/erosion — **RSA**
✗ Wrong options: HA: glenoid pain persists · TSA: incompetent cuff · arthroscopic release

RA + cuff tear — **RSA good outcomes**, even with **reparable** tear · options = TSA + cuff repair or RSA · **TSA > HA** in RA (pain, elevation, ROM) · **etanercept/DMARDs/prednisone: no effect** on RSA outcome · RSA complications in RA **≈ CTA** (not more) · no interposition arthroplasty in elderly

RSA outcomes — 10-y survival **~90–95%** · complications: **scapular notching**, infection, instability, nerve injury

Young patients – nonoperative care & arthroscopic debridement (CAM procedure)

Young OA first step — **NSAIDs, cortisone injection, PT / ROM stretching** before any surgery
✗ Wrong options: TSA · fusion · resurfacing · arthroscopy as first step

Arthroscopic debridement — **temporising** · improves pain + ROM · better with **age <50**, little JSN, concentric joint, added subacromial procedure · **~20% convert to arthroplasty within 1 y**
✗ Wrong options: 45%

Arthroscopic debridement predictor	Effect
Nonconcentric joint space / moderate–severe degeneration	**Poor**
Posterior glenoid wear, ↑ retroversion	**Poor** (↑ failure)
Bipolar (head + glenoid) OA	Poor
Preoperative stiffness, loose bodies	**Good**
Prior infection / instability without degeneration	No effect

Technique — treat **both glenoid and humeral head aggressively**
✗ Wrong options: ream glenoid + spare humerus

CAM procedure — young patient · GH debridement, chondroplasty, synovectomy, loose body removal, **humeral osteoplasty (goat's beard excision)**, capsular release, subacromial + subcoracoid decompression, **axillary nerve decompression**, biceps tenodesis

– – Active space – –

Inferior humeral osteophyte

axillary nerve entrapment → **teres minor** fatty infiltration
✗ Wrong options: infraspinatus/supraspinatus = suprascapular · subscapularis = upper + lower subscapular · teres major = lower subscapular ± thoracodorsal

6 Complications

Procedure	Typical complications
TSA	**Glenoid / humeral loosening**, infection, fracture, nerve injury, cuff tear (most common group)
RSA	**Scapular notching**, infection, instability, nerve; overall > TSA
HA	Glenoid wear/pain, **↑ revision**
Arthroscopy (pain pump, RF)	**Chondrolysis**

X vs Y

	Primary OA	RA	CTA
Glenoid wear	**Posterior**	**Central, medialisation**	**Superior**
Osteophytes	**Large, inferior**	**Few**	–
Head position	**Posterior subluxation**	Medialised	**Superior migration**
Bone	Sclerosis	**Osteopenia**, erosions	Osteopenia, acetabularisation
Cuff	Tear 5–10%	Tear 25–50%	Torn by definition
Arthroplasty	**TSA**	TSA (+ repair) or RSA; HA if no glenoid	**RSA**



RA: osteopenia, central wear, few osteophytes

	TSA	RSA
Design	Concave glenoid + convex humerus	**Convex glenoid (ball) + concave humerus**
Cuff	Must be intact	Deficient OK
10-y survival	92–95%	~90–95%
Distinct	↑ IR, ↑ glenoid loosening	Scapular notching, ↑ complications

How it's asked

"72 y, OA X-ray, neutral ER — where is glenoid wear?"	→ **posterior**
"Inflammatory vs osteoarthritis at arthroplasty — more likely?"	→ **medialisation of joint line**
"Walch B2 description?"	→ **biconcave, posterior subluxation + erosion, 10° retroversion** ✗ Wrong options: 25° = C2 · anteversion = D · deep central = A2 · monoconcave 70% = B3
"Chance of full-thickness supraspinatus tear at TSA for primary OA?"	→ **<10%**
"TSA vs HA for OA with intact cuff — HA gives?"	→ **↑ revision surgery**
"74 y, end-stage OA, posterior erosion, good cuff strength?"	→ **TSA**
"Best candidate for anatomic TSA?"	→ **60 y with 35% partial articular cuff tear**
"80 y, drop arm, Goutallier IV, GH arthritis — NOT appropriate?"	→ **arthroscopic cuff repair**
"75F, can't elevate arm, cuff arthropathy, most reliable surgery?"	→ **RSA** ✗ Wrong options: HA · resurfacing · debridement · fusion
"Old cuff repair, weak belly-press, ER 90°, end-stage OA?"	→ **RSA**
"Post-Bankart arthropathy, supraspinatus fatty atrophy, posterior erosion?"	→ **RSA**
"RA on etanercept, 2 cm cuff tear — true statement?"	→ **RSA has good outcomes in this cohort**
"Predictor of poor outcome after arthroscopic debridement?"	→ **nonconcentric joint space**
"40 y labourer, OA, no prior treatment?"	→ **NSAIDs, cortisone injection, PT**
"35 y weightlifter, new pain + stiffness, OA X-ray — next step?"	→ **anti-inflammatory drugs + range-of-motion stretching program** ✗ Wrong options: TSA · HA + interposition · resurfacing · debridement first

– – Active space – –

"Inferior humeral osteophyte neuropraxia — fatty infiltration of?"	→ **teres minor** (axillary nerve)
"True about arthroscopy for OA?"	→ **↑ failure with posterior glenoid wear** ✗ Wrong options: 45% conversion at 1 y · >50 y better · bipolar no effect · spare humerus

– – Active space – –

GLENOHUMERAL INTERNAL ROTATION DEFICIT (GIRD)

– Shoulder & Elbow · 18 / 60 –

1 Basics

Epidemiology

Who	**overhead athletes**, esp. **baseball pitchers**; also javelin
Definition	loss of glenohumeral **IR of throwing shoulder vs contralateral** · ↑ ER · ↓ total arc (symptomatic)
Course	deficit worsens with ↑ repetitions over a season + a career; may be **painless**

Definition & thrower adaptations (IR loss, total arc, posterior capsule)

Characteristic motion change	↓ **IR** in throwing arm · pitchers: ↓ **IR** + ↑ **ER** ✗ Wrong options: ER, abduction, adduction, forward elevation

Adaptation (dominant throwing shoulder)	Direction
Humeral head retroversion	↑ (✗ Wrong options: anteversion)
Glenoid retroversion	↑ (✗ Wrong options: anteversion)
Posterior capsule / PIGHL	**Tight + thickened**
Anterior capsule / AIGHL	**Lengthened, attenuated** (✗ Wrong options: stretch it)
ER at 90° + scapular plane	↑
IR	↓

Retroversion mechanism	humerus retroverted in utero → **de-rotates until maturity** → throwing stress on physis **halts de-rotation** → ↑ retroversion → more ER, less IR · bony changes in those who **began pitching before skeletal maturity**
Crockett CT	athletes: ↑ humeral head retroversion, ↑ glenoid retroversion, ↑ ER at 90° + scapular plane, ↓ IR · not in non-athletes
Why ER gain	**arc of acceleration ↑ → velocity ↑** · short IR arc + tight posterior capsule → **deceleration harder** → overuse injury
Posterior capsule thickening	from **repetitive eccentric stress in follow-through**

Total arc of motion concept

Situation	Total arc (IR + ER)
Adaptive, asymptomatic thrower (ERG ≥ GIRD)	**Normal** vs other side; normal kinematics
Thrower vs **average population**	↑ **overall arc** (ER gain > IR loss)
Pathological GIRD (GIRD > ERG)	↓ vs contralateral → deranged kinematics

Total arc	should be **equal between shoulders** · ~**180°** (total motion concept)

– – Active space – –

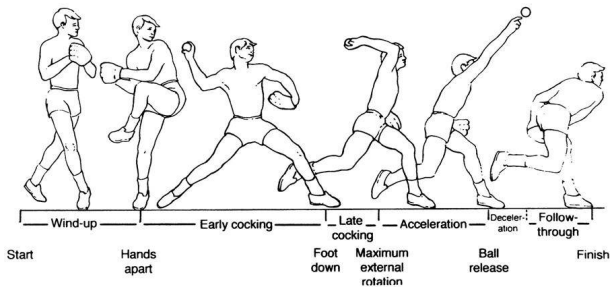
Source nuance	"tight posterior capsule with **normal** total arc" (OB) · "**↑** total arc vs average population" (OB) · "**↓** total arc" in symptomatic GIRD (Orthobullets, OB) – read the stem: adaptation vs pathology
Injury risk ✓	**↓ ER vs non-dominant arm** independently linked to injury (esp. with pre-existing IR deficit) ✗ Wrong options: ↑ ER (compensatory) · ↑ IR (uncommon) · ↑ horizontal adduction · ↓ flexion
Newer view	loss of total arc + **failure to gain ER** may matter more than IR loss alone

2 Mechanism

Cause	**repetitive throwing**; during **late cocking + early acceleration**
Capsular constraint mechanism	tight capsule → humeral head translates **OPPOSITE** to tight region

Tight region	Position	Head translation
Posterior capsule	Flexion	**Anterosuperior**
Posteroinferior capsule (PIGHL)	**ABER / cocking**	**Posterosuperior** ✓

Posterosuperior shift ↑ clearance for greater tuberosity but → internal impingement + **peel-back** of biceps anchor (biceps vector shifts posterior)

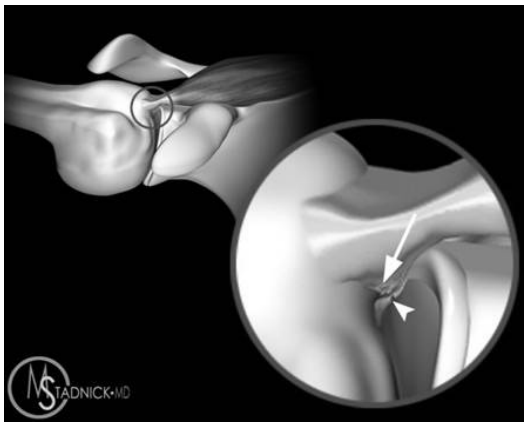


Throwing phases

Throwing phases	wind-up → early cocking → **late cocking (max ER)** → acceleration → deceleration → follow-through · "5 distinct phases" (OB)
Pathologic cascade	↓ velocity + command → posterior stiffness, **slow to loosen up** → posterior pain in late cocking / early acceleration (no mechanical symptoms) → **"SLAP event"** (posterosuperior labrum + biceps anchor fail in tension) → **surgery after SLAP event**; stretching works before it

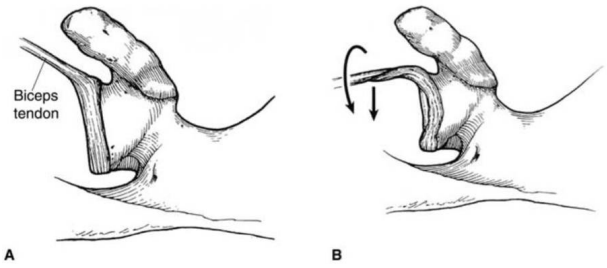
Internal impingement & associated lesions (SLAP, articular-sided cuff tears)

Most closely linked pathology	**internal impingement** ✗ Wrong options: subacromial · HAGL · Hill-Sachs · biceps tendinitis
Internal impingement	in **ABER (late cocking)** greater tuberosity + **posterosuperior glenoid** pinch **articular side of posterior supraspinatus + infraspinatus** and posterosuperior labrum



Internal impingement

Cuff tear in throwers	**articular-sided partial**, **SS–IS junction** (posterior SS + anterior IS) · (✗ Wrong options: bursal-sided · full-thickness (not described)) · general population = anterior supraspinatus
Tear mechanism	**compressive** (GT vs glenoid) + **tensile** (repetitive eccentric load, hyper-ER twisting)
Lesion most associated	**SLAP tear** (type II, posterior extension) ✗ Wrong options: Bankart · HAGL · anterior capsular contracture
SLAP risk	throwers with GIRD **25% more likely** to have SLAP · **peel-back** in late cocking



Peel-back mechanism

Cause of internal impingement	**posterior capsular contracture (GIRD)** or **anterior capsular laxity**
Chondral injury	**posterosuperior** ✗ Wrong options: anteroinferior = anterior instability
Associated conditions	GH instability · internal impingement · articular-sided partial cuff tears · SLAP · **ipsilateral elbow pain** (altered throwing)

SICK scapula

Letter	Meaning
S	Scapular malposition (hangs lower)
I	**Inferior medial border prominence**
C	**Coracoid pain** + malposition (pec minor contracture → tendinitis)
K	Dys**K**inesis of scapular movement

– – Active space – –

3 Presentation

Symptoms	vague / **posterior** shoulder pain · may be painless · **↓ velocity + command** · "dead arm" · arm heavy in later innings · slow to loosen up
Exam	**stabilise scapula** for true GH rotation · ↓ IR + ↑ ER at 90° abduction · **↑ sulcus sign** (stretched CHL + rotator interval)
Internal impingement exam	deep **posterior** pain at **90° abduction + max ER**; negative apprehension
Subacromial impingement (wrong option)	**anterolateral** pain with IR in abduction

4 Diagnosis & Classification

Diagnostic criteria

Criterion	Value
Clinically relevant GIRD	**IR loss >25°** vs contralateral (Orthobullets, OB) · **20–25°** (OB, Little Leaguer's) – read the options
Range reported	up to **50°**
Pathological	**GIRD > ERG** (↓ total arc)
No	**<15°** loss

Key radiology findings

X-ray	AP + lateral; **usually normal**
CT	**↑ glenoid retroversion**
MRI	**ABER view** shows associated lesions (articular-sided cuff tear, labrum) · often normal · not needed initially for isolated posterior tightness

Little Leaguer's shoulder (proximal humeral epiphysiolysis)



Little Leaguer's shoulder – widened physis

X-ray	**widened proximal humeral physis** + metaphyseal changes · **17% normal**
Risk	pitchers throwing **breaking pitches**, ignoring pitch counts/rest, year-round pitching

GIRD link GIRD exam findings at diagnosis → **increased likelihood of recurrence (3×)**, mean **6.2–7.6 months** · **pain after 2 months PT** = independent predictor of recurrence

Treatment stop throwing (**~3 months** no pitching) → PT (restore motion, kinetic chain) → return-to-throw · **no breaking balls until skeletal maturity**

5 Management

GIRD in thrower

- First line → **rest from throwing + posterior capsular stretching** *(sleeper + cross-body + periscapular/pec minor + cuff)*
- Fails extensive PT → **arthroscopic posteroinferior capsular release** *(some advocate anterior stabilisation)*
- After "SLAP event" → **surgery (SLAP repair)**

Treatment – posterior capsular stretching (sleeper/cross-body) & scapular work

Initial ✓ **PT directed at posteroinferior capsule stretching** + periscapular strengthening, **regardless of pathology** · then cuff + scapular stabilisation

Regimen rest from throwing + PT **6 months** (Orthobullets)

Emphasise **sleeper + cross-body stretches + periscapular strengthening**

Sleeper stretch lying on affected side, shoulder **90° abduction** (Orthobullets, OB) · **90° forward flexion** (OB), elbow 90°, push forearm to table, **scapula stabilised** → stretches **posterior capsule**
✗ Wrong options: anterior · anteroinferior · rotator interval · MGHL



Sleeper stretch

– – Active space – –

Stretch / exercise	Target	Restores IR?
Sleeper stretch	Posteroinferior capsule	**Yes**
Roll-over sleeper	Arm flexed **60°**, body rolled forward **30°**	Yes
Cross-body adduction	Posterior capsule	Yes
Doorway stretch	–	–
Pec minor stretch (rolled towel, push shoulders back)	SICK scapula, coracoid pain	No (adjunct)
Rows (retraction) · punches (protraction)	Scapula	No
Humeral head depression, ball on wall	Closed-chain cuff	No

Coracoid pain + GIRD **pec minor stretching + sleeper stretches**
✗ Wrong options: weighted ball program · subacromial steroid · arthroscopy

Outcomes **~90%** respond (young throwers) · ~10% older throwers fail → arthroscopic release

Internal impingement + partial posterior SS tear **NSAIDs + posterior capsular stretching + cuff strengthening** · refractory → SLAP repair / cuff debridement
✗ Wrong options: cuff repair

✗ Wrong initial option	Why
Arthroscopic posteroinferior release	Only after failed PT
Anterior(-inferior) capsular stretch / release	Anterior capsule already **lax/lengthened**
Selective ER stretching	ER already normal / ↑
Posterior labral repair + capsulorrhaphy	**Posterior instability** (linemen, weightlifters)
Capsulolabral imbrication / plication	**MDI**
Subacromial steroid	Pathology is **intra-articular**
Sling	Comfort only
Cessation 3–6 months alone	Does not address contracture; deconditioning
↑ deltoid / latissimus weights	↑ stiffness
Change pitch mix	Pitch selection not a risk factor for GIRD
MRI arthrogram first	Not needed for isolated tightness

Operative criteria

Procedure	Indication
Arthroscopic posteroinferior capsule release (± PIGHL)	**Failure of extensive PT** only
Anterior stabilisation	Alternative (controversial)
Repair thinned cuff	Significant thinning (transcuff or takedown)

Posterior capsule release vs anterior stabilization

– – Active space – –

Technique	electrocautery via **posterior portal**, camera **anterior** · release posteroinferior capsule + **posterior band IGHL** from **9 to 6 o'clock** at glenoid rim until cuff fibres seen · **shaver widens gap** (prevents recurrence) · gentle manipulation
Result	immediate gain **~65° IR**

X vs Y

	GIRD / internal impingement	**Subacromial impingement**	**Adhesive capsulitis**
Pain	**Deep posterior**, ABER	**Anterolateral**, IR in abduction	Global
Motion	↓ IR, ↑ ER	Normal	**Global active + passive ↓**
Cuff tear side	**Articular**	Bursal	–

	GIRD thrower	**Anterior instability**
Anterior capsule	Lax (microtrauma)	Torn / Bankart
Lesion	**SLAP**, articular cuff	**Bankart, HAGL**
Chondral injury	**Posterosuperior**	Anteroinferior

How it's asked

"Motion decreased in thrower's dominant arm?"	→ **internal rotation**
"Dominant thrower adaptations?"	→ **tight posterior capsule with normal total arc** ✗ Wrong options: humeral anteversion
"Pitcher, ↓ velocity, posterior pain, slow to loosen up – exam?"	→ **IR loss >25°**
"Side-lying stretch, arm 90°, push forearm to table – diagnosis + finding?"	→ **GIRD; >25° IR loss** ✗ Wrong options: adhesive capsulitis · <15°
"ROM finding most associated with injury risk?"	→ **decreased ER** vs non-dominant
"12-year-old, GIRD + widened physis – clinical course?"	→ **↑ recurrence** ✗ Wrong options: elbow pain unlikely · non-breaking pitches
"Pitcher, 30° IR loss vs average population?"	→ **thickened posterior capsule** ✗ Wrong options: ↓ total arc · humeral anteversion · anteroinferior chondral injury
"Sleeper + prone IR exercises – diagnosis?"	→ **GIRD**

– – Active space – –

"Pitcher, 30° IR loss, positive O'Brien, normal MRI – emphasise?"	**→ sleeper + cross-body + periscapular strengthening**
"Total arc 150°, scapula hangs low – initial management?"	**→ aggressive PT, posterior capsular stretching + scapular strengthening**
"IR stretch at 90° flexion, scapula stabilised – stretches?"	**→ posterior capsule**
"Pitching staff, mean GIRD 45° – intervention?"	**→ posterior capsular stretching**
"IR deficit 25°, negative O'Brien/Neer/Hawkins – next step?"	**→ posterior capsular stretching program** ✗ Wrong options: MRI · arthroscopy
"ER 110° vs 100°, IR 30° vs 70° – best treatment?"	**→ posterior capsular stretching** ✗ Wrong options: selective ER stretching · thermal shift
"Late-cocking posterior pain, IR 25° vs 60°, O'Brien painful – next?"	**→ posterior capsular stretching** ✗ Wrong options: SLAP repair · subacromial injection
"Home exercise to improve IR deficit?"	**→ sleeper stretch** ✗ Wrong options: rows · punches · towel pec stretch
"↓ IR + inferomedial scapular prominence + partial posterior SS tear – treatment?"	**→ NSAIDs + posterior capsular stretching + cuff strengthening**
"Pitcher, figure of GIRD – initial treatment?"	**→ PT for posteroinferior capsule stretching** ✗ Wrong options: anteroinferior stretch/release
"GIRD + coracoid tenderness, normal MRA – next?"	**→ pec minor + sleeper stretches**
"IR 5° vs 15°, ER 115° vs 90° – next?"	**→ posterior capsular stretches** ✗ Wrong options: 3–6 months rest · anterior stretches
"Humeral translation in cocking with posterior contracture?"	**→ posterosuperior**
"GIRD linked most closely to?"	**→ internal impingement**
"Range of motion in pitchers with internal impingement?"	**→ ↓ IR, ↑ ER**
"Javelin thrower, posterior pain at 90° abduction + max ER?"	**→ internal impingement**
"Arthroscopy: posterior SS + posterosuperior labral lesion – throwing phase?"	**→ late cocking**
"Lesion most associated with internal impingement?"	**→ SLAP tear** ✗ Wrong options: Bankart · HAGL · full-thickness tear
"Sleeper + cross-body stretches – associated EXCEPT?"	**→ bursal-sided partial cuff tear**
"Deep posterior pain at max ER, 30° less total arc – MRI?"	**→ articular-sided SS/IS partial tear**

"Dead arm, total arc −45° – arthroscopic finding?"	→ **SLAP lesion** ✗ Wrong options: glenoid anteversion · bursal tear	– – Active space – –

– – Active space – –

GLENOHUMERAL JOINT ANATOMY, STABILIZER, AND BIOMECHANICS

– Shoulder & Elbow · 19 / 60 –

1 Basics

Scapulohumeral rhythm

GH : ST ratio (TO)	**2:1** ✗ Wrong options: 1:2 · 2.5:1 · 3:1
First 30° elevation (TO)	largely **glenohumeral**; scapulothoracic small + inconsistent → after 30° both move **simultaneously**
180° abduction	**120° GH (⅔) + 60° scapulothoracic (⅓)**
Scapular plane	**30° anterior** to coronal plane
Abduction needs ER	clears greater tuberosity from acromion → **IR contracture → cannot abduct >120°**

Osteology

Structure	Key numbers
Humeral head	Spheroidal in 90% · diameter **43 mm** · **inclination 130°** to shaft · tuberosities = cuff attachments
Glenoid	**Pear-shaped** · upward tilt **5°** · covers only **25–30% of head** (TO)
Coracoid	**"Lighthouse"** of deltopectoral approach · coracobrachialis, pec minor, short head biceps
Acromion	3 ossification centres: **pre- (tip), meso- (mid), meta-acromion (base)**
Acromiohumeral interval	**7–8 mm**

Glenoid & humeral version

	Glenoid	Humeral head
Orthobullets	**5° retroversion** avg (scapular body axis); **7° retro → 10° ante**	**~20° retroversion** (transepicondylar axis)
TO	**Near 0°**, more often **slight retroversion**; <10° either way	**Retroverted 26° ± 11°**

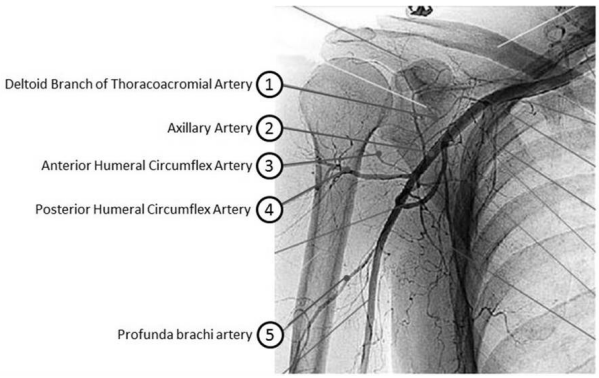
Humeral retroversion	**20°** (Orthobullets) · **26° ± 11°** (TO) – read the options
False (TO)	**"both glenoid and humeral head anteverted"**

Humeral head blood supply

Main supply	**posterior humeral circumflex artery** (current literature) → injury most likely → **AVN**

Why PHCA? **AHCA disrupted in ~80%** of proximal humerus fractures yet AVN infrequent

Ascending branch of AHCA runs **parallel to lateral aspect of LHB** in bicipital groove → **arcuate artery** = intraosseous continuation into head; do not injure when plating



Arteriogram: 4 = posterior humeral circumflex artery

✗ Arteriogram distractor	Label
Deltoid branch thoracoacromial	1
Axillary	2
Anterior humeral circumflex	3
Profunda brachii	5

Labral anatomy & blood supply

Composition **fibrocartilage** (TO)

Function **50% of socket depth** · cavity-compression · ↑ depth, ↑ articular surface area, **centralises head**, **seal → intra-articular pressure** (TO)

Embryology (TO) labrum forms **~8th week**; LHB + capsule insertion defined by **9th week**

Blood supply **suprascapular** · **circumflex scapular** (TO) / anterior humeral circumflex (Orthobullets) · **posterior humeral circumflex**; from **capsule + periosteal vessels**, not underlying bone

Poorest blood supply **anterosuperior labrum** (TO)
✗ Wrong options: posterosuperior

Biceps attachment (TO) to labrum, **mainly posterior**

Weak links anterior labrum anchors **IGHL → Bankart** · superior labrum anchors **biceps → SLAP**

– – Active space – –

2 Mechanism

Static vs dynamic stabilisers

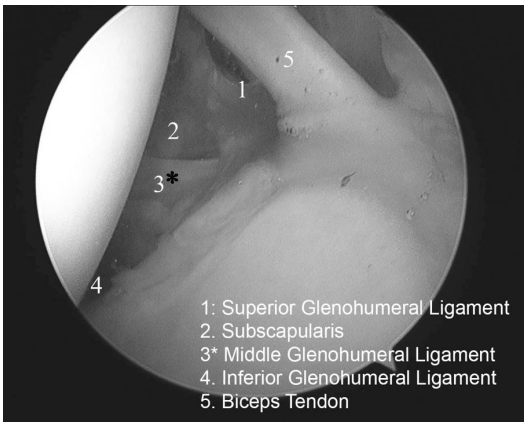
Static	Dynamic
Glenohumeral ligaments	**Rotator cuff** (main)
Glenoid labrum	Rotator interval
Articular congruity + version	**Long head biceps** (humeral head depressor)
Negative intra-articular pressure (release → head subluxes **inferiorly**)	Periscapular muscles

Bony articulation **inherently unstable**

Posterior capsule thin (**<1 mm**), **no ligaments**

Glenohumeral ligaments – function by arm position (IGHL, MGHL, SGHL, CHL)

Arm position	Anterior restraint	Inferior restraint	Posterior restraint
0° (side) + adduction		**SGHL / CHL**	
45° abduction + ER	**MGHL**		**MGHL**
90° abduction + ER	**Anterior band IGHL**	**Anterior band IGHL**	**Posterior band IGHL**
90° flexion, adduction, IR	Anterior band IGHL		**SGHL / CHL**



Arthroscopic view: 1 SGHL, 2 subscapularis, 3 MGHL, 4 IGHL, 5 biceps

Anterior static stability (TO) **anterior band IGHL = main restraint** (90° abduction + ER) · **MGHL** (45° abduction + ER) · **SGHL** (arm at side)

Hammock (TO) at **90° abduction** IGHL = hammock: **anteroinferior band = anterior** support, **posteroinferior band = posterior** support

Abduction + ER main stabiliser (TO) **IGHL**

>45° abduction, anterior + posterior stress (TO) **anterior inferior GHL**
✗ Wrong options: SGHL · CHL · anterosuperior labrum

– – Active space – –

SGHL anterosuperior labrum → humerus · **inferior translation at 0° abduction / adduction** (neutral rotation) → **greatest stress = inferiorly translated, arm adducted** · restrains **ER of adducted arm** · prevents **anteroinferior translation of LHB** (biceps pulley)

MGHL **anterior (and posterior) translation in midrange ~45° (45–60°) abduction + ER** · from labrum, crosses **subscapularis obliquely (~45°)** → **just medial to lesser tuberosity** · do not confuse with subscapularis

Anterior band IGHL **anterior/inferior translation at 90° abduction + maximal ER** (**late cocking**) · anchors anterior labrum → **Bankart weak link**

Posterior band IGHL **posterior subluxation at 90° flexion + IR** (forward flexed, adducted) · restraint in **posterior instability** (Kim test) → must repair · **tightness → internal impingement** + ↑ shear on superior labrum (SLAP)

Superior band IGHL **most important static stabiliser** (Orthobullets); **100% ↑ strain with SLAP lesion**

CHL coracoid → **rotator cable** · limits **posterior translation in flexion, adduction, IR** · limits **inferior translation + ER in adduction**

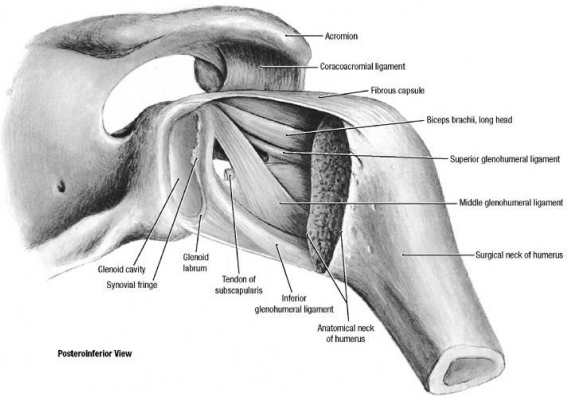
Primary restraint to inferior translation **CHL** (OB; Bigliani: inferior capsule + IGHL also contribute) · **SGHL/CHL** at 0° (Orthobullets, OB) – read the options

Ligament anatomy (TO)

Ligament	Origin → insertion (TO)	Note
CHL	**Proximal ⅓ of horizontal coracoid** → variable: superficial fibres **greater tuberosity**; only **15–50%** to lesser tuberosity	Blends with supraspinatus + subscapularis
SGHL (Flood ligament)	**Superior glenoid tubercle** → **fovea capitis** just above lesser tuberosity	Key **stabiliser of LHB** in groove
MGHL	Just beneath/medial to SGHL → **just medial to lesser tuberosity**	Blends with subscapularis
IGHL	Anterior + posterior bands + **axillary pouch** (reinforced by **fasciculus obliquus**) → humeral neck like a **collar** under articular margin	Hammock

Foramina (TO) **Weitbrecht** = between SGHL + MGHL · **Rouvière** = between MGHL + IGHL

IGHL anterior band origin **2–4 o'clock** (one TO) · **3–5 o'clock** (another TO); posterior band **7–9 o'clock** – read the options



Glenohumeral ligaments (Netter)

– – Active space – –

Stabilisers by abduction, neutral rotation (TO)

Abduction	Anterior	Posterior	Inferior
Rest	Supraspinatus	Posterior capsule	**Superior capsule, CHL, SGHL**, supraspinatus, biceps
0–45°	Subscapularis, **MGHL**, IGHL	Posterior capsule, IGHL	IGHL, MGHL
45–90°	**IGHL**, posteroinferior capsule, biceps (IR), cuff, deltoid	Posterior capsule, IGHL, biceps (ER)	IGHL, MGHL
>90°	**IGHL, axillary pouch**, subscapularis, infraspinatus	Posterior capsule, IGHL	–

Glenoid inclination also listed among stabilisers in the TO table

Rotator interval contents

Definition triangular space between **upper border of subscapularis** + **anterior margin of supraspinatus** (TO); bridged by capsule

Boundary	Structure
Base (medial)	**Coracoid** (lateral base)
Superior	Anterior edge **supraspinatus**
Inferior	Superior border **subscapularis**
Apex (lateral)	**Transverse humeral ligament**

Contents **CHL + SGHL** + LHB + thin capsule; CHL + SGHL = most constant structures of fibrous capsule
✗ Wrong options: MGHL

Layers (TO) 4 layers incl. **superficial CHL** · supraspinatus + subscapularis fibres · **deep CHL**

Arthroscopy area between **biceps + subscapularis tendon** = **rotator interval**; biceps deep to interval

Contracture **adhesive capsulitis** → release · **laxity** → **sulcus sign** (inferior laxity in adduction) → **RI closure / CHL imbrication** improves inferior stability (Harryman)

Dynamic stabilisers – concavity compression & rotator cuff

Rotator cuff **main dynamic stabiliser**; **compresses humeral head into glenoid**; works most in **midrange**, not extremes

Concavity compression **most important midrange stabilising mechanism**; depends on **depth of concavity + magnitude of compressive force**
✗ Wrong options: isometric ligaments · capsule tension · deltoid

✗ Distractor	Actual role
SGHL, MGHL	**Static** stabilisers
Teres major	Adducts + IR; not a significant stabiliser
Deltoid	Abducts; not main stabiliser

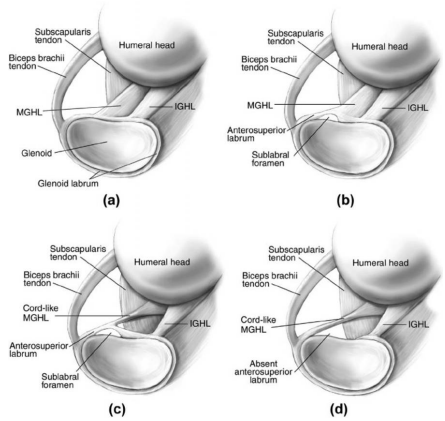
Free body analysis of deltoid

Arm 90° abduction A (deltoid lever) = 3 cm, B = 30 cm → (A × D) − (B × 0.5W) = 0 → **D = 5W**

4 Diagnosis & Classification

Normal variants (Buford complex, cord-like MGHL) – do not repair

Variant	Frequency	Description
Normal (labrum on rim + flat/broad MGHL)	Most common	Cord-like MGHL quoted in **86%** (Orthobullets)
Sublabral recess (sulcus) (TO)	**Most common** variant	**11–1 o'clock** at biceps–labral complex; smooth, **<2 mm**; follows rim **medially**; **does not extend posterior to biceps anchor**
Sublabral foramen (hole) (TO)	**~12%** (Orthobullets) · **11%** (TO)	Anterosuperior labrum detached **1–3 o'clock**, anterior to biceps anchor; smooth, **<1.5 mm**
Sublabral foramen + cord-like MGHL		
Buford complex	**~1.5%**	**Cord-like MGHL + absent anterosuperior labrum** (MGHL attaches to base of biceps anchor)
Meniscoid labrum	1%	



Anterosuperior labral variants: (a) normal, (b) sublabral foramen, (c) foramen + cord-like MGHL, (d) Buford

Buford (TO) **normal variant**, 1.5% · cord-like **MGHL** (✗ Wrong options: SGHL) + absent **anterosuperior** (✗ Wrong options: anteroinferior) labrum · not abnormal · **should not be repaired** → ↓ ROM

Repairing Buford to glenoid **painful loss of ER** (and elevation) · **loss of ER with arm at side** > at 90° abduction (MGHL tighter at side)

Sublabral foramen vs tear (TO) true anterosuperior tear **propagates superiorly into biceps anchor or inferiorly into IGHL** attachment

Tubercle of Assaki (TO) ridge of subchondral bone at **centre of glenoid** with focal **cartilage thinning** → do not mistake for cartilage defect

Subacromial pseudospur (TO) prominence of acromial angle at **CA ligament** attachment → mimics caudal osteophyte; also at deltoid tendon attachment → mimics enthesophyte

– – Active space – –

Key radiology findings

Sublabral recess	**best seen with arthrography** (TO)
AHI	**7–8 mm**; normal on X-ray but ↓ on supine MRI (arm weight removed) → **multiple-tendon cuff tear**
HAGL on MRI	tear of **humeral attachment of IGHL**

Acromial morphology

Type	Shape
I	**Flat**
II	**Curved**
III	**Hooked**

HAGL lesion

Mechanism	**abduction + hyperextension + ER** (e.g. 15-year-old wrestler)
HAGL	humeral avulsion of IGHL; **much less common than Bankart** (anteroinferior labral tear)

✗ Distractor	Why
Lesser tuberosity avulsion	Very rare; **12–13-year-olds**
Subscapularis tear	Same mechanism, **older** patients

5 Management

Arthroscopic vs open repair – external rotation preservation

Advantage of arthroscopic	**preservation of ER** (spares subscapularis; no anterior capsular incision)
Not advantages	↓ healing time · outpatient (both) · faster return (recovery identical) · ↓ recurrence (arthroscopic **slightly higher or equal**)

Shoulder arthrodesis position

Plane	Angle
Abduction	**15–20°**
Forward flexion	**20–25°**
Internal rotation	**40–50°**

Other sources	**30-30-30** (Orthobullets arthrodesis note) · ~40° abd / 35° FF / 20° IR (OB) – read the options

X vs Y

Ligament	Position tested	Restrains
SGHL	**Arm at side (0°/adducted)**	**Inferior** translation, ER
CHL	Adducted · flexion + adduction + IR	**Inferior** + ER · **posterior**
MGHL	**45° abduction + ER**	**Anterior** (± posterior)
AIGHL	**90° abduction + max ER**	**Anterior / inferior**
PIGHL	**90° flexion + IR** (adducted)	**Posterior**

	Sublabral recess	Sublabral foramen	Buford complex
Position	**11–1 o'clock** (biceps anchor)	**1–3 o'clock** (anterosuperior)	Anterosuperior
Labrum	Superior labrum not attached to cartilage	Detached from rim	**Absent**
MGHL	Normal	Normal / cord-like	**Cord-like**
Width	<2 mm	<1.5 mm	–
Repair?	No	No	**No (→ loss of ER)**

How it's asked

"Most important static stabiliser at 45° abduction + ER?"	→ **MGHL** (TO)
"Main stabiliser in abduction + ER?"	→ **IGHL** (TO)
"At 90° abduction, main stabiliser against anterior dislocation?"	→ **anterior-inferior GHL** (TO)
">45° abduction, restraint to anterior + posterior stress?"	→ **anterior inferior GHL** (TO)
"Primary static restraint to anterior translation at 90° abduction?"	→ **IGHL complex** ✗ Wrong options: bony shape · negative pressure
"SGHL under greatest stress?"	→ **inferiorly translated, arm adducted (5°)**
"Asterisk structure (crosses subscapularis) – function?"	→ **prevents anterior translation at 45° abduction** (MGHL)
"Labelled ligament restraining anterior translation at 45° abduction + ER?"	→ **MGHL (D)**
"Primary restraint to inferior translation?"	→ **coracohumeral ligament** ✗ Wrong options: MGHL · subscapularis · CAL
"90° abduction, full ER – resists anterior translation?"	→ **anterior band of IGHL complex**
"Arrowed structure, posterior portal – function?"	→ **resists anterior + posterior translation at 45° abduction** (MGHL)
"Posterior instability, + Kim test – structure to repair?"	→ **posterior band IGHL** ✗ Wrong options: AIGHL · CHL

– – Active space – –

"Besides biceps, rotator interval contains?"	→ **CHL + SGHL** ✗ Wrong options: MGHL
"Arthroscopic area between biceps and subscapularis?"	→ **rotator cuff interval**
"Dynamic stabiliser compressing head into glenoid?"	→ **rotator cuff muscles** ✗ Wrong options: teres major · deltoid
"Most important midrange stabilising mechanism?"	→ **concavity compression**
"Buford complex?"	→ **normal variant: cord-like MGHL + absent anterosuperior labrum** (TO)
"Repairing Buford complex causes?"	→ **loss of external rotation**
"Probe on cord-like MGHL repaired to glenoid — result?"	→ **loss of ER with arm at side**
"Asterisk, posterior portal, anterior structure crossing subscapularis?"	→ **MGHL**
"15-year-old wrestler, abduction-hyperextension-ER, MRI?"	→ **tear of humeral insertion of IGHL (HAGL)** ✗ Wrong options: lesser tuberosity avulsion · Bankart
"AVN after proximal humerus fracture — artery?"	→ **posterior humeral circumflex (4)**
"Advantage of arthroscopic anterior stabilisation?"	→ **preservation of external rotation**
"False about glenoid labrum?"	→ **posterosuperior labrum has poorest blood supply** (TO) (anterosuperior is poorest)
"GH : ST motion ratio?"	→ **2:1** (TO)
"False about glenoid/humeral version?"	→ **both anteverted** (TO)

INTERNAL IMPINGEMENT

– Shoulder & Elbow · 20 / 60 –

1 Basics

Epidemiology

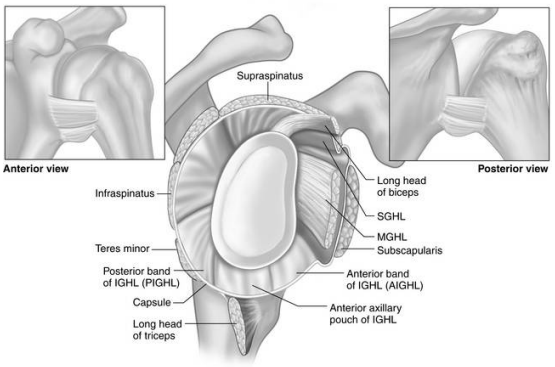
Who	major cause of shoulder pain in **throwing + overhead athletes**; very common in **elite baseball pitchers** · also swimming, volleyball, javelin
Definition	repetitive contact of **undersurface (articular side) of posterior rotator cuff** + greater tuberosity against **posterosuperior glenoid/labrum**
Impinged structure	**posterior rotator cuff** ✗ Wrong options: anterior cuff · GH ligaments · biceps
Contact	may be **physiologic or pathologic** in throwers

Anatomy – glenohumeral stability & cuff force couples

Restraint	Structures
Static	**GH ligaments**, labrum, articular congruity + version, negative intra-articular pressure
Dynamic	Rotator cuff, biceps, periscapular muscles

Force couple plane	Balance
Coronal	**Inferior cuff (IS, TM, subscapularis)** vs superior moment of **deltoid**
Transverse	**Subscapularis (anterior)** vs **IS + TM (posterior)**

Cuff role dynamic stability, centres head → stable fulcrum; goal of cuff repair = restore equilibrium



Glenohumeral ligaments – posterior band of IGHL

– – Active space – –

Ligament	Role in internal impingement
Posterior band IGHL (PIGHL) + posterior capsule	**Tight + thickened** ✓ (target of sleeper stretch)
Anterior band IGHL	**Lax** (✗ Wrong options: tight)
Anterior axillary pouch, MGHL, SGHL	Not implicated
Superior band IGHL	Most strain in **SLAP** lesions

2 Mechanism

Pathomechanics – late cocking, abduction–external rotation & posterosuperior contact

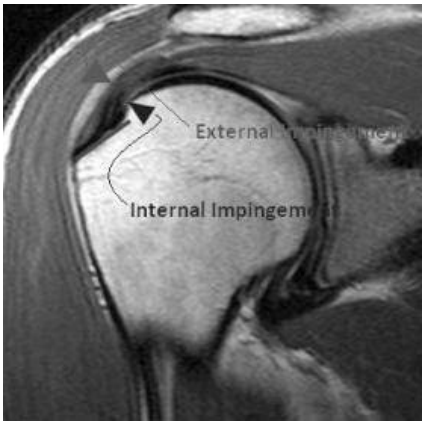
Position — **abduction + maximal ER**
✗ Wrong options: adduction/IR · abduction/IR · flexion/ER

Phase — **late cocking** (+ early acceleration); max ER
✗ Wrong options: wind-up · early cocking · deceleration · follow-through

Contact — **posterior articular supraspinatus** (± anterior infraspinatus) between greater tuberosity + **posterosuperior glenoid**

Peel-back — posterosuperior labrum peeled back by biceps → SLAP

Internal vs external — internal = **articular side** · subacromial ("external") = **bursal side**



Internal vs external impingement

Posterior capsular contracture & GIRD – stretching

Most clearly associated cause — **posterior capsular contracture** (✓ primary process)
✗ Wrong options: anterior capsular laxity (variable, less direct) · CA arch stenosis (subacromial) · Bennett lesion (no causal link)

Contracture effect — blocks normal anteroinferior translation in late cocking → **obligatory posterosuperior translation** → posterosuperior labral + articular cuff pathology

Etiology — tight **posterior band IGHL** + **anterior micro-instability** · multifactorial: altered biomechanics, subtle instability, posterior contracture, scapular dyskinesia

Primary etiology — **posterior capsular contracture** (OB, Orthobullets) · **GH instability** commonly primary (one OB explanation) – read the options

Anterior labrum — may become **thinned** (obligate translation → anterior micro-instability)

Associated conditions — **GIRD**, SLAP tears, **SICK scapula** + dyskinesia, biceps tendinitis, anterior instability

3 Presentation

– – Active space – –

Symptoms diffuse **posterior shoulder** pain (posterior deltoid), deep-seated · worse with throwing, esp. **late cocking + early acceleration** · ↓ velocity

Inspection proximal humeral retroversion · tender along **infraspinatus**

ROM **↑ ER + ↓ IR** · IR loss **>20° at 90°** (Orthobullets) · stabilise scapula · total arc **preserved** (Orthobullets, OB) · total arc **↓** (one OB stem) — read the stem

Strength rotator cuff weakness often (may be 5/5)

Isolated finding ↓ IR with **no other positive provocative test**
✗ Wrong options: sulcus sign · Speed's · ↓ ER · ↓ abduction

Test	Technique	Positive
Apprehension test (internal impingement)	Max ER + abduction + extension	**Posterior** pain, relieved in neutral
Whipple	Forward flexion + adduction + scapular retraction, resist	Pain → **partial supraspinatus tear**
Sulcus sign	Downward pull, arm adducted	**Inferior instability** (superior GH laxity)
Speed's	90° FF, supinated, resist downward	**Biceps tendinitis**

4 Diagnosis & Classification

Classification **none formal**

Key radiology findings

X-ray complete shoulder series; usually normal · **Bennett lesion** on AP



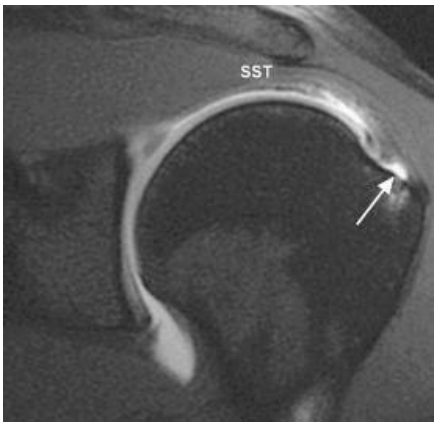
Bennett lesion

Bennett lesion **mineralisation / exostosis of posteroinferior glenoid** = **traction spur** of posteroinferior capsule (PIGHL) · formed in **deceleration + follow-through** · **baseball pitchers** · **22%** of MLB pitchers, **none needed surgery**
✗ Wrong options: football, swimmers, rowers

MRI / MRA **PASTA** (partial articular supraspinatus–infraspinatus tendon avulsion), fraying, tear · signal at **greater tuberosity** (posterolateral head edema, **cysts**) + **posterosuperior labrum** · ↑ anterior capsular volume · thickened **PIGHL**

ABER view reproduces impingement position; dynamic humeral + glenoid sides

– – Active space – –



PASTA lesion

Associated lesions & diagnosis (articular-sided cuff tears, posterosuperior labral fraying)

Lesion (spectrum)	Detail
Articular-sided partial cuff tear	**Supraspinatus–infraspinatus interval** (✗ Wrong options: bursal = external)
Posterior + superior labral lesion	**SLAP with posterior extension**; posterosuperior fraying
Bennett lesion	Posterior capsule/glenoid hypertrophy + scarring (calcified)
Chondral injury	**Posterior glenoid**
Greater tuberosity	Cystic changes
Capsule	**Lax anterior**, **tight posterior**

Not part of it	subscapularis tear · bursal-sided tear · Bankart · Hill-Sachs · tight anterior capsule · lax posterior capsule · full-thickness tear (needs significant trauma)
Bennett + ↓ arc + deep pain + posterosuperior labral fraying	**internal impingement** ✗ Wrong options: SLAP alone · subacromial = Hawkins + · calcific tendinitis
Javelin thrower, failed PT, SLAP with posterior extension	**internal impingement** ✗ Wrong options: EDS/MDI · traumatic anterior instability · osteonecrosis

Asymptomatic MRI findings in pitchers

Common asymptomatic findings	partial cuff tear · SLAP / posterior-superior labral tear · **↑ humeral retroversion** · thick posterior capsule · posterolateral head cysts
Asymptomatic pitcher	**continue pitching** ✗ Wrong options: rest · posterior release · cuff repair · rotational osteotomy
GIRD definition	**IR deficit 15° + total arc deficit ≥10°** (OB) · **>20° IR loss** (Orthobullets) · **>25°** (Orthobullets GIRD note) – read the options

5 Management

Internal impingement

- First line → **stop throwing + PT (posterior capsule stretching)**
- Failed extended PT + PASTA <50% / Bennett / peel-back → **arthroscopic debridement**
- Failed PT + partial >50% / full-thickness / unstable labrum → **repair (arthroscopic or mini-open)**
- Persistent posterior contracture / anterior instability → **add posterior release or anterior plication**

Nonoperative

Regimen — break from throwing until pain-free → supervised return with proper mechanics · **sleeper + cross-body stretches**, cuff strength balancing, scapular stabilisation, kinetic chain · **6 months** stretching

Stretch target — **posteroinferior capsule** · **posterior band IGHL**
✗ Wrong options: superior, anterior, anteroinferior, inferior

Prevention ✓ — **posterior capsular stretching** ↓ shoulder + elbow injury risk
✗ Wrong options: cuff strengthening · periscapular strengthening (less direct) · SLAP / partial cuff repair

Outcome — correlates with **compliance**; surgery results **unpredictable** → only after extended failed PT

Operative criteria

Procedure	Indication (after failed nonop)
Arthroscopic debridement (cuff ± labrum)	**Partial tear <50%** · **Bennett lesion** · **peel-back labral lesion**
Cuff ± labral repair (arthroscopic / mini-open)	**Partial >50%** or full-thickness · **unstable labral tear**
Posterior capsule release vs anterior stabilisation	Persistent posterior contracture or anterior instability (adjunct)

Techniques

Debridement — EUA + diagnostic arthroscopy (intra-articular + subacromial) · shaver · **accelerated rehab**, shorter immobilisation

Cuff repair — arthroscopic addresses labrum too · bursectomy · **no acromioplasty** if no bursal pathology · footprint abrasion · **portal of Wilmington** · partial: **in situ trans-tendinous** (pulley; functionally shortens tendon) or complete the tear → single/double row

Posterior release — cautery / shaver · **axillary nerve risk** at inferior border of infraspinatus

Anterior stabilisation — **capsular plication** most common

6 Complications

Progression to full-thickness tear — small risk after debridement alone

Return to play — **delayed / worse after cuff repair** in throwers

– – Active space – –

Axillary nerve injury posterior release, **inferior border of infraspinatus**

X vs Y

	Internal impingement	Subacromial (external) impingement
Cuff side	**Articular** (posterior SS/IS)	**Bursal**
Pain	**Posterior**, ABER, late cocking	**Anterolateral**, IR in abduction
Test	Posterior apprehension pain, ↓ IR	**Hawkins / Neer** +
Cause	Posterior capsular contracture, micro-instability	Hooked acromion, CA arch stenosis

	Bennett lesion	Hill-Sachs
Site	**Posteroinferior glenoid**	Posterosuperior humeral head
Nature	Traction spur (PIGHL)	Impaction fracture
Context	Pitchers	Anterior dislocation

How it's asked

"Posterosuperior labrum impinges on which structure?"	→ **posterior rotator cuff**
"Posteroinferior glenoid mineralisation — which athletes?"	→ **baseball pitchers**
"Phase of internal impingement?"	→ **late cocking**
"Arm position?"	→ **abduction + external rotation**
"Structure pathologically thickened (figure)?"	→ **posterior IGHL + posterior capsule** ✗ Wrong options: cuff = thinned · acromion · biceps · anterior labrum = thinned
"30° IR deficit — stretch which capsule?"	→ **posteroinferior**
"Pitcher, ↓ velocity, normal MRA, normal Hawkins — exam finding?"	→ **↓ internal rotation** ✗ Wrong options: sulcus · Speed's
"Sleeper stretch addresses?"	→ **posterior band of IGHL**
"Proven to ↓ shoulder and elbow injury?"	→ **posterior capsular stretching**
"Most clearly associated with internal impingement?"	→ **posterior capsular contracture** ✗ Wrong options: Bennett · anterior laxity
"25° IR deficit — tight structure (figure)?"	→ **A = posterior band IGHL** ✗ Wrong options: AIGHL, MGHL, SGHL, axillary pouch
"16-year-old pitcher, MRI posterolateral head edema + undersurface tear?"	→ **internal impingement**
"Characteristic lesion?"	→ **articular-sided rotator cuff tear**

"Javelin, failed PT, arthroscopy SLAP with posterior extension — association?"	→ **internal impingement**
"Late-cocking pain, ER 100°/IR 45° — MRI?"	→ **articular-sided supraspinatus tear** ✗ Wrong options: bursal · Bankart · Hill-Sachs
"Clinical + radiographic findings?"	→ **↑ ER/↓ IR + articular-side posterosuperior fraying + posterior capsule/glenoid scarring**
"Bennett lesion + ↓ arc + posterosuperior labral fraying — diagnosis?"	→ **internal impingement**
"Asymptomatic pro pitcher, MRI partial cuff + labral tear — advice?"	→ **continue pitching**

– – Active space – –

LATERAL EPICONDYLITIS (TENNIS ELBOW)

– Shoulder & Elbow · 21 / 60 –

1 Basics

Epidemiology

Frequency	**most common cause of elbow pain** · **1–3% of adults/y** · **dominant arm**
Demographics	**45–64 y** · ♂ **=** ♀ · up to **50% of tennis players** · labourers with **heavy tools**, repetitive gripping/lifting
Risk factors (TO)	**smoking**, **obesity**, repetitive wrist extension
Tennis risk factors	**poor swing technique (backhand)**, **heavy racket**, **incorrect grip size**, **high string tension**
Association	**radial tunnel syndrome** coexists: **5%** (Orthobullets, associated conditions) · **up to 15%** (Orthobullets, missed-diagnosis complication) – read the options

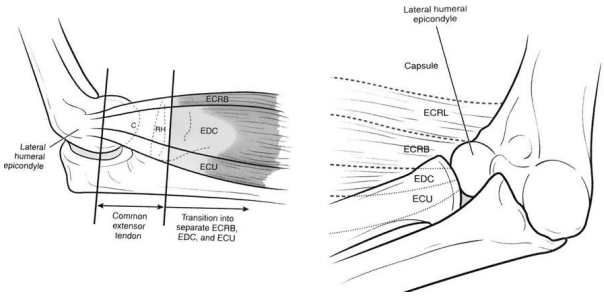
ECRB anatomy – origin, insertion & innervation

Most common tendon (TO) **ECRB**; **ECRB > ECRL** (TO); also ED, EDM, ECU, supinator can be involved
✗ Wrong options: ECRL · supinator · ED · ECU

Muscle	Origin	Insertion	Key fact
ECRB (extensor carpi radialis brevis)	**Lateral epicondyle** (anterior aspect); **13 × 7 mm diamond** superior to centre of epicondyle	**Base of 3rd MC**	**Deep branch radial n. (fibres of sixth + seventh cervical nerves, C6–C7)**; tendon just superficial to capsule; **deep + posterior to ECRL**
ECRL	**Lateral supracondylar ridge** (just below BR)	**Dorsal base 2nd MC**	**First fleshy fibres proximal to common extensor tendon**
Anconeus	**Same attachment site as ECRB**	–	–
ED, EDM, ECU	Lateral epicondyle (common extensor origin)	–	–

✗ Distractor	Belongs to
Largest cross-sectional area of elbow flexors	**Brachialis**
Lateral border of cubital fossa; greatest mechanical advantage of elbow flexors	**Brachioradialis**
Inserts on Lister's tubercle	**No muscle**

– – Active space – –



Common extensor origin: ECRB, EDC, ECU, ECRL

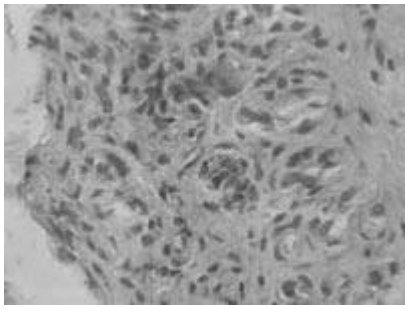
PIN	enters **supinator just distal to radial head** → radial tunnel syndrome
LUCL	**lateral epicondyle → supinator crest of ulna**

2 Mechanism

Overload	**eccentric overload** at common extensor origin → **microtear of ECRB origin** (± ECRL, ECU)
Triggers	**repetitive wrist extension + forearm pronation** · tenodesis effect to optimise grip → ECRB overuse · **backhand** in tennis · hammering
Underuse (TO)	**stress shielding** / pain-inhibited **underuse** of ECRB → weakening → **TRUE cause** ✗ Wrong options: "underuse cannot cause it"
Shear (TO)	shearing forces → **fibrocartilaginous ECRB enthesis** → weaker bone junction ✗ Wrong options: tensile

Histology – angiofibroblastic tendinosis

Lesion	**angiofibroblastic hyperplasia / dysplasia / tendinosis** = **fibroblast hypertrophy** (dense fibroblasts) + **disorganised collagen** + **vascular hyperplasia**
Inflammation	**NO inflammatory cells** → "**tendinosis**", tendonitis/epicondylitis = **misnomers**; **failed reparative process** (TO) ✗ Wrong options: macrophages · neutrophils
Also present	**normal tendon histology**, **hypercellularity** (TO), nerve sensitisation
✗ Histology distractors	abundant inflammatory cells · neutrophils around microvessels · cellular atypia · chondroblastic proliferation · primarily calcium deposition · osteoblastic rimming · reactive hyperaemia · cystic degeneration



Angiofibroblastic hyperplasia

Nirschl (OB)	1,213 patients, 88 operated; immature fibroblastic + vascular infiltration of ECRB origin; **97% improved** after excision + repair

– – Active space – –

Stages of lateral epicondylitis (TO)

Stage	Features
1	**Acute inflammatory** response; may resolve completely
2	**Angiofibroblastic hyperplasia** (fibroblasts, vascular hyperplasia, disorganised collagen), hypercellularity → **tendinosis**; **most common stage at presentation**
3	**Structural failure** → partial or complete **rupture**
4	Stage 2/3 + **fibrosis**, soft matrix calcification, **hard osseous calcification**

3 Presentation

Symptoms lateral elbow pain with **resisted wrist extension**, **gripping**; **↓ grip strength**

Tenderness **point tenderness at ECRB origin**, few mm **distal to tip** of lateral epicondyle

Neuro exam differentiates from **entrapment syndromes**

Provocative test	Manoeuvre
Cozen (TO)	**Resisted wrist extension, elbow extended** (± forearm pronated) → lateral pain
Maudsley	**Resisted extension of 3rd (long) finger** → lateral epicondyle pain
Passive stretch	**Maximal wrist flexion** / passive wrist flexion in pronation
✗ Wrong option: radial tunnel	**Resisted supination** (arm + wrist extended); tenderness **3–4 cm distal + anterior** to epicondyle

Rule out **LUCL insufficiency** (**chair test, push-up test**) · OCD / arthritis (↓ ROM) · **posterolateral plica** (pain/clicking on passive flexion in pronation + supination)

Differential of lateral elbow pain in athletes

Differential posterolateral plica · **PLRI** · **radial tunnel** · occult fracture · **cervical radiculopathy** · **capitellar OCD** · triceps tendinitis · radiocapitellar OA · **shingles**

Tennis player, continuous elbow pain (TO) source gives no key; Orthobullets causes = **incorrect grip size**, poor (backhand) technique, heavy racket, high string tension

4 Diagnosis & Classification

Diagnosis **clinical** (symptoms + exam)

Key radiology findings

X-ray (AP + lateral) usually **normal**; **calcification near epicondyle (up to ½)**; rarely changes management

MRI **not needed**; **↑ signal at ECRB origin (up to 90%)**, thickening/thinning, oedema · **does not correlate with severity** → does not dictate management · use when diagnosis unclear / refractory

US **thickened, hypoechoic ECRB**; **most useful tool in experienced hands** (operator-dependent)

– – Active space – –



MRI: increased signal at ECRB origin

5 Management

Lateral epicondylitis

- First line → **activity modification + eccentric PT / OT ± counterforce strap**
  (NSAIDs, ice; injections second line)
- Persistent → **injection (PRP better at 12 mo than steroid)**
- Failed ≥12 mo nonop + clear diagnosis → **ECRB release + debridement (open = arthroscopic)**

Nonoperative treatment – eccentric therapy, steroid & PRP injections

Natural history	resolves with time; recovery **6–12 mo**; **no consistent difference between nonop treatment modalities and placebo** / observation
Success	**80–90%** at 1 y, only **2–4%** need surgery (Orthobullets, OB) · **up to 95%** (Orthobullets prognosis, TO) · **>95%** (OB) – read the options
Eccentric exercise	**first line**; **eccentric > concentric** (TO); ↓ pain (grip gain unclear); beneficial effect on **tendon biology**
Occupational therapy (OB)	activity modification + **eccentric strengthening** + pain management = best initial
PT at 1 y	**PT > steroid injection**; **steroid (± PT) = worst outcomes**
NSAIDs	short-term benefit only; not an inflammatory condition; ↑ **GI side effects**

Injection	Evidence
Corticosteroid	**Short-term (2–6 wk)** ↓ pain + ↑ grip vs placebo (TO); **not beyond 6 wk** · RCT vs placebo: **equivalent** pain, grip, PROMs · **worst at 1 y** · **skin/muscle atrophy**, **tendon attenuation + rupture** · second line / not recommended
PRP	Orthobullets: **better than steroid** for long-term pain, healing, function · OB: **↓ pain + ↑ PROMs at 12 mo vs steroid**, **no ↑ strength/function**, no faster return to work, not better at 6 wk (steroid better short term) · TO (Cochrane): **no patient-important benefit vs placebo** – read the options
PRP type	**Leukocyte-rich** promising (✗ Wrong options: leukocyte-poor best); heterogeneous preparations
Others	Botulinum toxin, autologous blood, hyaluronate, acupuncture, ionto/phonophoresis, **nitroglycerin patch**

– – Active space – –

Steroid mechanism	analgesic on **neuropeptides (CGRP, substance P)** ↑ in tendinopathy; **suppresses fibroblast collagen + ECM** production
ESWT	**no difference vs placebo at 6 mo**; + stretching = stretching alone (TO)
Orthoses (TO)	**inelastic, nonarticular, proximal forearm strap** → may ↓ pain + ↑ grip at **3 wk**; bracing ≤6 wk may help ADLs · conflicting: not better than sham; may be inferior short term to steroid + topical NSAIDs
Tennis modifications	slower surface, more **flexible racquet**, **lower string tension**, **larger grip**

Operative criteria

Criterion	Detail
Indication	Failed **prolonged nonop (12 mo)** · **clear diagnosis** (isolated LE) · intra-articular pathology
Contraindication	**Inadequate nonop trial** · **noncompliance** with nonop
Poor-outcome / surgery predictors	**Depression, anxiety, poor coping (greatest)** · ipsilateral **radial tunnel** · **prior injection** · **workers' compensation**

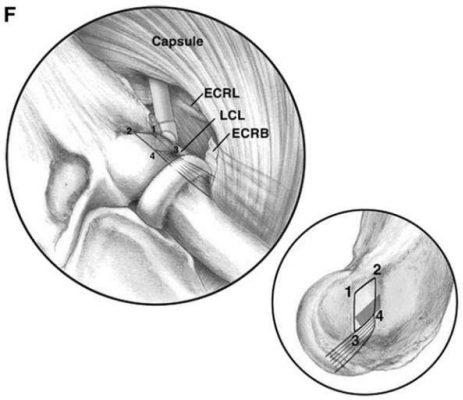
ECRB release & debridement

Open	incision over common extensor origin → **lift ECRL off ECRB** → **Nirschl scratch test** → excise degenerate tissue → **decorticate** epicondyle → **watertight capsule repair** → side-to-side closure → reattach (anchor / tunnel) ✗ Wrong options: synovial fistula
Arthroscopic	visualises + treats intra-articular pathology · resect lateral capsule **anteriorly** · release ECRB where muscle begins · decorticate · **do not pass mid-radial head**
Outcome	**open = arthroscopic** (satisfaction, return to work, complications) · **90–100% improve**, **up to 40% persistent pain**

6 Complications

Iatrogenic LUCL injury after debridement

Cause	debridement **posterior to equator of radial head** → **LUCL** injury → **PLRI**
Safe vs unsafe zones	anterior + superficial resection safe · **too distal → PIN** at risk
Presentation	lateral pain, **clicking/catching**, can't push up from chair / do push-ups → **chair push-up test +**
PLRI mechanism	instability with **elbow extension + supination + valgus**
Chronic iatrogenic PLRI (3 mo)	**LUCL reconstruction with palmaris longus or gracilis allograft** ✗ Wrong options: primary repair (chronic) · revision ECRB debridement · MCL procedures (milking manoeuvre/thrower)



Arthroscopic ECRB release zone relative to LCL

Complication	Detail
Persistent pain	**Up to 40%**
LUCL injury → PLRI	Beyond radial head equator
Missed radial tunnel	**Up to 15%**
Radial nerve / PIN injury	Distal dissection
HO	↓ with **thorough irrigation after decortication**
Stiffness	Esp. with intra-articular procedure
Synovial fistula	Capsule not repaired watertight
Infection, missed PLRI	–

X vs Y

	Lateral epicondylitis	Radial tunnel syndrome	PLRI
Tenderness	**ECRB origin**, just distal to epicondyle	**3–4 cm distal + anterior** (supinator)	Lateral
Painful test	**Resisted wrist extension** (elbow extended), long-finger extension	**Resisted supination**, resisted 3rd-finger extension	**Chair / push-up test**, clicking
Structure	ECRB	PIN	**LUCL**

How it's asked

"Most common tendon in tennis elbow?"	→ **ECRB** (TO)
"Microtear of which muscle?"	→ **ECRB** (TO) ✗ Wrong options: ED, ECU, ECRL
"Backhand pain, resisted middle-finger extension painful – attachment?"	→ **ECRB** (TO) ✗ Wrong options: BR, EDC, ECRL
"Carpenter, pain hammering – tendon?"	→ **ECRB** (TO; key not provided by source)
"Hammer pain, resisted wrist extension elbow extended – attachment?"	→ **ECRB** ✗ Wrong options: distal biceps, BR, ECRL, supinator

– – Active space – –

"Shares origin with the angiofibroblastic tendon?"	→ **anconeus** ✗ Wrong options: BR, annular ligament, FCU, PL
"Origin + insertion of implicated muscle?"	→ **lateral epicondyle, 3rd metacarpal** ✗ Wrong options: supracondylar ridge / 2nd MC = ECRL
"Labelled extensor origin (figure)?"	→ **B = ECRB** ✗ Wrong options: A ECRL, C ECU, D EDM, E EDC
"True about structure involved, golfer?"	→ **innervated by C6–C7 fibres** ✗ Wrong options: largest flexor CSA = brachialis · 2nd MC · cubital fossa border = BR · first fleshy fibres = ECRL
"True about histopathology?"	→ **inflammatory cells not seen** (TO)
"NOT true about lateral epicondylitis?"	→ **abundant macrophages** (TO) (underuse, hypercellularity, paucity of inflammatory cells = true)
"Region with asterisks on histology?"	→ **angiofibroblastic dysplasia**
"Histology of resected tissue?"	→ **angiofibroblastic tendinosis** ✗ Wrong options: chondroblastic · inflammation · calcium · no normal tendon
"3 mo pain, NSAIDs + band failed — next?"	→ **eccentric exercises** (TO) ✗ Wrong options: PRP, steroid, ESWT
"Squash player 3 mo — next step?"	→ **eccentric exercises** ✗ Wrong options: PRP, steroid, ultrasonic aspiration, surgery
"Golfer, short duration, X-rays normal — next?"	→ **eccentric PT program** ✗ Wrong options: steroid · radial tunnel injection · MRI · percutaneous tenotomy
"At 1 year, PT vs steroid?"	→ **PT more improvement than steroid**
"Steroid vs PRP vs placebo?"	→ **no consistent difference from placebo**
"PRP vs prior steroid — expected outcome?"	→ **↓ pain at 12 months** ✗ Wrong options: equal PROMs · faster return · better at 6 wk · better strength
"Most appropriate initial management (pathophysiology-based)?"	→ **OT: activity modification + eccentric strengthening + pain management** ✗ Wrong options: steroid, NSAIDs, surgery, PRP
"Post-arthroscopic release clicking, chair push-up pain at 3 mo — surgery?"	→ **LUCL reconstruction with palmaris / gracilis allograft** ✗ Wrong options: primary repair · MCL

"Excessive debridement beyond which point → instability?"	→ **C – equator of radial head** (posteriorly) ✗ Wrong options: distal = PIN
"Continuous elbow pain in tennis player?"	→ key not provided; risk factors = **incorrect grip size**, poor technique, heavy racket, high string tension

– – Active space – –

– – Active space – –

LATERAL ULNAR COLLATERAL LIGAMENT INJURY (PLRI)

– Shoulder & Elbow · 22 / 60 –

1 Basics

Primary elbow stabilisers (coronoid)

Primary constraint (TO) release alone → laxity · **secondary** – release alone insufficient, ↑ laxity only after primary divided

Primary constraints (TO)	Secondary constraints (TO)
Ulnohumeral articulation (coronoid)	**Radiohumeral articulation** (radial head)
Anterior MCL (AMCL)	Common **flexor-pronator** tendon
Lateral collateral ligament complex (LCLC)	Common **extensor** tendon
	Capsule

Coronoid (TO) **main stabiliser** of elbow; transection affects stability most · **loss of ≥50% → instability**
✗ Wrong options: anterior band MCL · LCL · radial head

Main osseous stability (TO) trochlea ↔ ulnar olecranon articulation; LCL + MCL blend into capsule

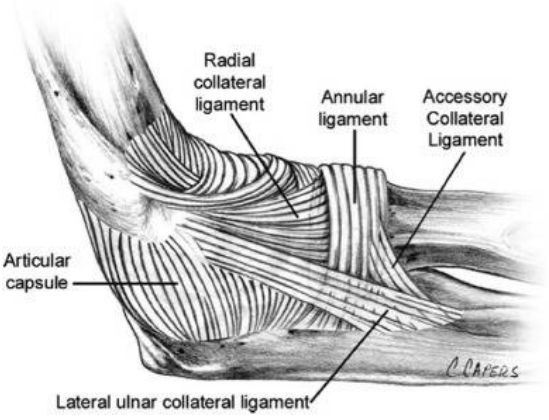
LUCL anatomy & failure site (humeral avulsion)

LCL complex (4) **radial collateral (LRCL)** · **annular** · **accessory lateral collateral** · **lateral ulnar collateral (LUCL)**

LUCL **primary stabiliser to varus + external rotation** (supination) stress; **most robust** of LCL complex; prevents ER subluxation of ulna on humerus

Origin → insertion **lateral epicondyle → tubercle of supinator crest (crista supinatoris)** of ulna

Failure site **humeral origin (lateral epicondyle)**, just anterior + distal to it; common extensor origin may also avulse; **radioulnar articulation intact**
✗ Wrong options: ulnar attachment



Lateral ligament complex: RCL, annular, accessory, LUCL

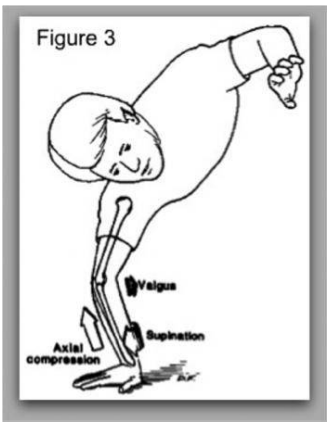
✗ Distractor (labelled diagrams)	Role
Annular ligament	Stabilises radial head in pronation/supination; does not need repair; not part of LUCL recon
Radial collateral / accessory collateral	Less important for PLRI
Orbicular ligament	Much less important laterally
Anterior band MCL	**Primary valgus** restraint (medial)
Posterior band MCL	Disrupted in **VPMRI**
Transverse ligament of Cooper	Coronoid → olecranon; **no stability role**
Posterior capsule	Contributes but does not cause PLRI
Anconeus	Does not need repair

✗ Location distractor	Site
Supinator crest	**LUCL insertion**
Radial tuberosity	Biceps insertion
Medial epicondyle	**MUCL origin**
Sublime tubercle	**MUCL insertion**

2 Mechanism

Definition PLRI = **posterolateral subluxation/dislocation of radiocapitellar + ulnohumeral joints**; ulna **supinates (ER) away from humerus**, radial head subluxes **posteriorly on capitellum** in **supination + extension**

Traumatic (most common) **elbow dislocation**: FOOSH → **supination + axial load + valgus** (posterolateral) + extension → LCL complex fails first → **anterior capsule / coronoid** → semilunar notch externally rotates off trochlea



Mechanism: axial compression, valgus, supination

Iatrogenic lateral elbow surgery (**lateral epicondylitis / ECRB debridement**, arthroscopic or open) going **posterior to equator of radial head** → excessive resection → **stay anterior to equator**

Chronic attenuation **cubitus varus malunion** → lateral thrust + abnormal **triceps vector** stretch LUCL

– – Active space – –

Radial head fractures usually **valgus posterolateral rotatory load** → concomitant LUCL injury common (avulsed from epicondyle)

O'Driscoll stages (circle of Horii)

Stage	Disruption	Result
1	**LUCL** ± partial/total LCL complex	**PLRI** (pain with varus stress)
2	+ **anterior capsule**	Incomplete posterolateral dislocation
3a	All soft tissue incl. **posterior MCL**, **except anterior bundle**	Dislocates posteriorly pivoting on **anterior bundle**
3b	**Complete MCL**	Full dislocation

Sequence **lateral → medial**

3 Presentation

Symptoms **pain** (vague, poorly localised) · **mechanical: clicking, catching, snapping, clunking, locking** with extension · **frank dislocation is not the most common presentation**

Classic **pushing off armrests to rise from chair** (axial load + supination + valgus) · difficulty with **triceps dips**, **push-ups**

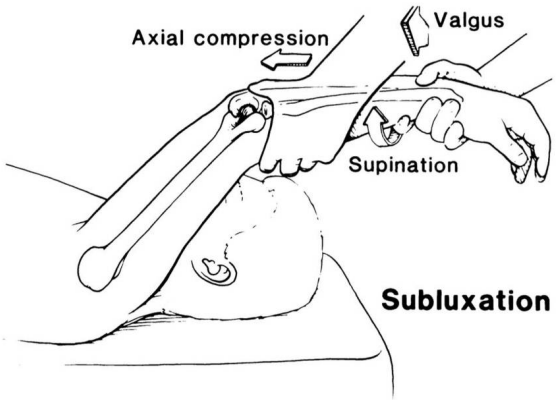
History prior **elbow dislocation** ("never felt right"), or prior **tennis elbow surgery**

Exam often benign; **tenderness over LUCL**, **varus instability** (no firm endpoint); full ROM

Diagnosis – lateral pivot-shift, chair push-up & clinical signs

Lateral pivot-shift (O'Driscoll) supine, arm **overhead** (shoulder fully ER), forearm **supinated**, **valgus + axial load**, elbow from **full extension → flexion** (to **40°** – Orthobullets; to 90° – OB)

Pivot-shift findings subluxed in extension → **reduces with flexion (triceps tension)** with **clunk**; max rotatory/posterior displacement **~40° flexion**; **skin dimple** proximal to radial head · awake = **apprehension**, anaesthetised = **clunk** (more reliable) · analogous to ITB in ACL-deficient knee



Lateral pivot-shift: supination, valgus, axial compression

– – Active space – –

Test	Technique / positive
Posterolateral rotatory drawer	Supine, overhead, supinated; finger under radial head → **posterior force + ER → radial head subluxes** behind capitellum
Apprehension	Overhead, supinated, valgus while flexing
Chair rise (chair push-up)	Pain/apprehension pushing up with forearms supinated (**= positive pivot shift**)
Table-top relocation	
Floor / prone push-up	**Can't push up with forearm supinated**

✗ Wrong option: valgus (MUCL) tests	Technique
Valgus stress	Valgus at **30° flexion**, medial pain
Milking manoeuvre	Elbow 90°, supinated, thumb pulled posteriorly
Moving valgus stress	Ranged **70–120°**, supinated, valgus
Symptom	**Pain late cocking / acceleration** = valgus instability

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral) rule out fractures, confirm **concentric reduction**; often **normal / little value** in PLRI; may show **impaction defect of posterolateral capitellum**

Fluoroscopy / stress views during pivot-shift → **radial head subluxation** (radiocapitellar + ulnohumeral)

MRI identifies **acute LUCL avulsion**; poor in recurrent/attenuated LUCL (**oblique course**); detects LUCL pathology in only **50%**

Elbow instability types

Instability	Cause / association
PLRI	**LUCL** – after dislocation; lateral pivot shift
Posteromedial rotatory (VPMRI)	**Varus** + **anteromedial coronoid facet** fracture
Valgus	**MUCL** – throwers, overuse
Varus	Rare; **lateral gapping**
Longitudinal forearm	**Essex-Lopresti**

– – Active space – –

	VPMRI	VPLRI
Radial head	**No** fracture	**Fracture**
Coronoid	**>15%, anteromedial facet**	**<15%, tip**
MCL	**Posterior band** ruptured; anterior band intact (on AM facet)	**Anterior band** ruptured
LCL	LCL complex (LUCL) avulsion	LCL complex (LUCL) avulsion
Exam	Valgus stress, moving valgus, milking	**Varus stress, chair rise, lateral pivot shift**

5 Management

LUCL injury / PLRI

- Acute simple dislocation, stable after reduction → splint 5–7 d → **early ROM**
- Unstable after reduction / entrapment / complex → **ORIF + LUCL repair**
- Mild chronic, low-demand → **brace + extensor strengthening, avoid varus**
- Chronic PLRI, failed nonop → **LUCL reconstruction with graft**

Acute dislocation – splinting position

Splint posterior splint **5–7 days** · **90° flexion** (Orthobullets) · **within stable arc, no specific angle** (OB) – read the options

Ligaments disrupted	Forearm position
LCL disrupted, MCL intact (usual)	**Full pronation** (tightens lateral structures)
LCL + MCL	**Neutral**
Isolated MCL	Supination – **not in PLRI** (LCL always torn)

Then early active ROM; full pronation/supination from 90° → full flexion; ↑ **extension by 30°/week in full pronation**; **full supination in extension after 6 wk**

✗ **Wrong options:** sling only, no immobilisation · splint at 45°/90° in supination

Operative criteria

Procedure	Indication
Open reduction + fracture fixation + LUCL repair	**Osteochondral fragment / soft-tissue entrapment** blocks concentric reduction · **complex dislocation** (fractures) · **acute instability** (e.g. unstable 0–60° after reduction)
LUCL reconstruction with graft	**Chronic PLRI** (tissue **attenuated**, not amenable to repair)
Bracing, extensor strengthening, avoid gravity varus	**Mild chronic PLRI, low-demand**
MCL reconstruction / hinged ex-fix	Only if lateral restoration fails – **rare**

Chronic ✗ traps	direct repair, isolated plication (less reliable) · **thermal shrinkage** (long-term effect unclear) · continued bracing after failed nonop · ECRB debridement · RCL reconstruction · MUCL (Tommy John, docking) reconstruction
Most common lateral ligament needing repair after posterolateral dislocation	**LUCL** (reliably torn, off humerus)

Reconstruction & postoperative protocol (avoid varus, pronation, splinting position)

Approach	**posterior midline** or **Kocher**
Graft	**palmaris longus** (most common), gracilis, triceps fascia; auto- or allograft
Configuration	tunnel at **supinator crest → "Y" tunnel** in humerus, tied over lateral column; graft must cover **> posterior 25% of radial head** (sling) · **+ capsuloligamentous plication**
Tensioning	**neutral rotation + 45° flexion** (Orthobullets)
Fixation	**docking** (humeral Y tunnels) · interference screws both sides · overlay (sutures over bone)



Docking LUCL reconstruction: Y tunnel humerus, supinator crest tunnel

Coronoid	**large fragment → screw from dorsal ulna** (ACL-type guide) · **small → excise + suture plication of anterior capsule** to tip
Post-op	**locked hinge brace**; **avoid varus** and **shoulder abduction** (gravity varus) **6–12 wk**; ROM with extension block → full extension + supination by **6–8 wk**
Forearm in rehab	**full pronation** during ROM / extension exercises until 6 wk (least stress on lateral repair); forearm most stable in pronation
Post-op avoid	**varus + supination** ✗ Wrong options: valgus + pronation

Ligament compromised	Extension exercises with forearm
LUCL (lateral)	**Pronation**
MUCL (medial)	**Supination**
Both	**Neutral**

– – Active space – –

Radial head arthroplasty + lateral repair	extension exercises in **pronation** ✗ Wrong options: avoid motion 6 wk → stiffness

Iatrogenic injury after lateral epicondylitis surgery

Presentation	after ECRB release/debridement: ↑ lateral pain, **catching/locking**, painful rising from chair, **+ pivot shift**
Injured	**LUCL** ✗ Wrong options: annular · orbicular · LRCL · anterior MCL
Prevention	keep debridement **anterior to equator of radial head**

6 Complications

Recurrent instability	**3–8%** (Orthobullets) · **<10%** after repair (OB) ✗ Wrong options: >10%
Cutaneous nerve injury	↓ with **posterior midline** approach
Others	infection · **↓ ROM**

X vs Y

	PLRI (LUCL)	Valgus instability (MUCL)
Patient	After dislocation / lateral surgery	**Thrower**
Symptom	**Clicking pushing off chair**	**Pain late cocking / acceleration**
Test	**Lateral pivot shift**, drawer, chair rise	**Milking**, moving valgus, valgus at 30°
Splint/rehab forearm	**Pronation**	**Supination**
Post-op avoid	**Varus, shoulder abduction**	Valgus

How it's asked

"Ligament D injury – characteristic symptom + sign?"	→ **painful clicking pushing off chair armrests; lateral pivot shift** ✗ Wrong options: late cocking · milking
"Volleyball player, old dislocation, catching rising from chair, clunk in supinated extension, normal imaging?"	→ **posterolateral rotatory instability** ✗ Wrong options: posteromedial · valgus · longitudinal
"Apprehension flexing supinated elbow with axial + valgus, posterolateral capitellum defect – disrupted structure?"	→ **LUCL** ✗ Wrong options: posterior band MCL · ligament of Cooper
"Chronic PLRI, + chair rise – manoeuvre to assess degree?"	→ **radiocapitellar reduction as supinated overhead elbow flexes from extension with valgus** ✗ Wrong options: subluxation on flexion · milking · moving valgus

"Supine, extension → flexion with valgus, supination, axial load — tests?"	→ **lateral ulnar collateral ligament**
"Positive pivot shift after dislocation — labelled ligament?"	→ **LUCL (C)** ✗ Wrong options: annular · anterior band UCL
"Axial load + valgus + supination in extension — injured ligament?"	→ **LUCL (E)** ✗ Wrong options: annular A
"Primary stabiliser to varus + ER, deficient after dislocation?"	→ **LUCL (E)** ✗ Wrong options: RCL · accessory
"Posterolateral dislocation — structure most likely needing repair?"	→ **LUCL (1)** ✗ Wrong options: annular · anconeus
"Most common rupture site of injured ligament?"	→ **humeral origin, lateral epicondyle (A)** ✗ Wrong options: supinator crest · sublime tubercle
"Old open dislocation, clicking with extension, can't do dips/push-ups — reconstruction?"	→ **LUCL reconstruction with palmaris graft** ✗ Wrong options: Tommy John · RCL
"Chronic PLRI failed nonop — treatment?"	→ **free tendon graft reconstruction + capsuloligamentous plication** ✗ Wrong options: direct repair · thermal shrinkage
"12 wk after dislocation, instability rising from chair — minimum procedure?"	→ **LCL reconstruction** ✗ Wrong options: LCL repair · hinged ex-fix
"Post-op told to avoid varus — ligament reconstructed?"	→ **LUCL (E)**
"Unstable 0–60° after reduction, repair planned — true?"	→ **avoid shoulder abduction post-op** ✗ Wrong options: ulnar-side avulsion · recurrence >10% · anteromedial coronoid
"Radial head arthroplasty + lateral repair — extension exercises?"	→ **with forearm pronated** ✗ Wrong options: supinated = MUCL · neutral = both
"Simple posterolateral dislocation reduced, no fracture — immobilise?"	→ **splint within stable arc in full pronation**
"After tennis elbow surgery, catching + PLRI — injured ligament?"	→ **lateral ulnar collateral** ✗ Wrong options: annular · orbicular
"ECRB debridement, + pivot shift — labelled structure injured?"	→ **LUCL (E)**
"Transection affecting elbow stability most?"	→ **coronoid** (TO)

– – Active space – –

LITTLE LEAGUE ELBOW

– Shoulder & Elbow · 23 / 60 –

1 Basics

Medial epicondyle apophysitis – pathogenesis & diagnosis

Definition — spectrum of **medial elbow** injuries in **adolescent pitchers**: **medial epicondyle stress fracture / apophysitis** · **UCL injury** · **flexor-pronator mass strain**

Most common cause of medial elbow pain in adolescent pitcher — **Little League elbow (medial apophysitis)** (TO stem; answer not given by TO source – OB: apophysitis far commoner than other throwing elbow conditions)
✗ Wrong options: golfer's elbow · olecranon bursitis · capitellar OCD

Age rule — **younger / open physis → apophysitis or avulsion** > UCL sprain; medial epicondyle apophysis **fuses at 16–18 y**

Pathogenesis — **repetitive valgus loading → tension overload** of medial structures; **repetitive flexor-pronator contraction stresses chondro-osseous origin → apophysitis**

✗ Pathogenesis distractor	Condition
Acute fragmentation of **entire capitellar ossific nucleus**	**Panner disease**
Microtraumatic **vascular insufficiency of capitellum** (compression + rotation)	**Capitellar OCD**
Rupture anterior band of anterior bundle UCL	UCL tear
Repetitive triceps contraction	Olecranon (posterior)

MCL (UCL) anatomy & biomechanics (TO)

MUCL components (TO) — **anterior oblique ligament (AOL / anterior bundle)** = strongest, **primary valgus restraint** · **posterior oblique / posterior bundle** · **transverse** ligament

Anterior bundle (TO) — **medial epicondyle → sublime tubercle**; primary valgus stabiliser **30–120°** (when throwing valgus is highest)

Bands of AOL (TO) — **reciprocal**: **anterior band tight in extension**, **posterior band tight in flexion**

Band (TO)	Primary range
Anterior band	Up to **90°** (one TO) · **20–120°** (another TO) – read the options
Posterior band	**>90°**, taut **90–120°**
POL	**>120°**

Other valgus stabilisers (TO) — **radial head** + **flexor-pronator mass** (dynamic)

Degeneration in pitcher's elbow (TO) — **posterior band of anterior bundle** (max force in **acceleration at 90–100°** flexion)
✗ Wrong options: anterior band · posterior bundle · anteromedial olecranon

Posteromedial olecranon (TO) repetitive microtrauma → **osteophytes at posteromedial** olecranon
✗ Wrong options: anteromedial

LCL complex (TO) LCL, annular, LUCL

2 Mechanism

Risk factors – pitch counts

Risk factor	Threshold
Pitches per game	**>80**
Months of competitive pitching	**>8 months/year**
Fastball speed	**>85 mph**
Pitching with **arm fatigue / pain**	Yes
Showcases	Yes
Breaking pitches (curveball, slider) before puberty	Yes (✗ Wrong options: changeup)

Not risk factors playing another position after pitching · **2 games/week** (allowed) · **delayed ossification** of medial epicondyle apophysis · 65–75 mph · 6–7 months/year

Age (y)	Max pitches / game	Max games / week
8–10	50	2
11–12	65	2
13–14	**75**	2
15–16	90	2
17–18	105	2

3 Presentation

Triad **medial elbow pain** in throwing arm · **↓ throwing speed (velocity), accuracy, distance**

Exam tenderness **medial epicondyle** (inferior aspect) · **pain with valgus stress** · **valgus instability = more severe** (check through flexion → extension)

Valgus tests for MUCL (TO)

Test (TO)	Technique	Accuracy
Valgus stress test	Elbow **20–30° flexion** (**unlocks olecranon**) + **forearm supinated**, humerus ER, valgus	Sens **66%**, spec **60%**
Milking manoeuvre	Pull **thumb** → valgus; **forearm supinated**, **elbow 90°**	
Moving valgus stress (O'Driscoll)	Upright, shoulder **90° abduction + max ER**, elbow max flexion → extend to 30° under constant valgus; **pain 70–120°** (late cocking → early acceleration)	Sens **100%**, spec **75%**

AOL integrity test (TO) **valgus stress at 20° flexion**
✗ Wrong options: full extension · varus · milking in pronation

– – Active space – –



Milking manoeuvre: thumb pulled, forearm supinated, elbow 90°

4 Diagnosis & Classification

Key radiology findings

X-ray AP + lateral often normal · **physeal widening** of medial epicondyle · **fragmentation / avulsion**; **compare contralateral** elbow



Medial epicondyle physeal widening and fragmentation

MRI more sensitive; **oedema of medial epicondyle apophysis**; differentiates **bony/physeal vs ligament** injury; confirms **UCL insufficiency**



MRI: oedema of medial epicondyle apophysis

First step — **rule out MCL injury**: **medial widening on stress X-ray / dynamic US**, or valgus instability on exam → **no widening + tender + physeal widening/ oedema = apophysitis**

– – Active space – –

✗ Diagnosis distractor	Clue
Displaced medial epicondyle avulsion	Displaced fragment → ORIF
MUCL tear	Medial widening on stress / instability
Valgus extension overload	**Olecranon (posteromedial) osteophytes**
Ulnar neuritis	Typical throwing condition but much less common than apophysitis

5 Management

Adolescent thrower, medial elbow pain

- Apophysitis / stress fracture / strain → **rest + activity mod + PT**
  (↓ innings, minimal immobilisation, coach + parent education)
- Displaced medial epicondyle avulsion → **ORIF (cannulated screws)**
- UCL disruption + insufficiency → **UCL reconstruction (palmaris)**

Treatment – rest & activity modification

Mainstay — **rest, activity modification, ↓ innings pitched per week, PT** (also for acute medial pain + valgus pain after a pitch)

PT aims — strengthen **dynamic stabilisers** (flexor-pronator), shoulder + elbow ROM, **core**; gradual return with proper mechanics + safe pitching rules

Immobilisation — **minimal** (maintain ROM); no long-arm cast 6–8 wk → **stiffness**

Avoid fragment excision — preserve medial epicondyle (**ligament + soft-tissue attachments**, stability)

No UCL reconstruction in Little League elbow — ligament **competent**

✗ Wrong option: arthroscopy / microfracture — for **capitellar osteochondral lesions**, not medial apophysitis

Capitellar OCD in young thrower

Lateral elbow tenderness, nondisplaced capitellar OCD — **cessation of throwing** (heals if activity curtailed)

Cast — only for **noncompliant**; **≤6 weeks**

Surgery — **loose bodies or cartilage flaps**
✗ Wrong options: excision/drilling for intact lesion · steroid injection

– – Active space – –

Operative criteria

Procedure	Indication
ORIF medial epicondyle	**Displaced medial epicondyle avulsion fracture**
UCL reconstruction (palmaris longus autograft)	**UCL disruption + insufficiency**

MCL (UCL) injury – reconstruction (TO)

UCL reconstruction indications (TO)
Acute complete rupture in **competitive thrower** wishing to remain active
High-grade distal tear
Chronic pain/instability after **≥4–6 wk** supervised nonop

Midsubstance tear + degeneration, wants to play (TO) — **MCL reconstruction**
✗ Wrong options: repair · brace/physio · arthroscopy

Nonop (TO) — partial tears; **"active" rest 2–3 months** (no throwing, no valgus stress) + **flexor-pronator strengthening**; also for athletes not returning to sport and asymptomatic in daily life

Paediatric UCL reconstruction — similar to adult; **palmaris longus autograft**

6 Complications

List — **ulnar nerve neuropathy** · continued pain + instability · loss of motion · **inability to return to same level**

X vs Y

	Little League elbow (medial)	Capitellar OCD (lateral)
Force	**Tension** (valgus, flexor-pronator pull)	**Compression + rotation** (radiocapitellar)
Pathology	Apophysitis / avulsion / stress fracture	**Microtraumatic vascular insufficiency** of capitellum
Tenderness	**Medial epicondyle**	**Lateral** elbow
Treatment	Rest + activity mod + PT	**Stop throwing**; surgery for loose body / cartilage flap

How it's asked

"Medial-sided elbow pain in adolescent pitcher most commonly due to?" → **Little Leaguer's elbow** (TO; answer not given by source)

– – Active space – –

"10-year-old pitcher: medial pain, ↓ effectiveness, ↓ distance — pathogenesis?"	→ **repetitive flexor-pronator contraction stressing chondro-osseous origin → apophysitis** ✗ Wrong options: capitellar fragmentation · UCL rupture
"12-year-old, tender medial, no widening on dynamic US — cause?"	→ **medial apophysitis** ✗ Wrong options: MUCL tear · valgus extension overload
"13-year-old, 85 pitches/game, 7 months/y, 75 mph — contributor?"	→ **pitches per game** (max 75 at 13)
"14-year-old, ↓ velocity, physeal widening — risk factor?"	→ **averaging 85 pitches/game** ✗ Wrong options: changeup · 65 mph · 6 months · delayed ossification
"6-year-old pitcher, lateral tenderness, nondisplaced capitellar lesion — initial?"	→ **cessation of throwing** ✗ Wrong options: cast 3 months · excision · drilling
"14-year-old acute medial pain after a pitch, MRI — initial treatment?"	→ **rest + activity modification** ✗ Wrong options: cast 8 wk · fragment excision · ORIF · reconstruction
"11-year-old, 3 straight days pitching, + moving valgus — initial?"	→ **rest + physical therapy** ✗ Wrong options: ORIF screws · reconstruction · cast ≥6 wk
"Degenerative changes in pitcher's elbow most common in?"	→ **posterior band of anterior bundle of UCL** (TO)
"Test for integrity of anterior oblique band of MUCL?"	→ **valgus stress at 20° flexion** (TO)
"Elbow position for valgus stress test?"	→ **20–30° flexion + supination** (TO)
"Athlete, MCL midsubstance tear with degeneration, wants to play?"	→ **MCL reconstruction** (TO)

– – Active space – –

LITTLE LEAGUER'S SHOULDER

– Shoulder & Elbow · 24 / 60 –

1 Basics

Definition

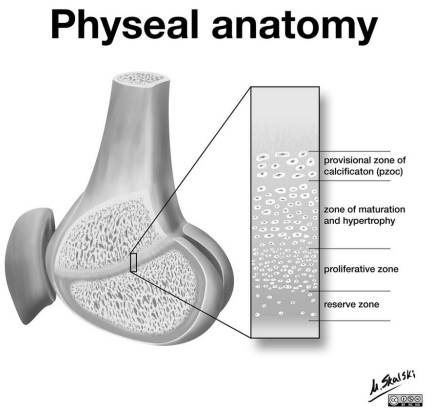
Little Leaguer's shoulder (LLS)	**proximal humeral epiphysiolysis** = overuse **Salter-Harris type I** stress injury of proximal humeral physis
✗ Wrong option: SH II	LLS does **not** produce a Salter-Harris II fracture

Epidemiology

Who	**skeletally immature overhead athletes**: **youth baseball pitchers** (dominant arm); occasionally **tennis**, football, racket sports
Sex	♂ **>** ♀
Age	**11–16 y** (Orthobullets, OB) · **11–14 y**, peak proximal physeal growth (OB)
Throwing	**10%** of paediatric shoulder pain is throwing-related

Physeal zone affected

Zone	**hypertrophic zone** = **weakest** part of growth plate
✗ Zone distractors	metaphysis (not a physeal zone) · provisional zone of calcification · **proliferative** · reserve



Physeal zones – LLS injures the hypertrophic zone

2 Mechanism

Forces repetitive **rotational (torsional) + distraction** stresses (± shear) on unossified physeal cartilage; **rotation > distraction** in importance

Throwing phase	Physeal stress
Late cocking (max ER, just before acceleration)	**Extreme rotatory torque ~400% greater** than physis tolerates → **commonest cause**
Deceleration	Forward arm motion vs cuff → **excessive eccentric** physeal stress
✗ Wrong option: early cocking	Not the peak

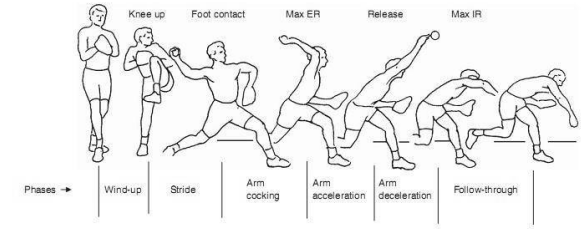


Fig. 1. Pitching phases and key events (adapted from Fleisig et al.,[12] with permission). **ER** = external rotation; **IR** = internal rotation; **max** = maximum.

Pitching phases — late cocking at max ER

Risk factors

Most important **number of pitches thrown** + **frequency of pitching** (innings, throwing days, pitches/innings per week–month–year, **multiple teams**, year-round play)

Breaking pitches **implicated** (Orthobullets) · **refuted / not worse than fastballs** (OB) — read the options; "% curveballs = most important factor" is wrong

Mechanics / body size poor throwing kinematics part of accepted theory (OB) · **no clear relationship** with weight/height or mechanics (another OB)

3 Presentation

History **↓ pitch velocity**, **↓ accuracy / control**; recent ↑ pitching (camp → season, two "select" teams)

Pain diffuse arm/shoulder pain **with throwing**, **anterolateral/lateral**; worse in **late cocking / deceleration**; resolves with rest → may progress to ADL / rest pain

Exam **point tenderness lateral proximal humerus over physis** · pain with rotation · **GIRD** (e.g. 30° IR difference) · ROM often full · no atrophy, no neuro deficit

– – Active space – –

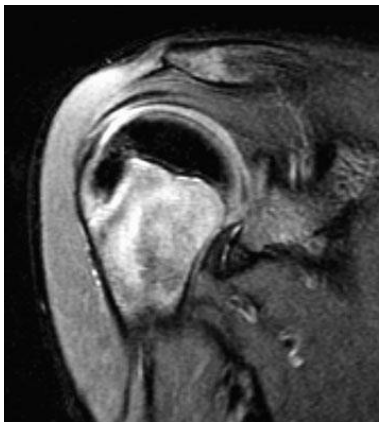
4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray views	**AP in external rotation**, scapular Y, axillary
Subtle cases	**AP in ER** (changes clearest **anterolateral physis**) + **contralateral comparison**
X-ray findings	**Widened (lateral) proximal humeral physis vs contralateral** · metaphyseal sclerosis · demineralisation/lucency · osteopenia · metaphyseal calcification · **fragmentation** · cystic change
Normal X-ray	**17%**
MRI	**Metaphyseal oedema adjacent to physis**; rules out **labral tear**, **partial articular-sided cuff tear** (rare concomitant)
MRI necessity	**Not needed** for diagnosis when X-ray classic



Widened proximal humeral physis vs normal side



MRI – oedema around proximal humeral physis

✗ View distractor	Used for
Distal humeral axial	Paediatric **medial epicondyle** fracture displacement
West Point axillary	**Bony Bankart**, glenoid bone loss
Zanca	**AC joint** (AP, 15° cephalic tilt, OB)
Swimmer's	Lateral **C-spine** when C7 not seen

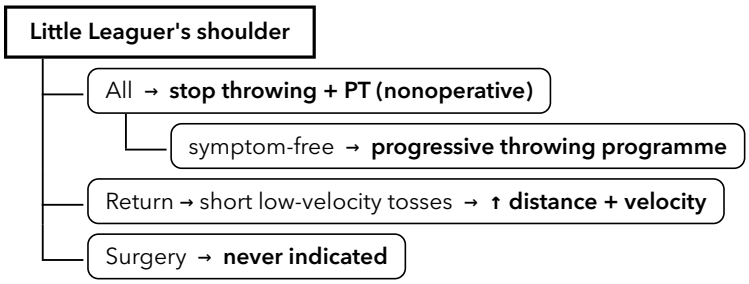
Diagnosis – physeal widening, imaging & throwing phase

Classic vignette — 11–16-yo pitcher, lateral shoulder pain with throwing, ↓ velocity, physeal tenderness, **widened physis → LLS**

15-yo, pain on throwing, ↓ IR, physeal widening (TO) — **Little League shoulder**
✗ Wrong options: adhesive capsulitis · pectoralis major tear · subacromial impingement

✗ Differential distractor	Why wrong
Septic arthritis	No infection
SLAP tear / internal impingement	Not this picture
External impingement	Subacromial bursitis, cuff tear
Chronic retracted supraspinatus tear	Age > 40
Fallen cortical fragment in empty cyst	**UBC** pathological fracture
Acromion tip stress fracture	LLS = **proximal humeral physeal** stress fracture
Deltoid strain	MRI oedema is **bony metaphyseal**

5 Management



Treatment – cessation of throwing & graded return

Mainstay — **cessation of throwing / pitching** until **completely asymptomatic → PT → progressive throwing programme**

Rest duration — **3 months** (Orthobullets; "minimum 3 months", OB) · **2–3 months** (OB) · **6–12 wk** (OB) · **6–8 wk** then progressive throwing (OB) · brief immobilisation + no pitching **for remainder of season** (OB) – read the options

PT — **rotator cuff strengthening**, **posterior capsule stretching**, core strengthening

– – Active space – –

Throwing programme start **only after symptom resolution**; short, low-velocity tosses → slowly ↑ distance + velocity

✗ Wrong management	Why
In situ screw fixation, ORIF (tension band), **CRPP**	Surgery not indicated; risks physis
Arthroscopic debridement / Bankart / labral repair	No labral pathology
Subacromial corticosteroid	Not impingement
Biopsy	Changes classic, not neoplastic
CT / MRI arthrogram as next step	Diagnosis already made
Continue playing with ↓ pitch count / breaking balls	Physis must heal
PT + continue pitching as tolerated	Risks growth arrest
Sling 4 wk + NWB	Unnecessary; ADLs allowed
Avoid throwing **until skeletal maturity**	Excessive

Prognosis **large majority return to pre-injury level**
✗ Wrong options: "unlikely to reach pre-injury level"

Prevention

Measures **proper mechanics (pitching coach) · discourage breaking balls until skeletal maturity · enforce pitch counts + rest days · avoid year-round pitching · ≤ 2 games/week**

Age (y)	Pitches/game	Max games/week
8–10	52	2
11–12	**68**	**2**
13–14	76	2
15–16	91	2
17–18	106	2

Recurrence risk factors (IR deficit, length of rest)

GIRD **3× more likely** recurrent symptoms; ↑ physeal injury risk

Inadequate rest **length of activity cessation** (e.g. only 1 month) = recurrence risk

> 2 games/week recurrence risk

Not recurrence risks pitch count within age limit (68 at 12 y) · avoiding breaking balls · pitching coach

6 Complications

Premature physeal closure **growth arrest** + **angular deformity** of proximal humerus

Cause **continuing to pitch** / not managed properly

X vs Y

	Late cocking	Deceleration
Arm	**Maximal external rotation**	Forward motion opposed by cuff
Physeal force	**Rotatory torque** (~400% > tolerance)	**Eccentric** stress
MCQ answer	**Commonest cause** of LLS	Also loads physis

	Orthobullets	OB explanations
Breaking balls	**Implicated**; discourage until maturity	**Refuted**; not worse than fastballs (avoidance still advised for recurrence)
Rest	**3 months**	6–8 wk · 6–12 wk · 2–3 months · remainder of season
Age	11–16	11–14 (also 11–16)

How it's asked

"15-yo, throwing pain, ↓ IR, physeal widening — diagnosis?"	→ **Little League shoulder** (TO)
"12-yo pitcher, lateral tenderness, AP vs contralateral — diagnosis?"	→ **Little Leaguer's shoulder** ✗ Wrong options: SLAP · internal/ external impingement · septic
"View that helps in subtle cases?"	→ **AP shoulder in external rotation** ✗ Wrong options: West Point · Zanca · distal humeral axial · Swimmer's
"12-yo throwing 100 pitches, anterolateral pain — imaging?"	→ **widened lateral proximal humeral physis** ✗ Wrong options: SH II · acromion stress fracture
"Physeal zone affected?"	→ **hypertrophic zone** ✗ Wrong options: proliferative
"Throwing phase causing LLS?"	→ **late cocking** ✗ Wrong options: early cocking
"14-yo pitcher, ↓ velocity, normal X-ray, MRI metaphyseal oedema?"	→ **proximal humeral epiphysiolysis** ✗ Wrong options: partial articular cuff tear · SLAP
"13-yo, MRI-confirmed physeal stress injury — treatment?"	→ **brief immobilisation + no pitching for remainder of season** ✗ Wrong options: until skeletal maturity · screw fixation
"15-yo pitcher, 3 months pain — next step?"	→ **stop throwing 6–8 wk → progressive throwing** ✗ Wrong options: MRI arthrogram · CT · biopsy
"12-yo on two select teams, physeal widening — management?"	→ **rest from throwing** ✗ Wrong options: steroid · biopsy · labral repair

– – Active space – –

"Next best step?"	**→ stop throwing until asymptomatic + PT** ✗ Wrong options: MRI · CRPP · continue activity
"Lateral shoulder pain worse in late cocking — plan?"	**→ no throwing 2–3 months + PT + progressive return** ✗ Wrong options: just ↓ pitch count · sling + NWB
"Team record for complete games — advice and outcome?"	**→ stop pitching, gradual return; likely back to pre-injury level**
"True statement about LLS?"	**→ IR deficit ↑ recurrence** ✗ Wrong options: proliferative zone · early cocking · % curveballs most important
"Rested 1 month, now 68 pitches, 1 game/wk, coach, no breaking balls — recurrence risk?"	**→ length of activity cessation**

LUXATIO ERECTA (INFERIOR GLENOHUMERAL JOINT DISLOCATION)

– Shoulder & Elbow · 25 / 60 –

1 Basics

Epidemiology

Incidence — **very rare: ~0.5%** of all shoulder dislocations (anterior + posterior far more common)

Definition — **inferior** GH dislocation; head trapped **under coracoid + glenoid**

Anatomy – static glenohumeral ligaments

Ligament	Restraint
SGHL	**Inferior** translation at **0° abduction** (neutral rotation)
MGHL	Anterior + posterior translation in **mid-abduction (~45°)** + ER
Anterior band IGHL	**Anterior/inferior** translation at **90° abduction + max ER** (late cocking)
Posterior band IGHL	**Posterior** subluxation at **90° flexion + IR**
Superior band IGHL	"Most important static stabiliser" (Orthobullets)

2 Mechanism

Energy — typically **high-energy**

Mechanism — **hyperabduction** → proximal humerus levered on **acromion** → tears inferior capsule/labrum → head disengages **inferiorly**

Soft tissue — variable tearing of static GH ligaments

3 Presentation

Clinical presentation – arm locked in abduction

Hallmark — arm **fixed in abduction, overhead** (e.g. **140°**); **can't lower arm → inferior** dislocation
✗ Wrong options: anterior · posterior · superior · lateral

Clinical diagnosis — possible on exam alone (unique posture); X-rays still for concomitant injury

– – Active space – –



Luxatio erecta — arm locked overhead

Associated injury	Rate / note
Neurological	**Up to 60%** — highest NV injury of all shoulder dislocations
Vascular	**Up to 39%** (axillary artery)
Proximal humerus fracture	Esp. **greater tuberosity**
Rotator cuff tear	**Extremely common** (older + young)
Anterior capsule / labral tear	Common

NV exam **pre- AND post-reduction**: radial + brachial pulse, **axillary nerve** + distal neuro exam

4 Diagnosis & Classification

Key radiology findings

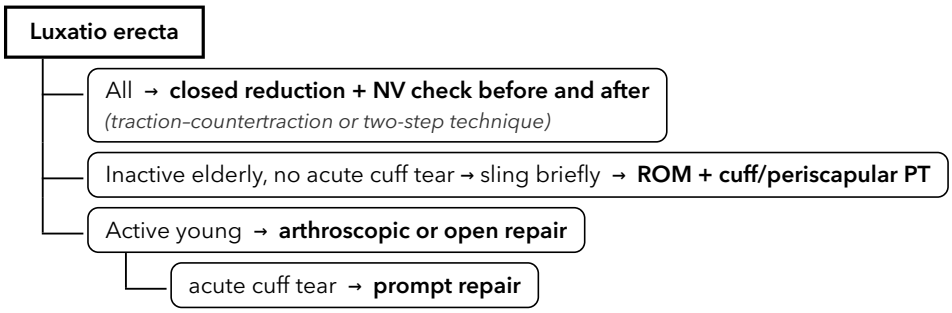
X-ray complete shoulder series → **inferior dislocation, arm fully abducted**



Inferior glenohumeral dislocation

MRI **after relocation** (soft-tissue injury common) → capsulolabral pathology, **cuff tears extremely common** · **prompt in young** to avoid missed cuff tear

5 Management



Closed reduction techniques

Traction–countertraction	as for anterior dislocation
Two-step technique	stand at head · **push humerus laterally + pull medial epicondyle superiorly** → head rotates **inferior → anterior** around glenoid (arm moves from abduction to **adduction against chest**) → then any anterior-dislocation reduction
Post-reduction	brief immobiliser → ROM (if cuff intact) → periscapular + cuff strengthening

Operative criteria

Option	Indication
Closed reduction + immobilisation	**Inactive elderly**; no acute traumatic cuff tear
Arthroscopic / open repair (repair vs reconstruction)	**Active younger** patients
Prompt cuff repair	**Acute traumatic cuff tear** (delay → retraction → irreparable)

Arthroscopy advantage addresses **capsulolabral damage + traumatic cuff tear** together

6 Complications

Complication	Key fact
Axillary nerve palsy	**Most common nerve palsy**; may resolve with reduction · persists → **EMG at 6–12 wk** (prognosis)
Brachial plexopathy	High energy + head displacement; usually resolves after reduction + observation
Axillary artery thrombosis	**May occur late**
Rotator cuff tear	Common in old + young → prompt MRI in young

– – Active space – –

X vs Y

	Luxatio erecta (inferior)	Anterior / posterior dislocation
Frequency	**0.5%**	**Much more common**
Arm fixed in abduction/ flexion	**Yes** (overhead)	**No**
NV injury	**Highest** (neuro ≤60%, vascular ≤39%)	Lower
Reduction	Two-step → convert to anterior	Standard techniques

How it's asked

"Shoulder locked at 140° abduction, can't lower arm — direction?"	→ **inferior** (luxatio erecta)
"Dislocation with highest neurovascular injury rate?"	→ **luxatio erecta**
"Most common nerve palsy?"	→ **axillary nerve**
"Persistent axillary palsy — when EMG?"	→ **6–12 weeks**
"Young patient after reduction — next?"	→ **MRI** (rotator cuff tear common)
"Late vascular complication?"	→ **axillary artery thrombosis**
"Two-step reduction converts inferior to?"	→ **anterior** dislocation

MEDIAL EPICONDYLITIS (GOLFER'S ELBOW)

– Shoulder & Elbow · 26 / 60 –

1 Basics

Epidemiology

Frequency **5–10× less common than lateral epicondylitis**

Demographics ♂ = ♀ · **dominant arm 75%**

Age 30s–60s, peak **30s–40s** (Orthobullets) · **4th–6th decades** / 40–60 y, peak working years (OB) – read the options

Sports repetitive **wrist flexion + forearm pronation** at ball release: **golf**, pitching, javelin, **bowling**, weightlifting, racquet sports

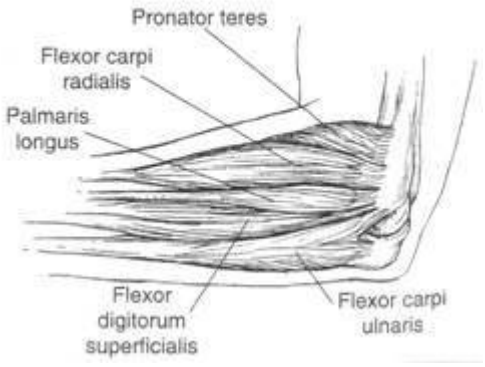
Tennis risks late ball strike (racquet head behind elbow) · poor forehand mechanics · no vibration dampeners

Occupation lifting **>20 kg**, forceful grip, vibration (plumbers, carpenters, construction)

Anatomy – common flexor tendon (CFT)

CFT **3 cm** long · anterior medial epicondyle + anterior bundle MCL · fibres parallel to MCL · ulnar head PT confluent with anteromedial capsule

Flexor-pronator mass	Nerve
Pronator teres	Median
FCR	Median
FDS	Median
Palmaris longus	Median
FCU	**Ulnar**



Flexor-pronator origin at the medial epicondyle

Muscles affected classically **PT > FCR** · newer studies: **all CFT muscles except palmaris longus** · 5 tendons form one common tendon
✗ Wrong options: number involved matters

– – Active space – –

Valgus role	flexor-pronators offload the **anterior band of MUCL** (primary static valgus restraint; deep to PT + FCR) · dynamic valgus support in **early acceleration** + wrist flexion at release

2 Mechanism

Pathology	repetitive **eccentric overload** → microtrauma at flexor-pronator origin
Stages	peritendinous inflammation → **angiofibroblastic hyperplasia** (as in lateral epicondylitis) → breakdown / fibrosis / calcification · **inflammation uncommon**

Associated conditions

Ulnar neuropathy	local inflammation involves ulnar nerve
UCL insufficiency	rule out, esp. throwers
Occupational cases (84%)	CTS · lateral epicondylitis · cuff tendinitis

3 Presentation

History	insidious medial pain · worse with wrist/forearm motion + **gripping** · **late cocking / early acceleration** · ± ulnar digit paraesthesia · ↓ grip strength · rarely acute blow → CFT avulsion
Tenderness	**5–10 mm distal + anterior to medial epicondyle** (PT/FCR origin)
Chronic	flexion contracture

Diagnosis – provocative testing

Test	Condition
Resisted forearm pronation + wrist flexion (fist clenched)	**Medial epicondylitis**
Resisted supination + wrist extension	Lateral epicondylitis
Moving (dynamic) valgus stress	MCL injury
Milking manoeuvre	MCL origin apprehension/ instability/pain
Pinch grip	AIN palsy
Elbow flexion test (max flexion, pronation, wrist extension 30–60 s), Tinel, 2PD	Ulnar neuritis

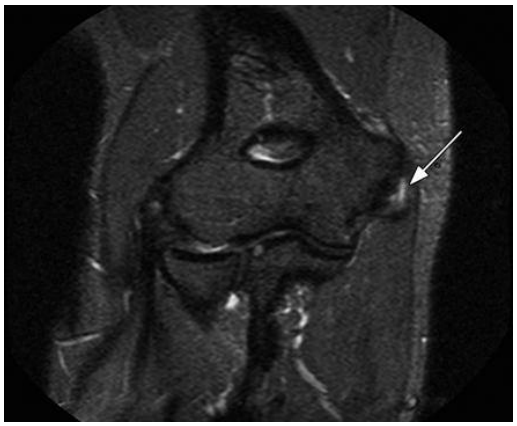
UCL vs flexor-pronator pain	UCL pain **posterior + distal** to medial epicondyle

4 Diagnosis & Classification

Key radiology findings

X-ray	usually normal · **25% calcification** of CFT or UCL · posteromedial osteophytes · stress views for valgus instability

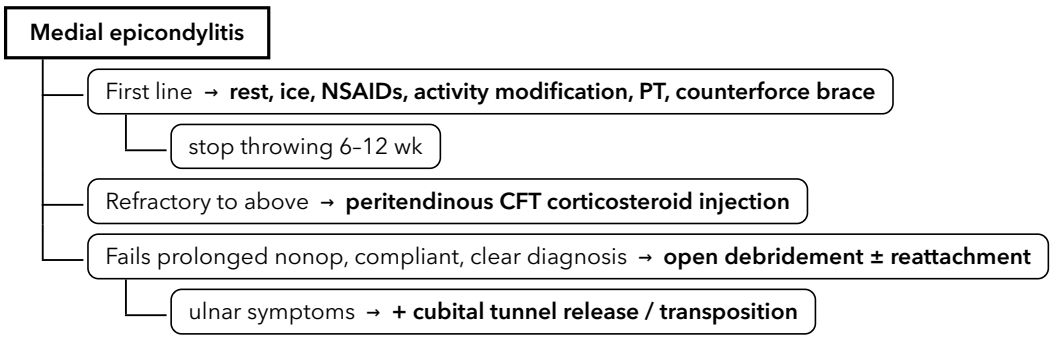
US	**>90%** sensitivity/specificity/PPV/NPV · dynamic · hypoechoic/anechoic degeneration
MRI	standard of care · **↑ T2 signal** in CFT (tendinosis/disruption) + peritendinous oedema · use for UCL/osteochondral injury, loose bodies, flexor-pronator rupture, unclear source



MRI: T2 hyperintensity at the CFT origin

EMG/NCS	only if ulnar nerve symptoms (not routine)
Differential	MCL injury · cubital tunnel · fracture · cervical radiculopathy · triceps tendinitis · **herpes zoster**

5 Management



Nonoperative treatment – injection

Prolonged nonop	appropriate: surgery **less predictable than lateral** epicondylitis
Modalities	counterforce brace / kinesiology tape · therapeutic US beneficial · **ESWT**: no definitive recommendation, equivocal (Orthobullets, OB) · ESWT "~90% pain relief" (one OB explanation) – read the options
Next step after 2 mo brace + NSAIDs + PT + rest	**corticosteroid into CFT peritendinous area** (short-term benefit) ✗ Wrong options: arthroscopic debridement · cubital tunnel injection · EMG without ulnar symptoms · acupuncture, not shown to help
Steroid complications	skin **depigmentation** (dark skin) · subcutaneous **atrophy** · tendon weakening · **ulnar nerve injury** · avoid multiple injections

– – Active space – –

Operative criteria

Criterion	Value
Failed nonop in **compliant** patient	**Up to 6 mo** (Orthobullets) · **4–6 mo** · **at least 6 mo** (OB)
Symptoms	Severe, affecting quality of life
Diagnosis	Clear
Exception	**Elite athlete with definite tendon tear** → earlier

Surgical outcomes & ulnar nerve involvement

Procedure — **open medial approach** · **PT–FCR interval** · excise pathologic tissue + side-to-side repair · ± epicondyle microfracture · **reattach flexor-pronator** if proximal origin involved · ± cubital tunnel release/transposition
✗ Wrong options: arthroscopic

Reattachment — debridement **with or without** reattachment → equivalent outcomes (Arevalo; complications 4.3%, RTS/RTW 81–100%)

Outcome — **pain and function both improve** · good/excellent **80%** (Orthobullets, < lateral) · **80–90%** (OB) · **87%** at 7 y (Gabel & Morrey)

Worst prognostic factor — **degree of ulnar neuropathy** (only parameter correlating with outcome; mild/none 24/25 good · moderate–severe 3/5 fair/poor)
✗ Wrong options: prolonged nonop · no reattachment · number of tendons · ESWT

Key exam finding — **pronation weakness at 90°**

Rehab — sling **1–2 wk** (avoid wrist volar flexion) · ROM after 2 wk · strengthening **6–8 wk** · **RTS 3–6 mo**

6 Complications

MABC neuropathy — avulsion/transection → if seen intra-op, **transpose into brachialis**

Others — ulnar nerve injury · infection

Prognosis — harder to treat + less studied than lateral epicondylitis

X vs Y

	Medial epicondylitis	Lateral epicondylitis
Frequency	5–10× less common	More common
Origin	Flexor-pronator (**PT, FCR**)	–
Provocation	**Resisted pronation + wrist flexion**	Resisted supination + wrist extension
Surgical success	**~80%**, less predictable than lateral	–
Associated nerve	**Ulnar**	–

	Medial epicondylitis	UCL injury
Tenderness	5–10 mm distal + anterior to epicondyle	**Posterior + distal** to epicondyle
Test	Resisted pronation + flexion	**Milking**, moving valgus stress

How it's asked

"Provocative test positive in medial epicondylitis?"	→ **resisted forearm pronation + wrist flexion with clenched fist** ✗ Wrong options: supination/extension = lateral · dynamic valgus/milking = MCL · pinch grip = AIN
"Masters tennis player, 2 mo PT, brace, NSAIDs, rest fail – next?"	→ **corticosteroid injection into CFT peritendinous area** ✗ Wrong options: arthroscopy · cubital tunnel injection · acupuncture · EMG
"Bowler-carpenter, 7 mo failed nonop, open debridement + reattachment – 1-y outcome?"	→ **pain and functional outcome scores both expected to improve**
"Surgical golfer's elbow – factor linked to inferior outcome?"	→ **degree of ulnar nerve involvement** ✗ Wrong options: prolonged nonop · no reattachment · number of tendons · ESWT
"Nerve injured at medial elbow surgery – intra-op fix?"	→ **MABC, bury in brachialis**

– – Active space – –

MEDIAL ULNAR COLLATERAL LIGAMENT INJURY

– Shoulder & Elbow · 27 / 60 –

1 Basics

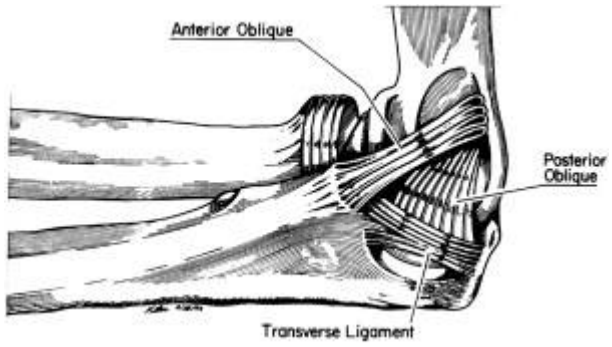
Epidemiology

Definition	attenuation / rupture of elbow UCL → **valgus instability** in **overhead throwers**
Who	originally **javelin** throwers → now mostly **baseball pitchers**; ↑ in high school + amateur pitchers; also water polo, wrestlers
Skeletal maturity	**uncommon in skeletally immature** → they get **little leaguer's elbow** (medial epicondyle) · isolated MUCL tear = **young skeletally mature** pitcher
Risk factors	**exceeding pitch count / innings limits** · short rest, no off-season · **higher pitch velocity** · poor mechanics · **kinetic chain deficits** (shoulder + core weakness, ↓ shoulder motion)

Anatomy & biomechanics (anterior bundle, isometry, posterior bundle)

Carrying angle	**6–11°** valgus (UCL note) · ♂ 5–10° / ♀ 10–15° (anatomy note) · ~11° ♂ / ~13° ♀ (exam note) – read the options
Stability	split evenly **osseous vs soft tissue**; UCL = primary valgus restraint **30–120° flexion** (+ flexor-pronator, capsule)

UCL component	Facts
Anterior oblique ligament (AOL) = anterior bundle	**Strongest, most important valgus stabiliser**; origin **anteroinferior medial epicondyle → sublime tubercle**, **2.8 mm distal** to ulnar articular margin; mean length **54 mm**; primary **30–120°**; greatest resistance at **~70°**
Posterior oblique ligament (POL) = posterior bundle	**Greatest change in tension** flexion → extension; **tight in flexion**; fan-like **capsular thickening**; inserts **sigmoid notch**; not subdivided into bands; secondary stabiliser at 30°
Transverse ligament	**No stability** – originates + inserts on **ulna** (same bone)



UCL complex – anterior oblique, posterior oblique, transverse ligament

Terminology trap	AOL = "anterior bundle" = older "anterior band"

AOL bands — **2 bands**: anterior + posterior (Orthobullets, most OB) · **3 bands**: anterior, central, posterior (2 OB explanations) · anterior, posterior, **oblique** (1 OB) – read the options

AOL band	Strain / tautness	Primary role
Anterior band	**Isometric** through ROM (✗ Wrong options: isokinetic); **taut 0–85°**; tightens at end-range **extension**; most vulnerable **in extension**	**Most important valgus restraint**; primary at **30°, 60°, 90°**
Posterior band	**↑ strain in flexion** (✗ Wrong options: isometric); **taut beyond 55°**; vulnerable **in flexion**	Primary at **120°**

Posterior bundle range — resists from **60° to full flexion** (one OB) · beyond **90°** (others)

Axis of rotation / isometric point — at UCL **origin = anteroinferior medial epicondyle**
✗ Wrong options: anterosuperior · posteroinferior · sublime tubercle = insertion

Flexion	Main valgus restraint
0–20° (extension)	**Bony: olecranon in fossa + radiocapitellar articulation**
~25°	Olecranon **unlocked** → UCL takes over
30–120°	**AOL (anterior band)**
>90° / max flexion	**POL**

Radial head — important **secondary** stabiliser in extension + flexion

Dynamic stabilisers — **flexor-pronator mass** (**FCU, FDS**) – dynamic, not static

2 Mechanism

Acute trauma — often with **elbow dislocation**

Overuse — **repetitive valgus microtrauma** → rupture of **anterior band** of AOL

Iatrogenic — **excessive olecranon osteophyte resection** puts MCL at risk

Associated — **traction ulnar neuritis** · **posteromedial olecranon impingement (VEO)** · elbow arthritis

Throwing phase of maximal valgus stress (late cocking/early acceleration)

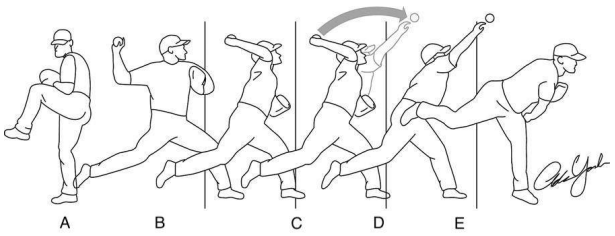
Max valgus — **late cocking → early acceleration**: forearm lags behind arm → valgus while elbow depends on **anterior band**; forces near **tensile limit** of UCL

Single best phase — **late cocking** (Orthobullets, several OB) · **early acceleration** (other OB: "highest in early acceleration") – choose **late cocking/early acceleration** if offered; else whichever of the two is listed

– – Active space – –

Forces valgus up to **64 N·m** · **500 N** compression at radiocapitellar joint · elbow **110° → 20°** · **3000°/s**

Phase	Key activity / pathology
Wind-up	Benign for elbow
Early cocking	Peak **deltoid**; cuff active
Late cocking	**Max shoulder ER ("layback")**, high torque → **max elbow valgus**; FDS + FCU protect
Early acceleration	Peak **triceps**; valgus + rapid extension
Late acceleration	Peak **pec major, latissimus, serratus anterior**
Ball release	Not a phase; end of acceleration
Deceleration	**Eccentric** contraction of all muscles, **highest torque**, "most harmful" → **SLAP, biceps, brachialis, teres minor** (✗ Wrong options: UCL)
Follow-through	Rebalance; elbow flexors prevent hyperextension; minimal UCL stress



Throwing phases – A wind-up, B early cocking, C late cocking, D late acceleration, E deceleration

Figure labels vary one figure: **C = late cocking** · another: **B = late cocking** (A early cocking, D deceleration) – read the figure

Valgus ↓ by **flexor-pronator contraction** · **forearm pronation** · **scapular protraction/retraction (trunk-scapular kinesis)** · **reduced velocity** · correct mechanics

Valgus ↑ by **↑ glenohumeral IR torque** · poor mechanics

3 Presentation

History acute **"pop"** + pain + can't throw · chronic: **loss of velocity ("zip")**, **loss of control/accuracy**

Pain medial/posteromedial in **late cocking + acceleration** · posteromedial in **deceleration** = **valgus extension overload**

Ulnar nerve paraesthesia ring + small (nerve **stretch** in late cocking) → check subluxation

Exam tenderness at/near **MCL origin** (posteromedial = VEO) · flexor-pronator integrity · **palmaris longus present?** (graft) · throwers may **lack full extension** · examine shoulder + kinetic chain

Throwing tolerance pain-free light throwing / several innings, no trauma → partial injury; complete rupture unlikely

4 Diagnosis & Classification

Diagnosis – moving valgus stress, milking manoeuvre & MRI T-sign

Test	Technique	Positive / accuracy
Valgus stress test	Elbow **20–30°** flexion (unlocks olecranon) + **humerus ER** + valgus	**50% sensitive**
Milking manoeuvre	**Pull thumb**, forearm **supinated**, elbow **90°**; supine/seated/standing	**Apprehension, instability or pain at MCL origin**
Moving valgus stress test	Milking position; valgus while ranging full arc (or valgus at full flexion → **quickly extend**)	Pain/apprehension **70–120°**; **100% sensitive, 75% specific**; **most strongly confirms**



Milking manoeuvre – valgus via thumb with forearm supinated

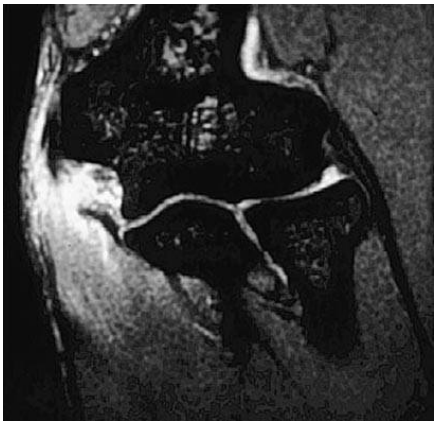
Pain in mid-arc (70–120°) **anterior bundle MCL · pain at terminal extension = posterior compartment / olecranon osteophytes**

✗ Exam distractor	Indicates
Pain with **resisted wrist flexion** (± pronation)	**Medial epicondylitis / flexor-pronator strain**
Froment's sign	Ulnar neuropathy
Hawkins	Shoulder impingement
Lateral pivot shift, **chair test**, **push-up sign**, PLRI test	**PLRI (LUCL)**
Painful **snapping** medially	**Ulnar nerve or triceps subluxation**

– – Active space – –

Key radiology findings

Modality	Fact
X-ray AP + lateral	Often **normal**; loose bodies, **UCL calcification**, **posteromedial osteophyte** (VEO)
Oblique views	Olecranon
Stress X-rays (gravity/ manual), both elbows	**Medial opening >3 mm = diagnostic**
MRI	**Gold standard / mainstay**; thickened ligament (chronic), calcification, midsubstance tear, proximal/distal avulsion, partial **undersurface** tear
Plain MRI	**57% sensitive, 100% specific**
MR arthrogram	**Diagnostic**, more accurate: **92% sensitive, 100% specific** (✗ Wrong options: "lacks specificity")
"Capsular T-sign"	**Contrast extravasates** around distal (undersurface) tear at sublime tubercle
Dynamic US	Laxity under valgus; operator dependent



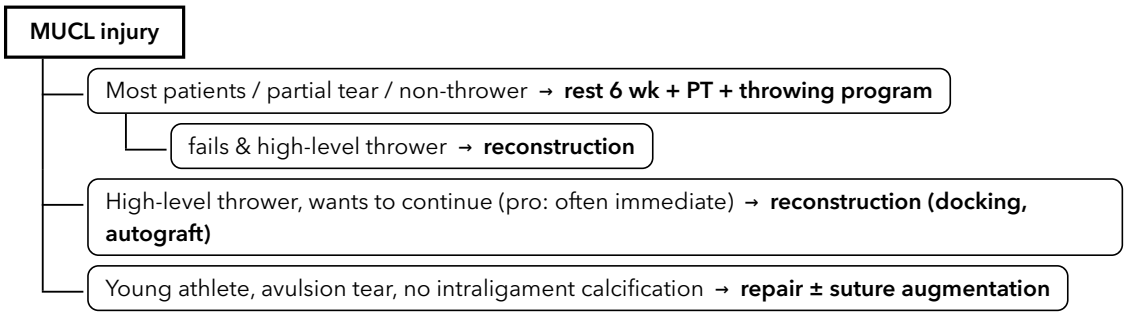
MR arthrogram – capsular T-sign at sublime tubercle

MRI labels (water polo Q) 1 = **flexor-pronator origin** · 2 = **UCL** · 3 = **RCL**

Differential diagnosis

Diagnosis	Discriminator
MUCL injury	**Moving valgus +**, milking +, loss of velocity
Medial epicondylitis / flexor-pronator strain	Pain on **resisted wrist flexion/pronation**
Ulnar neuropathy (isolated)	Doesn't explain valgus instability
Valgus extension overload	**Pain at terminal extension**; posteromedial osteophyte
Little leaguer's elbow (medial epicondyle stress fracture)	**Skeletally immature**

5 Management



Non-thrower partial injury – nonoperative care

First line — most patients + **partial tears**: **rest / activity avoidance 6 wk** + NSAIDs → **flexor-pronator strengthening** (secondary dynamic stabilisers) + mechanics, peri-scapular + core → **progressive throwing program**

Outcome — **42%** return to pre-injury sport at avg **24 wk** · most heal without surgery in non-high-level athletes

Adolescent acute partial (14-y basketball) — **activity avoidance 6 wk**
✗ Wrong options: stretching (full ROM) · arthroscopy (unless loose body) · static ex-fix (brace OK) · rheumatology work-up

Persistent at 6–12 wk — advanced imaging (MRI, MR arthrogram, US, stress X-rays)

Operative criteria

Procedure	Indication
UCL reconstruction (Tommy John)	**High-level throwers wanting to continue** · **failed nonop partial tears** willing to rehab · **midsubstance** tears · attritional tears with **calcification** (excision leaves void)
UCL repair	**Young athletes, avulsion-type tears** (proximal/distal), no intraligament calcification; ± high-strength suture augmentation
✗ Wrong option: repair	**Midsubstance tear** (reconstruction superior)

Repair history — early repairs poor → replaced by reconstruction; newer augmented repair promising

Reconstruction technique (docking vs Jobe) & return to play

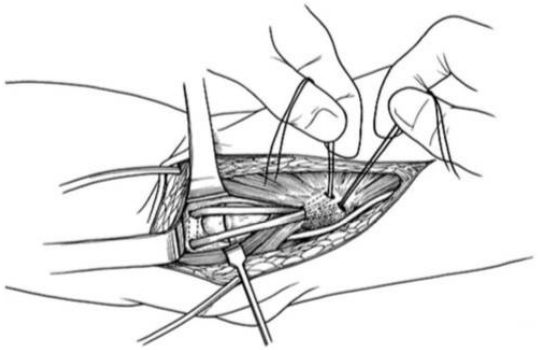
Goal — anatomic reconstruction of **anterior bundle** (medial epicondyle → sublime tubercle); **none stronger than native**

Approach — **flexor-pronator muscle-splitting** (✓ ↓ morbidity)
✗ Wrong options: **detachment** (original Jobe; flexor-pronator avulsion risk) · **medial epicondylectomy** in throwers

– – Active space – –

Graft — **ipsilateral palmaris longus autograft** most common/standard · gracilis autograft or allograft · PL vs gracilis both successful (PL possibly ↑ high-level return; gracilis does not ↓ laxity) · revision **autograft 1.6% vs allograft 2.6%**

Technique	Humerus	Ulna	Notes
Original Jobe (1986)	Tunnels	Tunnel	**Flexor-pronator tenotomy**, **figure-of-8**, **routine submuscular ulnar nerve transposition → abandoned**
Modified Jobe	**2 connected tunnels, "Y"**	Single tunnel (2 angled holes, sublime tubercle)	**Muscle-splitting**, no routine transposition; graft sutured **figure-of-8**
Docking	**Single socket**; sutures tied over bony bridge	**Two converging tunnels**	**Best**: **↑ return to sport, ↓ complications**, less nerve handling, **better graft tensioning**
Hybrid interference screw (DANE TJ)	Docking socket	**Single longitudinal socket + interference screw**	↓ iatrogenic fracture; docking + ulnar screw = **95% native strength**
Cortical button (Endobutton)	–	Suspensory	Mostly **revision**



Docking technique

Docking vs Jobe — **~15% more excellent results** · **3× fewer complications** · **3–5% less ulnar neuropathy** · RTP **>90%** vs **80–85%** (figure-of-8)

Biomechanics — docking + cortical button **stronger** than figure-of-8 + interference screw

Calcified attritional UCL in high-level thrower — **reconstruction (modified Jobe)**
✗ Wrong options: original Jobe · repair ± augmentation · arthroscopic debridement + repair

Return to play	Rate
Modern reconstruction	**~90%** (Orthobullets) · **85–95%** (OB) – answer **80–95%**
All athletes (systematic review)	**86%**
Collegiate	**95%**
Surgery overall good results	84–94%
Revision reconstruction	**Worse**

Rehab timeline — early wrist/elbow/shoulder ROM + core/hip · strengthening **4–6 wk** · **no valgus stress until 4 mo** · throwing program **4 mo** · **competition 9–12 mo**

Ulnar nerve management & postoperative neuritis

Pre-op ulnar status	At UCL reconstruction
No symptoms	**No routine decompression/transposition** (in situ release at most)
Symptoms (cubital tunnel)	Decompression ± transposition
Subluxation	**Transposition**

Routine transposition — avoid → ↑ **post-op ulnar neuropathy**, worse outcomes

Midsubstance tear + cubital tunnel symptoms + subluxation — **autograft reconstruction + ulnar nerve transposition**
✗ Wrong options: repair · reconstruction without nerve

Ulnar nerve in forearm — **between FCU and FDP**
✗ Wrong options: medial to brachial artery / most volar in carpal tunnel = median

Postoperative rehabilitation – forearm position

Repair	Elbow extension exercises with forearm in
MUCL reconstruction	**Supination**
LUCL repair (± radial head replacement)	**Pronation**
Combined medial + lateral	**Neutral**

6 Complications

Ulnar neurapraxia — **most common**; **3–26%** (Orthobullets, OB) · "<10%" (one OB) – read the options; **treat by observation** (most resolve weeks–months)

↓ Neurapraxia by — muscle-splitting · **more superior exiting proximal humeral tunnels** · selective transposition · docking

MABC injury — crosses **distal** incision

Fracture — ulna or medial epicondyle; **small bone bridge** → ORIF / larger fixation

Stiffness — **HO** → early therapy; HO excision ~**6 mo**

Others — failure to regain pre-injury level (esp. **revision**) · infection · flexor-pronator avulsion (detaching approach)

Not MUCL complications — PLRI / radial head + coronoid tip fractures (terrible triad, LCL)

X vs Y

	Anterior band (AOL)	Posterior band (AOL)	POL (posterior bundle)
Strain	**Isometric**	↑ in flexion	**Greatest change**
Taut	**0–85°**	**>55°**	**>90°** / max flexion
Role	**Primary valgus restraint**	Primary at 120°	Secondary; max flexion; cubital tunnel floor

– – Active space – –

	Original Jobe	Modified Jobe	Docking
Flexor-pronator	**Detached (tenotomy)**	**Split**	**Split**
Ulnar nerve	**Routine submuscular transposition**	Not routine	Minimal handling
Humeral fixation	Figure-of-8 tunnels	**Y-tunnels, figure-of-8**	**Single socket, tied over bridge**
Outcome	Abandoned	Good	**Best RTP, fewest complications**

How it's asked

"Primary valgus stabiliser in mid-flexion?"	→ **AOL of MUCL** ✗ Wrong options: ulnohumeral · radiocapitellar · POL · flexor-pronator
"All protect elbow from valgus EXCEPT?"	→ **↑ glenohumeral IR torque** ✗ Wrong options: flexor-pronator contraction · pronation · scapular motion · ↓ velocity
"Native MCL kinematics?"	→ **posterior bundle = greatest tension change** ✗ Wrong options: posterior bundle isometric
"Most static valgus stability?"	→ **anterior band of UCL** ✗ Wrong options: flexor/pronator = dynamic
"Most important static valgus stabiliser in thrower?"	→ **anterior band of anterior bundle**
"Structure B (posterior bundle) – true?"	→ **most change in tension flexion → extension** ✗ Wrong options: subdivided into bands
"Javelin thrower – ligament, arc, insertion?"	→ **AOL, 30–120°, sublime tubercle** ✗ Wrong options: POL / sigmoid notch
"Isometric component?"	→ **anterior band of anterior bundle** ✗ Wrong options: isokinetic
"UCL anatomy – true (figure)?"	→ **AOL greatest resistance at 70°** ✗ Wrong options: both bands isometric · POL tightest in extension
"Axis of rotation for repair?"	→ **anteroinferior medial epicondyle** ✗ Wrong options: sublime tubercle
"Greatest stress on MUCL – phase?"	→ **late cocking** ✗ Wrong options: wind-up · early cocking · ball release · follow-through
"Pitcher ruptures MUCL – phase?"	→ **early acceleration** (when late cocking not offered)

"Forces approaching tensile limit?"	→ **late cocking / early acceleration**
"Water polo, lost velocity — phase + MRI label?"	→ **late cocking (C) + UCL (2)**
"Milking-positive pitcher — phase reproducing symptoms?"	→ **late cocking (figure B)** ✗ Wrong options: deceleration = labrum/biceps
"Highest valgus load phase (pro pitcher pre-reconstruction)?"	→ **early acceleration**
"Medial pain, positive milking, lost velocity — phase?"	→ **late cocking** (figure C)
"Valgus via thumb at 90° + moving valgus pain — phase?"	→ **late cocking**
"Pitcher lost 10 mph, normal EMG — exam that best confirms?"	→ **moving valgus stress test** ✗ Wrong options: resisted wrist flexion · Froment · Hawkins · pivot shift
"MR arthrogram — diagnosis?"	→ **MCL rupture**
"Medial pain after 3 innings, milking test reproduces?"	→ **MCL injury** ✗ Wrong options: ulnar nerve / triceps subluxation = snapping
"Pitcher, pain in acceleration, lost velocity — structure?"	→ **anterior bundle UCL**
"Valgus through passive ROM painful 70–120°?"	→ **anterior bundle MCL** ✗ Wrong options: olecranon osteophytes = terminal extension
"Positive moving valgus, lateral X-ray — labelled structure?"	→ **anterior oblique band (C)**
"ER humerus + supinate + valgus, apprehension 70–120°?"	→ **MUCL** ✗ Wrong options: LUCL · annular · anterior capsule · triceps
"Skeletally mature 17-y pitcher, overused, ulnar paraesthesia, valgus pain?"	→ **attenuation of anterior band of AOL** ✗ Wrong options: medial epicondyle stress fracture · isolated ulnar neuropathy · flexor-pronator
"Capsular T-sign — true?"	→ **evaluate by pulling thumb, forearm supinated, elbow flexed** ✗ Wrong options: MRA lacks specificity · all isometric · 0–20° · deceleration
"Failed rest + interval program, UCL tear on MRI?"	→ **open reconstruction with ipsilateral palmaris longus** ✗ Wrong options: ECRB release · OATS · flexor-pronator reattachment · osteophyte excision
"Lowest complication + best outcome technique?"	→ **muscle-splitting + docking** ✗ Wrong options: routine transposition · detachment

– – Active space – –

"Best return to sport, minor leaguer?"	→ **docking + ulnar nerve in situ release** ✗ Wrong options: classic Jobe · submuscular transposition · epicondylectomy
"Calcified UCL, failed rest — surgery?"	→ **modified Jobe reconstruction** ✗ Wrong options: original Jobe · repair
"Jobe vs docking?"	→ **docking: higher RTP, fewer complications** ✗ Wrong options: 20% neuropathy regardless
"Pro pitcher, no ulnar symptoms — procedure?"	→ **autograft docking reconstruction** ✗ Wrong options: allograft · figure-of-8 · nerve decompression
"Return to preinjury throwing after docking?"	→ **80–95%**
"Midsubstance tear + cubital tunnel symptoms + subluxation?"	→ **autograft reconstruction + ulnar nerve transposition**
"Most common complication — nerve lies where?"	→ **between FCU and FDP** (ulnar)
"Surgical management true (no ulnar symptoms)?"	→ **routine ulnar transposition not indicated** ✗ Wrong options: repair midsubstance · detachment equivalent · docking worse
"Most likely post-op complication — management?"	→ **observation** (ulnar neurapraxia)
"14-y basketball, acute partial UCL, full ROM?"	→ **activity avoidance 6 wk** ✗ Wrong options: static ex-fix · arthroscopy
"Extension exercises after MUCL vs LUCL?"	→ **supination for MUCL · pronation for LUCL**

– – Active space – –

MULTIDIRECTIONAL SHOULDER INSTABILITY (MDI)

– Shoulder & Elbow · 28 / 60 –

1 Basics

Definition & epidemiology

MDI	symptomatic instability in **≥2 planes** (anterior, posterior, inferior) from **capsular redundancy**; **one plane must be inferior** (TO, OB)
Age	peaks **2nd–3rd decades**
Who	overhead athletes: **swimmers, volleyball, gymnasts, throwers**, dancers
Generalised laxity	in **45–75%** of MDI patients undergoing surgery

Torn loose / worn loose / born loose (TO)

Type (TO)	Mechanism	Example
Torn loose	Trauma	Traumatic dislocation (TUBS)
Worn loose	**Microtrauma / overuse**	Overhead athletes → **MDI**
Born loose	**Lax ligaments / capsule**	**Collagen disorders: Ehlers-Danlos, Marfan → MDI**

TUBS vs AMBRI (Matsen) (TO)

TUBS	AMBRI
Traumatic dislocation → recurrent instability	**A**traumatic (minimal trauma)
Unidirectional	**M**ultidirectional
Bankart lesion most common	**B**ilateral (frequently)
Surgery mainstay	**R**ehabilitation first
–	**I**nferior capsular shift for nonresponders

Pathoanatomy

Cause (TO)	**inferior capsular redundancy** ✗ Wrong options: Bankart · Hill-Sachs · cuff tear
Hallmarks	**patulous inferior capsule** (IGHL anterior + posterior bands) on MRI · **rotator interval deficiency**
Add-on lesions (TO)	from traumatic events: **Bankart** (anteroinferior labrum) · **Kim lesion** (posteroinferior labral avulsion) · glenoid erosion · Hill-Sachs, chondral, SLAP – **less often** than in unidirectional; rarely change treatment

– – Active space – –

Glenohumeral stability

Static restraints	Dynamic restraints
Glenohumeral ligaments	**Rotator cuff** – compresses head into glenoid (primary role)
Glenoid labrum	Biceps
Articular congruity + version	**Periscapular muscles**
Negative intra-articular pressure (release → head subluxes inferiorly)	

MDI = static deficiency → therapy builds **dynamic** stabilisers to compensate

3 Presentation

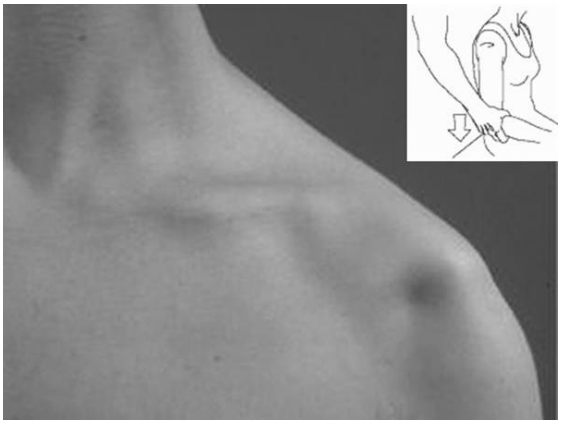
Symptoms — **pain > instability** often · weakness · **paraesthesia / transient numbness** · crepitus · instability **during sleep** · atraumatic, self-reduced dislocations, never needed ED

Impingement signs <20 y — think **MDI** (TO)

Voluntary dislocator — "popping shoulders in and out" → **PT**; **contraindication to surgery**

Diagnosis – sulcus sign, generalised laxity & capsular redundancy

Test	MDI finding
Sulcus sign	**≥2+**; inferior traction at **0° and 45° abduction** (TO) → dimple below acromion; **most specific** for MDI (**2 cm → up to 97% specificity**)
ER with arm adducted (and at 90° abduction)	**↑ ER** = **incompetent rotator interval**
Anterior + posterior **load and shift**	**≥2+** both directions; less specific
Apprehension / relocation	Common in **both** traumatic + MDI
Jerk / Kim tests	Posterior traumatic **and** MDI
Neer / Hawkins	Impingement in **<20 y** → MDI



Sulcus sign – dimple below acromion with inferior traction

Sulcus grade (TO)	Acromiohumeral interval
1+	<1 cm
2+	1–2 cm
3+	>2 cm

2+/3+ sulcus + ↑ ER at side (TO) **pathologic rotator interval** + patulous inferior capsule

Sulcus reflects **patulous inferior capsule + rotator interval** (TO, Orthobullets) · "incompetent rotator interval" (OB)

Beighton score (generalised laxity)

Item (9 points)	Points
Palms to floor, knees straight	1
Genu recurvatum	2
Elbow hyperextension	2
Small-finger MCP hyperextension (passive >90°)	2
Thumb to ipsilateral forearm	2
Not included	**Excessive forearm supination**

Cut-offs generalised laxity **>4/9** (Orthobullets) · **>6** associated with **connective tissue disorder** (Marfan, Ehlers-Danlos; OB)



MCP hyperextension – generalised ligamentous laxity

✗ Exam distractors cross-body adduction (**AC joint**) · O'Brien (**SLAP**) · **axillary webbing** (**pec major rupture**) · Jobe's (**supraspinatus**)

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray	Trauma series (**AP-IR, AP-ER, true AP, axillary, scapular Y**); **often normal**
MR arthrogram	Needed to assess **capsular volume**; **patulous inferior capsule** / capsular redundancy (e.g. posterior), **intact labrum**
MRI add-ons	Bankart (traumatic anterior) · Kim lesion (traumatic posterior) · glenoid bony erosion (chronic anterior)
Arthroscopy	**Drive-through sign**: scope passes easily between head + glenoid at **anterior band of IGHL** (also laxity)

Diagnostic criteria

MDI requires	
Instability in **≥2 planes**	One **inferior**
Sulcus ≥2+	± ↑ ER at side
Load and shift ≥2+	Anterior + posterior

Differential diagnosis

Diagnosis	Clue
Anterior / posterior (traumatic) instability	TUBS, Bankart, **glenoid bone loss**, inverted pear
Long thoracic nerve palsy	Serratus weakness, **medial winging**
Buford complex	Normal variant: **absent anterosuperior labrum + cordlike MGHL** to base of biceps anchor
GIRD	IR deficit **25°** (OB) · **>20°** (TO, Orthobullets) vs contralateral → sleeper stretches
Little leaguer's shoulder	**Salter-Harris I** proximal humeral physis, excessive pitch counts
C-spine disease, brachial plexitis, TOS	Neurologic

5 Management

MDI

- All patients first → **dynamic stabilisation PT (cuff + periscapular)**
  - still within PT trial (e.g. 8 wk) → **continue PT**
- Failed extensive PT + pain/instability limiting ADLs/sport → **capsular shift / plication + RI closure**
- Voluntary dislocator → **no surgery**
- Collagen disorder / refractory → **allograft capsular reconstruction**

Initial treatment – prolonged rotator cuff & periscapular strengthening

First line	**all patients**, incl. bilateral, voluntary, collagen disorders: **dynamic stabilisation PT → rotator cuff + periscapular** (serratus anterior, rhomboids) strengthening, proprioception, humeral head control, scapulohumeral rhythm + **home exercises**
Early rehab	**closed kinetic chain exercises** → safe co-contraction of scapular + cuff muscles
Duration	**3–6 months** (Orthobullets, TO, OB) · **≥6 months** (several OB) · **6–12 months** (one OB) – read the options
✗ Initial distractors	**immobilisation / sling** (not effective; comfort only) · posterior capsular stretching (= GIRD) · bracing · steroid injections · labral repair · HAGL repair · thermal capsulorrhaphy · open/arthroscopic shift before PT trial
Outcome	most patients satisfactory with nonop (pain + recurrence)

Operative criteria

Surgery for MDI	
Indications	**Failure of extensive PT** (sole indication, TO) · pain + instability interfering with ADLs / sport
Contraindication	**Voluntary dislocators**
Procedure of choice	**Inferior capsular shift** (open or arthroscopic)
Results	Good, but **less predictable** than traumatic anterior stabilisation

Surgery – capsular shift & rotator interval closure

Inferior capsular shift	capsule shifted **superiorly** → addresses patulous inferior capsule
Capsular plication	loose capsule to labrum, **balanced anterior + posterior** → ↓ capsular volume
Rotator interval closure	plicate **SGHL to MGHL** (anterosuperior capsule) → **eliminates sulcus sign** / inferior subluxation · greatest effect = **↓ ER with arm at side (0° abduction)** (~10°) ✗ Wrong options: restricts ER in cocking · IR at 90° · flexion/cross-body
Capsule rule	tight **anterior** capsule → ↓ **ER** · tight **posterior** capsule → ↓ **IR**
Do not eliminate sulcus	SLAP repair · Bankart repair (anteroinferior only) · supraspinatus repair · SAD
Open approach	deltopectoral; subscapularis tenotomy vs split; also fix any labral pathology
Allograft capsular reconstruction	rare; refractory, collagen disorders

– – Active space – –

Arthroscopic vs open similar outcomes; arthroscopic advantages

Post-op rehab	
0–4/6 wk	Sling / immobiliser
6–10 wk	ADLs, **45° limit** abduction + ER
10–16 wk	Gradual ROM
>16 wk	Strengthening
>10 months	**Contact sports** (only after normal strength + motion)

Recurrent instability with glenoid bone loss – bony augmentation

Clues to bone loss multiple recurrences · long history · instability in **mid-range (45–60° abduction + ER)**

Glenoid bone loss	Management
10%	Arthroscopic repair still OK
15% + labral tear, first-time (contact athlete)	**Surgical stabilisation** (✗ Wrong options: prolonged PT)
>20–25%	**Bony augmentation (Latarjet**: coracoid + conjoint tendon) – arthroscopic/open Bankart fail

✗ Bone loss distractors arthroscopic Bankart · subscapularis advancement · open capsular shift · hemiarthroplasty (no arthritis)

6 Complications

Thermal capsulorrhaphy – complications & failure

Status **historical / contraindicated**; abandoned for high complication + failure rates

Failure **high failure** for **anterior, posterior and MDI** at 2 + 5 y · recurrence peaks **2–3 wk** post-op (capsule weakest) · good 1-y result does not protect from later recurrence

Capsule revision finds capsule **thin + patulous** · **capsular necrosis / attenuation / insufficiency**
✗ Wrong options: thick fibrotic

Shrinkage depends on **total thermal energy + rate of delivery** · denatured tissue **does heal**
✗ Wrong options: fixed for peak temp

Revision after thermal failure **open revision** → capsular attenuation = **contraindication to arthroscopic revision**
✗ Wrong options: arthroscopic

Axillary nerve injury **heat-induced**; can't actively raise arm, full passive ROM, normal X-ray

Chondrolysis heat from **monopolar RF / laser** → progressive pain, ROM never regained, joint space loss



Glenohumeral chondrolysis after thermal capsulorrhaphy

✗ Chondrolysis distractor	Actually causes
Supraspinatus tear	Cuff tear arthropathy, superior head migration
Subscapularis dehiscence	Open deltopectoral → **anterior instability**
Anterior humeral circumflex coagulation	**Osteonecrosis** (main head supply)
Non-compliance with PT	Stiffness, pain (✗ Wrong options: chondrolysis)

Surgical complications

Recurrence	**most common** after arthroscopic or open capsulorrhaphy
Subscapularis deficiency	after **open anteroinferior shift**; **+ lift-off**, excessive ER; late **anterior subluxation** on axillary view
Loss of motion	over/asymmetric tightening → ↓ ER; treat **Z-lengthening subscapularis**; rare
Axillary nerve	iatrogenic neurapraxia → observe; **abduction + ER moves nerve away** from glenoid
Capsulorrhaphy arthropathy	**posterior glenoid wear**, posterior head subluxation, **glenoid retroversion**, **IR contracture**; **Putti-Platt, Magnuson-Stack**

X vs Y

	TUBS (traumatic)	AMBRI (MDI)
Onset	Trauma	**Atraumatic**
Direction	Unidirectional	**≥2 incl. inferior**
Lesion	**Bankart**, bone loss	**Capsular redundancy, RI deficiency**
Side	Unilateral	**Bilateral** often
Treatment	**Surgery**	**Rehab first** → inferior capsular shift

	MDI	GIRD / internal impingement
Exam	Sulcus, ↑ ER, load-shift both ways, Beighton +	**IR deficit ≥25°** vs other side
Treatment	**Cuff + periscapular strengthening**	**Posteroinferior capsular (sleeper) stretching**

– – Active space – –

How it's asked

"Cause of MDI?"	→ **inferior capsular redundancy** (TO) ✗ Wrong options: Bankart · Hill-Sachs · cuff tear
"Atraumatic recurrent dislocator, hyperextensible MCP – etiology?"	→ **capsular redundancy** ✗ Wrong options: inverted pear · LTN palsy · Bankart · Buford
"Connective tissue feature EXCEPT?"	→ **excessive forearm supination** (not Beighton)
"Gymnast, self-reduced dislocations, normal MRA – exam finding?"	→ **↑ ER in adduction** ✗ Wrong options: cross-body · O'Brien · axillary webbing · Jobe
"Most specific exam for MDI?"	→ **sulcus sign** ✗ Wrong options: apprehension · load-shift · jerk · Kim
"Voluntary popper, 2 cm bilateral sulcus – initial treatment?"	→ **cuff + periscapular strengthening** ✗ Wrong options: posterior stretching · plication · thermal · HAGL repair
"Swimmer, 2+ load-shift, 1.5 cm sulcus – next?"	→ **dynamic stabilisation therapy** ✗ Wrong options: bracing · labral repair
"Bilateral MDI volleyball, impingement signs?"	→ **physical therapy** ✗ Wrong options: injections · staged surgery
"17-y ♀ MDI – initial?"	→ **PT + home exercises** ✗ Wrong options: sling · thermal
"Swimmer, 2+ translation, elbow hyperextension, thumb to forearm?"	→ **PT cuff + scapulothoracic** ✗ Wrong options: open shift · thermal · sling
"13-y pitcher, Beighton signs, sulcus, 10° IR deficit?"	→ **MDI; periscapular training** ✗ Wrong options: GIRD sleeper stretches · little leaguer's shoulder
"Who gets prolonged PT rather than surgery?"	→ **17-y swimmer, Beighton 5 + sulcus** ✗ Wrong options: 15% bone loss dislocator · cuff tears
"Dancer, grade III sulcus, 8 wk PT done?"	→ **continue PT** ✗ Wrong options: shift · Bankart · Latarjet
"Rotator interval closure – true?"	→ **restricts ER at 0° abduction**
"Failed nonop MDI – successful surgery includes?"	→ **rotator interval closure** ✗ Wrong options: SLAP repair · capsular advancement into Hill-Sachs · IGHL repair
"Procedure that eliminates sulcus sign?"	→ **rotator interval closure** ✗ Wrong options: Bankart · SLAP · SAD

"Contraindication to arthroscopic revision?"	→ **capsular attenuation from thermal capsulorrhaphy** ✗ Wrong options: 10% bone loss · ALPSA · GLAD · SLAP + Bankart
"Can't actively elevate after thermal capsulorrhaphy, normal X-ray?"	→ **axillary nerve injury** ✗ Wrong options: chondrolysis
"Stiff painful shoulder 3 y after MDI scope, joint space loss?"	→ **monopolar RF thermal capsulorrhaphy** (chondrolysis)
"Thermal shrinkage — true?"	→ **high failure for anterior, posterior and MDI** ✗ Wrong options: thick fibrotic capsule · no healing
"6-y recurrent instability, mid-range apprehension, 3D-CT bone loss?"	→ **bony glenoid augmentation** ✗ Wrong options: arthroscopic Bankart · hemiarthroplasty

– – Active space – –

NEUROPATHIC (CHARCOT) JOINT OF SHOULDER

– Shoulder & Elbow · 29 / 60 –

1 Basics

Epidemiology

Incidence	**very rare** (~70 cases reported)
Demographics	mean age **~50** · ♂:♀ **2:1**
Sites	shoulder · elbow · foot & ankle
Syrinx → Charcot	**25% of syrinx patients** develop neuropathic arthropathy, **80% in upper limb** (Orthobullets, OB) · **25% of all Charcot joints** due to syringomyelia (Orthobullets associated conditions) – read the options

Syringomyelia – most common cause & MRI work-up

Cause	Note
Syringomyelia	**#1 cause of upper-limb (shoulder) Charcot**; **cervical syrinx = 75%** of Charcot shoulders; monoarticular (**shoulder > elbow**); Hatzis: syrinx in **5/6** shoulders
Arnold-Chiari malformation	**#1 cause of syringomyelia**
Hansen's disease (leprosy)	**2nd** most common cause in upper limb
Neurosyphilis (tabes dorsalis)	Usually **knee**; can be **polyarticular**; no longer the commonest
Diabetes	**#1 cause in foot & ankle**; reported in shoulder but less common; always rule out
Chronic alcoholism	Reported (Hatzis: 1/6)
Others	MS, ESRD, adhesive/TB arachnoiditis, meningomyelocele, traumatic SCI, familial sensory neuropathy
✗ Wrong option: **uremia**	**No documented association** (OB)

Pathophysiology

Syrinx	**fluid-filled cavity in spinal cord** → damages **decussating fibres of lateral spinothalamic tract** → **loss of pain + temperature**
Dissociated anaesthesia	pain/temperature lost · **proprioception + motor preserved** (early)
Syrinx enlarges	dorsal column + **anterior horn** → **areflexia**, weakness, atrophy
Neurotraumatic theory	↓ sensation/proprioception → **repetitive microtrauma** + abnormal joint pressures
Neurovascular theory	dysregulated reflexes → **↑ blood flow (hyperaemia)** → **↑ osteoclastic resorption**

Genes & lab markers

Molecular	**RANK/RANKL/OPG** pathway implicated
Labs	**ESR + WBC may be ↑** → hard to separate from osteomyelitis · can be **normal** (OB stem: CBC, ESR, CRP normal)
Histology	synovial hypertrophy · **detritic synovitis** (cartilage + bone fragments in synovium)

3 Presentation

History	trauma as inciting event in **30%**
Symptoms	swollen shoulder · **50% painless** · loss of function · instability · progressive **painless weakness** + ↓ ROM
Inspection	**swollen, warm, erythematous** (intermittent) → **mimics infection**
Motion	**loss of motion = most common finding (90%)** · crepitus · laxity/ mechanical instability
Neuro	↓ strength · **cape-like** sensory + temperature loss (back + arms) · asymmetric reflexes · **areflexia late**
Most common findings (OB)	**pain, loss of motion, instability**

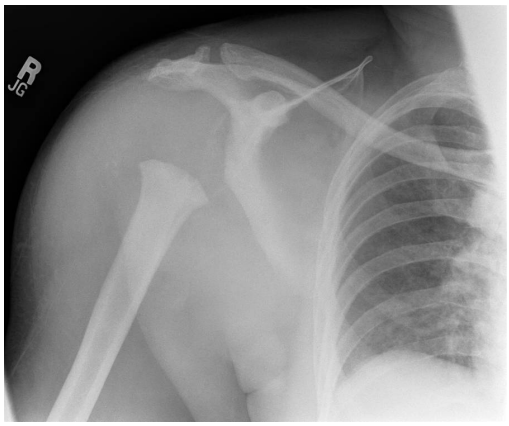
4 Diagnosis & Classification

Key radiology findings

X-ray (AP + scapular Y)	**gold standard / initial** diagnosis

Stage	X-ray finding
Early	Degenerative changes **mimic OA**
Late	**Superomedial flattening of humeral head**
Late	**Periarticular soft-tissue calcifications**
Late	**Glenoid sclerosis**
Late	**Extensive bone resorption** (humeral head "amputated"), fragmentation
Late	Destruction of **both sides** of joint → subluxation/dislocation

– – Active space – –



Charcot shoulder – resorption of humeral head, destruction both sides of joint

MRI cervical spine **next step in every Charcot shoulder** → syrinx (signal ≥ CSF on T1)



Cervical syrinx on MRI

CT if concern for osteomyelitis: intraosseous gas, cortical destruction, sequestra

Scan	Charcot	Osteomyelitis
Technetium bone scan	**Hot**	Hot
Indium-labelled WBC scan	**Cold (negative)**	**Hot (positive)**

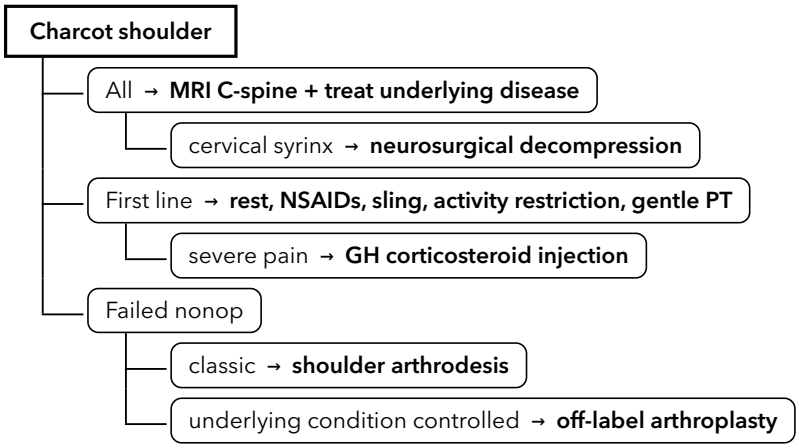
Best test to separate from osteomyelitis **indium WBC scan**

Differential diagnosis

✗ Distractor (OB)	Why not
AVN (loss of humeral head blood supply)	**Sclerosis + flattening** of head only
Metastasis (distant primary)	Variable; no obliteration of **both sides** of joint
Cartilage tumour	↑ calcification throughout lesion
Bone-forming tumour	↑ ossification throughout lesion
Osteomyelitis / septic joint	Same swelling, warmth, ↑ WBC/ESR, hot Tc scan → **indium scan hot**

Others synovial chondromatosis · soft-tissue sarcoma · tumoral calcinosis · Winchester syndrome · **Gorham's disease** · **Milwaukee shoulder**

5 Management



Nonoperative

First line rest, NSAIDs, **sling immobilisation** (slows ligamentous laxity), activity restriction, treat cause, gentle PT/passive stretching · **50% improve**

Corticosteroid injection severe pain · **glenohumeral** most effective · temporary **80%** pain ↓ (case reports)

Operative criteria

Procedure	Indication	Outcome / technique
Neurosurgical decompression	**Cervical syrinx present**	Slows progression, improves function + bone (**glenoid regrowth**); posterior fossa decompression, craniotomy, **syringoperitoneal shunt**, laminectomy
Arthrodesis	Severe pain, **failed nonop**; historically **only** option	**S-shaped incision** scapular spine → over acromion → anterolateral arm
Arthroplasty (hemi, TSA, rTSA, resurfacing)	Failed nonop **+ underlying condition treated**	**Off-label** (Charcot = **strict FDA contraindication**, loosening) · **70%** improved function at 5 y

Fusion position ("30-30-30") **20–30° abduction · 20–30° forward flexion · 20–30° internal rotation** → hand to mouth

No emergent ORIF / I&D not indicated (afebrile, normal labs) → **C-spine MRI first**

– – Active space – –

6 Complications

Complication	Risk factor
Infection	Surgery **without managing underlying condition**
Upper-limb DVT	Any surgery
Acromial stress fracture	**Reverse TSA** for Charcot
Progressive glenoid erosion	**Hemiarthroplasty**

Missed diagnosis often missed → **inappropriate surgery** (Hatzis)

X vs Y

	Charcot shoulder	Osteomyelitis	AVN humeral head
Pain	Often **painless** (50%)	Painful	Painful
Warm, red, swollen	Yes (intermittent)	Yes	No
X-ray	**Resorption + destruction both sides**, calcifications	Destruction, sequestra	**Sclerosis + flattening** (head only)
Indium WBC	**Cold**	**Hot**	–
Next step	**MRI C-spine**	Aspirate/CT	MRI shoulder

Charcot cause	Classic joint
Syringomyelia	**Shoulder** (upper limb)
Diabetes	**Foot & ankle**
Tabes dorsalis	**Knee** (polyarticular)

How it's asked

"Chronic shoulder pain, X-ray of destroyed/fragmented humerus – most common cause?"	→ **syringomyelia** ✗ Wrong options: diabetes · syphilis · alcoholism · uremia
"Most common cause of upper-extremity neuropathic arthropathy?"	→ **syringomyelia** ✗ Wrong options: Hansen's = 2nd · diabetes = foot · neurosyphilis = knee
"Painless progressive weakness, SCI, normal CBC/ESR/CRP, Charcot X-ray – early management?"	→ **cervical spine MRI** ✗ Wrong options: HIV test · repeat labs · emergent ORIF · emergent I&D
"Painless stiff shoulder, intermittent warmth, destroyed joint – work-up will show?"	→ **fluid-filled cavity in spinal cord** ✗ Wrong options: AVN · metastasis · cartilage/bone tumour
"Tract damaged by syrinx?"	→ **decussating fibres of lateral spinothalamic tract** (pain + temperature)

"Most common cause of syringomyelia?"	→	**Arnold-Chiari malformation**
"Test to differentiate Charcot from osteomyelitis?"	→	**indium WBC scan (cold in Charcot)**
"Syrinx found — treatment?"	→	**neurosurgical decompression**
"Shoulder fusion position?"	→	**20–30° abduction, flexion, IR each**
"Complication of rTSA in Charcot shoulder?"	→	**acromial stress fracture**

– – Active space – –

– – Active space – –

OLECRANON STRESS FRACTURE

– Shoulder & Elbow · 30 / 60 –

1 Basics

Epidemiology & location

Frequency	**rare**; predominantly **throwing athletes** (**baseball pitchers**), other high-level athletes; overuse injury
Location	predominantly the **olecranon epiphyseal plate** (persistent apophysis); line propagates **transverse or oblique**
Age	adolescent pitchers; may persist into **late adolescence** (**delayed closure of olecranon epiphysis**, 16–20 y; other elbow physes closed)
Associated throwing injuries	**UCL tear** · **medial epicondyle avulsion fracture**

2 Mechanism

Theory (3 accepted)	Note
Posteromedial olecranon impingement	–
Triceps traction force	Repetitive forceful triceps contraction
Valgus extension overload	Part of **valgus-extension overload syndrome**

Torg & Moyer	triceps contraction + **varus and valgus torques → olecranon epiphysis separates from epiphyseal plate**

3 Presentation

Pain	**posteromedial olecranon pain** of throwing arm; **worse with throwing** (after several innings), **better with rest**
Exam	mild swelling, **localised tenderness over proximal olecranon** (medial, distal to epicondyle); full ROM; no mechanical symptoms
Valgus stress test	**instability → associated UCL injury**; **normal** in isolated stress fracture
Late picture	acute worsening + **inability to throw** after failed nonop

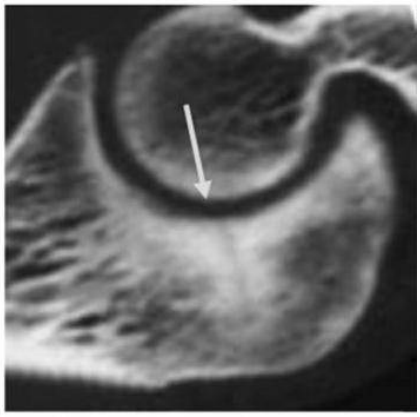
4 Diagnosis & Classification

Key radiology findings

Modality	Detail
X-ray	AP, lateral, oblique; optional **valgus stress** + **contralateral comparison**
X-ray findings	**Physeal widening, delayed fusion, fragmentation**; medial joint space widening if UCL injured
CT	Early confirmation; **best on lateral (sagittal) views**; **originates at articular surface → runs dorsal-proximal**
MRI	Early confirmation; **suspected UCL tear** → coronal **T2 fat-sat**: **T-sign** = UCL tear



Lateral X-ray – olecranon stress fracture through persistent physis



CT – fracture from articular surface running dorsal-proximal

Differential

✗ Distractor	Why wrong (11-yo pitcher, posteromedial tenderness, normal valgus stress)
Olecranon bursitis	No swelling/infection
OCD capitellum	No clicking / mechanical symptoms
UCL insufficiency	**Valgus stress normal**
Medial epicondylitis	Tenderness distal to epicondyle, over olecranon

– – Active space – –

5 Management

Olecranon stress fracture

- First line → **NSAIDs + rest ± splint (4–6 wk), no throwing up to 3 months**
  - delayed union / failed rest → **ORIF**
- Undisplaced → **single large compression (cannulated) screw**
- Displaced → **treat as olecranon fracture (screw ± TBW or plate)**

Nonoperative

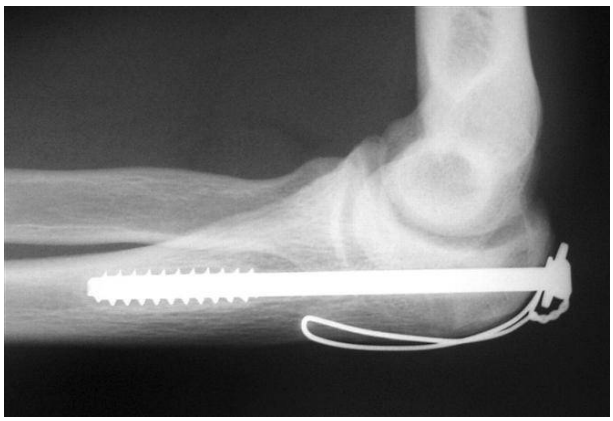
Regimen	short-term **NSAIDs**, **rest ± temporary splint**; **4–6 wk rest**, **removal from sport up to 3 months**; progressive ROM; **avoid valgus loading (throwing)**
Adjuncts	**electrical bone stimulation**; **vitamin D + calcium**

Operative criteria – compression screw fixation

Indication	Fixation
Delayed union / failed rest (e.g. **9 months**, or **two 6-wk rest periods**)	**Large compression screw** (intramedullary **cannulated**)
Undisplaced	**Single large screw** (e.g. **7.0 mm**)
Displaced	Screw **± tension band wire**, or plate

Plate fixation option once displaced (OB) · **not indicated** for this injury (another OB) – read the options

✗ Next-step traps elbow arthroscopy · open osteophyte removal · ulnar nerve transposition · **cast + strengthening** after failed rest · physiotherapy · **long-arm cast** (stiffness; only reasonable briefly at onset) · **hinged-elbow brace** (no healing benefit)



Compression screw + tension band construct, healed

Outcomes

Rettig (7.0 mm screw, 5 pitchers)	union mean **15.4 wk**; return to throwing mean **29.4 wk**
Charlton & Chandler (16–20 y, delayed epiphyseal closure)	UCL intact in all; ORIF with **TBW, screw ± autograft** → all **asymptomatic at 32 months** despite **> 30 months** pre-op pain

Surgery reliable results when rest fails

X vs Y

	Olecranon stress fracture	UCL insufficiency
Pain	**Posteromedial olecranon**	–
Valgus stress test	**Normal** (unless UCL also injured)	**Positive / instability**
Imaging	Physeal widening, CT fracture line	Medial joint widening; MRI **T-sign**

	Undisplaced	Displaced
Fixation	**Single large compression screw**	Screw ± TBW or plate (as olecranon fracture)

How it's asked

"19-yo pitcher, posterior elbow pain 9 months despite nonop, acute worsening — next?"	→ **internal fixation with compression screw** ✗ Wrong options: arthroscopy · osteophyte removal · ulnar transposition · cast
"11-yo pitcher, posteromedial pain, tender over proximal olecranon, normal valgus stress — diagnosis?"	→ **olecranon stress fracture** ✗ Wrong options: UCL insufficiency · OCD · medial epicondylitis · bursitis
"15-yo pitcher, two 6-wk rest periods failed, other physes closed — next?"	→ **cannulated screw fixation** ✗ Wrong options: plate · hinged brace · long-arm cast · physio
"Mechanism theories?"	→ **posteromedial impingement, triceps traction, valgus extension overload**
"First-line treatment?"	→ **NSAIDs + rest ± brief splint, no valgus loading**
"Indication for ORIF?"	→ **delayed union**

– – Active space – –

OS ACROMIALE

– Shoulder & Elbow · 31 / 60 –

1 Basics

Definition

Os acromiale	**unfused secondary ossification centre** of the acromion (**failure of fusion of anterior acromial apophysis**, OB) → impingement, cuff tendonitis

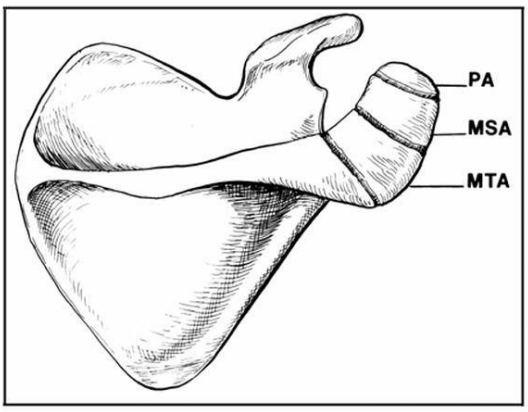
Epidemiology

Prevalence	**8%** (Orthobullets, OB) · **1–15%** (another OB) – read the options
Bilateral	**60%** (Orthobullets) · **33%** (Sammarco skeletons, OB) – read the options
Race	**Black > White** (**13.2% vs 5.8%**)
Sex	♂ **>** ♀ (**8.5% vs 4.9%**)
Usually incidental	found on imaging for other reasons, often **trauma**

Anatomy – failed fusion of meso- and meta-acromion

Ossification centre	Position	Origin of
Pre-acromion	**Tip**	**Anterior deltoid + coracoacromial ligament**
Meso-acromion	**Mid**	**Middle deltoid**
Meta-acromion	**Base**	**Posterior deltoid**
Basi-acromion	(4th centre, OB)	–

Number of centres	**3** (Orthobullets, OB) · **4: pre-, meso-, meta-, basi-** (another OB) – read the options
Fusion age	ossify medially toward clavicular facet **by 18 y** (OB) · **15–18 y** (Hurst, OB)
Most common site	**junction of meso- and meta-acromion** = commonest **symptomatic** form ✗ Wrong options: meso–pre · meta–pre
Blood supply	**acromial branch of thoracoacromial artery** → preserve at surgery



Ossification centres: pre- (PA), meso- (MSA), meta-acromion (MTA)

Pathological mechanism

✓ **Mechanism** **failure of fusion between ossification centres**

✗ Mechanism distractor	Condition
Failure of mesenchymal segmentation	**Tarsal coalition**
Defect in chondrocyte proliferation	**Achondroplasia**
Glycine substitution, abnormal collagen cross-linking	**Osteogenesis imperfecta**
Failure of cartilaginous differentiation	**Congenital radioulnar synostosis**

Associated conditions

Impingement + rotator cuff disease **meso-acromion** → cuff tendonitis + **full-thickness tears in 50%**

3 Presentation

Impingement pain anterior fragment **flexes / displaces downward** with deltoid contraction + arm elevation → ↓ subacromial space

Synchondrosis pain **motion at nonunion site**

Symptoms **superior shoulder pain**, **overhead activity**, **lying on affected side**; tenderness **anterolateral / superior acromion**

Trigger **trauma** can make a previously silent os symptomatic

Diagnostic injection **1% lidocaine** into site when pain source unclear (Spiegl)

4 Diagnosis & Classification

Key radiology findings

Modality	Use
Axillary lateral X-ray	**Best view** for diagnosis
CT	Nonunion site; **degenerative change (cysts, sclerosis, hypertrophy)**
Axial MRI	Shows os (superior pain + overhead vignette)

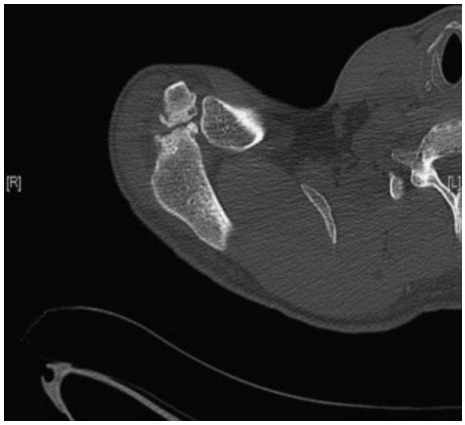
Os vs fracture os: **rounded edges + corticalisation of entire fragment** · fracture: acute

Fused os acromiale may be mistaken for a free ossicle (Sammarco)

– – Active space – –



Axillary lateral showing os acromiale



CT — unfused acromial fragment

✗ **Diagnosis distractor**	
Acute acromial fracture	Rounded corticated edges = os
Type IV AC separation	Posterior clavicle
Deltoid avulsion	—
AC OA / AC separation / PTRCT / SLAP	MRI shows os

5 Management

Os acromiale

- Incidental / mild symptoms → **observe, NSAIDs, PT, subacromial steroid**
  - fails 6–12 months → **surgery**
- Symptomatic os + impingement → **two-stage fusion**
- Impingement ± cuff tear, os incidental + nontender → **ASD + acromioplasty**
- Small symptomatic pre-acromion / failed ORIF → **fragment excision**

Nonoperative

Regimen	observation, **NSAIDs**, activity modification, therapy, **subacromial corticosteroid → majority**
Duration	surgery only after **6–12 months** of nonop

Operative criteria – treatment of symptomatic os acromiale (fusion)

Procedure	Indication	Detail
Two-stage fusion	**Symptomatic os + impingement**, failed nonop	**Stage 1: fuse ± bone graft** (rigid fixation: **TBW, sutures, cannulated screws**) → **Stage 2: acromioplasty**
Arthroscopic SAD + acromioplasty	**Impingement ± cuff tear, os incidental + nontender**	–
Fragment excision (open/arthroscopic)	**Symptomatic pre-acromion, small fragment · salvage after failed ORIF**	Arthroscopic: less periosteal + deltoid detachment; **better results with pre-acromion**

Direct excision of large fragment — **deltoid dysfunction** → prefer fusion

Warner (15 shoulders) — **cannulated screws + tension band + autograft** → union in **all but one**; 8 had cuff tears

Harris — commonest concurrent procedure **cuff repair (59%)**, then **distal clavicle excision (25%)**; all techniques improve outcomes

Abboud — meso-os + cuff pathology satisfactory in **4 of 8**; **80%** satisfactory overall vs **22%** in **workers' compensation**

44-yo, failed PT after cuff repair, symptomatic os + impingement — ✗ Wrong options: observation · more PT · ASD alone (os not incidental) · excision alone

6 Complications

Deltoid weakness — after **fragment excision**

Persistent pain / weakness

Prognosis — **poorer outcomes after cuff repair** with **meso-os acromiale**; os may worsen outcome of unrelated shoulder surgery (Hurst)

X vs Y

	Two-stage fusion	ASD + acromioplasty	Excision
Os status	**Symptomatic, tender** + impingement	**Incidental, nontender**	Small **pre-acromion** / failed ORIF
Deltoid	Preserved	Preserved	**Risk of dysfunction** (large fragments)

How it's asked

"27-yo, shoulder pain, X-ray shows os – cause?" — → **failure of fusion meso- and meta-acromion**
✗ Wrong options: meso–pre · acute fracture · type IV AC · deltoid avulsion

– – Active space – –

"23-yo, superior shoulder pain overhead + lying on side, axial MRI?"	→ **os acromiale** ✗ Wrong options: AC OA · PTRCT · SLAP
"Most common location?"	→ **junction of meso- and meta-acromion**
"Pathological mechanism?"	→ **failure of fusion between ossification centres** ✗ Wrong options: segmentation = coalition · glycine = OI
"44-yo, failed PT after cuff repair, symptomatic os with impingement – treatment?"	→ **two-stage fusion** ✗ Wrong options: ASD · excision only
"Os incidental + nontender, impingement ± cuff tear?"	→ **arthroscopic SAD + acromioplasty**
"Best view?"	→ **axillary lateral**
"Complication of excision?"	→ **deltoid weakness**

OSTEOCHONDRITIS DISSECANS OF ELBOW

– Shoulder & Elbow · 32 / 60 –

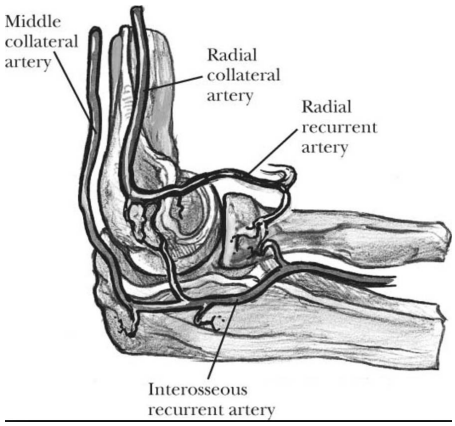
1 Basics

Presentation & population (capitellum, gymnasts/throwers)

Definition	localised injury → **separation of articular cartilage + subchondral bone** of the **capitellum**
Population	**gymnasts + throwing athletes** (pitchers) · repetitive overhead or **upper-limb weight bearing** · also cheerleaders ✗ Wrong options: swimmers/divers · linemen · rugby · cyclists
Age	**>10 y** (adolescent) (Orthobullets, OB) · "typically **>13 y**" (one OB explanation)
Sex	**boys > girls**
Site	**capitellum** of **dominant** arm (most common) · occasionally **radial head** (OB) ✗ Wrong options: "capitellum or lateral trochlea"
Juvenile vs adult	**juvenile OCD better prognosis**

Anatomy – blood supply

Capitellum	supplied by **2 end arteries**: **radial recurrent** + **interosseous recurrent** → vulnerable to ischaemia



Capitellar blood supply – radial recurrent + interosseous recurrent arteries

Pathophysiology

Mechanism	**repetitive compression** (lateral compression + axial loading) of **immature capitellum** → **vascular insufficiency** + **repetitive microtrauma** → osteochondral necrosis → separation
Throwers	**repetitive valgus overload of radiocapitellar joint**, **late cocking** phase
Panner's + OCD	likely a **continuum** (OB)

– – Active space – –

3 Presentation

Pain insidious, **activity-related lateral elbow pain**, dominant arm · posterolateral pain with throwing

Sign	Timing
Loss of extension (mild, ~15°)	**Early** sign = typical early presentation
Catching / locking / grinding / clicking / popping	**Late** – loose bodies; **poorer prognosis**
Lateral tenderness, mild **effusion**, ± crepitus	Any
Painful click on passive rotation, painful pronation/supination	Unstable lesion

Instability testing normal
✗ Wrong options: MUCL laxity

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral; slightly flexed-elbow view) confirms diagnosis: **well-defined subchondral lesion** in capitellum · radiolucency, **flattening**, sclerosis, fragmentation, loose bodies · initial films may be **negative**



Capitellar OCD – flattening and lucency

MRI best for **size, extent of oedema, cartilage status** · OCD: **separation of cartilage from capitellum + chondral fissuring** · stable: **intact cartilage + subchondral signal change**, T2 no signal around lesion

MR arthrogram contrast **outlining margins** of lesion = **unstable** fragment → **poor** prognostic sign
✗ Wrong options: good

Os supratrochleare dorsale normal variant, incidental
✗ Wrong options: prognostic

Classification – radiographic & arthroscopic

– – Active space – –

Type	Findings	Image
I	**Intact cartilage**; bony stability may or may not be present	
II	**Cartilage fracture** with bony collapse or displacement	
III	**Loose bodies** in joint	

Stability criteria (Takahara)

Stable (heals with rest) – all of	Unstable (surgery better) – any of
Open capitellar physis	**Closed** capitellar physis
Localised flattening or radiolucency of subchondral bone	**Fragmentation**
Good elbow motion	**Restricted motion ≥20°**

5 Management

Capitellar OCD

- Type I stable → stop activity 3–6 wk → **graded return over 6–12 wk**
- Unstable type I / stable type II → **arthroscopic microfracture or drilling**
- Large, incompletely displaced → **arthroscopic reduction + fixation**
- Unstable type II / type III → **arthroscopic debridement + loose body excision**
- Large type II–III engaging radial head → **OATS**
  - uncontained → **size-matched fresh allograft**

Stable lesions – activity cessation

First line — **cessation of offending activity** ± immobilisation · **>90% success**

Duration — **cessation of activity 3–6 weeks**, followed by slow progression back to activities over next **6–12 weeks** (Orthobullets, OB) · "**minimum 6 wk**" / "**6 wk**" (other OB explanations) – read the options

– – Active space – –

Return to play resumption when **asymptomatic** + examination shows **restoration of painless range of motion**
✗ Wrong options: wait for complete radiographic healing (residual X-ray abnormality persists even when asymptomatic)

No participation until symptoms resolve + motion recovers
✗ Wrong options: continue until worse

Avoid casting 6–8 wk too long → stiffness

Avoid immediate throwing programme · no surgery before trial of nonop (stable, no loose body)

Nonop healing >50% of stable lesions heal with activity restriction; **smaller lesions without cyst-like change** heal better

Operative criteria

Procedure	Indication	Post-op
Arthroscopic microfracture / drilling (extra- or transarticular)	**Unstable type I** · **stable type II** · symptomatic lesion with **intact articular surface** (in situ drilling)	Protected ROM; strengthen 2 mo; throw/WB **4–6 mo**
Fixation (arthroscopic reduction)	**Large**, **incompletely displaced** (partial separation)	As above; results variable
Arthroscopic debridement + loose body excision	**Unstable type II** · **type III** · mechanical symptoms	Early ROM ± brace; no throwing/WB **3 mo**
OATS (autograft/allograft)	**Large type II–III engaging radial head**; uncontained → fresh allograft	Strengthen 3 mo; throwing 5–7 mo
Lateral column osteotomy	**Investigational**, salvage (offloads lateral compression)	–

Surgery overall failed extended nonop · mechanical symptoms · loose bodies on imaging · unstable lesions

OATS (single plug) **93%** graft incorporation, high return to sport, low donor morbidity · large unstable lesions: **fixation/reconstruction > simple excision**

Microfracture among the most commonly performed + studied

Prognosis – skeletal maturity

Best predictor **skeletal age / open physes**
✗ Wrong options: symptom duration · gender · activity-related symptoms · anatomic variant

Healing most heal in **6–18 months** of nonop (open physes)

Poor nonop healing **radial head enlargement** · skeletal age difference · closed physis

Long term **~50%** residual ADL symptoms (Takahara: 50% nonop, 46% fragment removal) · **up to 50% arthritis**

6 Complications

Complications **stiffness** · pain · **failure to return to sport** · **arthritis**

X vs Y

	OCD of capitellum	Panner's disease
Age	**>10 y** (adolescent)	**<10 y** (first decade)
Lesion	**Focal, well-defined subchondral** lesion	**Entire capitellar ossific nucleus** fragmented; irregular epiphysis
MRI	Cartilage separation + chondral fissuring	–
Loose bodies	Common	Rare
Course	Variable; arthritis up to 50%	**Benign, self-limiting**, normal regrowth, no sequelae
Surgery	May be needed	**Contraindicated**

Distractor (X-ray / stem)	Diagnosis
Calcification at common extensor origin	**Lateral epicondylitis**
Widened medial epicondyle physis	**Little Leaguer's elbow**
Incarcerated medial epicondyle fragment	Medial epicondyle fracture
Olecranon osteophyte, posterior pain in extension	**Valgus extension overload**
Medial pain + instability	UCL (Tommy John)
After dislocation	PLRI
Posterior pain in extension	Olecranon stress fracture
Lateral pain, rarely effusion/mechanical	Plica

How it's asked

"13-y pitcher, lateral pain, effusion, painful click on rotation?"	→ **OCD** ✗ Wrong options: Tommy John · PLRI · stress fracture · plica
"Population for capitellar OCD?"	→ **gymnasts + throwing athletes**
"Characteristic of elbow OCD?"	→ **MRI: cartilage separation from capitellum + chondral fissuring** ✗ Wrong options: whole ossific nucleus · normal regrowth · <10 y · MUCL laxity
"Typical early presentation?"	→ **loss of extension** ✗ Wrong options: catching/locking = late · boys <10 = Panner · MR arthrogram outline = poor sign
"14-y gymnast, can't fully straighten, popping – which X-ray?"	→ **capitellar flattening + lucency** ✗ Wrong options: extensor calcification · physeal widening · olecranon osteophyte
"12-y gymnast, 15° extension loss, 3 × 7 mm stable lesion, no loose bodies?"	→ **no gymnastics until symptoms resolve + motion recovers**

– – Active space – –

"Little League pitcher, catching, lateral pain — initial management?"	→ **rest; resume when asymptomatic + painless full ROM** ✗ Wrong options: wait for X-ray healing · drilling · osteotomy
"11-y pitcher, lacks terminal extension, first visit?"	→ **stop pitching 3–6 wk + slow return** ✗ Wrong options: medial epicondyle ORIF · OATS · debridement
"Type I OCD on MRI, full ROM — initial treatment?"	→ **cessation 3–6 wk then progression over 6–12 wk** ✗ Wrong options: casting 6–8 wk · throwing programme
"Intact cartilage + subchondral signal change?"	→ **cessation of activity for 6 wk** ✗ Wrong options: microfracture · fixation · OATS
"Unstable type I or stable type II?"	→ **microfracture / drilling**
"Large lesion engaging radial head?"	→ **OATS**
"16-y cheerleader, closed physes — largest prognostic factor?"	→ **skeletal age**

– – Active space – –

PECTORALIS MAJOR RUPTURE

– Shoulder & Elbow · 33 / 60 –

1 Basics

Epidemiology

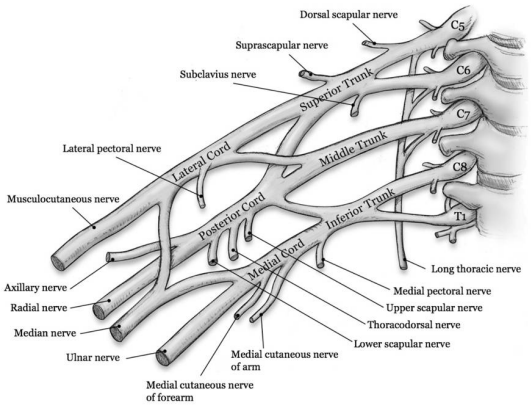
Incidence	rare (**<1/100,000/y**) but rising · **75%** of reported cases since 1990
Demographics	**almost exclusively ♂, 20–40 y** · **weightlifters**, **bench press** · almost no reports in ♀ (TO) ✗ Wrong options: equal ♂/♀ · old gym-goers
Risk	**anabolic steroids** / supplements
Other sports (TO)	rugby, football, wrestling, water skiing, boxing
Most common site	**tendinous avulsion at humeral insertion (65%)** > **musculotendinous junction (27%)** (TO) · intramuscular rare (direct trauma, stab) ✗ Wrong options: "MTJ most common" · muscle belly
Most common portion	**sternocostal (sternal) head** (✓ TO "all except" answer) · clavicular head often intact

Anatomy – sternal head & pectoral nerve innervation

	Clavicular head	Sternocostal head
Origin	**Medial half of clavicle** (+ proximal sternum)	Sternum, **costal cartilages 1–6** (TO), external oblique aponeurosis
Size	Smaller; may look confluent with deltoid	**Larger, >80% of volume**
Tendon layer at insertion	**Anterior**	**Posterior** (lower sternal fibres)
Nerve	**Lateral pectoral (C5–C7)**	**Lateral + medial pectoral** (dual, C5–T1)
Rupture	Fails last	**Fails first**, often isolated

Innervation	**lateral pectoral (C5–C7)** = clavicular + upper sternocostal · **medial pectoral (C8–T1)** = lower sternocostal + pectoralis minor (TO, Orthobullets)
Insertion	**lateral lip of bicipital groove** (humeral shaft lateral to groove), anterior + posterior layers (TO)
Action	**adduction + internal rotation** · forward elevation/flexion lesser (chiefly clavicular head) (TO) · not predominant forward flexion = **anterior deltoid + coracobrachialis**
Thoracoappendicular muscles	pec major, pec minor, subclavius, serratus anterior

– – Active space – –



Brachial plexus: lateral pectoral (lateral cord) and medial pectoral (medial cord)

✗ Nerve distractor	Supplies
Suprascapular (C5–C6)	Supraspinatus, infraspinatus
Thoracodorsal (C6–C8)	Latissimus dorsi
Medial brachial cutaneous (T1)	Sensation medial arm
Upper/lower subscapular	Subscapularis
Axillary / musculocutaneous	Deltoid, teres minor / biceps, brachialis

2 Mechanism

Mechanism – eccentric bench press

Mechanism excessive tension on **maximally eccentrically contracted** muscle · **downward (deceleration) phase of bench press**
✗ Wrong options: upward · max elevation · bar on chest

Position **final 30° of humeral extension** (inferior sternal fibres at maximal stretch) · extension + abduction + ER · TO: **sudden forceful abduction + ER of contracted muscle**

Failure sequence **inferior sternocostal fibres** (insert more posteriorly) → superior sternocostal → **clavicular head last**

3 Presentation

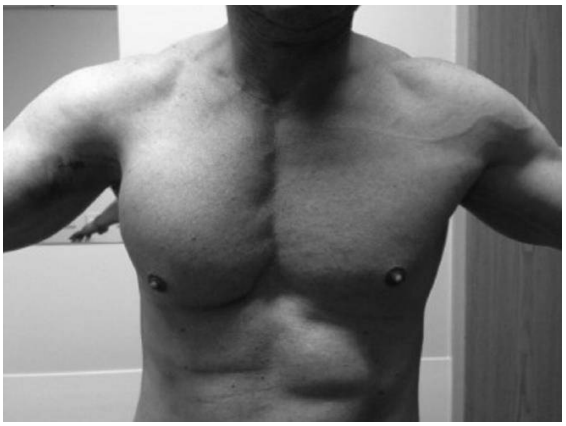
History **sudden pop / tearing** with resisted adduction + IR · pain, weakness

Inspection **swelling + ecchymosis anterolateral chest wall and/or proximal medial arm** · **loss of anterior axillary fold**, muscle retracted medially · palpable defect accentuated by **resisted adduction** · **"dropped nipple" sign**

Ecchymosis location **anterior brachium/upper arm → humeral avulsion** · **chest wall** → mid-substance / MTJ tear (less benefit from surgery)

Weakness **adduction + internal rotation** (± forward flexion); ER, abduction normal; apprehension negative

– – Active space – –



Loss of axillary fold, chest asymmetry, upper arm ecchymosis

✗ Exam-sign distractor	Lesion
Popeye	Proximal LHB tear
Medial winging	Long thoracic nerve / serratus
Belly-press	Subscapularis
ER lag	Infraspinatus

✗ Wrong option: axillary fold subscapularis and biceps injuries do **not** change the axillary fold

4 Diagnosis & Classification

Diagnosis – MRI

X-ray trauma series (true AP, scapular Y, axillary) · usually **normal** · lost pec shadow or bony avulsion

MRI **investigation of choice / gold standard** · complete vs partial · tear site + retraction (muscle belly vs bone–tendon junction) · needs **dedicated chest-wall sequence** (standard shoulder MRI misses it) · **T2 acute**, **T1 chronic** · intact clavicular head may mask partial sternal tear

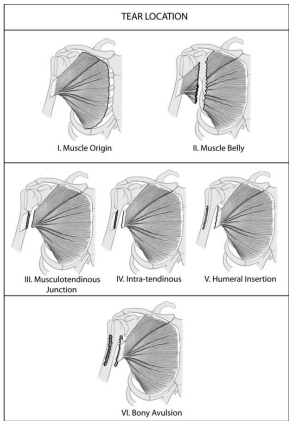
EMG no role without nerve injury

– – Active space – –

Modified Tietjen classification

Type	Description
I	Contusion / sprain
II	Partial tear
III	**Complete** tear, by location ↓
III-A	Muscle origin
III-B	Muscle belly
III-C	**Musculotendinous junction**
III-D	Intratendinous
III-E	**Tendon avulsion off humerus (no bone)**
III-F	Bony avulsion off humerus

Tietjen (TO) sprain · partial · complete (origin, belly, MTJ, tendon) + added **bony flake avulsion** + **intratendinous**



Modified Tietjen tear locations

5 Management

Pectoralis major rupture

- Low-demand, elderly, sedentary → **sling, ice, NSAIDs, PT**
  - muscle belly / low-grade partial / chest-wall tear → **nonop**
- Acute complete avulsion or MTJ, young/active → **open primary repair**
- Chronic, can't mobilise → **allograft reconstruction**
  - primary repair still possible years later

Operative criteria

– – Active space – –

Situation	Treatment
Avulsion from humeral insertion (complete pecto-humeral tear)	**Primary repair** (✓ surgery indication)
Acute tear, young athlete / labourer	**Primary acute repair** = gold standard
MTJ tear	Repair (Orthobullets, OB) · therapy may be considered (one OB) · may need augmentation/reconstruction (one OB) – read the options
Muscle belly / partial / chest-wall ecchymosis	**Nonop**
Chest asymmetry alone, no deficit	not an indication
Chronic, inadequate excursion	**Reconstruction (allograft)**

Treatment – acute repair, chronic allograft reconstruction & nonoperative deficits

Primary repair — **open, deltopectoral** · reattach to humerus lateral to bicipital groove · **suture anchors or buttons** (TO) · transosseous with cortical trough · cortical button · anchors (PEAK/all-suture) · all comparably excellent; some evidence **button + trough > anchors**

Acute vs delayed — acute easier, less scar, avoids later reconstruction/allograft
✗ Wrong options: delayed repair after failed nonop

Best long-term result (sternocostal tear) — **open repair of sternocostal portion**
✗ Wrong options: PT/NSAIDs · steroid injection · LHB repair · clavicular repair

Repair outcome — reliable strength, return to sport, satisfaction; improves regardless of tear site · most can be repaired primarily, **even chronic**
✗ Wrong options: majority need allograft

Chronic reconstruction — release adhesions superficial + deep (**protect medial + lateral pectoral nerves**) ± fascial release · graft: **Achilles allograft (most common)** (Orthobullets) · **semitendinosus weave** (OB) · gracilis weave · inferior to primary repair, better than nonop

Chronic weakness 2 y after PT, young labourer — **allograft reconstruction (semitendinosus weave)**
✗ Wrong options: latissimus transfer · electrical stimulation · arthrodesis · arthroscopic repair

Nonop deficits — **weakness of adduction + IR** (esp. isokinetic adduction, peak torque) · cosmetic defect · delayed recovery · poor satisfaction · ↓ return to sport

Nonop protocol — sling in adduction + IR · immediate PROM · AAROM/AROM over 6 wk · strengthening/unrestricted at **2–3 mo**

Tendon transfer — not for acute tear (pec major is a **donor** for subscapularis insufficiency)

– – Active space – –

6 Complications

Complication	Fact
Persistent pain	**Most common**
Re-rupture	**5–7%**; fails at **suture–tendon interface**
Residual weakness	–
Cosmetic deformity	–
Nerve injury in reconstruction	Medial + lateral pectoral during deep release

X vs Y

	Pec major tear	LHB rupture	Subscapularis tear
Sign	**Lost anterior axillary fold**, dropped nipple	**Popeye**	**Belly-press**, ↑ passive ER
Weakness	**Adduction + IR**	–	–
Axillary fold	**Lost**	Normal	Normal

	Sternocostal head	Clavicular head
Torn	**Most often, first**	Last, often intact
Nerve	Lateral + medial pectoral	Lateral pectoral

How it's asked

"Pec major tear – all false EXCEPT?"	→ **sternal head most commonly ruptured** (TO) ✗ Wrong options: old gym-goers · MTJ most common · equal sexes
"Bench press pop, upper medial arm pain, lost anterior fold?"	→ **pectoralis major tear** (TO) ✗ Wrong options: LHB · axillary fold tear · cuff
"Weightlifter, pop, MRI retracted muscle – nerve?"	→ **lateral + medial pectoral** ✗ Wrong options: axillary · musculocutaneous · subscapular · suprascapular
"Labelled plexus – innervation of most commonly torn portion?"	→ **A + E (lateral + medial pectoral)**
"True about this injury?"	→ **sternal head torn more than clavicular** ✗ Wrong options: ♀ · allograft needed · flexion loss · belly
"Bench press phase at greatest risk?"	→ **downward deceleration**
"38 y weightlifter, MRI arrow?"	→ **pectoralis major tear** ✗ Wrong options: biceps · pec minor · subscapularis

"When is surgery indicated?"	→ **avulsion from humeral insertion** ✗ Wrong options: never · after failed PT · asymmetry only · chest-wall ecchymosis
"30 y lifter, lost axillary contour, weak IR — definitive?"	→ **primary acute surgical repair**
"Bodybuilder, axillary + anterior brachium ecchymosis, MRI — best long-term?"	→ **open repair of sternocostal portion**
"Young electrician, lost fold worse with adduction, normal X-ray?"	→ **open repair of pec major avulsion** ✗ Wrong options: subscap repair · LHB tenodesis · brace
"Chronic tear 2 y after PT, weak — treatment?"	→ **allograft reconstruction (semitendinosus weave)**
"Nonop treatment leaves weakness of?"	→ **adduction + internal rotation**
"Repair to red label; preop sign?"	→ **lost axillary fold + chest asymmetry + arm ecchymosis (Figure C)** ✗ Wrong options: Popeye · winging · belly-press · ER lag
"42 y lifter, 3 days, MRI typical tear — next step?"	→ **primary repair** ✗ Wrong options: EMG · sling + PT · reconstruction · tendon transfer

– – Active space – –

PHASES OF THROWING

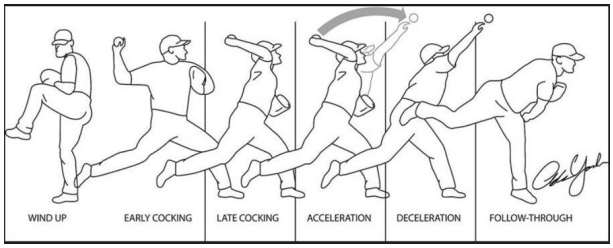
– Shoulder & Elbow · 34 / 60 –

1 Basics

The phases

Number of phases **5**: wind-up · cocking (early + late) · acceleration · deceleration · follow-through (Orthobullets, TO) · **4**: wind-up · cocking · acceleration · deceleration/follow-through (one TO) – read the options

Timing whole throw **~2 s**; wind-up + acceleration ≈ **75% (1.5 s)** (Orthobullets) · **wind-up + cocking = 80% (~1500 ms)**, acceleration **2% (50 ms)**, deceleration **18% (350 ms)** (TO) – read the options



Phases – wind-up, early cocking, late cocking, acceleration, deceleration, follow-through

Phase	Mechanics	Peak muscle activity	Associated pathology
Wind-up (500–1000 ms, TO)	Body rotates, hips + shoulders 90° to target; "controlled fall"; minimal shoulder force	**Rotator cuff inactive**	–
Early cocking	Arm loading; max valgus not yet reached	**Deltoid**	–
Late cocking (500–1000 ms, TO)	**Max shoulder ER** ("loaded", apprehension position), **high torque**; anterior capsule coiled	**Supraspinatus, infraspinatus, teres minor** (rotator cuff); subscapularis late cocking → early acceleration	**MUCL injury** (**max elbow valgus**) · **internal impingement** · **GIRD** · **SLAP (peel-back)**
Acceleration (**50 ms**, shortest, TO)	Max ER → **ball release (~ear level)**; elastic capsular recoil + IR muscles	Early **triceps** → late **pec major, latissimus dorsi, serratus anterior**	Enormous elbow valgus at release (TO)
Deceleration (350 ms, TO)	Eccentric braking; centre of gravity over plantar foot	**Eccentric contraction of all muscles**; **highest torque**	**Most harmful**: **SLAP**, **biceps**, **brachialis**, **teres minor**
Follow-through	Body rebalances, stops forward motion	Muscles **return to resting**	–

Most harmful phase **deceleration** (Orthobullets, TO) · "**late cocking + deceleration** put greatest force on shoulder" (TO)

Rotator cuff greatest stress **deceleration** (eccentric slowing of arm)

Minimal throughout short head of biceps + coracobrachialis

Forces & kinematics (TO)

Parameter (TO)	Value
Shoulder ER	**>160°**
IR angular velocity	**>6000°/s**
IR torque	**>60 N·m**
Anterior joint force, end of cocking	**>350 N**; **AIGHL + anteroinferior capsule** under greatest strain
Distraction force, deceleration	**>500 N**

IR muscles in acceleration (TO) subscapularis, pec major, latissimus dorsi, teres major

Deceleration brakes (TO) eccentric **ER cuff**, **scapular stabilisers**, **posterior deltoid**; elbow flexors limit extension; spine = force attenuator

Scapular kinematics & muscle activation

Scapula + humerus must work **in concert** for GH stability

Scapular rotation must **rotate in cocking + acceleration** → elevate / clear acromion → prevents cuff impingement

Retraction / protraction **maximal retraction in late cocking** (acromion starts to elevate) · **protraction in acceleration + ball release** (**serratus anterior, long thoracic nerve**)

✗ Scapula distractors max retraction at ball release · protracts in late cocking · fixed during throw (must be dynamic) · no effect on concavity-compression
✗ Wrong options: – centres head → ↑ concavity-compression

Kibler – 5 scapular roles stable GH articulation · retraction + protraction · **acromial elevation** · base for muscle attachment · **link in kinetic chain**

Kinetic chain (TO) **~½ of kinetic energy** transferred through **scapulothoracic** joint

SICK scapula (TO) **S**capular malposition · **I**nferior medial border prominence · **C**oracoid pain + malposition · dys**K**inesis → anterior shoulder pain, coracoid tenderness, AC pain, ↓ forward flexion

Dynamic medial elbow stabilisers (FDS, FCU)

Local dynamic elbow stability **FDS + FCU** in **cocking + acceleration** → dynamic compression; **protect UCL**
✗ Wrong options: FDP · ECRL/ECRB · FCR

Implication strengthen forearm flexors to protect UCL

2 Mechanism

Phase of maximal valgus stress – late cocking/early acceleration

Max elbow valgus **late cocking** (TO, Orthobullets, OB) → **MUCL** injury; MCQs combine as **late cocking/early acceleration**

Medial elbow pain + valgus stress test reproduced in **late cocking**
✗ Wrong options: early cocking (max valgus not yet) · acceleration/ deceleration (more force at shoulder)

– – Active space – –

Varus instability	**LUCL** ✗ Wrong options: valgus

Throwing athlete adaptations (TO)

Shoulder	**↑ ER range** (anterior capsule/IGHL stretch) → anterior instability + secondary impingement
IR:ER strength	normal **3:2**; exaggerated in throwers → ER weakness ↑ injury risk
Bony	↑ **humeral retrotorsion** (↓ anteversion) · ↑ **glenoid retroversion**
Elbow	breakdown of medial stabilisers (MCL, capsule, flexors) → **↑ carrying angle**; anterior capsular strain, posterior impingement, flexor strain → **fixed flexion deformity**

3 Presentation

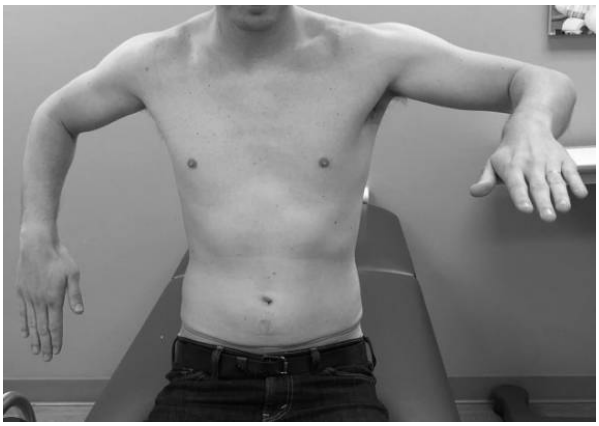
Internal impingement & GIRD (TO)

Internal impingement (TO)	late cocking (**ABER**): posterior cuff undersurface pinched between **greater tuberosity/humeral head and posterosuperior glenoid → posterosuperior labral tear** (hallmark) + **articular-sided (PASTA) supraspinatus** tear
GIRD (TO)	adaptive **loss of IR** in throwing shoulder; caused by **late cocking** (repetitive cocking)
GIRD pathology (TO)	**posteroinferior capsule + posterior band IGHL contracture** (thicker posterior capsule) ± posterior cuff tightness → ↑ GH contact pressure + impingement
GIRD symptoms (TO)	stiffness · prolonged warm-up · **loss of velocity ("dead arm")** · vague **posterior** pain · late cocking provocative
GIRD associations (TO)	posterior labral tears · posterosuperior PASTA tears · **SLAP (peel-back in ABER)** · scapular dyskinesia · **UCL tears**
Sulcus sign in GIRD (TO)	**~⅔ of pitchers**; rotator interval laxity (**CHL + SGHL**)
Cuff tears in throwers (TO)	mostly **degenerative partial/full** from impingement (**internal > external**) · minority **acute traumatic** from **deceleration** overload

4 Diagnosis & Classification

Diagnostic criteria – GIRD

GIRD	
Definition	IR loss **>20°** / **≥20°** vs contralateral (TO) · **25°** (OB, MDI topic) – read the options
Measurement	**Supine, 90° abduction + 90° elbow flexion**; max passive IR = point **just before scapula lifts**; goniometer; also TRM
Posterior tightness (Tyler)	Lateral decubitus, 90° abduction, neutral rotation, scapula stabilised → adduct; **1 cm loss ≈ 5° IR loss**
Posterior impingement sign (Meister)	Max passive **ABER (late cocking) → deep posterior pain** = posterosuperior labral / PASTA



Left GIRD – IR deficit on 90/90 exam

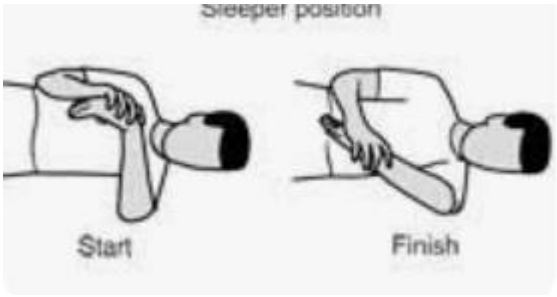
5 Management

Throwing shoulder with GIRD

- First line → **posterior capsular stretching + scapular strengthening**
  - sleeper stretch, cross-body stretch, passive prone stretch
- Severe GIRD + ↓ TRM, failed extensive nonop → **arthroscopic posteroinferior capsular release**

Throwing athlete rehabilitation (TO)

Mainstay (TO)	**posterior capsular stretching** + **scapular** strengthening
Sleeper stretch (TO)	lie on **affected side**, shoulder + elbow **90°**, scapula stabilised → other arm pushes forearm to floor (**IR**); **30 s × 5**



Sleeper stretch

Cross-body stretch (TO)	standing/sitting, shoulder flexed 90° → other hand pulls arm across chest (**horizontal adduction**); **30 s × 5**
Not appropriate for GIRD (TO)	**latissimus dorsi stretch** (✓ sleeper, ✓ cross-body, ✓ passive prone shoulder stretch)
Stretching benefits (TO)	↑ IR · ↑ TRM · ↑ **acromiohumeral distance**
Surgery (TO)	only after exhausted nonop; SLAP repair + articular cuff tear treatment **controversial** (variable RTP); **arthroscopic posteroinferior capsular release** for severe GIRD

Follow-through & injury prevention (TO)

Long follow-through (TO)	**decreases** chance of muscle injury ✗ Wrong options: increases injury · ↑ fatigue · no relation

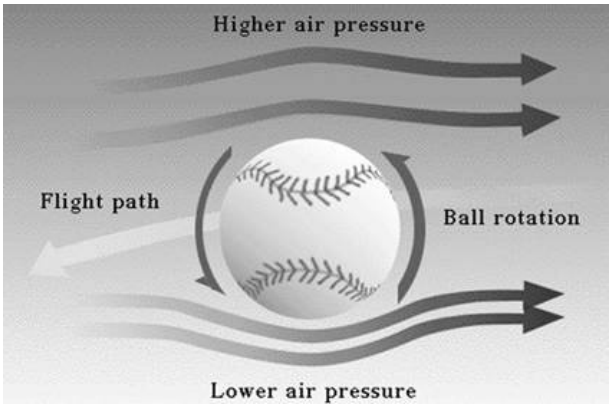
– – Active space – –

Why dissipates deceleration forces; forward torso rotation reacquires energy → ↓ burden on serratus + scapular stabilisers

Ball-flight physics (Magnus effect) (TO)

Magnus effect (TO) **spinning ball** drags boundary layer of air → **high pressure** where boundary layer opposes passing air · **low pressure** where they move together → **lift (Magnus) force** high → low → **curved flight path**

Sports (TO) **tennis**, cricket, soccer, baseball, golf
✗ Wrong options: sprints · swimming · basketball



Magnus effect – spin creates pressure differential and curved flight

X vs Y

	Late cocking	Deceleration
Shoulder position	**Max ER**, ABER	Rapid IR, arm slowing
Muscles	Cuff (supra, infra, teres minor) peak	**Eccentric all**; cuff greatest stress
Elbow	**Max valgus → MUCL**	Elbow flexors (brachialis)
Shoulder pathology	**Internal impingement, GIRD, SLAP peel-back**	**SLAP, biceps, teres minor**
Label	High torque	**Highest torque, most harmful**

	Early cocking	Acceleration
Peak muscle	**Deltoid**	**Triceps → pec major, lat, serratus**
Scapula	–	**Protracts** (serratus)
Duration	Part of 500–1000 ms	**50 ms – shortest**

How it's asked

"Scapula during throwing – true?" → **must rotate in cocking + acceleration to prevent cuff impingement**
✗ Wrong options: max retraction at release · protracts in late cocking · fixed · no concavity-compression effect

"Peak muscle in early cocking (figure)?"	→ **deltoid** ✗ Wrong options: subscapularis = late cocking/early acceleration · supraspinatus/teres minor = late cocking
"Most important dynamic medial elbow stabilisers in cocking + acceleration?"	→ **FDS + FCU**
"Phase of maximum valgus at elbow in bowling?"	→ **late cocking** (TO) ✗ Wrong options: acceleration · deceleration
"Phase causing GIRD?"	→ **late cocking** (TO)
"Medial epicondyle tenderness + valgus stress test – phase reproducing pain?"	→ **late cocking**
"Valgus instability after seasons of pitching – structure + phase?"	→ **MUCL; late cocking (C)** ✗ Wrong options: LUCL = varus
"Young athlete, IR deficit on exam – exercise NOT appropriate?"	→ **latissimus dorsi stretch** (TO) ✗ Wrong options: sleeper · cross-body · passive prone
"Magnus effect important in?"	→ **tennis** (TO) ✗ Wrong options: sprints · swimming · basketball
"Long follow-through in bowling?"	→ **decreases muscle injury** (TO)
"Most harmful / highest torque phase?"	→ **deceleration**
"Shortest phase?"	→ **acceleration (50 ms)** (TO)

– – Active space – –

POSTERIOR LABRAL TEAR

– Shoulder & Elbow · 35 / 60 –

1 Basics

Epidemiology

Definition	**reverse Bankart** lesion / attenuation of posterior capsulolabral complex
Frequency	relatively **rare** vs anterior labral pathology
Athletes	**weightlifters (bench press)** · **football linemen (blocking)** · **wrestlers** · swimmers · gymnasts · **baseball batters**

Anatomy

Posterior labrum	**fibrocartilage** · cavity-compression · **50% of socket depth** · **posterior stability** · **anchors PIGHL**
Static restraints to posterior translation (flexion, adduction, IR)	**PIGHL**, augmented by **SGHL**

✗ Distractor	Actual restraint role
Supraspinatus	**Dynamic**; prevents **inferior** instability
MGHL	Static; anterior + posterior at **45° abduction**
Subscapularis	**Dynamic**; posterior translation in **ER**
AIGHL	Static; **anterior** at **90° abduction + ER**

Most important factors in posterior instability (Bäcker)	**capsular lesions + IGHL rupture**

2 Mechanism

Mechanism & provocative position (flexion–adduction–internal rotation)

Most common mechanism	**repetitive microtrauma** to posterior capsulolabral complex
Provocative position	**posteriorly directed force** with arm **flexed, ADducted, internally rotated** (net posterior vector)
✗ Wrong option: flexion + ABduction + ER	anterior instability → **anteroinferior** labral tear
Batter's shoulder	**acute posterior subluxation** on **adducted lead shoulder** while swinging a bat → acute posterior labral tear ✗ Wrong options: recurrent anterior events · anabolic steroids
Posterior capsular tightening	→ **GIRD** ✗ Wrong options: cause of batter's shoulder

Glenoid retroversion	**~17° → 6× more likely** posterior instability vs ~7°
Kim lesion	**incomplete, sometimes concealed avulsion of posteroinferior labrum**
First-time posterior dislocation	**less likely to recur** than first-time anterior

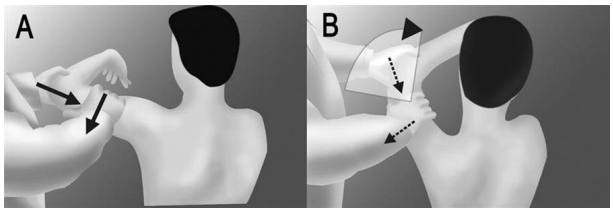
3 Presentation

Symptoms	**vague, deep, nonspecific posterior shoulder pain** (most common) · **↓ strength** (bench press, **batting power**) · clicking/popping · instability less common
Aggravated by	posterior force: **pushing heavy doors, bench press, push-ups, blocking, grappling**
Throwing pain	**late cocking** (Orthobullets) · late cocking/early acceleration = **anterior** instability (OB) – read the options
Exam	posterior joint-line tenderness; **full ROM**; large patients hard to examine

Diagnosis – jerk/Kim tests & MRI

Test	Technique	Positive / meaning
Posterior apprehension	**90° flexion + adduction**, stabilise scapula anteriorly, push humerus posteriorly	Pain / instability
Posterior load and shift	Arm at side; posterior force on proximal humerus	Graded translation (below)
Jerk (Jahnke) test	**90° abduction + full IR** (arm flexed, IR), **axial load while adducting across body** (push elbow posteriorly, scapula anteriorly)	**Clunk** = head subluxes off posterior glenoid → **posterior labral tear**
Kim test	**90° abduction + 45° forward flexion**, axial load on elbow, **posteroinferior** force	**Pain → posteroinferior labral tear** (most sensitive)
Push-pull test	(OB)	Pain from posterior labrum

Jerk/Kim accuracy **highly sensitive + specific** (Orthobullets) · sensitivity/specificity **"under question"** (OB) – read the options



Kim test: abduction with forward flexion, axial + posteroinferior force

Load-and-shift grade	Translation
0	None
1	To **edge** of glenoid
2	**Over edge**, **spontaneously relocates**
3	Over edge, **does not relocate**

– – Active space – –

✗ Exam distractor	Indicates
Forward elevation in scapular plane	**Impingement**
ER + abduction	**Anterior apprehension**
Flexion + abduction	Nonspecific
O'Brien active compression	SLAP / labrum

4 Diagnosis & Classification

Key radiology findings

X-ray — **true AP + scapular Y + axillary**; **axillary required** (posterior dislocation missed on AP alone); often **normal**

Chronic X-ray (axillary) — **glenoid retroversion** · **posterior glenoid erosion / bone loss**

MRI — **study of choice**; **intra-articular contrast (MR arthrogram)** ↑ sensitivity · alternative **CT arthrogram**

MR arthrography for labrum — **sensitivity 89%, specificity 91%, accuracy 90%** (Bencardino); standard MRI often misses labral lesions → **next step after failed nonop + labral test**
✗ Wrong options: diagnostic arthroscopy · CT · stress views

SLAP on MRA — coronal oblique: deep cleft between superior labrum + glenoid extending below biceps anchor

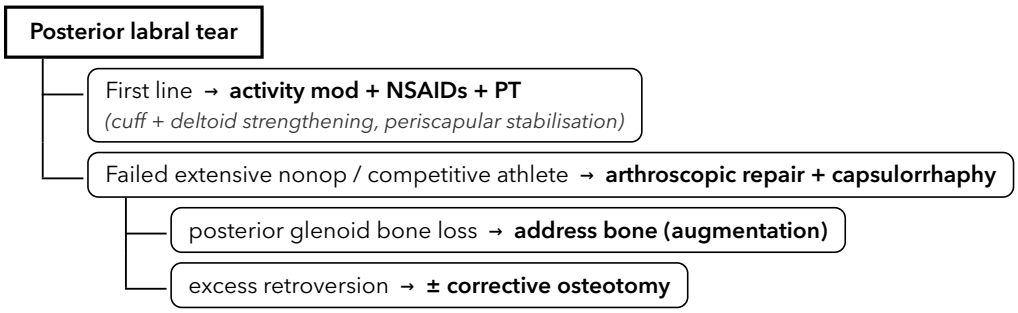
Incidental — posterior labral tears common on MRI of **asymptomatic throwers** → treat only if history + exam fit



Axial MR arthrogram: posterior labral tear

✗ Diagnosis distractor (bodybuilder, MRI arthrogram)	
Pectoralis major rupture · supraspinatus PT tear · SLAP · LHB tendinitis	Not on MRI

5 Management



Treatment – therapy vs arthroscopic repair, recurrence & postoperative restrictions

Nonop	**first line**; cuff + deltoid strengthening, **periscapular stabilisation**
Thrower with anterior follow-through pain, end-of-game, full ROM, small posterior labral contrast	**rotator cuff deconditioning → PT with throwing program** ✗ Wrong options: labral repair · debridement · capsular release · IR stretching (ROM is full)
Operative indication	**failed extensive nonop** (or competitive athletes)
Arthroscopic > open	avoids extensive posterior dissection; **more reliable return to play**
Suture anchors + capsulorrhaphy	**fewer recurrences + revisions** than non-anchored repair
Probe posterior labrum	to rule out subtle **Kim lesion**
Post-op	**avoid adduction + internal rotation** (stress posterior repair) → failure risk ↑ if not avoided

Post-op restriction	Repair
Avoid adduction + IR	**Posterior labrum**
Avoid ER	**Subscapularis**, **anterior labrum**
No restriction (stiffness risk)	Calcific tendinitis decompression, distal clavicle resection

Outcome	generally good; **overhead throwers** return less reproducibly

Posterior glenoid bone loss

Posterior glenoid bone loss	Failure of isolated capsulolabral repair
11%	**10.4×**
15%	**24.4×**

"Sub-critical" bone loss	even small posterior loss → **recurrence after isolated arthroscopic repair** ✗ Wrong options: "not critical, no bony procedure"
Glenoid width vs version	**width more predictive** of recurrence than version
Obligate translation	tightening posterior tissues → anterior translation = **therapeutic** ✗ Wrong options: causes anterior instability
Corrective osteotomy	for excessive retroversion; **does not obviate bone block** when bone loss present

– – Active space – –

6 Complications

Complications – axillary nerve injury

Axillary nerve (posterior branch)	within **1 mm of inferior capsule + glenoid rim** → injured by **suture passage at posteroinferior glenoid**
Posterior branch gives	**superolateral brachial cutaneous** nerve + **nerve to teres minor** → **lateral arm burning/numbness** + **teres minor ER weakness**
Overtightening posterior capsule	**anterior subluxation** or **coracoid impingement**
Recurrence	↑ if **posterior bone loss** not addressed

X vs Y

	Posterior labral tear (reverse Bankart)	Anterior labral tear (Bankart)
Position	**Flexion + adduction + IR**, posterior force	**Abduction + ER**, anterior force
Who	Linemen, lifters, wrestlers, batters	–
Painful activity	**Bench press, push-ups, blocking**	Late cocking, spiking, overhead serve
Test	**Jerk, Kim, load-and-shift**	Apprehension, relocation
Ligament	**PIGHL** (+ SGHL)	**AIGHL**
Post-op avoid	**Adduction + IR**	**ER**

	Jerk test	Kim test
Position	90° abduction + IR, adduct across body with axial load	90° abduction + 45° flexion, axial + posteroinferior force
Positive	**Clunk**	**Pain**
Best for	**Posterior** labral tear	**Posteroinferior** labral tear

How it's asked

"Bodybuilder, vague deep pain, weak bench press, MR arthrogram – etiology?"	→ **posterior labral tear** ✗ Wrong options: pec major · SLAP · LHB tendinitis
"Old 'pop', never reduced, axial MRI posterior labrum – painful activity?"	→ **heavy bench press** ✗ Wrong options: late cocking · volleyball spike = anterior · curl = SLAP
"Wrestler, + Kim + jerk – structure augmenting PIGHL in flexion-adduction-IR?"	→ **superior glenohumeral ligament** ✗ Wrong options: MGHL · AIGHL · subscapularis
"Wrestler, posterior labral tear MRI – position increasing incidence?"	→ **flexion, ADduction, internal rotation**

"Baseball player, ↓ batting power, + Kim/jerk — etiology?"	→ **acute posterior subluxation while batting** ✗ Wrong options: posterior capsular tightening = GIRD
"Ex-pitcher, failed nonop, + O'Brien — next diagnostic step?"	→ **MRI-arthrography** ✗ Wrong options: diagnostic arthroscopy · CT
"Lineman, pain with blocking, axial MRI — manoeuvre reproducing pain?"	→ **flexion, adduction, internal rotation** ✗ Wrong options: ER + abduction = apprehension
"Flexed IR arm pushed posteriorly, girdle anteriorly, clunk — injured structure?"	→ **posterior labrum (C)** (jerk test)
"Quarterback, anterior pain in follow-through, end of game, small posterior labral contrast — initial?"	→ **PT with throwing program**
"Post-op: avoid adduction + IR — initial diagnosis?"	→ **posterior labral tear (Figure C)**
"NFL tackle, posterior glenoid bone loss on CT — true?"	→ **capsulolabral repair alone may lead to recurrence** ✗ Wrong options: obligate anterior instability · version more predictive than width · osteotomy obviates bone block
"Burning lateral arm pain after arthroscopic posterior labral repair — nerve?"	→ **axillary** ✗ Wrong options: suprascapular · radial

– – Active space – –

POSTERIOR SHOULDER INSTABILITY & DISLOCATION

– Shoulder & Elbow · 36 / 60 –

1 Basics

Epidemiology

Incidence	**2–5%** of unstable shoulders (Orthobullets) · **2–12%** of instability (OB) – read the options
Missed	**50%** of traumatic posterior dislocations in ED undiagnosed · Hawkins & Neer: 41 locked dislocations, **50% with fracture**, 20 missed, mean **1 y** to diagnosis
Prior dislocation	up to **50%** of patients evaluated for posterior instability report a prior posterior dislocation · frank dislocation in only **2–5%** of posterior labral tears
Risk factors	**ligamentous laxity** · **glenoid retroversion** / hypoplasia (less common) · not excessive glenoid **anteversion** = anterior instability
Seizures + electric shock	tetanic contraction pulls head out; posterior dislocation unlikely without such force · anterior **still more common** with seizures (Orthobullets, OB) · posterior "more common after seizure" / **as common as anterior in epilepsy** (OB) – read the options
Atraumatic (esp. posterior) dislocation	warrants **neurological workup**

Anatomy – posterior restraints

Restraint	Role
Posterior band IGHL	**Primary restraint in IR**; most important at **90° flexion + IR**
SGHL + CHL	Primary **static** stabiliser to posterior subluxation in **flexion, adduction, IR** · also inferior translation of adducted arm + ER
Subscapularis	**Primary dynamic restraint against posterior subluxation** (and in ER)
Labrum	Deepens glenoid **50%**

Restraint conflict posterior band IGHL (OB, Orthobullets) vs SGHL (OB, Orthobullets) for flexion + IR – read the position in the stem

✗ Restraint distractor	Role
Anterior band IGHL	Anterior/inferior at **90° abduction + max ER** (late cocking)
MGHL	Anterior + posterior at **~45° abduction** in ER
Anterior labrum	Anterior; IGHL anchors here → Bankart
Superior labrum	**Biceps anchor**

Dynamic stabilisers by direction posterior labral tear → **subscapularis** · anterior labral tear / HAGL → **infraspinatus + teres minor**

– – Active space – –

Tubercle of Assaki (TO)

Tubercle of Assaki (TO) ridge / **focal elevation of subchondral bone at centre of glenoid** · **normal variant** · **focal thinning of overlying cartilage** · not a cartilage defect
✗ Wrong options: "depression of humeral head cartilage"

2 Mechanism

High-risk position arm **flexed, adducted, internally rotated** + **axial load**
✗ Wrong options: abduction + ER = anterior · hyperabduction = **luxatio erecta**

Type	Mechanism	Presentation
Acute (dislocation)	Trauma (~50%), **seizure, electric shock**	Dramatic
Chronic (instability)	**Repetitive microtrauma** → labral tear, incomplete avulsion, erosion, **patulous posterior capsule**	Insidious, vague **pain > instability**

At-risk athletes **football linemen** (blocking), linebackers, **weightlifters/bench press**, push-ups, weighted dips, rugby, swimmers, overhead athletes · stretches posteroinferior GHL

Classification acute vs chronic · voluntary vs involuntary

Associated lesion	Association
Posterior band IGHL avulsion	Acute subluxation
Reverse Bankart	Detached posteroinferior capsulolabral complex
Reverse Hill-Sachs	**Locked**, hard-to-reduce dislocation
Posterior labral cyst / posterior glenoid rim fracture	Chronic reverse Bankart
Lesser tuberosity fracture	Acute posterior dislocation
Large capsular pouch	Chronic instability (MR arthrogram)

Reverse Hill-Sachs (TO) **anteromedial** humeral head · Hill-Sachs = **posterosuperolateral**
✗ Wrong options: posterolateral

Cuff tears **not** commonly associated with posterior dislocation

3 Presentation

Acute dislocation **prominent posterior shoulder + coracoid** · **locked in IR + adduction** · comfortable in "sling position"

Most consistent sign **lack of ER** (mechanical block: anterior head defect on posterior glenoid) · ER loss differential: adhesive capsulitis, GH OA

Chronic instability may look normal · pain with **flexion, adduction, IR** · slipping when pass-blocking, pain bench pressing/push-ups

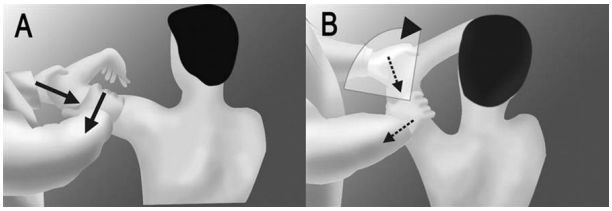
– – Active space – –

Recurrent posterior instability – Kim lesion/labral tear, jerk & Kim tests, restraints

Kim lesion (TO) — **incomplete + concealed avulsion of posteroinferior labrum** (superficial labrum intact, deep labrum detached) in posterior or multidirectional posteroinferior instability · usually **diagnosed at arthroscopy** · needs **arthroscopic repair**
✗ Wrong options: ALPSA = anterior labral periosteal sleeve avulsion · meniscal root avulsion

Posterior labral tear site — **posteroinferior quadrant**: **6–9 o'clock right**, **3–6 o'clock left**

Test	Technique	Positive / use
Jerk	90° abduction/flexion + IR, elbow bent · **axial load** along humerus → adduct into forward flexion (bring back to extension → reduces)	**Clunk** ± pain · posterior labral tear in **mid-range** glenoid
Kim	Seated, 90° abduction → **flex to 45°** + axial load at elbow + posteroinferior force (Orthobullets) · arm elevated **125°** + IR, downward + posterior force (OB)	**Pain** (± clunk) · more sensitive for **posteroinferior** tears
Jerk + Kim	Combined	**97% sensitive** for posterior labral tear
Posterior stress	Stabilise scapula, push posteriorly	Pain common, **not specific**
Load & shift	Supine, neutral rotation, 40–60° abduction + flexion	Grade translation



Kim test

Load & shift grade	Translation
1+	Apparent, not to rim
2+	**To glenolabral rim**
3+	**Over** rim
4+	Complete dislocation

– – Active space – –

✗ Test distractor	Assesses
Apprehension / relocation (abd + ER)	**Anterior** instability, Bankart
Hornblower; ER weakness at 90° abduction	**Teres minor** (axillary n.)
Jobe	Supraspinatus
Speed / Yergason	Biceps
O'Brien	SLAP
Belly-press, lift-off, IR lag	Subscapularis
Sulcus	Rotator interval
Pinch grip	AIN

✗ Lesion distractor	Feature
Bankart / Hill-Sachs	Anterior instability, apprehension
GIRD	Throwers, **tight posteroinferior capsule**, ↓ IR → posterior impingement, **anterior subluxation**
AC separation	Piano key sign
SLAP	Overhead pain, dead arm
PASTA	Partial articular supraspinatus avulsion
Pec major tear	↓ adduction + IR, lost axillary fold

4 Diagnosis & Classification

Locked posterior dislocation – diagnosis (axillary/Velpeau view, loss of ER, seizures, electrocution)

AP — **unreliable** · **"lightbulb" sign** (head internally rotated) · humeral head **overlaps glenoid**



Lightbulb sign on AP

Axillary lateral — **best view**; mandatory (needs **20–30° abduction**) · **more sensitive than trans-scapular** · any suspected dislocation → full trauma series (AP, scapular Y, axillary)

– – Active space – –

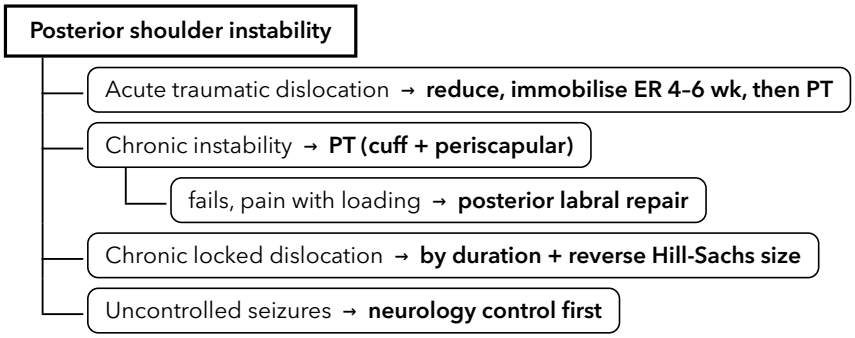
Can't abduct (pain/ spasm)	**Velpeau view** ✗ Wrong options: Zanca = AC/clavicle · Stryker notch = Hill-Sachs · MRI not first-line in trauma · EUA is for locked dislocation
Scapular Y	humeral articular surface **posterior** to glenoid
CT	chronic dislocation (**>2–3 wk**): size of **reverse Hill-Sachs** + bone loss · posterior glenoid bone loss (% diameter)
MRI / MR arthrogram	chronic instability without dislocation · posterior labral tear, reverse Hill-Sachs, cuff · **Kim lesion**: incomplete avulsion/cyst with intact chondrolabral junction · **posteroinferior capsulolabral separation**



MR arthrogram: posterior labral tear

Next step pattern	any "light bulb" / overlapping head / ER to neutral after seizure or fall → **axillary view** ✗ Wrong options: PT for frozen shoulder · sling · cuff repair · EMG · injection
Glenoid hypoplasia	**posteroinferior glenoid deficiency**, enlarged distal clavicle, glenoid indentation · **usually bilateral** → X-ray other side · rarely syndromic · **PT first** · glenoid osteotomy variable · early OA ✗ Wrong options: echo / abdominal US / skeletal survey unless Holt-Oram, Apert stigmata

5 Management



Chronic locked dislocation – reverse Hill-Sachs management (McLaughlin, arthroplasty)

Source	Closed reduction	McLaughlin / lesser tuberosity transfer	Hemiarthroplasty
Orthobullets	All acute (most reduce spontaneously) · disimpaction + graft if **<3 wk**	Chronic **<6 mo** + RHS **<40%**	**>6 mo**, RHS **>40%**, severe head arthritis, **head collapse on reduction**
Hawkins & Neer (OB)	**<6 wk + <20%**	**6 wk–6 mo + 20–45%**	**>6 mo or >45%** (glenoid normal)
OB (other)	**<3 wk + <25%** under GA; stable if IR to abdomen → 6 wk orthosis in slight extension + ER	Defect **<~40%**	Larger; **>50%** or several months

TSA — significant **glenoid arthritis** + an HA indication · glenoid intact → no glenoid component

Closed reduction after ~3 wk — exceedingly difficult · may **displace unrecognised surgical neck fracture**

McLaughlin — **subscapularis** into anteromedial defect · **modified (Neer/Hawkins)** = **lesser tuberosity + subscapularis** (used by most) · deltopectoral

Modified McLaughlin candidate — young man, **locked posterior dislocation after electric shock / seizure**
✗ Wrong options: AVN · RA = arthroplasty · subscap insufficiency = repair or pec transfer · anterior Hill-Sachs = graft/remplissage

6-wk locked dislocation, 38 y — **open reduction ± lesser tuberosity transfer** (CT first)
✗ Wrong options: PT · closed manipulation · cuff repair · HA

12-wk locked post-seizure, large RHS + lesser tuberosity fracture, 60 y — **hemiarthroplasty + subscapularis repair** · younger → **osteochondral allograft** + subscap repair

6 mo after motorcycle fall, 52 y — **hemiarthroplasty** most reliable
✗ Wrong options: PT · MUA · capsular release/plication

HA version — **anatomic version (20–30° retroversion)** · not excessive anteversion (re-dislocation, per source)

7 mo locked, 63F, light bulb — first **further radiographs (axillary)** → then HA

Open complications — stiffness · **AVN** · OA



CT: locked posterior dislocation with anteromedial reverse Hill-Sachs

– – Active space – –

Uncontrolled epilepsy, 9th dislocation, humeral + glenoid bone loss	**neurology + social work to control seizures first** · later: **posterior bone block** (iliac crest autograft or **distal tibial allograft**) ± modified McLaughlin / head grafting · HA ± glenoid reconstruction for chronic locked ✗ Wrong options: urgent HA · reconstruction before control · anterior procedure alone

Operative criteria

Procedure	Indication
Posterior labral repair (open/ arthroscopic)	**Recurrent instability despite PT** · pain loading arm in forward flexion (bench, blocking) · **negative Beighton**
Capsular shift + rotator interval closure	**Positive Beighton** (laxity) · shift less desirable in throwers · interval closure controversial
Posterior bone block	Posterior glenoid bone loss **≥11%** in contact athlete (**subcritical** by anterior standards)
Opening-wedge glenoid osteotomy	Excessive **congenital retroversion** (limited data)
Modified McLaughlin	Chronic locked, RHS <40%, <6 mo
HA / TSA	RHS >40%, >6 mo, collapse; TSA if glenoid arthritis

Treatment – therapy vs arthroscopic capsulolabral repair & postoperative protocol

Acute reduction	immobilise **10–20° ER, elbow at side, 4–6 wk** → PT (cuff + periscapular) + avoid high-risk position
After reduction, avoid	**cross-body adduction at 90° flexion + IR** (tensions SGHL + posterior IGHL)
Chronic instability first line	**PT** (cuff + periscapular; in-season athlete) · esp. **generalised laxity** · ± brief immobilisation, steroid injection
Failed rehab, positive jerk, grade 2 load-shift, lineman	**arthroscopic posterior capsulolabral repair** (arthroscopic ≈ open) ✗ Wrong options: thermal · open anterior capsulorrhaphy · injection · ER immobilisation for recurrent
Lineman, clunk, bench-press pain	**posterior labral tear → arthroscopic labral repair** ✗ Wrong options: thermal capsulorrhaphy · SLAP debridement/tenodesis
Thermal capsulorrhaphy	**contraindicated** · breaks collagen cross-links at **65–75 °C** · capsular necrosis, **chondral damage**, axillary nerve injury, recurrence
Technique	**lateral decubitus** · **high lateral / posterosuperior portal** for anchor trajectory
Results	open **80–85% success at 5–7 y** · arthroscopic similar (shorter follow-up)
Post-op	sling in **neutral rotation** (ER sling) or standard, up to 6 wk · **avoid flexion, adduction, IR** · full contact/heavy labour **6 mo** ✗ Wrong options: no elbow flexion = biceps tenodesis · avoid abd + ER = anterior/HAGL repair · PROM ≥6 wk = cuff repair · unrestricted ROM
Posterior glenoid bone loss	**lower critical value than anterior**; **11%** of glenoid diameter ↑ recurrence after isolated labral repair → **bone block in contact athletes** ✗ Wrong options: modified McLaughlin = chronic dislocation · only with large RHS · isolated repair better · augmentation doesn't change recurrence

First-time traumatic dislocation, young athlete with recurrent subluxations	**arthroscopic capsulolabral repair reasonable** · recurrence up to **90%** in young active · immobilisation (IR or ER) may not reduce recurrence · arthroscopic ≈ open with modern anchors if no glenoid loss/engaging lesion

6 Complications

Rank	Complication after labral repair
1	**Stiffness** (↓ **IR**)
2	**Recurrence** (7–50%) · ↑ with glenoid bone loss ≥11%, uncontrolled seizures
3	**DJD**
Other	Adhesive capsulitis · **overtightening → anterior subluxation / coracoid impingement** · **axillary / suprascapular** nerve

Not after posterior repair	anterior instability not expected · acromial stress fracture not typical

X vs Y

	Posterior instability	Anterior instability
Position	**Flexion, adduction, IR**	Abduction + ER
Symptom	**Pain** > instability	Instability, apprehension
Labral lesion	**Reverse Bankart / Kim** (posteroinferior)	Bankart, ALPSA
Head lesion	**Reverse Hill-Sachs (anteromedial)**	Hill-Sachs (posterosuperolateral)
Tests	**Jerk, Kim**, load & shift	Apprehension, relocation
Dynamic stabiliser	**Subscapularis**	Infraspinatus + teres minor
Post-op avoid	**Flexion, adduction, IR**	Abduction + ER
Critical glenoid bone loss	**Lower (~11%)**	Higher

	Jerk test	Kim test
Positive	**Clunk**	**Pain**
Best for	Posterior (mid-range)	**Posteroinferior**

How it's asked

"63F, 7 mo after fall, ↓ ER, light-bulb AP – next?"	**→ further radiographs (axillary view)** ✗ Wrong options: PT for frozen shoulder · arthroplasty first · ORIF
"Seizure 1 wk ago, ER to neutral, sling – next?"	**→ axillary lateral radiograph**
"Posterior dislocations as common as anterior in?"	**→ epilepsy patients**
"Drunk, woke with pain, 'cuff tear' sling 6 wk – next?"	**→ axillary radiograph**
"Acute posterior dislocation – shoulder locked in?"	**→ internal rotation**

– – Active space – –

"Electrician, no axillary possible due to spasm?"	→ **Velpeau view** ✗ Wrong options: EUA · MRI · Zanca · Stryker notch
"Exam finding of fixed posterior dislocation?"	→ **lack of external rotation** ✗ Wrong options: apprehension · relocation · sulcus · jerk
"ICU head injury, ER −30°, overlap on AP?"	→ **posterior shoulder dislocation**
"Light-bulb sign — predisposing factor?"	→ **seizures** ✗ Wrong options: ↓ IR = GIRD · inflammatory · anteversion · anterior IGHL injury
"Epileptic teen, 9th dislocation, can't afford levetiracetam, bone loss on CT?"	→ **social work + neurology to control seizures**
"Reverse Hill-Sachs location?"	→ **anteromedial** humeral head (TO)
"NOT true about tubercle of Assaki?"	→ **depression of humeral head cartilage** (TO)
"Who benefits from modified McLaughlin?"	→ **locked posterior dislocation after electric shock**
"52 y, locked posterior dislocation 6 mo — most reliable?"	→ **hemiarthroplasty**
"Seizure 12 wk ago, no ER, large RHS + lesser tuberosity fracture?"	→ **HA + subscapularis repair** ✗ Wrong options: sling · closed reduction · labral repair · LT transfer
"38 y, posterior dislocation 6 wk old?"	→ **open reduction ± lesser tuberosity transfer**
"61 y, seizure 5 mo, half head lost?"	→ **HA in anatomic version** ✗ Wrong options: anteversion · TSA · closed reduction · LT transfer
"After surgery for posterior dislocation, most common complication besides stiffness?"	→ **instability (recurrence)**
"Kim's lesion is?"	→ **incomplete, concealed avulsion of posteroinferior labrum** (TO)
"Lineman, posterior pain after block in flexion + IR?"	→ **posterior labral tear** ✗ Wrong options: Bankart · SLAP · HAGL · AC
"Lineman subluxes blocking, positive jerk — pathology?"	→ **Kim lesion (reverse Bankart)** ✗ Wrong options: Hill-Sachs · GIRD
"Positive jerk + Kim test?"	→ **posterior labral tear** ✗ Wrong options: SLAP · ALPSA · PASTA
"After posterior dislocation, 'arm will pop out' — positive test?"	→ **Kim test** ✗ Wrong options: Hornblower · Jobe · apprehension · Speed

"Arthroscopy shows posteroinferior capsulolabral tear — preop exam?"	→ **painful clunk with arm elevated 125° + IR, downward/posterior force (Kim)**
"Basketball player, posterior labral tear on MR arthrogram — test?"	→ **jerk test** ✗ Wrong options: O'Brien · Speed · Neer · apprehension
"Lineman slipping when pass blocking, no apprehension?"	→ **posterior labral tear**
"Primary restraint to posterior displacement in flexion + IR?"	→ **posterior band IGHL** ✗ Wrong options: anterior band · MGHL · anterior/superior labrum
"Football player, axial load flexed + IR = pain + clunk?"	→ **jerk test** ✗ Wrong options: Hornblower · Speed · apprehension · O'Brien
"Powerlifter, pain bench + push-up, normal exam — MRI?"	→ **posteroinferior capsulolabral separation**
"Subscapularis is critical stabiliser in which lesion?"	→ **posterior labral tear (C)** ✗ Wrong options: SLAP · HAGL · spinoglenoid cyst · Bankart
"Lineman, Kim test pain — matching MRI?"	→ **posterior labral tear (B)**
"Mechanism of posterior labral tear + reverse Hill-Sachs?"	→ **axial load on flexed, adducted, IR shoulder**
"Quarterback after reduction of posterior dislocation — avoid?"	→ **cross-body adduction at 90° flexion + IR**
"Swimmer, clicking, glenoid hypoplasia on X-ray — next?"	→ **physical therapy** ✗ Wrong options: echo · abdominal US · skeletal survey · osteotomy
"Lineman, failed rehab, positive jerk, grade 2 load-shift — best?"	→ **arthroscopic posterior capsulolabral repair**
"16 y, first dislocation + 2 subluxations — true?"	→ **arthroscopic capsulolabral repair is reasonable**
"Lineman, bench pain, clunk — MRI + surgery?"	→ **posterior labral tear; arthroscopic labral repair**
"Pitcher, positive Kim, labral repair — likely complication?"	→ **decreased internal rotation (stiffness)** ✗ Wrong options: posterior instability = 2nd · DJD = 3rd
"Post-op protocol after posterior labral repair?"	→ **avoid flexion, adduction, IR**
"Rugby player, posterior tear + 11% glenoid loss, bone block — true?"	→ **subcritical bone loss is an indication in contact athletes**

– – Active space – –

QUADRILATERAL SPACE SYNDROME

– Shoulder & Elbow · 37 / 60 –

1 Basics

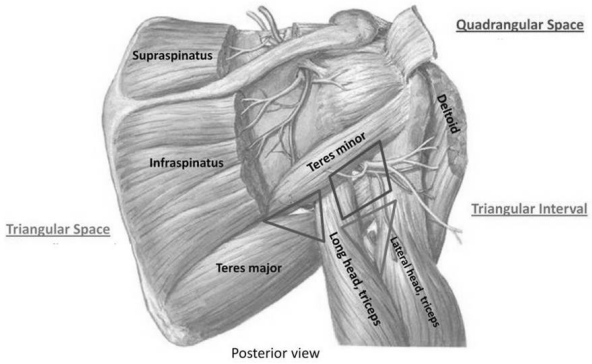
Epidemiology

Definition	compression of **axillary nerve + posterior humeral circumflex artery (PHCA)** in the quadrilateral (quadrangular) space → **posterolateral shoulder pain**
Incidence	**rare**; often **misdiagnosed as subacromial impingement**
Age	**20–40 y** (Orthobullets) · **20–35 y** (TO) – young adults
Side	**dominant shoulder** most commonly (TO)
Risk factors	**overhead athletes** (basketball, baseball pitchers), contact/throwing sports, **weightlifting** (muscle hypertrophy)

Anatomy – quadrilateral space

Boundary	Structure
Superior	**Teres minor** (inferior border)
Inferior	**Teres major** (superior border)
Medial	**Long head of triceps** (lateral border)
Lateral	**Surgical neck of humerus** (medial cortex)
Contents	**Axillary nerve** (C5 root, **posterior cord**) + **PHCA** (+ vein, TO)

Location posterior compartment of arm; **lateral to triangular space**, **medial to triangular interval** (with lateral + medial triangular spaces, TO)



Quadrangular space, triangular space and triangular interval

Axillary nerve supplies	**deltoid + teres minor** + sensation **lateral arm** (upper lateral brachial cutaneous)
Long head of triceps	**can have axillary innervation** (case report, OB)
Teres minor motor nerve	2 variants of fascial envelopment; nerve takes **highly angulated course at the fascial sling → stout fascial sling** may compress it (OB)

2 Mechanism

Etiology – causes of compression

Cause	Note
Fibrous bands	**Oblique fibrous bands = most common cause** (TO); compress nerve + artery in abduction/ER
Muscular hypertrophy	Weightlifting after a break (OB)
Paralabral cyst	**Most commonly inferior labral tears** (Orthobullets, TO) · **posterior labrum** ("occasionally", TO) – read the options
Trauma	Scapular fracture, shoulder dislocation
Masses	Benign or malignant

Position of max compression — **late cocking phase** of throwing = **abduction + external rotation** (TO)

Throwers — repetitive motion irritates nerve → dysfunction; usually **self-limited** (OB)

3 Presentation

Symptoms

Pain — **poorly localised posterior/lateral shoulder pain**; **worse at night**, overhead activity, late cocking/acceleration

Paraesthesia — **non-dermatomal**, lateral shoulder + **lateral arm**

Weakness — **external rotation** weakness, esp. **arm abducted 90° (throwing position)**; **5/5 ER with arm at side** (OB)

Aggravated by — active + resisted **abduction + ER**; forward flexion (OB)

Examination

Palpation — **point tenderness over quadrilateral space** (near teres minor insertion, TO)

Atrophy — **teres minor + deltoid** (✓ **deltoid atrophy**, TO)
✗ Wrong options: supraspinatus atrophy · positive apprehension · positive dynamic labral shear

Provocation (TO) — **arm abducted + externally rotated for 1 min** reproduces symptoms
✗ Wrong options: abducted + internally rotated

Detecting weakness (TO) — weakness/atrophy **difficult to detect unless significant deltoid involvement**

Neuro exam — usually **normal**; mild sensory change in axillary distribution

Typical negatives (OB) — full ROM, negative Hawkins/Neer, belly press/lift-off, O'Brien, normal radiographs, no cuff/labral tear on MRA

4 Diagnosis & Classification

Diagnosis – axillary nerve compression & teres minor atrophy

Diagnosis	**clinical** (point tenderness ± teres minor atrophy); MRI may show compression

Key radiology findings

Test	Finding
X-ray (AP, lateral, axillary)	**Usually normal**; rules out other pathology
MRI	**Fatty atrophy of teres minor** (✓ OB; T1 streaked white + atrophied) · deltoid fatty infiltration · compression of space · **inferior paralabral cyst** with labral tear
Arteriogram	**PHCA occlusion in abduction + ER, normal with arm at side** (TO)
EMG	**Confirms diagnosis**; axillary nerve involvement
Lidocaine block	Confirms diagnosis (see Management)

Selective teres minor atrophy on MRI	**highly suggestive** of QSS (TO)
Sanders	teres minor fatty atrophy in **3% of shoulder MRIs** (OB)
Next step to confirm (OB)	**subclavian arteriogram** with arm **adducted + abducted/ER**
MRI ✗ distractors	↓ intra-capsular volume (= **adhesive capsulitis** on arthrogram) · ↑ capsular volume · infraspinatus atrophy (suprascapular) · latissimus atrophy
Isolated teres minor atrophy (OB)	**common**, unclear aetiology (axillary nerve variants); **clinically distinct** from QSS

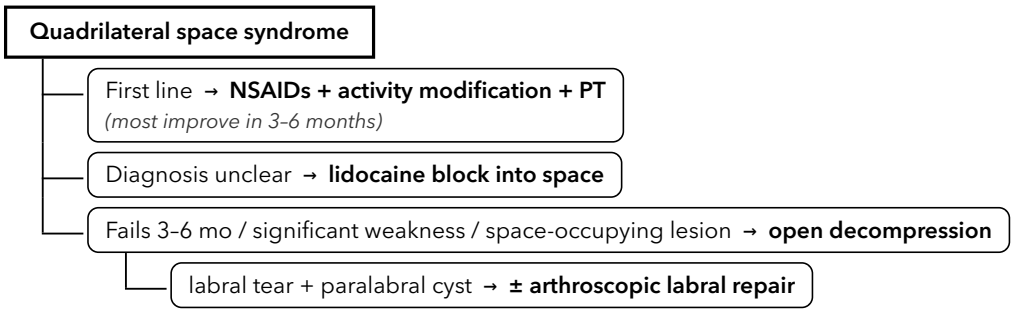


Fatty atrophy of teres minor on sagittal MRI

Brown classification (TO)

Type	Cause	Who
Vascular QSS	**Repetitive trauma to PHCA** in abduction + ER	**Professional/collegiate overhead athletes**
Neurogenic QSS	**Fixed anatomic anomalies – fibrous bands most common**	

5 Management



Nonoperative

Regimen	NSAIDs, activity restriction, physiotherapy (GH mobilisation + strengthening, **posterior capsule stretching**, massage)
TO regimen	activity modification, NSAIDs, **cortisone injections**, PT for **rotator cuff strengthening** – indicated initially for **all** patients
Outcome	**most improve with 3–6 months** nonoperative

Diagnostic lidocaine block

Technique	plain lidocaine into quadrilateral space; start **2–3 cm inferior to standard posterior arthroscopy portal**
Positive	**no point tenderness + pain-free full ROM** after injection

Operative criteria

Indication	Procedure
Failed nonop (3–6 mo)	Open release of quadrilateral space
Significant weakness + functional disability	Open release
Space-occupying lesion (paralabral cyst)	Decompression **± arthroscopic labral repair**
Isolated teres minor atrophy	**Release of fascial sling** to teres minor → ↓ pain, **Hornblower's** weakness resolves (OB)

Open quadrilateral space decompression

Approach	**lateral decubitus**; **posterior approach**, 3–4 cm incision over space; reflect posterior border of deltoid superolaterally; fat between teres minor + major
Technique	find axillary nerve using **humeral neck** as reference · **avoid cutting PHCA** · free fibrous bands (**lysis of fibrous bands**) · confirm release in **abduction + ER**
Post-op	sling for comfort · **early pendulums** (avoid new adhesions) · supervised full active ROM
Results	good results in overhead athletes after failed nonop (McAdams, 4 athletes, OB); usually symptom relief (TO)

6 Complications

Prognosis long-standing cases → **atrophy/weakness of teres minor + deltoid**

X vs Y

Lesion	Atrophy	Weakness
QSS (axillary n.)	**Teres minor** ± deltoid	**ER with arm abducted 90°**
Suprascapular notch	**Supraspinatus + infraspinatus**	Abduction + ER **at side**
Spinoglenoid notch (cyst)	**Infraspinatus only**	**Isolated ER at side**
Chronic cuff tear	Infraspinatus (retracted tear on MRI)	ER
Subscapularis	Rare (**upper + lower subscapular** nerves)	IR

QSS vs subacromial impingement QSS often **misdiagnosed as impingement**; in QSS **Hawkins + Neer negative**, MRI shows **teres minor atrophy**

✗ Wrong options (other diagnoses) Parsonage-Turner · long thoracic neuritis · internal impingement · external impingement · scapular dyskinesia

How it's asked

"Posterior shoulder pain, worse in abduction + ER, paralabral cyst in quadrilateral space — exam finding?"	→ **deltoid atrophy** (TO) ✗ Wrong options: supraspinatus atrophy · apprehension · dynamic labral shear
"True about quadrilateral space syndrome?"	→ **most common cause is oblique fibrous band** (TO) ✗ Wrong options: anterior pain · abducted + internally rotated · middle-aged diabetic female
"MRI in chronic QSS?"	→ **fatty atrophy of teres minor** ✗ Wrong options: infraspinatus · latissimus · loss of capsular volume
"Pitcher, pain + lateral arm paraesthesia, T1 fatty teres minor + deltoid — diagnosis?"	→ **quadrilateral space syndrome** ✗ Wrong options: Parsonage-Turner · long thoracic neuritis · internal/external impingement
"EMG shows QSS — muscle affected along with deltoid?"	→ **teres minor** ✗ Wrong options: teres major · supraspinatus · subscapularis
"Weightlifter, weakness of resisted ER at 90° abduction only, isolated teres minor atrophy?"	→ **QSS** ✗ Wrong options: suprascapular notch · spinoglenoid notch · scapular dyskinesia
"Next step to confirm QSS?"	→ **subclavian arteriogram, arm adducted + abducted/ER**

"Pitcher, weak ER at 90° abduction — muscle and aetiology?"	→ teres minor; axillary nerve at quadrilateral space
"Boundaries of quadrilateral space?"	→ teres minor above, teres major below, long head triceps medial, surgical neck humerus lateral

– – Active space – –

– – Active space – –

REVERSE SHOULDER ARTHROPLASTY

– Shoulder & Elbow · 38 / 60 –

1 Basics

Concept & history

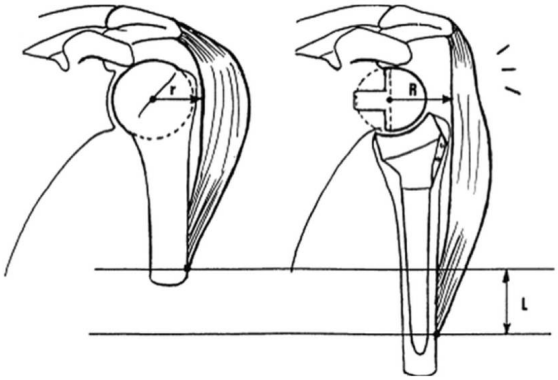
Design	**convex glenoid hemisphere (glenosphere)** + **concave humeral cup** (semiconstrained)
History	early reverse designs failed (**COR too lateral** → torque → glenoid loosening) · **Grammont 1985** = first successful reverse (TO) · popular in Europe; North America since **1990**
Grammont's 2 innovations (OB)	**large glenoid hemisphere with no neck** + **small, almost horizontal humeral cup at 155°** (non-anatomic valgus) covering **< half the glenosphere**
Indications summary	**cuff tear arthropathy (CTA)**, comminuted **4-part proximal humerus fractures**, **failed arthroplasty**

2 Mechanism

Biomechanics – medialised/inferior centre of rotation & deltoid tensioning

COR shift	**medial + inferior (distal)** (✓ TO, OB) ✗ Wrong options: superior · lateral · ↑ lateral humeral offset · posterior · proximal
Standard prosthesis without cuff (TO)	deltoid contraction → humeral head **subluxes superiorly**; reverse **corrects the abnormal vector** and re-creates a **fulcrum**
Medial COR	↑ **deltoid moment arm** (less force for same torque) · ↓ **torque/shear at glenoid interface** (✓ glenoid stability; force vector passes through component) · ↓ articular shear stress
Distal COR	**lengthens/tensions deltoid** (↑ resting length + tone) · humerus lowered relative to acromion · recruits **anterior + posterior deltoid** as abductors
Grammont advantages (OB)	**medialisation of COR + lengthening deltoid + ↑ deltoid moment arm** ✗ Wrong options: lateralisation · shortening deltoid
Matched radius of curvature	glenosphere + humeral cup → **stable fixed fulcrum** for elevation; shear converted to **compression**
Motions gained	**forward elevation + abduction** ↑ · **IR, ER (at side or at 90°), adduction** not significantly improved → **latissimus dorsi transfer** for ER ✗ Wrong options: not normal
↑ deltoid moment (OB)	**place glenosphere more inferiorly** (✓) · anything moving COR medial/inferior or lengthening deltoid ✗ Wrong options: constrained poly (no effect) · short stem (no effect) · **smaller NSA (varus) shortens deltoid**

– – Active space – –



Grammont biomechanics: medial COR lengthens deltoid lever arm (R > r) and deltoid (L)

3 Presentation

Pseudoparalysis	**inability to actively elevate arm** with **free passive ROM**, **no neurological lesion**; irreparable cuff tear (± GH arthritis)
Pseudoparalysis without arthritis (TO)	loss of fulcrum when cuff fails → head translates superiorly → deltoid loses lever · **first line = rehabilitation** (recruit accessory muscles) → **RSA only after PT fails**
Anterosuperior escape	**incompetent coracoacromial arch** → humeral head escapes into subcutaneous tissue (esp. after **HA + acromioplasty**); **anterosuperior bulging** on attempted elevation
Failed HA (cuff failure)	pseudoparalysis, lateral deltoid pain, **no crepitus**, near-normal passive ROM · weakness of ER at side ✗ Wrong options: infection → passive ROM also ↓

4 Diagnosis & Classification

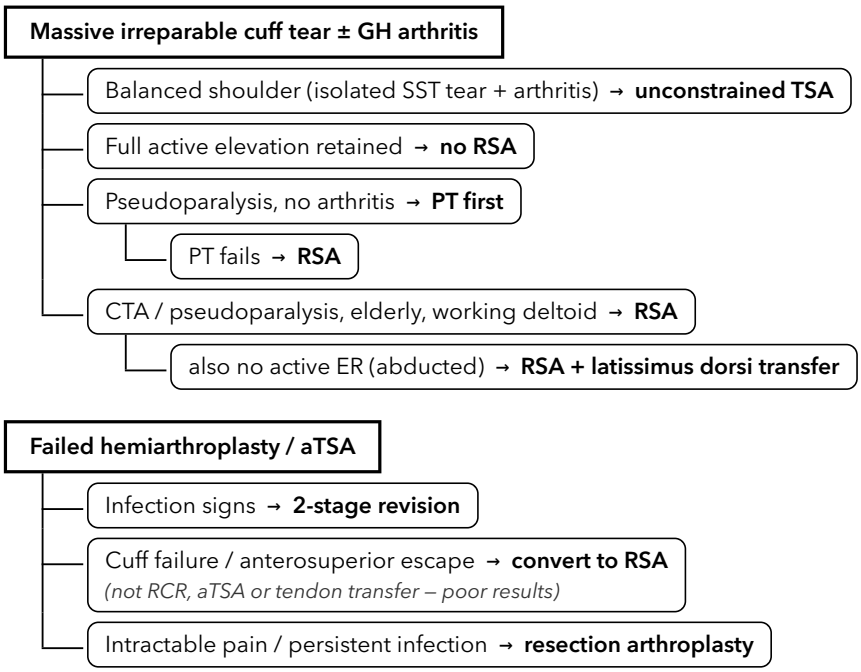
Key radiology findings

Investigation	Purpose / finding
True AP (Grashey)	Extent of arthritis, **superior migration of humerus**, acetabularisation of acromion
Axillary lateral	**Posterior glenoid wear**
Scapular-Y	
CT	**Glenoid version + bone stock** (if axillary inadequate), osteopenia
MRI	**Cuff integrity + fatty infiltration**

Anterosuperior migration on X-ray	cuff tear + **CA ligament deficiency** (OB)
Hamada ≥3	**acetabularisation of acromion** ✗ Wrong options: not linked to post-op acromial fracture
RA glenoid	**central wear, glenoid medialisation** ✗ Wrong options: posterior wear = primary OA

- - Active space - -

5 Management



Indications – cuff tear arthropathy, pseudoparalysis, failed arthroplasty or cuff repair, fracture

Indication	Detail
Cuff tear arthropathy	Severe GH arthritis + **superior escape** + massive cuff tear; **best results**
Pseudoparalysis	Irreparable cuff tear (± arthritis, TO)
Anterosuperior escape	Incompetent CA arch
Proximal humerus fracture in elderly	**3- or 4-part, age >70** · head-splitting · osteopenia (GT poor healing)
Cuff insufficiency "equivalent"	**Tuberosity nonunion/malunion** after trauma or arthroplasty
Failed arthroplasty	Prior HA or aTSA with **cuff insufficiency**; options exhausted
Failed cuff repairs	Irreparable tear after ≥2 repairs
Rheumatoid arthritis	**Only if glenoid bone stock sufficient**
Others (OB)	Proximal humeral tumours, B2 glenoid

Patient factors — **low demand** · **physiological age >70** ("exclusively >70 y", OKU 9) · **sufficient glenoid bone stock** · **working deltoid (intact axillary nerve)**

Not ideal — **40-y labourer** (glenoid loosening in young/active) · 40-y painful malunion · arthritis with **active overhead motion** · intact cuff → **aTSA** · 30-y locked posterior dislocation

Not indications (TO) — **massive cuff tear + arthritis with full active elevation** · **isolated supraspinatus tear + arthritis** → balanced → **unconstrained TSA**

Revision options – wrong (OB) — revision SST repair after 2 failed (doesn't address subscap) · **HA in subscap + SST insufficiency → instability + anterosuperior escape** · **latissimus transfer = younger, no arthritis** · cortisone/PT (doesn't restore function)

HA best results	marked glenoid bone loss, younger, **RA with forward elevation >90°**
HA failure reasons	**progressive glenoid degeneration + cuff deficiency** (✓ cuff failure; not advanced glenoid wear when pseudoparalytic without crepitus)
Instability after HA	elderly fall → **rotator cuff injury** → **convert to RSA** ✗ Wrong options: labrum
Humeral stem loosening	most commonly linked to **infection**
Sirveaux (OB)	80 shoulders: **96% no/minimal pain**, 3 revisions; ↑ forward elevation at 44 mo

Proximal humerus fracture – RSA vs alternatives

Option	Elderly 3/4-part fracture
Nonoperative	**Majority** of geriatric fractures
rTSA	**Best surgical option** in osteoporotic elderly; **better outcomes + lower reoperation than HA**; doesn't rely on tuberosity healing
HA	Outcome **depends on tuberosity healing** (cuff function)
ORIF	High demand + expected tuberosity healing; elderly → **= nonop function, ↑ complications**
IMN	2/3-part; ≈ ORIF; **high failure in osteoporosis**
PHR	**Last resort** – no bone stock; failed revisions, tumour

Fracture RSA – greater tuberosity healing

GT fixation	↑ **forward elevation + ER** (✓ OB); **anatomic GT repair** → ↑ ER, function, satisfaction vs resection/malunion · not mandatory (deltoid still powers elevation)
No effect on outcome	press-fit vs cemented stem · inlay stem · LTO (LT already fractured) · suprascapular decompression
Humeral height in fracture	calcar/tuberosity; if calcar missing → **pectoralis major insertion 5.6 cm** below top of humeral head

– – Active space – –

Contraindications – deltoid dysfunction & axillary neuropathy

Contraindication	Note
Deltoid denervation / axillary nerve dysfunction	**Absolute**; separate permanent vs temporary
Global deltoid deficiency	Absolute · **partial = relative** (reasonable results)
C5–C6 root avulsion · posterior cord palsy	→ complete deltoid palsy
Acromion deficiency	Contraindication (Orthobullets) · **relative** (OB)
Glenoid osteoporosis / insufficient bone stock	
Active infection (glenoid osteomyelitis)	
Complete brachial plexus injury · prior arthrodesis	
Severely impaired deltoid · isolated SST tear · full active elevation (TO)	

NOT contraindications
RA (sufficient glenoid stock; excellent short/midterm results) · **pseudoparalysis** (= indication) · **prior infection** (if not active) · **prior partial acromioplasty** · **absent GH joint-space narrowing** · **age >70** · **incompetent CA arch** (= indication)

Deltoid wasting + scapular dysrhythmia
not a candidate
✗ Wrong options: "candidate due to CTA" · "not candidate due to age/ massive tear"

Operative criteria

Factor	Requirement
Deltoid	**Functioning** (intact axillary nerve)
Age	**Physiological >70**, low demand
Glenoid	**Sufficient bone stock**
Cuff	Irreparable / deficient (pseudoparalysis)
Failed nonop	PT, injections
Fracture	**3/4-part, >70 y**, osteoporosis

Approach

	Deltopectoral	Anterosuperior (superolateral)
Deltoid	**Preserved**	**Anterior deltoid split/ violated**
Subscapularis	**Take-down needed**	**Preserved**
Glenoid	**Inferior pole exposed** (✓ baseplate position)	↑ exposure; **risk excess height / superior tilt**
Instability	Extensive capsular release → ↑ instability (TO: ↑ vs superolateral)	↓ (anterior stabilisers kept)
Axillary nerve	**↓ risk**	**↑ risk** (distal branches)
Notching	Not associated	**↑ notching** (insufficiently inferior baseplate)
Other	Extensile; **LD transfer possible**; poor posterior glenoid access; stiffness (immobilise for subscap)	Easier humeral axial prep + **GT fixation**

Technique

Humerus osteotomy **0–30° retroversion (typically 20°)**; more retroversion → ↑ post-op ER · LHB tenotomy/tenodesis · ream + broach as TSA
✗ Wrong options: no effect on notching

Glenoid excise labrum, **circumferential capsular release**, expose **inferior glenoid** subperiosteally · central guidewire by best bone stock · **baseplate as inferior as possible + inferior tilt** (↓ loosening + notching) · navigation

Screws **superior → coracoid base**, **inferior → scapular body** · unicortical screws to coracoid may ↓ acromial fracture; 2 screws may suffice

Glenosphere size **36 mm** females/smaller · **40 mm** large men

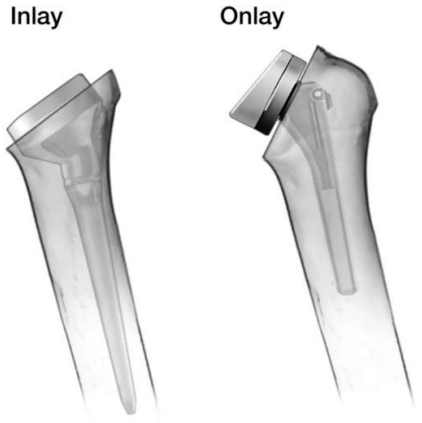
Newer prostheses (OB) **locking peripheral screws + 15° inferior tilt** · NSA **155° → 143° or 135°** → ↓ notching, ↓ instability, ↓ need for subscap repair

Humeral component design (inlay vs onlay)

	Inlay (Grammont)	Onlay
Stem	Straight, horizontal inlay tray	Curved
Bone	**More metaphyseal reaming → ↑ GT fracture risk**; ↑ bony contact	**Preserves proximal bone**
Cut		**More varus cut** → preserves GT, **minimises cuff damage**
Offset		**↑ humeral offset / lateralisation**
ROM		**↑ adduction, extension, ER**
Conversion		**Convertible to/from HA + aTSA**
Scapular spine fracture	**↓ risk** (✓ prevention, OB)	**↑ risk** (humeral lateralisation)

TO trap "onlay has more **valgus** cut and more cuff damage" = **FALSE** (onlay = **varus** cut, less damage)

– – Active space – –



Inlay vs onlay humeral components

Latissimus dorsi transfer with RSA

Indication	**loss of active elevation + ER** → RSA restores elevation, **LD restores ER**; primary or revision (✓ **failed HA, no active ER with arm abducted**)
✗ Wrong options	deltoid dysfunction (RSA contraindicated) · brachial plexus injury/ arthrodesis · CTA with **active ER >50°** (posterior cuff functioning → RSA or CTA head alone) · AVN + partial infraspinatus tear (nonop → resurfacing) · OA with 10° retroversion (TSA, ream anterior glenoid)

Rehabilitation

Early	sling; passive / active-assisted only
Sling duration	**3 wk if subscap NOT repaired** · **6 wk if repaired**; limit passive ER + active IR
Avoid	**pushing out of chair** (extension + adduction + IR) · TO: avoid these **8–12 wk**
Early ROM	not ↑ dislocation vs 6 wk immobilisation (OB)
Subscap re-tear	**anterior instability → early exploration + repair**

Outcomes

Best indication	**CTA** (failed arthroplasty = **highest complications**)
Deterioration	radiographic after **6 y**, clinical after **8 y**
Survivorship	**~90% at 10 y**

Surgeon learning curve & postoperative surveillance

Learning curve	early complications ↓ after **18–45 cases** (✓ 40; ✗ Wrong options: 10, 75, 100, 200)
Asymptomatic notching	**observe, routine follow-up with serial X-rays** (✓) · (✗ Wrong options: aspirate for C. acnes · sling for acromial fracture · revise baseplate) · IR to **L5 = excellent** for RSA

6 Complications

Overview – which is "most common"

Claim	Source
Scapular notching = most common (radiographic) complication	OB, TO, Orthobullets incidence
Dislocation = most common complication	OB (2 questions) – read the options
Instability/dislocation = most common cause of **early failure / early revision**	Orthobullets, TO, OB
Glenoid loosening = most common **mechanism of failure / long-term** complication	Orthobullets, TO
Glenoid loosening/fracture = most common after **revision RSA**	OB

TO ranking (early) **notching > instability > acromion/scapular spine fracture > infection > axillary neurapraxia**; overall complication rate **~15%**

Complications ∝ indication highest for **failed shoulder arthroplasty** (dislocation, loosening, ↓ function); low for CTA / failed RCR

Scapular notching – superior baseplate, neck-shaft angle & lateralisation

Definition erosion of **axillary border of inferior scapular neck** (scapular pillar) from **medial rim of humeral cup in adduction**; unique to RSA

Incidence **44–96%** Grammont (Orthobullets) · **51–96%** (OB) · **68%** (Lévigne, OB) · **62%** (Lévigne, 326 pts, OB); appears early (**1–2 y**), variable progression

Most common technical error **superior placement of glenoid component** (✓) → **inferior** scapular notching
✗ Wrong options: "superior notching"

↑ Notching	↓ Notching (protective)
Superior baseplate placement (upper half)	**Inferior placement / inferior overhang** – **greatest reduction**, least adverse effect
Superior tilt of glenosphere	**Inferior tilt** (more even force, impingement-free arc)
155° NSA (Grammont; medialises humeral component)	**Varus NSA (135°)** – **most important factor** (Gutiérrez)
Medialised COR / medialised glenosphere	**Lateralised offset** (bone/metal augment) – **protective** (✓ TO "all except")
Superior glenoid erosion (E2) → base plate in valgus	**Larger glenosphere** (OB)
Anterosuperior / superolateral approach	↓ **poly depth** (↑ ROM, ↓ notching)
↓ **scapular neck length** · CTA with narrow AHD · **infraspinatus FI grade 3/4** · high BMI · ↑ activity · duration	

Gutiérrez (sawbones) **NSA** = largest effect on **inferior impingement** → then glenosphere position → lateral offset → glenosphere size · **COR offset** = largest effect on **ROM** · **15° inferior tilt** → most uniform compression, least micromotion

No effect on notching humeral retroversion · stem length · horizontal humeral cut · retentive liner · cemented stem · locking screws · over-reaming

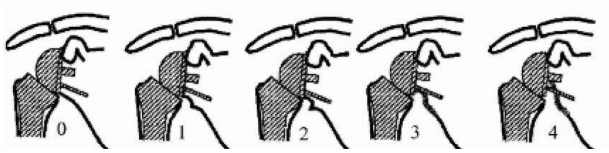
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Source conflicts — **inferior/posterior tilt**: protective (Orthobullets, TO, most OB) · **minimal effect, may ↑ notching** (OB7681; low position = biggest effect) · **glenosphere size**: larger ↓ notching (OB) · **no effect** (OB214573) – read the options

Consequences — **↓ ROM, strength, Constant scores** (✓ with preop superior erosion, Lévigne) · "not linked to detrimental outcomes" (same OB item) · particulate debris → osteolysis → **implant failure**; association with **baseplate failure** · rarely needs revision · not ↑ dislocation · not ↑ aseptic loosening

Baseplate lateralisation — ↓ notching but ↑ **baseplate loosening** (more stress)

Sirveaux grade	Extent
0	No notch
1	**Limited to scapular pillar**
2	**Contact with inferior screw** of baseplate
3	**Beyond inferior screw**
4	**Under baseplate, approaching central peg**



Sirveaux grades 0–4 of scapular notching

Instability/dislocation – risk factors & positions to avoid

Incidence — **2–3.4%** (Orthobullets) · **3.4% avg, 2–14%** (TO) · **2–4%** (OB) · **2–3%** (OB) · **1.5–31%** (OB) – read the options

Direction (TO) — **exclusively anterior / anterolateral**

Position — **extension + adduction + internal rotation** (pushing out of chair, tucking shirt) · TO keys: **adduction + extension** and **adduction + IR** · not abduction + ER (native shoulder; deltoid "wrap effect" stabilises RSA)

✗ Position distractors flexion + abduction (RCR) · flexion + IR + adduction (**posterior labral repair**) · ER + abduction (**anterior labral repair**) · IR at 90° abduction (face/mouth hygiene)

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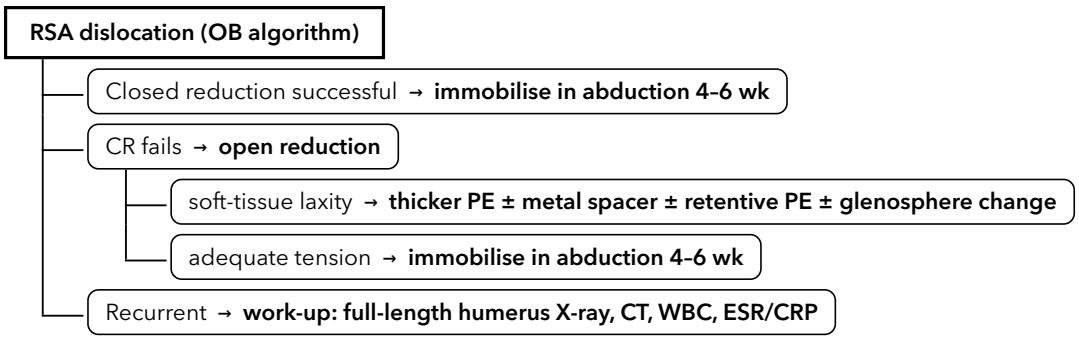
Risk factor	Note
Irreparable / inadequate subscapularis	**Strongest** (Orthobullets); ↑ with medialised prosthesis (OB) · **less of an issue with newer implants** (OB4801) – read the options
Proximal humeral bone loss / loss of tuberosities	
Failed prior arthroplasty / revision	**Most significant** in revision vignette
Proximal humeral nonunion / malunion	Tuberosity failure
Fixed preop GH dislocation	
Male sex · BMI >30 / obesity	(✗ Wrong options: female)
Previous shoulder surgery (open, extensive capsulotomy)	Not arthroscopic / mini-open biceps tenodesis
Axillary nerve injury · Grammont-style implant	
Deltopectoral (vs superolateral) · subscap **FI > Goutallier 3** (TO)	
Inadequate humeral length (↓ compression) → shallow socket	**Main biomechanical** parameters (TO)
✗ Wrong option: **rheumatoid arthritis**	Not a risk factor

Stability determinants (TO) soft-tissue balance (humeral length), glenosphere size, humeral joint-line inclination, humeral version, metaglene position

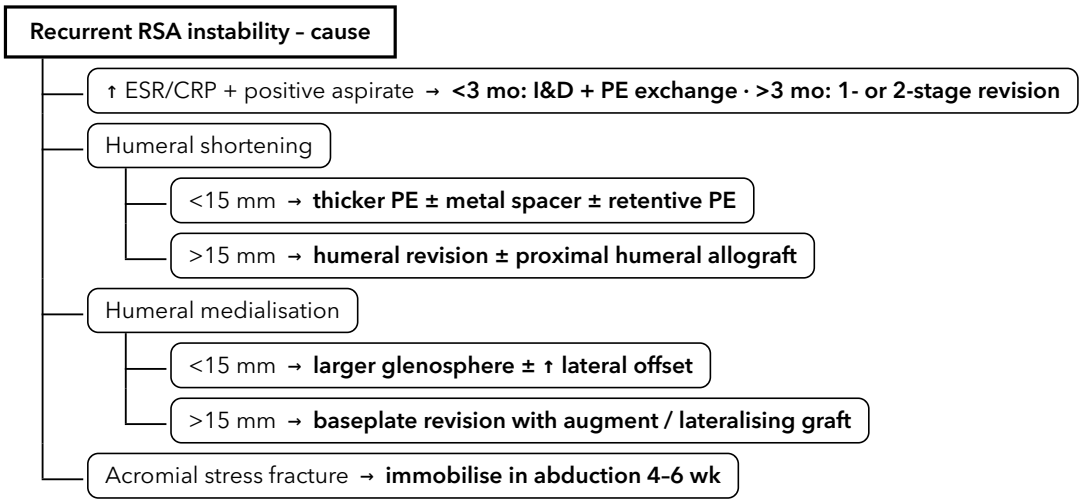
Prevention **lateralise baseplate/glenosphere** (✓) · ↑ glenosphere size · **eccentric baseplate** · lateralise/distalise humeral component · **constrained implant**
✗ Wrong options: valgus stem (medialises humerus, ↓ deltoid wrapping) · subscap tenotomy without repair

Treatment **immediate closed reduction + sling** (OB) · CR can be tried but **redislocation usual → open reduction + correct the surgical error** (TO) – read the options

✗ Dislocation treatment distractors EMG (deltoid weakness) · I&D + poly exchange (**acute infection**) · I&D + antibiotic spacer (**chronic infection**) · ORIF (periprosthetic fracture)



– – Active space – –



Acromial & scapular spine stress fractures

Incidence — **4%** (Orthobullets, OB) · **4–5%** / **~5%** (OB) · **4–8%** (OB) · **10.2%** with **lateralised glenosphere** (Levy, TO)

Mechanism — ↑ deltoid tension/lever arm → stress through acromion (deltoid origin) → **stress/fatigue fracture**; propagates from **superior or posterior screw** at base of acromion/scapular spine; **2–48 mo** post-op (TO) · OB: "most often occur during implantation" – read the options

✓ **Glenoid** — **universally stable glenoid implant** (TO)
✗ Wrong options: "fall on abducted shoulder"
✗ Wrong options: loose glenoid

Risk factor	Source
Female sex (up to 100% of series) · age 70–80	Orthobullets, OB, TO
Osteoporosis	✓ OB
Medialised preop COR (✓ OB) → **RA** (central wear)	Orthobullets, OB
Superior baseplate screw (✓ TO) · posterior screw	TO, OB
Lateralised glenosphere / glenoid · superior glenoid placement	TO, OB
Onlay humeral component (humeral lateralisation)	OB
↑ NSA (larger deltoid moment arm)	OB
Acromial thinning (end-stage CTA acetabularisation) · inflammatory arthritis · revision · massive cuff tear · prior surgery	TO, OB

Deltoid length conflict — **less change in deltoid length post-op** = risk (OB215540) · **distalisation / over-tensioning** = risk; superior baseplate ↓ risk (OB217851, OB218963) – read the options

Not risk factors — male · **Hamada ≥3** · RSA for fracture · age <50 · **inferior** baseplate screw

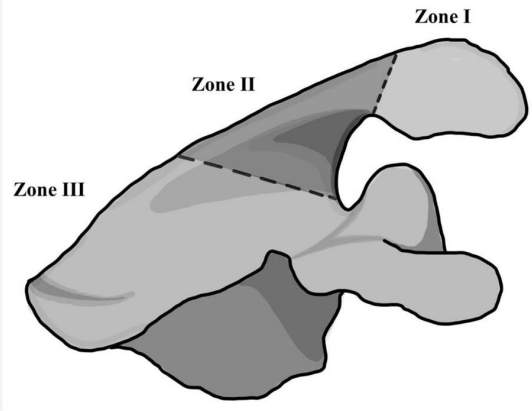
Presentation — acute **superior shoulder pain** + loss of function, often **atraumatic** or after fall; **missed on X-ray in up to 20% → CT**

Outcome — **inferior to RSA without fracture**; **elevation** most limited · better than preop · chronic pain **not** usual · **↑ revision**
✗ Wrong options: ER

Treatment — **nonop: sling (abduction pillow) 6 wk** · nonop union only **40–50%** · **ORIF / tension band** → ↑ union · TO: types I–II nonop (late ORIF if symptomatic nonunion); **type III → ORIF** if displaced, loss of function, nonunion
✗ Wrong options: nonunion <5%

Rare sequel — new instability (glenosphere orientation change, ↓ deltoid tension)

Levy type (Orthobullets)	Location	Periprosthetic scapular fracture (TO)
1	Acromion **anterior** to posterolateral corner	**Anterior acromion**
2	**Mid acromion** (base → posterolateral corner)	Acromion **posterior to AC joint, anterior to spine**
3	**Acromial base**	**Scapular spine**



Levy zones of acromial fracture

Periprosthetic fracture after a fall

Fall 8 mo post-RSA → acromial fracture — ✓ **osteoporosis is a risk factor**
✗ Wrong options: incidence 15% (≈5%) · chronic pain likely · nonunion <5% nonop · equal revision rate

Infection & complications after revision

Incidence — **1–2%** deep infection; large **subacromial dead space**

Organisms — **C. acnes + staphylococci** · C. acnes → **vancomycin + clindamycin**

Risk factors — **younger age (<65) + male = greatest** (✓ 45-y male RSA) · **primary reverse > aTSA** · **arthroplasty for trauma** · failed arthroplasty
✗ Wrong options: BMI/obesity, race, ASA, smoking, RA, diabetes

Treatment — **2-stage revision = gold standard** · acute → I&D + poly exchange · chronic → I&D + antibiotic spacer

Revision RSA — complication rate **11–36%** after failed arthroplasty; **glenoid loosening/ fracture ↑ (~25% at 5 y)** (✓) – high metal : small glenoid bone ratio; **periprosthetic glenoid** fractures commonest · also dislocation/recurrent instability
✗ Wrong options: humerus

Glenoid loosening treatment — **staged**: fill glenoid with **autograft** + **hemiarthroplasty** → await incorporation → new glenosphere

– – Active space – –

Nerve injury during glenoid preparation

Most commonly injured	**axillary nerve** (✓ label B) ✗ Wrong options: radial (A), musculocutaneous (C), median (D), ulnar (E)
Neurapraxia	**0.5–1%**; risks **anterosuperior approach** (deltoid retraction/levering) + **humeral lengthening** · direct laceration during glenoid implantation · **usually transient → observe**
Lädermann	subclinical EMG changes (mainly axillary) at 3.6 wk; **8 of 9 resolved <6 mo**; arm lengthening implicated

Baseplate failure – central peg error

Scenario	RSA + **iliac crest graft behind baseplate** → early baseplate migration → **central peg error** (peg in graft, not native glenoid; peg too short) (✓) ✗ Wrong options: tilt · dissociation · small glenosphere (larger = instability option)
Glenosphere dissociation	rare; intra-op failure to engage glenosphere on baseplate

X vs Y

	Anatomic TSA	**Hemiarthroplasty**	**Reverse TSA**
Cuff	**Must be functional**	Tuberosity healing needed	**Not needed** (deltoid)
Anterosuperior escape	**Contraindicated**	Performs poorly	✓ **Treatment**
Fracture elderly		Old standard; ↑ revision	**Best**
Infection risk	Lower		**Higher** (dead space)
Scapular spine fracture			**Highest (~5%)**

	Grammont 155°	**135° (varus) / lateralised**
Humeral component	Medialised	↑ lateral offset
Notching	**↑**	**↓**
Instability	↑ (↓ deltoid wrap)	↓
Deltoid length		Varus shortens deltoid (OB)
Acromial stress		Lateralisation ↑ (TO, OB)

Technical change	**Notching**	**Dislocation**	**Acromial fracture**	**Baseplate loosening**
Inferior baseplate	**↓**		↑ (distalisation, OB)	not ↑
Superior baseplate	**↑**	no direct effect	↓ (OB)	
Lateralised glenosphere	**↓**	**↓**	**↑**	**↑**
Larger glenosphere	↓	↓		
Superior screw			**↑**	

How it's asked

"Which is NOT an indication for RTSA?"	→ **massive cuff tear + arthritis with full active elevation** (TO) ✗ Wrong options: CTA · pseudoparalysis without arthritis · CTA + pseudoparalysis
"Most appropriate RSA candidate?"	→ **75-y man, painful arthritis + massive irreparable cuff tear** ✗ Wrong options: 40-y labourer · active overhead motion · failed HA from glenoid wear
"76-y, high-riding head, FF 80°, normal belly press?"	→ **RSA** ✗ Wrong options: TSA · RCR · arthrodesis
"71-y massive tear + arthritis + FF 40° vs 45-y failed RCRs vs intact cuff?"	→ **71-y, FF 40°**
"75-y, 2 failed RCRs, FE 60°, +lift-off, stage 4 FI?"	→ **reverse TSA** ✗ Wrong options: HA → anterosuperior escape · LD transfer
"Failed aTSA 3 y, FE 80°, deltoid intact — best function?"	→ **revision to RSA** ✗ Wrong options: tendon transfer · CTA hemi
"HA + acromioplasty, can't elevate >40°, anterosuperior bulge?"	→ **convert to RSA** ✗ Wrong options: TSA contraindicated
"HA 6 y, cuff atrophy, no infection?"	→ **convert to RSA** ✗ Wrong options: 2-stage · resection · aTSA · RCR
"HA dislocation after fall, can't elevate, anterosuperior escape?"	→ **rotator cuff; convert to RSA** ✗ Wrong options: labrum
"76-y osteoporotic 3-part fracture, low demand, surgery chosen?"	→ **rTSA** ✗ Wrong options: ORIF · IMN · HA · PHR
"Pseudoparalytic HA, no crepitus — reason for failure?"	→ **rotator cuff failure** ✗ Wrong options: glenoid wear · infection · acromial fracture
"Deltoid denervation is a contraindication to?"	→ **reverse TSA** ✗ Wrong options: arthrodesis · ACDF
"Deltoid wasting, can't flex/ER — RSA candidate?"	→ **no — deltoid dysfunction**
"Contraindication to RSA?"	→ **axillary neuropathy** ✗ Wrong options: prior infection · pseudoparalysis · prior acromioplasty · no joint-space narrowing

– – Active space – –

"NOT a contraindication to RSA?"	→ **rheumatoid arthritis** ✗ Wrong options: C5–6 avulsion · acromial deficiency · glenoid osteomyelitis · posterior cord palsy
"Malunion + arthritis + IR lag — contraindication to best surgery?"	→ **global deltoid deficiency** ✗ Wrong options: RA with bone stock · age >70 · incompetent CA arch
"Biomechanical advantage of RSA over standard arthroplasty / hemiarthroplasty?"	→ **COR more medial** (TO) ✗ Wrong options: superior · lateral · ↑ lateral offset
"Early reverse designs failed — why?"	→ **COR too lateral** → glenoid torque + loosening
"Grammont advantages?"	→ **medial COR, lengthened deltoid, ↑ deltoid moment arm**
"Directional COR change improving kinematics?"	→ **medial**
"Technique increasing deltoid moment?"	→ **glenosphere more inferior** ✗ Wrong options: smaller NSA · constrained poly · short stem
"Motion gaining most besides abduction?"	→ **forward elevation** ✗ Wrong options: ER · IR · adduction
"Scapular notching risk — all EXCEPT?"	→ **lateralised offset** (TO) ✗ Wrong options: superior placement · superior tilt · superior glenoid erosion
"Technical error causing scapular notching?"	→ **superior glenoid component placement**
"Superior baseplate → which complication?"	→ **inferior scapular notching**
"Prosthetic factor most reducing notching (Grammont)?"	→ **inferior positioning of glenoid component** ✗ Wrong options: inferior/posterior tilt · cemented stem · locking screws
"Notching 2 y post-op — implicated factor + sequela?"	→ **preop superior glenoid erosion; ↓ ROM, strength, Constant**
"Superior tilt on X-ray — most at risk of?"	→ **scapular notching** ✗ Wrong options: acromion fracture · baseplate failure · dissociation
"Modification minimising notching?"	→ **inferior baseplate** ✗ Wrong options: smaller glenosphere · retentive liner · horizontal cut
"Notching at 4 y — what happened at surgery?"	→ **glenosphere too superior** ✗ Wrong options: too much inferior tilt · shallow poly · over-lateralisation
"Modification MOST increasing notching risk?"	→ **155° NSA humeral component** ✗ Wrong options: inferior baseplate · larger glenosphere · lateral offset · retroversion

"155° prosthesis, baseplate upper half — future concern?"	→ **scapular notching**
"Prevent most common complication intra-op (OB = dislocation)?"	→ **lateralised baseplate** ✗ Wrong options: 4 bicortical screws · ↑ stem angle · drain · subscap tenotomy
"Baseplate in upper 50% — most at risk?"	→ **scapular notching** ✗ Wrong options: dislocation · baseplate loosening
"Step preventing notching seen at 2 y?"	→ **varus humeral NSA** ✗ Wrong options: smaller glenosphere · superior glenosphere · medialising
"Most common instability position in RTSA?"	→ **adduction + extension** (TO)
"Most common mechanism of RTSA dislocation?"	→ **adduction + internal rotation** (TO)
"Position to avoid after RSA?"	→ **extension, internal rotation, adduction** (pushing out of chair) ✗ Wrong options: ER + abduction · flexion + IR + adduction
"Dislocation risk factors EXCEPT?"	→ **rheumatoid arthritis** ✗ Wrong options: malunion · bone loss · failed arthroplasty · fixed dislocation
"Medialised reverse in pseudoparalysis — true?"	→ **instability ↑ with irreparable subscapularis** ✗ Wrong options: contraindicated in pseudoparalysis · LD transfer allows deltoid dysfunction · superior + lateral COR
"Revision RSA for failed aTSA — most common cause of early revision?"	→ **instability** ✗ Wrong options: notching · infection · acromion fracture
"Dislocation 2 wk after revision lateralised RSA — biggest risk?"	→ **revision arthroplasty** ✗ Wrong options: early ROM · female · mini-open tenodesis
"Management of most common post-op complication after RTSA?"	→ **immediate closed reduction + sling** ✗ Wrong options: EMG · I&D + poly exchange · spacer
"True about scapular fracture after RSA?"	→ **superior baseplate screw is a risk factor** (TO) ✗ Wrong options: loose glenoid · fall on abducted arm · medialised glenosphere
"Acromial fracture after fall at 3 mo, nonop — long-term counselling?"	→ **inferior to RSA without fracture** ✗ Wrong options: ER most limited · persistent pain likely
"Which procedure has highest risk of scapular spine fracture?"	→ **reverse TSA** (~5%)

– – Active space – –

"Onlay RSA, atraumatic scapular spine fracture at 1 y — prevention?"	→ **inlay humeral implant** ✗ Wrong options: ↑ NSA · avoid inferior screw · thinner liner
"RA patient, RSA — elevated risk of?"	→ **acromial stress fracture** ✗ Wrong options: dislocation · glenoid loosening · poly wear
"Atraumatic acromial fracture at 6 mo — risk factor?"	→ **medialised preop COR** ✗ Wrong options: male · Hamada ≥3 · RSA for fracture · age <50
"Acromial fracture after fall — true?"	→ **osteoporosis is a risk factor** ✗ Wrong options: 15% incidence · nonunion <5%
"Complications most frequent when RSA done for?"	→ **failed shoulder arthroplasty**
"After revision RSA — significantly ↑ risk of?"	→ **glenoid component loosening** (~25% at 5 y)
"Most likely to get infection after shoulder arthroplasty?"	→ **45-y male after reverse TSA** ✗ Wrong options: diabetic · obese · RA
"Nerve injured during glenoid preparation?"	→ **axillary nerve (B)**
"Baseplate on iliac crest graft migrates at 2 wk?"	→ **central peg error**
"RSA + latissimus dorsi transfer — patient?"	→ **failed HA, no active ER with arm abducted** ✗ Wrong options: CTA with ER >50° · brachial plexus injury
"Fracture RSA — improves FE + ER?"	→ **greater tuberosity fixation** ✗ Wrong options: press-fit · inlay · LTO
"Surgeon learning curve for RSA?"	→ **18–45 cases** (✓ curve at 40)
"Asymptomatic notching at 9 mo?"	→ **routine 1-year follow-up with X-rays** ✗ Wrong options: aspiration · revision
"Humeral prosthesis in RSA — all true EXCEPT?"	→ **onlay has more valgus cut + more cuff damage** (TO) (✓ inlay ↑ GT fracture, ✓ onlay ↑ ER, ✓ onlay convertible)

– – Active space – –

REVISION TOTAL ELBOW ARTHROPLASTY

– Shoulder & Elbow · 39 / 60 –

1 Basics

Epidemiology

Lifetime revision rate	**~13%** · much **higher than TKA/THA**
TEA survival	**92% (5 y)**, **81% (10 y)**, **71% (15 y)**, **61% (20 y)**
Shift in indication	originally **RA**; with anti-rheumatics (DMARDs) → now mostly **post-traumatic** (distal humerus fractures)
Risk factors for TEA failure	**smoking** · **medical comorbidities** (RA, hypothyroidism, uncontrolled DM, iron-deficiency anaemia, obesity) · **non-compliance with activity restrictions** · **non-constrained** (dislocation) or **highly constrained** (loosening) systems

Failure modes & revision rates by indication

Mode of failure	Rate	Key facts
Aseptic loosening	**15%**	↑ with **cementation + highly constrained** designs; **anterior impingement** of coronoid into anterior humeral flange in max flexion · **most common in OA**
Component failure	**15%**	Mostly **bushing wear** (linked/semiconstrained); **younger** patients or **pre-existing deformity**
Infection	**~8% → ~3%** since DMARDs (Orthobullets) · **3–5%** (OB) – read the options	**Higher than other joint arthroplasties**
Periprosthetic fracture	**5%**	Intra-op (preparation, reaming, implantation) · post-op (non-compliance, ground-level falls)
Instability	–	**Most common failure of unconstrained (unlinked) TEA** – late ligament insufficiency or malposition

Revision rate by indication	**traumatic < inflammatory arthritis < OA** (lowest → highest)
OA patients	**aseptic loosening = most common complication** (higher demand) ✗ Wrong options: infection · fracture
RA patients	↑ **tendon + ligament insufficiency**
Design	unconstrained/unlinked · **semiconstrained/linked** (e.g. **Coonrad-Morrey**) = **best longevity + function** · constrained
Polyethylene / bushing wear	linked + semiconstrained · not the main failure of unconstrained · **component fracture uncommon**
Gill & Morrey (OB)	RA, Coonrad-Morrey, 10–15 y: **97%** no/mild pain · arc **28–131°** · **92.4% survival** · 14% reoperation · 86% good/excellent

– – Active space – –



Aseptic loosening of TEA

Infection – organisms, risk factors & staged revision

Organism	Fact
S. aureus (GPC clusters, catalase +, **coagulase +**)	**Most common (~40%)**
S. epidermidis (GPC clusters, catalase +, **coagulase –**)	**2nd (30–35%)**; **worst outcomes / lowest reimplantation** – **biofilm / encapsulation → most difficult to eradicate**
Cutibacterium (P.) acnes (anaerobic aerotolerant GP bacillus)	**Most common in TSA**, not common in TEA
GPC chains, α-haemolytic, optochin-sensitive	S. pneumoniae (pneumonia)
GPC chains, β-haemolytic, bacitracin-sensitive	**Group A strep** → necrotising soft-tissue infection

Infection risk factors **RA on steroids** · **hypothyroidism** (independent) · **diabetes** · **thin soft-tissue envelope** · **prior elbow infection** · **prior elbow surgery** · **psychiatric illness**

Not risk factors female gender · Nordic heritage · **isolated +RF with –anti-CCP** · HbA1c <6.0%
✗ Wrong options: RA

Genes & lab markers

Test	Value / meaning
TSH ↑ + free T4 ↓	**Hypothyroidism** = independent PJI risk (TSH normal 0.5–5.0; free T4 normal 0.7–1.9)
ESR, CRP, WBC	Usually ↑ in PJI (WBC may be normal)
IL-6, alpha-defensin	**Not studied** in elbow PJI
Synovial WBC	**No accepted cut-off** for elbow
Aspiration culture	**+ = chronic infection**; **very high PPV, very low NPV → negative aspirate does not rule out**
Intra-op histology	**Most effective** diagnostic test; **specificity 93%**, **NPV 90%**, **sensitivity 51%**
Intra-op cultures	**Negative in 10%** of infected TEA; chronic **~15%** vs acute **3%**

No definitive test diagnosis needs **high clinical suspicion**

– – Active space – –

3 Presentation

Symptoms **squeaking / crepitus** (bushing wear) · pain · swelling · stiffness

Red flags for infection **pain at night or rest** · **progressive loss of motion without cause** · **early unexplained loosening** · erythema, **sinus tract**, persistent wound drainage

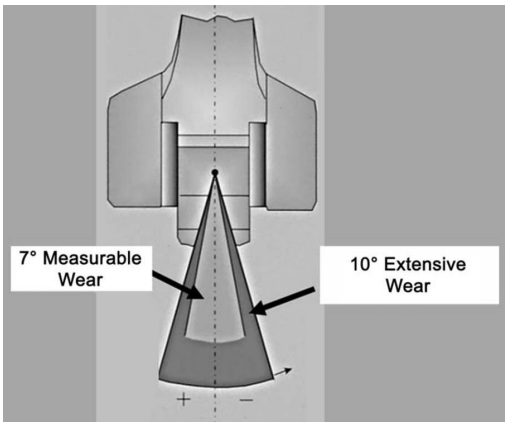
Exam diffuse tenderness, swelling, **deformity** (significant bushing wear), painful limited ROM

4 Diagnosis & Classification

Key radiology findings

X-ray **AP + lateral of humerus, elbow, forearm** → loosening, periprosthetic fracture, **bushing wear**

Bushing wear angle angle between ulnar and humeral components · **>10° = mild–moderate wear** (Orthobullets text) · figure labels **7° measurable, 10° extensive** — read the options



Bushing wear angle

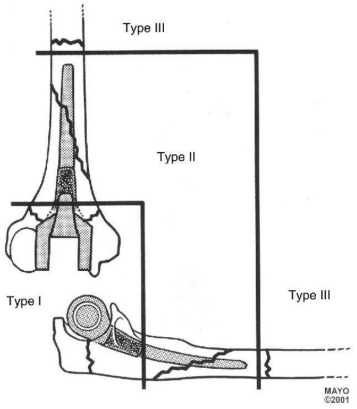
CT periprosthetic **osteolysis / loosening**; **cement mantle intact or broken** in fracture

MRI **metal subtraction + contrast** → abscess / soft-tissue infection

– – Active space – –

Mayo (O'Driscoll & Morrey) classification of periprosthetic fracture

Type	Location / features	Treatment
I	**Periarticular** – humeral condyle or olecranon; osteolysis + muscle distraction	**Undisplaced** → immobilisation / soft-tissue repair (**fibrous union OK**) · **Displaced** → ORIF with heavy non-absorbable sutures or **tension band** · intact extensor + well-fixed stem → **fragment excision**
II1	Along stem, **well-fixed implant**	**ORIF + component retention ± strut allograft** (unicortical screws + cerclage cables)
II2	Along stem, **loose implant, good bone stock**	**Revision with long-stem** ± strut allograft + impaction grafting (± locking plate / cerclage)
II3	Along stem, **loose implant, severe bone loss**	Revision with **extensive allograft**; often **resection arthroplasty**
III	**Distal to prosthesis**	Well-fixed → **humerus: immobilise**, **ulna: ORIF (plate)** · loose → treat as **II2** · CT/X-ray to confirm implant + cement mantle



Mayo classification of periprosthetic TEA fracture

5 Management

Failed TEA

- Bushing wear, stable implants, no infection → **I&D + bushing exchange (retain)**
- Acute PJI (<90 d) → **I&D + component retention (63%)**
- Chronic PJI → **explant + 2-stage revision (90%)**
  - unfit / severe bone loss / poor soft tissue → **resection arthroplasty**
- Aseptic loosening / Mayo II2 → **single-stage long-stem revision**

Operative criteria

Procedure	Indication	Outcome
Nonop (immobilise 2–4 wk → brace)	**Type I** with stable prosthesis · **Type III humerus** with stable implant (→ Sarmiento) · unfit for surgery	–
I&D + bushing exchange, retain	**Squeaking/crepitus**, stable implants, no infection · **acute PJI <90 d**	Bushing exchange **75% good at 5 y** · acute PJI **63%** eradicated · chronic only **31%**
ORIF, retain ± excision ± strut	**Type I olecranon** · **II1** · **III ulna**	**80–90%** no complications
Single-stage revision ± ORIF/ allograft	**Type II2** · **aseptic loosening**	Chronic infection only **66%** eradicated
Explant + 2-stage revision	**Infected TEA** · Mayo II2	Chronic infection **90%** eradicated
Resection arthroplasty	**Salvage**: resistant PJI, can't tolerate multiple surgeries, **severe bone loss**, poor soft tissue · **II3** not reconstructable	**71%** eradication; **lowest MEPS** function

Chronic P. acnes (6 mo), well-fixed, good bone stock, healthy
2-stage revision
✗ Wrong options: IV antibiotics alone · arthroscopic I&D · I&D + poly exchange · single stage

Technique pearls

Bushing exchange
prior approach · ulnar nerve **explore/decompress if symptomatic**, else identify + protect · **uncouple** components → inspect integrity, loosening, rotation → exchange bushing

ORIF
humerus: **posterior paratricipital** muscle-sparing · ulna: posterior + subcutaneous border extension · **radial nerve crosses lateral IM septum 14 cm proximal** to lateral epicondyle · Type I with extensor disruption → **tension band, K-wires dorsal to implant** engaging cement + figure-of-8

Single-stage revision
remove all loose implant + cement · **long stem in all**, **2 cortical widths beyond fracture** · **cement stem BEFORE struts** (extravasation under strut → no incorporation, nerve irritation) · humerus: **longer posterior + shorter anterior strut** · ulna: dorsal strut under **anconeus** or **FCU**
✗ Wrong options: flexion impingement

2-stage (infection)
posterior triceps-sparing · humeral removal: burr + flexible osteotomes, slap hammer; stuck → **posterior humeral cortical split** · well-fixed ulna → **extended olecranon osteotomy** (repair with cables) · **antibiotic spacers/ beads** in both canals; no cement in subcutaneous tissue · ± static ex-fix · **≥6 wk IV culture-specific antibiotics** → reimplant when markers normalise, **long stem + antibiotic cement**

2-stage (fracture)
stage I: union with **ICBG + plate** · stage II: **longer stem + impaction graft**

Resection arthroplasty
thick flaps; nerves in unpredictable places; remove all cement; stabilise with heavy sutures/wires if needed

– – Active space – –



Resection arthroplasty

6 Complications

Complication	Rate / risk factors	Treatment
Persistent deep infection	**~10% after 2-stage**, **~30% after single stage** · **polymicrobial**, **S. epidermidis**, **RA on steroids**, **extensive bone loss**	**Resection arthroplasty**
Ulnar nerve palsy	**Cement extravasation**, **not visualising nerve**	–
Radial nerve palsy	**Cement extravasation**, **long plates** impinging	–
Symptomatic hardware	**ORIF of periprosthetic ulna** fractures	Remove after union

Prognosis depends on **cause of failure**, **comorbidities**, remaining **bone stock**

X vs Y

	Unconstrained (unlinked)	Semiconstrained (linked)	Highly constrained
Main failure	**Instability**	**Bushing / poly wear**, loosening	**Aseptic loosening**
Longevity	–	**Best** (Coonrad-Morrey)	–

	Single-stage revision	Two-stage revision	Resection arthroplasty
Chronic infection eradication	**66%**	**90%**	**71%**
Persistent infection	~30%	~10%	–

How it's asked

"Chronic P. acnes TEA, 6 mo, well-fixed, good bone, healthy – treatment?" → **two-stage revision arthroplasty**

"RA, cemented semiconstrained TEA 8 y, ↑ ESR/CRP, lysis – hardest organism to eradicate?" → **Staphylococcus epidermidis**
✗ Wrong options: S. viridans, E. coli, Vibrio, C. difficile

"History item most linked to TEA PJI (TSH 10.6, free T4 0.23)?"	→ **TSH and free T4 levels** = hypothyroidism ✗ Wrong options: female · Nordic · HbA1c 5.9% · +RF with −anti-CCP
"Draining TEA at 12 wk — organism with worst reimplantation prognosis?"	→ **gram-positive cocci in clusters, catalase-positive, coagulase-negative** = S. epidermidis ✗ Wrong options: coagulase + = S. aureus, most common
"Most common failure of unconstrained TEA?"	→ **instability** ✗ Wrong options: poly/bushing wear · humeral loosening · component fracture
"TEA revision rates lowest → highest?"	→ **traumatic < inflammatory arthritis < osteoarthritis**
"TEA for OA — most common complication (figure)?"	→ **aseptic loosening (Figure A)** ✗ Wrong options: fracture · infection · normal X-ray
"Most effective test for TEA infection?"	→ **intra-op histology** ✗ Wrong options: negative aspirate excludes

– – Active space – –

ROTATOR CUFF ARTHROPATHY

– Shoulder & Elbow · 40 / 60 –

1 Basics

Epidemiology

Demographics	♀ **>** ♂ · **7th decade** · **dominant** shoulder
Risk factors	**rotator cuff tear** · **RA** · **crystal-induced arthropathy** · **haemorrhagic shoulder** (haemophilia, elderly on anticoagulants)
History	term **cuff tear arthropathy** coined by **Neer (1983)**

Defining features

3 common features (all patients)	**non-functioning rotator cuff** · **GH degenerative changes** · **superior migration of humeral head**
Full picture	+ **subchondral osteoporosis** · **humeral head collapse** · **femoralisation** of head · **acetabularisation** of coracoacromial arch
Why diagnosis matters	definitive treatment depends on whether **cuff functions**

2 Mechanism

Pathogenesis – loss of force couples/concavity compression & anterosuperior escape

Concavity compression	cuff compresses head into glenoid → concentric rotation → **primary mechanism centring head against upward pull of deltoid**
Force coupling	muscle balance centres the net force vector on glenoid → stable **fulcrum** for deltoid + periscapular muscles · loss → **instability, eccentric wear, CTA**
Supraspinatus + infraspinatus	depress head caudally; dynamic stabilisers
Pathway	cuff deficiency → **loss of dynamic compression** → abnormal joint forces → **superior migration** → **mechanical cartilage wear** → **acetabularisation of acromion**

✗ Concept trap	Truth
Absent supraspinatus → centring impossible	**Subscapularis + infraspinatus** can still centre head
Cuff repair restores stability after concavity lost	**False** – cannot fully restore once **effective glenoid concavity lost**
Intact biceps prevents migration	**False** – biceps is **not a head depressor**
Cuff vector optimally depresses head	**False** – not optimally oriented

Mechanical factors	loss of concavity (compression effect), ↓ ROM + function, head migration, instability ± recurrent dislocation

Nutritional factors **loss of watertight joint space** → ↓ joint fluid → **cartilage atrophy** (↓ water + GAG) + **subchondral collapse** (disuse osteoporosis)

Crystal-induced synovial degradation proteins destroy cuff + cartilage → end-stage **calcium phosphate crystal** deposits

Pseudoparalysis mechanism

Pseudoparalysis inability to **actively forward flex/elevate** with **full passive ROM** · cuff **can't DEPRESS** head → deltoid pulls humerus up → **anterosuperior subluxation / escape** · deltoid lever arm ↓
✗ Wrong options: can't elevate

Injection test **no change** in active elevation after GH local anaesthetic → **pain is NOT the cause** → mechanical (RSA)

✗ Pseudoparalysis distractor	Why wrong
Posteroinferior / inferior subluxation	Deltoid pulls **anterosuperiorly**
Deltoid atrophy	Not the classic chronic-tear mechanism
Suprascapular nerve palsy	Weak supra + infraspinatus but **cuff still checkreins** head → no pseudoparalysis
C5 radiculopathy	Weak deltoid + cuff, **paraesthesias**, no pseudoparalysis
Fibroblastic capsular thickening	**Adhesive capsulitis** (passive ROM lost)

✗ Pathogenesis distractor	Diagnosis
Haemosiderin, synovial hyperplasia, cytokines	**Haemophilic arthropathy** (can cause massive cuff tears)
GPC proliferation, PMNs	**Septic arthritis** (P. acnes = **gram + bacillus**, may have **normal markers**)
Microcirculation occlusion, osteocyte death	**AVN** (subchondral lucency/collapse, no superior migration)
T-cell autoimmune	**RA** (**central** glenoid erosion, periarticular osteopenia)

3 Presentation

Symptoms pain incl. **night pain** · subjective weakness + stiffness

Inspection **supra/infraspinatus atrophy** · **anterosuperior escape** (head prominence anteriorly on elevation) · **subcutaneous effusion** (capsular fluid leak, "geyser")

ROM ↓ active ± passive · **crepitus** GH/subacromial · **pseudoparalysis**

Acute-on-chronic low-energy injury → **acute decompensation** of compensated chronic tear → pain + disability

Test	Finding	Structure
ER lag sign	Can't hold passive ER, elbow 90°	**Massive infraspinatus** tear
Hornblower	Can't ER / hold ER at **90° abduction + 90° elbow flexion**	**Teres minor** dysfunction
Belly press	Normal → subscapularis intact	Subscapularis

4 Diagnosis & Classification

Key radiology findings

Views AP, axillary, **Grashey (true AP)**

X-ray **↓ acromiohumeral interval** · **acromial acetabularisation** · **femoralisation** of head · **asymmetric superior glenoid wear** · **lack of osteophytes** · **osteopenia** · **"snowcap" sign** (subchondral sclerosis) · **anterosuperior escape**



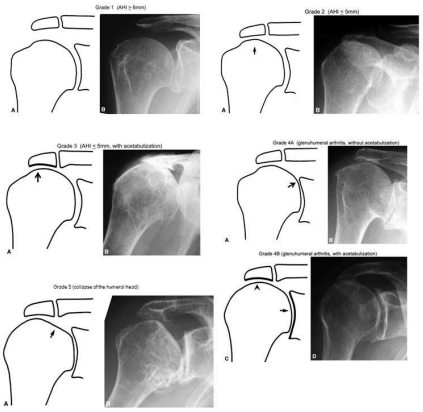
Cuff tear arthropathy, high-riding head

MRI irreparable tear: **massive fatty infiltration** + **severe retraction** · **not needed** if X-ray already shows anterosuperior escape

Labs normal WBC/ESR/CRP (vs septic, RA)

Hamada classification

Grade	Findings
1	**AHI ≥6 mm**
2	**AHI ≤5 mm**
3	AHI ≤5 mm + **acetabularisation of acromion**
4A	**GH arthritis without acetabularisation** (AHI <7 mm)
4B	**GH arthritis with acetabularisation** (AHI ≤5 mm)
5	**Humeral head collapse**

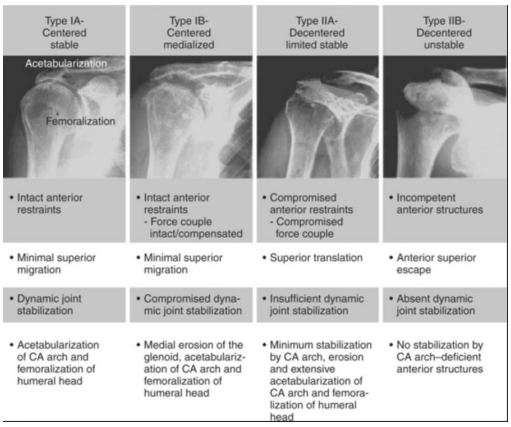


Hamada classification

Hamada versions stage 2+ AHI **≤5 mm**, stage 5 **head collapse** (Orthobullets) · AHI **<7 mm**, stage 5 **osteonecrosis** (TO) — read the options

Seebauer classification

Type	Pattern	Features
IA	**Centred, stable**	Intact anterior restraints; minimal superior migration; dynamic stabilisation; **femoralisation + acetabularisation**
IB	**Centred, medialised**	Intact/compensated anterior restraints; compromised stabilisation; **medial glenoid erosion**
IIA	**Decentred, limited stability**	Compromised anterior restraints; **superior translation**; minimal CA-arch stabilisation
IIB	**Decentred, unstable**	Incompetent anterior restraints; **anterosuperior escape**; no dynamic or CA-arch stabilisation



Seebauer classification

5 Management

Rotator cuff arthropathy

- Function preserved / first visit → **NSAIDs, steroid injection, PT (deltoid, scapular, cuff)**
- Pain only, intact CA arch + deltoid, young active → **hemiarthroplasty**
- Pseudoparalysis / elderly >70 / anterosuperior escape → **reverse TSA**
  - + ER pseudoparalysis (Hornblower +, ER lag) → **+ latissimus dorsi transfer**
  - deltoid / axillary nerve dysfunction → **no RSA**
- Infection / poor soft tissue → **resection arthroplasty (salvage)**

Nonoperative care when function is preserved

Indication **first-line**; esp. **good active elevation** (e.g. 150°), first presentation

Regimen activity modification · **NSAIDs** · **subacromial steroid injection** · PT (**deltoid**, scapular, cuff strengthening) · not straight to surgery

Acute decompensation rest, PT, injection often settle the painful phase

– – Active space – –

Operative criteria

Procedure	Indications / facts
Arthroscopic debridement	**Controversial**, unpredictable; **preserve CA arch** (✗ Wrong options: acromioplasty · CA ligament release)
Hemiarthroplasty	**Anterior deltoid preserved** + **CA arch intact** (else **anterosuperior / subcutaneous escape**) · **younger, active** · relieves pain (good–excellent **~75%**) but **not function** (elevation **40–70°**)
Reverse TSA	**Pseudoparalytic CTA** · **elderly (>70)**, low demand · **anterosuperior escape** · needs **functioning deltoid (axillary nerve)** + **good bone stock** · improves **pain + function**
Latissimus dorsi transfer	**ER pseudoparalysis** · with RSA
Pectoralis major transfer	**IR deficiency / subscapularis insufficiency** (± aTSA for massive anterior tear + arthritis)
Resection arthroplasty	**Salvage** – chronic osteomyelitis, infection, poor soft tissue
Glenoid resurfacing / anatomic TSA	**Contraindicated** – superior glenoid **shear / edge loading** ("**rocking horse**") → **glenoid loosening** (#1 cause of aTSA revision)
Cuff repair	Only if no end-stage changes, active, **minimal fatty infiltration**; **not with supraspinatus atrophy**
SCR / tendon transfer	Younger, irreparable tear, must follow strict rehab
Ream-and-run	**Young labourer, GH arthritis, intact cuff** (✗ Wrong options: CTA)

Reverse arthroplasty for pseudoparalysis & outcomes

Biomechanics moves **centre of rotation inferior + medial → ↑ deltoid mechanical advantage / lever arm →** deltoid powers elevation

Outcome **improves pain and range of motion even in setting of preoperative pseudoparalysis** · low loosening with modern constructs · durability > hemi

Can't elevate after successful local anaesthetic → RSA
✗ Wrong options: biceps tenolysis, open debridement, hemi (pain-reducing only)

Young vs old younger: cuff repair, **SCR**, or hemi · **older, less active: RSA**

Implant options convertible platform stem, central post or screw baseplate, **short cementless stem** (osteopenia) all OK · **severe osteoporosis / low deltoid tuberosity index → cemented stem** · **metal-backed polyethylene glenoid (aTSA) = contraindicated**

RSA with latissimus/teres major transfer for loss of external rotation

Indication for LDT with RSA **pseudoparalysis of ER** (deltoid restores elevation but **not ER**) from infraspinatus + teres minor dysfunction

ER dysfunction criteria **+ Hornblower** · **ER lag >15° vs contralateral** · **teres minor fatty degeneration ≥ stage 2** (Goutallier / Fuchs)

Not LDT indications negative Hornblower · <5° ER lag · **IR pseudoparalysis** · supraspinatus retraction medial to glenoid · isolated supraspinatus tear
✗ Wrong options: RSA either

RSA contraindications **axillary nerve neurotmesis** · **severe deltoid dysfunction** (e.g. stroke)

Post-traumatic arthritis, normal motor — TSA, no transfer

Pectoralis major transfer for subscapularis insufficiency – nerve risk

Technique — upper part or whole pec major tendon → near **lesser tuberosity**; route **deep to conjoint tendon, superficial to musculocutaneous nerve**

Nerve at risk — **musculocutaneous** (tension neurapraxia) → ↓ **elbow flexion** (biceps, brachialis)

✗ Wrong option: weakness	Nerve
Elbow extension	Radial
Shoulder ER	Suprascapular (infraspinatus), axillary (teres minor)
Shoulder adduction	Medial/lateral pectoral, thoracodorsal, lower subscapular
Shoulder abduction	Suprascapular, axillary

Traumatic subscapularis tear mechanism

Mechanism — fall, grasping to stop fall → **forced external rotation** against resistance
✗ Wrong options: IR · extension · abduction · flexion

6 Complications

Implant choice & complications (superior glenoid wear, metal-backed glenoid)

Scapular notching — humeral poly abuts **inferior glenoid in adduction** → bone loss, osteolysis, loosening, **lower outcome scores**

Notching	Factor
↑ Risk	**CTA with superior glenoid wear** → glenosphere too **superior** / **superior tilt** · **anterosuperior (deltoid-splitting) approach**
↓ Risk	**Lateralised humeral implant** with **lower neck-shaft angle 132.5°** (vs Grammont **155°**) · **~10° inferior tilt** · **inferior-offset glenosphere** with **~3 mm inferior overhang** · deltopectoral approach



Scapular notching after RSA

– – Active space – –

Acromial / scapular spine stress fracture mechanism = **deltoid tensioning by implant design** · **~4–5%**, mean **9 mo** · ↑ **onlay humeral component**, **lateralisation**, **osteoporosis**, **female**, medialised pre-op COR · **nonop union 40–50%**
✗ Wrong options: bisphosphonate mechanism · C. acnes · superior tilt (= notching)

Complication	Note
Instability	**More common after hemi**, rare after RSA
Infection	C. acnes → indolent infection
Neurovascular injury	Musculocutaneous with pec major transfer
Deltoid dysfunction	–
Glenoid loosening	aTSA in CTA (rocking horse)

X vs Y

	Hemiarthroplasty	Reverse TSA	Anatomic TSA
Patient	Younger, active	**Elderly >70**, low demand	Not in CTA
Needs	**Intact CA arch + anterior deltoid**	**Functioning deltoid (axillary n.)**, bone stock	Functioning **supraspinatus + subscapularis**
Pain	Relieved (~75%)	Relieved	–
Function	**Poor (40–70°)**	**Improved**, even pseudoparalysis	Glenoid loosening

	CTA	RA	AVN	Septic (P. acnes)
Glenoid wear	**Superior**, asymmetric	**Central** erosion	–	–
X-ray	↓ **AHI**, acetabularisation, **no osteophytes**, snowcap	**Periarticular osteopenia**	Subchondral lucency / **head collapse**, no superior migration	–
Labs	Normal	↑ inflammatory markers	–	**May be normal**

How it's asked

"Remote cuff repair, normal labs, high-riding head – pathogenesis?"	→ **loss of dynamic compression → abnormal forces → mechanical cartilage wear** ✗ Wrong options: haemosiderin · GPC · AVN · T-cell
"Untreated progression of cuff tear on MRI (figure)?"	→ **cuff tear arthropathy (Figure E)** ✗ Wrong options: chondrosarcoma · OA · dislocation · fracture
"True about CTA before RSA (Hamada 4a)?"	→ **concavity compression is primary centring mechanism vs deltoid**

"Full passive ROM, can't flex, no change after injection — pathomechanics?"	→ **anterosuperior subluxation of humeral head** ✗ Wrong options: posteroinferior · deltoid atrophy · SSN · failure to elevate
"Pseudoparalysis after ladder fall, acetabularisation — process?"	→ **inability of cuff to depress head** ✗ Wrong options: capsulitis, SSN, C5, inferior subluxation
"75 y, subcutaneous fluid, 80° active / 170° passive, CTA X-ray — next?"	→ **reverse TSA** ✗ Wrong options: biopsy · aspiration · anatomic TSA · syrinx MRI
"75 y, high-riding head, severe arthritis — treatment?"	→ **reverse shoulder arthroplasty** ✗ Wrong options: resurfacing · hemi · TSA · debridement
"30° active elevation, AHI 0 mm — best function?"	→ **reverse shoulder arthroplasty**
"Can't raise arm even after injection relieved pain?"	→ **reverse TSA**
"82 y fall, acute-on-chronic cuff tear, failed 4 mo — surgery?"	→ **reverse TSA** ✗ Wrong options: cuff repair · ream-and-run · hemi · aTSA
"72 y, walks + housework, CTA — implant (figure)?"	→ **reverse TSA (Figure F)**
"True about best option for CTA, 70° elevation?"	→ **improves pain + ROM even with pseudoparalysis** ✗ Wrong options: needs CA arch · pain only · high loosening
"Pain on top of shoulder 9 mo after RSA, CT fracture — mechanism?"	→ **deltoid tensioning by implant design** ✗ Wrong options: bisphosphonates · ↓ lateralisation · C. acnes · superior tilt
"RSA + latissimus transfer most appropriate?"	→ **70 F, pseudoparesis of elevation + ER, acetabularisation**
"Pseudoparalysis + Hornblower +, belly press normal — treatment?"	→ **RSA + latissimus dorsi transfer** ✗ Wrong options: aTSA + pec transfer = anterior tears
"Indication for concurrent latissimus transfer?"	→ **pseudoparalysis of external rotation** ✗ Wrong options: axillary neurotmesis · negative Hornblower · IR
"82 F, 1 mo pain, elevates to 150° — management?"	→ **trial of anti-inflammatory medication or cortisone injection and/or deltoid strengthening**
"Iatrogenic nerve injury after pec major transfer — weak movement?"	→ **elbow flexion** (musculocutaneous)
"Traumatic subscapularis tear mechanism?"	→ **forced external rotation**

– – Active space – –

"Risk factor for scapular notching?"	**→ CTA with superior glenoid wear** ✗ Wrong options: lateralised 132.5° · 10° inferior tilt · inferior offset · deltopectoral
"Implant feature contraindicated in CTA?"	**→ metal-backed polyethylene glenoid component** ✗ Wrong options: convertible stem · central post · medialised COR · short cementless stem

ROTATOR CUFF TEARS

– Shoulder & Elbow · 41 / 60 –

1 Basics

Epidemiology

FT tear prevalence	**28% >60 y · 65% >70 y** (Orthobullets, OB) · ~1/3 >60 y, >60% at 70 y (OB) · ~30% >60 y (OB) – read the options
Risk factors	**age** · **smoking** · **hypercholesterolaemia** · **family history**
Burden	leading cause of shoulder disability; ~**250,000 RCR/y** (USA); **2nd most common reason manual labourers miss work**
Age pattern	mostly **>40 y**; degenerative/attrition in elderly · acute trauma in young

Genes & lab markers

Inherited / metabolic	**family history** + **hypercholesterolaemia** = risk factors (no gene named in sources)
Indolent post-RCR infection labs (OB)	WBC borderline (11.0) · **ESR ↑ (40)** · **CRP mildly ↑ (1.5)** · Gram stain negative · culture positive at **day 7–21 → Cutibacterium acnes**

Natural history – asymptomatic prevalence & tear progression

Fact	Value
Asymptomatic tears (FT + PT) **>60 y** on MRI/US	**30–55%** (Tempelhof US **30%** · Sher MRI **54%** = 28% FT + 26% PT)
Asymptomatic ≥60 y (Orthobullets)	MRI **55%** · US **23%** of asymptomatic
Asymptomatic → symptomatic	**50% in 2–3 y** (Orthobullets) · **30–50% in 2–3 y** (OB) – read the options
Symptomatic FT tear enlarges	**50% at 2 y** (Orthobullets) · ~**50% within 2–3 y** (OB) · **20–50% at ~2 y, >50% at 5 y** (OB)
Asymptomatic tear enlarges	~**1/3** within 2–3 y (OB)

Progression rules	**larger initial tears progress faster** (**linear correlation size ↔ rate**) · **FT > PT** (faster, more atrophy) · ↑ pain = marker of progression · ↑ fatty atrophy / ↓ muscle quality ↔ ↑ pain
✗ Progression traps	FT tears don't progress · symptoms unrelated to size · pain independent of muscle quality · **all** asymptomatic tears become symptomatic (subset never painful) · "10–15% FT enlarge by 2 y" · "20% symptomatic in 3 y" · PT tears progress faster
Incidental FT tear, minimal symptoms (72 y, 7 mm)	**observation** ✗ Wrong options: repair · debridement · tenotomy
Atraumatic FT tear + PT	**subjective functional improvement** (↑ PROMs) · ~**75% avoid surgery at 2 y** · ↑ ROM (forward elevation) · minimal pain long-term · PT aim = strengthen surrounding muscles, **not tendon healing**

– – Active space – –

Anatomy – footprint, rotator cable, biceps pulley, innervation & CA ligament

Function — **dynamic stability** by **balancing force couples** → stable fulcrum; **coronal**: inferior cuff (IS, TM, SSc) vs deltoid's superior moment · **transverse**: SSc (anterior) vs IS + TM (posterior); goal of treatment = restore equilibrium

Footprint	Value
Supraspinatus ML width	**12.7 mm** (Orthobullets) · **12.1–12.7 mm** (Dugas, Ruotolo; OB) · **14–16 mm** (OB) · ~12–13 mm (OB) – read the options
Infraspinatus · teres minor · subscapularis	13.4 · 13.9 · **17.9 mm** (SSc widest)
AP dimension	**20 mm** (SS + anterior IS)
Max insertional length × width	23 × 16 mm (OB)
Articular cartilage → medial footprint	**1.6–1.9 mm**
Tendon thickness	10–12 mm
Usual tear site	**15 mm posterior to biceps**, near **SS/IS junction**

Footprint ↔ % tear — **6–7 mm** exposed ≈ **50%** PT tear

5 histologic layers — **articular side = half the strength of bursal side → most tears articular-sided** · I CHL fibres (1 mm) · II dense parallel fibres (3–5 mm) · III loose 45° bundles (3 mm) · IV loose CT + CHL · V capsule (2 mm)

Bursal vs articular tissue — bursal = tendon bundles → greater deformation + tensile strength · articular = tendon + ligament + capsule complex

Blood supply — subscapular, suprascapular, humeral circumflex aa (layers II–III); **bursal > articular** (hypovascular); critical zone next to lateral SS insertion

Rotator interval — capsule + **LHB** + **SGHL** + **CHL**; gap between SS leading edge and SSc superior border

Rotator crescent — thin sheet = distal SS + IS insertions (asterisk on arthroscopy)

Rotator cable — **thickened CHL** near its **avascular zone**, **perpendicular to SS fibres**, spans SS + IS insertions · **>2.5× thicker** than crescent · "**suspension bridge**" · cable disruption → destabilises tendon, tear propagates

Biceps pulley (medial sling) — **SGHL + CHL** (rotator interval tissue) · SSc = **anterior wall of bicipital groove** → SSc tear tears pulley → **medial LHB subluxation**

CA ligament — extra-articular · suspension function · restrains anterior translation (with CHL) · contains head in cuff arthropathy · resection → **↑ GH translation** (anterior/inferior)

✗ CA resection distractors ↑ passive IR · ↑ axillary recess volume · ↓ AC joint forces · ↓ LHB resting tension

Nerve	Muscles
Upper + lower subscapular	**Subscapularis** (lower also **teres major**)
Suprascapular	Supraspinatus + infraspinatus
Axillary	**Teres minor** + deltoid
Thoracodorsal (middle subscapular)	Latissimus dorsi
Musculocutaneous	Biceps, brachialis, coracobrachialis
Lateral pectoral · medial pectoral	Pec major (± minor) · pec major + minor
Long thoracic · dorsal scapular	Serratus anterior · rhomboids + levator scapulae

Biceps subluxation on MRI → innervation of injured muscle? **subscapular nerves** (upper + lower) – SSc tear

Sagittal MRI marked muscle **infraspinatus** (TO, FNB 2022/2023) · **teres minor** (OB, sagittal T1 = muscle quality)

Lateral subacromial portal view through FT tear glenoid, labrum, **biceps tendon** visible

MGHL anterior restraint at **45° abduction**; **deep to SSc** (can be mistaken for it); **absent MGHL** → suspect **SSc tear** (MGHL translates medially with retracted SSc)

Posterior lever push **IR arm + posterior stress on head** → tensions SSc → shows **SSc insertion on lesser tuberosity**

2 Mechanism

Etiology

Mechanism	Features
Chronic degenerative (intrinsic = primary aetiology)	Older; **SIT**; large tears extend to superior SSc
Chronic impingement (extrinsic)	Starts **bursal surface** or intratendinous
Acute avulsion	**SSc in young** (hyperabduction/ER, fall) · **SIT >40 y with dislocation**
Iatrogenic	Failed repair, esp **SSc after open anterior surgery** (open Bankart)

Disease continuum subacromial impingement → subcoracoid impingement → calcific tendonitis → **RCT** → cuff arthropathy

Associated AC pathology · proximal biceps subluxation/tendonitis · **internal impingement** (throwers; PT tears)

PT tear causes (OB) age-related metabolic/vascular degeneration · subacromial impingement · **anterior instability** · internal impingement · repetitive microtrauma · acute trauma less often · **articular side = poorer blood supply**

– – Active space – –

Articular vs bursal origin	articular = **intrinsic** degeneration · bursal = **extrinsic** (acromial spurs, AC arthritis); degenerative tears "start bursal side" (one OB explanation) vs articular commoner (Orthobullets, TO) – read the options

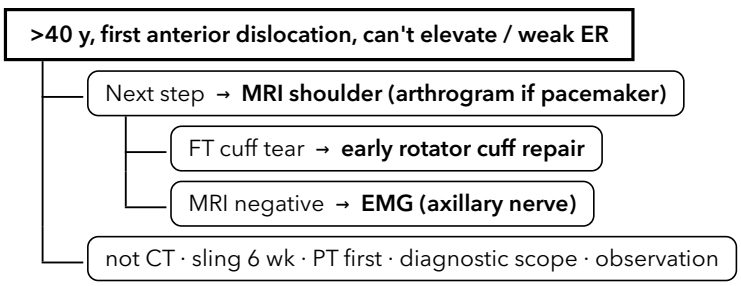
Throwing athletes – mechanism & management

Phase	**deceleration** (label E): cuff = principal decelerator → **eccentric tensile failure**
Site	**articular side of posterior SS or IS**; **posterior pain after ball release**
Overhead-athlete cuff dysfunction	primary impingement · primary tensile overload · secondary impingement/tensile overload from GH instability; microtrauma (overhead) vs macrotrauma (contact)
FT tear in thrower	**must be repaired** ✗ Wrong options: debrided
Adolescent pitcher, undersurface PT tear, never treated	**cessation of all throwing 6 wk** → cuff + periscapular strengthening → slow supervised return, pitching mechanics ✗ Wrong options: ↓ pitch count · continued play · arthroscopic/mini-open repair

Cuff tear after shoulder dislocation in older patients

Rule	**>40 y** anterior dislocation → **rotator cuff tear** (pre-existing degeneration, ↓ elasticity) » recurrent instability; often **missed**
Which tendon	**supraspinatus** commonest (TO, OB) · **SIT** (SS, **IS**, TM) > SSc · SSc tears more with **posterior** dislocation or young hyperabduction/ER
Incidence	**35–86%** >40 y (OB) · **54%**, rising to **100% >70 y** (Simank) · **15%** >40 y, **up to 40% >60 y** (OB) – read the options
Bimodal instability	peak <40 y (Bankart, HAGL) · 2nd peak >60 y (cuff tears, fractures); Hill-Sachs common, usually not addressed
Clue (TO)	pain lifting in **scapular plane (scaption)** after 6 wk immobiliser → **US/MRI** for cuff ✗ Wrong options: LHB entrapment · loose body
Axillary nerve	**7–10%**; far less common than cuff tear; usually too mild for gross weakness; **lateral arm sensation intact** → nerve palsy/radiculopathy unlikely

(✗ Wrong options: **Distractors** – posterior Bankart (posterior dislocation)) · HAGL (only ~11% of first dislocations >35 y) · SLAP (throwers, no link to anterior dislocation) · SSN injury (uncommon)



Arthrogram	dye leaks **GH → subacromial space** (geyser) = cuff tear; normal = dye contained in capsule

Treatment	**rotator cuff repair**; improved pain + function vs nonop; delay → atrophy + retraction ✗ Wrong options: hemiarthroplasty · ORIF · TSA · RTSA · coracoid transfer · labral repair · PT alone
Most likely to need surgery (52 y)	**supraspinatus** ✗ Wrong options: labrum · GH ligaments · humeral head · biceps
Massive but completely repairable	**acute tear after dislocation, 45 y** (no retraction/atrophy yet) · (✗ Wrong options: RA (attrition, steroids) · prior repair (revision) · SS fossa atrophy · superior humeral migration (→ arthropathy)) · chronic massive repairs retear **~50%**

3 Presentation

Pain	insidious, deltoid region, **overhead** activity · **night pain** = poor predictor for nonop
Weakness	**active ROM lost, passive ROM preserved** · acute pain + weakness after trauma

Physical examination – external rotation lag sign

Muscle	Weakness	Test	Technique / positive
Supraspinatus	Elevation in Jobe position	**Jobe / empty can**, **drop arm**	Abducted, IR, **30° forward in scapular plane, thumb down** → resist pressure · drop test: lowers slowly → sudden drop
Infraspinatus	ER at **0° abduction**	**ER lag sign**	**Elbow 90°, passive ER → can't actively maintain**
Teres minor	ER at **90° abd + 90° ER**	**Hornblower**	90° abd, 90° elbow flexion, 90° ER → **arm falls into IR**
Subscapularis	IR at 0° abduction	**Belly press**, **lift-off**, **bear hug** (upper border), **IR lag**, **↑ passive ER**	Belly press: **elbow drops back** · lift-off: can't lift hand off back

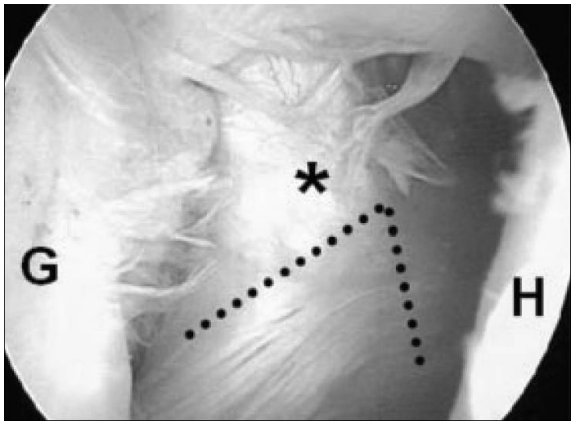
✗ Wrong options (non-cuff tests)	Yergason (resisted supination, elbow 90°) = biceps · Jerk = posterior labrum · Wright = **TOS** · relocation = anterior instability · Hawkins/Neer = impingement · O'Brien = SLAP

Subscapularis tears – belly-press/lift-off tests, biceps instability

Who	**young**, acute avulsion, **hyperabduction/ER** (fall, wave, door handle) · iatrogenic after **open Bankart** · degenerative = superior portion; largest cuff muscle, prevalence under-recognised
Symptoms	**anterior shoulder pain** (mimics **subcoracoid impingement**)
Signs	**↑ passive ER** (e.g. 90° vs 45°, 100° vs 70°) · **↓ active IR** · belly press, lift-off, bear hug, IR lag
✗ ROM traps	↑ passive flexion/extension/abduction · ↑ active IR · ↑ passive IR
MRI	SSc discontinuity off **lesser tuberosity** + **LHB out of groove (medial subluxation/dislocation)** = near-pathognomonic
LHB dislocated medially → also injured?	**subscapularis**; rare lesser tuberosity injury ✗ Wrong options: MGHL · Bankart (not biceps stabilisers)

– – Active space – –

Biceps dislocation routes	beneath SSc · within SSc · extra-articular
Arthroscopic findings (all of the above)	uncovered lesser tuberosity · retraction to glenoid · avulsed SGHL · medial biceps subluxation
Godenèche (218 scopes)	pathologic LHB with intact SGHL/CHL in only **11%**; even then **medial sheath wall torn in 92%** → SGHL/CHL lesion = LHB nearly always unstable
Comma sign	**SGHL + CHL** complex attached to **superolateral corner of retracted SSc** (Lo & Burkhart) · trace it **inferiorly** → SSc; chronic SSc retracts medially, scarred to deltoid fascia



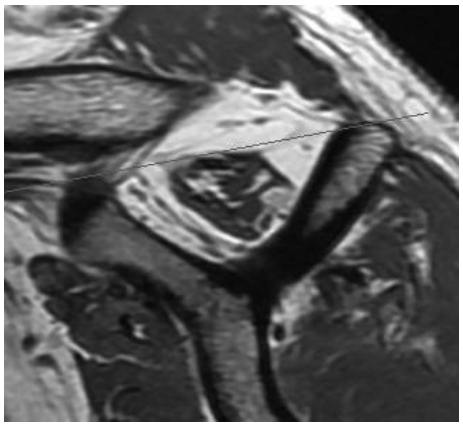
Comma sign: SGHL/CHL complex (asterisk) on superolateral SSc border

4 Diagnosis & Classification

Key radiology findings

Modality	Finding / fact
X-ray	True AP, AP IR/ER, axillary, **outlet view** (acromion); GT cysts, calcific tendonitis, CHL calcification, **type III acromion**, **AHI <7 mm** = proximal migration (chronic RCT)
Arthrogram	When **MRI contraindicated** (pacemaker); dye **GH → subacromial** = tear; MR arthrogram ↑ sens/spec
MRI	**Diagnostic standard**; size, shape, retraction; **fatty atrophy best on sagittal**; **medial biceps subluxation = SSc tear**; humeral head cysts in almost all chronic RCT
ABER MRI	↓ effacement of articular surface → contrast enters defect → **↑ sensitivity for PT (PASTA) tears**
Ultrasound	Dynamic, cheap, confirms injections; **user dependent**; = MRI accuracy for cuff

Tangent sign	sagittal oblique: line **scapular spine top → coracoid top**; SS belly **below line = positive → SS fatty atrophy → irreparable** / poorer outcome after repair (TO) ✗ Wrong options: SSc atrophy · retraction medial to glenoid · biceps dislocation



Tangent sign on sagittal MRI

PASTA on coronal T2 **articular-sided SS tear**; intra-articular contrast delineates

Lesion	Imaging / definition
PASTA	Partial articular supraspinatus tendon avulsion
ALPSA	Labroligamentous complex **medial + inferior**, rolled under **intact periosteum**
GLAD	Nondisplaced anteroinferior labral tear + **glenoid cartilage** avulsion
HAGL	IGHL avulsed from **humeral neck**
Bankart	Anterior scapular periosteum **disrupted**
SLAP	Superior labrum anterior → posterior to biceps root

Diagnostic criteria – irreparable cuff

Marker	Threshold
Size	**>5 cm** (Cofield) or **≥2 tendons**
Fatty atrophy	**Goutallier 3–4**
AHI	**<6 mm** (TO) · **<5–7 mm** (OB) · <7 mm = proximal migration (Orthobullets)
Tangent sign	**Positive**
Retraction	Medial to glenoid
Clinical	**Chronic pseudoparalysis** · **profound ER weakness** · long tear duration
Pattern	**Posterosuperior (SS + IS) > anterosuperior (SS + SSc)**

Pseudoparalysis can't actively elevate **>90°** (some authors **>45°**, TO), **full passive**; **no improvement after lidocaine** · **SSc + SS complete tear → 80% pseudoparalysis**

Anatomic classification

Type	Features
SIT (SS, IS, TM)	**Majority**; subacromial impingement; degenerative older or **dislocation >40 y**
Subscapularis	More prevalent than thought; **subcoracoid impingement**; acute avulsion young (hyperabduction/ER) or **iatrogenic failed repair**

– – Active space – –

Cuff tear size

Size	cm
Small	0–1
Medium	1–3
Large	3–5
Massive	**>5** or **≥2 tendons** (Orthobullets) · ">2 tendons" (one OB)

Cuff tear shape

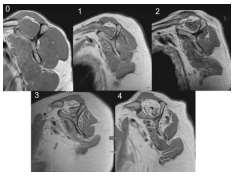
Shape	Repair
Crescent	Little retraction, mobile ML → direct to bone, minimal tension
U-shape	Apex at/medial to glenoid → **margin convergence** side-to-side first (↓ strain on lateral margin)
L-shape	One leaf more mobile → margin convergence
Massive immobile	U or longitudinal → **interval slide**

Ellman classification of partial-thickness tears

Grade	Depth
I	**<3 mm (<25%)**
II	**3–6 mm (25–50%)**
III	**>6 mm (>50%)**
Location	**A** articular · **B** bursal · **C** intratendinous

Snyder also classifies PT tears by location + depth

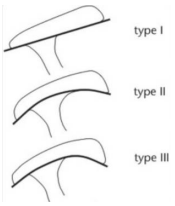
Goutallier classification of fatty atrophy

Grade	Fat vs muscle	MRI
0	Normal	
1	**Some fatty streaks**	
2	More muscle than fat	
3	Equal fat and muscle	
4	**More fat than muscle**	

Lafosse classification of subscapularis tears (TO)

Type	Lesion	Surgical approach
I	**Partial**, superior third (articular)	Debridement or single-row
II	Complete, superior third	Single-row
III	Complete, **superior two-thirds**	Double-row / footprint restoration
IV	Complete, **entire tendon**, head centred (concentric)	**Extensive 3-sided release** + robust repair
V	Complete + **head subluxation** (eccentric; coracoid impingement)	Often open repair or **pec major transfer**

Acromial morphology & impingement (TO)

Bigliani type	Shape	Note
I	Flat	
II	Curved	**Most common type**
III	**Hooked**	↑ anterior + inferior extension → **most associated with impingement + RCT**
IV	Convex (upturned), 3%	no impingement

Parameter	View	↑ RCT risk
Acromion type	**Outlet**	Type III
Acromial slope	**Outlet**	Normal 23–45°; **small angle** (steep)
Acromial tilt	**Outlet**	**↑ tilt** (keyed); **>35°** (normal 29–34°) · ">8.5 mm" (other TO copy) · "lower tilt angle → ↑ tears" (one TO text) – read the options
Lateral acromial angle	True AP (TO) · oblique coronal MRI (Banas, TO)	Normal 64–99°; **<70°** → high likelihood FT tear
Acromion index	**True AP** (✗ Wrong options: outlet)	**>0.7** (deltoid vector more superior)
Critical shoulder angle	–	**>35°**
Acromiohumeral interval	**Outlet**	**<7 mm**

Direction summary ↑ RCT with **type III**, **↑ AI**, **↑ tilt**, **↑ CSA**, **↓ AHD**, **↓ LAA**

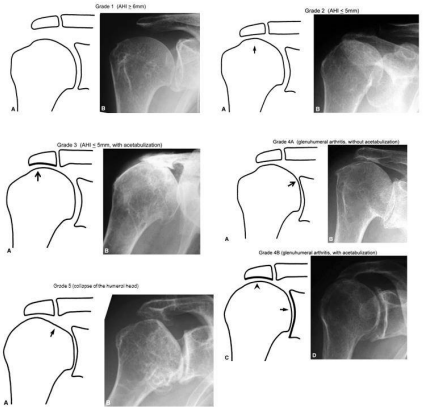
Bursal tears invariably associated with severe degenerative acromial change (friction/rubbing)

– – Active space – –

Cuff tear arthropathy – radiographic signs (TO)

Sourcil (eyebrow) sign — **sclerosis/erosion of acromion undersurface**; superiorly migrated head abuts acromion → **rotator cuff tear arthropathy** (anterosuperior escape) · sens/spec/PPV/NPV all **poor**
✗ Wrong options: calcific tendinitis · SS tendinitis · adhesive capsulitis · SLAP · instability

Hamada stage	Finding
1	AHI **>6 mm** (TO) · ≥6 mm (OB figure)
2	AHI **<7 mm** (TO) · **≤5 mm** (OB figure) – read the options
3	+ **acetabularisation** of acromion
4a	GH arthritis **without** acetabularisation
4b	GH arthritis **with** acetabularisation
5	Humeral head **osteonecrosis** (TO) · **collapse** (OB figure)



Hamada grades of cuff tear arthropathy

5 Management

Rotator cuff tear

- Asymptomatic / incidental → **observation**
- Atraumatic FT or PT → **PT + NSAIDs + subacromial CSI, 3–6 mo**
  - fails → **arthroscopic debridement or repair**
- Acute traumatic FT (young; >40 after dislocation) → **early repair**
  (do not delay for PT; better within 3 wk – 3/4 mo)
- Massive irreparable → **partial repair / SCR / transfer / RTSA**

Operative criteria

Indication	Procedure
Low-grade PASTA (<50%, <5–7 mm footprint) + impingement, failed nonop	**SAD + cuff debridement**
Bursal >3 mm (>25%)	Release, debride degenerative tissue + **repair**
Articular >50% (>6 mm (OB) · >7 mm (Orthobullets, OB) exposed footprint)	**Completion + repair**
PASTA, young, acute, **≥25% healthy bursal tissue**	**In situ repair** (bursal tissue left intact)
Intrasubstance tear, intact footprint	**Trans-tendon repair**
Acute (<3 mo) traumatic tear, <60 y	Repair (relative indication)
FT or PT failing nonop	Repair (relative)
Massive irreparable, intact SSc, no arthritis	SCR / LD / lower trapezius transfer
Massive + GH arthritis, **intact deltoid**	**RTSA**
Relative CI to repair	Advanced fatty infiltration/atrophy · significant GH arthritis

Operative vs nonoperative indications (acute traumatic vs atraumatic tears)

Nonop regimen
PT + NSAIDs + subacromial CSI = **first line**; avoid overhead; aggressive **cuff + scapular stabiliser** strengthening **3–6 mo**

Partial tear, improving, never had formal PT (70 y)
formal PT
✗ Wrong options: repair · debridement · RTSA · repeat MRI unless unexplained progression

PASTA, full strength, 6 wk (carpenter)
physical therapy first
✗ Wrong options: PRP · repair · SLAP repair · SAD

Failed 12 wk PT + injection, PASTA
repair or debridement
✗ Wrong options: repeat injection · continued PT · capsular release (full passive ROM) · rotator interval release (for adhesive capsulitis or retracted SSc)

Low-grade PASTA + impingement, 5/5 strength, failed nonop
SAD + debridement
✗ Wrong options: SR/DR repair · labral repair · tenodesis

Subacute FT SS tear, weakness, type 1 acromion, no fatty change (47 y)
rotator cuff repair alone
✗ Wrong options: repeat CSI · porcine SIS xenograft · + acromioplasty · + remplissage/tenodesis/DCE

AAOS points
exercise + NSAIDs for PT tears · **routine acromioplasty not required** · **non-cross-linked porcine SIS xenograft not to be used** · workers' comp → less favourable outcome

Purpose of repair
restore muscle function; then ↓ pain, prevent irreversible atrophy

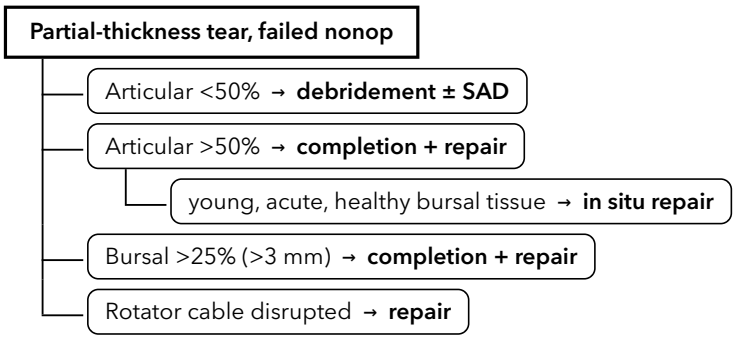
Massive tear SS + SSc, 40 y, 5 mo, no atrophy
rotator cuff repair; attempt repair before salvage; surgery **within 6 mo** → better outcome
✗ Wrong options: debridement/SAD/DCE · pec major transfer · 6 wk more PT · SCR

– – Active space – –

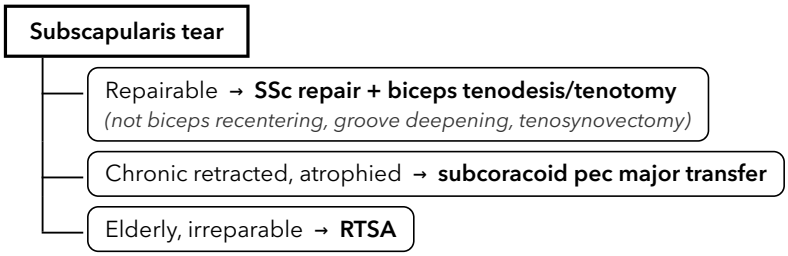
Traumatic FT tear, young — outcome driver	**time to surgical repair** (repair within **3–4 mo**; acute repair **within 3 wk** best) · (✗ Wrong options: BMI · age · male gender) · higher preop ASES → ↑ VAS improvement only
Atraumatic FT tear — strongest predictor of surgery	**patient expectation of PT** (also higher activity, non-smoker) · pain not related to anatomic severity · pain ↔ comorbidities, lower education, race

Partial-thickness tears – PASTA, bursal vs articular, debridement vs completion & repair

PASTA (TO)	**articular side**; far more common than bursal or intratendinous
Articular vs bursal	articular **more common** · bursal **more symptomatic** (subacromial bursitis) · **bursal = higher risk of progression** · articular fares better → higher % tolerated before repair
Repair thresholds	**bursal >25% (>3 mm)** · **articular >50%** ✗ Wrong options: bursal <25% · articular >25%
Articular >50% tear but poor lateral tissue	**complete the tear → repair** ✗ Wrong options: fascial graft · allograft · trans-tendon · biopsy
Bursal tear 4 mm deep, 1.5 cm AP, softening	**convert to FT + repair with suture anchors** ✗ Wrong options: debride + acromioplasty · abort · incidental
Footprint exposed 3 mm, bursal intact	**debridement + SAD** (<5 mm exposure ≈ <50% thickness; ✗ Wrong options: completion · transtendinous repair · tenotomy · synovectomy)
6 mm ML width tear	**50%** of SS tendon
Rotator cable involved	**bursal 4 mm tear involving anterior cable → repair** · intact cable 3–4 mm → less likely to need repair · articular <7 mm → debridement · SSc upper-border fraying without retraction → no repair (but often-missed tear)
High-grade PASTA (>50%), failed 9 mo PT + PRP (tennis, 51 y)	**tear completion + repair to footprint** ✗ Wrong options: isolated DCE (negative cross-body) · more nonop · SAD ± tenodesis · trans-tendon + DCE



Subscapularis repair – biceps, 3-sided release (TO)



Biceps with SSc tear unstable → **tenodesis or tenotomy** (recentering unpredictable; prevents recurrent biceps pain/failure) · tenodesis **low in groove**; address biceps **first** (rubs/fails repair)

SSc tear after open Bankart (6 mo) **subscapularis repair** + biceps tenotomy/tenodesis
✗ Wrong options: PT · tenolysis → cosmetic deformity · pec major (tendon repairable)

Which lesion → RCR + biceps tenodesis? **SSc tear with dislocated biceps**; no significant GH arthritis
✗ Wrong options: Hill-Sachs · GT fracture · bony Bankart · Bankart

Chronic retracted atrophied SSc (55 y carpenter) **pectoralis major transfer** under conjoined tendon · Achilles allograft alternative for degenerated SSc
✗ Wrong options: fusion = salvage · CSI · arthroscopic/open repair

Lafosse IV – release from coracoid base (TO) **coracohumeral ligament**
✗ Wrong options: CA ligament · transverse humeral ligament · conjoined origin

3-sided release (TO)	Structure
Anterior (subcoracoid space)	**CHL** + adhesions to **base of coracoid** – key for lateral excursion
Superior	Rotator interval; free **comma tissue** from coracoid base
Posterior (intra-articular)	Capsulotomy from **capsule + MGHL** to **6 o'clock**

Repair steps (TO) biceps first → **70° scope, posterior portal** for lesser tuberosity → traction stitch at comma via accessory anterolateral portal → anchors **inferior first → superior** → tie in **internal rotation** · III/IV → modified double-row

Arthroscopic vs open SSc repair arthroscopic superior motion + pain (Orthobullets)

Repair technique – single vs double row, transosseous-equivalent & failure mode

Mini-open deltoid-splitting; ↓ deltoid avulsion vs open; immediate PROM; active ~6 wk · arthroscopic = equivalent results

Arthroscopic vs open (medium tear) **no difference in retear rate** or function · arthroscopic → earlier rehab (no deltoid to protect) · mini-open → worse early pain/function only

Interval slides **anterior** = release SS from rotator interval (incise CHL) · **posterior** = release SS from IS, then margin convergence

Double-row (TOE) medial **mattress** + lateral **simple** → **more anatomic footprint**, larger contact area

DR vs SR	Finding
Retear	**DR lower** (esp tears/retraction **>3 cm**); SR higher (Millett meta-analysis, 7 level I RCTs)
Healing rate	DR higher
Biomechanics	DR **higher ultimate tensile load**; = peak-to-peak elongation, stiffness, conditioning elongation
Function, pain, time to heal (short/mid term)	**No difference** (esp <3 cm)
Long term	DR may improve function (one RCT)
Cost	DR more (more anchors)
Blood loss	No difference

– – Active space – –

Lowest retear risk — **Goutallier 1 + DR TOE**
✗ Wrong options: Goutallier 3–4 · SR

Two DR constructs (B, C) — equivalent retear; SR (A) possibly higher; **A and B = equivalent outcome scores at 1 y**

Vascularity — **anchor drilling** (holes in GT) + **peribursal tissue** → main supply to repair; ↑ with exercise

✗ Wrong options (false repair "truths") — motion restriction ↑ blood flow · DR better clinical results · SAD ↑ healing · non-healing always → poor outcome (**~half don't heal**, outcomes preserved)

GT trough (cancellous) — **not stronger** repair (animal models)

Failure mode — **suture pull-out through tendon = tissue failure** (most common, early) → gap formation → poor tendon–bone healing

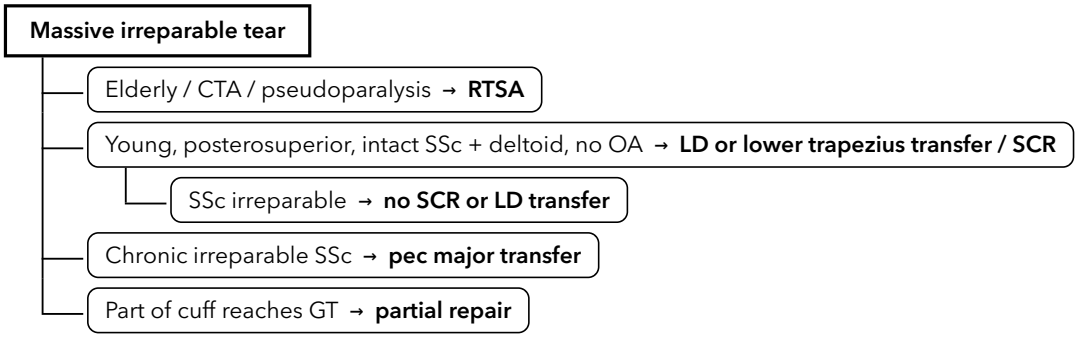
Less common failure	Note
Anchor breakage	Biocomposite fatigues less than metal
Anchor pull-out	Poor BMD, improper placement
Eyelet abrasion suture rupture	Metal anchors (sharper)
Knot failure	
Infection	~1.9%

Causes of failed RCR — failure to heal (**19–94%**: poor tissue, vascularity, bone) · surgical (deltoid, infection, foreign body, stiffness, nerve) · diagnostic (missed interval, LHB, SSc) · technical (excess tension, anchor placement); arthroscopic RCR complication rate ~10%

Traumatic retear 3 wk post-op, 1 cm, <65 y — **revision RCR** (healing >95%)
✗ Wrong options: PT · CSI · sling · debridement (for chronic irreparable)

Revision RCR vs RTSA — RTSA favoured by older age, **static proximal humeral migration**; also aetiology of retear, tissue/MRI quality

Massive/irreparable tears – partial repair, SCR & tendon transfers



Partial repair — SS can't reach GT, posterior cuff can → **partial repair** (balances coronal + axial force couples) · pain options: acromioplasty + CA release, **reverse acromioplasty (tuberoplasty)**, biceps tenotomy · ✗ RTSA in 48 y · other young options: SCR, transfers, **subacromial balloon spacer** (none superior)

Pseudoparesis vs pseudoparalysis — **active FE >90° (e.g. 100°) after lidocaine injection → pseudoparesis** (pain) · no change → pseudoparalysis · arthritis/retraction occur in both

Young (35 y), pseudoparalysis, AHD <6 mm, Goutallier 3, minimal arthritis – least preferred (TO) — **RSA** (✓ LD, ✓ lower trapezius, ✓ SCR)

SCR (TO, Mihata)	Detail
Mechanism	Passive constraint to **superior head translation**; space-occupier + tether; restores force couples
Graft	**Folded fascia lata 6–8 mm** · acellular dermal **3–4 mm**; anchors **superior glenoid rim → medial GT**
Indications	Failed nonop massive irreparable **SS + IS**, **Hamada 1–2**, **intact/reparable SSc**, **functional deltoid**; intolerable pain; pseudoparalysis
Contraindications	**Irreparable SSc** (subscapular nerves) · **Hamada ≥3** · GH OA
Best candidate	**59 y, SS + IS, normal X-ray**

SCR caveat — irreparability confirmed only after full diagnostic arthroscopy (Hartzler & Burkhart); short-term data only, graft failure risk

Transfer	Indication	Key facts
Pectoralis major (subcoracoid, **under conjoined**)	**Chronic irreparable SSc**	Replicates SSc vector; restores IR; **4–6 wk rigid immobilisation**; not posterosuperior
Latissimus dorsi (± teres major)	**Irreparable posterosuperior (SS + IS)**, **young labourer**, Goutallier ≥3	To cuff/SSc/GT; brace **6 wk, 45° abd, 30° ER**; restores ER + elevation
Lower trapezius (± **Achilles allograft**)	Irreparable posterosuperior, middle-aged labourer	More anatomic; **superior outcomes vs LD** (Orthobullets); restores ER

Transfer counselling — pain relief more reliable than function (Omid & Lee)

LD transfer – best candidate — **young labourer, massive posterosuperior tear + SS atrophy/fatty infiltration** · **some active elevation (90–100°)** · intact SSc + deltoid, no OA (e.g. 50 y, FE 100°, ER −10°)

LD transfer – worse outcomes — **positive lift-off / SSc deficiency** (strongest; Gerber, Irlenbusch) · **deltoid weakness/detachment** · **nonsynergistic** (out-of-phase) action · stage 3–4 fatty degeneration, poor tendon · **female** · preop elevation **<90°** · **complete loss of ER** · superior escape · salvage after failed repair (vs primary)

Not a worse-outcome factor — **absence of CA ligament** · other prior surgery

LD relative CI — SSc deficiency · deltoid deficiency · pseudoparalysis · advanced GH arthritis · CTA (stays painful) · >80 y pseudoparalytic · acute tear in young (repair first) · thoracodorsal palsy · disability claimant

Belly press elbow drops back — SSc dysfunction → **↓ LD transfer benefit**
✗ Wrong options: ↑ passive IR · drop sign = SS · Hornblower = TM · axillary fold ecchymosis = pec major rupture

Posterosuperior deficiency + intact belly press → transfer? — **latissimus dorsi**
✗ Wrong options: pec major = SSc · pec minor · conjoined = instability · LHB

– – Active space – –

LD nerves at risk	**radial nerve** deep (anterior) to LD tendon, **~3 cm medial to insertion → IR arm** at tenotomy; injury → **EPL (extensor pollicis longus)** + wrist/digit extensors · posterior branch **axillary** (deep fascia of posterior deltoid, tendon passage)
Lower trapezius best for	**45 y manual labourer, irreparable posterosuperior, loss of ER** ✗ Wrong options: pitcher SLAP · irreparable SSc → pec major · GH OA → TSA · CTA → RTSA
Retear 2 mo after massive repair (rough PT), ER lag	**revision repair SS/IS**; failure: intrinsic (age, size, tissue, systemic disease, smoking) + extrinsic (construct, aggressive rehab) ✗ Wrong options: SSc revision · LD first · interval release · anatomic TSA
RTSA mechanism	**lengthens deltoid lever arm**; COR **medial + inferior**; convex glenoid + concave humerus; ↓ shear · survival **90–95% at 10 y** · most reliable for 65 y painter with SS + IS atrophy, no OA ✗ Wrong options: dermal augmentation · SCR · partial repair · trapezius transfer
Other grafts (Orthobullets)	xenograft (bovine dermis/intestine; mixed) · allograft (human skin/fascia; good short term) · synthetic (foreign body reaction) · **lateral acromionectomy = contraindicated** (historic)

Concomitant procedures (acromioplasty, distal clavicle excision, suprascapular nerve release)

Add-on to RCR	Effect
Acromioplasty / SAD	No difference in PROs, symptomatic retear, ROM, strength, pain · **lower long-term reoperation** (Level 1, esp **type 2–3** acromion; OB) · "routine acromioplasty not recommended" (AAOS, OB) – read the options
Distal clavicle excision	**No effect on 2-y/long-term outcomes**, ROM, revision
SSN release	**No difference** in pain or function; SSN dysfunction in **8–27%** of massive tears (traction from retraction → fatty infiltration)
LHB tenotomy vs tenodesis	Negligible difference
CA ligament release	↑ anterior/inferior GH translation

No long-term effect either	mini-open vs arthroscopic · starting PT at 6 wk · SR vs DR
Negative predictors of RCR outcome	smoking · severe obesity · tear size · fatty infiltration · **age 70+** · pre-existing arthropathy · **workers' comp**

Postoperative rehabilitation – early vs delayed motion

Healing	tendon-to-GT healing **8–12 wk** = rate-limiting step · protect ≥**6 wk**, ~**8 wk** for large 2-tendon tear
Protocol	**early PROM** (prevents adhesive capsulitis) → **AROM at 6 wk** → strengthening **≥2–3 mo** · no active strengthening at 1 mo ✗ Wrong options: isometric ER · eccentric SS · resistance · deltoid (can't isolate from cuff)

EMG — **passive forward flexion = minimal cuff activation** · active, active-assisted, isometric, **pulley-assisted** all activate cuff (avoid early)

Early (<6 wk) vs delayed (6 wk sling)	Result
Functional outcomes	**Equivalent**
Terminal ROM / stiffness at 1 y	**No difference**
Retear / healing	No difference (Cuff & Pupello, 68 pts) · **delayed may ↑ healing**
Short-term PROs	Not better with early; early may ↑ pain

PT not available first 6 wk — **won't affect ROM at 1 y**
✗ Wrong options: peer-to-peer · home band strengthening · strength testing · prophylactic CSI

Scapular plane forward elevation, prone retraction, 2 mo post-RCR (TO) — **lower trapezius** (LT > MT + SA) · SA activity ↑ with elevation angle; **LT ↑ steeply at end range** · deficit = endurance/control/timing, not weakness → low–moderate repetitive resistance

Trapezius part (TO)	Exercise
Upper	Shrug ± retraction; overhead shrug
Middle	Overhead retraction against wall
Lower	**Prone scapular setting** (retract + depress, lift arm off table)
Lower + middle	Prone scapular retraction, arm to ~120°

Return to work — **full duty 6–10 mo** · light duty after progressive strengthening (earliest **8–10 wk**) · labourers later + lower outcomes than desk jobs; most return to full · PROM PT starts 2–6 wk

Analgesia — NSAIDs improve pain at 1 wk (Orthobullets)

– – Active space – –

6 Complications

Healing & retear risk factors (age, fatty infiltration, workers' comp, injections)

Retear rate — **10–90%** · massive chronic ~**50%**
✗ Wrong options: <5%

Factor	Effect
Age >65	**Highest risk of non-healing** (✗ Wrong options: protective); >70 → inferior outcomes
Goutallier grade (3–4)	**Greatest retear risk** (vs age, rehab, technique, gender)
Tear **>5 cm**, multiple tendons, retraction **medial to glenoid**	↑ retear
Diabetes · smoking · severe obesity	↑ retear
Poor post-op compliance	↑ retear
Concomitant AC / biceps procedure	↑ retear (Orthobullets)
Workers' comp / work-related	**↓ functional improvement + ↓ satisfaction**, later RTW
Frequent CSI	**↑ revision surgery** (frequency-dependent)
Not risk factors	Female gender · DR technique · delayed rehab · hand dominance · BMI/age/sex (young traumatic tear)

Timing of surgery — repair **<3 mo** from acute tear → better · operative time >45 min → not better (longer = overnight stay, SSI)

Complications – infection, positioning & regional anaesthesia

Suprascapular nerve (TO) — on **undersurface of SS + IS**; **~2.9 cm from superior glenoid rim**, **1.8 cm from scapular spine → no sharp dissection >1 cm medial to glenoid rim** · tethered at supraspinous fossa + notch → **tension on lateral SS advancement** (and medial retraction) · injured during **posterosuperior cuff mobilisation** in massive RCR, posteriorly directed glenoid-neck sutures (anterior labral repair), glenoid osteotomy
✗ Wrong options: thoracoacromial a. · axillary n. · PCHA

Infection — **<1%** (Orthobullets) · ~**1.9%** (OB) · **0.006–3.4%** (OB) – read the options; S. aureus, strep, **C. acnes** (formerly *P. acnes*)

C. acnes — **delayed/indolent** (10 wk), healed portals, mild ↑ ESR/CRP, Gram negative → anaerobe, cultures positive **7–21 days**
✗ Wrong options: Pseudomonas (water-borne, not shoulder)

Pre-op steroid injection — **within 1 month → highest infection risk** (**65–75% ↑**) · 1–3 mo ↑ · **>3 mo = no ↑ risk** → delay surgery ≥3 mo · not dose, location, number, steroid type

6 injections, surgery 3 mo later — **↑ revision, no ↑ infection**

Pneumothorax — after **interscalene/supraclavicular block** or arthroscopy · risk = **smoking/COPD** (hyperinflated lungs, high pleural dome) · dyspnoea also from block **diaphragm paralysis** → CXR
✗ Wrong options: DM · revision · obesity · SSc repair

Beach chair — **LFCN palsy** up to **2%**; resolves <6 mo; padding, seatbelt, slight reverse Trendelenburg; **BMI >35** risk
✗ Wrong options: femoral · obturator · pudendal (perineal post/hip scope) · sciatic

Others — **deltoid detachment** (open) · AC pain · axillary nerve injury · stiffness (early PT/ROM doesn't reduce at 1 y)

X vs Y

	Articular-sided (PASTA)	**Bursal-sided**
Frequency	**Far more common**	Less
Cause	Intrinsic degeneration, weaker hypovascular side	Extrinsic (acromial spur, AC arthritis)
Symptoms	Less	**More** (bursitis)
Progression	Lower	**Higher**
Repair threshold	**>50%** (>6–7 mm footprint)	**>25%** (>3 mm)

	Supraspinatus/SIT tear	**Subscapularis tear**
Patient	Older, degenerative; >40 y dislocation	Young, hyperabduction/ER; post-open Bankart
Pain	Lateral, overhead, night	**Anterior**
Signs	Jobe, drop arm, ER lag	**↑ passive ER**, belly press, lift-off, bear hug, IR lag
MRI	Tear 15 mm behind biceps	**Medial LHB subluxation**
Impingement	Subacromial	**Subcoracoid**
Nerve	Suprascapular	**Upper + lower subscapular**

	Pseudoparalysis	**Pseudoparesis**
Active elevation	**<90°**, full passive	Limited by pain
After lidocaine	**No change**	**Improves >90°**

	SCR	**LD transfer**	**Pec major transfer**	**RTSA**
Tear	SS + IS	SS + IS	**SSc**	Any massive + CTA
Patient	Younger	**Young labourer**	Younger	**Elderly**, low demand
Needs	Intact SSc, deltoid, Hamada 1–2	Intact SSc + deltoid, some elevation	Under conjoined	**Intact deltoid**

– – Active space – –

	Dislocation <40 y	Dislocation >40 y
Main lesion	Bankart, HAGL	**Cuff tear** (SS, SIT)
Problem	Recurrent instability	**Weakness**; redislocation low
Sling 6 wk		Unnecessary

How it's asked

"Sagittal MRI, orange arrow – muscle?"	→ **infraspinatus** (TO)
"Sagittal T1 arrow – muscle?"	→ **teres minor**
"Thick CHL band perpendicular to SS near avascular zone?"	→ **rotator cable** ✗ Wrong options: crescent · rotator interval · MGHL
"CA ligament resection results in?"	→ **↑ GH translation**
"SS footprint ML width?"	→ **14-16mm** (OB) · 12.7 mm (Orthobullets) – read the options
"6 mm ML articular tear = what %?"	→ **50%**
"Comma sign / biceps pulley ligaments?"	→ **CHL + SGHL** ✗ Wrong options: MGHL · CA · IGHL
"Lateral subacromial portal through FT tear – arrow?"	→ **biceps tendon**
"Medial LHB subluxation – nerve to injured muscle?"	→ **upper + lower subscapular** ✗ Wrong options: suprascapular · axillary · thoracodorsal
"Posterior lever push shows?"	→ **SSc insertion (structure #5)**
"Posterior lever push + which finding supports SSc tear?"	→ **absent MGHL**
"Lafosse IV – release from coracoid base?"	→ **coracohumeral ligament** (TO)
"SSc tear arthroscopic findings?"	→ **all of the above** (uncovered LT, retraction, avulsed SGHL, biceps subluxation)
"Abnormal belly press after motorcycle fall?"	→ **subscapularis tear**
"Repaired SSc – pre-op abnormal test?"	→ **lift-off** ✗ Wrong options: jerk · Wright · Jobe · Hornblower
"Truck driver held door, MRI SSc tear – test?"	→ **belly press**
"Can't lift hand away from back – diagnosis?"	→ **subscapularis rupture**
"SSc tear + biceps out of groove, failed 3 mo nonop?"	→ **SSc repair + biceps tenodesis**
"Biceps treatment with SSc rupture?"	→ **tenodesis or tenotomy** ✗ Wrong options: recentering
"Chronic retracted atrophied SSc, 55 y?"	→ **pectoralis major transfer**
"Unstable LHB dislocated medially – also injured?"	→ **subscapularis**

"Weakness 6 mo after open Bankart, MRI SSc rupture?"	→ **subscapularis repair**
"Which image → RCR + biceps tenodesis?"	→ **SSc tear with dislocated biceps (A)**
"SSc tear exam finding?"	→ **↑ passive ER** ✗ Wrong options: ↑ active IR
"Medial biceps subluxation — provocative test?"	→ **bear hug (D)**
"Surfer, SSc tear — injured pulley ligaments?"	→ **CHL + SGHL**
"PASTA occurs on?"	→ **articular side** (TO)
"Coronal T2, carpenter fell from ladder — lesion?"	→ **PASTA** ✗ Wrong options: ALPSA, HAGL, SLAP, GLAD
"Bursal tear 4 mm deep, softening?"	→ **convert to FT + suture-anchor repair**
"MRI + US undersurface SS defect?"	→ **articular-sided SS tear**
"Poor tissue, >6 mm footprint PT tear?"	→ **complete tear + repair**
"PASTA, 3 mm footprint exposed?"	→ **debridement + SAD**
"True statement about PT tears?"	→ **bursal-sided progress more**
"Which PT tear needs repair over debridement?"	→ **bursal 4 mm involving anterior cable**
"High-grade PASTA failed 9 mo?"	→ **tear completion + repair**
"Lateral portal sees intra-articular structures in pitcher?"	→ **rotator cuff tendon tear**
"Phase of eccentric cuff failure?"	→ **deceleration (E)**
"16 y pitcher, undersurface tear, untreated?"	→ **stop all throwing 6 wk**
"50 y, pain after 6 wk immobiliser post-dislocation?"	→ **rotator cuff tear** (TO)
"65 y, pacemaker, arthrogram dye in subacromial space?"	→ **rotator cuff repair**
"Can't abduct 10 days after dislocation — confirm with?"	→ **MRI shoulder** ✗ Wrong options: EMG · CTA
"60 y, can't elevate + excess ER after dislocation — next?"	→ **MRI** (EMG only if negative)
"8 wk after dislocation, can't elevate — cause?"	→ **rotator cuff tear** ✗ Wrong options: axillary · spinal accessory
"52 y, 3/5 empty can + ER, intact sensation — next?"	→ **MRI shoulder**
"45 y, SS tear 6 wk after dislocation?"	→ **rotator cuff repair**
"Massive tear most amenable to complete repair?"	→ **45 y with anterior dislocation**
"70 y, 12 wk weak after dislocation, sensation intact?"	→ **MRI shoulder; rotator cuff tear**
"Expected injury after dislocation at 52 y?"	→ **rotator cuff tear (B)**
"65 y dislocation — structure most likely injured?"	→ **infraspinatus** (SIT) ✗ Wrong options: axillary · SSN · SSc

– – Active space – –

"Structure most likely needing surgery after dislocation at 52 y?"	→ **supraspinatus**
"Asymptomatic tears >60 y?"	→ **30–55%**
"72 y, incidental 7 mm FT tear on lipoma MRI?"	→ **observation**
"True about progression?"	→ **larger tears progress faster**
"Nonop chronic tears — true?"	→ **linear correlation size ↔ progression**
"SS tear most commonly located?"	→ **15 mm posterior to biceps**
"Atraumatic FT tear + PT — likely outcome?"	→ **subjective functional improvement**
"PASTA, carpenter, 6 wk, full strength?"	→ **physical therapy**
"47 y subacute crescent SS tear, type 1 acromion?"	→ **rotator cuff repair** ✗ Wrong options: + acromioplasty
"Low-grade PASTA + impingement after failed nonop?"	→ **SAD + cuff debridement**
"Strongest predictor of surgery, atraumatic FT tear?"	→ **patient expectation of PT**
"40 y skier, massive SS + SSc tear, 5 mo, no atrophy?"	→ **rotator cuff repair**
"70 y partial bursal IS tear improving with home exercise?"	→ **formal PT**
"PASTA failed 12 wk PT + CSI?"	→ **repair or debridement**
"Traumatic FT tear — factor most affecting outcome?"	→ **time to surgical repair**
"True about RCR biology?"	→ **bone anchor drilling ↑ vascularity**
"Repair failure at 2 mo, biocomposite DR?"	→ **suture pull-out from tissue**
"Fall 3 wk after repair, 1 cm retear, 52 y?"	→ **revision RCR**
"Early failure mechanism?"	→ **tissue failure**
"DR biomechanical advantage?"	→ **higher ultimate tensile load**
"DR vs SR — true?"	→ **↓ retear with DR** ✗ Wrong options: function · pain · healing time
"Arthroscopic vs open, 2.8 cm tear?"	→ **no difference in retear rate**
"Lowest retear risk?"	→ **Goutallier 1 + DR TOE**
"Constructs A (SR) vs B (DR) at 1 y?"	→ **equivalent outcome scores**
"DR vs SR clinically?"	→ **similar short-term function**
"Figure C (DR) vs B (SR)?"	→ **improved healing + lower retear**
"Workers' comp outcomes?"	→ **less functional improvement + lower satisfaction**
"73 y smoker — highest non-healing risk?"	→ **73 years of age (>65)** ✗ Wrong options: smoking · 4 wk timing · workers' comp · dominance
"Goutallier 4, DR, delayed rehab — retear risk?"	→ **Goutallier grade**

"CSI variable linked to infection?"	→ **injection within 1 month**
"Work-related cuff injury outcome?"	→ **poorer outcomes**
"6 CSIs, surgery in 3 mo?"	→ **↑ revision, not infection**
"Labourer — return to full duty?"	→ **6–10 months**
"Prone scapular-plane elevation exercise targets?"	→ **lower trapezius** (TO)
"Large tear, mini-open transosseous — rehab?"	→ **early PROM, AROM at 6 wk**
"No PT visits for 4 wk post-op?"	→ **won't affect ROM at 1 y**
"Early PROM vs 6 wk immobilisation (RCTs)?"	→ **equivalent functional outcomes**
"Minimal cuff EMG activity?"	→ **passive forward flexion** ✗ Wrong options: pulley-assisted
"1 mo after 2-tendon DR repair?"	→ **continue PROM, no strengthening**
"Early vs delayed ROM — true?"	→ **no difference in terminal ROM**
"SCR contraindication?"	→ **irreparable subscapularis** (TO)
"35 y pseudoparalysis, AHD <6 mm — least preferred?"	→ **reverse shoulder arthroplasty** (TO)
"Positive tangent sign predicts?"	→ **irreparable SS from fatty atrophy** (TO)
"Optimal LD transfer candidate?"	→ **36 y labourer, massive tear + SS atrophy**
"NOT a worse-outcome factor after LD transfer?"	→ **absence of CA ligament**
"Best LD indication by ROM?"	→ **50 y, FE 100°, ER −10°**
"47 y landscaper, ER lag + Hornblower, intact SSc?"	→ **latissimus dorsi transfer**
"Retear 2 mo post-op, ER lag?"	→ **revision SS/IS repair**
"50 y irreparable — best isolated LD candidate?"	→ **man with active elevation 90°**
"SS won't reach GT, posterior cuff does?"	→ **partial repair**
"Pre-op factor → poor LD result?"	→ **positive lift-off**
"Nerve 3 cm medial to LD insertion — lost muscle?"	→ **EPL** (radial)
"Patch-like lateral pain, ER weak, belly press intact — transfer?"	→ **latissimus dorsi**
"LD transfer indicated?"	→ **30 y carpenter, massive tear + fatty infiltration (A)**
"Finding that ↓ thoracodorsal-innervated transfer benefit?"	→ **elbow drops back on belly press**
"Pseudoparesis over pseudoparalysis?"	→ **FE 100° after injection**
"Best SCR candidate?"	→ **59 y, SS + IS, normal X-ray (A)**
"Lower trapezius transfer candidate?"	→ **45 y labourer, irreparable posterosuperior, loss of ER**

– – Active space – –

"SCR contraindicated if muscle innervated by label D torn?"	→ **upper subscapular nerve (D)**
"65 y painter, painless pseudoparalysis — mechanism of best procedure?"	→ **lengthening deltoid lever arm** (RTSA)
"No effect on 2-y RCR outcome?"	→ **concomitant distal clavicle resection**
"SSN release with massive repair?"	→ **no difference in function or pain**
"RCR + acromioplasty vs without?"	→ **lower long-term reoperation** ✗ Wrong options: ROM · PROs · strength · retear
"Mobilising posterosuperior cuff — structure at risk?"	→ **suprascapular nerve** (TO)
"10 wk post-RCR, cultures positive day 12?"	→ **Cutibacterium acnes**
"Pneumothorax after interscalene — risk factor?"	→ **35-pack-year smoking**
"Beach chair lower-limb nerve palsy?"	→ **lateral femoral cutaneous**
"Infraspinatus (label 10) tear — exam?"	→ **can't maintain passive ER, elbow 90° (ER lag)** ✗ Wrong options: Hornblower = TM · belly press = SSc · Jobe = SS · Yergason = biceps
"Acromial parameter ↑ RCT risk?"	→ **increased acromial tilt** (TO) ✗ Wrong options: type 2 · ↑ LAA · ↑ AHD
"Most common acromion type for impingement?"	→ **type III** (TO) (type II = commonest overall)
"Acromion index NOT measured on outlet view?"	→ **acromion index** (true AP) (TO)
"Sourcil sign?"	→ **rotator cuff tear arthropathy** (TO)

SCAPULAR WINGING

– Shoulder & Elbow · 42 / 60 –

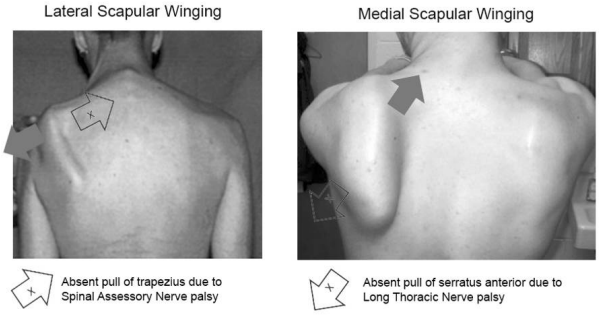
1 Basics

Definition & types

Definition dysfunction of **scapular stabilising muscles** → imbalance + abnormal scapular motion; rare, debilitating

Named by direction of the **superomedial corner** of the scapula

	Medial winging	Lateral winging
Muscle	**Serratus anterior**	**Trapezius** (rarely rhomboids)
Nerve	**Long thoracic (C5, C6, C7)**	**Spinal accessory (CN XI)**
Unopposed pull	**Medialising retraction (rhomboids)** + **elevation (trapezius)**	**Lateralising protraction** (serratus, pec major + minor)
Who	**Young athletes**; **more common**	**Iatrogenic** (neck surgery)
Commonest cause	**Repetitive stretch/traction** (>50%)	**Iatrogenic** (**up to 70%**)



Lateral (trapezius/CN XI) vs medial (serratus/LTN) winging

Anatomy – scapulothoracic stabilisers

Scapulothoracic articulation **not a true joint**; attached via **AC joint** ligaments; **suction** by serratus anterior + subscapularis holds scapula to thorax

– – Active space – –

Scapula attachment for **17 muscles**

Muscle	Origin → insertion	Action	Nerve
Serratus anterior	**Ribs 1–8** (Orthobullets) · top **8–9 ribs** (OB); 7–10 slips → anteromedial (vertebral) border	**Protraction** (+ upward rotation, stabilises vertebral border)	**Long thoracic (C5–7)**
Trapezius	Superior nuchal line, EOP, nuchal ligament, **C7–T12** spinous → lateral ⅓ clavicle, acromion, spine	**Upward rotation + elevation**	**Spinal accessory (CN XI)**
Rhomboid major	**T2–T5** spinous → medial border below minor	**Retraction** (+ elevation, OB)	**Dorsal scapular**
Rhomboid minor	**C7–T1** spinous → medial border at base of spine	Retraction	**Dorsal scapular**
Levator scapulae	**C1–C4 transverse processes** → medial border at spine level	Elevation + downward rotation	**C3–C4 cervical plexus** + dorsal scapular

Serratus parts upper: downward rotation (glenoid down), stabilises superior scapula · middle: protraction · lower: **upward rotation + abduction**

Serratus blood supply long thoracic artery (superior) · **thoracodorsal artery** branches (inferior)

Trapezius parts upper: upward rotation + elevation · middle: **retraction + adduction** · lower: downward rotation + depression

Long thoracic nerve

Roots **ventral rami C5, C6, C7 · C4 contribution 13% · no C7 in 8%**
✗ Wrong options: C8–T1 · C4–6 · C6–8

Course **first nerve to branch** off brachial plexus · posterior to axillary vessels + plexus · **superficial to serratus**, branch to each slip

Vulnerability **long course, small diameter, little connective tissue** · easily injured in **axillary dissection**

Crow's foot where LTN crosses the dominant inferior **serratus branch of thoracodorsal artery**

Spinal accessory nerve

CN XI spinal + cranial parts → accessory trunk → **jugular foramen** → crosses **lateral to IJV** → **SCM** → **posterior triangle** → trapezius · also ventral rami **C2–C4** (OB)

Vulnerability **superficial course in posterior triangle**

Dorsal scapular nerve – rhomboid & levator

Medial scapular muscles innervated by **dorsal scapular nerve** (TO)
✗ Wrong options: median · ulnar · musculocutaneous · superior/inferior subscapular · thoracodorsal

Dorsal scapular nerve (TO) **anterior ramus of C5** → **levator scapulae + rhomboids** · OB: **C4 + C5 (mostly C5)**; less commonly causes winging

Medial-border origins (TO) posterior surface: **levator scapulae, rhomboid major + minor** (deep extrinsic) · anterior surface: **teres major** (TO) · **teres minor = lateral border** (TO)

Attachments (TO) levator → **transverse processes of upper cervical** · rhomboids → **spinous processes of upper thoracic**

Fossa muscles (TO) supraspinatus, infraspinatus, subscapularis from their fossae

Nerve (TO, OB)	Origin	Muscles	Winging?
Dorsal scapular	**C5 anterior ramus**	**Levator, rhomboids**	Less common
Suprascapular	**Superior trunk** (C5–6)	Supraspinatus, infraspinatus	No
Upper (superior) + lower (inferior) subscapular	**Posterior cord** (C5–6)	Subscapularis, **teres major**	No
Thoracodorsal	**Posterior cord** (C6–8)	**Latissimus dorsi**	No
Axillary	**Posterior cord**	Deltoid, teres minor	No
Long thoracic	C5–7	Serratus	**Medial**
Spinal accessory	CN XI	Trapezius	**Lateral**

2 Mechanism

Medial winging – causes

Cause	Detail
Traction (>50%)	**Repetitive stretch = most common** (gradual) — head tilted away during overhead activity, **throwing, swimmers, weightlifters, volleyball**, prolonged abduction · acute high-energy (motorcycle)
Compression – acute	Blunt chest/head/neck trauma (contact sports, **MMA blows**), sudden shoulder depression (fall)
Compression – chronic	**Scalenes** · **subcoracoid** (coracoid–1st/2nd rib; carrying loads) · inflamed bursae · anteroinferior scapular border
Iatrogenic (10%)	**Radical mastectomy / axillary node dissection**, thoracic surgery (**flail chest fixation**), **chest tube**
Penetrating	Chest wall
Neuralgic amyotrophy (Parsonage-Turner)	**Isolated LTN palsy in 22%**; no trauma, antecedent immune/inflammatory insult
Mechanical	**Serratus avulsion** · displaced **inferior pole scapula fracture**
Muscle inhibition	Underlying GH pathology (cuff, labrum — Kibler dyskinesia), fatigue from repetitive tensile use

Most common cause conflict **repetitive traction** (Orthobullets, OB) · **Parsonage-Turner = most often associated** (OB6211) · Parsonage-Turner **"very rarely"** (OB7315) — read the options; nerve = **long thoracic** either way

Vastamäki (OB) 92 isolated serratus palsies: **exertion 40%**, trauma 24%, surgery/anaesthesia 11% · **infection → better recovery**, **surgery/anaesthesia → poorer**

– – Active space – –

Lateral winging – causes

Iatrogenic (most common)	**cervical lymph node biopsy**, **radical neck dissection**, posterior cervical surgery, carotid endarterectomy
Traumatic	traction (**sudden lateral neck flexion**, MVA), blunt (**deep tissue massage**), penetrating neck, fall in elderly

3 Presentation

Medial winging – long thoracic nerve/serratus anterior

Symptoms	**vague, nonspecific shoulder girdle pain + fatigue** (base of neck, scapula, deltoid) · spasms · weak **lifting away from body/overhead** · discomfort sitting against chair · subjective instability (no stable platform) · ± traction plexus symptoms
Deformity	**inferomedial scapula elevates, protrudes posteriorly + medially**; scapula superior + medial with **inferior pole rotated medially** · **worse with forward flexion / wall push-up**
Mechanism of deformity	**compensatory overpull of rhomboids (major + minor) + levator scapulae** (✓ OB) · unopposed **trapezius** pulls medially (antagonist)
Exam	abduction often **≤90°** · weak **forward flexion + abduction** · wasting of anterior scalene triangle (SCM atrophy)
Scapular compression test	examiner pushes scapula anteriorly + stabilises chest during flexion → **positive = ↑ motion** with support
Scapular assistance test (OB)	upward pressure on **inferomedial angle** during elevation → **↓ pain + ↑ arc** = positive
✗ Exam distractors	ER lag/hornblower (teres minor) · **belly press** (subscapularis) · **apprehension** (anterior instability) · **Wright's** (TOS)

Lateral winging – spinal accessory nerve/trapezius & Eden-Lange transfer

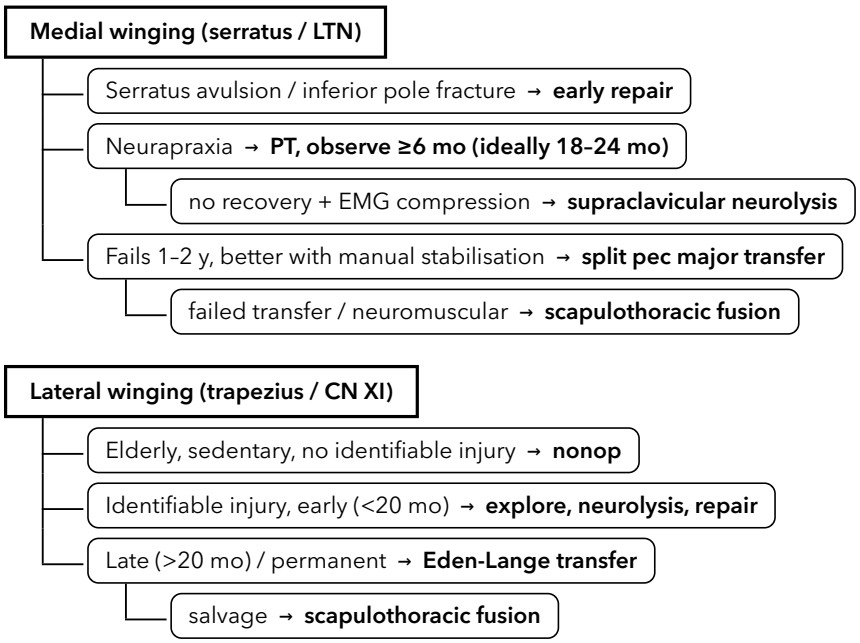
Symptoms	**similar to medial** + **impingement** (CA arch translates inferiorly as scapula depresses)
Deformity	**superomedial scapula drops + protrudes posterior and lateral**; **inferior angle rotates laterally** · **worse with abduction + resisted ER**
Exam	**drooping/depressed shoulder girdle** · **trapezius atrophy** (flattened superior trapezius) · weak flexion + abduction · **manual scapular stabilisation ↓ pain**

4 Diagnosis & Classification

Key radiology findings

EMG medial	LTN palsy vs **mechanical** (serratus avulsion); **serial every 3 mo** to follow recovery
EMG lateral	isolated CN XI vs wider neurological injury; **repeat every 3–6 mo**
EMG → which nerve	medial winging + abnormal EMG = **long thoracic (C5–7)**; trapezius denervation = **spinal accessory**

5 Management



Nonoperative – medial

Regimen	observation, PT (**serratus anterior strengthening**, stretching), avoid heavy lifting · modified thoracolumbar brace (**poor compliance, little benefit**)
Duration	**≥6 months**, ideally **18 mo–2 y**
Recovery	majority resolve **by 2 y** (Orthobullets; **within 24 mo**, Martin, OB) · **within 1 y** (OB) · **~25% need surgery** (OB)

Operative criteria

Procedure	Indication
Early serratus repair	**Mechanical avulsion** / inferior pole fracture + symptomatic winging
LTN neurolysis	Failed nonop **≥6 mo** + EMG **compression** (distal latency, denervation)
Split pec major transfer	Failed nonop **1–2 y** + **relief with manual scapular stabilisation**
Surgery (OB)	Symptomatic serratus winging **>1 y + EMG denervation** → fusion, fascial sling, muscle transfer
Eden-Lange	CN XI injury **diagnosed late (>20 mo)**
CN XI exploration/ repair	**Identifiable injury diagnosed early** (<20 mo)
Scapulothoracic fusion	**Diffuse neuromuscular disorders**, **failed muscle transfer**; not first choice; goal **pain relief**

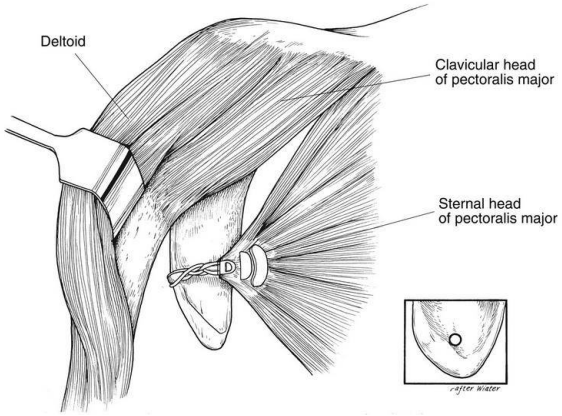
LTN neurolysis

Technique	**supraclavicular decompression** where nerve traverses scalenes
Outcome	**98%** pain relief + winging resolution after failed nonop · **better strength (flexion, abduction) than muscle transfer**

– – Active space – –

Split pectoralis major transfer

Technique	**sternal head** of pec major → **inferior (inferomedial) border/pole of scapula**; direct (± bone block) or with **hamstring/allograft** extension · **most effective** for medial winging · dynamic stabiliser
Predictor of success	**symptom relief with preop manual scapular stabilisation**
Outcome	persistent **abduction weakness** common
Complications	transfer failure at scapula · **breast asymmetry** · infection · adhesive capsulitis



Split pectoralis major transfer

Nerve transfer – medial

Donors	**lateral branch of thoracodorsal nerve → LTN** · **medial pectoral + sural graft → LTN**; preserves muscle biomechanics · nerve/muscle transfers **mixed results** (OB)

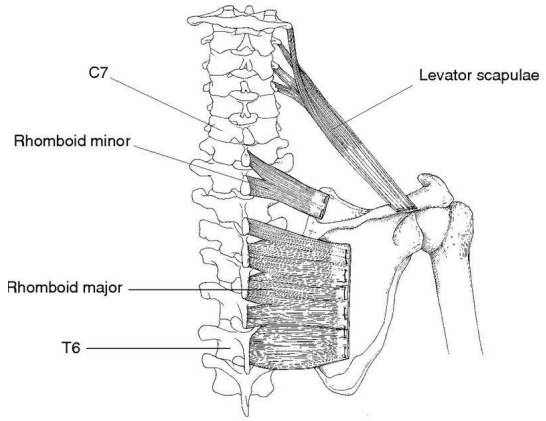
Lateral – nonoperative

Role	**controversial** (most injuries = iatrogenic direct → surgery) · nonop for **elderly, sedentary, no identifiable injury** · **less effective than for serratus palsy**
Poor prognosis with nonop	**can't raise arm above shoulder** at presentation · **dominant arm**
Delay	marked delay in recognition common

Eden-Lange transfer

Classic	**levator scapulae → lateral scapular spine** + **rhomboid minor + major → infraspinatus fossa** (lateralises: medial border → lateral)
Triple tendon transfer	all three to **scapular spine** (Rm medial → RM → LS lateral) to reconstruct trapezius
Results	**reliably restores ROM + function** · **better than scapulothoracic fusion** (OB)

(✗ Wrong options: **Distractors** – split pec major (= serratus/medial)) · long head triceps / infraspinatus transfers (rarely used) · latissimus transfer · levator only



Eden-Lange: levator, rhomboid minor and major transferred laterally

Scapulothoracic fusion

Technique	anterior scapula to posterior ribs (wire cables, plates/screws)
Outcome	**~20° gain in abduction** · **82% satisfied at 5 y**
Complications	nonunion · **pleural effusion** · adhesive capsulitis · symptomatic hardware

X vs Y

	Medial winging	**Lateral winging**
Scapula	Inferior pole **medial**, scapula elevated	**Drooping**, inferior angle **lateral**, depressed
Worse with	**Forward flexion / wall push-up**	**Abduction + resisted ER**
Atrophy	–	**Trapezius**
Nonop	**Effective** (most recover)	**Less effective**
Muscle transfer	**Split pec major (sternal head)**	**Eden-Lange (levator + rhomboids)**
Nerve transfer	Thoracodorsal lateral branch → LTN	

Nerve injury	**Result**
Long thoracic	**Medial winging**
Spinal accessory	**Lateral winging**
Thoracodorsal	Latissimus weakness, no winging
Suprascapular	Supra/infraspinatus wasting, no winging
Upper/lower subscapular	Subscapularis/teres major, no winging
Axillary	Deltoid/teres minor, no winging
C8–T1	No winging

How it's asked

"Long thoracic nerve injury causes?"	→ **medial scapular winging** ✗ Wrong options: lateral · teres minor/ infraspinatus atrophy
"Swimmer, medial winging, abnormal EMG – roots?"	→ **C5, C6, C7**

– – Active space – –

"MMA fighter, blows to torso, medial winging – deficient muscle?"	→ **serratus anterior (D)** ✗ Wrong options: trapezius · teres major · latissimus · infraspinatus
"Medial winging + abnormal EMG – nerve?"	→ **long thoracic** ✗ Wrong options: suprascapular · axillary · thoracodorsal
"Chest wall stabilisation for flail chest, inferior angle translates medially?"	→ **long thoracic nerve** ✗ Wrong options: spinal accessory · C8–T1 · subscapular
"Quarterback, medial winging, PT – additional finding?"	→ **inferomedial scapular support ↓ pain, ↑ ROM (scapular assistance test)**
"Most common condition causing winged scapula – nerve?"	→ **long thoracic**
"Clinical photo of medial winging – nerve/muscle?"	→ **long thoracic / serratus anterior** ✗ Wrong options: dorsal scapular/ trapezius
"Nerve to muscle not functioning in medial winging?"	→ **long thoracic** ✗ Wrong options: dorsal scapular · thoracodorsal · lower subscapular
"Swimmer, wall push-up winging, EMG – nerve on plexus diagram?"	→ **A = long thoracic** ✗ Wrong options: medial pectoral · thoracodorsal
"True of wall push-up winging?"	→ **compensatory overpull of rhomboids + levator scapulae** ✗ Wrong options: C8–T1 · Eden-Lange · node biopsy · retraction deficit
"Winging after neck exploration, trapezius denervated?"	→ **lateral winging, spinal accessory injury**
"72-y fall, lateral winging – nerve?"	→ **cranial nerve XI** ✗ Wrong options: XII · C3–5
"Cervical node biopsy, winging at rest + at 90° flexion?"	→ **spinal accessory**
"Posterior cervical surgery, lateral winging – best restores function?"	→ **rhomboids + levator transfer (Eden-Lange)** ✗ Wrong options: split pec · fusion
"Node biopsy 2 y ago, trapezius denervated – surgery?"	→ **lateral transfer of levator + rhomboid minor + major** ✗ Wrong options: pec transfer · levator only · latissimus
"75-y after ENT node biopsy, winging, untreated – nerve?"	→ **spinal accessory**
"Medial scapular muscles innervated by?"	→ **dorsal scapular nerve** (TO) ✗ Wrong options: subscapular · thoracodorsal · musculocutaneous

SCAPULOTHORACIC DYSKINESIS

– Shoulder & Elbow · 43 / 60 –

1 Basics

Definition	**abnormal scapular motion** during coordinated scapulohumeral movement → impingement + dysfunction · rare
Who	athletes, **baseball pitchers** · sequela of prior shoulder injury disrupting **scapular stabiliser activation**

Aetiology

Causes (multifactorial)	neurologic injury · thoracic **kyphosis** · **periscapular muscle fatigue** · **poor throwing mechanics** · pain (shoulder, neck)

2 Mechanism

Pathoanatomy	scapulothoracic **power imbalance → scapular protraction** → altered GH mechanics → ↑ stress on **anterior capsule + posterosuperior labrum**
Injury risk	labrum · rotator cuff (**articular-surface tears**) · capsule
Thrower's cascade	**posteroinferior capsular tightness** + scapular dyskinesis → articular-side cuff tear + **posterosuperior labral tear**

SICK scapula

SICK muscular **fatigue syndrome** of throwers

Letter	Meaning
S	**Scapular malposition**
I	**Inferior medial border prominence**
C	**Coracoid pain + malposition** (tender coracoid = **pectoralis minor contracture**)
K	dys**K**inesis of scapular movement

GIRD **loss of internal rotation** (e.g. 15°) = posterior capsule tightness

3 Presentation

Symptoms	shoulder pain/dysfunction **worse with arm elevation** · **loss of throwing velocity** · posterosuperior ± anterior pain
Exam	**scapulothoracic crepitus** · scapula **lower + protracted** · **symptoms relieved by scapular stabilisation** (diagnostic)
Associated	coracoid tenderness · IR deficit · positive relocation test

– – Active space – –



Wall push-up – protracted, winged scapula

4 Diagnosis & Classification

Diagnosis **clinical**

Key radiology findings

X-ray (complete shoulder series) **usually unremarkable**

MRI often no intra-articular lesion



Scapular asymmetry with arm abduction

5 Management

- **Scapular dyskinesis**
  - Mainstay → **NSAIDs, PT, local injections**
  - Rehab (Kibler, proximal → **distal)**
    - Stage 1 → **full scapular motion + coordinate scapula with trunk + hips**
    - Stage 2 → **strengthen scapular muscles**
    - Stage 3 → **shoulder flexibility, closed-chain, sport-specific**

Rehabilitation – kinetic chain, pectoralis minor & posterior capsule stretching

Principle	**proximal → distal** (kinetic chain); progress by **function, not timetable**; acute → recovery → maintenance phases
Initial stage	**coordination of scapular motion with trunk + hip movements** ✗ Wrong options: isometric · isokinetic · closed chain · axial loading (later)
PT emphasis	**core** strengthening · scapular stabilisers (**serratus anterior, trapezius**) · rotator cuff · proper mechanics in throwers
SICK / GIRD initial regimen	**stretch posterior capsule** (IR deficit) **+ pectoralis minor** (coracoid tenderness)

✗ Regimen distractor	Why wrong
Posterior capsule + **subscapularis strengthening**	Misses pec minor (coracoid pain)
Posterior capsule + **plyometrics**	Misses pec minor (coracoid pain)
Anterior capsule stretching (any)	IR deficit = **posterior** capsule tightness

How it's asked

"Pitcher, posterior pain, scapular dyskinesis, normal MRI – emphasise first in rehab?"	→ **coordination of scapular motion with trunk + hip movements** ✗ Wrong options: isometric · isokinetic · closed chain · axial loading
"Pitcher, 15° IR loss, coracoid tenderness, + relocation, normal MRI – initial regimen?"	→ **stretch posterior capsule + pectoralis minor** ✗ Wrong options: subscapularis strengthening · plyometrics · anterior capsule stretching
"SICK scapula – coracoid pain due to?"	→ **pectoralis minor contracture**
"Diagnostic exam finding?"	→ **symptoms relieved by scapular stabilisation**
"Scapular position in dyskinesis?"	→ **lower + protracted**

– – Active space – –

SHOULDER ARTHRODESIS

– Shoulder & Elbow · 44 / 60 –

1 Basics

Goal

Aim	**stable base** for the upper limb → optimise **hand + elbow** function
Ideal patient	**flail shoulder** (deltoid + cuff paralysis) with **functional hand, wrist + elbow**; young, esp. **labourer**

Anatomy

GH fusion surface	small → decorticate **GH articular surfaces + humeral head–acromion undersurface** to ↑ fusion area
Fixation strength	only **glenoid fossa + base of coracoid** strong enough for screws

3 Presentation

Symptoms	**specific to underlying condition**; symptomatic GH dysfunction

4 Diagnosis & Classification

Key radiology findings

Test	Purpose
X-ray AP, lateral, axillary	Bone stock for fusion, deformity
CT	**Glenoid bone loss** (esp. failed arthroplasty)
EMG	When **neurological status of scapular muscles** is ill-defined

5 Management

Indications – deltoid and rotator cuff paralysis

Indication	Note
Paralytic disorders	**Deltoid + cuff paralysis** (post-traumatic), **brachial plexus palsy**, paralysis in infancy
Irreparable deltoid + cuff deficiency with arthropathy	
Failed TSA / failed revision arthroplasty	Salvage
Tumour resection	Bone deficiency
Chronic infection	Painful ankylosis
Recurrent instability	Failed previous repairs (severe, refractory)

Rühmann (OB) 43 cases: paralysis **74%**, OA/humeral head necrosis 35%, infection 16%, instability

Contraindications

List **ipsilateral elbow arthrodesis** · **contralateral shoulder arthrodesis** · **no functional scapulothoracic motion** · **trapezius, levator scapulae or serratus anterior paralysis** · **Charcot** in acute inflammatory stage (**Eichenholtz 0–2**) · **elderly** · **progressive neurological disease**

Operative criteria – who benefits vs alternatives

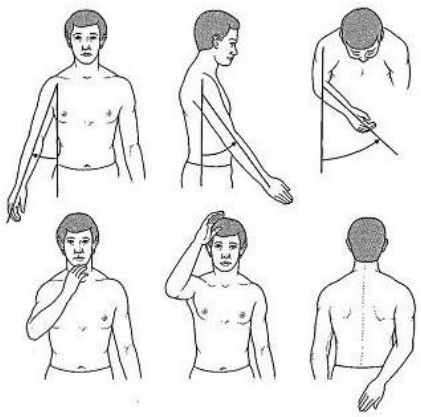
Patient	Best option
Flail shoulder (deltoid + cuff paralysis), hand works	**Arthrodesis** ✓
No cuff function	No **TSA** (contraindicated)
No deltoid function	No **reverse TSA** (contraindicated)
Chronic lateral winging	Eden-Lange (levator + rhomboids)
Irreparable cuff tear, overhead motion	Latissimus dorsi transfer
Young AVN	Hemiarthroplasty
Recurrent MDI after arthroscopy	Therapy, revision arthroscopic/open stabilisation
Infected TSA removed (70-y)	Treat infection → **2nd-stage TSA / HA / reverse**
Cleidocranial dysostosis	**Scapulothoracic** (✗ Wrong options: glenohumeral) arthrodesis

Fusion position

Goal **hand to mouth** (feeding) + **groin** (hygiene)

Source	Abduction	Forward flexion	Internal rotation
Orthobullets "30-30-30"	**20–30°**	**20–30°**	**20–30°**
OB typical	**~40°**	**~35°**	**~20°**
Rühmann range (OB)	20–60°	20–40° (anteversion)	0–50°; **no ER**
Orthobullets GH anatomy note	15–20°	20–25°	40–50°

Positions conflict 30-30-30 (Orthobullets) · 40-35-20 (OB) – read the options



Arthrodesis position: reach mouth, head, back

– – Active space – –

Technique

Approach	**S-shaped incision**: scapular spine → over acromion → anterolateral arm
Steps	resect cuff from proximal humerus · **tenodese biceps** · decorticate glenoid, humeral head + acromial undersurface · position 30-30-30
Fixation	**10-hole 4.5 mm pelvic reconstruction plate** along **scapular spine → acromion → humeral shaft** · **compression screws across GH into glenoid fossa** · screw into **base of coracoid**

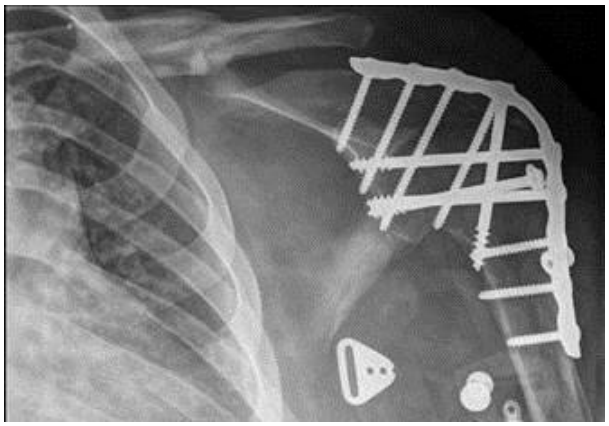


Plate fixation of shoulder arthrodesis

Postoperative care

0–6 wk	**thermoplastic orthosis** from day 1
6 wk	**sling** if no radiographic loosening
3 mo	**mobilisation + thoracoscapular strengthening** if no loosening
Recovery	**6–12 months**; marked ↑ pain + function in selected patients

6 Complications

List	**infection** · **nonunion** · **malposition** · **prominent hardware** (pain) · **humeral shaft / periarticular fracture**

How it's asked

"45-y, deltoid + cuff paralysis 5 y after motorcycle injury, hand/wrist work?"	→ **shoulder arthrodesis** ✗ Wrong options: Eden-Lange · TSA · reverse TSA · latissimus transfer
"Who benefits most from GH arthrodesis?"	→ **30-y labourer with deltoid + cuff paralysis** ✗ Wrong options: AVN · MDI · infected TSA removed · cleidocranial dysostosis
"Position of shoulder fusion?"	→ **30° abduction, 30° flexion, 30° IR** – hand to mouth (OB: ~40/35/20)

– – Active space – –

SHOULDER ARTHROSCOPY: INDICATIONS & APPROACH

– Shoulder & Elbow · 45 / 60 –

1 Basics

Indications

Indications diagnostic · loose bodies · **cuff repair/debridement** · labral/SLAP/ instability repair · **subacromial decompression** · AC joint / **distal clavicle resection** · **suprascapular nerve release** · capsular release (contracture) · synovectomy · **biceps tenotomy/tenodesis**

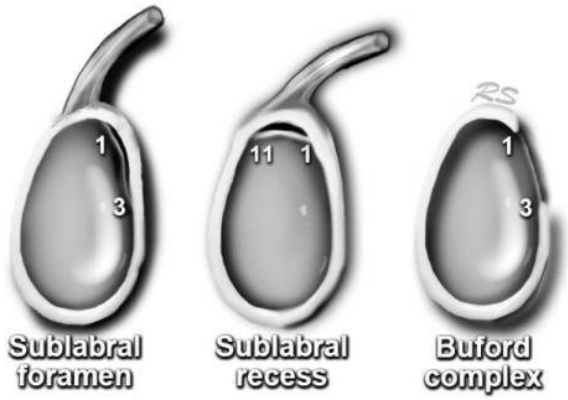
Diagnostic scope – normal anatomy & variants

Diagnostic scope **30° scope via posterior portal**: biceps, SS/IS/TM, **rotator interval**, MGHL/ IGHL, **subscapularis recess (loose bodies)**, anterior labrum, glenoid, humeral head

Rotator interval **biceps tendon + superior edge of subscapularis + glenoid**

Anterosuperior labrum + MGHL **widest anatomic variability**

Variant	Labrum	MGHL
Attached labrum	Attached	**Broad – most common**
Sublabral hole (foramen)	Sublabral hole	**Cord-like**
Buford complex	**Absent** anterosuperior labrum	**Cord-like**



Sublabral foramen, sublabral recess, Buford complex

Normal bare areas **central glenoid** + **posterior humeral head**
✗ Wrong options: pathology

– – Active space – –

5 Management

Positioning

	Beach chair	Lateral decubitus
Orientation	Upright, **anatomic**	Glenoid parallel to floor (non-anatomic)
Traction	**None → ↓ neurapraxia**	**10–13 lb** (avoid over-distraction) → **traction neurapraxia** (NV + soft tissue)
Arm	Mobile; arm holder possible	**30–60° abduction + 20–30° forward flexion**
Conversion to open	**Easy** (deltopectoral; no re-prep)	Must reposition/redrape
Bleeding	**↓ venous pressure + bleeding**	–
Cerebral perfusion	**↓ (hypoperfusion, CDEs, stroke, death)**	**Better**
Hypotension / bradycardia	**↑ risk**	No ↑
Posterior portals	**Difficult** (head in the way)	Easy
Labrum	Less capsular distortion	**Traction accentuates labral tears**
Anaesthesia	**Regional + sedation possible**	Not ideal if regional not tolerated
5 o'clock portal	**Safer** (Gelber)	**Axillary + MCN closer**

Lateral position (TO) **vacuum beanbag + kidney rest**; **10–13 lb** traction; **30–60° abduction, 20–30° forward flexion**

Lateral modification (TO) tilt patient **20–30° posteriorly** → glenoid parallel to floor → **less traction** (↓ plexus neurapraxia), **labral tears accentuated** (pulled away from bed), **better access to inferior third** of labrum/capsule

Beach chair advantages (TO) faster positioning · **↓ neurapraxia (no traction)** · less capsular distortion · arm mobility · easy open conversion

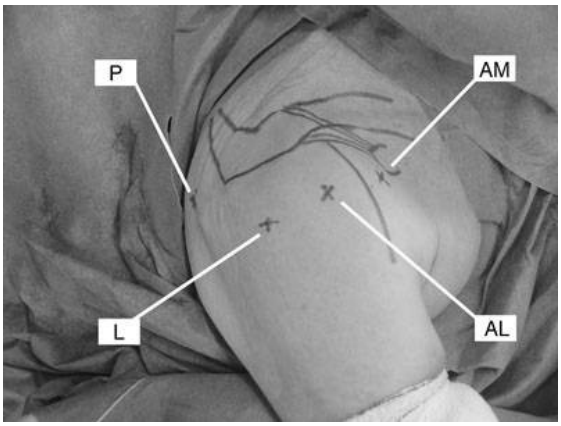
Beach chair disadvantages (TO) difficult posterior portals · **↓ cerebral perfusion with hypotensive anaesthesia → stroke, death**

✗ Wrong options: **EXCEPT (TO)** – "beach chair → **more** brachial neurapraxia" is **false**

Primary portals

Portal	Location	Function	Passes / risk
Posterior	**2 cm inferior, 1 cm medial** to posterolateral acromion; aim at **coracoid tip**	**Primary viewing**; **first portal**	Between **infraspinatus (SSN)** + **teres minor (axillary)** or through IS
Anterior	**Lateral to coracoid**, anterior to AC joint; spinal needle under vision	Viewing + SAD	Between **pec major (pectoral nn.)** + **deltoid (axillary)**
Lateral	**1–2 cm distal** to lateral acromion	**SAD**	Through **deltoid (axillary)**

– – Active space – –



Portal sites: posterior (P), lateral (L), anterolateral (AL), anteromedial (AM)

Secondary portals

Portal	Location	Function	Risk
Anteroinferior (5 o'clock)	Slightly inferior to coracoid	**Anterior labral repair anchors**	**Axillary (> 7 o'clock risk, Meyer)** + MCN
Posteroinferior (7 o'clock)	Under vision, spinal needle	**Posterior labral repair anchors**	**Axillary nerve** (low posterior)
Neviaser (supraspinatus)	**Just medial to lateral acromion**	Anterior glenoid view, **SLAP repair**	Through **supraspinatus (SSN)**; SSN if into **suprascapular notch**
Port of Wilmington (posterolateral)	**Just anterior to posterolateral acromion**	**Posterior SLAP + cuff lesions**	–

Trans-subscapularis portal **avoid** in either position (Gelber)

Placement secondary portals under direct vision from posterior portal with **spinal needle**

6 Complications

Portal-related nerve injury

Structure	Anatomy	At risk when
Axillary nerve	Quadrangular space → deep surface deltoid, **~7 cm below acromion**	Posterior portal **too inferior** (**7 o'clock**) → pain, weakness, ↓ **lateral arm/shoulder** sensation
Suprascapular nerve	Supraspinatus → infraspinatus fossa	Posterior portal **too medial**; Neviaser into notch
Cephalic vein	Deltopectoral groove	Anterior portal **too lateral** – **structure most at risk** (Lo, Meyer)
Musculocutaneous nerve	Enters muscle **2–8 cm distal to coracoid tip**	Anterior portal **too inferior**
Phrenic nerve	–	**Interscalene block**

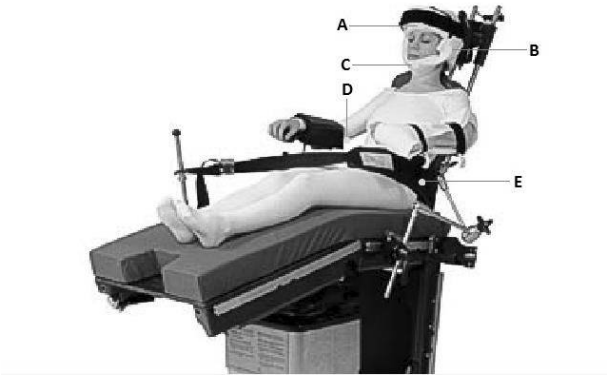
Lo (outside-in) all NV structures **> 20 mm** from all portals

– – Active space – –

No neural structure at risk	lateral superior portal · posterior superior portal

Positioning complications (beach chair, lateral decubitus, obesity)

Pressure point (beach chair)	Nerve	Deficit
A – forehead (tight face mask)	**Supraorbital** (terminal **CN V**)	Paraesthesia **forehead + anterior scalp**
B – mastoid process (mask straps)	**Great auricular (C2–C3)**	Paraesthesia **ear, posterior auricular area, angle of mandible**
C – chin strap	–	**Skin breakdown lip/chin**
D – contralateral elbow	**Ulnar**	Ulnar neurapraxia
E – lateral abdominal belt / ASIS	**LFCN**	Burning/numbness **anterolateral thigh**, above knee, **no weakness**



Beach chair pressure points A–E

Neck position	Complication
Hyperextension (± rotation)	**Stroke** + **CN XII (hypoglossal)** palsy – traction + compression by angle of mandible (✗ Wrong options: face-mask compression)
Hyperflexion	**Spinal cord ischaemia → paraplegia**

Not reported with beach chair	CN III (ptosis, mydriasis, diplopia) · **AIN** (FPL + index/long FDP) · **femoral** (quadriceps, anterior thigh, medial leg)

Cerebral hypoperfusion

CN VI palsy post-op	**lateral rectus** paralysis (can't abduct eye) → cerebral hypoperfusion in beach chair
↓ Risk	**lateral decubitus** · **regional anaesthesia + sedation** (fewer CDEs) · **SBP > 90 mmHg**
Increases or no effect (not protective)	**general anaesthesia (↑ CDE)** · **SBP < 70 mmHg** (↑) · open vs arthroscopic · metal vs bioabsorbable anchors
Stroke	also reported after lateral decubitus (risk small in both)

LFCN palsy (obesity)

Path	LFCN crosses near **ASIS** → compressed by belt / waist flexion / pannus

Risk factors	**obesity (BMI > 30; highest-risk option BMI > 35)** · **prolonged operative time** · **inadequate ASIS padding**
Not risk factors	age > 50 · type of surgery (open/arthroscopic/arthroplasty) · **diabetes** · foam between pannus + thighs (= **protective**)
Prevention	hips **centred at bed break**, **hip flexion ~45°** (less waist flexion), **padding under belt** + between pannus and thighs
Holtzman (397 pts)	LFCN palsy **5.5%**, mostly **transient**; weight + BMI significant
Levy protocol (BMI > 30)	LFCN **5 → 1** case (OB) · **none, even BMI > 40** (another OB)
Obese patients	↑ nerve palsies; post-op ↑ **DVT, infection, wound breakdown, hardware failure, mortality**

X vs Y

	5 o'clock (anteroinferior)	7 o'clock (posteroinferior)
Use	**Anterior** labral anchors	**Posterior** labral anchors
Nerve	**Axillary (more risk, Meyer)** + MCN	**Axillary**

	Sublabral foramen	Buford complex
Labrum	**Sublabral hole**	**Absent** anterosuperiorly
MGHL	Cord-like	**Cord-like**

How it's asked

"Lateral arm numbness + weakness after arthroscopy – portal?"	→ **posterior inferior (7 o'clock)** ✗ Wrong options: Neviaser = SSN · anterior = MCN · lateral/posterior superior = none
"All true about positioning EXCEPT?"	→ **beach chair has more brachial neurapraxia** (TO)
"Lateral decubitus traction?"	→ **10–13 lb**, 30–60° abduction, 20–30° flexion (TO)
"Beach chair pad at mastoid – deficit?"	→ **great auricular: ear, posterior auricular, angle of mandible** ✗ Wrong options: CN III · CN XII · AIN · femoral
"PACU, can't gaze left after beach chair – prevention?"	→ **lateral decubitus position** ✗ Wrong options: GA · SBP < 70 · open · metal anchors
"Obese, burning outer thigh above knee, no weakness – pad where?"	→ **lateral abdominal belt / ASIS (LFCN)**
"Anterolateral thigh numbness after beach chair – biggest risk factor?"	→ **BMI > 35** ✗ Wrong options: age · diabetes · surgery type
"Hypoglossal palsy after beach chair – cause?"	→ **neck hyperextension**

– – Active space – –

"Absent anterosuperior labrum + cord-like MGHL?"	**→ Buford complex**
"Cephalic vein at risk – portal error?"	**→ anterior portal too lateral**

– – Active space – –

SHOULDER EXAM

– Shoulder & Elbow · 46 / 60 –

1 Basics

Exam components

Sequence inspection · palpation · ROM · neurovascular · provocative tests · always **compare both sides**

Inspection skin, scars, symmetry, swelling, atrophy, hypertrophy, **scapular winging**

Palpation C-spine spinous processes, SC joint, clavicle, **AC joint**, acromion, coracoid, scapular spine · paraspinals, periscapular, deltoid, **greater tuberosity (cuff insertion)**, **LHB in groove**

Range of motion

Plane	Normal
Forward elevation	**180°**
Abduction (scapula stabilised)	**90°**
ER at side	**80°**
ER at 90° abduction	Compare sides
IR to vertebral level	**T4–T8**
IR at 90° abduction	Compare sides

ROM rule active vs passive, both sides; normal relative to contralateral

Neurovascular

Root	Motion	Muscle	Reflex
C4	Scapular stabilisation (winging)	Upper serratus anterior	–
C5	Shoulder abduction · elbow flexion palm up	Deltoid · biceps	**Biceps**
C6	Elbow flexion thumb up · wrist extension	Brachioradialis · ECRL	**Brachioradialis**
C7	Elbow extension · wrist flexion	Triceps · FCR	**Triceps**
C8	**Finger flexion, grip, thumb extension**	**FDS**	–
T1	Finger abduction	Interossei (ulnar)	–

Sensation C4–T1 dermatomes · **pulses** brachial, radial, ulnar

– – Active space – –

Anterior band of IGHL – restraint in abduction/external rotation

Ligament	Restrains
Anterior band IGHL	**Anterior translation at 90° abduction + full ER** (moves anteriorly)
MGHL	Anterior translation at **45° abduction**
SGHL + CHL	**Inferior translation of adducted arm** + ER in adduction
Posterior band IGHL	**Posterior translation in IR**

4 Diagnosis & Classification

Impingement & biceps tests (Hawkins, Speed)

Impingement syndrome (TO) anterior pain in **60–120° abduction** (painful arc) · greater tuberosity slides under acromion, bursa smooths it · causes: **subacromial bursitis**, **hooked acromion**, cuff pathology (supraspinatus tendinitis, calcific tendinitis, partial tear), **GT fracture** (TO)

Test	Technique	Positive / meaning
Neer sign	Stabilise scapula, **passive forced forward elevation**	Anterior pain (GT under acromion **70–110°**) · needs **full ROM** · sens **75–86%**, spec **50%** · also + in stiffness, OA, instability, bone lesions
Neer test	Repeat Neer after **subacromial lidocaine**	**Marked pain relief** = subacromial impingement
Hawkins-Kennedy	**Flex 90°, elbow 90°, forcible IR** (GT under CA ligament)	Pain · **sensitive, not specific** · sens **75–92%**, spec **45%** · = **subacromial bursitis** (TO)
Scapular assistance (TO)	Push **inferomedial scapular angle** during elevation (upward rotation + posterior tilt)	**Relief** of impingement pain + ↑ arc
Internal impingement	90° abduction + max ER + extension (late cocking)	**Posterior** throwing pain · relieved by relocation / elbow in front of body



Hawkins-Kennedy test

Neer test injectate — **5 mL 1% lidocaine** (TO) · **4 cc 1% lidocaine + 4 cc 0.5% bupivacaine + 2 cc steroid** (Orthobullets) — read the options

– – Active space – –

Biceps test	Technique	Positive
Speed	Resisted forward elevation, **elbow extended, forearm supinated**	Pain in **bicipital groove**
Yergason	Elbow 90°, pronated → **resisted supination**	Groove pain = biceps tendon pathology
Popeye	Bulge in arm	**LHB proximal rupture**
Groove tenderness	Palpation	Inflamed LHB or **SLAP**

Speed + / empty can – — **bicipital tendinitis** (TO)
✗ Wrong options: supraspinatus tendinitis · SLAP · impingement

Rotator cuff strength & lag tests (lift-off, hornblower, ER lag)

Muscle	Tests
Subscapularis	**Lift-off**, **belly-press / Napoleon**, **bear hug**, **IR lag**, ↑ passive ER
Supraspinatus	**Jobe empty can**, **drop arm (Codman)**
Infraspinatus	**ER lag sign** (specific), drop sign (Hertel), ER resistance stress
Teres minor	**Hornblower (Patte)**, ER strength at 90° abduction

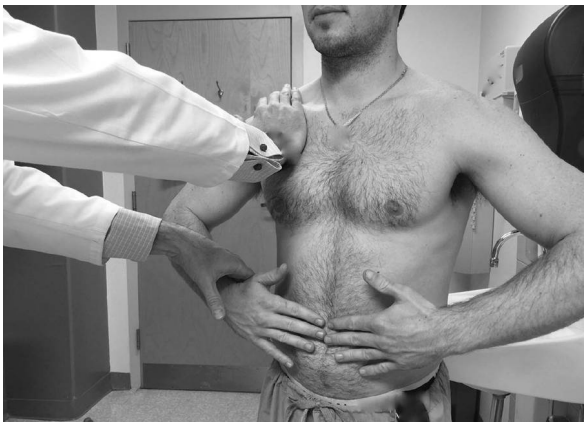
Lift-off (Gerber) (TO) — dorsum of hand on lumbar spine (IR) → can't lift off = **subscapularis tear** · specific for **IR deficit** (TO) · best for **inferior subscapularis** · more accurate if hand reaches opposite scapula · confounded by other muscles
✗ Wrong options: abduction · ER deficit · anterior instability



Lift-off test

Belly-press (Gerber) / Napoleon (TO) — palm presses abdomen, elbow in front of trunk · weak → **elbow drops behind trunk** (shoulder extension) = belly-press · **wrist flexes** = **Napoleon sign** · best for **superior subscapularis**

– – Active space – –



Belly-press test

Bear hug (Burkhart & De Beer) (TO)	palm on **opposite shoulder**, elbow in front, examiner pulls hand away **anteriorly** · can't resist = positive · **most sensitive**: detects tears **<50%** (upper third) missed by lift-off/belly-press (TO) · positive if **≥20% weaker** than other side (Orthobullets)
Most sensitive + specific subscap test	**IR lag sign** (Orthobullets, OB) · **bear hug** most sensitive (TO) — read the options
IR lag sign	behind patient, elbow 90°, shoulder 20° elevation + 20° extension, lift hand off lumbar spine to near-max IR, release wrist → lag
Not subscap testing	**isolated IR at side** (pec major + latissimus contribute)
Subscapularis insertion	**lesser tuberosity → positive bear hug after trauma = lesser tuberosity avulsion** ✗ Wrong options: GT fracture = drop arm · superior migration = CTA · dislocation can't be positioned · AC separation
Jobe empty can (TO)	**90° abduction in scapular plane (30° forward), IR thumb down**, resist downward push · **weakness = supraspinatus tear** (TO) · **pain only** = subacromial bursitis/irritation, tear only with true drop arm (Orthobullets)
Drop arm (Codman) (TO)	arm elevated overhead → lower slowly · sudden pain or uncontrolled lowering = **supraspinatus tear** · passive 90° scapular plane → drop = **most specific for full-thickness tear (98%)** (Orthobullets)
ER resistance stress (TO)	arms at side, elbows 90°, **ER 60°**, resist examiner's IR push → pain/ weakness = **infraspinatus + teres minor**
ER lag sign (TO)	seated, elbow 90°, shoulder **20° elevation in scapular plane**, full ER → hold · angular drop = **infraspinatus tear** (specific) ✗ Wrong options: subscapularis · teres minor · adhesive capsulitis
Drop sign (TO)	same as ERLS but **90° elevation** in scapular plane · infraspinatus · ERLS slightly more sensitive + specific · not the Orthobullets "drop sign" (= drop arm, supraspinatus)
Hornblower (Patte) (TO)	arm **90° abduction in scapular plane**, elbow 90° → attempt ER against resistance · can't = hornblower posture · **most accurate for teres minor** · TO option: "test for **infraspinatus and teres minor**" · Orthobullets: hold 90° abd + 90° ER → **arm falls into IR** · also + in **posterior supraspinatus** tears

– – Active space – –



Positive hornblower's sign

Teres minor **axillary nerve**, ER

✗ Distractor	Tests
Jobe	Supraspinatus weakness ± impingement
Belly-press, IR lag	Subscapularis
Hawkins	Impingement
"Jones test"	not supraspinatus test

Lesser tuberosity fracture – subscapularis testing

Bear hug after ATV accident can't hold palm on opposite shoulder vs anterior pull → **lesser tuberosity fracture** on X-ray

Labral tests (Porcelli)

Test	Technique	Positive
O'Brien (active compression)	**90° flexion, elbow extended, 10–15° adduction, pronated (thumb down / full IR)** → resist downward force · repeat supinated/ER	**Deep GH pain pronated, not supinated = SLAP** · **superficial AC pain = AC joint**
Crank	Abducted, passive rotation + axial load	Click/pain = labrum
Modified dynamic labral shear (TO)	Standing, **abduct 120°** scapular plane, elbow 90°, ER → lower arm keeping ER	**Painful click posterior joint line 120°→90°** = SLAP
Uppercut (TO)	Neutral rotation, elbow 90°, supinated fist → punch to chin vs resistance	Pain = **SLAP**
Porcelli (TO)	See below	**Posterior labral tear**

Porcelli test (TO) Morey 2016 · **posterior labral tear** · **sens 100%, spec 99%**, inter-examiner reliability **>0.8** · outperformed jerk, Kim, load-and-shift · **not yet validated**
✗ Wrong options: SLAP · subscapularis · PASTA

Porcelli technique (TO) part 1: **90° flexion, 10–15° adduction, max IR (thumb down)**, examiner behind stabilises scapula, resisted elevation → note pain/strength · part 2: same, examiner's **thumb on posterior joint line** pushing anteriorly (prevents posterior subluxation) · **pain ↓/gone in part 2 = positive** · pain persists = **AC joint** pathology

– – Active space – –

Instability tests

Direction	Test	Technique / positive
Anterior	**Apprehension** (Jobe, TO)	Supine, **90° abduction + full ER** (behind coronal plane, elbow 90°) → apprehension
Anterior	**Relocation**	Posterior force on head → apprehension relieved
Anterior	**Anterior release**	Release hand in abd/ER → subluxation · **"3 in 1"** (apprehension + relocation also +)
Anterior	Anterior load & shift / drawer	Supine, **40–60° abd** (TO drawer: 60°, neutral rotation, axial load) · grade = laxity, pathologic only if symptomatic
Posterior	**Jerk (Jahnke)**	Sitting/standing, **flex 90°, IR, adducted**, axial load at elbow → posterior jerk; abduct → reduces (TO) · clunk or pain
Posterior	**Kim**	Seated, 90° abduction → flex to 45° + axial load + posteroinferior force · **pain**
Posterior	**Posterior stress**	Flexion, adduction, IR + posterior force → pain + instability
Posterior	Posterior load & shift / drawer	Increased translation vs other side
Posterior	**Loss of ER**	Locked in IR = posterior subluxation/dislocation
Inferior / MDI	**Sulcus**	Pull arm down (TO: at **0° and 45° abduction**) · persists at **2+/3+ with ER at side = pathologic rotator interval**

Translation grade (drawer / load & shift)	Finding
1+	**To glenoid rim**
2+	**Over rim, reduces spontaneously**
3+	**Dislocates / locks out**, no spontaneous reduction

Sulcus grade	Acromiohumeral interval
1+	**<1 cm**
2+	**1–2 cm**
3+	**>2 cm**

AC joint & other tests

Test	Meaning
Cross-body adduction	AC joint pain
AC tenderness	AC pathology
Wright's	**TOS**: abduct + ER, head turned away → **radial pulse lost**
Medial winging (wall push at 90° flexion)	**Serratus anterior / long thoracic nerve**
Lateral winging	**Trapezius / spinal accessory (CN XI)**
Axillary webbing defect	**Pectoralis major rupture**

X vs Y

	Lift-off	Belly-press	Bear hug
Subscap portion	**Inferior**	**Superior**	Upper third, partial <50%
Position	Hand on lumbar spine	Palm on abdomen	Palm on opposite shoulder
Positive	Can't lift off	Elbow drops back / wrist flexes (Napoleon)	Can't resist anterior pull

	ER lag sign	Drop sign (TO)	Hornblower
Elevation	**20°** scapular plane	**90°** scapular plane	**90°** abduction
Muscle	**Infraspinatus**	Infraspinatus	**Teres minor**

	Neer	Hawkins
Manoeuvre	Passive forced elevation, scapula fixed	90° flexion + **forced IR**
Sens / spec	75–86% / 50%	75–92% / 45%

	O'Brien – SLAP	O'Brien – AC
Pain site	**Deep** in GH joint	**Superficial** over AC
Forearm	Pronated, not supinated	Pronated, not supinated

How it's asked

"Full-thickness teres minor tear – positive test?"	→ **Hornblower's sign** ✗ Wrong options: Jobe · belly-press · IR lag · Hawkins
"90° abduction + full ER – ligament resisting anterior translation?"	→ **anterior band IGHL** ✗ Wrong options: CHL · SGHL · MGHL · posterior band
"ATV crash, positive bear hug – X-ray?"	→ **lesser tuberosity fracture**
"Correct statement for test in figure (arm in hornblower posture)?"	→ **Hornblower's test for infraspinatus and teres minor** (TO) ✗ Wrong options: Jones · O'Brien · IR lag
"Muscle tested in figure (hand behind back)?"	→ **subscapularis** (lift-off) (TO)
"Lift-off test is specific for?"	→ **internal rotation deficit** (TO)
"ER lag sign suggests?"	→ **infraspinatus tear** (TO)
"Anterior pain on IR, positive lift-off – muscle?"	→ **subscapularis** (TO)
"Diagram: 90° flexion + forced IR?"	→ **Hawkins-Kennedy test** (TO) ✗ Wrong options: Neer · empty can · ER lag

– – Active space – –

"Positive Hawkins sign indicates?"	→ **subacromial bursitis** (TO) ✗ Wrong options: cuff tear · subscap tear · brachial plexus
"Positive Speed + negative empty can?"	→ **bicipital tendinitis** (TO)
"Porcelli test detects?"	→ **posterior labral tear** (TO)

SHOULDER HEMIARTHROPLASTY

– Shoulder & Elbow · 47 / 60 –

1 Basics

Definition

Hemiarthroplasty (HA)	humeral articular surface replaced with **stemmed humeral component**; glenoid left native
Most common indication	GH arthritis with **inadequate glenoid bone stock** for TSA
Mechanics	relies on **rotator cuff compression** for motion + strength; **CA arch** = barrier to proximal migration
Use	less common since RSA; still for young patients/laborers, CTA, osteonecrosis, complex proximal humerus fractures

4 Diagnosis & Classification

Key radiology findings

Modality	Use
True (Grashey) AP	**30–40° oblique** to coronal plane; extent of DJD, fracture pattern
Axillary	**Posterior glenoid wear**; fracture displacement
CT	**Glenoid version + glenoid bone stock**; unclear fracture pattern
MRI	**Rotator cuff** status

↓ Acromiohumeral distance superior head migration → **cuff deficiency** → evaluate cuff (MRI / CT arthrogram)



Axillary view – posterior glenoid wear

– – Active space – –

5 Management

Indications & limitations (cuff integrity, osteonecrosis, CTA head, ream-and-run)

Indication	Detail
Primary arthritis	**Cuff deficient** · glenoid stock inadequate · **high glenoid loosening risk (young, active laborers)**
Rotator cuff arthropathy	**HA > rTSA if forward flexion > 90°**
Osteonecrosis	**Without glenoid involvement**
Proximal humerus fracture	3-part with poor bone · **4-part** · **head-split** · major articular destruction

Contraindication	Note
Infection	–
Neuropathic (Charcot) joint	e.g. syrinx
Unmotivated patient	–
CA ligament deficiency	CA arch + cuff deficient → **superior (anterosuperior) escape**
RA · capsulorrhaphy arthropathy	Listed as contraindications (one OB)

RA / capsulorrhaphy arthropathy contraindication (OB) · RA results "satisfactory in most, ↓ ROM" (TO) · ream-and-run works in capsulorrhaphy arthropathy (OB) – read the options

Outcomes by diagnosis

Best results (TO) **humeral head AVN / osteonecrosis**: consistent pain relief in **90%**, near-normal ROM

Less good (TO) **RA, OA, glenoid dysplasia, post-traumatic arthrosis**: satisfactory in most, **↓ ROM**

Most influential factor on function **status of the rotator cuff**
✗ Wrong options: CA ligament · glenoid wear · inferior osteophyte · AC arthritis

Irreparable cuff **TSA contraindicated** → HA (limited function) or **RSA**

Fractures **excellent pain relief** in majority; outcome **inversely proportional to age + time from injury to surgery**

HA vs TSA for GH OA

Statement	Truth
HA outcomes with eccentric glenoid wear	**Inferior to TSA** ✓
HA function overall	Inferior to TSA
HA revision rate	**Equal or greater** than TSA (mid–long term); higher with eccentric wear
HA outcomes eccentric vs concentric	**Worse with eccentric** wear
Glenoid	**Progressive erosion** may occur

Extended (CTA) head

Theoretic advantage **metal-to-bone articulation with the acromion** → greater arc of smooth metal contact with glenoid + acromion

Evidence unproven; results **comparable to standard HA**; **no other biomechanical advantage**
✗ Wrong options: superior stability · fixed fulcrum · ↑ deltoid moment arm · ↑ offset

Ream-and-run

Definition HA + **concentric glenoid reaming** (**non-prosthetic glenoid arthroplasty**) with **nubbed spherical reamer** → single smooth concavity

Advantage avoids **prosthetic glenoid failure** → **activity beyond that recommended for TSA**

Ideal patient **young, active, motivated** (e.g. 42-yo weightlifter)

Works in OA, **capsulorrhaphy arthropathy**, **post-traumatic arthritis**
✗ Wrong options: contraindicated

Rehab **therapy is critical**
✗ Wrong options: "TSA needs more therapy"

Outcomes excellent 2-year functional + radiographic results

Technique

Approach **deltopectoral**

Head cut start at **medial insertion line of supraspinatus**

Humeral stem version

Version	Result
30° retroversion ✓ (broach + stem, TO)	**Ideal**; lateral fin **slightly posterior to bicipital groove**
Excess anteversion	**Anterior dislocation**
Excess retroversion (e.g. 45°, TO)	**Posterior dislocation** (✗ Wrong options: anterior instability)
✗ Wrong options: neutral · 10° anteversion · neutral–10° retroversion	Wrong (TO)

– – Active space – –

45° retroversion (TO) FNB 2024 item, answer not provided by source; options: anterior instability · limit IR · best function · shaft fracture

Height, head size & fixation

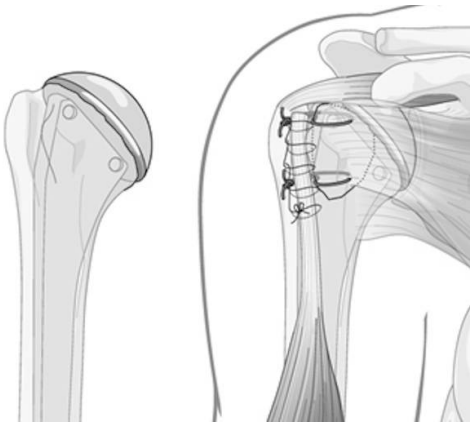
Greater tuberosity height	**7–8 mm below top of head** (Orthobullets) · **6–8 mm** (OB) – read the options
Why	maintain **cuff + biceps tension**; recreate **medial calcar** contour
Height landmark	top of head → **upper border of pectoralis major = 56 mm**; cement prosthesis proud
Head size	contralateral X-ray or **measure resected head**; too large → **overstuffing**
Fixation	**cemented = standard of care** (better QoL, ROM, strength vs uncemented)

Tuberosity malposition after fracture hemiarthroplasty

Importance	**tuberosity migration = commonest cause of failure** in fracture HA; non-anatomic placement → variable outcomes, **rotation deficit**
Technique	secure tuberosities **to each other + to shaft**; **autograft from resected head** → ↓ pull-off, ↑ healing

Malposition	Consequence
GT too proximal (above head)	**GT impingement on acromion** in abduction
GT too distal	**Over-tensioned cuff**, stiffness, ↓ motion
LT too lateral (close to GT)	**↑ Subscapularis tension → ER deficit**

✗ GT position traps does **not** change nonunion risk · does **not** cause overstuffing



Tuberosity fixation around hemiarthroplasty

Rehab

Fracture HA	**early passive motion 6–8 wk** until healed → **strengthening once tuberosity healed**

6 Complications

Complication	Detail	Consequence / treatment
Progressive glenoid arthrosis	Young, active	**Conversion to TSA**
Tuberosity displacement / malunion	Commonest in fracture HA	Reposition + bone graft
Overstuffing	Head too large	→ **stiffness, accelerated glenoid arthritis**
Subcutaneous (anterosuperior) escape	**Cuff + CA arch deficient**	**RSA > anatomic TSA**



Anterosuperior escape after hemiarthroplasty

Failed hemiarthroplasty – glenoid erosion & conversion to TSA/RSA

Painful hemiarthroplasty

- Infection excluded? (WBC 1900, 20% PMN, cultures −ve 3 wk) → **aseptic**
- Intact cuff + glenoid wear → **keep stem, revise head + add glenoid (TSA)**
- Pseudoparalysis + superior escape → **convert to RSA**

Glenoid arthritis, intact cuff, well-fixed stem	**modular**: retain stem, **revise head + prosthetic glenoid** ✗ Wrong options: MRI · two-stage spacer · explant stem · RSA
Cuff failure + superior escape (pseudoparalysis)	**conversion to RSA** ✗ Wrong options: MRI (cuff failure implied) · cuff repair + TSA · fusion if glenoid stock good · revision HA · PT
Why not TSA for cuff-deficient failed HA	poor function, **continued superior migration**, **glenoid loosening (rocking-horse)**
Cuff arthropathy after HA	bony erosion, superior head migration, **acromion + AC joint erosion**

– – Active space – –

X vs Y

	Hemiarthroplasty	Anatomic TSA	Reverse TSA
Glenoid	Native	Prosthetic	Prosthetic (reverse)
Cuff needed	**Yes**	**Yes**	No
Best for	**AVN, 4-part fracture**, young/laborer, poor glenoid stock	OA, **eccentric glenoid wear**	Cuff deficiency, **superior escape**

How it's asked

"Best results of hemiarthroplasty?"	→ **humeral head AVN** (TO)
"Recommended stem position?"	→ **30° retroversion** (TO)
"Lesser tuberosity placed too close to GT – problem?"	→ **ER deficit** (subscap over-tension)
"GT above superior head after fracture HA – risk?"	→ **tuberosity impingement** ✗ Wrong options: nonunion · overstuffing
"Function after HA or TSA most affected by?"	→ **rotator cuff integrity**
"Extended CTA head advantage?"	→ **metal-to-bone articulation with acromion**
"66-yo – hemiarthroplasty most indicated for?"	→ **humeral head osteonecrosis** ✗ Wrong options: Charcot · CA ligament deficiency · RA · capsulorrhaphy arthropathy
"42-yo with GH OA – HA vs TSA?"	→ **HA inferior with eccentric glenoid wear**
"42-yo weightlifter – ream-and-run?"	→ **allows activity beyond TSA limits**
"Painful HA 10 y, intact cuff, aseptic aspirate – next?"	→ **revise head + prosthetic glenoid, retain stem**
"Pseudoparalysis + superior escape after fracture HA?"	→ **convert to RSA**
"Contraindication to HA?"	→ **CA ligament deficiency** (superior escape)

SHOULDER IMAGING

– Shoulder & Elbow · 48 / 60 –

1 Basics

MRI physics – field strength

Field strength stronger magnet → ↑ intrinsic **signal-to-noise ratio**

3 T vs 1.5 T **3 T = 9× proton energy** of 1.5 T → **1.5 T ≈ 11% of 3 T**
✗ Wrong options: 33% · 66%

Sequence	TR / TE	Bright	Dark	Shoulder use
T1	**Short TR, short TE**	**Fat**	Fluid, bone, ligament, marrow, fibrocartilage	**MR arthrogram**; **Hill-Sachs**
T2	**Long TR, long TE**	**Fluid** (inflammation), marrow	Bone, ligament, muscle, fibrocartilage	**Cuff pathology** – full-thickness tear
STIR	Fat saturation (suppression)	Fluid + oedema ↑	Fat ↓	**Oedema vs fatty infiltration** of cuff muscles

CT vs MRI

	CT	MRI
Best for	**Cortical + trabecular bone**, **bony defects**	**Soft tissue**, **bone contusion / trabecular microfracture**
Cost	**↑ Radiation**	–
Artefact	**No magnification artefact** (unlike X-ray)	–

4 Diagnosis & Classification

Radiographic views (Zanca, West Point)

Complete trauma series	Focus
Shoulder AP	GH joint space, DJD
Grashey (true AP)	GH joint space, DJD, **proximal migration of humerus**
AP in ER / AP in IR	**Hill-Sachs**
Axillary lateral	**Anterior + posterior dislocation**, acromion
Velpeau	Axillary modification **if can't abduct arm**
Scapular Y lateral	**Anterior + posterior dislocation**

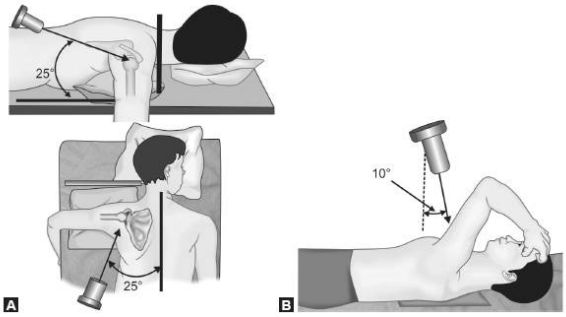
– – Active space – –

Additional view	Focus
Supraspinatus outlet	**Acromion morphology** – type I flat, II curved, **III hooked** (→ impingement + cuff pathology)
Zanca	**AC joint** – AC disease, **distal clavicle osteolysis**, distally projecting osteophytes
Stryker notch	**Hill-Sachs**
West Point axillary	**Anteroinferior glenoid – bony Bankart**, proximal humerus fracture
Garth	Anteroinferior glenoid – **bony Bankart**
Hobbs	**Anterior + posterior SC dislocation**
Serendipity	**Anterior + posterior SC dislocation**

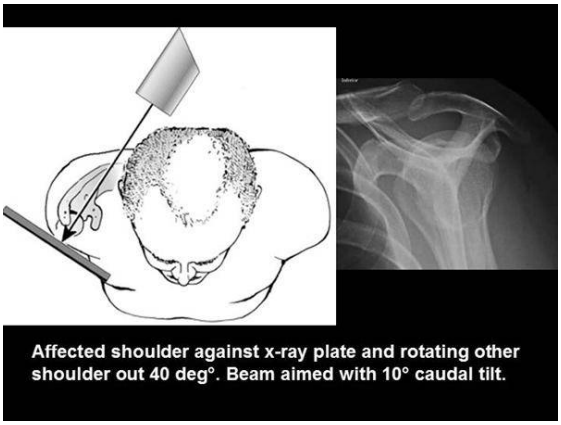
Zanca technique (TO) beam angled **10–15° superiorly (cephalad)** + **kV ↓ to 50%** of standard

Bony Bankart (TO) **West Point axillary** = popular view for **glenoid bone defect**
✗ Wrong options: IR/Stryker = Hill-Sachs · serendipity · Greenspan

Hill-Sachs (TO) **internal rotation view + Stryker notch view**



West Point prone axillary (A) and Stryker notch (B) positioning



Supraspinatus outlet view technique

Acromion morphology (outlet view)

Type	Shape	Significance
I	**Flat**	–
II	**Curved**	–
III	**Hooked**	**Impingement + cuff pathology**

X-ray views to remember (TO)

View	Use (TO)
Axillary	Lateral view of shoulder
West Point axillary	**Bankart lesion**
IR view + Stryker notch	**Hill-Sachs**
Zanca	**AC joint**
Serendipity	**SC joint** dislocation
Greenspan	**Radial head + capitellum** fractures
Jones (AP) elbow	Reduction of **supracondylar humerus** fracture
Oblique + PA wrist in ulnar deviation	**Scaphoid** fracture
Brewerton	**Metacarpal head** fractures
Robert's	**Thumb CMC** joint
Carpal tunnel view	**Hook of hamate** fracture
Ball catcher's	**RA hand erosions**
Judet (obturator + iliac oblique)	**Acetabular** fractures
Frog leg	Intra-op lateral hip
Von Rosen	**CDH**
Axial / skyline / Merchant	**Patellar subluxation**
Mortise (AP ankle in 15° IR)	Ankle injuries
Canale	**Talar neck** fractures
Harris (hindfoot)	**Calcaneus** fractures
Broden	Intra-op **calcaneal** fracture assessment
Swimmer's	Lateral C-spine with shoulder pulled down → **C6, C7, T1**
Ferguson	**20° caudocephalic AP lumbar** → L5 transverse process vs sacrum (**far-out syndrome**)

Key radiology findings – CT

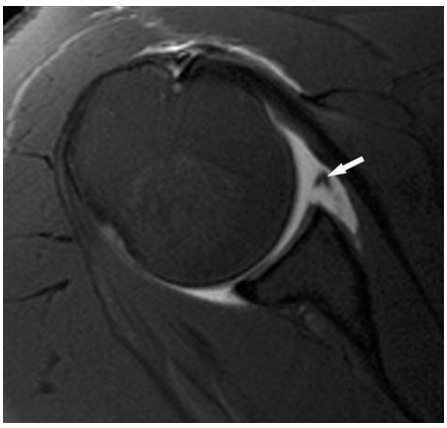
CT plane	Best for
Axial	**Reverse Hill-Sachs**
Coronal	**Fractures**
Sagittal	**Anteroinferior glenoid insufficiency** (bone loss)
3D reconstruction	**Glenoid version** for TSA

Key radiology findings – MRI & MR arthrogram

ABER position sequence beyond coronal/sagittal/axial · hand **behind head** · **tensions anteroinferior GHL + labrum, relaxes capsule** → **Bankart**, **partial/full-thickness cuff tears**, **internal impingement**
✗ Wrong options: true 90–90

– – Active space – –

MR arthrogram percutaneous **dilute gadolinium** into joint → **labral + ligament** pathology: **Bankart**, **SLAP**, **GLAD**, **ALPSA**, **HAGL**



MR arthrogram, Bankart lesion

X vs Y

Lesion	Best X-ray view	Best cross-sectional
Hill-Sachs	**AP in IR / ER**, **Stryker notch**	T1 MRI
Reverse Hill-Sachs	–	**Axial CT**
Bony Bankart	**West Point**, **Garth**	**Sagittal CT** (glenoid insufficiency)
Soft-tissue Bankart / SLAP / HAGL	–	**MR arthrogram**, ABER
AC joint / DCO	**Zanca**	–
SC dislocation	**Serendipity**, **Hobbs**	–
Cuff tear	Grashey (proximal migration)	**T2 MRI**; STIR (fatty vs oedema)

How it's asked

"1.5 T proton energy as % of 3.0 T?"	→ **11%** (3 T = 9×)
"Best view for AC joint?"	→ **Zanca view** (TO) ✗ Wrong options: serendipity · Greenspan · Brewerton
"Best view for bony Bankart lesion?"	→ **West Point axillary view** (TO) ✗ Wrong options: IR + Stryker notch · serendipity · Greenspan
"Hill-Sachs best seen on?"	→ **IR view / Stryker notch** (TO)
"SC joint dislocation view?"	→ **serendipity** (or Hobbs)
"Can't abduct arm for axillary view?"	→ **Velpeau**
"Glenoid version before TSA?"	→ **3D CT**
"Reverse Hill-Sachs?"	→ **axial CT**
"MRI position for internal impingement / anteroinferior labrum?"	→ **ABER**

"Distinguish cuff muscle oedema vs fatty infiltration?" → **STIR**

– – Active space – –

– – Active space – –

SHOULDER PERIPROSTHETIC FRACTURE

1 Basics

Epidemiology

Incidence	**0.6–3%** of shoulder arthroplasty (Orthobullets, OB) · **1.5–3%** in stemmed anatomic TSA (OB) · age **50–85**
Share of complications	**20% of all complications** (intra-op fractures, Orthobullets) · overall arthroplasty complication rate **11%**; post-op periprosthetic fractures = **6.6%** (reverse) vs **1.2%** (anatomic) of complications
Timing	**most are intra-operative**, at the index implant (anatomic + reverse); stemless: few post-op reports
Location	**adjacent to stem tip** · **greater tuberosity** (esp. **revision** TSA)
Risk factors	**osteoporosis** · **aggressive reaming** · endosteal notching · incorrect reaming entry (ideal **9 mm lateral** to anatomic centre of rotation) · **press-fit > cemented** stems · revision · soft-tissue contracture · instability history · ♀ · post-traumatic arthritis · ↑ comorbidity · osteonecrosis (**steroid** > alcohol, post-traumatic)

2 Mechanism

Intra-operative

Revision TSA highest risk: greater exposure, subacromial/cuff scarring → ↑ retraction + torque

Intra-op step (revision)	Share of intra-op fractures
Removing previous implant	**Up to 81%**
Reaming / broaching	Up to 31%
Trialing / insertion	Up to 19%
Exposure (forceful retractor)	Up to 15%

Implant removal metaphyseal-filling → GT fracture (cement removal, interface disengagement) · diaphyseal-engaging → GT fracture (**metaphyseal stress shielding**) · prevent: clear lateral soft tissue, implant-specific tools, **controlled corticotomy**

Primary TSA lower risk · **endosteal notching from reaming** · **excessive humeral ER during exposure** · cortical breach, esp. osteopenic bone

Post-operative

Fall (FOOSH) **stem tip = stress riser** → most post-op fractures **at/adjacent to stem tip** · fatigue fracture through stress riser

Atraumatic	**loosening** → implant "rattling" weakens cortex
Nonunion risk ↑	prosthesis **disrupts endosteal blood supply** · **stem distracts fragments** (no end-to-end healing)
Nonunion factors	osteoporosis · **RA** · ♀ · **displacement >2 mm** · **cement at fracture site**

3 Presentation

History	fall on outstretched hand · **prodromal / worsening pain** = mechanical loosening
Exam	**varus/valgus deformity**, ecchymosis, shortening, diffuse tenderness · can't move elbow/shoulder (**pain**) · document full NV exam
Radial nerve	check status; palsy common with distal humerus fracture, **mostly neurapraxia** ✗ Wrong options: surgery for it

4 Diagnosis & Classification

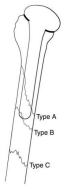
Key radiology findings

X-ray	orthogonal humerus: **AP, axillary lateral, Grashey, scapular Y** · fracture at **stem tip**, usually **oblique** · decide on **stem stability** + **zones of lucency** (no ingrowth / osteolysis)
CT (metal suppression)	concern for **loose stem** (pattern, prodromal pain) or comminution · morphology, bone stock, cuff muscle quality, glenoid version (if revising)



Loose humeral stem — axillary lateral

Wright and Cofield classification

Type	Fracture	Diagram
A	Centred near **stem tip**, extends **proximally** ("proximal to stem tip" — OB)	
B	Centred near **stem tip**, extends **distally** (at level of tip)	
C	**Distal to stem tip** (into distal metaphysis)	

– – Active space – –

Wright series (Mayo) 499 arthroplasties → **9 fractures** (1 A, 5 B, 3 C) · 3 splinted → **failed to heal**

5 Management

Post-operative periprosthetic humerus fracture

- Loose stem (any type) → **revision arthroplasty + supplementary fixation**
  (poor stock → long stem bypassing 2–3 cortical diameters; good stock → shorter stem + fixation)
- Stable stem, type A/B → **ORIF: hybrid locking plate + cerclage cables**
  (long spiral/oblique A/B → bracing possible, union <50%)
- Type C → **fracture brace (nonop)**
  - fails / nonunion → **ORIF ± allograft strut**

Intra-operative fracture

- Type A → **stem spanning fracture by 2 cortical diameters**
- Type B / C → **proximally porous-coated long stem, 2–3 cortical diameters**
- Type C unsuitable for long stem → **ORIF**

Operative criteria

Situation	Treatment
Long oblique / spiral A or B, stable stem · **type C**	Immobilisation (coaptation → **functional brace**); **union <50%**
Intra-op type A	Longer stem spanning **2 cortical diameters** ± supplementary fixation
Intra-op type B or C	**Proximally porous-coated long stem**, **2–3 cortical diameters**; distal cement
Post-op transverse A/B, stable stem · C failed nonop · can't tolerate nonop	**ORIF**
Loose prosthesis, any type	**Revision arthroplasty + supplementary fixation**
Poor bone stock (loose)	**Long stem** bypass ≥2–3 cortical diameters ± allograft
Good bone stock (loose)	**Short stem** (ends proximal to original) + fixation; tip doesn't distract fracture

Source nuance long spiral **type B**, stable stem: bracing possible (Orthobullets) · **ORIF hybrid plate + cables** best for active patient (OB) – read the stem

Stable-stem fractures – ORIF with plate/cerclage & radial nerve risk

Best for stable stem A/B **ORIF hybrid locking plate + cerclage cables**
✗ Wrong options: revision (stem well fixed) · **lag screw + neutralisation** · nonop near stem tip → nonunion



Hybrid locking plate and cerclage cables

Construct	plate placed **eccentrically** → locking screws skive past stem (bicortical) · **cables proximal** to stem tip · **screws distal** · usually **4.5 mm LC-DCP** · alternatives: dual plates, **3.5 mm attachment plates** · deficient cortex → **cortical strut allograft** + cables (2 hemicylinders = "sarcophagus")
Approach	**anterolateral** (extend deltopectoral: coracoid → DP groove → lateral biceps) · biceps–brachialis interval, **split brachialis** (medial = musculocutaneous, lateral = radial) · **posterolateral** for distal fracture with stable stem
GT / calcar fracture	suture or cable fixation
No nonop with sling / hanging arm cast	bracing is the nonop method; active patient wants fast return → ORIF
No APC (allograft-prosthetic composite)	not with good bone quality + no bone loss
Evidence	Kurowicki: 7 type B ORIF without revision → **all united** · Andersen: ORIF 17/17, revision 18/19 united · stable IM stem + supplementary fixation > loose stem

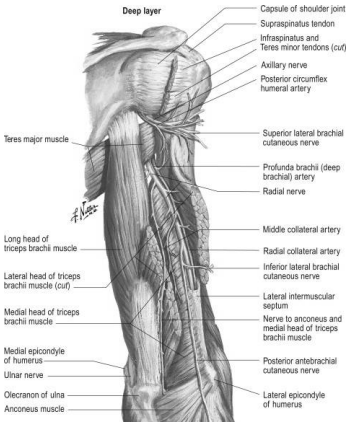
Radial nerve and cerclage

Landmark	Distance to radial nerve at spiral groove
Coracoid	~13 cm
Distal lesser tuberosity	~10 cm
Superior latissimus insertion	~8 cm
Inferior latissimus insertion	**~4 cm**

Cable 4 cm below inferior LD insertion → iatrogenic radial nerve injury
✗ Wrong options: median (medial septum) · axillary (proximal) · subscapularis · brachial artery (superficial, anteromedial)

Safe zone	cables **just inferior / proximal to inferior edge of LD insertion**

– – Active space – –



Radial nerve in spiral groove

Long stem technique

Porous-coated long stem	proximal coating = metaphyseal stability · **pack cement distally**, **avoid cement extrusion into fracture** · strut + cables if deficient · **olecranon fossa** may prevent it

6 Complications

Complications	**nonunion** · delayed union · **GH stiffness** · **radial nerve injury** (fracture or **iatrogenic cerclage**)
Prognosis	high union when treated appropriately; union **2–8 months** · **significant ↓ shoulder ROM**

X vs Y

	Stable stem	**Loose stem**
Clues	No lucency, no prodromal pain	Lucent zones, **prodromal pain**, "rattling"
Type A/B	**ORIF hybrid plate + cables**	**Revision long stem** + fixation
Type C	Brace → ORIF if fails	Revision

How it's asked

"Fall, stable rTSA stem, type B fracture – best treatment?"	→ **ORIF hybrid locking plate + cerclage cables** ✗ Wrong options: revision cemented/ cementless long stem · lag screw · nonop
"Cable placed 4 cm inferior to latissimus insertion – complication?"	→ **iatrogenic radial nerve injury** ✗ Wrong options: median · axillary · brachial artery
"Active 69-year-old, T-score −0.45, long spiral B, stable stem – treatment?"	→ **ORIF plate + screws/cerclage cables** ✗ Wrong options: sling · hanging cast · revision · APC

"Loose stem with any fracture type?"	→ **revision arthroplasty + supplementary fixation**
"Intra-op fracture — stem should bypass by?"	→ **2 cortical diameters** (2–3 for type B/C)
"Why high nonunion near stem tip?"	→ **disrupted endosteal blood supply** + stem distracts fragments
"Most common step causing intra-op fracture in revision?"	→ **implant removal** (up to 81%)
"Type C fracture, stable stem?"	→ **fracture brace** ✗ Wrong options: sling

– – Active space – –

– – Active space – –

SHOULDER PROSTHETIC JOINT INFECTION

– Shoulder & Elbow · 50 / 60 –

1 Basics

Epidemiology

Significance	**major cause of revision within first 2 years** after shoulder arthroplasty
Incidence	**~4% primary TSA + RSA** (Orthobullets, OB) · 3–4% · 0–5% · up to 4–5% (other OB explanations) – read the options · **~15% revision** TSA/RSA
aTSA vs rTSA	**no difference** in primary shoulders without prior surgery · ↑ in primary **rTSA after ≥1 prior non-arthroplasty operation** (Florschütz)

Risk factors & prevention (smoking, steroid injection, nasal decolonisation)

Risk factor	Note
Male	**2.5×** PJI rate · ↑ C. acnes colonisation
Arthroplasty for fracture / trauma	**3×** vs elective
Revision surgery	~15%
Intra-articular steroid injection within 3 months	Injection **4 months** (or 15 months) before = **not** a risk factor
Smoking	↑ infection + fracture + loosening
Young age	–
Postoperative haematoma	–
Coagulopathy, **ESRD**, **lupus**, **RA**, uncontrolled comorbidities	Medical comorbidities

Pre-op smoker	**smoking cessation programme next** (smoking = suggested contraindication to TSA/RSA) ✗ Wrong options: PT trial (pseudoparalysis CTA) · tramadol (= 1st line knee OA) · CBC/ESR/CRP (infection unlikely) · rheumatology referral
Complication risk (patient)	younger age · ↑ BMI · RA · diabetes · smoking

C. acnes colonisation correlates **male** · **native hair at surgical site** · **humeral loosening**
✗ Wrong options: age >70 · CKD · female · glenoid lucent lines

✓ Validated prevention	Detail
MRSA (methicillin-resistant Staphylococcus aureus) screening + nasal decolonisation	If nares colonized → application of **5% povidone-iodine solution to each nostril** pre-operatively (**10 s/nostril, 1 h pre-op**) · or **2% mupirocin BID × 5 days**
IV antibiotic prophylaxis	Within **30 min** of incision, continue **24 h**
Penicillin allergy prophylaxis	**Clindamycin for C. acnes** (Orthobullets, OB) · **vancomycin, not clindamycin** (OB) – read the options
Skin prep	**Chlorhexidine (ChloraPrep)** · hydrogen peroxide prep
Hair removal	**Electric clippers**
Intra-op	Frequent **saline irrigation** · **dilute povidone-iodine** before closure
Closure / dressing	Dermabond · **silver-impregnated** dressings

Not validated	Note
Axillary hair shaving	No meaningful ↓
Sterile helmets/hoods	May ↑ infection
Laminar flow	Not validated

2 Mechanism

C. acnes – microbiology, prolonged culture & antibiotic choice

Most common organism **Cutibacterium acnes** (formerly Propionibacterium acnes) · others: S. aureus, S. epidermidis

Description **gram-positive, anaerobic rod** (OB) · **gram-positive rod, facultative anaerobe** (OB) · "gram-positive, facultative, aerotolerant, anaerobic rod" (Orthobullets) · **ferments lactose → propionic acid** · non-spore-forming

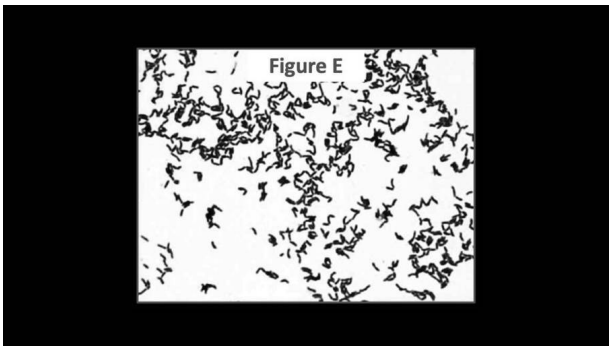
Habitat **axilla**, dermal **sebaceous glands**; commensal skin flora

Biofilm forms in **18–90 h** on implant + synovium (>> planktonic) → **aspiration only 17% sensitive**

Behaviour indolent, low-grade, low virulence · usually **no swelling, erythema, fever, pus** · inflammatory markers **often normal**

Culture mean incubation **7–21 days** · usually negative in first **3–5 days** · may take **up to 3 weeks**

– – Active space – –



C. acnes – gram-positive rods

✗ Organism distractor	Description
S. aureus	GP cocci, facultative anaerobe, **coagulase +** · commonest in **hip/knee** PJI
S. epidermidis	GP cocci, facultative anaerobe, **coagulase –**
E. coli	GN rod, facultative anaerobe · rare in shoulder
Klebsiella	GN **encapsulated** rod
N. gonorrhoeae	GN diplococci
C. diphtheriae	GP **club-shaped**, nonmotile, noncapsulated bacillus
Streptococci · fungi	Rare · exceedingly rare

Biofilm & timing

Biofilm (all bacteria) on implant **within 4 weeks** · **15% cells + 85% polysaccharide (glycocalyx)** → adherence + immune protection · no safe removal → **explant if >4 wk**

Source **direct** (sinus tract, wound dehiscence) · **haematogenous** (dental work, gallbladder) in longstanding infection-free joint

3 Presentation

Symptoms **persistent pain (most common)** · **draining sinus (2nd)** · systemic (fever, chills, sweats) · **stiffness**

Sinus tract to joint **diagnostic**

Typical C. acnes stem months–years after arthroplasty, **gradually ↑ pain**, normal WBC/ESR/CRP, humeral lucency

4 Diagnosis & Classification

Classification – time of onset

	Acute	Chronic
Timing	**<3–6 wk** (CDC **<90 days**)	**>3–6 wk** (CDC **>90 days**)
Biology	Confined to joint; no bone–implant invasion; less biofilm	**Biofilm**; invades **prosthesis–bone interface**
Treatment	**I&D + retention (DAIR)**	**2-stage revision**

DAIR window — **<4 wk** (Orthobullets treatment) · **within 5–6 wk** (OB) · <3–6 wk (Orthobullets classification) – read the options

Key radiology findings

X-ray — normal early · osteopenia · **periprosthetic lucencies** · pseudosubluxation of head · effusion · **endosteal scalloping** · bony resorption

Humeral vs glenoid loosening — **glenoid loosening = most common cause of TSA failure**, not specific for infection · **humeral loosening → suspect PJI**



Glenoid radiolucency after TSA

US / MRI — fluid collections (US: loculated away from GH joint)

Bone scan — not routinely (efficacy not shown)

PET — adjunct; **98% sensitivity + 98% specificity**

Genes & lab markers

Marker	Fact
WBC	Not sensitive/specific; **normal in C. acnes**
CRP	Peaks **2–3 d** post-op → normal **14–21 d**; may be normal in C. acnes
ESR	Peaks **5–7 d** → normal **90 d**; may be normal
ESR/CRP overall	**Insufficient sensitivity/specificity** for PJI
Alpha-defensin	Investigational
Synovial WBC	**>50,000**
Crystals	**Don't exclude** infection
Gram stain	Often **negative**; + in ~75%

Diagnostic work-up – aspiration

Synovial tissue culture — **gold standard** (+ ~80%) · **≥4 specimens** at surgery · **aerobic + anaerobic + broth** media

Hold cultures — **14–28 days** (Orthobullets) · **14–21 days / 2–3 weeks** (OB) – read the options · fungal **4 wk** · mycobacterial **8 wk**

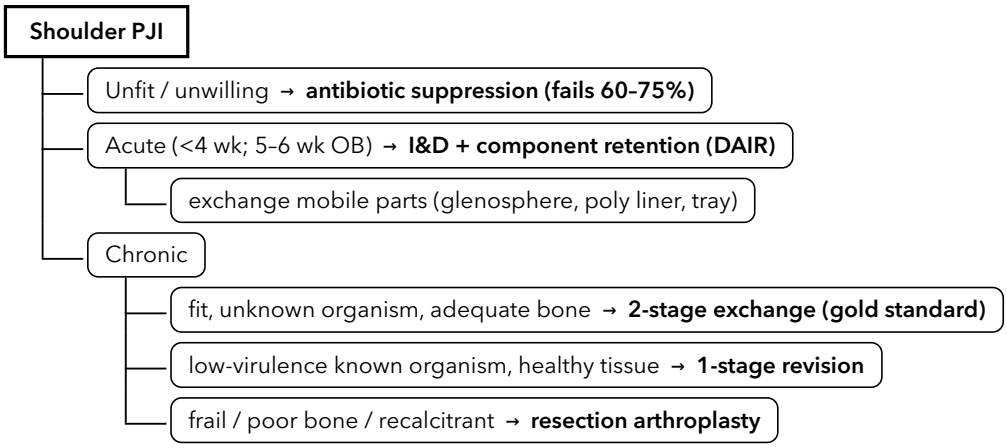
Aspiration — **low sensitivity + low NPV · dry tap common → does not rule out · + S. aureus culture = high specificity + PPV**

Fluoroscopic / US guidance — ensures intra-articular placement · **no ↑ PPV/NPV or yield**, operator-dependent, ↑ cost

– – Active space – –

Antibiotics	stop **2 weeks before aspiration** ✗ Wrong options: current antibiotics don't affect yield
No growth at 5 days	**observe cultures additional 2 weeks** before revision ✗ Wrong options: revision · arthroscopic I&D

5 Management



Antibiotics

C. acnes drug of choice	**penicillin** (low MIC) · penicillin G, amoxicillin ✗ Wrong options: TMP-SMX · meropenem · vancomycin (not 1st line)
Penicillin allergy	**ceftriaxone or clindamycin**
Duration	prolonged IV; guided by cultures + ID consult

Mechanism of action	Antibiotic
Binds transpeptidase (PBP) → inhibits cell wall synthesis	**Penicillin** ✓
Reversibly binds **50S**	Macrolides
Irreversibly binds **30S**	Aminoglycosides
Blocks **30S–50S initiation complex**	**Linezolid**
Binds cell wall, disrupts peptidoglycan **cross-linking**	**Vancomycin**

Operative criteria

Option	Indications	Outcome
Antibiotic suppression	Severely ill, unwilling, multiple comorbidities	**Fails 60–75%**
I&D + retention	**Acute** infection	**50% failure**
Single-stage revision	Uncommon; **low-virulence** organism with good sensitivity; healthy patient + soft tissue	Up to **90%**; ↓ stay, cost, antibiotic time · consider **RSA** (cuff sacrificed)
2-stage exchange	**Gold standard**; medically fit; **unknown organism**; adequate bone stock; eradication before stage 2	**60–90%**; some keep spacer
Resection arthroplasty	Frail/complex, **insufficient bone stock**, recalcitrant	Poor function; **pain relief >50%**; **tuberosity preservation** predicts better result
Arthrodesis	Rare (bone stock poor)	Pain ↓
Amputation	**Life-threatening** infection	–

Spacer antibiotic-PMMA; local antibiotics, **pseudocapsule**, maintains soft-tissue tension, ↓ pain, allows PT · stem from **Steinmann pin + chest tube**



Resection arthroplasty

Resection **antero-superior subluxation** if tuberosities + cuff removed

Arthrodesis S-shaped incision · resect cuff, biceps tenodesis, decorticate glenoid/head/acromion undersurface · **10-hole 4.5 mm pelvic recon plate** · **30° abduction, 30° flexion, 30° IR** · complications: infection, nonunion, malposition, prominent hardware, humeral shaft fracture

Approach **deltopectoral** for all

6 Complications

Complications **failure to eradicate** (inadequate clearance after suppression, I&D, 1-stage) · recurrent infection (2-stage) · humerus/glenoid fracture · poor bone stock

– – Active space – –

X vs Y

	Shoulder PJI	Hip/knee PJI
Commonest organism	**C. acnes**	**S. aureus / S. epidermidis**
Inflammatory markers	**Often normal** (C. acnes)	↑ with S. aureus / MRSA
Aspiration value	Less clear: low sensitivity, dry taps	Clearer
Gold-standard treatment	2-stage	2-stage

How it's asked

"rTSA, drainage at 4 wk, I&D + poly exchange, cultures negative at 10 d — MOA of best antibiotic?"	→ **binds transpeptidase, inhibits cell wall synthesis (penicillin)** ✗ Wrong options: 50S macrolide · 30S aminoglycoside · linezolid · vancomycin
"Normal labs, painful aTSA, aspiration — hold cultures 2–3 wk for?"	→ **gram-positive, anaerobic bacteria** ✗ Wrong options: GP aerobic cocci · GN rods · diplococci · club-shaped
"18-mo TSA, lucent lines, normal labs, no growth at 5 d — next step?"	→ **observe cultures additional 2 weeks** ✗ Wrong options: arthroscopic I&D · revision
"Independent correlate of P. acnes colonisation?"	→ **native hair at surgical site** ✗ Wrong options: age >70 · CKD · female · glenoid lucent lines
"aTSA 2 mo ago, infection confirmed — organism characteristics?"	→ **gram-positive rod, facultative anaerobe** ✗ Wrong options: coagulase + / − cocci · GN rods
"C. acnes grown — antibiotic of choice?"	→ **penicillin** ✗ Wrong options: TMP-SMX · clindamycin = allergy only · meropenem · vancomycin
"Smoker with CTA planned for arthroplasty — next step?"	→ **smoking cessation programme**
"rTSA infection — NOT a risk factor?"	→ **steroid injection 4 months before surgery** (✓ risk: coagulopathy, fracture, male, smoking)
"Validated PJI prevention before rTSA?"	→ **5% povidone-iodine to each nostril if MRSA-colonised** ✗ Wrong options: axillary shaving · helmets · laminar flow · clindamycin for allergy
"rTSA 2 y, ↑ ESR/CRP — most common organism (Gram picture)?"	→ **gram-positive rod (C. acnes)** ✗ Wrong options: staph · strep · E. coli · fungus

"Shoulder aspiration — true statement?"	→ **positive S. aureus culture = high specificity** ✗ Wrong options: high sensitivity/NPV · dry tap rules out · fluoroscopy ↑ PPV/NPV · antibiotics don't matter
"Gold standard for chronic shoulder PJI?"	→ **2-stage exchange**
"Acute PJI within weeks of surgery?"	→ **I&D + component retention (DAIR)**

– – Active space – –

– – Active space – –

SLAP LESION

– Shoulder & Elbow · 51 / 60 –

1 Basics

Epidemiology

Incidence	uncommon: **<5%** of shoulder injuries · yet a common cause of pain + weakness in throwers
Who	**dominant shoulder of overhead/throwing athletes** · older patients after traumatic falls
Risk factor	**GIRD**
Associations	internal impingement · GIRD · **articular-sided** cuff tears · (subtle) instability · scapular dyskinesis · paralabral cyst · sublabral clefts, Buford complex

Anatomy – biceps anchor, Buford complex & vascularity

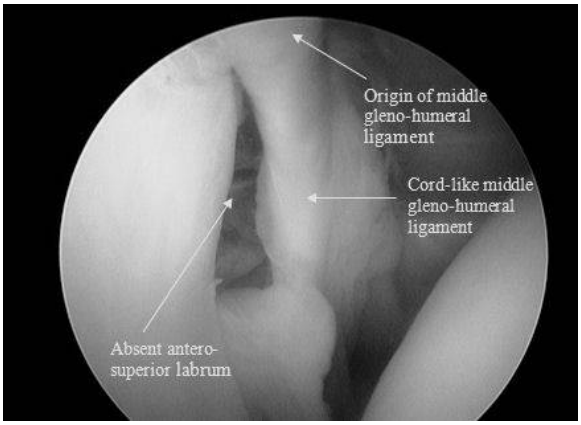
LHB attachment	**~50% to supraglenoid tubercle, ~50% to labrum** (predominantly posterior labrum) · tubercle **>6 mm medial** to glenoid surface
Posterior / posterior-dominant attachment	**~70%** of shoulders (✗ Wrong options: 15% · 30% · 100%) · most common: **posterior to 12 o'clock** (✗ Wrong options: "anchored anteriorly")

Tuoheti type	Biceps attachment	Share
I	Completely posterior	I + II **>70%**
II	Predominantly posterior	
III	Anterior + posterior	25%
IV	Anterior	5%

Biceps function	torsional + horizontal stability in **late cocking** · **poorest blood supply** in intra-articular course
Labrum	fibrocartilage; **chock block** to subluxation · superior labrum attaches more **medially** + anchors biceps = **weak link**

Labral blood supply — **suprascapular, circumflex scapular, posterior humeral circumflex** arteries via **capsule + periosteum** · **anterosuperior labrum = poorest** → degenerative tears, poor healing
✗ Wrong options: bone · single source

Variant	Feature	Pitfall
Sublabral recess / foramen	± thickened MGHL	Mistaken for tear on MRI
Buford complex	**Cord-like MGHL + absent anterosuperior labrum**; **normal variant**, **1.5%**	**Do not repair → significant loss of ER** + pain
Meniscoid labrum	**1%**	–
Superior MGHL attachment	On glenoid	–



Buford complex – arthroscopic view

Arthroscopy tell — erythema + tearing **under** labrum = tear (vs normal recess)

2 Mechanism

Mechanisms — **repetitive overhead activity** · FOOSH with **tensed biceps** · traction on arm

Thrower pathoanatomy — tight **posterior IGHL** → GH contact shifts **posterosuperiorly** → ↑ shear on superior labrum → **"peel-back"**

Consequence — SLAP **↑ strain on anterior band IGHL** → ↓ stability

3 Presentation

History — "pop" during overhead activity or trauma · **lag** between injury + symptoms

Symptoms — **vague deep shoulder pain** · popping/clicking · weakness, fatigue, ↓ performance

Exam — bicipital groove tenderness · check **IR + total arc** vs contralateral · SS ± IS **atrophy** (suprascapular neuropathy from cyst) · apprehension positive in **85%**

– – Active space – –

Tests **no single specific test**

Test group	Tests
Biceps provocation	**Speed's**, **Yergason's**, Kim biceps load
SLAP	**O'Brien (active compression)**, **crank** (full abduction, load + rotate), **dynamic labral shear**, Kibler anterior slide

Related topics (distal clavicle excision, biceps dislocation)

Arthroscopic vs open DCE	advantage = **evaluate GH joint** · failed open DCE: **15/20 had SLAP** (9/15 good–excellent after arthroscopic treatment) ✗ Wrong options: fewer complications · infection · time
Arthroscopic DCE	sacrifices **inferior** AC ligament, **preserves superior** AC ligament
Medial LHB dislocation	**subscapularis tear** ✗ Wrong options: supraspinatus (LHB rupture) · transverse ligament · shallow groove · type I SLAP; type II SLAP can cause instability

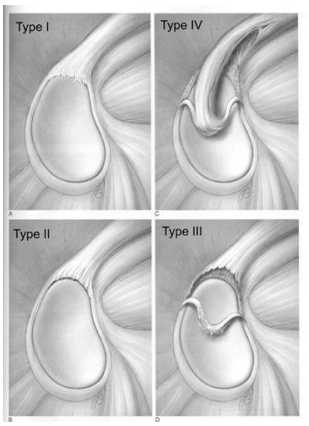
4 Diagnosis & Classification

Diagnosis – MR arthrogram

X-ray	usually normal
Study of choice	**MR arthrogram** · **gold standard = arthroscopy** ✗ Wrong options: US · EMG · labs · PT first when suspicious
MRI sign	T2 signal between superior labrum + glenoid, **lateral to glenoid rim, posterior to biceps**
Accuracy	sens **~50%**, spec **~90%**, ↑ with arthrogram (Orthobullets) · linear high signal on oblique coronal T2: sens **41%**, spec **86%**, accuracy 63% (OB) – read the options
Paralabral ganglion cyst	**highly specific for labral tear**; usually **spinoglenoid notch** ✗ Wrong options: suprascapular notch
Overdiagnosis	concern (↑ SLAP repairs in ABOS case logs)
Arthroscopic peel-back test	labrum peels back at **90° abduction + ER**

SLAP classification

Type	Orthobullets description (share)	Clock face (TO)	TO description
I	Labral + biceps **fraying, anchor intact** (11%)	11–1	Fraying superior labrum
II	Fraying + **detached biceps anchor** (**41%**, most common)	11–1	Detachment of biceps anchor
III	**Bucket handle, intact anchor** (33%); biceps separates from fragment	11–1	**Bucket handle, intact biceps**
IV	**Bucket handle + detached anchor** (15%); biceps stays with fragment	11–1	Bucket handle with **biceps extension**
V	II + **anteroinferior extension (Bankart)**	**11–5**	Bankart + SLAP
VI	II + **unstable flap**	11–1	Flap tear of bucket-handle SLAP
VII	II + **MGHL injury**	**11–3**	Extension into MGHL
VIII	II + **posterior extension**	**11–7**	Posterior extension
IX	**Circumferential**	**11–11**	Pan-labral
X	II + posteroinferior extension (**reverse Bankart**) (Orthobullets)	**11–1+**	SLAP + **noncontiguous** posterior labral tear (TO)



SLAP types I–IV

Systems **Snyder** = I–IV · **Maffet** added V–VII · high inter/intra-observer variability

Type X wording "posteroinferior extension / reverse Bankart" (Orthobullets) · "noncontiguous posterior labral tear" (TO) – read the options

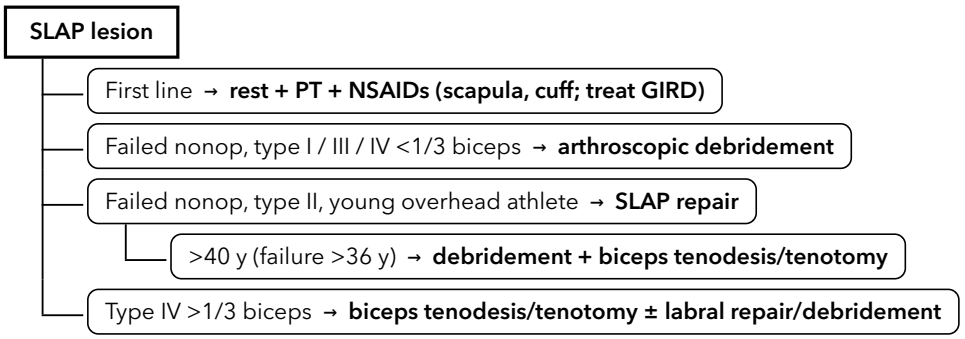
Classification & variants (type III, Andrews lesion) (TO)

Dynamic labral shear + tear 11–1 o'clock, intact biceps (TO) **type III** SLAP
✗ Wrong options: type II = biceps anchor detached · Bankart

Andrews lesion (TO) **anterior + superior labral detachment without extension posterior to biceps** footplate · **throwers**
✗ Wrong options: SLAP II · posterosuperior · anterior + posterior · posterior IGHL avulsion

- - Active space - -

5 Management



Treatment – nonoperative vs repair vs tenodesis (age, type IV) & return to sport

Initial ✓ **rest + rehab for scapular mechanics + cuff strengthening** (+ NSAIDs; address GIRD)
✗ Wrong options: aggressive **concentric biceps** strengthening (aggravates) · surgery first

Labral tear in thrower initial: **posterior capsular stretching + cuff strengthening**

Operative criteria

Situation (failed nonop, severe symptoms)	Procedure
Type I, III, IV <1/3 biceps	**Arthroscopic debridement** (III → debridement ± repair of fragment)
Type II, young overhead athlete	**Labral repair** (most amenable to repair)
Type II, >40 y · biceps fraying **>50%**	**Debridement + biceps tenotomy/tenodesis** (✗ Wrong options: repair (stiffness))
Type IV with >1/3 (>30%) biceps	**Biceps tenodesis** ± labral repair/debridement
Type V (II + Bankart) · II + posterior extension	**SLAP + Bankart repair** (active patient)
Paralabral cyst + SLAP II	**Cyst decompression + SLAP repair**

Age thresholds tenodesis consensus **>40 y** (Orthobullets, OB) · repair failure **>36 y** (Orthobullets, OB) · stiffness risk **>35 y** (OB) – read the options

Tenodesis in young viable even **<25 y** athletes (recent studies)

Repair failure factor **age >36**
✗ Wrong options: recreational swimming · male · sedentary job · type II

Return to sport

Group / treatment	Rate
Same level after repair (overall)	**60–70%**
Return to prior play after repair	**50–60%**
Non-pitchers returning to baseball after repair	**65–85%**
Nonop: return to play	**~40%**
Nonop: return to prior play	**20–30%**

Pitchers vs non-pitchers — **non-pitchers return at higher rates** after repair · RTP similar for repair vs tenodesis · elite throwers: variable, unpredictable

Repair technique

Steps — debride flaps · **decompress cysts** · accessory/percutaneous portals for anchors near rim · **sutures anterior to biceps → stiffness**

Tenodesis — all-arthroscopic or mini-open; tenotomise near labrum

Postoperative rehabilitation after SLAP repair

Period	Allowed	Avoid
Wk 0–4	**Passive + active-assisted flexion in scapular plane**; passive **0–90° FE**	**Passive ER at 90° abduction** (~4 wk), extremes of abduction + ER, **resisted biceps**
Wk 4–6	Active ROM, isometrics; **isotonic cuff** 4–6 wk	**Resisted elbow flexion until 6 wk**
Wk 6–12 (to 8)	Functional + light strengthening	–
Wk 8–16 / 12+	**Sport-specific** strengthening	–
~6 months	Return to sport	–

Limit after repair — **abduction + ER** – max tension on repair
✗ Wrong options: IR

✗ Rehab traps — immediate full active ROM · **6 wk sling, no motion** (stiffness) · cuff strengthening by wk 2 · eccentric biceps at wk 2 · latissimus pull-downs

6 Complications

Stiffness

Most common after SLAP repair (~78%, Orthobullets) · risk: incidental SLAP repaired in **>45 y** during cuff repair · **post-surgical adhesive capsulitis**

Treatment — early pendulum/AAROM from wk 1 → **capsular release** if fails >6 months PT

Suprascapular nerve – anchor placement & paralabral cyst

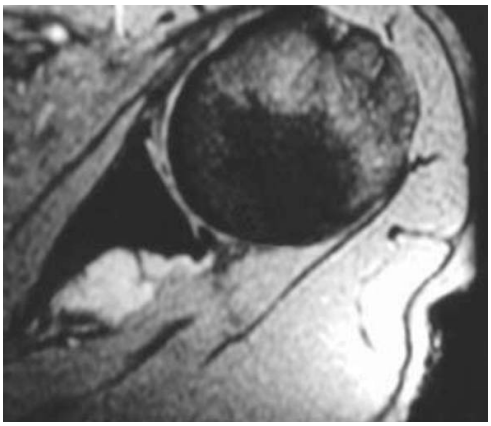
Mechanism — deep drilling / anchors from **anterior or anterosuperior portal** · instrumentation **medial to glenoid** · overdrilling glenoid

Distance — SSN **1.5–2.0 cm** from glenoid cortex at scapular spine level

Origin — SSN from **superior (upper) trunk**
✗ Wrong options: middle trunk (no branches) · C5–7 roots = long thoracic · posterior/medial cord

✗ At risk distractors — axillary nerve, thoracoacromial artery, thoracodorsal nerve, superior thoracic artery

– – Active space – –



Spinoglenoid paralabral cyst

Cyst location	Muscles affected
Suprascapular notch	**SS + IS**
Spinoglenoid notch	**IS only**

Cyst management no structural lesion → rehab · with labral tear → **decompression + labral repair**

Failed repair

Causes chondral injury, hardware failure, **anchor pullout** · age **>36** → tenodesis better

X vs Y

	Buford complex	Sublabral foramen	SLAP II
Labrum	**Absent** anterosuperior	Present; recess/foramen beneath	Detached at biceps anchor
MGHL	**Cord-like**	± thickened	–
Key point	**Do not repair** (↓ ER)	MRI pitfall – not a tear	Repair / tenodesis

	SLAP III	SLAP IV	Andrews (TO)
Labrum	Bucket handle	Bucket handle	Anterosuperior detachment
Biceps	**Intact**	**Torn into**	Not posterior to footplate

Distractor lesion	Key feature
HAGL	Humeral avulsion of anterior IGHL
ALPSA	Capsulolabral periosteal sleeve displaced **medially** on glenoid neck (axial)
Bankart	Anteroinferior labrum, after anterior dislocation
Hill-Sachs	Anterior instability; engaging defects → surgery
Reverse Hill-Sachs	Posterior instability; surgery if **>20%** of articular surface

How it's asked

"MRI tear 11–1 o'clock, intact biceps, positive dynamic labral shear?"	→ **SLAP type III** (TO)
"Andrews lesion?"	→ **anterior + superior labral tear** (TO)
"Buford complex?"	→ **normal variant: cord-like MGHL + absent anterosuperior labrum**
"% posterior/posterior-dominant biceps attachment?"	→ **70%**
"True about the superior labrum (SLAP on MRI)?"	→ **anterosuperior portion most susceptible to vascular insufficiency** ✗ Wrong options: anterior anchor · single artery · suprascapular notch cyst · limit IR after repair
"FOOSH, overhead athlete, MRI superior labral signal?"	→ **SLAP tear** ✗ Wrong options: HAGL · ALPSA · Bankart
"Pitcher, O'Brien + Jobe relocation, total arc 180° – next?"	→ **MR arthrogram**
"47-year-old, type II SLAP, biceps fraying >50%, failed 8 months nonop?"	→ **debridement + biceps tenotomy/ tenodesis** ✗ Wrong options: SLAP repair
"Factor associated with SLAP repair failure?"	→ **age >36**
"Third-baseman, type II SLAP – counsel?"	→ **better return to same level than pitchers**
"38-year-old, O'Brien/Speed/Yergason +, initial treatment?"	→ **rest + rehab for scapular mechanics + cuff**
"Finding favouring tenodesis over repair?"	→ **type IV with 40% biceps involvement**
"Rehab after type II SLAP repair?"	→ **limited passive 4 wk → active to 8 wk → sport-specific to 12–16 wk**
"Avoid at 2 weeks after SLAP repair?"	→ **passive ER at 90° abduction**
"Immediate rehab after SLAP repair?"	→ **passive + active-assisted flexion in scapular plane**
"Anchor drilling from anterosuperior portal – structure at risk?"	→ **suprascapular nerve**
"SLAP + cyst on MRI – treatment?"	→ **supraglenoid cyst decompression + SLAP repair**
"Instrument passed medial to glenoid – nerve origin?"	→ **superior trunk**
"Arthroscopic DCE advantage?"	→ **evaluation of GH joint**
"Medial biceps dislocation cause?"	→ **subscapularis tear**

– – Active space – –

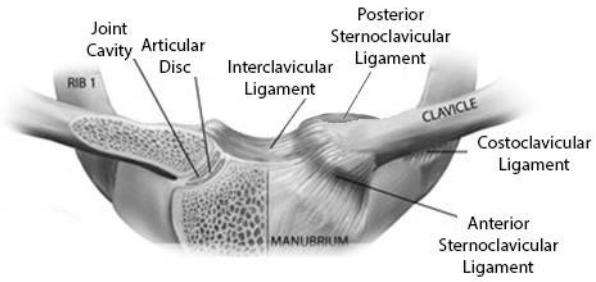
STERNOCLAVICULAR JOINT

– Shoulder & Elbow · 52 / 60 –

1 Basics

Joint anatomy

Type	**diarthrodial saddle** joint, **incongruous (~50% contact)**; **fibrocartilage**; contains **intra-articular disc** ✗ Wrong options: hyaline/synovial cartilage
Motion	arm elevation to **90° → 30° rotation** at SC joint



SC joint: disc, anterior/posterior SC, costoclavicular, interclavicular ligaments

Stabilising ligaments

Ligament	Function
Posterior SC capsular ligament	**Primary restraint to anterior–posterior stability**
Anterior SC ligament	Primary restraint to **superior displacement** of medial clavicle
Costoclavicular (rhomboid) – anterior fasciculus	Resists **superior rotation + lateral displacement**
Costoclavicular – posterior fasciculus	Resists **inferior rotation + medial displacement**
Intra-articular disc ligament	**Prevents medial displacement**; secondary restraint to superior displacement

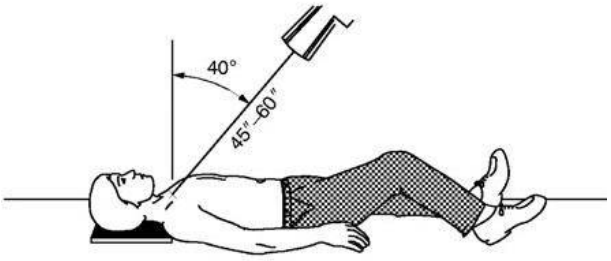
Medial clavicle anatomy

First to ossify	**clavicle**, **5th–6th week of gestation**
Last to fuse	**medial clavicular epiphysis**, closes **20–25 y** (last physis)
Before 25 y	most "SC dislocations" are **medial clavicular physeal fractures** → distinguish

4 Diagnosis & Classification

Key radiology findings

Serendipity view	**40° cephalic tilt** of sternum + clavicles; best plain radiograph



Serendipity view: 40° cephalic tilt

CT	**most sensitive + specific**

5 Management

Instability after medial clavicle resection – figure-of-eight reconstruction

Cause	medial clavicle resection (e.g. for **SC degenerative joint disease**) compromising **costoclavicular ligament → medial clavicular instability**: ↑ pain, **medial bump**, digit tingling
Treatment	**semitendinosus figure-of-eight graft** through **clavicle + manubrium** → best **peak load-to-failure**

(✗ Wrong options: **Distractors** – subclavius tendon transfer) · medial clavicular osteotomy · further medial resection · SC arthrodesis

X vs Y

	SC dislocation	Medial clavicle physeal fracture
Age	Adult (physis closed)	**<25 y** (physis closes 20–25)
Frequency <25 y	Less common	**Most injuries**

How it's asked

"Instability after medial clavicle resection (medial bump, tingling) – procedure?"	→ **semitendinosus figure-of-eight graft** ✗ Wrong options: subclavius transfer · further resection · arthrodesis
"Primary restraint to AP translation of SC joint?"	→ **posterior SC capsular ligament**
"Ligament preventing medial displacement of clavicle?"	→ **intra-articular disc ligament**
"Best plain view of SC joint?"	→ **serendipity view (40° cephalic tilt)**; **CT** most sensitive
"Last physis to close?"	→ **medial clavicle, 20–25 y**

– – Active space – –

SUBACROMIAL IMPINGEMENT

– Shoulder & Elbow · 53 / 60 –

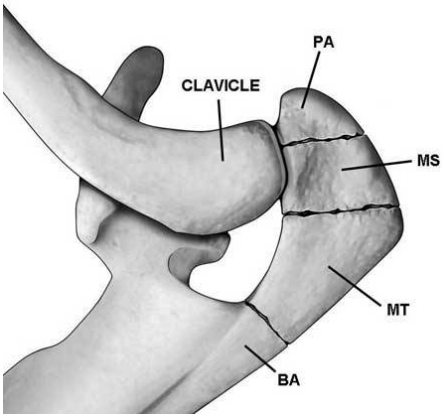
1 Basics

Epidemiology

Frequency	**most common cause of shoulder pain** · **44–65%** of shoulder disorders · with biceps tendinitis, among commonest causes

Anatomy – acromion ossification

Ossification centres	**pre-acromion** (tip) · **meso-acromion** (mid) · **meta-acromion** (base)
Os acromiale	failure of centres to fuse



Acromial ossification centres

Pathophysiology – metalloproteases

Extrinsic compression	cuff between humeral head and **anterior acromion**, **CA ligament**, **AC joint**
Intrinsic degeneration	supraspinatus attrition → can't balance head → **superior migration** → narrowed subacromial space
Bursitis	abutment → inflamed subacromial bursa → anti-inflammatory treatment rationale

Genes & lab markers

Marker	Condition
Metalloproteases (MMP) ↑	**Subacromial bursitis / impingement** ✓
TNF, **COX-1 + COX-2** ↑	Subacromial bursitis
TSH (hypothyroid), **HbA1c** (diabetes)	Wrong – tests for **adhesive capsulitis**
WBC	Wrong – tests for **septic bursitis**
ALP	Wrong – tests for osteoblastic tumours, Paget, hyperparathyroidism
AFP, PSA, CEA, CA-125	Wrong – tumour markers

Histology (tendinopathy) **disorganised collagen** · **mucoid degeneration** · inflammatory cells · bursa: high MMP + cytokines

2 Mechanism

Continuum (Neer) **impingement + bursitis (stage 1)** → tendinitis/fibrosis → partial → full-thickness → **massive tear** → **cuff tear arthropathy**

Associated conditions **hooked (type III) acromion** · **os acromiale** · **posterior capsular contracture** · **scapular dyskinesia** · tuberosity malunion · instability

3 Presentation

Pain **insidious** · **overhead activity** + lifting away from body · **night pain** = **poor predictor of nonop success**

Strength **usually normal**; full passive ROM

Test	Technique	Positive / meaning
Neer sign	Passive FF **>90°**	Pain
Neer test	Subacromial **lidocaine** then repeat	**Pain relieved**
Hawkins	**IR + passive FF to 90°**	Pain
Jobe (empty can)	Resisted pronation + FF 90°	**Supraspinatus** pathology
Painful arc	Abduction in scapular plane	Pain **60–120°**
Yocum	Hand on opposite shoulder, raise elbow	**Sensitive, nonspecific**
Internal impingement	ABER	Pain
Speed's	–	Biceps tendinitis

O'Brien pain not relieved in neutral equivocal → **not labral**; don't order MRA first

4 Diagnosis & Classification

Key radiology findings

View	Shows
True AP	**Acromiohumeral interval** – normal **7–14 mm**
30° caudal tilt	**Subacromial spur**
Supraspinatus outlet	**Acromial morphology** (Bigliani)
Axillary lateral	**Os acromiale** (best)

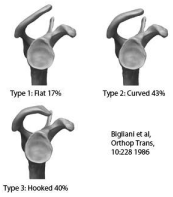
X-ray findings proximal humeral migration (CTA) · traction osteophytes · **CA ligament calcification** · GT cysts · **type III hooked acromion** · os acromiale

MRI degree of cuff pathology; subacromial/subdeltoid bursitis · CT arthrogram + US also image cuff accurately

MRI indications acute injury · failed nonop · **weakness** / suspected large tear
✗ Wrong options: first with benign exam

– – Active space – –

Bigliani classification of acromion morphology

Type	Shape	Diagram
I	**Flat**	
II	**Curved**	
III	**Hooked** → associated with impingement	

Reliability **poor inter-observer**

Acromial morphology – os acromiale

Types pre- · **meso-** · meta- os acromiale (named by unfused centre)

MRI (axial + sagittal) unfused segment ± **edema** in skeletally mature adult → **meso os acromiale**
✗ Wrong options: type I/III acromion



Os acromiale on outlet view

5 Management

Subacromial impingement (no cuff tear)

- First line → **PT (cuff + periscapular) + NSAIDs ± subacromial steroid**
  (scapular dyskinesia → integrated kinetic-chain rehab)
- Failed ≥4–6 months nonop → **subacromial decompression ± acromioplasty**
- Os acromiale → stage 1 fuse with graft → **stage 2 acromioplasty**

Nonoperative management – physical therapy & PRP

Mainstay **PT** (aggressive cuff strengthening + periscapular stabilisation), NSAIDs, subacromial injection

Benign exam (full strength, Jobe/Speed painful) **PT** before imaging
✗ Wrong options: AC injection · ASD · MRA · MRI

Outcome — **most improve with PT alone** (even with type III acromion) · but **tendinitis tends to recur**
✗ Wrong options: **"most fail nonop"**

PT vs surgery — PT **as effective** as subacromial decompression ± debridement (tendinitis, partial tears) · ASD + PT not better at 2 y

PRP — **no effect** on pain, function, ROM at **1 y** vs therapy alone · not harmful · does not prevent arthropathy · meta-analysis: insufficient evidence

Operative criteria

Procedure	Indication
Subacromial decompression / acromioplasty	Impingement **failed ≥4–6 months** nonop
Two-stage os acromiale	Fuse with bone graft → later acromioplasty (✗ Wrong options: direct excision → deltoid dysfunction)
Avoid acromioplasty + CA ligament release	**Massive irreparable cuff tear** (preserve CA arch → avoid anterosuperior escape)

Technique – acromioplasty

Step 1 — **anterior acromionectomy** (arthroscopic: limited by **anterior deltoid origin**, keep intact)

Step 2 — **anteroinferior acromioplasty** smooths undersurface (rasp open; shaver/burr arthroscopic) · open → meticulous **deltoid repair** to bone

Subacromial decompression outcomes (workers' compensation)

Lower satisfaction — **workers' compensation** (also longer return to work) · **anxiety + depression**
✗ Wrong options: dominant arm · male · smoker · age <60

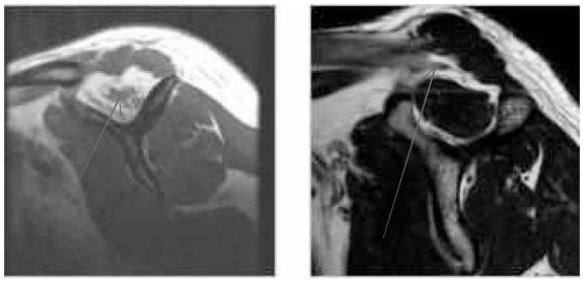
Massive cuff tears – irreparability & SCR contraindications (TO)

Irreparable (TO) — can't reach tuberosity despite release (size, retraction, atrophy, fatty infiltration)

Irreparability marker (TO)	Value
Massive tear	**>5 cm** (Cofield) or **≥2 tendons**
Pattern	**Posterosuperior (SS + IS)** > anterosuperior (SS + subscap)
Fatty atrophy	**Goutallier III–IV**
Acromiohumeral distance	**<6 mm**
Function	**Chronic pseudoparalysis** (can't elevate **>90°**; some say >45°)
MRI	**Positive tangent sign**

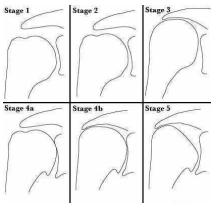
Tangent sign (TO) — sagittal oblique; line **top of scapular spine → top of coracoid** · SS belly **below line = positive → fatty atrophy, irreparable supraspinatus** / poor repair outcome
✗ Wrong options: subscapularis · retraction medial to glenoid · biceps dislocation

– – Active space – –



Tangent sign

Hamada classification of cuff tear arthropathy (TO)

Stage	Finding (TO wording)	Diagram
1	AHI **>6 mm**	
2	AHI <7 mm	
3	AHI <7 mm + **acetabularisation** of acromion	
4a	AHI <7 mm + **GH arthritis**, no acetabularisation	
4b	AHI <7 mm + acetabularisation + GH arthritis	
5	AHI <7 mm + **humeral head osteonecrosis**	

Hamada versions AHI **<7 mm**, stage 5 **osteonecrosis** (TO) · stage 2+ AHI **≤5 mm**, stage 5 **head collapse** (Orthobullets) – read the options

Superior capsular reconstruction (TO)

Concept (TO) passive restraint to **superior head translation** → restores force couples, ↓ subacromial impingement, ↓ pain

Graft (TO) folded **fascia lata 6–8 mm** or **acellular dermal 3–4 mm** · anchors on **superior glenoid rim** + **medial tuberosity**

SCR (TO)	Criteria
Indications	Failed nonop **massive irreparable** tear (SS + IS) · **Hamada 1–2** · **intact/reparable subscapularis** · **functional deltoid** · intolerable pain · pseudoparalysis
Contraindications	**Irreparable / torn subscapularis** ✓ · **Hamada ≥3** · GH OA

✗ Contraindication distractors (TO) Hamada I · pseudoparalysis · massive tear failed conservative treatment (= indications)

Young patient, massive irreparable posterosuperior tear (TO)

Option (TO)	Indication
Latissimus dorsi transfer	Retracted posterosuperior tear, **Goutallier ≥3** SS + IS; not if subscap insufficient or deltoid atrophy
Lower trapezius transfer	**More anatomical** for posterosuperior; promising
Pectoralis major transfer	**Anterosuperior** (SS + subscap) – most common
SCR	Young, no significant arthritis
Reverse TSA	**Older** patients with cuff arthropathy → **least preferred in 35-year-old** ✓

Tendon transfer prerequisites (TO) young, active, no significant GH arthritis, **intact subscapularis**

6 Complications

Deltoid dysfunction failed repair after open acromioplasty · **excessive acromionectomy** arthroscopically · **direct excision of os acromiale**

Anterosuperior escape acromioplasty + CA ligament release in **massive irreparable** tear

X vs Y

	Subacromial impingement	Adhesive capsulitis	Septic bursitis
ROM	**Full passive**	↓	–
Marker	**MMP, TNF, COX**	TSH, HbA1c (associations)	**WBC**

	Latissimus transfer (TO)	Pectoralis major transfer (TO)
Tear	**Posterosuperior**	**Anterosuperior**
Requires	Intact subscapularis, deltoid	–

How it's asked

"Insidious overhead pain, full ROM, MRI bursitis – elevated marker?"	→ **metalloproteases** ✗ Wrong options: TSH · HbA1c · WBC · ALP
"Markers contributing to subacromial bursitis pain?"	→ **metalloproteases** ✗ Wrong options: AFP, PSA, CEA, CA-125
"Partial SS tear, lidocaine relieved 90%, asks for PRP?"	→ **no effect on pain, function, ROM at 1 year**
"Type III acromion, Hawkins + empty can +, 5/5 – true about nonop?"	→ **most improve with PT alone** ✗ Wrong options: most fail · ASD better at 2 y · PRP better · never recurs

– – Active space – –

"Golfer, Jobe + Speed's painful, full strength — next?"	→ **PT for cuff strengthening** ✗ Wrong options: ASD · MRI · MRA · AC injection
"Lower satisfaction after ASD?"	→ **workers' compensation**
"Axial + sagittal MRI unfused acromion with edema?"	→ **meso os acromiale**
"Contraindication to SCR?"	→ **irreparable subscapularis tear** (TO)
"35-year-old, pseudoparalysis, AHD <6 mm, Goutallier 3 — least preferred?"	→ **reverse shoulder arthroplasty** (TO)
"Positive tangent sign predicts?"	→ **irreparable supraspinatus (fatty atrophy)** (TO)
"Best view for os acromiale?"	→ **axillary lateral**
"Night pain significance?"	→ **poor predictor of nonop success**
"Massive irreparable tear — avoid?"	→ **acromioplasty + CA ligament release** (anterosuperior escape)

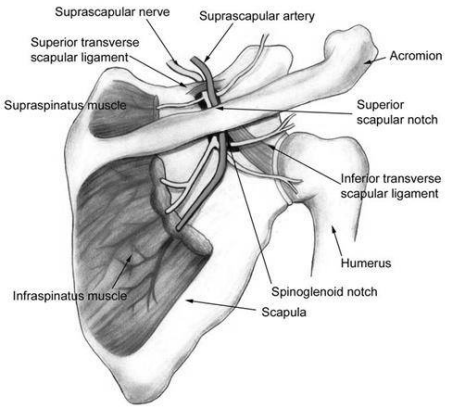
SUPRASCAPULAR NEUROPATHY

– Shoulder & Elbow · 54 / 60 –

1 Basics

Anatomy – suprascapular nerve course

Origin	**C5–C6**, **superior (upper) trunk** of brachial plexus → across **posterior triangle of neck** → superior border of scapula
Course	under **superior transverse scapular (suprascapular) ligament** at **suprascapular notch** → supraspinous fossa (**supraspinatus**) → around lateral border of scapular spine through **spinoglenoid notch** (under **spinoglenoid ligament**) → infraspinous fossa (**infraspinatus**)
Suprascapular ligament	from medial base of coracoid over notch · **artery ABOVE, nerve BELOW**
Function	**motor only** (SS + IS) · **no sensory** → no sensory deficit
Transglenoid fixation risk	nerve lies **1.5 cm medial** to glenoid labrum at spinoglenoid notch



Suprascapular nerve – sites of compression

Site	Level	Muscles weak
Suprascapular notch	**Proximal**	**Supraspinatus + infraspinatus**
Spinoglenoid notch	**Distal** (SS branch already given off)	**Infraspinatus only**

TO wording	"infraglenoid notch" compression → infraspinatus only (= spinoglenoid notch)
Atraumatic compression sites	**suprascapular + spinoglenoid notches** (fascial bands, ganglion cysts) ✗ Wrong options: scalenus anterior · cervical rib · conjoined tendon · subcoracoid

Epidemiology & associations

Who	**overhead athletes**: **volleyball** (classic), baseball pitchers/catchers, tennis · also weightlifters, rowers, linebackers
Associated	**SLAP tears** · posterior labral tears · internal impingement

– – Active space – –

2 Mechanism

Suprascapular notch – causes

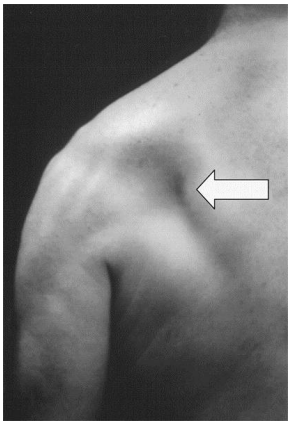
Causes	**ganglion cyst** (often with labral tear) · **transverse scapular ligament** entrapment/thickening · **fracture callus**

Spinoglenoid notch – causes

Causes	**posterior labral tear → cyst** · spinoglenoid ligament · spinoglenoid notch ganglion · **traction** (**45% of volleyball players**) · transglenoid fixation
Cyst mechanism	labral tear = **one-way valve** → synovial fluid exits joint, can't drain → paralabral cyst
Cyst frequency	ganglion cysts at **spinoglenoid notch** > **less commonly suprascapular notch** · cyst origin: transverse scapular ligament, scapular fibrous tissue, or GH joint
Labral source	**SLAP** most often, posterior labrum less (OB) · **posterior labral tear** (Orthobullets, OB) – read the options; posterior SLAP / **posterosuperior** labral tear fits both

3 Presentation

Symptoms	**deep, diffuse, posterolateral shoulder pain** · weakness, fatigue with overhead activity · ↓ velocity · no numbness
Suprascapular notch	tender notch · **supraspinatus weak**: abduction 90° + 30° FF + IR (**Jobe +**) · **infraspinatus weak**: **ER with elbow at side** · atrophy above + below spine
Spinoglenoid notch	**isolated weak ER with arm at side** · **atrophy below scapular spine** (posterior shoulder) · **supraspinatus normal** (empty can negative) · O'Brien + (associated SLAP)



Isolated infraspinatus atrophy

Most specific exam finding (spinoglenoid cyst)	**weakness of ER** ✗ Wrong options: impingement sign · apprehension · active compression · abduction weakness
ER in elevation	isolated IS palsy: **full ER with arm elevated** · combined SS + IS: **can't ER in forward elevation**

Distinguishing shoulder tests

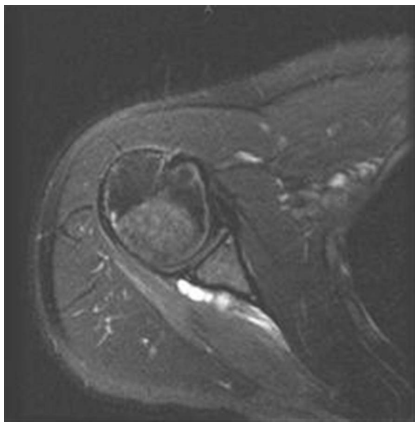
Test	Muscle / lesion	In SSN entrapment
Jobe / empty can	**Supraspinatus**	Abnormal at **suprascapular notch** ✓
ER with arm at side	**Infraspinatus**	Abnormal at **both** sites ✓
Hornblower (90° abduction + 90° ER, falls into IR)	**Teres minor** (axillary)	Normal
Belly press · bear hug · lift-off	**Subscapularis**	Normal
Anterior load and shift (supine, 40–60° abduction + FF)	Anterior instability	Normal
Kim test (seated, 90° abduction → 45° FF + axial + posteroinferior force)	**Posterior instability**	Normal
Jerk test	**Posterior labral tear**	May be + (cyst source)

4 Diagnosis & Classification

Key radiology findings

MRI identify **compressive mass / cyst** + associated **labral lesion** · cyst: well-defined, **low T1, high T2** · denervation: ↓ bulk + **fatty infiltration** (SS ± IS)

Paralabral cyst most commonly with **labral tears**



Spinoglenoid notch cyst



Suprascapular notch cyst

– – Active space – –

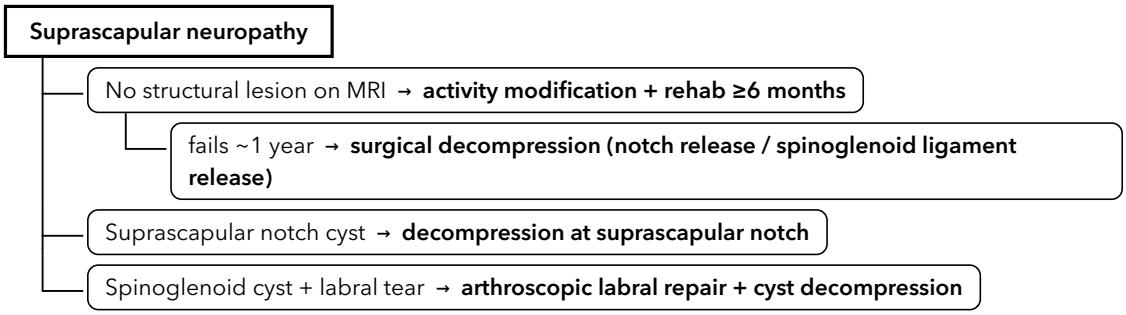
Diagnosis – EMG & compression sites

EMG/NCV	**diagnostic**; with MRI = investigations of choice · localises compression
Isolated ER weakness + atrophy, MRA normal (no cyst, no tear)	**EMG/NCV first** → found notch entrapment with early SS denervation ✗ Wrong options: cervical MRI first · subacromial steroid · notch release before EMG · CRP/ESR
C5 radiculopathy	weakness out of proportion; not when specific SSN pattern

Lesion (distractor)	Nerve / site	Deficit
Quadrangular (quadrilateral) space	**Axillary + PHCA**	**Teres minor** atrophy, deltoid (forward elevation, abduction)
Triangular interval	**Radial nerve + profunda brachii**	Elbow + wrist extension
Subscapular recess	Anterior to subscap, below coracoid	Loose bodies
Upper/lower subscapular nerves	Subscapularis	IR (lift-off)
Long thoracic (C5–7 roots)	Serratus anterior	Not affected by spinoglenoid cyst

Brachial plexus figure	suprascapular = **A (upper trunk)** · dorsal scapular · medial pectoral · lower + upper subscapular = wrong options

5 Management



Operative criteria

Situation	Treatment
No structural lesion	**Nonop ≥6 months** (posterior capsule stretching for spinoglenoid)
No lesion + failed **~1 year** nonop	Suprascapular notch release · or **spinoglenoid ligament release** (posterior approach)
Structural lesion (cyst) at suprascapular notch	**Nerve decompression at notch**
Labral tear + spinoglenoid cyst	**Arthroscopic labral repair + cyst decompression** ✓

Treatment – labral repair with cyst decompression

Young competitive athlete, posterior labral tear + spinoglenoid cyst	**cyst decompression + posterior labral repair** ✗ Wrong options: 6 wk rest · PT · suprascapular notch decompression · cuff repair + acromioplasty
Catcher, posterosuperior tear + large cyst, severe ER weakness	**arthroscopic repair + decompression** ✗ Wrong options: aspiration + steroid · rest · acromioplasty · rehab
Labral tear key	address tear (cause of cyst) · **no isolated cyst decompression** when tear present
Labral repair alone	not enough (doesn't decompress cyst, atrophy may progress — OB) · "good outcomes with labral repair alone; needle aspiration also used" (one OB explanation) — read the options; best answer = **repair + decompression**
Failed 6 wk PT with cyst + IS atrophy	operate ✗ Wrong options: another 6 wk PT · cervical MRI + EMG · tenodesis
Mechanical compression	surgical decompression usually required

6 Complications

Iatrogenic injury

Arthroscopic SSN release at suprascapular notch → injured nerve → weak **abduction, forward elevation + ER**, wasting **above + below scapular spine** = suprascapular notch level
✗ Wrong options: spinoglenoid = IS only · axillary · spinal accessory · dorsal scapular

SLAP repair	anchors/instruments **medial to glenoid** → SSN injury

X vs Y

	Suprascapular notch	**Spinoglenoid notch**
Muscles	**SS + IS**	**IS only**
Weakness	**Abduction (Jobe) + ER**	**ER at side** only
Atrophy	Above + below spine	**Below spine**
Usual cause	Cyst, **transverse scapular ligament**, callus	**Paralabral cyst (SLAP/posterior labrum)**, traction
Treatment (lesion)	Decompress at notch	**Labral repair + cyst decompression**

	Spinoglenoid cyst	**Quadrilateral space syndrome**
Nerve	Suprascapular	**Axillary**
Atrophy	**Infraspinatus**	**Teres minor** (± deltoid)
Test	ER at side weak	Hornblower; forward elevation/abduction weak

– – Active space – –

How it's asked

"Burning pain, SS + IS atrophy, weak abduction + ER — site?"	→ **suprascapular notch** (TO)
"Volleyball, isolated IS atrophy, axial MRI cyst — compression?"	→ **suprascapular nerve in spinoglenoid notch** ✗ Wrong options: quadrangular space · triangular space
"Volleyball, MRI cyst in suprascapular notch — weakness?"	→ **abduction + external rotation**
"SLAP + cyst on sagittal MRI — extra exam finding?"	→ **weak external rotation**
"Weak ER at side, atrophy below spine, O'Brien + — MRI?"	→ **SLAP tear + spinoglenoid notch cyst**
"Isolated IS atrophy on clinical photo + MRI — cyst location?"	→ **spinoglenoid notch** ✗ Wrong options: quadrangular space · subscapular recess
"Isolated ER weakness, normal flexion/abduction, 44 y?"	→ **spinoglenoid notch cyst** ✗ Wrong options: CN XI · Parsonage-Turner
"Normal empty can + belly press, weak ER — cause?"	→ **SSN entrapment at spinoglenoid notch**
"Suprascapular notch cyst — weakness on exam?"	→ **ER with arm at side** ✗ Wrong options: scapular protraction · IR
"Volleyball, posterosuperior pain — most specific exam finding?"	→ **weakness of external rotation**
"Weak ER at side, normal lift-off — nerve + site?"	→ **suprascapular nerve, spinoglenoid notch**
"Pitcher, O'Brien +, ER weak, empty can 5/5 — muscle at risk?"	→ **infraspinatus**
"Posterior labral tear + paralabral cyst — structure affected (MRI figure)?"	→ **infraspinatus** ✗ Wrong options: supraspinatus · subscapularis
"Spinoglenoid cyst — exam?"	→ **weak ER with arm adducted** ✗ Wrong options: Jobe · belly press · Hornblower
"Linebacker, weak ER at side, spinoglenoid cyst — arthroscopic finding?"	→ **SLAP tear** ✗ Wrong options: Bankart · PASTA · Hill-Sachs
"Cyst in suprascapular notch on MRI — which muscles?"	→ **supraspinatus + infraspinatus**
"Spinoglenoid cyst on MRI — which muscle weak (figure)?"	→ **infraspinatus (B) only**
"Weak ER at side, full ER when elevated — MRI?"	→ **spinoglenoid cyst + IS fatty atrophy**

"Weak forward elevation + ER at 0° abduction — commonest compression site (figure)?"	→ **suprascapular nerve (A)**
"Rower, IS atrophy, negative empty can — cause?"	→ **spinoglenoid notch cyst** ✗ Wrong options: Bankart · subscapularis tear
"Volleyball, suprascapular notch cyst — abnormal test?"	→ **Jobe's test** ✗ Wrong options: load-and-shift · bear hug · Hornblower · Kim
"Weightlifter, weak ER, posterior labral tear + cyst — treatment?"	→ **spinoglenoid cyst decompression + posterior labral repair**
"Catcher, cyst + posterosuperior labral tear — management?"	→ **arthroscopic repair + decompression**
"Jerk test +, labral tear + spinoglenoid cyst — procedure?"	→ **labral repair + cyst decompression** ✗ Wrong options: isolated decompression
"Posterosuperior tear, cyst, mild IS atrophy, failed 6 wk PT — next?"	→ **labral repair + cyst decompression** ✗ Wrong options: repair only · cervical MRI
"MRI cyst, abnormal EMG — nerve?"	→ **suprascapular**
"Tennis player, paralabral cyst, ER weakness — cause?"	→ **suprascapular nerve compression** ✗ Wrong options: C5 radiculopathy · IS tear
"Superior labral cyst — clinical finding?"	→ **atrophy of posterior shoulder** ✗ Wrong options: any sensory deficit
"Weak ER, no tear or cyst on MRA — next?"	→ **EMG/NCV**
"Atraumatic SSN compression sites?"	→ **suprascapular + spinoglenoid notches**
"After arthroscopic notch release: weak abduction + ER, wasting above and below spine?"	→ **suprascapular nerve at suprascapular notch**

– – Active space – –

THORACIC OUTLET SYNDROME

– Shoulder & Elbow · 55 / 60 –

1 Basics

Epidemiology

Definition compression of **brachial plexus ± subclavian vessels** between neck and axilla (first rib–clavicle); heterogeneous, potentially disabling

Incidence **1–2%** of population

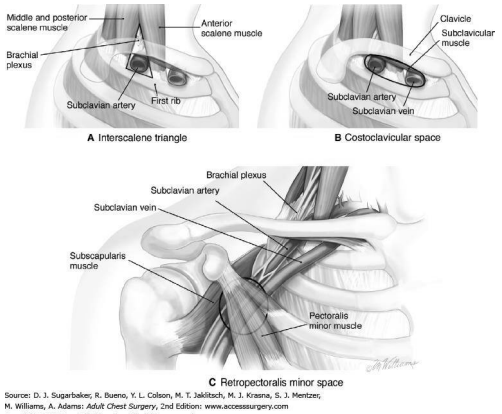
Demographics ♀ **>** ♂ **(3:1)**; **thin, long neck, drooping shoulders**; age **20–60**

Type	Frequency	Note
Neurogenic	**95% (most common)**	Lower plexus
Venous	**4%**	Paget-Schroetter; athletic ♂ > ♀
Arterial	**<1%**	Aneurysm, emboli; athletic ♂ > ♀

Anatomy – thoracic outlet spaces

Boundaries (OB) **clavicle, first rib, subclavius, costoclavicular ligament, anterior scalene**

Space	Borders	Contents	Test
Interscalene triangle (proximal)	**Ant: anterior scalene · Post: middle scalene · Inf: 1st rib**	**Plexus trunks + subclavian artery** (✗ Wrong options: vein (runs **beneath anterior scalene**))	**Adson**, supraclavicular pressure
Costoclavicular space (middle)	Ant: clavicle + **subclavius** · Post: 1st rib + scalenes · Med: **costoclavicular ligament** · Lat: upper scapular border	**Divisions + subclavian artery + vein**	**Costoclavicular manoeuvre**
Retropectoralis minor (subcoracoid; distal)	Sup: **coracoid** · Ant: **pectoralis minor** · Post: **ribs 2–4**	**Cords + axillary artery + vein**	**Wright**



Three spaces of the thoracic outlet

Most affected (OB) **subclavian artery, vein + lower trunk (C8–T1)**

2 Mechanism

Pathogenesis **anatomic predisposition + superimposed neck trauma** (acute or repetitive)

Split **soft tissue 70% · osseous 30%**

Soft tissue	Detail
Scalene abnormalities	**Anterior scalene hypertrophy** · plexus **through** anterior scalene · **middle scalene anterior insertion** on 1st rib
Scalenus minimus	Accessory; **30–50%** of TOS; C-transverse process → 1st rib **between subclavian artery + T1 root**
Fibromuscular bands	↑ stiffness, ↓ compliance; congenital band → lower trunk
Costoclavicular ligament	Abnormal insertion → **Paget-Schroetter**
Tumours	**Pancoast** (1–3% lung cancers; **lacks cough/haemoptysis/dyspnoea**), neuroblastoma, plexus schwannoma
Pectoralis minor	Abnormal

Osseous	Detail
Cervical rib	**<1%** population; from **C7**
Prominent C7 transverse process	
Clavicle / 1st rib	Displaced fracture, **hypertrophic callus**, malunion
AC / SC injury	Dislocation
Tumours	1st-rib metastasis (**breast, prostate, kidney**), osteoid osteoma

Cervical rib types

Type	Description
1	**Complete**, articulates with 1st rib or manubrium
2	Incomplete, **free bulbous tip**
3	Incomplete, **fibrous band** distally
4	**Short bar** (mm) beyond C7 transverse process

Overuse repetitive overhead lifting, **hyperabduction**; **weightlifting, rowing, swimming**

Vascular damage aneurysm, thrombosis, emboli, limb-threatening ischaemia

Paget-Schroetter syndrome

Definition **venous TOS** in **well-developed young athletes → upper-limb DVT**

Cause intermittent **subclavian vein** obstruction in **costoclavicular space** by **abnormal costoclavicular ligament** or **anterior scalene hypertrophy**

– – Active space – –

3 Presentation

Neurogenic

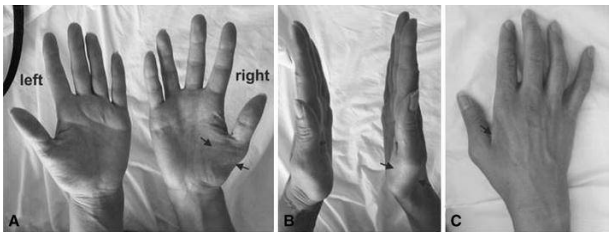
Pain	neck, **trapezius (92%)**, chest, shoulder, arm
Paraesthesia	**98%**; **non-radicular**, **wide plexus distribution** · **lower plexus (C8–T1) or combined (C5–T1) in 90% → ring + small fingers**, medial forearm
Other	weakness, numbness, **heaviness with overhead activity** · activity-related + **night symptoms** (↓ pressure → returning sensation felt as pain) · uni- or bilateral
Posture	rounded shoulders, ↑ kyphosis, depressed/downward-rotated scapula

Gilliatt-Sumner hand (TO)

Definition (TO)	rare, severe sign of **true neurogenic TOS**; **chronic lower-trunk (C8–T1)** compression by **cervical rib or congenital fibrous band**
Pattern	**preferential APB atrophy, relative hypothenar sparing** (TO, ✓) · atrophy of **APB + hypothenar + interossei** (Orthobullets, TO Elvey list) – read the options
Why APB (TO)	**T1-innervated** muscles hit harder (median + ulnar both get lower-trunk fibres)
Red flag (TO)	thenar wasting usually = CTS, but in TOS **+ medial forearm sensory change** + milder ulnar intrinsic involvement

(✗ Wrong options: **Distractors (TO)** – symmetric interossei/hypothenar wasting sparing thenar) · claw hand from ulnar compression at humerotricipital groove · thumb/index C6 sensory loss

Diagnosis (TO)	EDX (neurogenic changes) + MRI/US
Treatment (TO)	**surgical decompression** → stops progression of atrophy + pain; partial motor recovery possible
Differential (TO)	**CTS, ALS (motor neuron disease), Hirayama disease**



Gilliatt-Sumner hand: thenar (APB) wasting

Vascular

Venous	**episodic cyanosis + swelling**, distended veins, deep arm ache, **heaviness after activity**
Arterial	**unilateral Raynaud-type**: coolness + **pallor → cyanosis → erythema**, worse in cold; **hand** predominantly; ulcers/microemboli (rare)

Provocative tests & neurogenic TOS

All provocative tests high false-positive rate

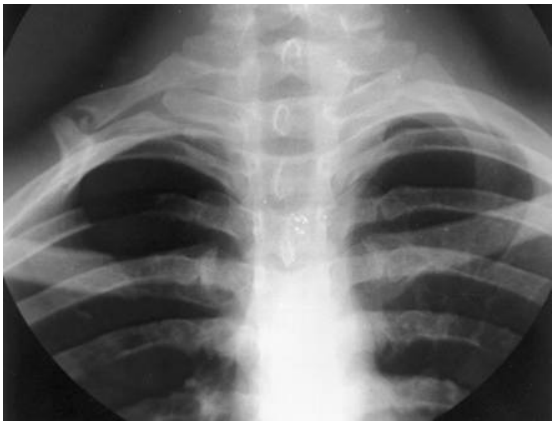
Test	Technique	Positive	Space
Supraclavicular pressure	Squeeze upper trapezius + anterior scalene **30 s**, arm at side	Pain/paraesthesia	Interscalene
Adson	Arm slightly abducted + ER (TO: abducted + **extended**), elbow extended; **neck extended + rotated TOWARD tested side**; inhale + hold	↓/absent radial pulse ± symptoms; **51% of normals** ↓ pulse; TO: **low sensitivity + PPV** (modified: tilt to opposite side / symptom aggravation)	**Interscalene**
Costoclavicular (military / "attention")	Shoulders retracted + depressed, chest out; shoulder extended ~30° **1 min**	↓ pulse ± symptoms	**Costoclavicular**
Wright (hyperabduction)	Neck rotated **away**; shoulder ER + **max abduction above head 1 min** (TO: 1–2 min)	↓ pulse (**7% of normals**) ± symptoms	**Retropectoralis minor**
Roos / EAST	Orthobullets: **abduct 90° + elbow 90°**, open/close hands **3 min** · TO: arms **abducted, ER, elevated overhead**, open/close fingers **1–3 min**	**Reproduction of symptoms**; can't complete 3 min; relief on dropping arms	**Entire outlet**
Cyriax release	Examiner behind, elevates shoulder girdle passively **3 min**	Symptoms reproduced (unloading)	Unloading
Halstead (TO)	Downward traction on arm, **neck hyperextended + rotated to opposite side**	**Radial pulse disappears**	
ULTT of Elvey (TO)	Abduct 90°, elbow extended, **wrist dorsiflexed**, then **ear to contralateral shoulder**	Ipsilateral symptoms = **neurogenic TOS**	Plexus stretch
✗ Wrong option: **Spurling (TO)**	Neck extension + lateral flexion + **axial compression**	**Radicular** pain	**Cervical radiculopathy**

– – Active space – –

Key radiology findings

Test	Use / finding
CXR + C-spine X-ray	**Cervical rib**, prominent C7 TP, low shoulder girdle, **Pancoast** (or normal)
CT	Osseous lesions, rib/clavicle malunion
MRI	Soft-tissue anomalies
EMG/NCV	Historically equivocal (normal until permanent damage) · early: **↓ NCV in medial antebrachial cutaneous + median motor to APB** (C8–T1)
Doppler US	Subclavian vein obstruction/thrombosis — **95% sensitive, 92% specific** for venous TOS
CTA/MRA · arteriography	**Embolic disease / aneurysm**
Venography	Suspected **subclavian/axillary vein thrombosis**

Normal neuro exam + EMG/NCS does **not** exclude TOS (✓ cervical rib vignette)



Bilateral cervical ribs

Diagnosis – cervical rib

Classic vignette **bilateral ring + small finger tingling**, worse **overhead**, normal neuro exam + EMG/NCS, cervical ribs on X-ray → **TOS**
✗ Wrong options: C5–6 / C6–7 disc · cubital tunnel · radial tunnel

5 Management

Thoracic outlet syndrome

- Neurogenic → **PT, activity modification, NSAIDs**
  - fails 6 mo / progressive atrophy → **decompression**
- Scalene contracture → **US-guided scalene block (lidocaine / Botox)**
- Vascular → **by lesion (see table)**

Nonoperative

First line	avoid provocative activity (overhead, ± change job) · NSAIDs, muscle relaxants · **PT**: core/back + shoulder-girdle strengthening, **posture**, relaxation · TENS
Worse results	**obese** · **workers' compensation** · **double crush** (carpal/cubital tunnel)
Anterior scalene block	**US-guided lidocaine or botulinum toxin**; successful block → **14% higher** good surgical outcome
Leffert (OB)	reserve surgery for **intolerable** symptoms (complications significant)

Operative criteria

Indication	
Failed nonop 6 months	Decompression
Progressive atrophy / worsening neuro deficit	Decompression
Embolic events · stenosis with pain + insufficiency · **subclavian aneurysm** · thrombosis with critical ischaemia	Vascular intervention

Decompression procedures

Procedure	**Indication**
1st rib resection + anterior + middle scalenectomy + neurolysis	**Most common**; **95% good outcomes**; combined transaxillary + supraclavicular
Isolated scalenectomy	**Upper plexus** symptoms · **no bony anomaly** · muscular/obese · **recurrence after 1st rib resection**
Isolated pec minor tenotomy	Symptoms reproduced at **retropectoralis minor** space
Cervical rib resection · band release · costoclavicular ligament resection · clavicle malunion ORIF	Per cause

Vascular intervention

Lesion	**Treatment**
Acute embolism	**IV heparin** ± embolectomy ± thrombectomy ± **tPA** + anticoagulation
Small-vessel embolism	**tPA + anticoagulation**
Large/proximal embolism	**Embolectomy + anticoagulation**
Mild stenosis	**Endovascular stent**
Subclavian aneurysm / severe stenosis or thrombosis + critical ischaemia	**Resection** ± repair / saphenous / arterial / synthetic graft
Chronic emboli + critical ischaemia	**Bypass**

– – Active space – –

Approaches

	Transaxillary	Supraclavicular	Posterior
Use	**Most common**; **best 1st rib exposure**	**Upper plexus**, scalenes, 1st rib neck, vessels	**Recurrent TOS**, **prior neck surgery**
Best for	1st rib, cervical rib, bands, **lower plexus (C7–T1)** neurolysis	**Isolated scalenectomy + arterial reconstruction**	Proximal plexus
Pros	**Cosmetic scar**, no NV retraction for rib		
Cons	**Plexus injury risk**	Poor 1st rib view (needs plexus/vessel retraction)	Extensive dissection → shoulder dysfunction; **LTN, dorsal scapular, accessory nerve** risk

6 Complications

Pneumothorax **one of the most common** after 1st rib resection

X vs Y

	Neurogenic TOS	Cervical radiculopathy	Cubital / carpal tunnel
Distribution	**Non-radicular, wide (plexus)**, C8–T1	**Radicular**	**Clear dermatomal/peripheral** distribution
Provocation	Overhead, Roos, Adson, Wright	**Spurling**	
EMG	Often normal early; MABC + APB motor ↓		

	Gilliatt-Sumner (TOS)	CTS
Thenar (APB) wasting	**Yes** (preferential)	Yes
Medial forearm sensation	**Abnormal**	Normal
Lesion	**Lower trunk C8–T1**	Median at wrist

How it's asked

"35-y, bilateral ring/small tingling overhead, normal EMG, X-ray shown?"

→ **thoracic outlet syndrome (cervical rib)**
✗ Wrong options: C5–6/C6–7 disc · cubital tunnel · radial tunnel

"Arms abducted, ER, overhead; open-close fingers 1–3 min — test?"

→ **Roos test** (TO)
✗ Wrong options: Adson · Wright · Spurling

"Gilliatt-Sumner hand in true neurogenic TOS?"	→ **preferential APB atrophy, relative hypothenar sparing** (TO) ✗ Wrong options: symmetric interossei sparing thenar · claw from ulnar at humerotricipital groove · C6 thumb/ index loss
"Shoulder ER + max abduction above head 1 min, neck turned away — test and space?"	→ **Wright test; retropectoralis minor space** ✗ Wrong options: Adson = interscalene · costoclavicular
"Most common complication of 1st rib resection?"	→ **pneumothorax**

– – Active space – –

TOTAL ELBOW ARTHROPLASTY

– Shoulder & Elbow · 56 / 60 –

1 Basics

Epidemiology & forms of elbow arthroplasty

Trend — **TEA for trauma** (elderly distal humerus fractures) = one of the **fastest-growing** indications; historically for **RA**

Forms — **TEA** · **hemi** (radiocapitellar, distal humeral) · **ulnohumeral distraction / interposition arthroplasty** · **olecranon fossa debridement** · radial head arthroplasty

Implant design (linked, semiconstrained, unlinked) & survivorship

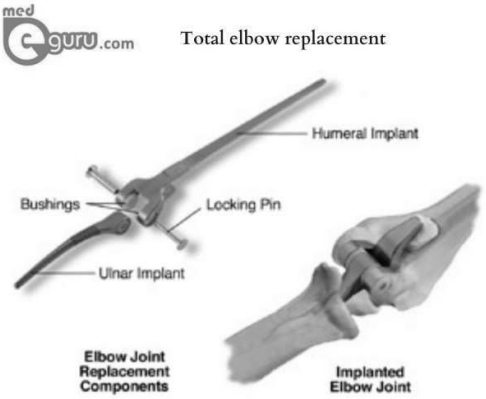
	Unconstrained (unlinked, resurfacing)	Semiconstrained (linked)	Constrained
Example	**Ewald capitello-condylar**	**Coonrad-Morrey**	**Dee**
Hinge	None	**"Sloppy hinge": 7° varus-valgus + 7° axial rotation**	**Rigid hinge**
Needs	**Competent collaterals + soft-tissue envelope**, good bone, **precise alignment**	**collaterals not needed**; radial head not needed	–
Use	–	**Most commonly used**	**No longer used**
Results	No proven superiority	**Best results** of all designs	**Highest loosening**
Commonest failure	**Instability** (5–10% dislocation)	Instability (Orthobullets, TO) · bushing wear (OB)	**Aseptic loosening**, then **bushing wear**

Semiconstrained laxity — ↓ stress at **bone–cement interface → ↓ aseptic loosening**, at the cost of **bushing wear**

Constrained — no coronal/rotational flexibility → **high shear stress → early loosening**; theoretically most stable

Unconstrained suits — humeroulnar disease with **intact collaterals + radiocapitellar joint**: OA, post-traumatic arthritis, acute intra-articular distal humerus fracture, distal humerus malunion, bony ankylosis

Unconstrained relative contraindication — **late-stage RA** (capsuloligamentous instability + bony erosion); RA → **linked**
✗ Wrong options: unlinked: tendon injury → instability



Semiconstrained (linked) TEA: humeral + ulnar implants, bushings, locking pin

Design pearls

Stems	humeral + ulnar stems → better fixation, ↓ loosening
Anterior humeral flange	extracortical; resists **posteriorly directed + rotational** forces; designs without flange → **early humeral loosening**; improper flange placement → humeral loosening (TO)
Radial head	**not needed for stability** in linked designs; often debrided/resected in RA

4 Diagnosis & Classification

Key radiology findings

Imaging	Purpose
AP + lateral elbow	Bone stock, **medial + lateral columns intact**, **canal diameter**
C-spine flexion–extension	**RA** patients (pre-op)
CT	Bone stock / fracture
True AP in full extension (post-op)	**Bushing wear**: humeral yoke–ulnar component angle **> 7° = wear**, **> 10° = extensive/complete** (TO) · **> 10°** concerning (Orthobullets)
Gruen zones (4)	Zone 1 periarticular, zones 2–4 = stem thirds; **diffuse lucency ≥ 3 zones**, **focal ≤ 2 zones**

Semiconstrained laxity (TO)	**7° total = 3.5° each direction**; **> 7° = partial wear**; **> 10° = complete wear**

– – Active space – –



Bushing wear — varus-valgus tilt of ulnar component on AP

Cement mantle quality (Morrey, TO)

Quality	Radiolucent zone	Cement beyond stem tip
Adequate	**< 1 mm**	Yes
Marginal	≥ 2 mm	Yes
Marginal	< 2 mm	No
Inadequate	**≥ 2 mm**	**No**

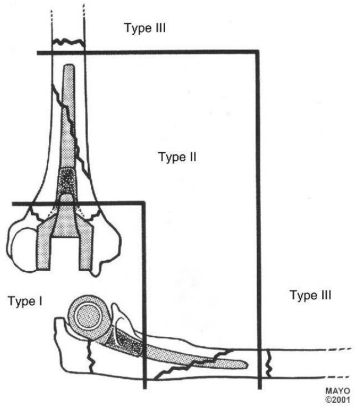
Humeral bone loss grading

Grade	Humeral bone stock
I	Articular surface damage only; **intact subchondral bone**
II	**Absent trochlea**; intact medial + lateral epicondyles
III	**Absent medial or lateral epicondyle**
IV	**Loss of both epicondyles** or resection **proximal to olecranon fossa**

Mayo (O'Driscoll & Morrey) periprosthetic fracture classification

Type	Characteristics	Treatment
I (**commonest**, TO)	**Periarticular — humeral condyle or olecranon**; osteolysis around hinge + muscle distraction	Undisplaced → **immobilise / soft-tissue repair** (fibrous union OK) · displaced → **ORIF heavy nonabsorbable suture or TBW**
II1	Along stem	**ORIF, retain components ± strut allograft**
II2	Along stem, **loose implant, good bone stock**	**Revision long-stem ± strut allograft + impaction grafting** (± locking plate / cerclage)
II3	Along stem, **loose implant, severe bone loss**	**Revision + extensive allograft**; often **resection arthroplasty**
III	**Distal to prosthesis tip**	Well-fixed → **immobilise humerus, ORIF ulna** · loose → treat as **II2**

Basis modelled on **periprosthetic femoral** fracture classification; site + implant fixation + bone stock



Mayo periprosthetic elbow fracture types I–III

5 Management

Operative criteria – RA, elderly distal humerus fracture, failed interposition

Indication	Criteria
Rheumatoid arthritis	**10–20%** of RA have elbow disease; **Larsen 3–5** (Orthobullets) · grade III–IV (OB) + pain, functional loss, instability; ideally **> 65 y**; synovectomy for less advanced disease
Primary OA (advanced)	**> 65 y**; **mid-arc pain** from ulnotrochlear cartilage loss
Fracture	**Physiologically elderly (> 70 y)**: acute complex **unreconstructable intra-articular distal humerus** fracture, missed fracture-dislocation, **poor bone**
Post-traumatic OA (advanced) · chronic instability	–
Failed interposition arthroplasty	End-stage arthritis + instability → **semiconstrained TEA**

Contraindications **absolute**: **active infection** (→ arthrodesis), **Charcot joint** · **relative**: poor neurological control (UE paralysis), **active patient < 65 y**, **olecranon osteotomy** · RA list (OB): active/**historical** infection, paralysis, won't follow instructions

Olecranon osteotomy contraindication: preserve **extensor mechanism**; trans-olecranon approach **jeopardises ulnar component stability** → never for TEA

Not contraindications central comminution · **ulnar nerve transposition** · radial head impaction · retrocapitellar comminution (radiocapitellar joint secondary)

- **Elderly unreconstructable distal humerus fracture**
  - Physiologically elderly, low demand, poor bone → **TEA (semiconstrained)**
    - approach → **direct posterior triceps-sparing**
  - not ORIF (unstable fixation), acute fusion or anterolateral approach

Fracture TEA vs ORIF **faster recovery + more predictable function**; elderly accommodate limits, fewer complications
✗ Wrong options: slower recovery

– – Active space – –

Anterolateral approach	brachialis/brachioradialis (radial nerve between) → **no access for TEA**
Prior radial head resection in RA	**semiconstrained > unconstrained** (both better in native elbows)
Failed interposition, 60-yo sedentary	**semiconstrained TEA** ✗ Wrong options: unconstrained fails with ligament instability · revision interposition · Outerbridge-Kashiwagi · arthroscopic osteocapsular arthroplasty

Survivorship & outcomes

Indication	Outcome
RA	**Longest survivorship** of all indications (> fracture, flail elbow, OA, post-traumatic); **92.4% revision-free at 10 y**; **> 90% up to 15 y**
RA complications	**14%** (Orthobullets) · **~15%** (OB) – triceps avulsion, deep infection, periprosthetic fracture, aseptic loosening (✗ Wrong options: "< 5%")
Primary OA	**10-y survival 80–85%**; more active → **↑ aseptic loosening + stem fracture vs RA**
Overall	**> 80% at 10 y**
Post-traumatic	Functional ROM, satisfaction at 5 y; **complications 27–43%**, **re-operation 25%**

Longest survivorship combination	**RA + semiconstrained**
Linked TEA, OA vs RA	OA **↑ aseptic loosening**; **instability equal** (linked)

Surgical approach & triceps insufficiency

Approach	Technique	Key fact
Triceps-reflecting (Bryan-Morrey)	Triceps + **anconeus reflected medial → lateral**; reattached to ulna with **nonabsorbable suture through bone tunnels**	**Triceps failure**; **↓ extension power up to 30%**; **delayed ROM**
Triceps tongue	Fascial tongue raised from olecranon proximally; release collaterals	Fractures or TEA
Triceps-splitting	Longitudinal split in continuity with forearm fascia; or **V-shaped turndown** (insertion intact)	Allows **extensor lengthening**
Triceps-sparing (e.g. Alonso-Llames, para-olecranon)	Mobilise medial + lateral triceps borders; insertion retained	**Best for acute fracture TEA**; **earlier ROM**, ↑ final ROM; **↓ ulnar visualisation**
Triceps "on"	Midline incision; release/protect **ulnar nerve**; release flexor-pronator + MCL, common extensors + LCL; elevate triceps off posterior humerus; dislocate ulnohumeral joint	**No triceps release → no insufficiency**
Trans-olecranon	Osteotomy, triceps stays on olecranon	**ORIF only**, never TEA

Triceps-off vs triceps-sparing	triceps-off: **delayed ROM**, ↑ triceps failure · **no difference**: cement mantle, ulnar nerve palsy · sparing: ↑ final ROM
Bryan-Morrey (1982)	initial 49 patients no triceps failure; later series ↑ failure

TO approaches	splitting/reflecting keep triceps attached to periosteum + forearm fascia (± olecranon wafer); **triceps-preserving (bilateral tricipital)** = exposure by ulnohumeral dislocation, **useful in severe RA**; no consensus on best approach
Post-op protection (triceps-reflecting)	**restrict active extension** early; **6–8 wk** before **resisted extension** ✗ Wrong options: active flexion · supination · passive motion

Technique

Positioning	**supine** (arm over chest, assistant; avoid ETT) or **lateral decubitus** over bolster (no assistant, ↓ manipulation)
Humerus	resect **olecranon fossa**, **keep medial + lateral columns**; **bone wedge behind anterior flange**; orient to **posterior flat surface of distal humerus**
Ulna	resect **olecranon tip** + **coronoid tip** (prevents impingement on flange → **axial pistoning + loosening**); orient **perpendicular to dorsal flat olecranon**; centre of **greater sigmoid notch**, rotation from **posterior olecranon cortex** (TO)
Alignment (TO)	**implant–medullary canal parallelism** (Figgie); COR displacement limit **5 mm**; axis: **anterior humeral cortex** (AP), **roof of coronoid fossa** (depth), **condylar plane** (coronal rotation)
Pistoning (TO)	check coronoid, protruding cement, HO on trial motion
Stability check (TO)	joint line should separate **≤ 1–2 mm** on distraction at **90° flexion + pronation**; **UCL + extensor mass** contribute most
Cement	modern cementing; semiconstrained commonest

Post-op care & lifelong lifting restrictions

Immobilisation	**all protocols acceptable** (immediate motion, or splint 60–90° for 2–5 d, 5–10 d, ≥ 10–14 d → active flexion + gravity-assisted extension); historically 4 wk; immediate motion safe
Evidence	**earlier ROM → fewer complications** · **prolonged immobilisation → higher implant survival** (Orthobullets); complications **lowest without immobilisation** in RA (OB)
Lifting limit	**5–10 lb** lifelong (Orthobullets, OB) · **≤ 5 lb** (OB) — read the options
Avoid after TEA	impact sport (golf, tennis) · immediate aggressive triceps strengthening · **immobilising wrist/hand 4 wk** (mobilise immediately) · avoiding pronation/supination (doesn't stress ulnohumeral implant)
Patient selection	> 65 y, low demand, able to obey restrictions ("none do"), **medical optimisation**

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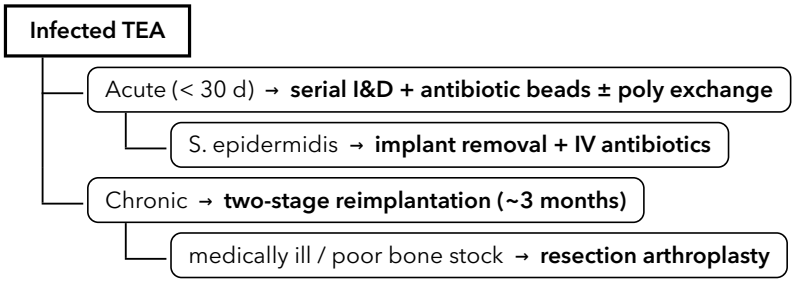
6 Complications

Complication	Rate / fact
Aseptic loosening	**Radiographic 17%, clinical 6%** (Orthobullets) · clinically significant **7–15%** (TO) · **< 10%, usually > 10 y** (OB); **commonest failure of constrained**
Infection	**8%** (Orthobullets) · **3–8%** (TO) · **5–8%**, within first 5 y (OB); **commonest early failure**
Instability	**7–19%**; **commonest failure of semiconstrained** (Orthobullets, TO) and unconstrained
Bushing wear	Constrained + semiconstrained; isolated exchange **1–6%**, mean **7.9 y**; **commonest cause of revision / mechanical failure of semiconstrained** (OB)
Periprosthetic fracture	**5–30%** of primary TEA; **trauma** commonest cause
Component fracture	Humeral **0.65%** · **ulnar 1.2%** (ulna > humerus); usually **ulnar**, late
Triceps insufficiency	**2–8%**; complete disruption **< 2%**
Ulnar neuropathy	Paraesthesia **< 40%**, resolves **2–6 wk**; radial nerve less often
Wound healing	↑ with **long-term steroids**

Complications – instability & component loosening (TO)

Unconstrained, 3 months post-op (TO)	**instability** ✗ Wrong options: ulnar loosening · infection · fracture
"True about TEA" (TO)	**loosening most commonly humeral component** ✓ ✗ Wrong options: instability commonest in semiconstrained (it is commonest in **unconstrained**) · loosening commonest in unconstrained · periprosthetic fracture humerus > ulna
Loosened component	**humeral** (TO answer; historically commonest revision cause, now **< 5%** in semiconstrained) · **ulnar** with current flanged designs (another TO) – read the options
Semiconstrained commonest failure	**instability** (Orthobullets, TO) · **bushing wear** (OB) · humeral loosening historically (TO) – read the options
Instability treatment (TO)	collateral reconstruction usually fails → **revision to semiconstrained hinge**
Constrained, commonest failure	**aseptic loosening** > bushing wear > infection (8%); instability = unconstrained/semiconstrained; ulnar neuropathy < instability

Infection



Acute window	**< 30 days** (Orthobullets) · I&D + **polyethylene exchange within 6 wk** if components stable (OB) — read the options
Organisms (TO)	**S. aureus + S. epidermidis**; **S. epidermidis** = encapsulating, **biofilm** → persistent → **implant removal + IV antibiotics**
Frail 78-yo RA, WBC 25,000, S. epidermidis	**resection arthroplasty** ✗ Wrong options: lifelong suppression (only indolent e.g. **C. acnes**) · I&D + poly exchange (not acute) · one-stage (**50%** success only) · two-stage at 3 wk (wait **~3 months**)
Early failure, semiconstrained post-traumatic	**infection** ✗ Wrong options: loosening > 10 y · bushing wear = intermediate · component fracture = late · ulnar neuropathy = not implant failure

Triceps insufficiency

Bryan-Morrey, 4 wk, "pop" on resisted extension	**triceps repair failure** ✗ Wrong options: ulnar component fracture · axle disengagement · periprosthetic fracture (rare without major trauma, late)
Can't extend at 6 wk after soft-tissue repair	**Bryan-Morrey** ✗ Wrong options: triceps-sparing · triceps-on · trans-olecranon · Alonso-Llames
Triceps roles (TO)	**sole overhead extension** + thick **soft-tissue cover** (↓ wound problems, infection); prevention = meticulous approach + repair

Bone loss & salvage

Causes	multiple revisions, fractures, osteolysis
≤ 8 cm distal humeral loss	**longer prosthesis with extended anterior flange**; severe → **endoprosthesis (total humerus)**
Salvage	flail elbow, amputation, arthrodesis

X vs Y

	Triceps-reflecting (Bryan-Morrey)	**Triceps-sparing**
Triceps	Detached, reflected with anconeus, repaired	**Left on olecranon**
ROM start	**Delayed**	**Early**
Triceps failure	**Higher**; ↓ extension power up to 30%	Low
Exposure	Good	↓ Ulnar visualisation
Best for	–	**Acute distal humerus fracture**

	TEA for RA	**TEA for primary OA**
Survivorship	**Longest** (92.4% at 10 y)	80–85% at 10 y
Patient	Low demand	**More active → ↑ loosening, stem fracture**
Design	**Linked** (✗ Wrong options: unlinked)	Linked or unlinked

– – Active space – –

How it's asked

"Unreconstructable distal humerus fracture, 78-yo — contraindication to TEA?"	→ **olecranon osteotomy** ✗ Wrong options: ulnar transposition · central comminution
"85-yo nursing-home resident, comminuted distal humerus — treatment?"	→ **TEA via direct posterior triceps-sparing approach** ✗ Wrong options: ORIF · fusion · anterolateral
"82-yo RA, distal humerus fracture — shortest recovery, best function?"	→ **TEA**
"RA, prior synovectomy + radial head excision, Larsen III?"	→ **semiconstrained TEA** ✗ Wrong options: unconstrained · synovectomy
"Failed interposition arthroplasty, end-stage + instability?"	→ **semiconstrained TEA**
"TEA survival in RA?"	→ **good at 15 years**
"Unconstrained TEA — relative contraindication?"	→ **late-stage RA**
"Longest implant survivorship?"	→ **RA + semiconstrained**
"Constrained TEA — commonest failure?"	→ **aseptic loosening** ✗ Wrong options: bushing wear
"Linked TEA, OA vs RA?"	→ **OA ↑ aseptic loosening**
"Most accurate pre-op counselling?"	→ **semiconstrained = most favourable** ✗ Wrong options: no extension loss with reflecting · RA complications < 5% · fracture TEA slower
"Post-traumatic OA — true about designs?"	→ **semiconstrained varus-valgus laxity ↓ bone–cement stress** ✗ Wrong options: constrained best · semiconstrained needs collaterals
"Bryan-Morrey, pop + can't extend at 4 wk?"	→ **triceps repair failure**
"Triceps-reflecting — restrict which motion?"	→ **active extension**
"Can't extend at 6 wk, soft-tissue repair failed — approach?"	→ **Bryan-Morrey**
"Immobilisation after triceps-sparing TEA?"	→ **any protocol acceptable**
"Triceps-off vs triceps-sparing?"	→ **delayed ROM** with triceps-off
"Lifelong instruction after TEA?"	→ **limit load to ≤ 5 lb**
"Frail RA, chronic S. epidermidis TEA infection?"	→ **resection arthroplasty**
"Semiconstrained TEA — commonest early failure?"	→ **infection**
"True about TEA?"	→ **loosening most commonly humeral component** (TO)

"Unconstrained TEA at 3 months — commonest complication?"	→ **instability** (TO)	– – Active space – –

– – Active space – –

TOTAL SHOULDER ARTHROPLASTY

– Shoulder & Elbow · 57 / 60 –

1 Basics

Principles

Definition	replace humeral head + **resurface glenoid**; **cemented all-polyethylene glenoid = standard of care**
vs THA/TKA	greater **ROM** · success depends on **soft tissues** (intact cuff, esp. subscapularis) · glenoid **less constrained → greater shear stress → mechanical loosening**
Requirements	**intact + functional rotator cuff** · adequate **glenoid bone stock + version**
Outcomes	**pain relief** = most predictable benefit (more than hemi) · preserved IR + ER · survival **93% at 10 y** (Orthobullets, OB) · ">90% at 10 y" (OB) · TSA/hemi for OA: good/excellent **95%** (Norris) · **worse for post-capsulorrhaphy arthropathy**
Humeral fixation	cemented or press-fit (porous-coated); both good longevity; humeral loosening **uncommon without infection**

Humeral anatomy & implant design (version, stemless, platform stems)

Parameter	Native value
Humeral head retroversion	Mean **~20°** (transepicondylar axis) (OB) · Boileau & Walch mean **17.9°** (−6.7 to 47.5°) · range **0–55°** depending on method (TO)
Neck-shaft angle (inclination)	**45° ± 5°** (range **30–50°**); arthritic ≈ 50° (flatter) (TO)
Humeral head height	Top of head **8–10 mm above top of GT** (TO, native)
Lateral humeral offset (LHO)	**Lateral base of coracoid → lateral margin of greater tuberosity** (TO)
Humeral offset	Centre of rotation → humeral canal axis; medial **4–14 mm**, AP **−2 to 10 mm** (TO)
Head thickness	**70% of radius of curvature** (OB)
Radius of curvature	Glenoid **22–28 mm** · head **23–28 mm** (TO)
Glenoid version	**0–3° retroversion** (OB) · **1.5° retroversion** (10.5° retro to 9.5° ante) (TO) · ~0°, <10° either way (TO)
Glenoid inclination	Superior **~4°** (4.2°; −7 to 20°) (TO)

LHO (TO)	**↓ LHO → ↓** lever arm of **deltoid + supraspinatus →** weak abduction · **↑ LHO → overstuffing → ↓** motion, **↑** poly wear · reconstruct within **4 mm** of normal
Retroversion axes (TO)	proximal: articular surface plane / COR–articular centre / GT–articular centre · distal: **trochlear axis**, transepicondylar line, forearm

Restoring COR may prolong glenoid fixation + ↓ poly wear (TO)

Humeral implant	Key fact
Stemless	**Completely metaphyseal (biological) fixation** · **more bone preserving** · **no osteotomy needed for proximal humerus malunion** (not constrained by stem) · not for 3/4-part fractures (→ stemmed RSA in elderly)
Stemmed	Cemented or uncemented porous-coated
Short stems	↑ **varus malalignment** (metaphyseal-diaphyseal mismatch)
Large-diameter, high-fill stems	↑ **proximal stress shielding**
Platform (convertible) stem	Modular conversion TSA → rTSA without stem removal → ↓ **operative time**, ↓ blood loss, preserves humeral bone, ↓ cost

Subscapularis failure **independent of implant design**

2 Mechanism

Rocking-horse phenomenon & glenohumeral mismatch

Rocking horse **eccentric glenoid loading** → lifting of opposite unloaded rim → **glenoid loosening** · causes: **superior migration of head (cuff deficiency)** · abnormal glenoid version (ant/post translation)

Minimised by (TO) **glenoid radius of curvature larger than humeral head radius** (**2–4 mm**) → normal translation without rim loading
✗ Wrong options: perfect congruence · larger head radius

Mismatch (TO) recommended **<4 mm** (2–4 mm larger glenoid) · "mismatch <4 mm ↑ loosening" (another TO explanation) – read the options · matched within 2 mm: no ↑ loosening reported · **too much mismatch** → ↑ translation, ↑ poly wear, fracture

Modern glenoids cemented all-poly, radius **2–6 mm larger** than head (TO)

Glenoid conformity	Effect
Conforming	**More stable** · **rim stress** + radiolucencies
Nonconforming	↑ **polyethylene wear**
Overall	Neither superior

Osteolysis **rare** in TSA: larger particle size → harder for macrophages to mount RANK–RANKL reaction (OB)

3 Presentation

Indications & contraindications (cuff status, brachial plexus injury, RA, capsulorrhaphy arthropathy)

Indications pain (esp. **night**) + ↓ ADL · **glenoid chondral wear to bone** (primary OA, inflammatory arthritis esp. **RA**) · posterior head subluxation · OA after primary/recurrent dislocation · complicated proximal humerus fracture not fixable (TO)

– – Active space – –

TSA vs hemi — TSA **preferred for OA + inflammatory arthritis** (better pain relief + forward elevation in RA) · **age <55: TSA > hemi** (outcomes, survival, satisfaction)

AAOS CPG (end-stage GH OA) — **TSA with all-polyethylene cemented glenoid** · (✗ Wrong options: humeral head resurfacing · CTA head · **metal-backed glenoid (↑ revision/ loosening)**) · hemi + ream-and-run = option but TSA preferred

Contraindication	Note
Upper / preganglionic brachial plexus palsy	**Absolute** (TO FNB) · root avulsion = flail arm, poor recovery
Active infection	"**Only absolute** contraindication" (other TO explanation) – read the options
Irreparable rotator cuff tear	→ **hemi or rTSA**; **rocking horse** → glenoid loosening
Rotator cuff arthropathy (proximal migration on X-ray)	Glenoid resurfacing contraindicated
Insufficient glenoid bone stock	**Glenoid erosion to coracoid → favours hemiarthroplasty**
Deltoid dysfunction	e.g. can't abduct past **30°**
Relative (TO)	**Charcot**, severe plexopathy, **intractable instability**, prohibitive comorbidity, cuff + deltoid dysfunction
Isolated supraspinatus deficiency	**Relative** only (TO)

NOT a contraindication	Why
Non-retracted isolated supraspinatus tear / 2-cm full-thickness SS tear	Reparable at TSA
Passive ER <10°	Release at surgery
Eccentric posterior glenoid erosion / retroverted glenoid	Eccentric reaming
Poor glenoid bone stock	Build up with graft (TO)
Inflammatory arthritis	**Indication** if bone stock adequate
Subchondral cysts	Normal end-stage OA finding
Age (young)	TSA favoured over hemi

Cuff tears at TSA — full-thickness in **5–10%** of TSA patients (Orthobullets) · **<5%** in primary OA (OB) – read the options · **+ impingement signs → pre-op MRI**

RA + good elevation (120°), no superior migration — **TSA cemented all-poly**
✗ Wrong options: rTSA · hemi · arthroscopic debridement = less advanced disease

Hemiarthroplasty indications (TO)	Hemiarthroplasty contraindications (TO)
Rough humeral surface, **intact glenoid cartilage**, enough glenoid arc	Recent sepsis
Insufficient bone for glenoid	**Neuropathic joint**
Fixed upward displacement (CTA, severe RA)	Paralytic disorder
History of **remote infection**	Cuff + deltoid deficiency
Heavy demands (occupation, sport, LL paresis)	Non-cooperative patient

Post-capsulorrhaphy arthropathy

Putti-Platt — subscapularis + anterior capsule divided; lateral stump → anterior glenoid neck; medial stump → tuberosity (**pants-over-vest**) → **limits ER** → incongruent contact loading → **post-capsulorrhaphy arthropathy**
✗ Wrong options: no coracoid transfer

Magnuson-Stack — middle subscapularis + piece of LT transferred **lateral to bicipital groove** (restrict ER) → **subscapularis lacerated** during biceps release at TSA → **↑ ER, anterior instability, + belly press**

4 Diagnosis & Classification

Key radiology findings

View	Look for
True AP (Grashey)	Extent of arthritis · **superior migration** (= cuff deficiency) · Iannotti best-fit circle
Axillary	**Posterior glenoid wear** · head subluxation
CT	**Glenoid version + glenoid bone stock**
MRI	Rotator cuff (**Goutallier** fatty change)

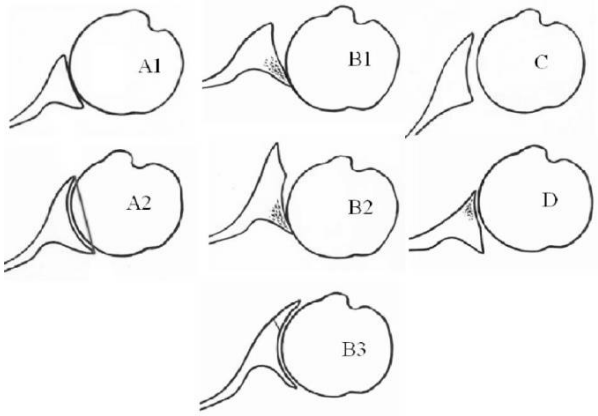
Cuff-deficiency signs — **acetabularization of acromion** → incompetent cuff → **rTSA** · disrupted **Shenton's line** of shoulder → cuff not intact

Infraspinatus atrophy on MRI — **independent risk factor for post-op superior cuff tear**

– – Active space – –

Walch classification of glenoid wear

Type	Description
A1	Concentric, centred head; minor central erosion
A2	Deeper central erosion; line joining rims transects head
B0	Pre-OA **posterior subluxation** of head
B1	Posterior narrowing, osteophytes, sclerosis, **no bone loss**
B2	**Biconcave**, **posterior rim erosion**, retroverted · max erosion **posteroinferior (7–8 o'clock)**
B3	**Monoconcave** posterior wear; head subluxation **>70%** or retroversion **>15°**
C1	**Retroversion >25°** regardless of erosion (dysplastic)
C2	Biconcave, posterior bone loss, posterior translation
D	**Anteversion** or **anterior** head subluxation (<40%)



Walch classification of glenoid wear

Commonest OA pattern (TO) **posterior subluxation → posterior glenoid wear → ↑ retroversion**

B2 vs A1 **B2 ↑ radiographic loosening** (TO)

5 Management

Implant choice

- **Glenohumeral arthritis**
  - Intact cuff + adequate glenoid → **aTSA (cemented all-poly glenoid)**
    - retroversion >15° → **augment or posterior bone graft**
  - Glenoid eroded to coracoid → **hemiarthroplasty**
  - Irreparable cuff / CTA / proximal migration → **rTSA (or hemi)**
  - Plexus palsy / active infection → **no TSA**

Approach

– – Active space – –

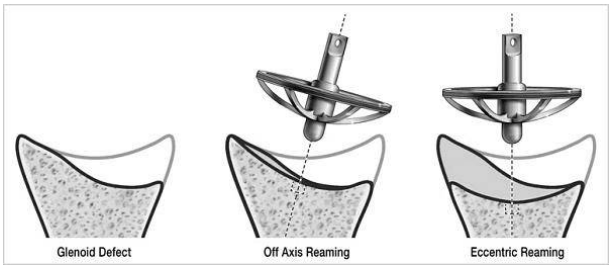
Deltopectoral	**cephalic vein** = interval landmark · detach **subscapularis + capsule** from anterior humerus → **dislocate anteriorly** · tight shoulder: release **upper half** (Orthobullets) · **upper 1/3** (OB) of **pectoralis major** tendon
Superior border of subscapularis (TO)	**biceps tendon** (just medial to pec major insertion → follow up to **rotator interval**) · biceps ruptured/dislocated → **base of coracoid** marks medial interval
Inferior border of subscapularis	landmark for **axillary nerve** (anterior → posterior under GH joint → quadrangular space)

Palpate cuff insertion before head cut – ✓ protects cuff

Subscapularis management & failure (peel, tenotomy, LTO; postoperative restrictions)

Takedown options	**peel** (off bone) · **tenotomy** (trans-tendon) · partial tenotomy · **lesser tuberosity osteotomy (LTO)**
Which is best?	**no technique superior**: no difference in strength, ROM/function, biomechanical strength or failure rates ✗ Wrong options: LTO lower failure · peel more ROM · tenotomy higher failure
LTO	interest due to **bone-to-bone healing** (biomechanical strength); high union rate · **nonunion** → very poor outcomes, multiple revisions
Most commonly ruptured muscle after TSA (TO)	**subscapularis** · factors: **multiple previous operations**, **overstuffing**, **aggressive physio**, tendon length compromise (poor technique)

Glenoid morphology & retroversion correction (eccentric reaming, bone graft, augments)



Eccentric reaming of the high anterior side

Why correct	failure → glenoid malposition → ↓ contact area → ↑ contact pressure → **accelerated loosening**

– – Active space – –

Operative criteria – glenoid retroversion

Glenoid retroversion	Orthobullets / OB	TO
<10–15° (e.g. **9°, 11°, 13°**)	**Eccentric reaming of anterior (high) side**	**Up to 15°** → eccentric reaming
>15° (e.g. **19°, 21°**)	**Posterior bone graft** (Orthobullets) · **augmented glenoid or bone graft** (OB; augments ↓ complications)	**15–25° → glenoid augments** (reaming perforates anterior vault)
>25°	Allograft reconstruction of posterior defect	**Bone graft**
Insufficient bone (to coracoid)	No eccentric reaming → posterior bone graft or hemi	–
Cuff-deficient / acetabularised acromion	**rTSA** (reaming irrelevant)	–

15° threshold more reaming compromises **subchondral bone** + risks **glenoid vault penetration** · excessive **medialisation** weakens fixation + alters mechanics

Eccentric reaming not for **humeral anteversion/retroversion** (address with head cut + stem) · humeral head collapse · **superior inclination 18°** (exceeds correctable)

Glenoid deficiency build up with **iliac crest autograft** or **resected humeral head** · **never cement** to fill

Axial CT: retroversion + bone loss, glenoid to be resurfaced **bone grafting of the glenoid** (posterior)
✗ Wrong options: infection work-up · cuff repair · acromioplasty

Component positioning

Component	Target
Glenoid version	**Neutral to 10° retroversion** / "neutral or slight retroversion" (TO FNB) · **neutral** (5° ante to 10° retro acceptable) (TO) · **neutral to slight anteversion** (OB, another TO) · "recreate neutral" (Orthobullets) – read the options
Glenoid inclination	**Neutral**; superior/inferior inclination → highest failure; retroverted glenoid loosens more than anteverted (TO)
Humeral stem	**25–45° retroversion** (Orthobullets, TO) · **20–40°** (one TO explanation)
Glenoid retroverted	**↓ stem retroversion** → avoid **posterior dislocation**
Stem alignment	**avoid valgus**
Head height (implant)	Top of head **5–8 mm above GT** (Orthobullets, TO)
Head size	**avoid overstuffing** → ↑ joint reaction force + cuff tension

Accurate glenoid positioning (TO) **most critical** + most challenging step: limited bone stock, variable anatomy, few landmarks

Glenoid component design

Better	Worse	Note
Convex (curved) back	Flat back	Flat back ↑ loosening
Pegged	Keeled	Biomechanically superior; studies: no difference or ↑ loosening with keel
All-polyethylene	**Metal-backed**	Glenoid too small for metal + PE; metal-backed ↑ loosening/revision
Cemented (Orthobullets: ↓ loosening)	Uncemented	One TO states **uncemented better** – read the options

Central peg non-keeled designs: central peg usually **not cemented** → not a loosening risk factor

Capsule & intra-operative fractures

Anterior capsular contracture (passive ER **<40°**) – **anterior release + Z-lengthening**

Stretched posterior capsule **plication** (volume reduction)

Intra-op GT fracture (minimal displacement) – standard stem + **suture fixation** + cancellous autograft

Intra-op shaft fracture remove → **long cemented stem + cerclage** · most common intra-op complication = **fracture**, humeral shaft **mid–distal diaphysis** (TO, Norris)

Rehabilitation & postoperative restrictions

Limiting factor early **protection of subscapularis repair** · restriction set by **strength of subscapularis repair**
✗ Wrong options: supraspinatus · dislocation · loosening · pec release · capsule · bone quality

Restrictions (≈6 wk) **no passive ER >30°** · **no active IR past plane of body** · **no pushing out of a chair** · passive/active-assisted only

Allowed early pendulums · passive FE to 90° · **passive IR to chest** · active elbow flexion

Progression ER isometrics → IR eccentric + isometric

Postoperative stiffness & opioid use

Stiff at 6 wk (FE 90°, ER 0°), X-ray normal, sling still on – **start PT for ROM now**
✗ Wrong options: more sling · aspiration · MRI · DVT duplex (DVT = swelling + pain)

Postoperative narcotic use	Factor
↓ use (protective)	**Increasing age**
↑ use	**Pre-op narcotic use** · unemployment
No correlation	Gender, smoking, alcohol

Orthopaedics highest narcotic-prescribing surgical specialty

– – Active space – –

Professionalism – impaired colleague

Attending smells of alcohol before TSA — **notify the department chairman** before case (chain of command)
✗ Wrong options: medical board first · resident does case · discuss afterwards · refuse to participate

Substance abuse — **leading cause of medical board sanctions**; physicians recover better than general population

6 Complications

Overview

Complication (TO)	Rate
Stiffness	66%
Glenoid loosening	**Commonest cause of failure (39%)**
Periprosthetic fracture (GT, shaft)	11%
Iatrogenic cuff tear (neck cut, overstuffing)	7.7%
Humeral loosening (infection, **RA, osteonecrosis**)	6.5%
Posterior instability	**6%**
Anterior instability (subscapularis failure)	**5%**
Neurovascular (arcuate artery)	4.8%
Infection	4.6%

Overall complication rate — **~15%** · order: glenoid loosening > instability > cuff tear > periprosthetic fracture > infection > modular dissociation > deltoid dysfunction (TO)

Most common complication / failure — **glenoid loosening** (TO, OB) · Orthobullets: newer evidence → **cuff failure may be more common** than glenoid loosening – read the options

Most common (TO)	Hemiarthroplasty	TSA	Reverse TSA
Complication	**Progressive glenoid erosion** (fracture cases: **tuberosity displacement**)	**Glenoid loosening**	**Scapular notching** > instability/dislocation
Cause of revision	Glenoid erosion	**Glenoid loosening**	Glenoid component loosening
Nerve	–	**Axillary (neurapraxia)**	Axillary

Glenoid loosening – most common complication & risk factors

Frequency — most common cause of TSA failure: **~30% of primary OA revisions** (OB) · **39%** (TO) · reoperation **2.9% primary vs 28% revision** TSA

Commonest complication at 5 y — **loosening of glenoid component**
✗ Wrong options: humeral loosening · acromial stress fracture · infection

✓ Risk factor	Not a risk factor
Glenoid morphology: **biconcave (Walch B2)**, **posterior glenoid wear**, excessive retroversion	**Age** (incl. >65)
Dysfunctional/deficient rotator cuff (rocking horse)	Inferior humeral head osteophytes
Insufficient glenoid bone stock	Subtle **anterior** subluxation (biconcave goes with **posterior**)
Excessive reaming removing subchondral bone	**Goutallier/cuff atrophy** (not correlated)
Flat-back glenoid	**Curved-back** (↓ loosening) · **pegged** component
Mismatch (see Mechanism)	Stemless humerus · diabetes (infection) · uncemented central peg
Soft-tissue instability · infection · aseptic osteolysis	✗ Wrong options: osteonecrosis · RA · OA (as cause of glenoid loosening)

Radiolucent lines — presence **does not correlate with symptoms** or clinical failure (3 + 7 y) · **progression does correlate** (50% by 3–4 y)

Treatment — glenoid removal (if bone stock) ± **bone graft** for later re-implantation · **convert to rTSA** (salvage) · glenoid morphology not fixable with aTSA → rTSA

Subscapularis failure

Commonest instability after TSA — **anterior** (subscapularis + anterior capsule pull-off) (Orthobullets, TO) · TO complication list gives posterior 6% vs anterior 5% – read the options
✗ Wrong options: "pull-off of rotator cuff" (= supraspinatus)

Anterior dislocation after TSA — **subscapularis rupture in all cases** (Wirth & Rockwood) · classic: **pushing up from chair** (fires subscapularis)

Mechanism — fall/catching self · **forced IR** or **excessive ER** (beyond normal **40–60°**) · forced ER withdrawal (hot pan) → LTO failure · late rupture with sport (golf, tennis)

Sign of subscapularis insufficiency	Note
↑ Passive ER	e.g. 90° at side (abnormal at 72 y), 110° vs 80°
Lift-off / IR lag	Can't hold hand off back
Belly-press (abdominal compression)	Weak
Bear hug	**≥20% weaker** than other side
IR weakness · can't tuck shirt / reach back pocket	–
Anterior instability with ER	(✗ Wrong options: instability with IR = posterior)
Axillary X-ray: anterior subluxation	Head not centred

– – Active space – –



Anterior subluxation of humeral head after subscapularis failure

Confirm — **ultrasound** · X-ray: displaced lesser tuberosity (failed LTO) · glenoid loosening/locked posterior dislocation **visible on X-ray** (normal X-ray → tendon rupture)

Consequence — loss of **anterior–posterior force couple** / concavity-compression → instability, early loosening

Timing / patient	Treatment
Acute (<6 wk), young/active	**Early exploration + revision repair** (± augmentation)
>6 wk	Anterior capsular reconstruction · tendon transfer · **rTSA**
Chronic / older	**Revision to rTSA** (most reliable)

Subscapularis innervation — **upper + lower subscapular nerves ← posterior cord**

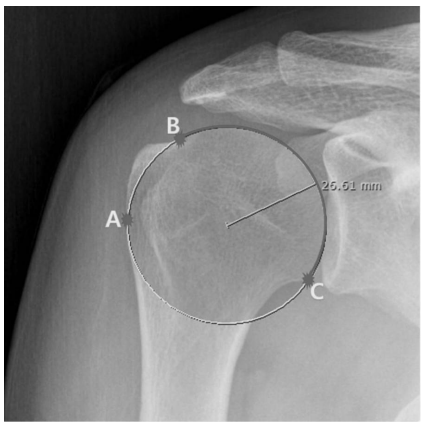
✗ Plexus distractor	Branches
Lateral cord	**Lateral pectoral**, **musculocutaneous**
Medial cord	Medial pectoral, **medial brachial cutaneous (T1)**, MABC
Superior trunk	**Suprascapular**, nerve to subclavius
C5 root	**Dorsal scapular** (rhomboids, levator scapulae)

Rotator cuff failure after TSA (overstuffing, falls, sizing)

Frequency — **one of the most common complications**; common long-term secondary dysfunction · **common reason for conversion to rTSA**

Cause	Mechanism
Excessive bone removal at humeral neck osteotomy (cut too distal / excessive retroversion)	**Iatrogenic** cut/detachment of cuff insertion (✗ Wrong options: 30° retroversion = normal · 45° inclination · deltoid retraction = axillary nerve)
Overstuffing (oversized head, ↑ LHO)	**Attritional supraspinatus + subscapularis** tears; ↓ motion
Fall / trauma	**Traumatic cuff tear** (SS 2/5, normal X-ray → MRI)
Undiagnosed pre-op cuff tear	Improper soft-tissue balancing
Pre-op infraspinatus atrophy	↑ **superior cuff tear**
Degeneration, malposition	Late tear

Iannotti best-fit circle Grashey AP: 3 points = **lateral cortex below GT flare + lateral + medial ends of anatomic neck** · resurfaced head **outside circle = overstuffing → component sizing** error



Iannotti best-fit circle for humeral head sizing

Anterior + superior escape **SS + subscapularis** tears (Jobe + belly-press +; Hornblower −) ← **overstuffing**

High-riding humerus after fall **cuff failure → convert to rTSA**

Late pain at night, can't do overhead **positive empty can test** (SS tear) · repair if possible, else **rTSA**

Catastrophic glenoid fracture humeral head **directly on glenoid bone** (normally gap = poly thickness; poly is radiolucent)

Cuff tear → glenoid loosening (cuff deficiency predisposes)

Infection – C. acnes & work-up

Risk factor	Not
Male (more sebaceous glands, more C. acnes; more TSA for post-traumatic/ CTA)	Female
Age <65	Age >65
Reverse TSA (vs anatomic)	Anatomic TSA
Inflammatory or post-traumatic arthritis	Primary OA
Corticosteroid injection within 3 months (e.g. 2 months)	Injection 4 months before

Rate C. acnes infection **1–2%** after primary TSA (Orthobullets) · **4.6%** (TO)

C. acnes (P. acnes) **gram-positive, facultative, aerotolerant, anaerobic rod**; ferments lactose → propionic acid · **commonest cause of late/indolent infection + implant failure** · **>50% of chronic infections**; **S. epidermidis 2nd (15%)**

C. acnes facts colonises **shoulder > hip/knee** · **men > women** bacterial burden · biofilm 18–90 h · presentation: early **pain + stiffness**, later swelling + redness · **usually no fever** (subtle)

Cultures hold **14 days** (OB) · **10–14 days** in anaerobic bottles, mean detection **6 days** (Orthobullets TSA note) · **14–28 days** (Orthobullets PJI note) — read the options · 16S rRNA PCR · **sonicate** explanted implant

– – Active space – –

Sampling — **arthroscopic tissue culture 100% sensitive + specific** vs fluoroscopic aspiration **17% sensitive**, 100% specific · aspiration may be **normal** · imaging + in only 24%

New pain after pneumonia / ↑ WBC + ESR — **haematogenous → radiographs + aspiration with Gram stain + culture** · well-fixed implants → I&D may avoid full revision
✗ Wrong options: wait 2 wk for repeat labs · PT/NSAIDs/stretching · MRI · emergent revision

Timing	Treatment
Early (<6 wk)	**Open I&D**
Late (>6 wk) / grossly infected (frozen >10 WBC/HPF, GPC in clusters)	**Remove components + antibiotic spacer → 2-stage** reimplantation · IV antibiotics **6 wk** (penicillin G, ceftriaxone, clindamycin, vancomycin) then PO **2–6 mo**

Avoid in 2-stage stage 1 — bone grafting glenoid (won't heal with infection; do at stage 2) · resection arthroplasty (poor function) · head exchange/arthroscopic debridement (won't eradicate)

Neurovascular injury during approach (musculocutaneous, axillary, radial, arcuate artery)

Structure	At risk when	Deficit / fact
Axillary nerve (most common)	Inferior capsular release, inferior subscapularis, labrectomy; **excessive deltoid retraction**	Lateral shoulder numbness, weak abduction; **10–25 mm** below glenoid rim (↑ with abduction + ER); ≤3 mm from inferior capsule
Musculocutaneous (lateral cord)	**Retraction on conjoint tendon / strap muscles** (short head biceps + coracobrachialis)	**Weak forearm supination + elbow flexion** (biceps)
Radial nerve	**Latissimus dorsi tenotomy** for transfer (with rTSA)	–
Arcuate artery (terminal branch of **anterior humeral circumflex**, main GT supply)	**Elevating biceps tendon from groove** (runs lateral side of groove)	Bleeding
Anterior humeral circumflex vessels	Distal dissection at **inferior subscapularis** (peel)	–
Suprascapular nerve	≤1 cm from glenoid rim	–

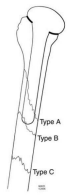
MCN distance from coracoid — **~3 cm** (OB) · **2.0 ± 0.5 cm** (Apaydin; closest with arm **abducted + IR**) · **3.1 cm** (Boardman) – read the options · enters conjoint **5.8 cm** below coracoid · 0.5 cm from labrum at 90° abduction

Nerve injury rate — **1–4%** of shoulder arthroplasty · 58% of iatrogenic injuries fail to recover (Carofino)

Axillary nerve + posterior humeral circumflex artery — pass beneath GH joint through **quadrilateral space**

Latissimus transfer — routine with **Grammont (medialised)** rTSA for ER deficit (+ Hornblower) · ↓ use with **lateralised** designs

Periprosthetic fracture – Wright & Cofield

Type	Location	Intra-op	Post-op	
A	Near **stem tip, extends proximally** (tuberosities)	Span with standard (2–3 cortical diameters) or long stem; transosseous sutures for tuberosity	**Nonop** if minimally displaced · significant overlap = loose stem → **long-stem revision**	
B	**Centred at stem tip**, extends distally	Span with long stem; cement distal canal (✗ Wrong options: cement escape); strut allograft + cerclage	**Revise to long-stem**	
C	**Distal to stem tip**	Long stem, or near olecranon fossa → plate ± cerclage, strut	**ORIF**; plate overlaps stem by **2 cortical diameters** (avoid stress riser)	

Acceptable alignment **≤20° flexion/extension · ≤30° varus/valgus · ≤20° rotation**

Humeral stem loosening more common in **RA + osteonecrosis → rule out infection** · uncommon with stemmed primary aTSA

X vs Y

	Anatomic TSA	Reverse TSA
Cuff	**Must be intact**	Deficient (CTA, pseudoparalysis, anterosuperior escape)
Design	Less constrained (shear → loosening)	Convex glenosphere + concave humerus; COR **inferior + medial** → longer deltoid lever
ER at side + IR behind back	**Better** (✓ reach behind back)	Worse
Return to activity	**Quicker**	Slower
Forward flexion / PROMs / revision / complications	**Similar**	Similar
Typical complication	Glenoid loosening, subscap failure	**Scapular notching**, instability, **acromial stress fracture (~4%)** (over-tensioning)
Dislocation	**Anterior** (subscapularis)	**Anterior**; mechanism **extension + IR + adduction** (rising from chair) (TO) · **hyperextension + ER** (another TO) – read the options

Post-op exam test	Muscle
Belly-press, lift-off, bear hug, IR lag, ↑ passive ER	**Subscapularis**
Jobe / empty can, drop arm / drop sign	Supraspinatus
ER lag sign	Infraspinatus
Hornblower	Teres minor
Yergason	Biceps
Loss of passive ER (complete)	**Locked posterior dislocation**

– – Active space – –

Late failure clue	Diagnosis
↑ Passive ER, + belly-press, **anterior** subluxation, early post-op	**Subscapularis failure**
High-riding humerus, + empty can	**Supraspinatus / cuff failure**
Head outside Iannotti circle, attritional SS tear	**Overstuffing (sizing)**
Progressive glenoid lucency, biconcave pre-op, normal labs/cultures	**Aseptic glenoid loosening**
Pain + stiffness, humeral lucency, normal or ↑ labs	**C. acnes infection**

How it's asked

"TSA absolute contraindication?"	→ **upper brachial plexus palsy** (TO) ✗ Wrong options: SS deficiency · poor glenoid bone stock · retroverted glenoid
"All-poly glenoid contraindicated in?"	→ **irreparable rotator cuff tear** ✗ Wrong options: RA · OA · osteoporosis · prior resurfacing
"Putti-Platt 20 y ago, painful stiff shoulder — diagnosis?"	→ **post-capsulorrhaphy arthropathy**
"Contraindication to TSA?"	→ **preganglionic brachial plexus injury** ✗ Wrong options: ER <10° · posterior erosion · 2-cm SS tear · inflammatory arthritis
"End-stage OA, AAOS CPG — next step?"	→ **TSA with all-polyethylene cemented glenoid** ✗ Wrong options: metal-backed · CTA head · resurfacing
"Most appropriate aTSA candidate?"	→ **72-y male, 9 months post TKA, debilitating pain, MRI shows intact rotator cuff** ✗ Wrong options: plexus palsy · can't abduct >30° · proximal migration · erosion to coracoid
"RA, FE 120°, no superior migration — most predictable?"	→ **TSA with cemented all-polyethylene glenoid**
"Favours hemi over TSA?"	→ **glenoid wear to level of coracoid** ✗ Wrong options: age · non-retracted SS tear · cysts
"aTSA vs rTSA counselling?"	→ **aTSA: increased ability to reach behind back** ✗ Wrong options: equal ER · better FF with rTSA · worse PROMs
"Primary OA, 7° retroversion — aTSA (Figure E) vs rTSA (Figure F)?"	→ **procedure shown in Figure E (aTSA) provides better post-operative range of motion** (similar complications + outcomes)
"Ideal glenoid component position?"	→ **neutral to 10° retroversion** (TO)

"Eccentrically ream the glenoid when?"	→ **retroversion 15°** (TO) · **13°** (OB) · **9°** (OB)
"Axial CT: retroversion + posterior bone loss, resurfacing planned?"	→ **bone grafting of glenoid**
"Walch B2 — maximum erosion?"	→ **posteroinferior (7–8 o'clock), area D**
"Glenoid augmentation needed for?"	→ **retroversion 21°** ✗ Wrong options: 11° · humeral retroversion · head collapse · acetabularization = rTSA
"Cause of glenoid loosening?"	→ **dysfunctional rotator cuff** (TO) ✗ Wrong options: osteonecrosis · RA · OA
"Rocking-horse effect minimised by?"	→ **glenoid radius larger than humeral head radius** (TO)
"Recommended mismatch?"	→ **<4 mm** (TO)
"Commonest complication 5 y after TSA?"	→ **glenoid component loosening** (TO)
"Pre-op feature most linked to early glenoid loosening?"	→ **biconcave glenoid / glenoid morphology** ✗ Wrong options: osteophytes · age · Shenton's line · cuff atrophy · pegged/curved back · stemless · diabetes
"Aseptic glenoid loosening 5 y post — true?"	→ **pre-op posterior glenoid wear is a risk factor** ✗ Wrong options: 2nd to infection · lucent lines correlate · central peg uncemented
"Most likely mode of failure to warn about?"	→ **glenoid loosening** ✗ Wrong options: cuff tear · humeral loosening · fracture · infection
"Most common muscle ruptured after TSA?"	→ **subscapularis** (TO)
"Landmark for superior subscapularis border?"	→ **biceps tendon** (TO)
"Early rehab restricted to protect?"	→ **subscapularis tendon** ✗ Wrong options: supraspinatus · dislocation · loosening
"Greatest influence on early restrictions?"	→ **strength of the subscapularis repair**
"Can't lift hand off back, weak IR 8 mo after TSA?"	→ **subscapularis insufficiency** ✗ Wrong options: subscapular nerve palsy · biceps · brachial neuritis
"Rose from chair at 3 wk → anterior dislocation — mechanism?"	→ **subscapularis rupture**
"Fall at 4 wk, ER 80° passive, subscap tear — other complaint?"	→ **instability with external rotation** ✗ Wrong options: with IR = posterior · lateral numbness = axillary

– – Active space – –

"Fall at 4 wk, ER 90° at side, normal X-ray?"	→ **rupture of subscapularis repair** ✗ Wrong options: glenoid loosening · locked posterior dislocation
"Repaired tendon's nerves arise from?"	→ **posterior cord** (upper + lower subscapular)
"Belly-press weak, ER 110°, displaced LT on X-ray?"	→ **lesser tuberosity nonunion**
"Prior Magnuson-Stack — failed repair after TSA gives?"	→ **subscapularis, increased ER, anterior instability, positive belly-press**
"Movement to avoid immediately after aTSA?"	→ **passive ER to 60°** ✗ Wrong options: IR to chest · FE 90° · pendulums · elbow flexion
"Failed LTO after slip — exam?"	→ **increased passive external rotation** ✗ Wrong options: ER lag · drop arm · Hornblower · Yergason
"Superior subscapularis technique long term?"	→ **no technique is superior / no differences are observed between techniques**
"Click, 80° ER, axillary X-ray anterior subluxation?"	→ **subscapularis failure** ✗ Wrong options: infection · SS tear (empty can 5/5) · axillary neurapraxia
"Withdrew arm from hot pan, failed LTO — exam?"	→ **positive bear hug test** ✗ Wrong options: loss of passive ER = locked posterior dislocation
"Golfer 3 y post aTSA, lift-off lag, off-centre head?"	→ **subscapularis rupture**
"FALSE about TSA?"	→ **most common instability is pull-off of rotator cuff** (TO; it's subscapularis)
"Manoeuvre risking cuff injury?"	→ **excessive bone removal with humeral neck osteotomy**
"Fall, SS 2/5, normal X-rays 2 y after TSA?"	→ **rotator cuff tear**
"Anterosuperior escape, + Jobe + belly-press — cause + risk factor?"	→ **SS + subscapularis tear; overstuffing** ✗ Wrong options: osteolysis (rare)
"Pre-op infraspinatus atrophy → risk of?"	→ **superior rotator cuff tear** ✗ Wrong options: acromial stress fracture = rTSA
"High-riding humerus after fall, 15 y post TSA?"	→ **rotator cuff failure → rTSA**
"24 mo post, night pain, can't go overhead — also present?"	→ **positive empty can test**
"Head larger than best-fit circle, attritional SS tear?"	→ **component sizing** (overstuffing)
"P. acnes after arthroplasty — incorrect?"	→ **usually associated with fevers**
"Pain after pneumonia, ↑ WBC/ESR, TSA 5 y?"	→ **obtain radiographs + aspirate the shoulder**

"Late pain post pneumonia, mild ↑ labs, normal X-ray?"	→ **shoulder aspiration with Gram stain + culture**
"Late infection with lucencies — organism?"	→ **Propionibacterium acnes** (S. epidermidis 2nd)
"Frozen >10 WBC/HPF, GPC in clusters — treatment?"	→ **remove components + antibiotic spacer** ✗ Wrong options: graft glenoid now · resection · head exchange
"Draining sinus 3 mo post aTSA — risk factor?"	→ **corticosteroid injection 2 months before surgery** ✗ Wrong options: anatomic · female · >65 · OA
"Retraction on strap muscles — weakness?"	→ **forearm supination** (MCN)
"rTSA + latissimus transfer — nerve at risk at tenotomy?"	→ **radial**
"Arcuate artery injured during?"	→ **elevating biceps tendon from groove**
"Inferior border of subscapularis protects?"	→ **axillary nerve**
"Conjoint tendon over-retraction injures?"	→ **musculocutaneous nerve from lateral cord**
"Stem + glenoid position in TSA?"	→ **stem 25–45° retroversion + glenoid neutral or slight retroversion** (TO)
"Lateral humeral offset = GT to?"	→ **lateral base of coracoid process** (TO)
"Average humeral head version?"	→ **20 degrees retroversion** ✗ Wrong options: 40°/60° retroversion · anteversion
"Stemless vs stemmed — true?"	→ **stemless less likely to need osteotomy for malunion** ✗ Wrong options: fractures · ↓ subscap failure · stemmed more metaphyseal
"Retained platform stem at revision to rTSA?"	→ **lower overall operative time**
"Stiff at 6 wk in sling, X-ray normal?"	→ **PT for range-of-motion exercises**
"Factor with ↓ post-op narcotic use?"	→ **age** ✗ Wrong options: prior narcotics ↑
"Attending smells of alcohol before TSA?"	→ **notify the department chairman**

– – Active space – –

TRAUMATIC ANTERIOR SHOULDER INSTABILITY (TUBS)

– Shoulder & Elbow · 58 / 60 –

1 Basics

Epidemiology

TUBS	**T**raumatic **U**nilateral dislocation, **B**ankart lesion, requires **S**urgery
Frequency	one of the most common shoulder injuries; **1.7%/year** general population; shoulder = most commonly dislocated major joint
High-risk groups	**military**, **contact/collision athletes** (rugby, American football commonest)
Hyperlaxity	global hyperlaxity (Ehlers-Danlos, collagen disorders) → OR **2.68** for anterior instability · **3× recurrence** · fewer capsulolabral lesions · more often atraumatic instability

Anatomy – static & dynamic restraints

Static	bony anatomy, capsule, GH ligaments, **labrum (+50% glenoid depth)**
Dynamic	rotator cuff, long head of biceps
Concavity-compression	cuff compresses head into glenoid concavity; **lost with anterior glenoid rim loss** → ↓ effective depth ✗ Wrong options: adhesion-cohesion · negative intra-articular pressure: both preserved in closed capsule

Ligament	Taut / restrains in	Direction
SGHL	**Arm at side (0° abduction)**	**Inferior**
MGHL	**45° abduction + ER**	Anterior (and posterior, OB)
Anterior band IGHL	**90° abduction + ER**	**Anterior – main anterior restraint**
Posterior band IGHL	90° abduction	Posterior

Main anterior stabiliser (TO)	**IGHL** ✗ Wrong options: MGHL · SGHL · anterior capsule · subscapularis
Main stabiliser above 45° abduction (TO)	**IGHL**
AIGHL glenoid attachment	usually to **labrum**; less commonly direct to glenoid neck

Pathoanatomy – Bankart lesion, IGHL & mechanism (abduction–external rotation)

– – Active space – –

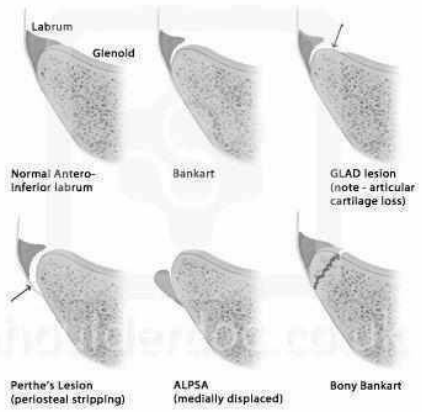
Bankart lesion — avulsion of **anteroinferior labrum + anterior band IGHL** from anteroinferior glenoid; labrum separated from the **bony glenoid**; "essential lesion" of traumatic anterior dislocation (TO)
✗ Wrong options: cartilage–labral junction only

Bankart frequency	Source
80–90% of TUBS	Orthobullets
85% (Rowe) · **87%** (Hintermann) · **97%** (Taylor 63/67) · 73.5% (Wolf)	OB studies

Most common surgical finding — **anteroinferior labral tear**
✗ Wrong options: anterosuperior · posterior · Hill-Sachs

Hintermann (212 dislocators) — Bankart 87% · ventral capsule insufficiency 79% · **Hill-Sachs 68%** · GHL insufficiency 55% · complete cuff tear 14% · posterior labral tear 12% · SLAP 7%

Lesion	Definition	Key point
Bankart	Labrum + IGHL off anteroinferior rim; **medial periosteal hinge undisrupted** (TO)	Most common
Perthes (TO)	Labrum **avulsed (not torn)**, attached by **intact but medially stripped periosteum**	May sit in **normal position** (neutral-arm MRI) → **hard to see arthroscopically**; stabiliser function lost → **recurrent dislocation**
ALPSA	"**Medialised Bankart**": labroligamentous complex **displaced medially (+ inferiorly)** on scapular neck, intact medial hinge, **fluid underneath**; anterior labrum absent on rim (TO)	**Heals medially** on glenoid neck · ↑ arthroscopic failure · common in **recurrent instability treated nonop**
GLAD	**Partial** anteroinferior labral tear + **adjacent glenoid articular cartilage defect**	↑ arthroscopic failure · debridement relieves pain (Neviaser)
HAGL	IGHL avulsed from **humeral** insertion	See HAGL block
Bony Bankart	Fracture of anteroinferior glenoid	Up to **49%** of recurrent dislocators
Kim lesion (TO)	Concealed tear at **posterior** glenoid chondrolabral junction + concealed deep labral detachment	**Posterior** instability; visible chondrolabral junction intact; probing → **labral fluctuance**; ± glenoid hypoplasia



Labral lesion spectrum – Bankart, GLAD, Perthes, ALPSA, bony Bankart

– – Active space – –

Association (TO)	Bankart, Perthes, HAGL, ALPSA, GLAD → all **recurrent anterior** instability · Kim → **posterior**
97% rule	**97%** of recurrent instability have **Bankart or ALPSA**
Mechanism	anteriorly directed force with arm **abducted + externally rotated** · posterior instability = **adduction + internal rotation**
Bankart vs position	AIGHL/anteroinferior labrum taut in **ER at 90° abduction** ✗ Wrong options: 45° = MGHL
Associated lesion in young dislocator	**Hill-Sachs** ✗ Wrong options: full-thickness cuff tear = elderly · reverse Hill-Sachs = posterior · coracoid fracture = rare · GT fracture = less common than Hill-Sachs
"Clunk" in 22-y basketballer	Bankart + **Hill-Sachs**

Combined anterior & posterior labral tears

Pan-labral tear	**270–360°** labral tear (anterior + inferior + posterior); anterior and/or posterior instability
Exam	positive **Jerk + Kim** (posterior) **+ apprehension + relocation** (anterior) → **270° tear** → anterior + posterior labral repair (arthroscopic)
✗ Wrong options (images)	bony Bankart / anterior Bankart (anterior repair only) · subscapularis tear (cuff repair) · Kim lesion (posterior only)
270° repair	**15% failure** (Mazzocca)
Military	high incidence of posterior and combined front-to-back tears → **unrecognised pan-labral tear** = cause of recurrence

HAGL lesion

HAGL	**humeral avulsion of the (inferior) glenohumeral ligament**, anterior band IGHL, **without subscapularis tear**; term coined by **Wolf (1995)**
RHAGL	reverse HAGL = posterior band IGHL → **posterior** instability
Frequency	**7–9%** of unstable shoulders (Wolf 9.3%: Bankart 73.5%, capsular laxity 17.2%) · Taylor **1/67** · Rhee 2% — **rare**
Patient	**slightly older** than Bankart (Orthobullets) · **older + more violent trauma** (OB) · female collegiate athletes · hyper-abduction + ER
MRI	best on **coronal** sequences: **J-shaped IGHL**, contrast **extravasating down humeral neck**; bony HAGL variant; "floating" AIGHL = HAGL + capsulolabral detachment
Associations	always with other lesions (Bankart, LHB tear, cuff tear); more severe anterior pain
Significance	"other end of the IGHL" → **missed HAGL = cause of failed Bankart repair** → higher recurrence if unrepaired
Treatment	**open repair = classic / gold standard** · arthroscopic possible but technically hard (**70° scope**, accessory portals) ✗ Wrong options: arthroscopic contraindicated classically

2 Mechanism

Hill-Sachs lesion

Hill-Sachs	**compression/impaction fracture of posterosuperior (posterolateral) humeral head** against **anteroinferior glenoid rim**; chondral impaction injury
Frequency	**80–100%** of traumatic dislocations · **25%** of traumatic subluxations · 68% (Hintermann) · 73% on X-ray (Edwards)

Most common associated intra-articular finding with Bankart
✗ Wrong options: cuff tear · SLAP · posterior labral tear · Buford

Arthroscopy	indentation on posterior head **beside the bare area** ✗ Wrong options: bare area = normal cartilage-free zone
Significance	clinically irrelevant **unless it engages**; **end-range (ABER) instability**, not mid-range (TO)
Glenoid bone loss (TO)	from anteroinferior rim fracture or **erosion** by humeral head; **6.8% loss per dislocation** (average)
Reverse Hill-Sachs	anteromedial head; **posterior** dislocation; lesser tuberosity fracture = posterior dislocation

3 Presentation

History	traumatic event; ask **formal reduction vs spontaneous**; instability feeling; pain from subluxation/excess translation
Apprehension	supine, **90° abduction + 90° ER** → apprehension; positive in **mid-range abduction = glenoid bone loss**
Relocation	posterior force on humeral head → ↓ apprehension
Load and shift	grades 0–III (table below); **sulcus sign** with arm at side

Diagnostic criteria – hyperlaxity

Criterion	Threshold
Beighton score	**>4**
Hyper-ER at side	**>85°**
Hyperabduction (Gagey)	**>105°**
Load–shift	**>2+ in ≥2 planes**

ISIS hyperlaxity	anterior (ER >85°) or inferior (Gagey) laxity = 1 point

- - Active space - -

4 Diagnosis & Classification

Key radiology findings

Modality / view	Use
Trauma series: true AP (Grashey) + scapular Y + axillary	Required; Grashey = beam ~45° off AP, end-on GH joint; **Y = humeral head over Y intersection** (scapular spine + coracoid)
Axillary	**Best for detecting dislocation**; bone loss visible here = already significant; Velpeau if can't abduct
West Point	**Glenoid bone loss / bony Bankart**
Stryker notch	**Hill-Sachs** (supine, hand on head, **10° cephalic**); poor for Bankart
AP in ER	Hill-Sachs (ISIS item)
Bernageau profile	Glenoid profile, "cliff sign"; **less accurate than 3D imaging**
CT ± 3D	Bony injury, **% glenoid bone loss**; **3D CT = gold standard**
CT arthrogram	**MRI contraindicated** (pacemaker, cochlear implant); poorer soft tissue
MRI	**Best for labral tear**; validated for bone loss
MR arthrogram	**Study of choice for soft tissue**: sens **86–91%**, spec **86–96%** (MRI 44–100% / 66–95%)

MRI positioning – ABER view

ABER **abduction + ER → humerus in coronal plane, glenoid in axial plane**

Shows **anteroinferior labrum**, **anterior band IGHL**, articular surface of supraspinatus, posterosuperior labral peel-back (sens 73%, spec 100%)

Glenoid bone loss – assessment & critical thresholds

Method	Fact
3D CT en face	**Gold standard**; **highest interrater reliability**; predictive error **similar to MRI** (Rerko)
Best-fit circle (TO)	Circle fitted to **posterior + inferior** glenoid ≈ original glenoid; or **contralateral glenoid** reference
Bare-spot method	Bare spot = glenoid centre (Burkhart); GBL = (B − A) / 2B (A = bare spot→anterior edge, B = →posterior edge)
MRI	Tends to **overestimate** GBL (✗ Wrong options: underestimates); less bone–soft tissue contrast; cuts must be en face
MR arthrogram	Not a substitute for CT (poorer bony definition)
X-ray (Bernageau, West Point)	Significantly less accurate

Worked example (OB) bare spot 9 mm anterior, 12 mm posterior → (12 − 9)/24 = **12.5%** + non-engaging HS → **Bankart alone**

Worked example (OB) 6.9 mm lost of 28.3 mm → **24.4%** = critical → **coracoid transfer**

Inverted pear — glenoid morphology with ↑ bone loss (**>25%**, TO/OB); **89% failure** after arthroscopic repair

Glenoid shape — normally pear-shaped (narrow superior); anteroinferior loss → **inverted pear**

Threshold	Meaning	Source
>20–25%	**Critical**; soft tissue alone fails (>2/3 failure) → bony procedure	Orthobullets, most OB
>25–30%	Osseous augmentation needed	OB (bone loss explanations)
20–30%	"Glenoid bone deficiency" → Latarjet; **>60% recurrence** with Bankart alone	OB
15–25%	Biomechanically unstable	OB
21% of glenoid length	Instability + ↓ ER after Bankart (Itoi)	OB
17.3%	Critical value (Shin: 43.9% vs 3.7% recurrence)	OB
>13.5%	**Subcritical** – critical may be this low; open Bankart ↑ recurrence	Orthobullets, OB
<15%	Soft-tissue stabilisation alone works	OB, TO

Bone-loss conflict — critical **20–25%** (Orthobullets) · **25–30%** (some OB) · **13.5–17.3%** subcritical (recent studies) — read the options

Strongest predictor of failure after arthroscopic Bankart — **glenoid bone loss**
✗ Wrong options: age at first dislocation · age at surgery · >3 anchors · on-track HS

Most predictive intra-op finding — **anterior glenoid deficiency 35%**
✗ Wrong options: non-engaging HS · loose body · bursitis · 10% PASTA

Mechanism lost with rim loss — **concavity-compression**

Hill-Sachs lesion & glenoid track (on/off-track)

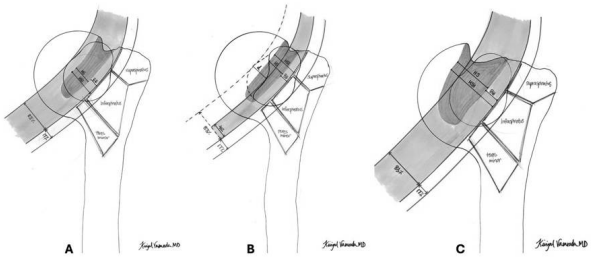
Bipolar concept — glenoid bone loss + Hill-Sachs together; bipolar > unipolar engagement risk

Glenoid track (TO) — area of posterior humeral articular surface in contact with glenoid along **posterior end-range (ABD + ER)**; width = medial margin of track → medial margin of cuff footprint

Formula	Value
GT = 0.83D − d	D = inferior glenoid diameter, d = anterior bone loss width (Orthobullets, TO, most OB)
GT = **0.84(D) − ABL**	OB (glenoid-track formula question)
84% vs 83% (TO)	**84% cadaveric** · **83% live at 90° abduction**
HSI = HS + BB	Hill-Sachs width + **bony bridge** to cuff insertion
HSI > GT	**Off-track = engaging** → recurrence
HSI < GT	**On-track = non-engaging**

– – Active space – –

Coefficient conflict **0.83** (Orthobullets, TO, most OB) · **0.84** (one OB; "HS >84% of glenoid width = off-track") – read the options



Glenoid track vs Hill-Sachs interval – on-track (A), off-track from glenoid loss (B) or large Hill-Sachs (C)

Medial extension **the more medial the HS extends, the more likely off-track** ✓

✗ Glenoid-track traps on-track engages · unipolar > bipolar risk · **GT only by CT** (MRI also usable) · HSI < GT engages

Dynamic arthroscopic exam (TO) assess engagement **after the Bankart repair**

Tyre & pothole glenoid track = tyre, HSI = pothole; off-track = tyre falls in (big HS or big glenoid loss)

Off-track predictive value 75% vs 8% recurrence (Shaha); **superior to >20% bone-loss cut-off**

Worked example	Calculation	Verdict
Kabaddi (TO): D 24, width 20, HSI 18	d = 4 → **16.6%**; GT = 19.92 − 4 = **15.92**	**Off-track + subcritical, elite contact → Latarjet**
Water polo (OB): D 29, d 2, HSI 19	GBL **6.9%**; GT = 24.07 − 2 = **22.07**	**On-track → isolated arthroscopic Bankart**
TO: 25% GBL, GT 18, HSI 21.3	Off-track + critical	**Latarjet**

Instability Severity Index Score (ISIS – Balg & Boileau)

Factor	Parameter	Points
Age at surgery	**<20 y**	**2**
Sport level	**Competitive**	**2**
Sport type	**Contact or forced overhead**	**1**
Hyperlaxity	Anterior or inferior	**1**
Hill-Sachs on AP in ER	Visible	**2**
Glenoid contour on AP	**Loss of contour**	**2**
Total		**10**

Score	Recurrence after arthroscopic Bankart	Advice
≤6	**~10%** acceptable (3 points → 5%)	Arthroscopic stabilisation
>6	**~70%**	**Open / Latarjet**

Latarjet trigger ISIS **>6** (Orthobullets summary, OB, TO) · ISIS **>4–6** (Orthobullets technique) – read the options

ISIS worked (OB) 18 ♀ competitive tennis (forced overhead), HS + glenoid contour loss = **9** (>70%) · competitive + contact + HS on ER AP, age 21, normal laxity (ER 80°, negative Gagey) = **5**

– – Active space – –

Anteroposterior translation (load & shift)

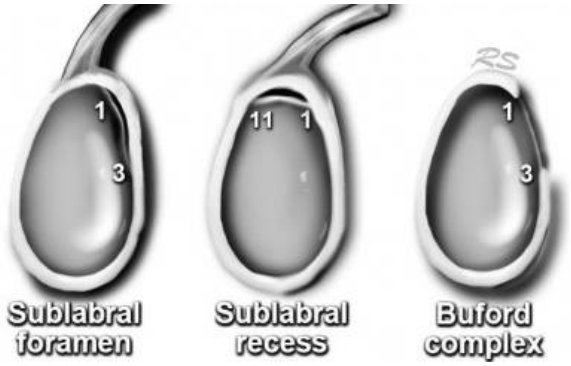
Grade	Translation
0	Normal
1+	**Up to glenoid rim**
2+	**Over rim, spontaneously reduces**
3+	Over rim, **does not reduce**

Sulcus sign

Grade	Acromiohumeral interval
1	**<1 cm**
2	**1–2 cm**
3	**>2 cm**

Normal variants mistaken for pathology (Buford complex)

Variant (TO)	Location	Feature	Prevalence
Sublabral foramen (hole)	**1–3 o'clock** (anterosuperior)	Unattached anterosuperior labrum	**11%** (TO) · ~12% (Orthobullets)
Sublabral recess	**12 o'clock** (biceps anchor)	Doesn't extend to 1–3 o'clock; **mimics SLAP**	–
Buford complex	**1–3 o'clock**	**Absent anterosuperior labrum + thick cord-like MGHL**	**~1.5%**



Sublabral foramen, sublabral recess, Buford complex

Buford (TO) normal variant; **not associated with recurrent dislocation**; do not repair (→ loss of ER)

– – Active space – –

5 Management

First-time traumatic anterior dislocation

- Reduce under sedation / IA lidocaine → **traction-countertraction (commonest)**
  - fails 5–10% → **blocks: biceps tendon, capsule, fracture fragments**
- Low-risk → **sling ≤1 wk + PT (cuff + periscapular)**
- Age >40 + persistent weakness → **MRI shoulder (cuff tear)**
- In-season athlete, no bone loss → rehab, RTP in-season → **surgery end of season**
- Young (<20–25) contact / military + Bankart on MRI → **early arthroscopic Bankart**

Recurrent instability – glenoid side (3D CT)

- GBL <13.5–15% + on-track → **arthroscopic Bankart (≥3–4 anchors)**
- GBL <15% + off-track → **Bankart + remplissage**
- Subcritical 13.5–25% → contact athlete / high ISIS → **Latarjet**
  - non-contact, off-track → **Bankart + remplissage**
- GBL >20–25% / inverted pear → **Latarjet (coracoid transfer)**
- Failed Latarjet / massive loss → **ICBG (Eden-Hybinette) or distal tibia allograft**

Recurrent instability – humeral side (Hill-Sachs)

- On-track / non-engaging (<20%) → **nothing extra**
- Off-track / engaging 20–40% (>25%) → **remplissage**
- >40% → **humeral head allograft / ICBG; hemiarthroplasty if >40–45% + arthritis**

Operative criteria

Procedure	Indications
Arthroscopic Bankart ± capsular plication	**First-time** dislocation + MRI Bankart in athlete **<25 y** · high-demand athletes · **recurrent (>1) dislocation** after nonop · **GBL <20–25%** · on-track (± remplissage if off-track)
Open Bankart ± capsular shift	GBL <20–25% · **revision after failed arthroscopic Bankart** without GBL >20% · **HAGL** · acute glenoid fracture · **hyperlax needing formal capsular shift** · contact athlete returning to sport
Latarjet / Bristow	**GBL >20–25%** (inverted pear) · **subcritical >13.5%** in high-risk · **ISIS >6** · contact athletes / athletes of consequence (climbers, big-wave surfers) · Europe uses aggressively
ICBG / distal tibia / distal clavicle graft	GBL >20–25% · **revision of failed Latarjet** · **less graft resorption** than Latarjet
Remplissage + Bankart	**Off-track / engaging** HS (**>25–40%**) with **GBL <20–25%** · subcritical GBL (arthroscopic-alone recurrence ~20%)
Hill-Sachs bone graft	Engaging HS **>40%**
Tendon transfer (conjoint / LHB)	Capsulolabral deficiency; with labral repair if **GBL <15%**

– – Active space – –

Bipolar matrix (TO)	Hill-Sachs on-track	Hill-Sachs off-track
GBL <15%	**Bankart**	**Bankart + remplissage**
GBL 15–25%	**Latarjet** (Bankart + remplissage if lower demand)	**Latarjet**
GBL >25%	Latarjet or ICBG	Latarjet or ICBG

TO algorithm: GBL	Hill-Sachs	Treatment (TO)
<10%	None / non-engaging	**Bankart**
<10%	Engaging, head loss <30%	**Remplissage**
<10%	Engaging, head loss >30%	**Latarjet**
10–25%	None / non-engaging	**ISIS <6 → Bankart · >6 → Latarjet**
10–25%	Engaging, head loss <20%	**Remplissage**
10–25%	Engaging, head loss >20%	**Latarjet**
25–40%	Any	**Latarjet** + ICBG to humeral head if head loss >30%
>40%	Any	**ICBG / allograft**

TO Hill-Sachs size rule non-engaging (**usually <20%**) → nothing · engaging **20–40%** → remplissage · engaging **>40%** → bone graft (Eden-Hybinette)

TO soft-tissue zone isolated Bankart if GBL **<15–20%** (stricter **13.5%** best-fit circle in high-demand athletes) + on-track

TO subcritical (15–20%) **contact athlete (rugby, kabaddi, wrestling) → Latarjet even at 15%** (Bankart failure ~30%) · non-contact → Bankart + remplissage

TO bone grafting triggers GBL >20–25% · **failed previous Bankart** · bipolar loss making an on-track HS off-track

Wrestler, 30 dislocations (incl. in sleep), Bankart + 40% HS (TO) → Latarjet
✗ Wrong options: Bankart · remplissage · McLaughlin

Recurrence risk – age, ISIS score & natural history

Age at first dislocation	Recurrence with nonop	Source
<20 y	**~90%** (80–90% teens; >90%)	Orthobullets, OB
14–18 y	**~90%**	TO
<25 y	**60–94%**; likely to re-dislocate	OB, TO
<30 y	↑ risk (significant)	OB
All comers	**33–67%**	OB

Most significant risk factor (TO, OB) **young age (<20)**; only consistent predictor regardless of immobilisation/exercise

– – Active space – –

Age cut-off conflict — **<20** (Orthobullets, ISIS, OB) · **<25** (TO, some OB) · **<30** (OB immobilisation question) · 21–30 (Kralinger) — read the options

Recurrence risk factors (TO)	
Modifiable	**Manual occupation**, overhead occupation, immobilisation after dislocation, **collision sport**, time to RTS, psychosocial (pain, **fear of reinjury**)
Non-modifiable	**GT fracture**, **Hill-Sachs**, **bony Bankart**, **axillary nerve palsy**, **age**, **gender**, **hypermobility**

Orthobullets risk factors for re-dislocation — **age <20 (highest)**, **male**, **contact sport**, **hyperlaxity**, **GBL >20–25%**, **GT fracture**

Hill-Sachs & recurrence — HS **not significantly associated** (TO) · HS = ISIS item / predictor after Bankart (OB) — read the options

Dominance — **non-dominant less likely to recur** (dominant limb used in high-risk activity; TO) · **no difference** dominant vs non-dominant (OB) — read the options

Not top risk factors (OB) — contralateral dislocation (seen in 25% of recurrent) · family history (34% recurrence risk) · patellar instability · dominance

Natural history (Hovelius, 25 y nonop) — 43% never re-dislocated · 15% re-dislocated then stabilised · 27% had surgery · immobilisation length no effect

Recurrent after PT, dislocates in sleep → discuss surgical stabilisation
✗ Wrong options: steroid · NSAIDs · change therapist · 6 wk immobilisation

First-time dislocation – nonoperative vs early arthroscopic Bankart

First-time dislocators — management controversial; **ASES: surgery for athletes 14–30 y at end of season** if **positive apprehension + bone loss**

Reduction techniques — **traction-countertraction (commonest)** · **Kocher**: arm at side in ER → forward flex → IR · **Hippocratic**: heel in axilla · **Stimson**: prone, weight hung from arm · scapular manipulation

Immobilisation — **no benefit >1 wk**; short (≤1 wk) = prolonged (≥3 wk) recurrence (41% vs 37%, Paterson) → **may stop sling at 1 wk**

ER immobilisation — early studies ↓ recurrence <40 y (labrum reduced to glenoid) → **subsequent studies refuted**; IR 24.7% vs ER 30.8% (Liavaag, NS)

Immobilisation conflict — no benefit / refuted (Orthobullets, OB) · **immobilisation decreases recurrence** (TO) — read the options

Bracing — **no evidence it ↓ recurrence**; limits ABD + ER in season

Return to play (nonop) — **faster than surgery**; 7–21 days (Orthobullets) · 2–3 wk (OB) · **3–4 wk** with full painless ROM + strength (OB) → can finish season

Nonop vs surgery recurrence — nonop **higher**: 88% vs 14% (Arciero) · 77% vs 11% (Bottoni); RTP without recurrence next season 40% vs 90% (Dickens)

Delaying surgery — further events → more labral damage, **more bone loss**, ↑ failure after Bankart; trend = **surgery after first dislocation**

Bony vs soft-tissue (first-time) — Latarjet **equal or lower** recurrence than Bankart

Water polo, 3rd dislocation, GBL <7%, HSI 19 < GT 22.07 → isolated arthroscopic Bankart
✗ Wrong options: remplissage → ↓ ER · Latarjet / DTA · humeral OATS

Arthroscopic vs open Bankart repair

	Arthroscopic	Open
Re-dislocation	**Similar** (RCTs) · slightly higher in some · 11% vs 8% (Harris) · 15.1% vs 7.7% (Rollick)	Similar
ROM	**Slightly better** (less pain, ER preserved)	More pain, ↓ ROM
Return to work/sport	Equal (RCT)	**Equal or slightly better** (some studies)
Arthritis	**No difference**	No difference
Approach	Arthroscopic, ≥3 (preferably 4) anchors	**Deltopectoral**; subscapularis **transverse split or tenotomy**; **inferior capsule shifted superiorly**

Arthroscopic advantages cosmesis, ↓ morbidity, ↓ ER loss, diagnostic confirmation

Contraindications to arthroscopic (Stein) significant GBL · attenuated capsulolabral tissue · engaging HS · **HAGL** (open = gold standard)

Isolated open Bankart + capsular shift **recurrent after arthroscopic repair, minimal GBL (6%), on-track, contact athlete** ✓

✗ Open Bankart distractors 20% GBL rugby (→ Latarjet) · 5% GBL off-track (→ Bankart ± remplissage) · 45% HS (→ humeral graft) · 18% GBL wrestler (→ bony)

Technique **fully mobilise labrum** before repair · **≥3 anchors** (<3 → failure) · "drive-through sign" pre-repair · knotless anchors

Too many anchors **postage-stamp fracture** through anchor holes

Recurrence after arthroscopic Bankart 13% (Ahmed) · 15.3% (Boileau) · 4% without bone defect vs **67%** with (Burkhart) · **89%** contact athletes with inverted pear

Bony Bankart – arthroscopic fixation

Small bony Bankart (<20%) **arthroscopic repair incorporating the fragment**; 95% RTS (Sugaya)
✗ Wrong options: ORIF · iliac crest · coracoid transfer

26-y accountant, 3rd dislocation, 10% bony defect, no prior surgery → arthroscopic bony Bankart repair
✗ Wrong options: ER immobilisation · remplissage · coracoid/iliac crest

ORIF large **acute** glenoid rim fractures only
✗ Wrong options: recurrent instability

Bony Bankart + deltoid weakness after reduction → bony Bankart stabilisation + observe axillary nerve (EMG changes in ~48%; surgery only if no recovery at **3–6 months**)

Fixation open: screws or suture (linear / bridge)

Imaging 3D CT > CT > MRI > X-ray for bone loss (MOON)

Unaddressed bony Bankart commonest cause of failed soft-tissue Bankart
✗ Wrong options: labral nonunion · HAGL · SLAP

Remplissage

Remplissage ("to fill") **infraspinatus + posterior capsule** sutured into Hill-Sachs (posterior capsulodesis + infraspinatus tenodesis) → **intra-articular HS becomes extra-articular** (TO) → can't engage rim in ER

– – Active space – –

Goal (TO)	✓ extra-articular HS ✗ Wrong options: more secure labrum · ↑ head diameter · ↑ glenoid track width
Effect	shortens HSI → **off-track → on-track**; ↑ rotation before engagement (**does not increase glenoid track**)
Indications	**off-track/engaging HS** + GBL **<20–25%** (<15% TO) · HS **>25%** (OB) / **25–40%** (Orthobullets) · chronic Bankart + engaging HS catching in **90/90 → arthroscopic Bankart + remplissage** ✗ Wrong options: remplissage alone · humeral graft alone
Technique	arthroscopic, percutaneous / posterolateral portal; **pass sutures before anterior repair, tie after**; knotted or knotless; **double-pulley**
Post-op	**avoid forward flexion + adduction** (tensions posterior capsule; abduction need not be avoided); brief ER immobiliser
Complications	**ER loss 5–15°**; recurrence **5% at 2 y**; IR loss <2°
Outcomes	vs **Latarjet (2 y): 1.4% vs 3.2% recurrence** despite greater bipolar loss · vs **Bankart alone: ↓ recurrence + revision** (MacDonald RCT) · Latarjet better ER

Moderate HS + labral tear after 2nd dislocation → arthroscopic labral repair + remplissage
✗ Wrong options: PT · labral repair alone · hemiarthroplasty – HS not >45% · thermal capsulorrhaphy

Connolly procedure	**infraspinatus + GT bone block** into HS · analogous **Neer-McLaughlin** = subscapularis + LT into **reverse HS** (posterior)
Humeral head allograft (OATS) (TO)	massive deep HS **>40%** where remplissage would severely limit ER
HS bone graft outcomes	small cohorts; early arthritis **16% at 4 y**; Constant 78.5–86.8; options: allograft, arthroplasty, rotational osteotomy

Latarjet/bone block – indications, mechanism & outcomes

Latarjet	transfer of **coracoid + conjoined tendon + CA ligament** to anteroinferior glenoid through **subscapularis split**; deltopectoral; open = arthroscopic outcomes (steep arthroscopic learning curve) ✗ Wrong options: detached

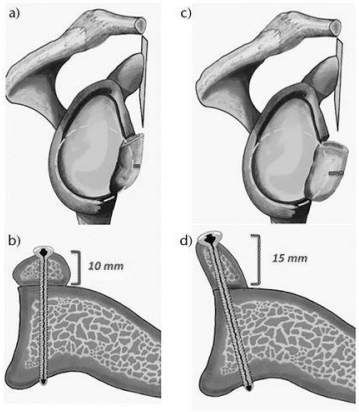
Triple effect	Mechanism
Bony	↑ glenoid surface/width, **lengthens glenoid track / articular arc**
Sling (TO)	**Conjoint tendon** (short head biceps + coracobrachialis) over lower subscapularis → **dynamic stability to anteroinferior capsule in ABD + ER**; brace across lower third of joint
Capsular	**CA ligament stump** repairs capsule

Triple effect question **all three (bony + sling + CA ligament)** ✓

Sling effect (TO) ✓ conjoint tendon dynamic stability
✗ Wrong options: CA ligament transfer = capsular · bone surface = bony · LHB tenodesis = DAS

– – Active space – –

	Traditional Latarjet	Congruent arc	Bristow
Coracoid orientation	**Undersurface on glenoid**; lateral surface = new track	**Rotated 90°**; **inferior surface** = track, articulates with head	**Distal ~1 cm tip**, orientation unchanged
Track gain	Less (~10 mm)	**Most: ~15 mm**	Less
Note	**Less fracture-prone** at screw insertion	Better for **severe bone loss**	Less common; **MC nerve tethering** (TO)



Traditional (a–b, ~10 mm) vs congruent arc (c–d, ~15 mm) Latarjet

Graft position flush, **no lateral overhang** (→ OA) · not **≥1 cm medial** or >1.5 cm migration (→ redislocation) · intra- vs extra-articular placement (intra-articular OA concern unproven)

Outcomes recurrence **0–8%** (<10%) · good–excellent **>90%** · RTS same level **75–90%** · RTS any level 80–100%

Return-to-sport conflict same level **75–90%** (OB) · 72.6% (Hurley) · 56–95% (Glogovac) · 49% contact athletes (Privitera) — read the options

– – Active space – –

Latarjet vs Bankart (bone loss) 4.9% vs 67% recurrence (Burkhart); Latarjet 2.7% (Rollick) with ↑ complications

Clinical scenario (OB)	Answer
Inverted pear + recurrent instability, failed arthroscopic Bankart	**Coracoid autograft (Latarjet)**
GBL **>20%**, rugby, 5 dislocations	**Open coracoid transfer**
GBL **30%** (best-fit circle), no prior surgery	**Open coracoid transfer** (✗ Wrong options: capsular shift = MDI · ORIF)
GBL **30%**, Beighton 2, HS	**Bony glenoid augmentation** (Latarjet / distal tibia)
GBL **33%**, lacrosse	Latarjet – **conjoint tendon sling**
Failed arthroscopic Bankart, **20% GBL**	**Latarjet – recurrence <10%**
23-y pro rugby, small HS, 15% GBL	**Latarjet** (✗ Wrong options: 33-y recreational with 10% → Bankart ± remplissage · linebacker large HS 6% → remplissage / humeral allograft · swimmer circumferential tear → repair)
≥10 dislocations, **24.4% GBL**, HS 15.4 mm	**Coracoid transfer**
Pole vaulter, off-track HS, significant GBL	**Coracoid transfer** (✗ Wrong options: remplissage + plication without labral repair · latissimus transfer with competent subscap)
Infantryman, **25% GBL**, off-track, failed repair	**Coracoid transfer** (✗ Wrong options: re-repair with ≥3 anchors · arthroplasty)
Chronic anterior dislocation + inverted pear	**Open reduction + glenoid augmentation** (if HS <20%)
Failed Bankart + **postage-stamp fracture**	**Bone block**
Failed arthroscopic Bankart, GBL >20–25% on CT	**Open capsulorrhaphy + bone block** (✗ Wrong options: arthrodesis)

Chronic dislocation (Sahajpal) glenoid defect **>20–30% → bony augmentation**; humeral defect engaging or **20–40% → address**; **>40% → hemiarthroplasty**

Postage-stamp fractures anterior rim fractures through anchor holes; male **≤25 y**, sport, **≥3 anchors**, **knot-tying anchors**, anchor-site osteolysis → **bone block**

Recurrence predictors after Bankart (OB) age <20, Hill-Sachs, glenoid lesion, hyperlaxity, off-track
✗ Wrong options: dominant side · SLAP · >5 preop dislocations · on-track

Distal tibia allograft (DTA) **true osteochondral graft**; **matches humeral head radius of curvature** + large articular surface (TO); functions mainly by **↑ glenoid track**; revision of failed Latarjet
✗ Wrong options: sling – differs from Latarjet

Iliac crest (Eden-Hybinette) (TO) autograft for large defects where **coracoid volume inadequate**; 89% healing at 1.4 y; screws or buttons; risks: hardware failure, subscapularis repair failure

Revising a failed Latarjet hardware removal + ICBG or DTA; electrocautery/nerve check to protect MC nerve (scarred, overlaps axillary nerve)

Dynamic anterior stabilisation (DAS, TO)	**LHB transferred to anterior glenoid through subscapularis split → sling effect**; for **chronic** anteroinferior instability with **poor labrum/GHL quality** and **limited bone loss**; also with SLAP, overhead athletes/throwers
DAS technique (TO)	LHB tenotomy, whipstitch · split subscap in middle · drill **3 o'clock**, 1.5–2.0 cm deep · tenodesis screw · add labral repair (bumper effect)
DAS contraindications (TO)	**significant bone defects** · LHB lesions · subscapularis lesion · prior LHB tenotomy/tenodesis; fills gap between soft-tissue and bone-block procedures ✗ Wrong options: alternative to Latarjet with bone loss
Conjoint tendon / LHB transfer (Orthobullets)	conjoint detached from coracoid tip (CA ligament preserved) or LHB; socket in anterior glenoid + suspensory fixation through subscap split; low recurrence with labral repair if **GBL <15%**; biceps possibly better; no ROM loss
Latarjet in epileptics	contraindicated with **uncontrolled seizures**

Seizure-related dislocation

Epileptic, >10 dislocations, uncontrolled seizures, GBL >20–25% → closed reduction + neurology referral
✗ Wrong options: no reduction attempt · reduction alone · stabilisation now · fusion

Rule	**exhaust medical (seizure) management before surgery**; high recurrence even with bony augmentation
Pathology	**large bony defects** (HS + bony Bankart) + degenerative change; muscle contraction + trauma; anterior = posterior frequency
Surgery once controlled	**bony procedure** (Latarjet / bone graft); soft tissue alone fails

Failed stabilisation with subscapularis deficiency – tendon transfer

Signs	**belly-press** weakness; post-open-repair subscap failure → **positive lift-off + excessive ER**
Irreparable	retraction + **atrophy** on MRI

29-y, failed arthroscopic + open Bankart, irreparable subscap → tendon transfer (latissimus dorsi or pectoralis major)
✗ Wrong options: PT · Latarjet · Magnuson-Stack · RTSA – age

Latissimus vs pec major	latissimus **better replicates subscap line of pull / IR moment arm**, better scores; pec major historically to lesser tuberosity; pec major not after failed subscap repair post-arthroplasty

Historical & thermal procedures (Putti-Platt, capsulorrhaphy)

Procedure	Technique	Problem
Putti-Platt	**Lateral advancement of subscapularis + medial advancement of capsule** (imbrication)	**Loss of ER**, **arthrosis** with ↓ ROM (TO)
Magnuson-Stack	Subscapularis advanced lateral to bicipital groove (± to GT)	↓ ER, posterior glenoid loading
Boyd-Sisk	Biceps transferred laterally + posteriorly	**High recurrence**
Bristow	Coracoid tip to anterior rim	**MC nerve injury** commoner (TO)

– – Active space – –

Putti-Platt (TO)	✓ arthrosis with compromised ROM · (✗ Wrong options: redislocation never reported · rotates HS by ↑ retroversion · MC nerve injury common (infrequent)) · Putti-Platt + Bristow → **IR contractures**
Post-capsulorrhaphy arthropathy	1970s–80s open repair → pain, stiffness, **lack of ER**, **posterior glenoid wear + retroversion**, IR contracture
Thermal capsulorrhaphy (TO)	used in **shoulder instability**; monopolar RF, introduced **1996** ✗ Wrong options: adhesive capsulitis · dural tear · stiff elbow
Thermal mechanism (TO)	**65–80°C** shortens capsule; collagen denaturation, ↓ mechanical properties, cell necrosis → fibroblast repair; stretches if loaded early; **grid/stripe** technique aids healing; ↓ capsular proprioceptors
Thermal outcomes (TO)	volume ↓ **26–36%** (vs plication **19%**); best in mild, unidirectional anterior instability **with** suture repair; **isolated → unacceptably high failure**
Thermal complications (TO)	**failure (commonest)**, **chondrolysis**, axillary nerve injury, capsular attenuation; now contraindicated

6 Complications

Recurrence

Commonest cause	**unrecognised glenoid bone loss** with soft-tissue-only repair (80% of failures) · missed off-track HS · **<3–4 anchors** · missed HAGL · missed pan-labral tear (military) · seizures
Anchor conflict	**<4 anchors** = risk (Orthobullets) · **<3 anchors** = risk (OB) — read the options
Age at surgery	not related to failure (one OB) · independent predictor (Ahmed, OB) — read the options

Bankart repair complications

Stiffness	**loss of ER** (overtightening; worse with remplissage) → treat **Z-lengthening of subscapularis**; overtightening → **post-capsulorrhaphy arthropathy** (older patients)
Subscapularis failure (open)	positive lift-off, excessive ER
Hardware	anchor pull-out; postage-stamp fracture
Chondrolysis	**thermal capsulorrhaphy** or **intra-articular pain pumps** (both now contraindicated)

Neurovascular injury – axillary nerve & artery (dislocation, capsular shift)

Most common nerve after dislocation

axillary (course around surgical neck, close to joint); typically **neurapraxia** (stretch)
✗ Wrong options: neurotmesis = penetrating/laceration · brachial plexus · MC

Axillary nerve injury rate	Source
Up to 5% (transient neurapraxia)	Orthobullets, OB (older cohort)
42% clinical + EMG (Visser)	OB
~48% EMG changes	OB
Any nerve injury **21–65%**; neuro deficit 13.5% (Robinson); 11.6% (Atef)	OB

Rate conflict

≤5% clinical (Orthobullets) vs **42–48%** EMG (OB) – read the options

Axillary nerve

C5–6 (upper trunk, posterior cord) → deltoid + teres minor; weakness esp. **30–100° abduction**; ↓ sensation **lateral proximal arm**

Timeline

neurodiagnostics after first **2–3 wk** (14–21 d if no recovery) · persistent weakness + sensory loss at **12 wk → EMG/NCS** · surgery (neurolysis, graft, transfer) if **no recovery 3–6 mo**
✗ Wrong options: observe more · repeat MRI · explore · triceps branch transfer before EDS

High-grade injury risk (Avis)

proximal humerus fracture, multiple nerves, neuropathic pain, **delayed reduction**

Surgical anatomy

axillary nerve anterior to subscapularis → drapes **inferolateral border → quadrilateral space**; **12 mm** (Orthobullets) · **12–16 mm** (OB) · 1–1.5 cm (Kang) from inferior glenoid / infraglenoid tubercle

Arthroscopic risk zone

6 o'clock, inferior to infraglenoid tubercle (also 4:30) during capsular shift/plication; plication **≤1 cm from rim safe**, >1 cm risk; anteroinferior portals riskiest
✗ Wrong options: spinoglenoid/suprascapular notch = suprascapular n. · 9 o'clock posterior · 12 o'clock biceps

Protective position

abduction + external rotation ↑ nerve–capsule distance
✗ Wrong options: adduction/IR – beach chair adduction + IR = risk

Open capsular shift

injury during **dissection of subscapularis off inferior capsule** / retractor on inferomedial subscap belly → **can't abduct, elbow flexion + supination intact = axillary (C5–6)**
✗ Wrong options: MC C5–7 · lower subscapular nerve

Axillary artery

1–2 cm from inferior glenoid · closest at **≥90° ER: 4–5 mm** (4.1 mm Stone) · ≥2 cm from coracoid · **greatest risk at inferior glenoid with arm in ER**
✗ Wrong options: thoracoacromial – at coracoid · MC – medial · radial – triangular interval · ulnar

– – Active space – –

Latarjet complications – musculocutaneous nerve, graft resorption & overhang arthritis

Complication rate up to **25%** (Orthobullets) · 15–30% (Gupta) · 30% (Griesser); higher than Bankart

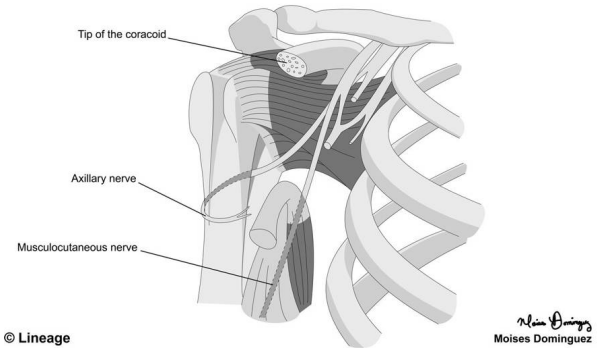
Complication	Fact
MC nerve injury	**Most common nerve** (Orthobullets, most OB); **instrumentation / aggressive medial retraction around conjoint tendon**, coracoid harvest; pierces coracobrachialis ~**5.6 cm** distal to coracoid tip
Axillary nerve injury	**During graft fixation**; 12 mm from infraglenoid tubercle; 2nd commonest
Graft lysis/resorption	Up to **90%** some resorption **within 6 months**; **superior** graft; late pain + apprehension at ~9 mo
Lateral overhang	**GH OA** (up to **38%**); 20% new arthritis (Mizuno)
Stiffness	**ER loss** (✗ Wrong options: IR · elbow flexion)
Nonunion, hardware (screw pull-out)	
Axillary artery	4–5 mm at 90° ER
Subscapularis rupture	**Low** – split, not detached

MC nerve **terminal branch of lateral cord**, **C5–7**; → coracobrachialis, biceps, brachialis; injury → **lateral forearm numbness (LACN)** + **↓ elbow flexion/ supination**

Latarjet nerve conflict **MC = most common** (Orthobullets, most OB) · **axillary = most common** (one OB; neuromonitoring: all clinical deficits axillary) – read the options

Resorption conflict common, **up to 90%** (Orthobullets, OB) · "uncommon overall", mostly first 6 mo (another OB) – read the options

Nerve injury management mostly **traction/contusion neurapraxia → observe 3–6 wk**; delayed EMG if persistent; all resolved by 6 mo (neuromonitoring)



Axillary and musculocutaneous nerves relative to coracoid tip

After Latarjet MC + axillary nerves move medially, **MC overlaps axillary** → revision risk; axillary artery position unchanged

✗ Nerve distractors lateral winging (spinal accessory) · medial winging (long thoracic) · IR weakness (subscapular) · thumb extension weakness (PIN/radial) · median, ulnar, radial (paths far)

Sling post-Latarjet protects **graft healing**
✗ Wrong options: subscap detached

Older patients – rotator cuff tear after dislocation

Rule young → recurrence · **old → rotator cuff tear**

Age	Cuff tear after dislocation
>40 y	**30%** (Orthobullets, OB) · "30% at 30 y" (one OB)
>60 y	**80%**

Tendons "**SIT**": **supraspinatus** (commonest), infraspinatus, teres minor; recurrent instability in elderly → **subscapularis** tears
✗ Wrong options: subscap – bear hug

Cuff tear vs axillary palsy in elderly cuff tear **far more common**; **deltoid firing + full passive ROM + intact sensation** → cuff tear → **suprascapular nerve** muscles (SS/IS)
✗ Wrong options: dorsal scapular · lateral pectoral · MC · axillary

Persistent pain/weakness ≥2 wk after dislocation (>40 y) → MRI shoulder
✗ Wrong options: EMG · C-spine MRI · steroid injection – harms later repair · PT first · CT glenoid

Test for commonest pathology **drop arm test** (supraspinatus)
✗ Wrong options: O'Brien – SLAP · apprehension – young/bone loss · bear hug – subscap · Spurling – C-spine

MR arthrogram **most sensitive (and specific) for full- + partial-thickness cuff tears**; MRI ≈ US

Treatment surgical repair → better pain + satisfaction; full-thickness tears don't heal nonop; **GT fracture** with dislocation >50 y (↑ recurrence)

Terrible triad of the shoulder

Shoulder terrible triad (TO) **anterior dislocation + rotator cuff tear + brachial plexus (mostly axillary) nerve injury**; cuff usually **complete** tear

EXCEPT (TO) **proximal humeral fracture**

Late complications – chondrolysis, osteonecrosis & dislocation arthropathy

Osteonecrosis (skeletally immature) rare; after fracture-dislocation or stabilisation → **observation** (revascularises)
✗ Wrong options: arthroplasty · grafting · capsular release

Chondrolysis progressive pain + stiffness ~1 y after Bankart with **heat (RF) probe** → joint space loss
✗ Wrong options: frozen shoulder · subscap failure

Dislocation arthropathy **age at first dislocation** (>25 y, Hovelius) + high-energy primary dislocation → severe OA; "**goat's beard**" inferior osteophyte; ↓ ER
✗ Wrong options: current age · dominance · diabetes · sex

After Bristow-Latarjet mild arthropathy 35%, severe 14%; first dislocation >23 y = risk

– – Active space – –

X vs Y

	Bankart	Perthes	ALPSA	GLAD	HAGL
Labrum	Off rim	Off rim, may look **in place**	**Medially displaced**, healed on neck	Partial tear	Intact on glenoid
Periosteum	Hinge intact	**Stripped, intact**	Sleeve intact medially	–	–
Cartilage	–	–	–	**Defect**	–
Failure point	Glenoid	Glenoid	Glenoid neck	Chondrolabral	**Humerus**
Best repair	Arthroscopic	Arthroscopic	Arthroscopic (↑ failure)	Arthroscopic	**Open**

	Young dislocator (<25)	Older dislocator (>40)
Main lesion	**Bankart** + Hill-Sachs	**Rotator cuff tear** (30% >40, 80% >60)
Main problem	**Recurrence**	Weakness, pseudoparalysis
Next test	MRI/MRA, 3D CT for bone	**MRI shoulder**
Nerve	Axillary neurapraxia	Axillary / plexus (less common than cuff)

	Remplissage + Bankart	Latarjet
Fixes	**Humeral side** (HSI ↓)	**Glenoid side** (GT ↑)
GBL	<15–25%	>20–25% or subcritical contact
Approach	Arthroscopic	Open (or arthroscopic)
ROM	**ER loss 5–15°**	ER loss; better ER than remplissage
Key risk	Stiffness	**MC nerve**, resorption, overhang OA

	Latarjet	Distal tibia allograft	Iliac crest (Eden-Hybinette)
Graft	Coracoid + conjoint + CA ligament	Osteochondral allograft	Tricortical autograft
Mechanism	**Triple effect**	**↑ glenoid track only**	Bony
Curvature	–	**Matches humeral head**	–
Use	Primary bony option	Massive loss, failed Latarjet	Large defect, coracoid inadequate, failed Latarjet

	MC nerve (Latarjet)	Axillary nerve
Cord / roots	Lateral cord, **C5–7**	Posterior cord, **C5–6**
When injured	Retraction / instrumentation **around conjoint tendon**	**Graft fixation**; capsular shift at 6 o'clock; dislocation
Deficit	**Lateral forearm numbness**, ↓ elbow flexion + **supination**	↓ abduction (deltoid), lateral arm numbness

	Anterior instability	Posterior instability
Position	**ABD + ER**	**ADD + IR**
Labral lesion	Bankart, ALPSA, Perthes, GLAD	**Kim lesion**, posterior labral tear
Humeral lesion	Hill-Sachs (posterosuperior)	**Reverse Hill-Sachs**, LT fracture
Ligament avulsion	HAGL	RHAGL
Humeral fill	Remplissage / Connolly (infraspinatus + GT)	**McLaughlin** (subscap + LT)

How it's asked

"Perthes lesion is?"	→ **associated with recurrent dislocation** (TO)
"Anteroinferior labral detachment with medially stripped intact periosteum?"	→ **Perthes** (TO) ✗ Wrong options: Bankart · GLAD · ALPSA
"Main anterior stabiliser of shoulder?"	→ **IGHL** (TO) ✗ Wrong options: MGHL · SGHL · anterior capsule
"Main stabiliser against anterior dislocation above 45° abduction?"	→ **IGHL** (TO) ✗ Wrong options: subscapularis
"MRI 2 months after dislocation – labrum displaced medially with periosteal sleeve?"	→ **ALPSA** (TO) ✗ Wrong options: Bankart · SLAP · HAGL
"Most common finding at surgery for traumatic anterior instability?"	→ **anteroinferior labral tear** ✗ Wrong options: Hill-Sachs
"17-y footballer, MRI after injury – arthroscopic finding?"	→ **Bankart lesion** ✗ Wrong options: cuff tear · SLAP · glenoid fracture · HAGL
"Labral tear + Hill-Sachs on MRA – position at injury?"	→ **abduction + external rotation**
"25-y basketballer, first dislocation – MRI shows?"	→ **anteroinferior labral tear** ✗ Wrong options: supraspinatus tear = >40 y
"Typical separation on MRI after anteroinferior dislocation?"	→ **anteroinferior labrum from bony glenoid** ✗ Wrong options: from cartilage · capsule from humerus
"Bony HAGL – when is this ligament most stressed?"	→ **anterior force at 90° abduction + ER** ✗ Wrong options: 0° inferior = SGHL · 45° = MGHL
"Structure torn, name, position of greatest contribution?"	→ **anteroinferior labrum, Bankart, ER at 90° abduction**
"Young dislocator with Bankart – most likely associated injury?"	→ **Hill-Sachs** ✗ Wrong options: cuff tear · reverse HS · coracoid · GT fracture

– – Active space – –

"Recurrent anterior + posterior instability, positive Jerk, Kim, apprehension – lesion needing both repairs?"	→ **270° pan-labral tear**
"Recurrent instability on MRA, surgeon recommends open treatment?"	→ **HAGL** ✗ Wrong options: GLAD · SLAP · ALPSA · PASTA
"Recurrent subluxation, coronal MRI – ligament injured?"	→ **inferior glenohumeral (HAGL)**
"Re-dislocation 1 month after arthroscopic Bankart – missed pathology?"	→ **HAGL**
"36-y diving catch, post-reduction MRI lesion?"	→ **HAGL** (older, violent trauma)
"Goal of remplissage?"	→ **make intra-articular Hill-Sachs extra-articular** (TO) ✗ Wrong options: ↑ glenoid track
"Rugby, 12% GBL, narrow off-track Hill-Sachs?"	→ **arthroscopic Bankart + remplissage** (TO) ✗ Wrong options: Eden-Hybinette · Latarjet
"Kabaddi, D 24 mm, width 20 mm, HSI 18 mm?"	→ **off-track; Latarjet** (TO) ✗ Wrong options: Bankart + remplissage – 16.6% subcritical, elite contact
"Associated intra-articular lesion with Bankart in rugby player?"	→ **Hill-Sachs** ✗ Wrong options: Buford – normal variant
"Best X-ray for Hill-Sachs?"	→ **Stryker notch** ✗ Wrong options: Grashey · Y · Zanca · axillary
"Ex-pro footballer, catching in 90/90, chronic Bankart + large HS?"	→ **arthroscopic Bankart + remplissage** ✗ Wrong options: remplissage alone
"Arrow on posterior head beside bare area after dislocation?"	→ **Hill-Sachs defect**
"Compressive lesion after anteroinferior dislocation – structure?"	→ **posterosuperior humeral head**
"12.5% GBL, non-engaging Hill-Sachs?"	→ **Bankart repair** ✗ Wrong options: remplissage · capsular shift alone · Latarjet · distal tibia
"Bankart + remplissage – post-op restriction?"	→ **avoid forward flexion + adduction**
"Glenoid track formula?"	→ **GT = 0.84(D) − ABL** (0.83D − d in most sources)
"Most increases glenoid track?"	→ **congruent arc Latarjet** (~15 mm) ✗ Wrong options: traditional · Bristow · remplissage
"True about glenoid track?"	→ **more medial HS → more likely off-track**

"Medium-large HS + labral tear after 2nd dislocation – best treatment?"	→ **arthroscopic labral repair + remplissage** ✗ Wrong options: thermal capsulorrhaphy
"Recurrence after arthroscopic repair in lineman – cause?"	→ **anteroinferior glenoid bone defect** (missed bony Bankart)
"Highest recurrence risk after arthroscopic Bankart?"	→ **18-y competitive tennis, HS + glenoid contour loss (ISIS 9)**
"Intra-op finding most predictive of recurrence?"	→ **35% anterior glenoid deficiency**
"Failed Bankart, glenoid rim loss – mechanism deficient?"	→ **concavity-compression**
"Water polo, 2 dislocations – greatest recurrence risk after labral repair?"	→ **inverted-pear glenoid** ✗ Wrong options: dominant arm · overhead athlete · age <25 · biceps extension
"Isolated labral repair would fail for which image?"	→ **anterior glenoid fracture / bone loss**
"Factor most linked to failure after arthroscopic Bankart, >10 dislocations?"	→ **glenoid bone loss** ✗ Wrong options: age at first dislocation · age at surgery
"Why 3D CT for bone loss?"	→ **highest inter-rater reliability** ✗ Wrong options: lowest predictive error · MRI underestimates
"Most significant risk factor for recurrence after first dislocation?"	→ **young age** (TO) ✗ Wrong options: bony Bankart · manual occupation · GT fracture
"Highest risk for recurrent instability?"	→ **age <20 at dislocation** ✗ Wrong options: contralateral · dominance · family history · patellar instability
"3 recurrences despite PT, last in sleep?"	→ **discuss surgical stabilisation**
"Competitive contact footballer, HS on ER AP, normal laxity – ISIS?"	→ **5 points**
"48-y first dislocation, 1 wk in sling, full strength?"	→ **stop immobilisation now – no added recurrence risk** ✗ Wrong options: ER/IR/neutral 2 more weeks
"Nonop first-time dislocator – true statement?"	→ **short immobilisation = prolonged** ✗ Wrong options: bracing helps · low risk · future events don't matter
"17-y footballer, first dislocation, small non-engaging HS?"	→ **nonop allows return this season** ✗ Wrong options: surgery faster · ER brace better
"Water polo, GBL <7%, HSI 19 < GT 22?"	→ **isolated arthroscopic Bankart**

– – Active space – –

"Advantage of arthroscopic over open stabilisation?"	→ **slightly better ROM** ✗ Wrong options: lower recurrence · better return to work/sport
"Arthroscopic vs open Bankart?"	→ **similar dislocation rates** ✗ Wrong options: more OA with open
"Isolated open Bankart most appropriate?"	→ **recurrent after arthroscopic repair, 6% GBL, on-track**
"Bony Bankart + deltoid weakness 3 days after reduction?"	→ **bony Bankart stabilisation** (observe axillary nerve)
"Accountant, 3rd dislocation, 10% glenoid defect on 3D CT?"	→ **arthroscopic bony Bankart repair**
"Acute dislocation, bony fragment 10% of glenoid?"	→ **arthroscopic repair incorporating fragment** ✗ Wrong options: ORIF · iliac crest · coracoid transfer
"25% glenoid loss, GT 18 mm, HSI 21.3 mm?"	→ **Latarjet** (TO)
"Wrestler, 30 dislocations, 40% posterolateral head loss?"	→ **Latarjet** (TO) ✗ Wrong options: McLaughlin
"Incorrect about DAS?"	→ **suitable arthroscopic alternative to Latarjet with bone loss** (TO)
"Latarjet sling effect?"	→ **conjoint tendon dynamic stability to anteroinferior capsule in ABER** (TO)
"Latarjet stabilises by?"	→ **bony support + conjoint sling + CA ligament capsular repair (1, 2 and 5)**
"Ideal patient for open reduction + glenoid augmentation?"	→ **chronic anterior dislocation + inverted-pear glenoid** ✗ Wrong options: large engaging HS · acute bony Bankart · cuff arthropathy
"Failed Bankart, inverted pear on CT?"	→ **coracoid autograft** ✗ Wrong options: Connolly · remplissage · humeral augmentation
"Rugby, 5 dislocations, >20% GBL?"	→ **open coracoid transfer**
"Failed arthroscopic Bankart, glenoid defect on CT?"	→ **open capsulorrhaphy + bone block**
"30% GBL, no prior surgery?"	→ **open coracoid transfer**
"Return to previous level after Latarjet?"	→ **75–90%**
"33% GBL, bone transfer – true?"	→ **conjoint tendon sling** ✗ Wrong options: subscap detached · axillary = retraction nerve · medial graft → OA · traditional > congruent arc
"30% GBL, Beighton 2, HS?"	→ **bony glenoid augmentation**
"Failed Bankart, 20% GBL – true?"	→ **recurrence <10% after Latarjet** ✗ Wrong options: revision arthroscopic first · Latarjet = DTA mechanism

"Latarjet most appropriate in?"	→ **23-y pro rugby, small HS, 15% GBL**
"19-y footballer, 24% GBL, HS 15.4 mm?"	→ **coracoid transfer**
"Pole vaulter, 5th dislocation, off-track, significant GBL?"	→ **coracoid transfer** ✗ Wrong options: latissimus transfer
"Infantryman, failed repair, 25% GBL, off-track?"	→ **coracoid transfer**
"Risk factor justifying bone block after failed Bankart?"	→ **postage-stamp fractures** ✗ Wrong options: SLAP · dominance · >5 dislocations · on-track
"Lateral forearm numbness after coracoid transfer – label?"	→ **musculocutaneous nerve** ✗ Wrong options: long thoracic · suprascapular · axillary · radial
"Nerve most at risk in Latarjet for 40% bony Bankart?"	→ **musculocutaneous**
"Lateral forearm numbness + supination weakness post-op – nerve and procedure?"	→ **musculocutaneous & Latarjet**
"Post-Latarjet neurapraxia finding?"	→ **lateral forearm numbness**
"True of Latarjet complications?"	→ **lateral graft overhang → GH OA** ✗ Wrong options: axillary = harvest nerve · IR stiffness · resorption uncommon
"Latarjet complication true?"	→ **lateral cord branch injury from aggressive retraction** ✗ Wrong options: medial graft → OA · resorption >2 y · subscap detached
"9 months post-Latarjet, new apprehension, CT superior graft loss?"	→ **coracoid graft resorption** ✗ Wrong options: nonunion · lateral placement
"Nerve most likely injured with anterior dislocation?"	→ **axillary**
"Most common neurological problem with simple dislocation?"	→ **neurapraxic axillary nerve injury**
"62-y, can't abduct 2 wk after dislocation, deltoid firing, sensation intact – nerve to injured structure?"	→ **suprascapular (label B)** – cuff tear
"Inferior capsular shift – arm position protecting the nerve?"	→ **abduction + external rotation**
"Neurapraxia after arthroscopic capsular shift in adduction + IR – site?"	→ **6 o'clock, inferior to infraglenoid tubercle**
"Structure at greatest risk at inferior glenoid with arm in ER during Latarjet?"	→ **axillary artery**
"Persistent abduction weakness + lateral arm numbness 3 months after dislocation, intact cuff?"	→ **electrodiagnostic studies**
"Can't abduct after open capsular shift, elbow flexion intact – nerve and roots?"	→ **axillary, C5–6**
"61-y, pain + weakness 6 wk after dislocation – next step?"	→ **MRI to evaluate rotator cuff** ✗ Wrong options: labrum · EMG · CT

– – Active space – –

"62-y, weakness 4 wk after dislocation?"	→ **MRI shoulder** ✗ Wrong options: injection · C-spine MRI · EMG · PT
"Provocative test for commonest pathology, 61-y after dislocation?"	→ **drop arm test**
"64-y, weakness 6 wk after dislocation, neck radiation?"	→ **shoulder MRI** ✗ Wrong options: EMG · C-spine MRI
"MR arthrogram after dislocation in 55-y – true?"	→ **most sensitive for partial-thickness cuff tears**
"Osteonecrosis after dislocation in skeletally immature – treatment?"	→ **observation**
"Pain + stiffness 1 y after Bankart with heat probe?"	→ **chondrolysis**
"Severe OA at 55 after single dislocation at 30?"	→ **age at time of dislocation**
"Epileptic, >10 dislocations, poor compliance, GBL?"	→ **closed reduction + neurology referral**
"Failed arthroscopic + open Bankart, belly-press weakness, atrophied retracted subscap?"	→ **tendon transfer** ✗ Wrong options: Latarjet · Magnuson-Stack · RTSA
"Arm position for humerus coronal + glenoid axial on MRI?"	→ **abduction + external rotation (ABER)**
"Terrible triad of shoulder EXCEPT"	→ **proximal humeral fracture** (TO)
"True of Putti-Platt?"	→ **arthrosis with compromised ROM** (TO) ✗ Wrong options: MC injury common · no redislocation
"Thermal capsulorrhaphy is used in?"	→ **shoulder instability** (TO)
"True about Buford complex EXCEPT"	→ **associated with recurrent dislocation** (TO)

– – Active space – –

TRICEPS RUPTURE

– Shoulder & Elbow · 59 / 60 –

1 Basics

Epidemiology

Incidence rare · **0.8%** of tendon ruptures

Demographics ♂ **2:1** · age **30–50** · competitive **weightlifting** (bench press), bodybuilding, **football**

Risk factors & diagnosis (flake sign, MRI)

Risk factor	Yes / No
Anabolic steroids	✓ (established link)
Local steroid injection	✓
Fluoroquinolones	✓
Renal osteodystrophy / renal disease, **hyperparathyroidism**	✓
OI, RA, type I DM, Marfan	✓
Chronic olecranon bursitis, prior triceps surgery	✓
Creatine supplements	**not associated**

Anatomy – lateral expansion & anconeus

Head	Origin
Long	**Infraglenoid tuberosity**
Lateral	Posterior humerus between teres minor insertion and **above spiral groove**, lateral border, lateral IM septum
Medial	Posterior humerus **distal to spiral groove**, medial humerus, medial IM septum

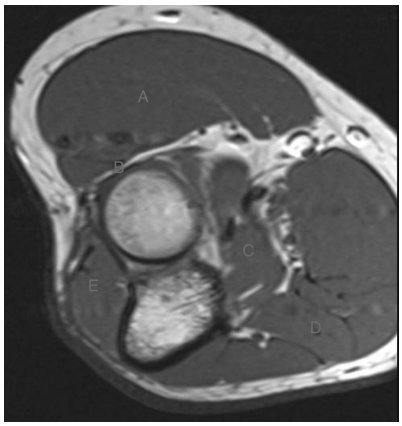
Triceps pennate · **only muscle of posterior arm compartment** · **radial nerve (C6–C8)** · primary elbow extensor

Insertion footprint **1.1 cm distal to olecranon tip** · width **1.9–4.2 cm** · triceps tendon proper (all 3 heads → olecranon) + **lateral triceps expansion** (posterior ulnar crest · **ECU fascia + anconeus deep fascia** · antebrachial fascia)

Anconeus **secondary elbow extensor**; terminal **15° extension** + supination · **compensates for torn triceps** → resisted extension may persist

Anconeus vs distractors brachioradialis, brachialis = flexion · LCL = varus stabiliser · pronator teres = pronation

– – Active space – –



Axial MRI: E = anconeus, A = brachioradialis, B = LCL, C = brachialis, D = pronator teres

2 Mechanism

Mechanism	**forceful eccentric contraction** (bench press, muscle-ups) · **FOOSH** · direct blow
Site	most often **osseous insertion** (medial or lateral head) · less often muscle belly / MTJ

3 Presentation

History	**painful pop**
Exam	posterior elbow pain, swelling, ecchymosis · **palpable defect** (not always)
Resisted extension	inability to extend against resistance **not always present**: **intact lateral expansion** or **anconeus** compensation · complete tear → can't extend **against gravity** (OB)
Modified Thompson squeeze	**prone**, elbow at table edge, forearm hanging → squeeze triceps → **no extension** = complete rupture of **triceps proper + lateral expansion**
Missed	**10/23** missed initially (van Riet) · **nearly 50%** misdiagnosed (Yeh)

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral)	**"flake sign"** on lateral = **pathognomonic** of triceps avulsion (olecranon tip fleck)



Flake sign: avulsed fleck at olecranon tip

MRI location + severity; confirms, classifies, guides treatment · **partial** = small fluid-filled defect in distal tendon · **complete** = **large fluid-filled gap**, paratricipital oedema · use when too painful to examine or partial suspected



MRI: distal triceps tear

✗ Flake-sign distractors distal biceps, brachialis, LUCL, MUCL tears · elbow arthritis · olecranon bursitis · anconeus avulsion (= **lateral epicondyle** fracture)

Descriptive classification

Feature	Options
Degree	Complete · partial · intact
Location	Muscle belly · MTJ · tendinous insertion · avulsion
Lateral expansion	Intact · torn

Formal system none

- - Active space - -

5 Management

Triceps rupture

- Partial, extends vs gravity / low-demand, poor health → **splint 30° flexion 4 wk**
- Acute complete → **primary repair (best within 3 wk)**
- Partial >50% + significant weakness → **primary repair**
- Chronic / delayed → **reconstruction ± tendon graft**

Operative criteria

Indication	Treatment
Acute complete tear (active, medically fit)	**Primary repair**
Partial >50% + significant weakness	**Primary repair**
Partial, extends against gravity	Splint
Low demand, poor health	Splint
Late (>3 wk)	Reconstruction, tendon graft

Approach posterolateral

Tear site	Technique
MTJ	**V-Y advancement** ± plantaris augmentation
Tendinous insertion	**Bunnell / Krackow** non-absorbable whipstitch → **transosseous tunnels**, periosteal flap, or **suture anchors**
Bony avulsion	Tension band · screw + washer

Tunnels vs anchors **no difference** in biomechanical strength or function · **transosseous: ↑ re-rupture + complications**

Post-op 30–45° flexion **2 wk** · active ROM at **4 wk** · no weightlifting **4–6 mo**

Outcome early diagnosis + repair → excellent; anatomic footprint repair

6 Complications

General elbow stiffness/weakness · **ulnar nerve injury** · repair failure

Repair-specific olecranon bursitis · flexion contracture · re-rupture

X vs Y

	Partial tear	Complete tear
Extension vs gravity	Preserved	Lost (unless anconeus / lateral expansion compensate)
Thompson squeeze	Extension present	**No extension**
MRI	**Small fluid-filled defect**	**Large fluid-filled gap**, paratricipital oedema
Treatment	Splint 30° × 4 wk · repair if **>50% + weak**	**Primary repair**

How it's asked

"Triceps rupture associated with all EXCEPT?"	→ **creatine supplements** ✗ Wrong options: anabolic steroids · local steroid · renal osteodystrophy · fluoroquinolones
"40 y weightlifter, pop on bench press, lateral X-ray fleck – MRI shows?"	→ **triceps tendon tear** ✗ Wrong options: distal biceps · brachialis · LUCL · MUCL
"44 y, FOOSH rollerblading, posterior pain, olecranon flake?"	→ **triceps rupture** ✗ Wrong options: arthritis · bursitis · LCL · anconeus avulsion
"Muscle-ups, positive Thompson squeeze — structure assisting resisted extension?"	→ **anconeus (E)**

– – Active space – –

VALGUS EXTENSION OVERLOAD (PITCHER'S ELBOW)

– Shoulder & Elbow · 60 / 60 –

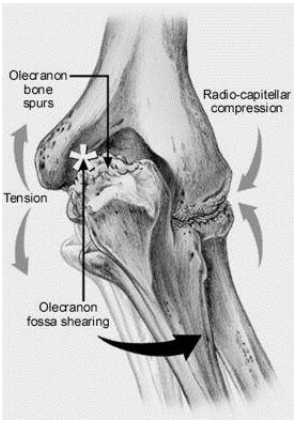
1 Basics

Epidemiology

Definition	**posteromedial elbow pain** from repetitive microtrauma in **throwing athletes**
Prevalence	up to **65%** of elite baseball players with symptomatic elbows
Who	**dominant arm** of throwers (pitchers, quarterbacks) · also swimmers, volleyball, gymnasts, racquet sports, golfers
Location	**posteromedial olecranon** (✓ label D) · posterior trochlea · olecranon fossa
Association	**cubital tunnel syndrome ~25%**

2 Mechanism

Forces of pitching	**medial tension (MCL)** · **lateral radiocapitellar compression** · **posterior extension overload** · **shear on medial olecranon tip + fossa**
Phases (OB)	**valgus stress in acceleration** (esp. with **UCL insufficiency**) + **extension impaction in deceleration**
Medial laxity	any cause → **compensatory ↑ stress posteromedially** → osteophytes / loose bodies
Sequence	repetitive impaction of olecranon into fossa → **chondromalacia/cartilage injury** → **osteophytes (posteromedial olecranon + humerus)** → **loose bodies** (fragmentation) · **capitellar osteochondral lesions** · **UCL attenuation** with repetitive strain

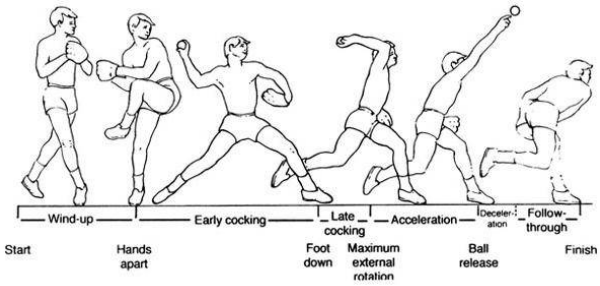


Valgus extension overload: medial tension, radiocapitellar compression, olecranon fossa shearing

3 Presentation

Diagnosis – posteromedial olecranon osteophytes, examination & throwing phase

Pain	**posteromedial**, at **full extension**; **deceleration / follow-through** (✓; after ball release) · rarely acceleration ✗ Wrong options: wind-up, early/late cocking
Function	**loss of terminal extension** (osteophytes block olecranon seating) · ↓ accuracy · pain after ~25 pitches
Exam	**tender posteromedial olecranon** · **flexion contracture** (e.g. 10°) · **crepitus** (loose bodies, synovitis) · **pain with forced extension**
Provocative tests	**arm bar test** (sustained extension) · **bounce test** (repeated terminal extension)
UCL assessment	check competency (failure → valgus instability); moving valgus stress may be painful **without instability** in VEO



Phases of pitching: VEO pain in deceleration/follow-through

4 Diagnosis & Classification

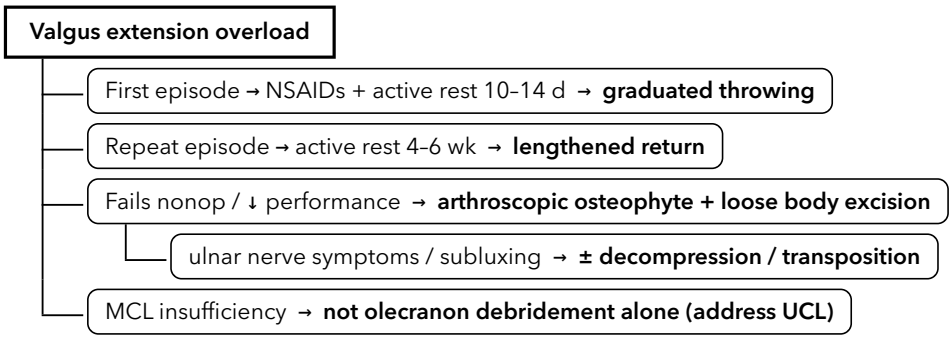
Key radiology findings

Modality	Finding
X-ray AP, lateral, oblique	**Posteromedial olecranon osteophyte** (AP key; medial olecranon) · loose bodies (capitellar fragmentation) · **MCL calcification** · **humeral hypertrophy** (↓ room in fossa)
CT	**Best osseous detail**; 3D for planning
MRI	Not needed for diagnosis · **associated MCL tear** (partial/complete) · capitellar OCD

Diagnosis	**clinical** (posteromedial pain ↑ with extension) + imaging of posteromedial osteophyte

– – Active space – –

5 Management



Nonoperative

First line NSAIDs, **stop throwing**, correct mechanics, ± steroid injection (**paucity of data**) · also **mid-season** or **end of career**

PT **flexor-pronator strengthening** (✓ next step) · shoulder, forearm + kinetic chain

Patient	Active rest	Then
First-time symptomatic	**10–14 days**	Graduated throwing
Repeat symptomatic	**4–6 weeks**	Lengthened gradual return

Operative criteria

Factor	Rule
Indication	**Persistent symptoms after failed (extended) nonop** · diminished performance
Relative contraindication	**MCL insufficiency** (for olecranon debridement alone)
Ulnar nerve release / transposition	**Significant ulnar symptoms** · nerve **subluxes** over medial epicondyle (transposition for symptomatic unstable nerve)

Arthroscopic osteophyte resection

Preferred arthroscopic (✓) or open **cheilectomy**; generally **excellent results** if no over-resection

Position lateral decubitus, prone or supine (**supine + articulating holder** if combined with **MCL reconstruction**)

Steps mark ulnar nerve course · diagnostic arthroscopy, remove loose bodies · **posterolateral viewing portal** · **direct posterior portal** (through triceps) · clear fibrous tissue (RF + shaver) · free fractured osteophyte (switch portals → remove via posterolateral) · **contour to native margin only** · **arthroscopic valgus stress test** (medial gapping = incompetent MCL)

Also chondromalacia debridement/drilling · lateral meniscoid lesion / posterolateral plica

Excessive olecranon resection – unmasking UCL insufficiency

Rule remove **osteophyte only, not native olecranon** → else **loss of bony restraint → ↑ MCL strain + valgus angulation**

Threshold	**≥9 mm** resection → **UCL rupture** under valgus (cadaveric; Orthobullets, OB) · **>3 mm** / past native margin → ↑ MCL stress + valgus instability (Orthobullets complications) – read the options
Result	**valgus elbow instability** (✓) · **MCL damaged** ✗ Wrong options: LUCL · RCL · annular · arcuate
Presentation	after debridement: **medial pain (late cocking)**, laxity + pain on valgus stress, **positive milking manoeuvre**, **negative arm bar**, new UCL tear on MRI
✗ Failure-cause distractors	accelerated return-to-throw · lateral instability (rare) · posterior bundle (dominant UCL structure = **anterior band of anterior bundle**) · anatomic variant · ulnar neuritis · capitellar OCD · olecranon stress fracture

6 Complications

Complication	Prevention
Valgus instability	Avoid over-resection (see thresholds)
Ulnar nerve injury	Mark nerve before medial portals · **"nick and spread"** · limit/remove **shaver suction** in posteromedial gutter

X vs Y

	VEO	UCL injury	Medial epicondylitis
Pain	**Posteromedial**	**Medial**	Medial, **5–10 mm distal to epicondyle**
Phase	**Deceleration / follow-through**	**Acceleration** (late cocking)	
Test	**Arm bar, bounce**; crepitus	**Valgus stress at 20–30°**, **milking manoeuvre**, moving valgus stress	**Resisted flexion/ pronation**
X-ray	**Posteromedial osteophyte**	No fracture, **no posteromedial osteophyte**	No VEO osteophytes
Contracture	**Yes**		Unlikely

– – Active space – –

Differential	Key feature
Acute UCL rupture (pop, ↓ velocity, MRA full-thickness)	**UCL reconstruction** in competitive thrower (✗ Wrong options: isolated ulnar transposition · osteophyte excision without osteophyte)
Partial UCL tear, non-thrower / low-risk sport	Nonop: activity modification + PT
MCL sprain (recreational thrower)	Medial pain **only when throwing**, ↓ speed + distance, no instability, normal X-ray
PLRI	Lateral; **pivot shift, chair rise**; LUCL; dislocation or lateral epicondylitis debridement
Capitellar OCD	Lateral pain, swelling, **catching/locking**
Ulnar nerve subluxation	Palpable in flexion; neuritis to hand
Pronator syndrome	**Volar forearm** pain on resisted pronation; not throwing-related
Medial epicondyle avulsion	**Skeletally immature**
Olecranon stress fracture	**Posterior** pain + swelling; lateral X-ray/MRI
✗ Wrong option: varus extension overload	Distractor — **valgus** EO (not varus) mimics MCL injury

How it's asked

"Typical location of osteophytes in VEO?"	→ **posteromedial olecranon (D)**
"Pitcher, posteromedial pain after ball release, normal milking — diagnosis + next step?"	→ **VEO; PT with flexor-pronator strengthening** ✗ Wrong options: UCL reconstruction · PRP · LUCL reconstruction · extension splint
"Pitcher, 10° flexion contracture, posteromedial tenderness, no instability?"	→ **valgus extension overload** ✗ Wrong options: medial epicondylitis · olecranon stress fracture · trochlear OCD · epicondyle avulsion
"Exam findings in VEO?"	→ **posteromedial pain, crepitus, flexion contracture** ✗ Wrong options: pivot shift · distal epicondyle tenderness · lateral locking
"Throwing phase most painful in VEO?"	→ **follow-through / deceleration** ✗ Wrong options: acceleration · late cocking
"Excessive normal olecranon resection → on return to play?"	→ **valgus elbow instability**
"Valgus instability 9 mo after osteophyte removal — structure?"	→ **medial collateral ligament** ✗ Wrong options: LUCL · annular
"Medial pain in late cocking + valgus laxity after olecranon debridement?"	→ **excessive olecranon resection**

"New UCL tear after VEO arthroscopy + accelerated rehab — cause?"	→ **excessive olecranon resection** ✗ Wrong options: accelerated protocol · posterior bundle
"Pitcher, pop, full-thickness UCL tear on MRA, wants to return?"	→ **UCL reconstruction** ✗ Wrong options: ulnar transposition · osteophyte excision
"Recreational thrower, medial pain only throwing, ↓ distance, normal X-ray?"	→ **MCL sprain** ✗ Wrong options: ulnar neuritis · pronator syndrome · medial epicondylitis · varus extension overload

– – Active space – –